National
Qualifications
2026

X813/76/12

Chemistry
Paper 1 — Multiple choice

TUESDAY, 12 MAY
9:00 AM – 9:40 AM

Total marks — 25

Attempt ALL questions.

You may use a calculator.

Instructions for the completion of Paper 1 are given on *page 02* of your answer booklet X813/76/02.

Record your answers on the answer grid on *page 03* of your answer booklet.

You may refer to the Chemistry Data Booklet for Higher and Advanced Higher.

Space for rough work is provided at the end of this booklet.

You must leave your answer booklet on your desk; if you do not, you could lose all the marks for this paper.

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Total marks — 25

Attempt ALL questions

1. The bonding and structure in fullerenes is

A ionic lattice
B metallic lattice
C covalent network
D covalent molecular.

2. The two hydrogen atoms in a molecule of hydrogen are held together by a

A hydrogen bond
B polar covalent bond
C pure covalent bond
D van der Waals force.

3. Which of the following statements is true?

A The oxygen atom is larger than the oxide ion.
B The magnesium atom is larger than the magnesium ion.
C The sulfide ion is smaller than the sulfur atom.
D The calcium ion is larger than the calcium atom.

4. What is the enthalpy change, in $kJ\,mol^{-1}$, for the following reaction?

$$Sc(g) \rightarrow Sc^{2+}(g) + 2e^{-}$$

A 633
B 1235
C 1868
D 2470

5. Which of the following is a redox reaction?

A $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
B $NaOH + HCl \rightarrow NaCl + H_2O$
C $NiO + 2HCl \rightarrow NiCl_2 + H_2O$
D $CuCO_3 + 2HCl \rightarrow CuCl_2 + H_2O + CO_2$

6. Iodide ions can be oxidised to iodine.

$$2I^-(aq) \rightarrow I_2(s) + 2e^-$$

Which of the following ions could be used to oxidise iodide ions to iodine?

A $SO_3^{2-}(aq)$

B $SO_4^{2-}(aq)$

C $Cr^{3+}(aq)$

D $Cr_2O_7^{2-}(aq)$

7. Oil molecules are more likely to react with oxygen in the air than fat molecules.
During the reaction the oil molecules

A are reduced

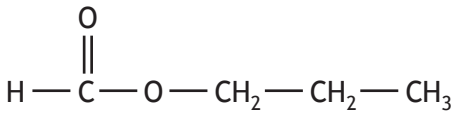
B form rancid products

C are hydrolysed

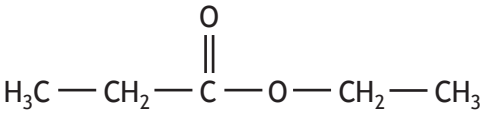
D become unsaturated.

8. A student was given samples of ethanol, propan-1-ol, methanoic acid and ethanoic acid.
Which of the following esters **could not** be made from these chemicals?

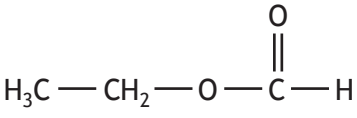
A



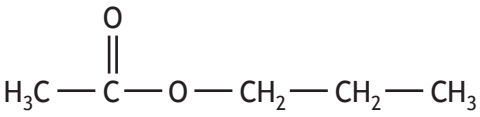
B



C



D



[Turn over

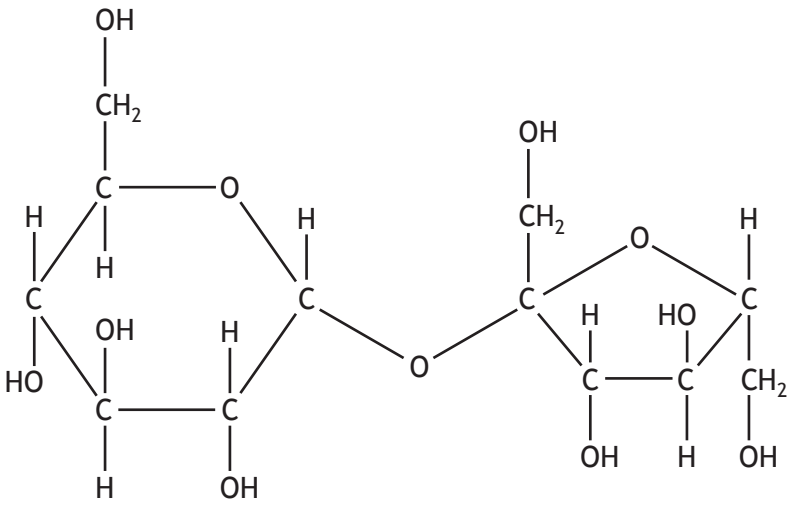
9. A compound with the formula CH_2O_2 must be

A an ester
B an alcohol
C an aldehyde
D a carboxylic acid.

10. Which process is used to convert ethanoic acid to ethanal?

A Addition
B Condensation
C Hydrolysis
D Reduction

11. Olestra is a calorie free fat made by reacting carboxylic acids with sucrose.
The structure of a sucrose molecule can be represented as shown.



How many carboxylic acid molecules can react with one molecule of sucrose?

A 3
B 5
C 8
D 11

12. Fats have higher melting points than oils because

A fats are more saturated
B oils are more saturated
C fats are less closely packed
D oils are more closely packed.

13. Propanal, C_3H_6O, can be converted to propanoic acid by heating with acidified dichromate solution.

$$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$$

$$C_3H_6O + H_2O \rightarrow C_3H_6O_2 + 2H^+ + 2e^-$$

How many moles of $Cr_2O_7^{2-}$ would react with one mole of propanal?

A $\frac{1}{3}$

B $\frac{2}{3}$

C 1

D 3

14. Which of the following is an isomer of 2,2-dimethylpentan-1-ol?

A $CH_3CH_2CH_2C(CH_3)_2CH_2OH$

B $(CH_3)_3CCH(CH_3)CH_2OH$

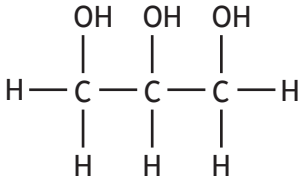
C $CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_2OH$

D $(CH_3)_2CHC(CH_3)_2CH_2CH_2OH$

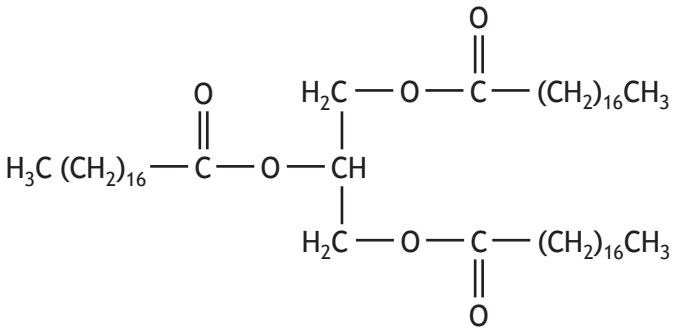
[Turn over

15. Which of the following could be made by reacting edible oils with glycerol?

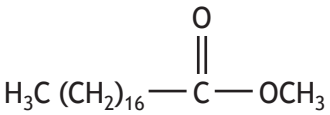
A



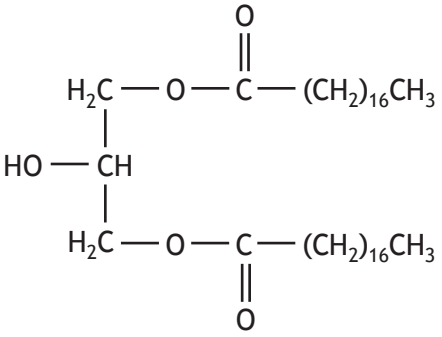
B



C



D



16. A two-step reaction is shown.

Step 1

$$S_2O_8^{2-}(aq) + 2I^-(aq) + 2Fe^{2+}(aq) \rightarrow 2SO_4^{2-}(aq) + 2I^-(aq) + 2Fe^{3+}(aq)$$

Step 2

$$2SO_4^{2-}(aq) + 2I^-(aq) + 2Fe^{3+}(aq) \rightarrow 2SO_4^{2-}(aq) + I_2(aq) + 2Fe^{2+}(aq)$$

The ion that is the catalyst in the overall reaction is

A $S_2O_8^{2-}(aq)$

B $I^-(aq)$

C $Fe^{2+}(aq)$

D $SO_4^{2-}(aq)$

17. A student added 0.1 mol of zinc to a solution containing 0.05 mol of silver(I) nitrate.

$$Zn(s) + 2AgNO_3(aq) \rightarrow Zn(NO_3)_2(aq) + 2Ag(s)$$

Which of the following statements about the experiment is correct?

A 0.05 mol of zinc reacts.

B 0.05 mol of silver is produced.

C Silver nitrate is in excess.

D All of the zinc reacts.

[Turn over

18. The following equilibrium exists as an orange solution.

$$\underset{\text{(yellow)}}{Fe^{3+}(aq)} + CNS^{-}(aq) \rightleftharpoons \underset{\text{(red)}}{[FeCNS]^{2+}(aq)}$$

Addition of **X** causes the reaction mixture to become red.

X could be

A $FeCl_3$

B $[FeCNS]Cl_2$

C NaCl

D $FeCl_2$

19. Which of the following is **not** a factor that affects the rate of a reaction?

A Activation energy

B Kinetic energies of reactant molecules

C Concentration of reactants

D Enthalpy change of reaction

20. Chromatography separates substances by making use of their differences in size and

A boiling point

B polarity

C reactivity

D colour.

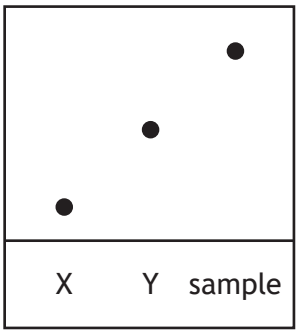
21. A chemist is attempting to make a fragrance compound by the following chemical reaction.

compound **X** + compound **Y** → fragrance compound

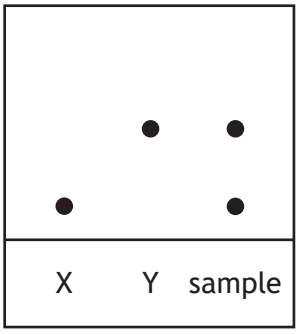
Compounds **X** and **Y** are mixed in a flask. After some time, a sample is removed from the flask and compared with pure samples of compounds **X** and **Y** using chromatography.

Which of the following chromatograms shows that a pure sample of the fragrance compound has been produced?

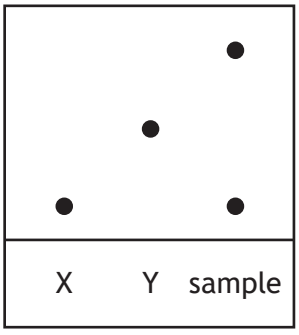
A



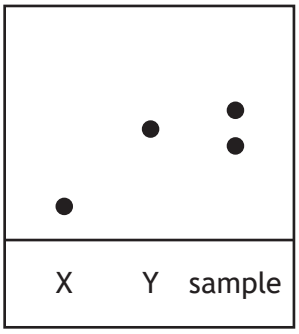
B



C



D



[Turn over

22. The vitamin C content of a carton of orange juice was determined by four students. Each student carried out the experiment three times.

	Concentration of vitamin C (mg/100 cm^3)		
	Experiment 1	**Experiment 2**	**Experiment 3**
Student A	28.0	24.0	23.0
Student B	21.4	21.6	21.8
Student C	21.9	22.0	21.9
Student D	22.0	21.5	21.9

The most reproducible results were obtained by

A student A

B student B

C student C

D student D.

23. Which of the following is the most appropriate apparatus to accurately prepare a 0.10 mol l^{-1} solution from a 1.0 mol l^{-1} solution?

A 1 cm^3 dropper and a 10 cm^3 measuring cylinder.

B 10 cm^3 volumetric flask and a 100 cm^3 conical flask.

C 25 cm^3 pipette and a 250 cm^3 volumetric flask.

D 50 cm^3 burette and a 500 cm^3 measuring cylinder.

24. The octane number is a measure of how easily hydrocarbons ignite.

	Hydrocarbon	Octane number
1	3-methylpentane	74.5
2	pentane	61.7
3	butane	93.6
4	2-methylpentane	73.4
5	hexane	24.8
6	methylcyclopentane	91.3

A student predicted that as the chain length of a hydrocarbon increases, the octane number decreases.

Which set of three hydrocarbons should have their octane numbers compared to provide the best test of this prediction?

A 1, 4, 6

B 1, 2, 4

C 3, 4, 5

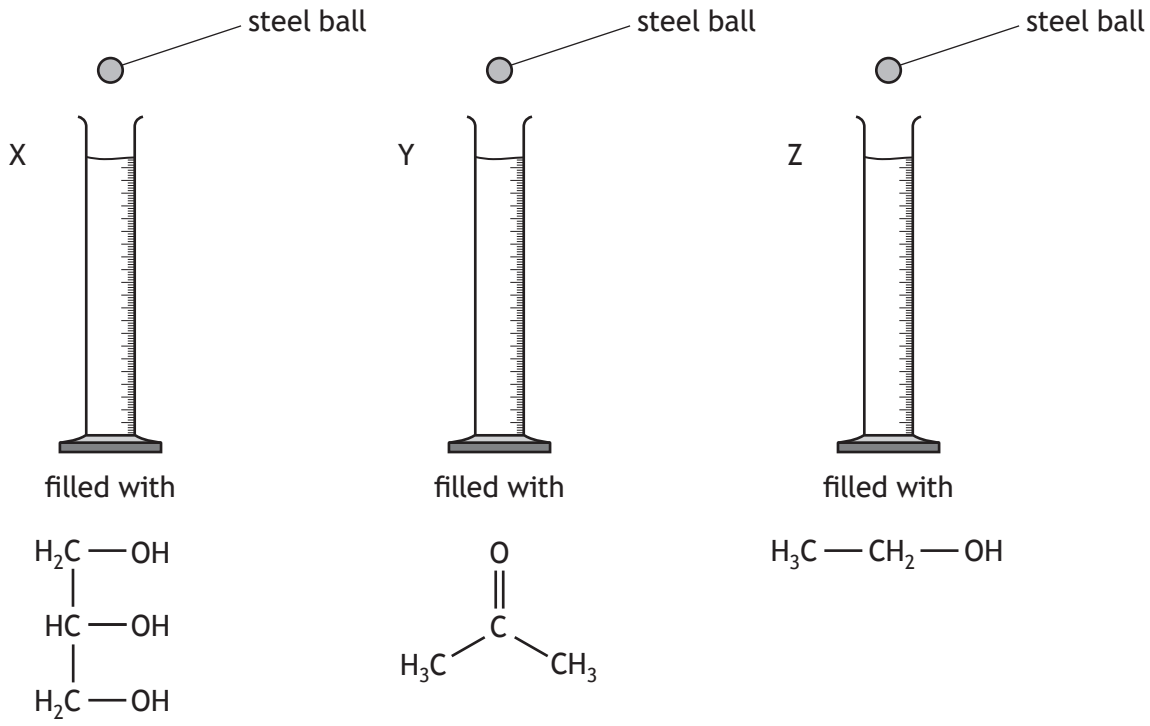
D 2, 3, 5

[Turn over

25. The type of forces between molecules can affect the viscosity of a liquid.

A steel ball will fall at different speeds through different liquids.

The more viscous a liquid the more slowly the ball falls through the liquid.



Which of the following shows the correct order for the speed of the steel balls falling through the liquids from the fastest to the slowest?

A Y, X then Z

B X, Z then Y

C Y, Z then X

D Z, Y then X

[END OF QUESTION PAPER]

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National Qualifications 2026

Mark

X813/76/01

Chemistry Paper 2

TUESDAY, 12 MAY

10:10 AM – 12:30 PM

X8137601

Fill in these boxes and read what is printed below.

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth

Day | Month | Year

Scottish candidate number

Total marks — 95

Attempt ALL questions.

You may use a calculator.

You may refer to the Chemistry Data Booklet for Higher and Advanced Higher.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers and rough work is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting. Any rough work must be written in this booklet. Score through your rough work when you have written your final copy.

Use **blue** or **black** ink.

Do not remove any exam materials. You must leave this booklet on your desk; if you do not, you could lose all the marks for this paper.

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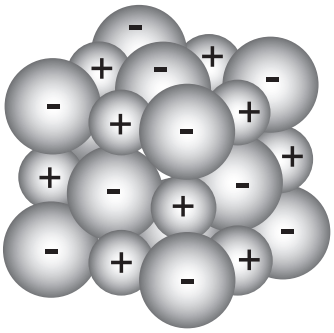
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Total marks — 95

Attempt ALL questions

1. (a) Sodium chloride is an ionic compound and has the structure shown.



(i) State the term used to describe the structure of sodium chloride. **1**

(ii) An ionic bond is a type of electrostatic attraction.

(A) Describe what is meant by the term electrostatic attraction. **1**

(B) Explain why sodium and chlorine form an ionic compound using electronegativity values from the data booklet. **1**

1. (continued)

(b) The boiling point of pure water is 100 °C. Dissolving an ionic solute in water raises the boiling point.

The increase in boiling point depends on the total number of solute particles.

The increase in boiling point (ΔT_b), in °C, can be calculated using the formula shown.

$$\Delta T_b = 0.51 \times c \times i$$

c = the concentration of the solution in mol l^{-1}

i = the number of particles in solution when one formula unit of solute dissolves

The values of i for some ionic solutes are shown in the table.

Ionic solute	Value of i
KCl	2
$CaBr_2$	3
$(NH_4)_3PO_4$	4

(i) State the value of **i** for lithium sulfate. **1**

(ii) Calculate the boiling point, in °C, of an aqueous solution of 0.35 mol l^{-1} of ammonium phosphate. **2**

[Turn over

2. Water, H_2O, and methane, CH_4, are covalent molecules.

The table shows the van der Waals forces present and boiling points for these molecules.

Molecule	Van der Waals forces present	Boiling point (°C)
CH_4	London dispersion forces	−162
H_2O	London dispersion forces Hydrogen bonds	100

(a) Explain **fully** how London dispersion forces are formed. **2**

(b) The boiling points of these two molecules, H_2O and CH_4, can be used to compare the strength of van der Waals forces.

Explain why H_2O and CH_4 are suitable for this comparison. **1**

(c) Explain **fully** why water is a polar molecule. **2**

3. Trends in melting points, ionisation energy, covalent radius and electronegativity can be related to the position of an element in the periodic table.

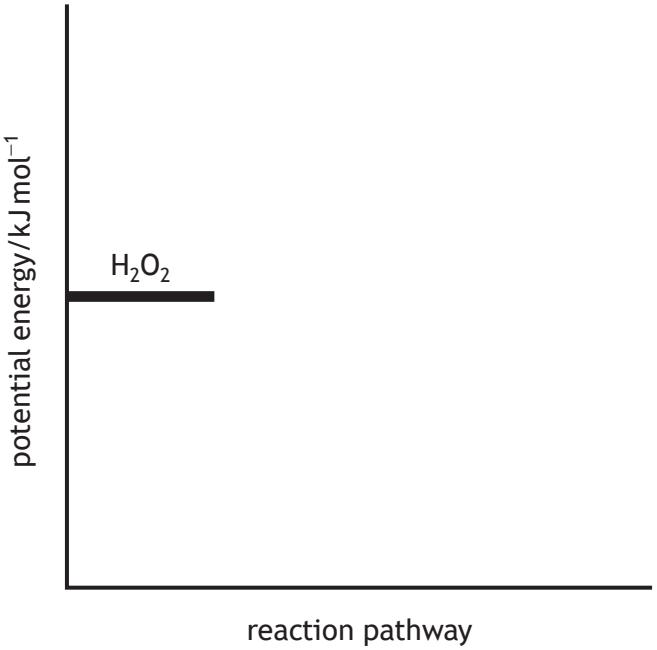
Using your knowledge of chemistry, comment on how the position of an element in the periodic table can be used to predict its physical and chemical properties. **3**

[Turn over

4. Hydrogen peroxide, H_2O_2, decomposes very slowly to produce water and oxygen.

$$2H_2O_2(aq) \rightarrow 2H_2O(\ell) + O_2(g)$$

(a) The activation energy, E_a, for this reaction is 75 kJ mol^{-1} and the enthalpy change, ΔH, is -98 kJ mol^{-1}.



(i) Use the activation energy and enthalpy change given above to complete the potential energy diagram for the reaction. **1**

(An additional diagram, if required, can be found on *page 32*.)

(ii) Calculate the activation energy, in kJ mol^{-1}, for the **reverse** reaction. **1**

4. (continued)

(b) Manganese dioxide catalyses the decomposition of hydrogen peroxide solution.

When a catalyst is used in a reaction more successful collisions occur and the rate of reaction increases.

Explain **fully** the way in which catalysts increase the number of successful collisions. **2**

(c) The balanced equation for the decomposition of hydrogen peroxide, H_2O_2, is shown.

$$2H_2O_2(aq) \rightarrow 2H_2O(\ell) + O_2(g)$$

$$GFM = 34 \text{ g}$$

Calculate the mass, in g, of hydrogen peroxide required to produce 60 cm^3 of oxygen gas. **2**

Take the volume of one mole of oxygen gas to be 24 litres.

[Turn over

4. (continued)

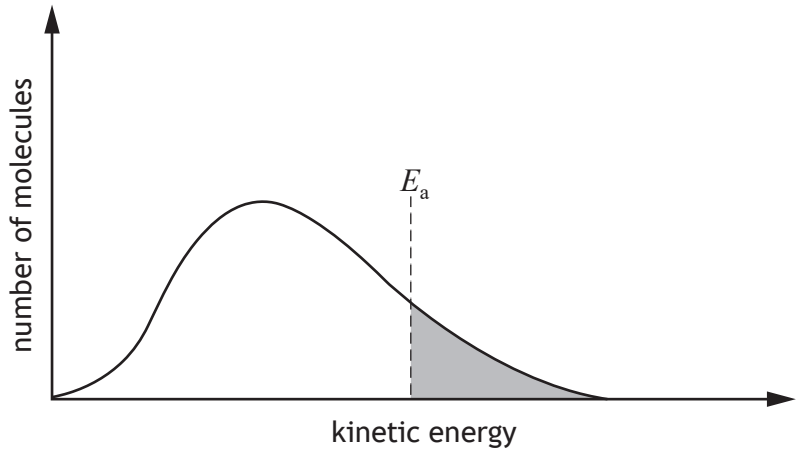
(d) Hydrogen peroxide, H_2O_2, should be stored in the dark as sunlight can provide sufficient energy to break the weak oxygen to oxygen single bond in the molecule creating free radicals.

(i) State the meaning of the term free radicals. **1**

(ii) Identify the free radical formed from hydrogen peroxide. **1**

(e) Increasing the temperature will increase the rate of a reaction.

The diagram shows the distribution of kinetic energies of molecules in a sample of a gas.



(i) Suggest what is represented by the shaded area in the diagram. **1**

(ii) Add a second curve to the diagram to show the distribution of kinetic energies at a higher temperature. **1**

(An additional diagram, if required, can be found on *page 32*.)

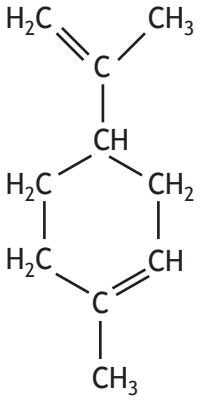
5. The peel of citrus fruits contains essential oils.

(a) Essential oils in citrus fruits have aromas.

State another property of an essential oil. **1**

(b) Limonene is an essential oil found in citrus peel.

A structure for limonene is shown.



(i) When limonene is heated, it can undergo decomposition to form isoprene units.

Draw a structural formula for isoprene. **1**

(ii) State the number of isoprene units that would be produced when one molecule of limonene decomposes. **1**

[Turn over

5. (continued)

(c) Citrus peel has been investigated as a possible source of biofuel for cars.

(i) The first step in producing the fuel is to extract essential oils, including limonene, from citrus peel.

One method uses hexane to dissolve the essential oils.

Explain why hexane is a suitable solvent for essential oils. **1**

(ii) An experiment was carried out to determine the enthalpy of combustion for limonene.

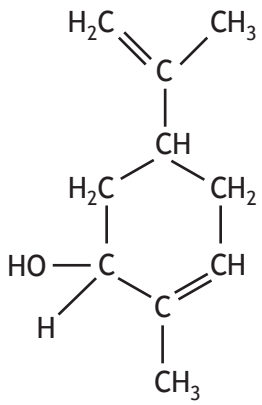
3.5 g of limonene ($GFM = 136$ g) was burned and the temperature rise in 500 cm^3 of water was 56 °C.

Calculate the enthalpy of combustion, in $kJ\,mol^{-1}$, of limonene using these experimental results. **3**

5. (continued)

(d) Limonene can undergo oxidation to produce carveol.

A structure for carveol is shown.



(i) Explain why carveol can be classified as a secondary alcohol. **1**

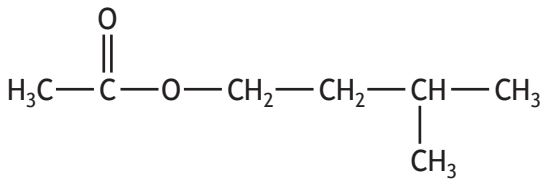
(ii) Carveol can be further oxidised to produce a ketone.

Draw a structural formula for the ketone produced when carveol is oxidised. **1**

[Turn over

6. Bananas contain a range of compounds.

(a) An ester responsible for the flavour of bananas is shown.



(i) Name the alcohol used to make this ester. **1**

(ii) This alcohol was passed over heated black copper(II) oxide.
Suggest what would be seen at the end of the reaction. **1**

6. (continued)

(b) Bananas produce the gas ethene, C_2H_4, ($GFM = 28$ g) which causes them to ripen.

To speed up the ripening of bananas, they are stored in a gas mixture containing 4% ethene and 96% nitrogen.

A gas mixture has a volume of 27 000 litres.

Calculate the mass of ethene required, in g, for this gas mixture. **2**

Take the volume of one mole of ethene gas to be 23 litres.

(c) When bananas are cut, the colour changes from yellow to brown.

Enzymes cause the breakdown of chemicals in the banana, resulting in this colour change.

(i) Name the class of compounds to which enzymes belong. **1**

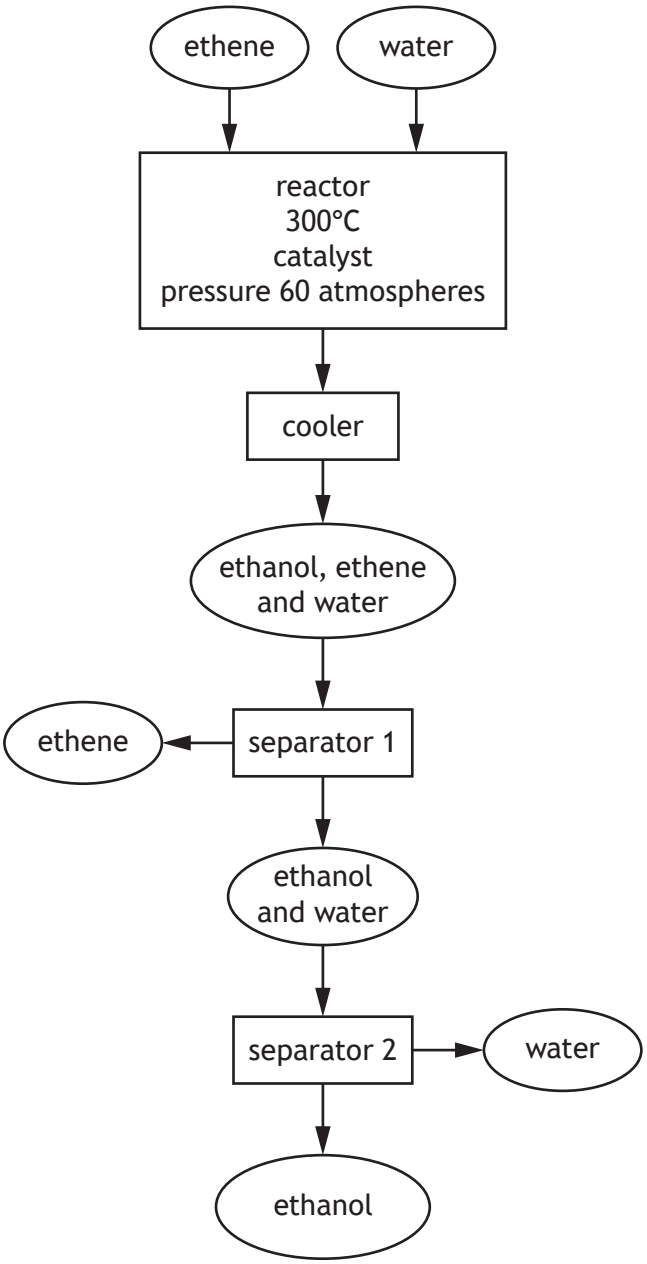
(ii) To stop cut bananas turning brown, they can be treated with lemon juice, denaturing the enzymes.

Explain **fully** what is meant by denaturing. **2**

[Turn over

7. Ethanol is an important industrial chemical.

(a) The flow diagram of an industrial process to produce ethanol is shown.



(i) **On the flow diagram**, draw an arrow to suggest one way to maximise profit in this industrial process. **1**

(ii) Name the process used in separator 2 to separate ethanol from water. **1**

7. (a) (continued)

MARKS DO NOT WRITE IN THIS MARGIN

(iii) In the reactor ethene reacts with water to form ethanol.

$$C_2H_4(g) + H_2O(g) \rightleftharpoons C_2H_5OH(g)$$

(A) The reaction taking place is called hydration.

State another name for this type of reaction. **1**

(B) Increasing the pressure in the reactor improves the yield of ethanol.

Explain **fully** why increasing pressure would shift the equilibrium to the right producing more ethanol. **2**

(C) Using mean bond enthalpies from the data booklet, calculate the enthalpy change, in kJ mol^{-1}, for this reaction. **2**

$$C_2H_4(g) + H_2O(g) \rightleftharpoons C_2H_5OH(g)$$

(D) The atom economy for this reaction is 100%.

Explain what this means. **1**

[Turn over

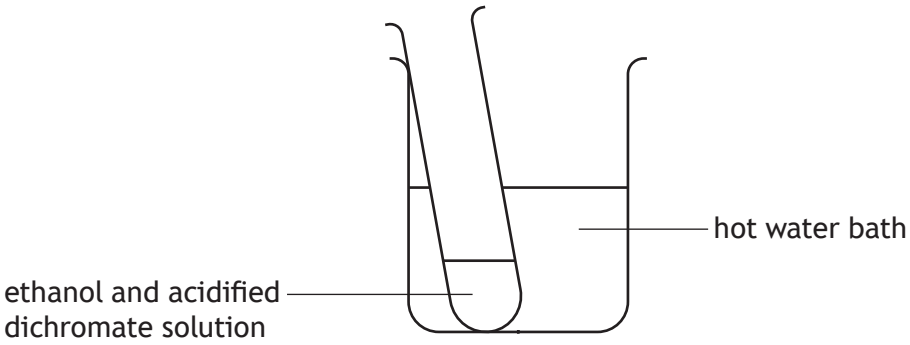
(b) Ethanol can be oxidised to compound **X**.

$$\underset{\text{ethanol}}{CH_3CH_2OH} \xrightarrow{\text{Step 1}} \underset{\text{ethanal}}{CH_3CHO} \xrightarrow{\text{Step 2}} \underset{\text{compound X}}{CH_3COOH}$$

(i) State why steps 1 and 2 can be described as oxidation. **1**

(ii) Name compound **X**. **1**

(iii) In the laboratory, acidified dichromate solution can be used to convert ethanol to compound **X**.



(A) Suggest why a water bath is a suitable method of heating the reaction mixture. **1**

(B) Describe the colour change that would be observed. **1**

7. (b) (iii) (continued)

(C) Complete the ion electron equation for the reaction. **1**

$C_2H_5OH \rightarrow CH_3COOH$

(D) The other reaction taking place is shown.

$Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^- \rightarrow 2Cr^{3+}(aq) + 7H_2O(\ell)$

Explain why the dichromate solution needs to be acidified. **1**

(E) State the classification of alcohols that do not react with acidified dichromate solution. **1**

(iv) Name another oxidising agent that could be used to convert ethanal to compound **X** in step 2. **1**

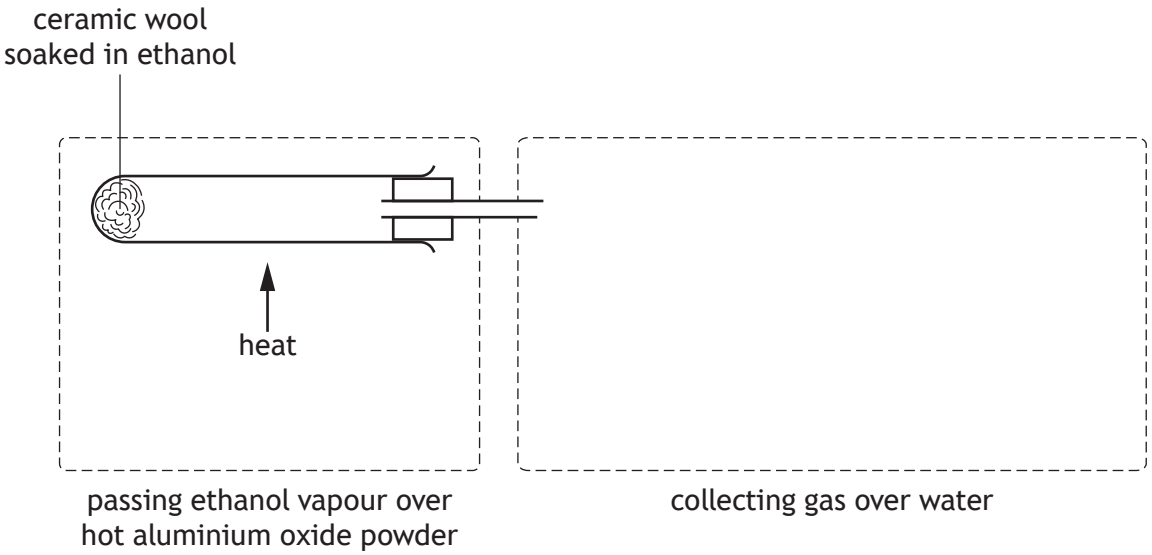
Turn over

7. (continued)

(c) Ethanol can be converted to ethene in the laboratory using a catalyst.

In this experiment, aluminium oxide powder is heated. This heat also causes the ethanol to turn to vapour.

The ethanol vapour passes over the hot aluminium oxide powder and forms ethene. The ethene gas is collected over water.



(i) Complete both parts of the diagram to show how ethanol vapour is passed over hot aluminium oxide powder and how the ethene gas is collected over water.

Your answer should include labels. **2**

(An additional diagram, if required, can be found on *page 33*.)

(ii) When bromine solution is added to the collected gas it rapidly decolourises.

State the term given to a chemical that rapidly decolourises bromine solution. **1**

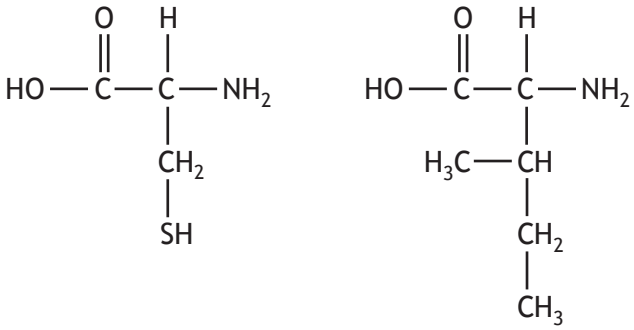
8. Snake venom contains proteins.

(a) Proteins are formed by joining together amino acids.

(i) Name the type of reaction that takes place when amino acids are joined. **1**

(ii) Two amino acids can join to form a dipeptide.

Two of the amino acids found in snake venom are shown.



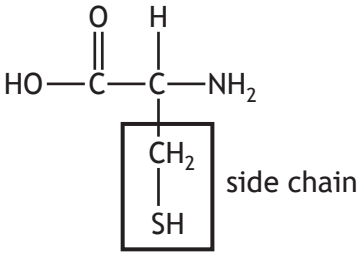
Draw a structural formula to show a dipeptide formed when these two amino acids join. **1**

[Turn over

8. (continued)

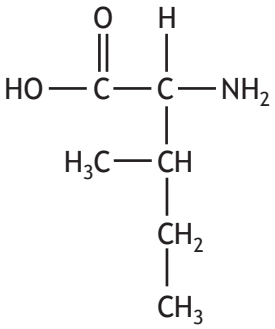
MARKS DO NOT WRITE IN THIS MARGIN

(b) Amino acids can be classified according to their side chains.



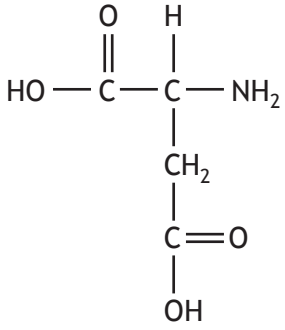
(i) Four types of side chain are **acidic**, **basic**, **hydrophobic** and **polar**.

(A) The structure of the amino acid isoleucine is shown.



Identify the class of amino acid to which isoleucine belongs. **1**

(B) The structure of the amino acid aspartic acid is shown.



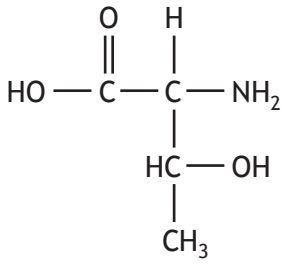
Suggest why aspartic acid can be classified as acidic. **1**

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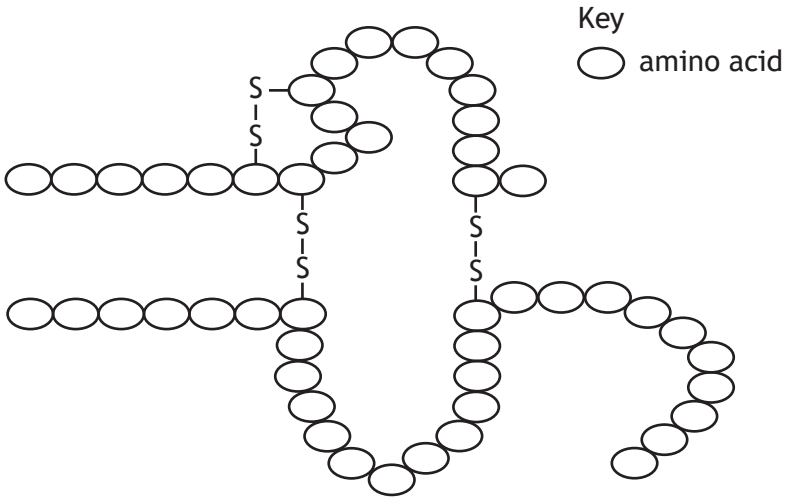
(ii) When amino acids form proteins, the proteins are stabilised by intermolecular forces that form between the side chains.

The structure of the amino acid threonine is shown.



Identify the strongest intermolecular force that would form between molecules of threonine. **1**

(c) Proteins that contain the amino acid cysteine can form disulfide bonds (— S — S —) between cysteine side chains. This makes the protein resistant to being broken down when heated.



Suggest why proteins forming disulfide bonds are resistant to heat. **1**

[Turn over

9. Liquids and powders that remove dirt and stains from clothes contain a variety of chemicals.

The chemicals in the liquids and powders include soaps, detergents, bleaching agents and enzymes.

Using your knowledge of chemistry, explain the action of the chemicals used to remove dirt and stains from clothes. **3**

10. Ammonia, NH_3, can be made using the Haber process.

(a) In the Haber process, nitrogen and hydrogen react together in the presence of an iron catalyst.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \qquad \Delta H = -92 \text{ kJ mol}^{-1}$$

(i) Complete the table to show the change in conditions required to increase the yield of ammonia. **1**

Condition	Increase/Decrease
Pressure	increase
Temperature	
Concentration of ammonia in the reaction mixture	

(ii) State the effect of the catalyst on the equilibrium position of this reaction. **1**

(iii) Under certain conditions, 500 kg of nitrogen reacts with excess hydrogen to produce 405 kg of ammonia.

$$\underset{GFM = 28 \text{ g}}{N_2(g)} + 3H_2(g) \rightleftharpoons \underset{GFM = 17 \text{ g}}{2NH_3(g)}$$

Calculate the percentage yield of ammonia under these conditions. **2**

[Turn over

10. (continued)

(b) The hydrogen needed for ammonia production by the Haber process can be made by reacting methane with steam.

$$CH_4(g) + H_2O(g) \rightarrow CO(g) + 3H_2(g)$$

Calculate the enthalpy change, in kJ mol^{-1}, for this reaction using the following information. **2**

$$CO(g) + \tfrac{1}{2}O_2(g) \rightarrow CO_2(g) \qquad \Delta H = -283 \text{ kJ mol}^{-1}$$

$$H_2(g) + \tfrac{1}{2}O_2(g) \rightarrow H_2O(g) \qquad \Delta H = -242 \text{ kJ mol}^{-1}$$

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g) \qquad \Delta H = -803 \text{ kJ mol}^{-1}$$

10. (continued)

(c) A new method for producing ammonia has recently been developed.

Lithium oxide is reduced to lithium using a laser. Lithium then reacts with nitrogen in the air to form lithium nitride.

(i) Lithium nitride, Li_3N, then reacts with water.

$$Li_3N(s) + 3H_2O(\ell) \rightarrow NH_3(g) + 3LiOH(aq)$$

$GFM = 34.7$ g $\quad GFM = 18.0$ g $\quad GFM = 17.0$ g $\quad GFM = 23.9$ g

Calculate the atom economy for the production of ammonia. **2**

(ii) Using this method, 0.0013 g of ammonia ($GFM = 17.0$ g) was produced in 78 seconds.

Calculate the time, in hours, required to produce 1 mole of ammonia. **2**

[Turn over

MARKS | DO NOT WRITE IN THIS MARGIN

11. The concentration of a sodium hydroxide solution can be determined by titration with oxalic acid.

(a) A student made an oxalic acid solution of accurately known concentration.

(i) State the term used to describe a solution of accurately known concentration. **1**

(ii) Calculate the mass, in g, of oxalic acid ($GFM = 90$ g) needed to make 250 cm^3 of 0.050 mol l^{-1} oxalic acid solution. **1**

(iii) A section of the student's notebook is shown with the experimental procedure they used to make the solution.

1. The sample of oxalic acid was dissolved in 25 cm^3 of deionised water in the beaker.
2. The solution was poured into the 250 cm^3 volumetric flask using a funnel.
3. The solution was made up to the graduated mark with tap water so that the top of the meniscus meets the line.
4. The flask was stoppered and then inverted several times to mix the solution.

The solution prepared by the student did not have an accurately known concentration.

Suggest **two** improvements the student could make to the procedure. **2**

11. (continued)

(b) The student titrated the sodium hydroxide solution with a new oxalic acid solution using an indicator.

(i) State why it was necessary to add an indicator to this titration. **1**

(ii) The burette was clamped vertically, rinsed with the new oxalic acid solution and filled up to above 0.0 cm^3 of the scale.

Suggest one other step required before the burette is ready to begin the titration. **1**

[Turn over

MARKS

DO NOT WRITE IN THIS MARGIN

(iii) Three 25.0 cm^3 samples of sodium hydroxide, NaOH, were titrated with a solution of 0.0517 mol l^{-1} oxalic acid, $(COOH)_2$.

The results are shown in the table.

Sample	Volume of oxalic acid used (cm^3)
1	27.4
2	27.0
3	26.9

(A) State why the volume of oxalic acid used in the calculation was taken to be 26.95 cm^3, although this is not the average of the three titre volumes in the table. **1**

(B) Calculate the concentration, in mol l^{-1}, of the sodium hydroxide solution. **3**

$$(COOH)_2(aq) + 2NaOH(aq) \rightarrow 2H_2O(\ell) + (COONa)_2(aq)$$

(C) The sodium salt of oxalic acid produced in the reaction has the formula $(COONa)_2$.

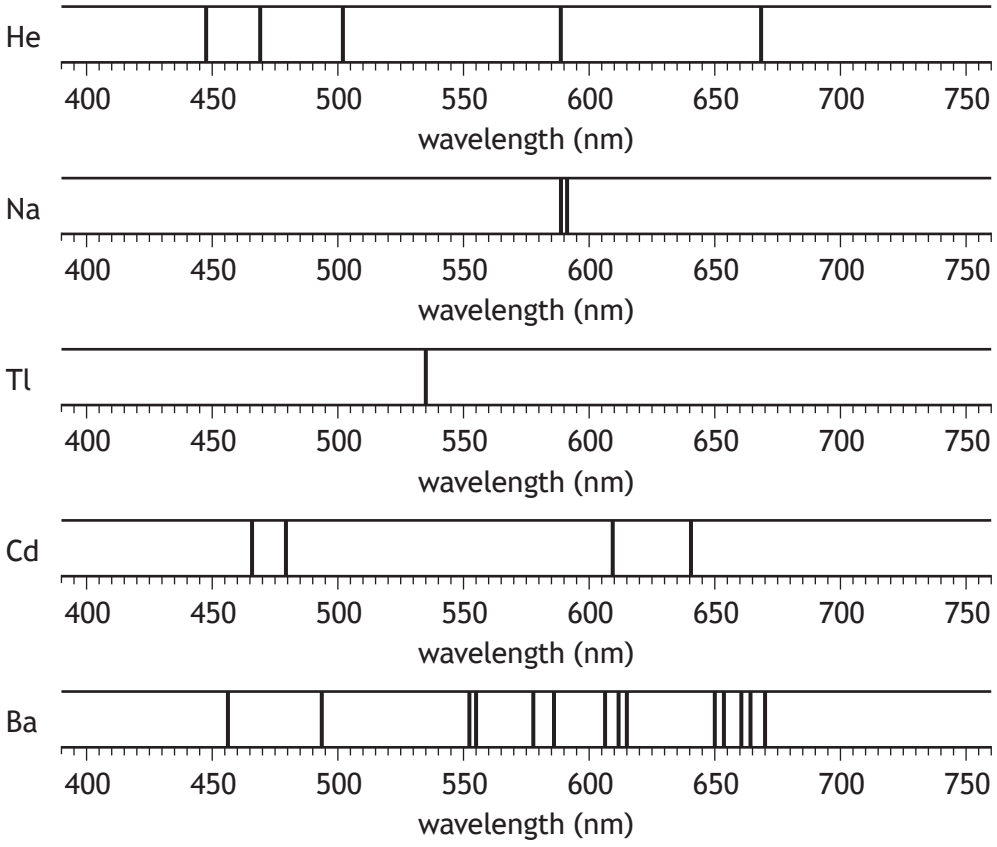
Draw a structural formula for the sodium salt of oxalic acid. **1**

12. Flame emission spectroscopy is a method of chemical analysis used to identify the elements present in a sample and determine their concentrations.

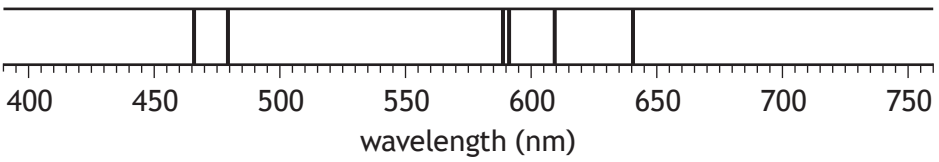
The sample is vaporised in a flame. Heat energy from the flame excites electrons in atoms to higher energy levels. When the excited electrons fall back to lower energy levels, light is emitted which is detected by the spectrometer.

Each element emits light as a pattern of characteristic wavelengths called an emission spectrum.

(a) The emission spectra for some elements are shown.



(i) A sample of water was analysed and the following emission spectrum was observed.



Name the elements present in this water sample. **1**

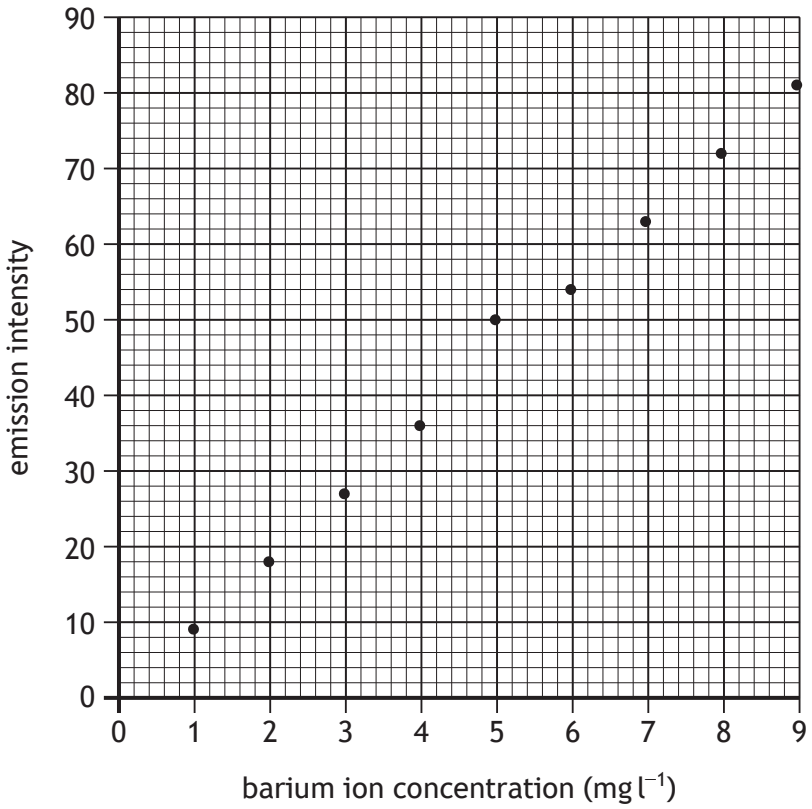
12. (a) (continued)

(ii) Thallium, Tl, shows only one line in this emission spectrum at 535 nm.

Use the colour wheel on *page 19* of the data booklet to suggest what colour flame thallium would produce. **1**

(b) The concentration of barium ions in a water sample can be determined using flame emission spectroscopy.

(i) The emission intensity was measured for a range of solutions with known concentrations of barium ions and a graph was produced.



Using the same procedure, the emission intensity of a sample was measured as 40.

Determine the concentration of barium ions, in mg l^{-1}, in the sample. **1**

12. (b) (continued)

(ii) State the term used to describe water containing high levels of dissolved metal ions. **1**

(c) The temperature of the flame is important. If it is too high, the element will ionise.

(i) Write the equation for the first ionisation energy of helium. **1**

(ii) There is no third ionisation energy value listed for helium in the data booklet.

Explain why this is the case. **1**

[END OF QUESTION PAPER]

National
Qualifications
2025

X813/76/12

Chemistry
Paper 1 — Multiple choice

THURSDAY, 1 MAY
9:00 AM – 9:40 AM

Total marks — 25

Attempt ALL questions.

You may use a calculator.

Instructions for the completion of Paper 1 are given on *page 02* of your answer booklet X813/76/02.

Record your answers on the answer grid on *page 03* of your answer booklet.

You may refer to the Chemistry Data Booklet for Higher and Advanced Higher.

Space for rough work is provided at the end of this booklet.

Before leaving the examination room you must give your answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

X8137612

Total marks — 25

Attempt ALL questions

1. Which of the following is a pure covalent bond?

A N–Br

B H–Cl

C O–H

D C–I

2. Elements which are strong reducing agents

A undergo reduction reactions

B are found on the right-hand side of the periodic table

C have low electronegativity values

D tend to form ions by gaining electrons.

3. $MnO_4^-(aq) + 5Fe^{2+}(aq) + 8H^+(aq) \rightarrow Mn^{2+}(aq) + 5Fe^{3+}(aq) + 4H_2O(\ell)$

How many moles of Fe^{2+} are required to react with 100 cm^3 of 0.5 mol l^{-1} $KMnO_4$ solution?

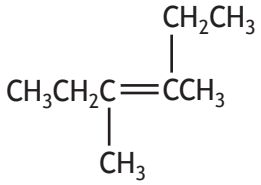
A 0.05

B 0.25

C 2.5

D 5

4. Which of the following is the correct systematic name for the structure shown?



A 2,3-diethylbut-2-ene

B 2-ethyl-3-methylpent-2-ene

C 4-ethyl-3-methylpent-3-ene

D 3,4-dimethylhex-3-ene

5. Which of the following is a tertiary alcohol?

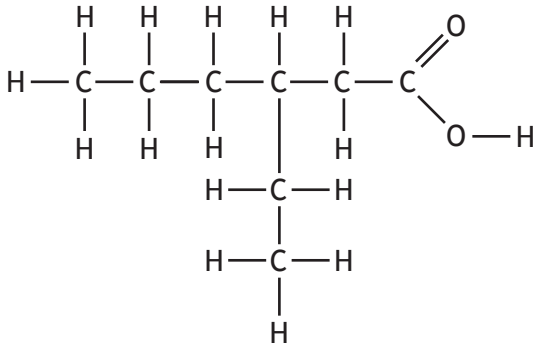
A 2-methylpentan-1-ol

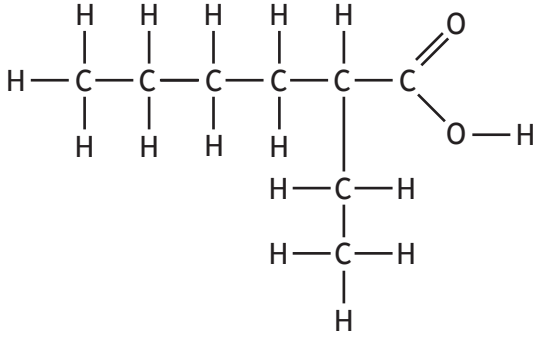
B 2-methylpentan-2-ol

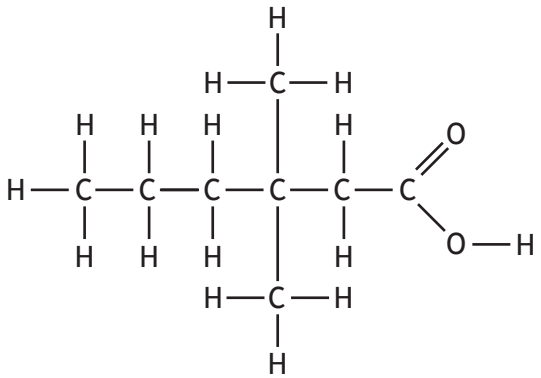
C 2-methylpentan-3-ol

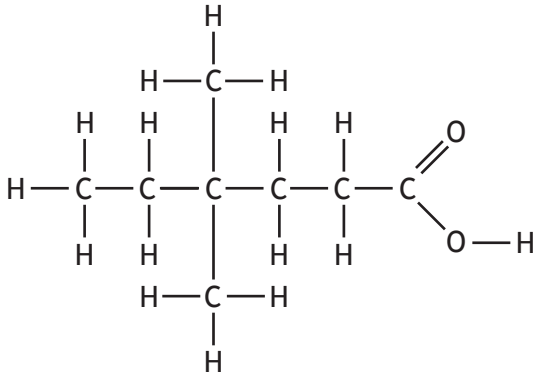
D 4-methylpentan-2-ol

6. Which of the following is the structure of 3-ethylhexanoic acid?

A 

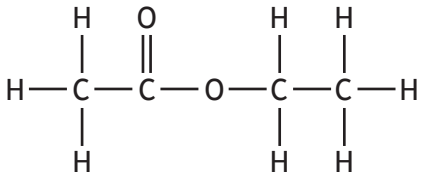
B 

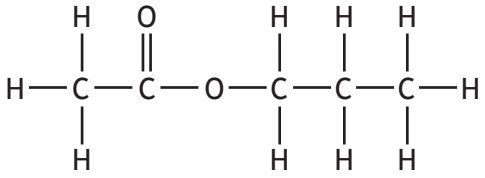
C 

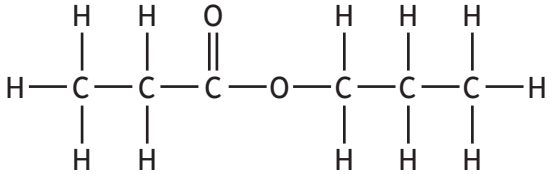
D 

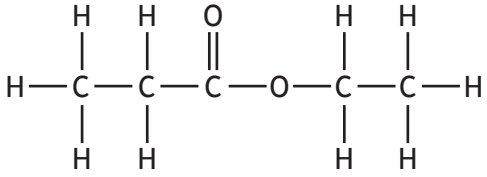
7. Hydrolysis of an ester gave an alcohol and a carboxylic acid, both of which had a gram formula mass of 60 g.

Which of the following is the structure of the ester?

A 

B 

C 

D 

8. What is the change in mass, in g, when one mole of butanal undergoes reduction?

A −16

B +16

C −2

D +2

9. 6-methylhept-5-ene-2-one is found in the sweat of some humans.

Which of the following is the correct formula for 6-methylhept-5-ene-2-one?

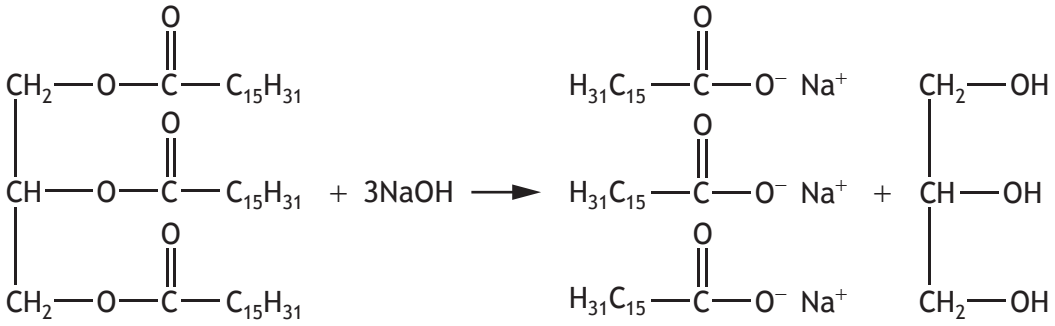
A $CH_3C(CH_3)CHCOCH_2CH_3$

B $CH_3CH(CH_3)CH_2CH_2CH_2COCH_3$

C $CH_3C(CH_3)CHCH_2CH_2COCH_3$

D $CH_3CH_2C(CH_3)CHCH_2COCH_3$

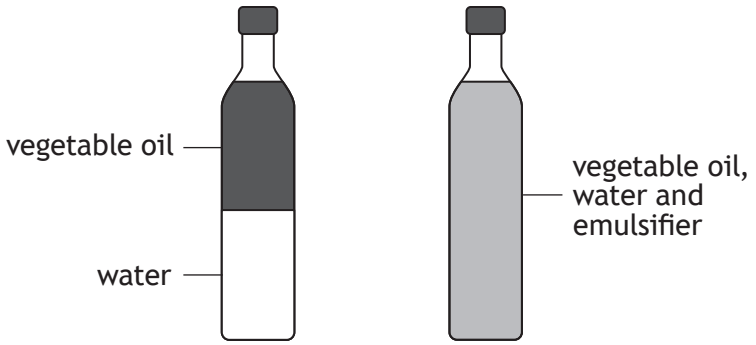
10. Saponification is the chemical reaction between edible fats and sodium hydroxide to produce soaps.



This reaction is an example of

A hydration
B hydrolysis
C esterification
D condensation.

11. An emulsion can be formed by shaking vegetable oil and water with an emulsifier.



In the emulsion formed, the fatty acid chains of the emulsifier are

A hydrophilic and dissolve in water
B hydrophobic and dissolve in water
C hydrophilic and dissolve in vegetable oil droplets
D hydrophobic and dissolve in vegetable oil droplets.

[Turn over

12. Which of the following is a terpene?

A $C_{13}H_{20}$

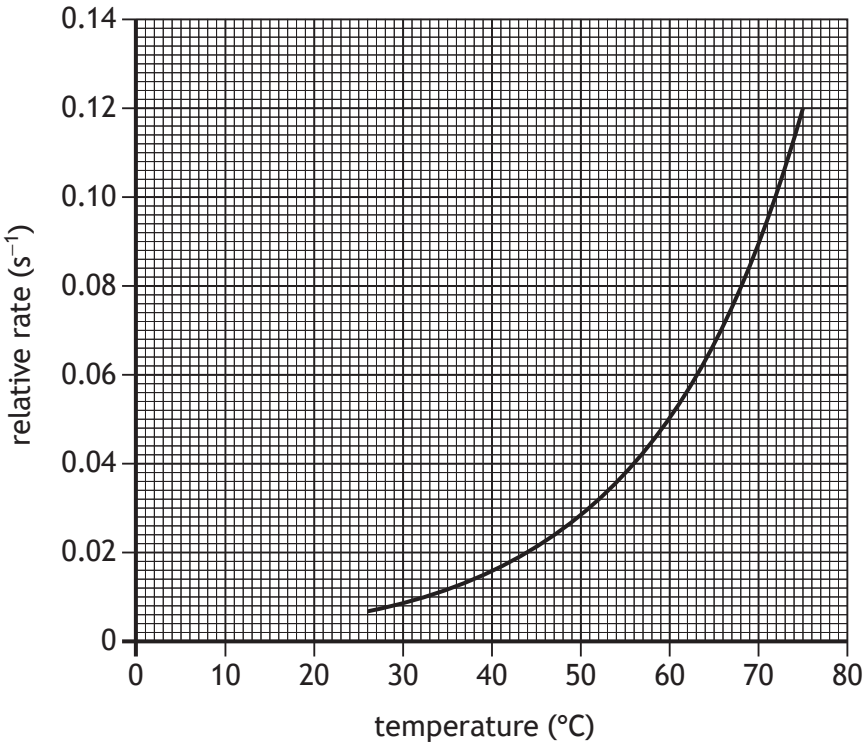
B $C_{15}H_{24}$

C $C_{17}H_{28}$

D $C_{19}H_{32}$

13. An investigation was carried out to determine the effect of changing temperature on the rate of a reaction.

The results from the investigation are shown in the graph.



Use the graph to determine the temperature rise, in °C, required to double the rate of the reaction.

A 0.04

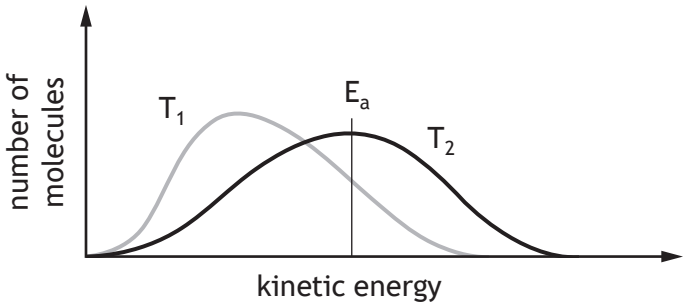
B 10

C 12

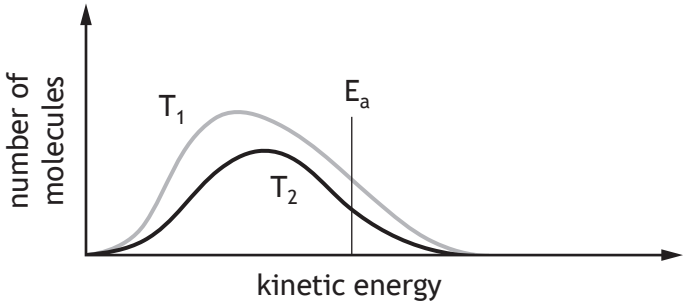
D 25

14. Which of the following diagrams shows the distribution of kinetic energies for the molecules of a reaction when the temperature of the reaction is increased from T_1 to T_2?

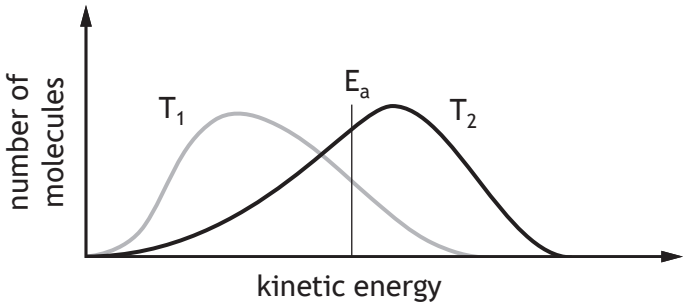
A



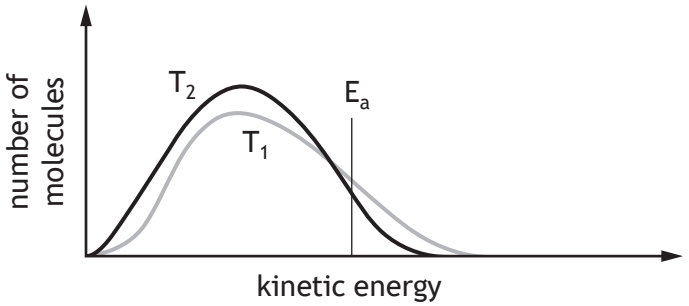
B



C



D



[Turn over

15. Some energy values associated with a chemical reaction are shown in the table.

Activation energy of forward reaction ($kJ\,mol^{-1}$)	Activation energy of reverse reaction ($kJ\,mol^{-1}$)
165	179

What is the enthalpy change for the forward reaction?

A −14 kJ

B +14 kJ

C −344 kJ

D +344 kJ

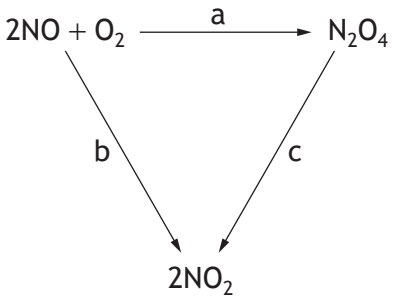
16. The equation for a reaction at equilibrium is shown.

$$A \rightleftharpoons B \qquad \Delta H = -100\ kJ\,mol^{-1}$$

Which of the following conditions will favour the formation of B?

	Reaction temperature	Activation energy of the forward reaction
A	high	low
B	low	high
C	high	high
D	low	low

17. Consider the reaction pathway shown.



What is the relationship between a, b and c?

A c = a + b

B c = a − b

C c = −a − b

D c = −a + b

18. According to Hess's law, which row in the table shows how equations **1** and **2** should be manipulated to calculate the enthalpy change for equation **3**?

Equation **1** $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(\ell)$

Equation **2** $2CO(g) + O_2(g) \rightarrow 2CO_2(g)$

Equation **3** $2CH_4(g) + 3O_2(g) \rightarrow 2CO(g) + 4H_2O(\ell)$

	Equation 1	**Equation 2**
A	unchanged	unchanged
B	multiplied by 2	reversed
C	unchanged	reversed
D	multiplied by 2	unchanged

19. Which of the following is a correct statement about a catalyst in a reaction?

A It increases the activation energy.

B It decreases the kinetic energy of reactant particles.

C It shifts the equilibrium to the right.

D It has no effect on the enthalpy change.

20. A two-step reaction is shown below.

$$A \xrightarrow[\text{step 1}]{} B \xrightarrow[\text{step 2}]{} C$$

Step 1 gave a yield of 50%. The overall yield of product C was 40%.

The yield for step 2 would be

A 10%

B 20%

C 80%

D 90%

[Turn over

21. A mixture of aluminium phosphate and aluminium sulfate is known to contain 5 moles of aluminium ions and 1 mole of phosphate ions.

How many moles of sulfate ions are present?

A 1

B 2

C 4

D 6

22. The electron affinity of an element is the energy change when one mole of gaseous atoms combines with one mole of electrons to form one mole of gaseous ions.

Which of the following equations represents the electron affinity of chlorine?

A $Cl(g) + e^- \rightarrow Cl^-(g)$

B $\frac{1}{2}Cl_2(g) + e^- \rightarrow Cl^-(g)$

C $\frac{1}{2}Cl_2(g) \rightarrow Cl^+(g) + e^-$

D $Cl(g) \rightarrow Cl^+(g) + e^-$

23. The following equilibrium exists in bromine solution.

$$\underset{\text{(red)}}{Br_2(aq)} + H_2O(\ell) \rightleftharpoons \underset{\text{(colourless)}}{Br^-(aq)} + 2H^+(aq) + \underset{\text{(colourless)}}{OBr^-(aq)}$$

The red colour of the solution would **fade** on adding a few drops of a concentrated solution of

A HCl

B KBr

C $AgNO_3$

D NaOBr.

24. When carrying out a titration to determine the mass of vitamin C in a tablet, which of the following will **not** help determine the end-point of the titration accurately?

A Adding the solution dropwise near the end-point.

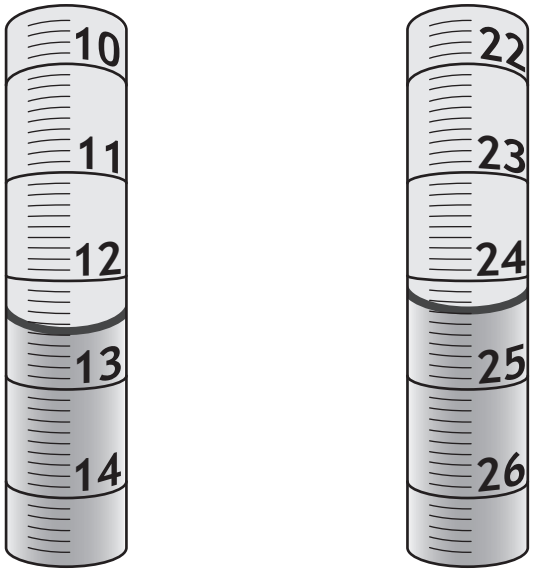
B Using a white tile below the conical flask.

C Using a dry conical flask.

D Swirling the flask.

25. A burette was used to carry out a titration.

The diagrams show the burette at the start and end of the titration.



start end

What volume of solution was added from the burette?

A 11.8 cm^3

B 12.2 cm^3

C 12.5 cm^3

D 24.3 cm^3

[END OF QUESTION PAPER]

FOR OFFICIAL USE

National Qualifications 2025

Mark

X813/76/01

Chemistry Paper 2

THURSDAY, 1 MAY

10:10 AM – 12:30 PM

X8137601

Fill in these boxes and read what is printed below.

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth

Day | Month | Year

Scottish candidate number

Total marks — 95

Attempt ALL questions.

You may use a calculator.

You may refer to the Chemistry Data Booklet for Higher and Advanced Higher.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers and rough work is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting. Any rough work must be written in this booklet. Score through your rough work when you have written your final copy.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

X8137601 01

Total marks — 95

Attempt ALL questions

1. Some metal ions are required for a healthy diet.

(a) Sodium, magnesium and calcium form positively charged ions.

The table shows the values for ionisation energies for these elements.

Element	Ionisation energies (kJ mol^{-1})			
	First	Second	Third	Fourth
Sodium	496	4562	6910	9543
Magnesium	738	1451	7733	10 543
Calcium	590	1145	4912	6491

(i) Write the equation for the first ionisation of sodium. **1**

(ii) Using the ionisation energies from the table, calculate the energy required, in kJ mol^{-1}, for the following reaction. **1**

$$Ca(g) \rightarrow Ca^{2+}(g) + 2e^-$$

(iii) Explain **fully** why the second ionisation energy of sodium is much higher than the second ionisation energy of magnesium. **2**

1. (a) (continued)

(iv) The Mg^{2+} ion and the Na^{+} ion have the same number of occupied energy levels.

Explain why the Mg^{2+} ion is smaller than the Na^{+} ion. **1**

(b) Potassium forms ionic bonds.

(i) State what is meant by the term ionic bond. **1**

(ii) Adults with low potassium levels may be advised to take tablets or eat potassium-rich foods.

An adult takes eight potassium tablets per day for seven days. Each tablet contains 0.0012 moles of potassium ($GFM = 39.1$ g).

A banana contains an average mass of 450 mg of potassium.

Calculate the number of whole bananas the adult would have to consume to provide the same mass of potassium provided by eight tablets per day for seven days. **2**

[Turn over

2. Information about four elements from the third period of the periodic table is shown in the table.

Element	Bonding	Structure
Sodium		lattice
Silicon	covalent	
Phosphorus		molecular
Chlorine	covalent	

(a) Complete the table to show the bonding and structure for each element. **2**

(An additional table, if required, can be found on *page 34*.)

(b) Covalent radius is half the distance between the nuclei of two covalently bonded atoms of an element.

(i) State the trend in covalent radius going from sodium to chlorine. **1**

(ii) Suggest why no covalent radius is listed in the data booklet for argon. **1**

2. (continued)

(c) Explain **fully** why the boiling points of the noble gases increase going down group 0.

In your answer you should name the van der Waals forces involved. **3**

[Turn over

3. Pure covalent bonding and ionic bonding can be considered as opposite ends of a bonding continuum.

Using your knowledge of chemistry, comment on how the position of a compound on the bonding continuum can be explained and how this can be used to predict its bonding and properties. **3**

4. Alcohols and carboxylic acids are used in many laboratory experiments.

(a) Alcohols can take part in combustion reactions.

(i) Balance the equation for the enthalpy of combustion of propan-1-ol. **1**

$$C_3H_7OH \quad + \quad O_2 \quad \rightarrow \quad CO_2 \quad + \quad H_2O$$

(ii) An experiment was carried out to determine the enthalpy of combustion of propan-1-ol. The results are shown in the table.

Volume of water heated (cm^3)	150
Initial temperature of water (°C)	21.2
Final temperature of water (°C)	35.1
Mass of propan-1-ol burned (g)	0.498
Mass of one mole of propan-1-ol (g)	60

Calculate the enthalpy of combustion of propan-1-ol, in $kJ\,mol^{-1}$, for this experiment. **3**

(iii) Name the pieces of laboratory equipment that are essential for carrying out this experiment. **2**

[Turn over

4. (a) (continued)

(iv) Suggest a reason why the enthalpy of combustion determined experimentally is lower than the value given in the data booklet. **1**

(b) Alcohols can undergo oxidation reactions.

(i) Oxidation reactions involve a change in the oxygen to hydrogen ratio.

Determine the oxygen to hydrogen ratio for the alcohol ethane-1,2-diol. **1**

(ii) A primary alcohol, **X**, with the formula C_4H_9OH can undergo oxidation to compound **Y**.

(A) Suggest a suitable oxidising agent for this reaction. **1**

(B) Compound **Y** can be oxidised to give product **Z**, which turns universal indicator red.

Draw a structural formula for product **Z**. **1**

(C) Alcohol **X**, C_4H_9OH, has an isomer that is a secondary alcohol.

Name the product that would be formed by oxidation of this isomer. **1**

4. (continued) **MARKS** DO NOT WRITE IN THIS MARGIN

(c) The concentration of a solution of ethanoic acid can be determined using sodium hydroxide solution.

(i) A solution of ethanoic acid was prepared using a piece of glassware that allows the volume to be made up to exactly 250 cm^3.

Name this piece of glassware. **1**

(ii) A titration was carried out to calculate the accurate concentration of the ethanoic acid solution.

(A) A burette was rinsed with deionised water and then filled with sodium hydroxide solution.

Suggest an improvement that could be made to this step in the titration procedure. **1**

(B) Four 25.0 cm^3 samples of the ethanoic acid solution were titrated with 0.105 mol l^{-1} sodium hydroxide solution.

The results are shown in the table.

	Titration			
	1	**2**	**3**	**4**
Initial reading (cm^3)	0.2	21.0	0.4	20.4
Final reading (cm^3)	21.0	41.5	20.4	40.7
Volume used (cm^3)	20.8	20.5	20.0	20.3

Use these results to calculate the average volume used in cm^3. **1**

4. (c) (ii) (continued)

(C) In another experiment, 25.0 cm^3 samples of ethanoic acid were titrated with 0.105 mol l^{-1} sodium hydroxide solution. The average volume of sodium hydroxide solution used was 19.8 cm^3.

One mole of sodium hydroxide reacts with one mole of ethanoic acid.

Calculate the concentration, in mol l^{-1}, of the ethanoic acid. **3**

(iii) An impure sample of ethanoic acid, CH_3COOH, contained traces of potassium ethanoate.

(A) Write the ionic formula for potassium ethanoate. **1**

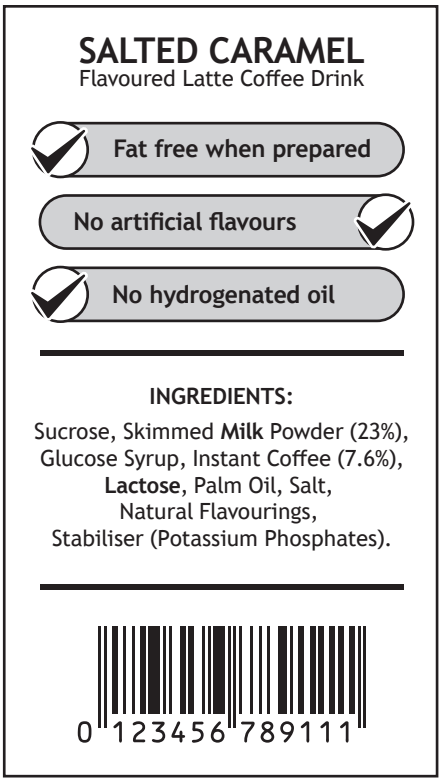
(B) A solution of the impure sample was prepared.

The concentration of ethanoic acid ($GFM = 60$ g) was found to be 0.45 mol l^{-1}.

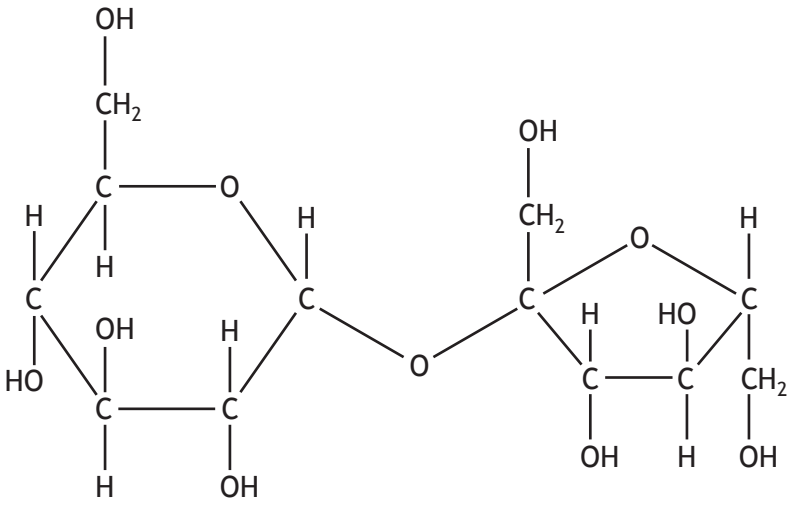
Calculate the mass, in g, of ethanoic acid in 200 cm^3 of this solution. **1**

5. The label from a packet of instant latte coffee drink is shown below.

MARKS DO NOT WRITE IN THIS MARGIN



(a) Sucrose is the main ingredient in the latte coffee drink. The structure of sucrose is shown.



Explain **fully** why sucrose is water soluble. **2**

5. (continued)

(b) Skimmed milk powder contains peptide molecules that are formed by joining together amino acid molecules.

Amino acids can be represented by the first three letters of their name; for example, Phe represents phenylalanine.

(i) Hydrolysis of a peptide molecule found in milk gave a mixture of three smaller peptide molecules with the following amino acid sequences.

Tyr-Pro-Phe

Gly-Pro-Phe

Iso-Pro-Gly-Pro

The original peptide molecule was formed from seven amino acids joined together.

Write the amino acid sequence for the original peptide molecule. **1**

_____-_____-_____-_____-_____-_____-_____

(ii) Phenylalanine is an amino acid that cannot be made in the human body.

State the term used to describe amino acids that the body cannot make. **1**

(c) Palm oil used in the latte coffee drink is an edible oil.

(i) State the class of compounds to which edible oils belong. **1**

(ii) When palm oil is hydrolysed, mixtures of fatty acids and another compound are produced.

(A) Name the other compound produced. **1**

5. (c) (ii) (continued)

(B) The table gives information on the fatty acids obtained on hydrolysis of palm oil.

Fatty acid	Formula
Palmitic acid	$C_{15}H_{31}COOH$
Oleic acid	$C_{17}H_{33}COOH$
Linoleic acid	$C_{17}H_{31}COOH$
Linolenic acid	$C_{17}H_{29}COOH$

Name the saturated fatty acid obtained on hydrolysis of palm oil. **1**

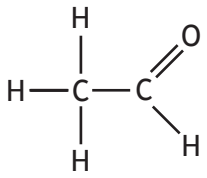
(iii) The latte coffee drink packet states it contains 'no hydrogenated oil'.

Hydrogenation takes place when unsaturated edible oils react with hydrogen.

State another name for this type of reaction. **1**

(d) Coffee contains many different compounds.

(i) Acetaldehyde gives coffee a strong fruity aroma.

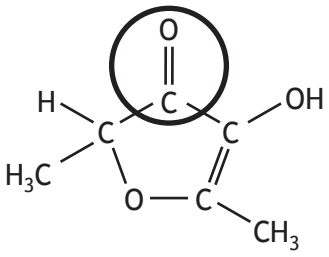


acetaldehyde

State the systematic name for acetaldehyde. **1**

5. (d) (continued)

(ii) Furaneol gives coffee a caramel flavour.



furaneol

Name the functional group circled in furaneol. **1**

(iii) Terpenoid molecules are also responsible for the aroma of coffee.

Terpenoids are classified by the number of isoprene units (2-methylbuta-1,3-diene) in the molecule.

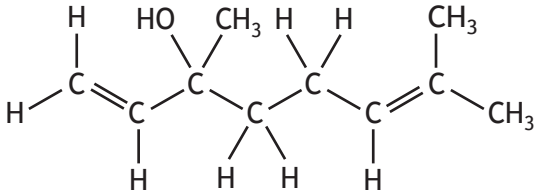
Class of terpenoid	Number of isoprene units
Hemiterpenoid	1
Monoterpenoid	2
Sesquiterpenoid	3
Diterpenoid	4
Sesterterpenoid	5

(A) Cafestol, $C_{20}H_{28}O_3$, is a terpenoid found in coffee.

Name the class of terpenoids to which cafestol belongs. **1**

5. (d) (iii) (continued)

(B) Linalool is a monoterpenoid found in coffee with the following structure.



Circle an isoprene unit on the linalool structure above. **1**

(An additional structure, if required, can be found on *page 34*.)

(e) One cup of the latte coffee drink contains 125.0 mg of caffeine. Consuming up to 5.7 mg of caffeine per kg of body mass per day is recommended to be safe for an adult.

Calculate the maximum number of cups of the latte coffee drink a 66 kg adult could safely drink without exceeding this recommendation. **1**

[Turn over

6. The reaction between iodide ions and hydrogen peroxide in the presence of thiosulfate ions, with starch as the indicator, can be used to determine the relationship between concentration and rate.

(a) Hydrogen peroxide, $H_2O_2(aq)$, reacts with iodide ions in acidic solution to form iodine.

$$H_2O_2(aq) + 2H^+(aq) + 2I^-(aq) \rightarrow 2H_2O(\ell) + I_2(aq)$$

Write the ion-electron equation for the reduction reaction taking place. **1**

(b) As iodine molecules are produced, they immediately react with thiosulfate ions, $S_2O_3^{2-}$, and are converted back to iodide ions.

$$2S_2O_3^{2-}(aq) \rightarrow S_4O_6^{2-}(aq) + 2e^-$$

$$I_2(aq) + 2e^- \rightarrow 2I^-(aq)$$

Combine the two ion-electron equations to give the overall redox equation. **1**

6. (continued)

(c) In the reaction, the mixture remains colourless until all the thiosulfate ions have been used up. This is shown by the sudden appearance of a blue/black colour.

A set of experiments were carried out to determine the relationship between concentration and rate. The concentration of iodide solution was varied by diluting the original concentration with water and keeping the total volume of solutions the same in each experiment. The time taken for the blue/black colour to appear was recorded.

The results of the experiments are shown.

	Experiment				
	1	**2**	**3**	**4**	**5**
Volume of iodide solution (cm^3)	25	20	15	10	5
Volume of water (cm^3)	0	5		15	20
Time to turn blue/black (s)	8	10	13	20	40
Rate of reaction (1/t) (s^{-1})	0.120	0.100	0.077	0.050	0.025

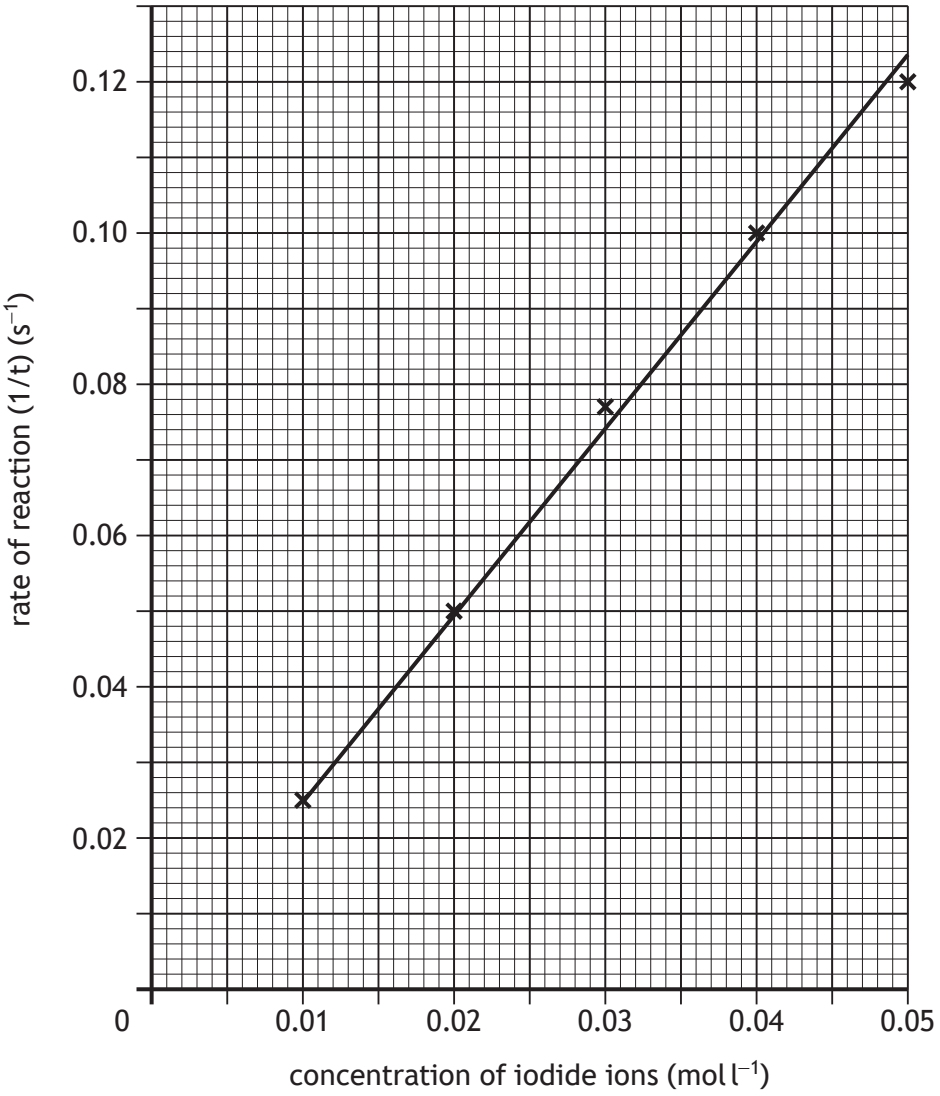
(i) Complete the table to show the volume of water needed for experiment 3. **1**

(ii) A dry beaker must be used for each experiment.

Suggest a reason why the beaker should be dry. **1**

[Turn over

(d) A graph of the rate of reaction against the concentration of iodide ions is shown for these results.



(i) Use information from the graph to state what happens to the rate of reaction when the concentration of iodide ions is doubled. **1**

(ii) Use information from the graph to calculate the reaction time, in seconds, when the concentration of the iodide ions was 0.025 mol l^{-1}. **1**

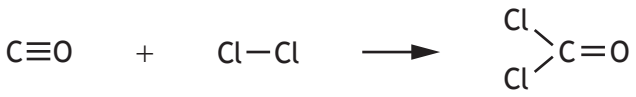
6. (d) (continued)

(iii) Explain **fully** why increasing the concentration causes an increase in reaction rate. **2**

[Turn over

7. Carbon monoxide is a gas used in industry.

(a) Carbon monoxide can be used to produce carbonyl dichloride, a chemical used in manufacturing plastics.



The enthalpy change, ΔH, for this reaction is -107.6 kJ mol^{-1}.

Using this data and the bond enthalpy values shown in the data booklet, calculate the bond enthalpy, in kJ mol^{-1}, of the C≡O bond. **2**

(b) Carbon monoxide can react with hydrogen to form methane.

$$CO(g) + 3H_2(g) \rightleftharpoons CH_4(g) + H_2O(g) \qquad \Delta H = -206 \text{ kJ mol}^{-1}$$

Cooling the reaction mixture from 400 °C to 80 °C moves the position of equilibrium to the right, increasing the yield of methane.

State **two** reasons why the yield of methane is increased. **2**

MARKS

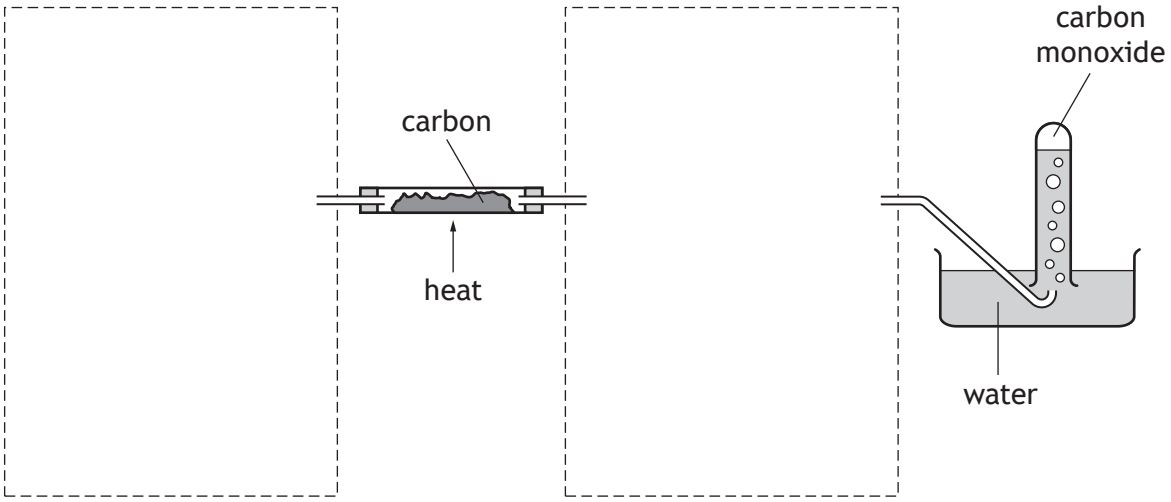
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(c) Carbon monoxide can be produced in several ways.

(i) One method of producing carbon monoxide is to react carbon dioxide with hot carbon.

The carbon dioxide is made by the reaction of dilute hydrochloric acid with solid calcium carbonate.

Unreacted carbon dioxide is removed by passing through a sodium hydroxide solution before the carbon monoxide is collected by displacement of water.



Complete the diagram to show how the carbon dioxide can be produced and how the unreacted carbon dioxide can be removed.

Your answer should include labels. **2**

(An additional diagram, if required, can be found on *page 34*.)

(ii) Carbon monoxide can also be produced by the reaction of zinc with calcium carbonate.

Zn	+	$CaCO_3$	→	ZnO	+	CaO	+	CO
$GFM =$ 65.4 g		$GFM =$ 100.1 g		$GFM =$ 81.4 g		$GFM =$ 56.1 g		$GFM =$ 28.0 g

Calculate the atom economy for the production of carbon monoxide. **2**

8. Metals can be extracted from their compounds by different methods.

(a) The extraction of tin from tin(IV) oxide, SnO_2, by heating with carbon has been investigated in a series of experiments.

(i) In one experiment the effect of temperature on the percentage yield of tin was investigated.

$SnO_2(s)$	+	$C(s)$	$\rightarrow$	$Sn(\ell)$	+	$CO_2(g)$
$GFM =$ 150.7 g				$GFM =$ 118.7 g		

At 850 °C, the percentage yield of tin is 64%.

Calculate the mass of tin oxide required, in g, to produce a mass of 100 g of tin at this temperature. **2**

(ii) In a second experiment, 25.2 g of tin(IV) oxide reacted with 3.0 g of carbon.

$SnO_2(s)$	+	$C(s)$	$\rightarrow$	$Sn(\ell)$	+	$CO_2(g)$
$GFM =$ 150.7 g		$GFM =$ 12.0 g				

Name the reactant in excess **and** calculate the number of moles of this reactant left unreacted. **2**

8. (a) (continued)

(iii) In a third experiment, 26.5 litres of carbon dioxide was produced.

$$SnO_2(s) + C(s) \rightarrow Sn(\ell) + CO_2(g)$$

Calculate the mass of tin(IV) oxide ($GFM = 150.7$ g) that reacted. **2**

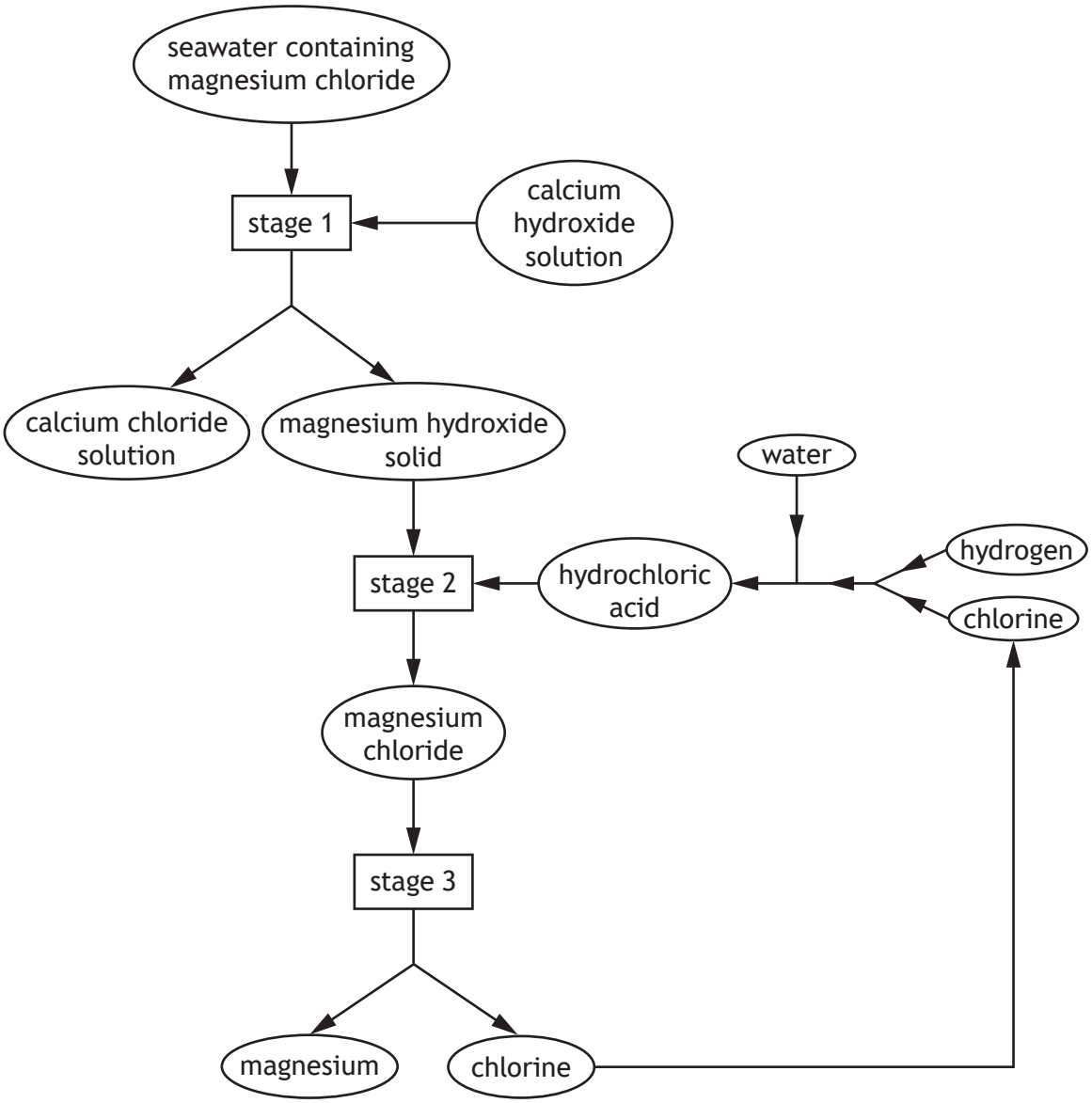
Take the volume of one mole of carbon dioxide gas to be 80 litres.

[Turn over

MARKS

DO NOT WRITE IN THIS MARGIN

(b) Magnesium metal can be extracted from magnesium chloride present in seawater, as shown in the flow diagram.



(i) Suggest how the magnesium hydroxide and calcium chloride produced in stage 1 can be separated. **1**

(ii) Chlorine produced in stage 3 is recycled back into stage 2.

From the flow diagram, suggest another way that profit is maximised. **1**

MARKS DO NOT WRITE IN THIS MARGIN

9. Hydrogen peroxide, H_2O_2, is highly reactive and can cause damage to proteins and fatty acids in the human body.

(a) Hydrogen peroxide, H_2O_2, can react to form hydroxyl radicals (HO•) starting a chain reaction. AH, BH and CH represent molecules found in the human body.

Reaction step	Name of step
$H_2O_2 \rightarrow 2HO\bullet$	initiation
$HO\bullet + AH \rightarrow HOH + A\bullet$ $A\bullet + BH \rightarrow AH + B\bullet$ $B\bullet + CH \rightarrow BH + C\bullet$	
$A\bullet + HO\bullet \rightarrow AOH$	termination

(i) Complete the table to show the name of the missing step. **1**

(An additional table, if required, can be found on *page 35*.)

(ii) The free radicals shown in the table are written as, for example, A•

State what is represented by the symbol • **1**

(b) When hydroxyl free radicals react with enzymes, they can break bonds between the protein molecule chains causing them to unfold and change shape.

State the term used to describe this change in shape. **1**

(c) Free radical scavengers are molecules that react to prevent chain reactions.

Give an example of a type of consumer product that has free radical scavengers added. **1**

[Turn over

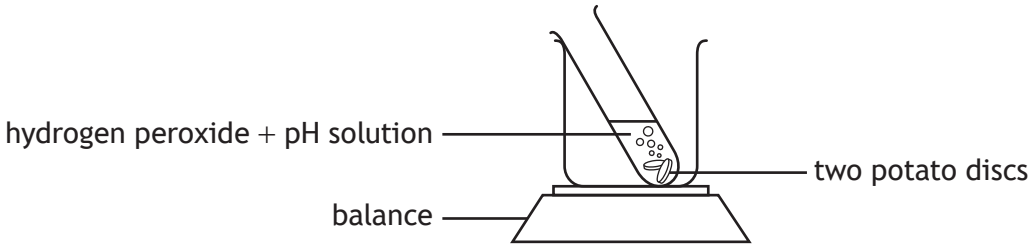
9. (continued)

(d) In some living organisms an enzyme called catalase breaks down hydrogen peroxide into water and oxygen.

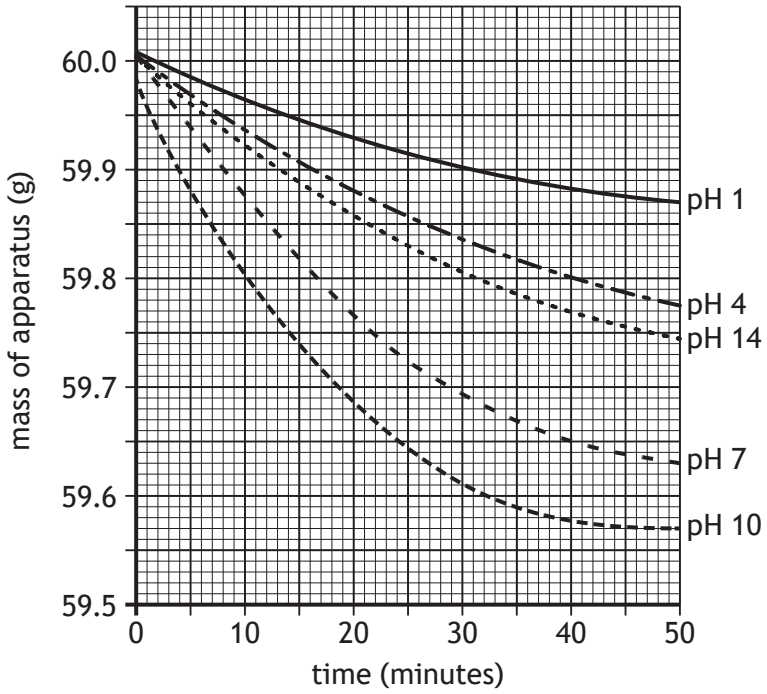
$$2H_2O_2(aq) \rightarrow 2H_2O(\ell) + O_2(g)$$

A student investigated how changes in pH affected the rate of reaction that was followed by measuring the mass lost over time.

Potato discs were placed in solutions with different pH values and the change in mass recorded.



The results are shown in the graph.



(i) Suggest why the mass of the apparatus decreased over time. **1**

9. (d) (continued)

(ii) In each experiment, two potato discs were added to 1.0 cm^3 of hydrogen peroxide and 15.0 cm^3 of pH solution in a boiling tube.

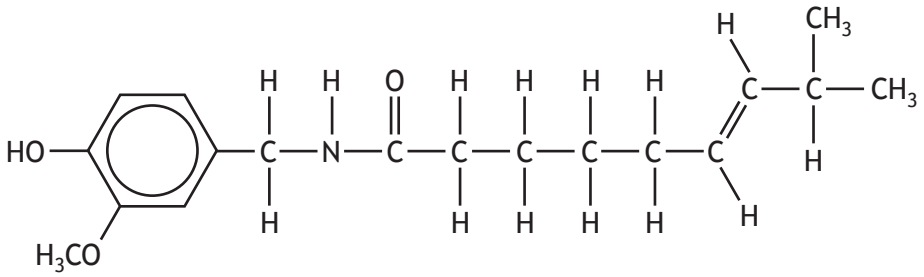
State another **two** variables that the student should have kept constant. **2**

(iii) Using the graph, describe the effect that changing pH has on the rate of the breakdown of hydrogen peroxide by catalase in potatoes. **1**

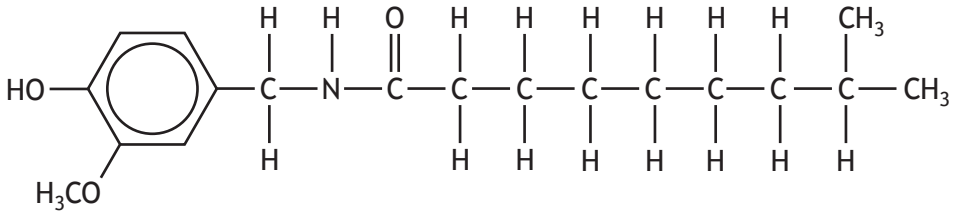
[Turn over

10. Chilli peppers produce a burning sensation in the mouth. Chemists have identified a group of compounds found in chilli peppers called capsaicinoids that cause this sensation.

(a) The structures of two capsaicinoids are shown below.



capsaicin (CAP)



dihydrocapsaicin (DHC)

(i) Circle the amide link in the CAP structure above. **1**

(An additional structure, if required, can be found on *page 35*.)

(ii) Describe a chemical test, and the expected result, that could be used to distinguish between CAP and DHC. **1**

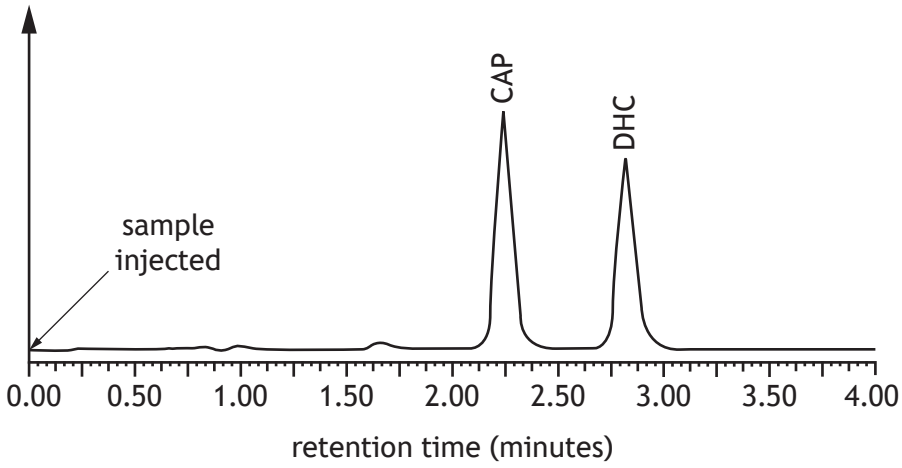
(b) Taking a drink of water has little effect on the burning sensation in the mouth produced by capsaicinoids.

Suggest why the burning sensation produced by capsaicinoids is not affected by drinking water. **1**

10. (continued)

(c) High pressure liquid chromatography, HPLC, can be used to separate different capsaicinoids present in chilli peppers.

The time taken for a substance to travel through the HPLC apparatus is known as its retention time and is the time between the sample being injected and the centre of the peak.



Determine the retention time, **in seconds**, of CAP. **1**

[Turn over

10. (continued)

(d) Creams containing CAP are sold for pain relief.

A 60 g tube of cream contains 0.025% CAP.

CAP costs £1930.32 for 1.0 kg.

Calculate the cost, in pence, of the CAP in one tube of cream. **2**

(e) Chilli peppers are a good source of vitamin C, $C_6H_8O_6$.

Complete the ion electron equation for the oxidation of vitamin C. **1**

$$C_6H_8O_6(aq) \rightarrow C_6H_6O_6(aq)$$

11. Sodium hydrogencarbonate, $NaHCO_3$, is a water-soluble solid and is present in indigestion tablets.

When sodium hydrogencarbonate is heated to 112 °C it breaks down and carbon dioxide gas is given off.

$$2NaHCO_3(s) \rightarrow Na_2CO_3(s) + H_2O(g) + CO_2(g)$$

Sodium hydrogencarbonate also reacts with acids to form salts. The hydrogencarbonate ions react with hydrogen ions from the acid to form water and carbon dioxide gas.

$$HCO_3^-(aq) + H^+(aq) \rightarrow H_2O(\ell) + CO_2(g)$$

Using your knowledge of chemistry, comment on how the mass of sodium hydrogencarbonate in an indigestion tablet could be determined by experiment. **3**

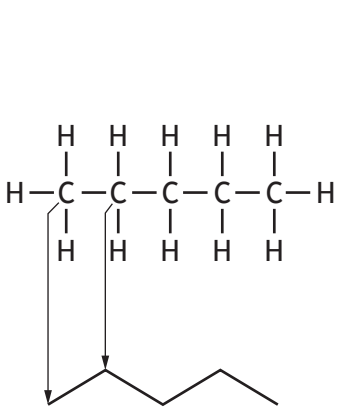
DO NOT WRITE IN THIS MARGIN

12. Chemists can represent the structures of carbon-chain molecules in a simplified way known as skeletal formulae.

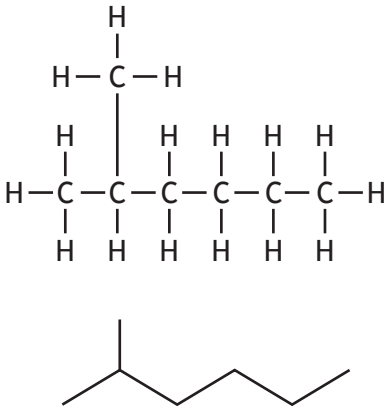
In a skeletal formula, the symbols for carbons, and for the hydrogens attached to carbons, are not shown. The carbon-chain structure is shown as a zig-zag line in which each bend represents a carbon atom. The ends of the zig-zag line also represent carbon atoms.

Each carbon atom is understood to have the correct number of hydrogen atoms attached.

Some examples are shown below.

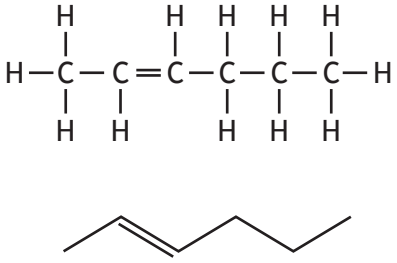


pentane



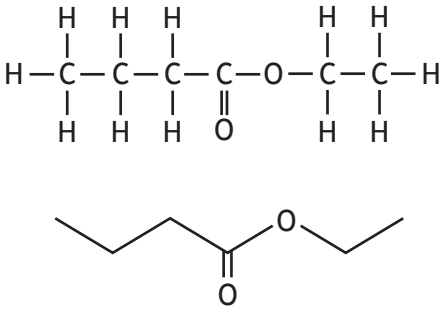
2-methylhexane

A double bond is shown by a double line in the zig-zag line.

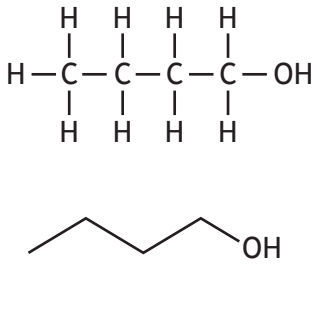


hex-2-ene

Other atoms, such as oxygen, are shown by their symbols attached to the appropriate part of the zig-zag line.



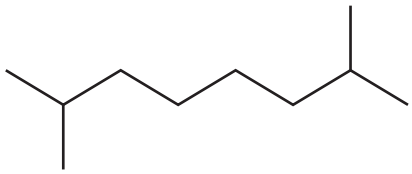
ethyl butanoate



butan-1-ol

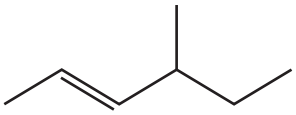
12. (continued)

(a) A skeletal formula is shown.



Name this molecule. **1**

(b) Draw the full structural formula for the molecule shown below. **1**



(c) Draw the skeletal formula for pentanoic acid. **1**

[END OF QUESTION PAPER]

National
Qualifications
2024

X813/76/12

Chemistry
Paper 1 — Multiple choice

THURSDAY, 23 MAY
9:00 AM – 9:40 AM

Total marks — 25

Attempt ALL questions.

You may use a calculator.

Instructions for the completion of Paper 1 are given on *page 02* of your answer booklet X813/76/02.

Record your answers on the answer grid on *page 03* of your answer booklet.

You may refer to the Chemistry Data Booklet for Higher and Advanced Higher

Space for rough work is provided at the end of this booklet.

Before leaving the examination room you must give your answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

X8137612

Total marks — 25

Attempt ALL questions

1. The difference between the first ionisation energies of sodium and chlorine is mainly due to the difference in the

A number of electrons

B number of neutrons

C number of protons

D mass of each atom.

2. Which line in the table is likely to be correct for the element francium?

	State at 30 °C	**Covalent radii (pm)**
A	solid	less than 238
B	liquid	less than 238
C	solid	greater than 238
D	liquid	greater than 238

3. Which of the following contains pure covalent bonds?

A CO_2

B H_2S

C PH_3

D CF_4

4. Iodide ions can be oxidised using acidified potassium permanganate solution.

The equations are

$$2I^-(aq) \rightarrow I_2(aq) + 2e^-$$

$$MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightarrow Mn^{2+}(aq) + 4H_2O(\ell)$$

How many moles of iodide ions are oxidised by one mole of permanganate ions?

A 1

B 2

C 5

D 10

5. In the reaction between potassium sulfite and iodine dissolved in potassium iodide, the species most likely to act as the oxidising agent is

A $I^-(aq)$

B $I_2(aq)$

C $SO_3^{2-}(aq)$

D $SO_4^{2-}(aq)$

6. Isovaleric acid is a compound found in wine and has the shortened structural formula $(CH_3)_2CHCH_2COOH$.

Which of the following is an isomer of isovaleric acid?

A 2-methylbutanoic acid

B 3-methylbutanoic acid

C 2-methylpentanoic acid

D 3-methylpentanoic acid

7. A compound has the molecular formula C_4H_7OH and reacts with acidified potassium dichromate.

The compound is

A a saturated secondary alcohol

B a saturated tertiary alcohol

C an unsaturated secondary alcohol

D an unsaturated tertiary alcohol.

8. The relative rate of a reaction which reached completion in 3 minutes 20 seconds is

A $0.005\ s^{-1}$

B $0.313\ s^{-1}$

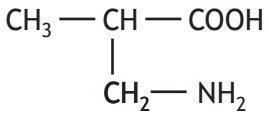
C $0.005\ min^{-1}$

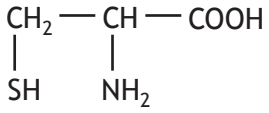
D $0.313\ min^{-1}$

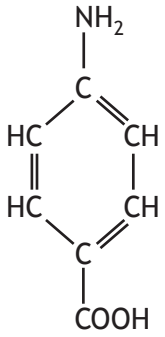
[Turn over

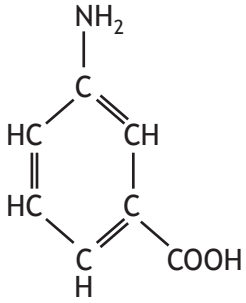
9. Some amino acids are called α-amino acids because the amino group is on the carbon atom next to the acid group.

Which of the following is an α-amino acid?

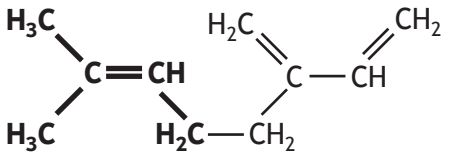
A 

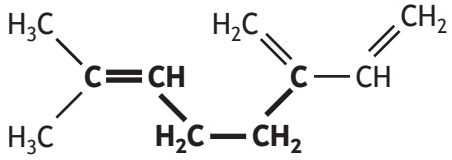
B 

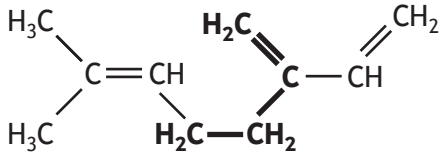
C 

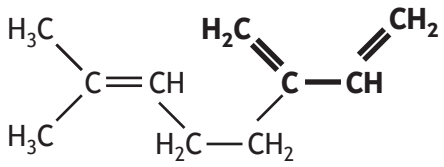
D 

10. Which of the following contains a correctly highlighted isoprene unit?

A 

B 

C 

D 

[Turn over

11. Edible oil molecules contain carbon to carbon double bonds that can undergo addition reactions with iodine.

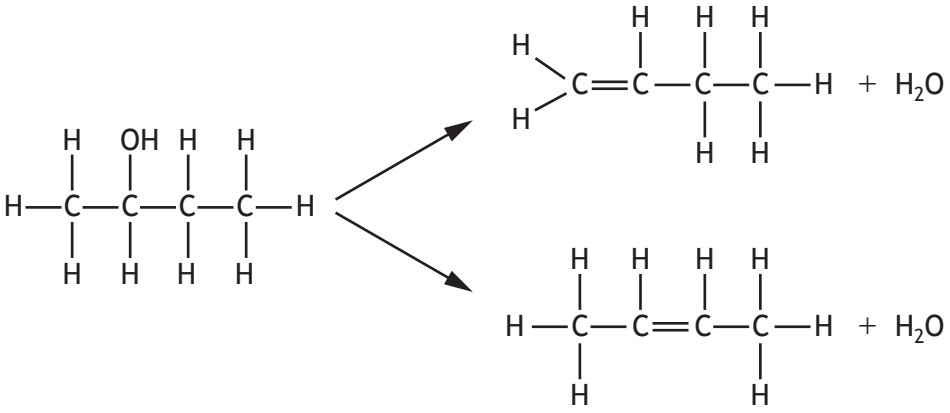
The iodine number of an oil is the mass of iodine, in grams, that will react with 100 g of the oil.

Which line in the table shows the oil that is likely to have the lowest melting point?

	Oil	Iodine number
A	corn	123
B	linseed	179
C	olive	81
D	soya	130

12. Dehydration is the removal of water from a single molecule.

Dehydration of butan-2-ol can produce two isomeric alkenes, but-1-ene and but-2-ene, as shown.



Which of the following alcohols can also produce, on dehydration, two isomeric alkenes?

A Propan-2-ol

B Pentan-3-ol

C Hexan-3-ol

D Heptan-4-ol

13. What type of reaction takes place when propan-1-ol is formed from propanoic acid?

A Condensation
B Hydrolysis
C Oxidation
D Reduction

14. The boiling points of the alkanes increase as the carbon chain length increases due to the increasing strength of

A covalent bonds
B hydrogen bonds
C London dispersion forces
D permanent dipole to permanent dipole interactions.

15. Which of the following is **not** a correct statement about the effect of a catalyst?
A catalyst

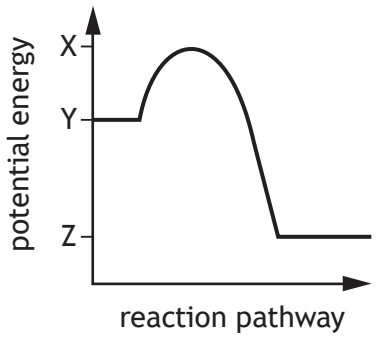
A increases the energy of molecules so that there are more successful collisions
B lowers the energy that molecules need for successful collisions
C provides an alternative reaction pathway to the products
D increases the rate of reaction.

16. 5 moles of ammonium phosphate, $(NH_4)_3PO_4$, contains how many moles of positive ions?

A 3
B 5
C 12
D 15

[Turn over

17. A reaction has the potential energy diagram shown.



The enthalpy change for the forward reaction is

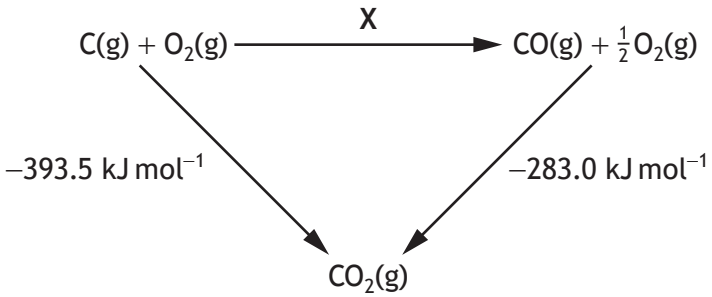
A X − Y

B Y − X

C Y − Z

D Z − Y

18. Consider the reaction pathway shown.



According to Hess's law, the enthalpy change, in kJ mol^{-1}, for reaction **X** is

A −676.5

B −110.5

C +110.5

D +676.5

19. The mean bond enthalpy of the N-H bond is equal to one third of ΔH for which of the following changes?

A $N(g) + 3H(g) \rightarrow NH_3(g)$

B $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

C $\frac{1}{2}N_2(g) + 1\frac{1}{2}H_2(g) \rightarrow NH_3(g)$

D $N(g) + 1\frac{1}{2}H_2(g) \rightarrow NH_3(g)$

20. The equation represents a mixture at equilibrium.

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$$

Which line in the table is true for the mixture after a further two hours of reaction under the same conditions?

	Rate of forward reaction	**Rate of back reaction**
A	decreases	decreases
B	increases	increases
C	unchanged	decreases
D	unchanged	unchanged

21. $C(s) + H_2(g) + O_2(g) \rightarrow HCOOH(\ell)$ $\Delta H = a$

$HCOOH(\ell) + \frac{1}{2}O_2(g) \rightarrow CO_2(g) + H_2O(\ell)$ $\Delta H = b$

$C(s) + O_2(g) \rightarrow CO_2(g)$ $\Delta H = c$

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(\ell)$ $\Delta H = d$

What is the relationship between a, b, c and d?

A $a = c + d - b$

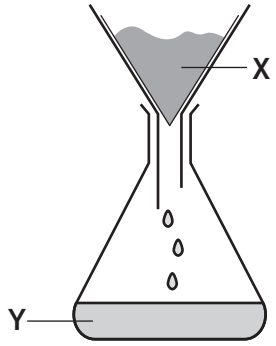
B $a = b - c - d$

C $a = -b - c - d$

D $a = c + b + d$

[Turn over

22. A filtration experiment was carried out to separate **X** and **Y**.



Which line in the table correctly describes **X** and **Y**?

	X	**Y**
A	solvent	residue
B	residue	filtrate
C	filtrate	precipitate
D	residue	solvent

23. Which piece of apparatus should be used to accurately measure 45 cm^3 of a solution?

A 50 cm^3 beaker

B 50 cm^3 pipette

C 50 cm^3 burette

D 50 cm^3 measuring cylinder

24. A student was carrying out a titration.

Which of the following would help the student to accurately observe the end-point?

A Repeating the titration

B Using a white tile under the flask

C Rinsing the flask between titrations

D Disregarding the rough titre

25. Carbon dioxide can be produced by the following reaction.

$$2HCl(aq) + K_2CO_3(aq) \rightarrow 2KCl(aq) + CO_2(g) + H_2O(\ell)$$

The most suitable method of isolating the carbon dioxide is by

A bubbling through calcium hydroxide

B collection over water

C evaporation

D distillation.

[END OF QUESTION PAPER]

FOR OFFICIAL USE

National Qualifications 2024

Mark

X813/76/01

Chemistry Paper 2

THURSDAY, 23 MAY

10:10 AM – 12:30 PM

X8137601

Fill in these boxes and read what is printed below.

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth

Day | Month | Year

Scottish candidate number

Total marks — 95

Attempt ALL questions.

You may use a calculator.

You may refer to the Chemistry Data Booklet for Higher and Advanced Higher.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers and rough work is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting. Any rough work must be written in this booklet. Score through your rough work when you have written your final copy.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

X8137601 01

Total marks — 95

Attempt ALL questions

1. Nitrogen and phosphorus are group 5 elements.

(a) The first ionisation energy of phosphorus is lower than the first ionisation energy of nitrogen.

(i) State what is meant by the term **first ionisation energy**. **1**

(ii) Write the equation for the first ionisation of phosphorus. **1**

(b) The melting points of phosphorus and nitrogen are shown.

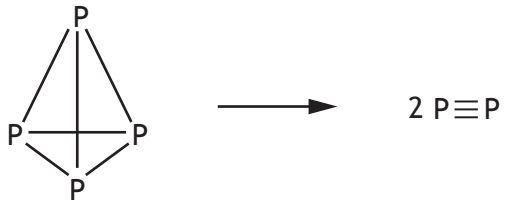
Element	Melting point (°C)
Nitrogen	−210
Phosphorus	44

Explain why the melting point of phosphorus, P_4, is much higher than that of nitrogen, N_2. **3**

In your answer you should refer to the intermolecular forces involved.

1. (continued)

(c) At very high temperatures the following reaction occurs.



Bond	Bond enthalpy (kJ mol^{-1})
P — P	201
P ≡ P	488

Use the bond enthalpy values in the table to calculate the enthalpy change, in kJ mol^{-1}, for the reaction. **2**

[Turn over

1. (continued)

(d) Phosphorus can be obtained by reacting calcium phosphate with silicon dioxide and carbon.

$$2Ca_3(PO_4)_2 + 6SiO_2 + 10C \rightarrow 6CaSiO_3 + 10CO + P_4$$

$GFM = 310.3$ g $\qquad\qquad$ $GFM = 124.0$ g

(i) 750 g of calcium phosphate, $Ca_3(PO_4)_2$, produced 115.5 g of phosphorus.

Calculate the percentage yield of phosphorus, P_4. **2**

(ii) Suggest a term that can be used to describe the action of carbon in this reaction. **1**

(iii) At 1500 °C, 280 g of carbon monoxide gas ($GFM = 28.0$ g) was produced and occupied a volume of 1450 litres.

Use this information to determine the molar volume, in litres, of carbon monoxide at this temperature. **2**

2. Air bags in cars are intended to prevent injuries in a car crash. When an air bag is activated, a series of reactions takes place.

(a) In the first reaction, sodium azide decomposes into sodium metal and nitrogen gas. The nitrogen gas inflates the air bag.

(i) An ignitor supplies the energy required for the reaction to occur.

State the term used to describe the minimum kinetic energy required by particles before a reaction can occur successfully. **1**

(ii) A typical car air bag contains 80 g of sodium azide, NaN_3.

Calculate the energy released, in kJ, when 80 g of sodium azide decomposes. **1**

$$NaN_3 \rightarrow Na + 1\tfrac{1}{2}N_2 \qquad \Delta H = -42.6 \text{ kJ mol}^{-1}$$

$GFM = 65.0$ g

(iii) Sodium azide is an ionic compound.

State the charge on the azide group ion. **1**

[Turn over

2. (continued)

(b) In the second reaction, sodium reacts with potassium nitrate, KNO_3, forming metal oxides and more nitrogen gas.

$$Na + KNO_3 \rightarrow K_2O + Na_2O + N_2$$

Balance this equation. **1**

(c) In the third reaction, the metal oxides react with silicon dioxide, SiO_2, to form less harmful by-products.

Silicon dioxide has a melting point of 1713 °C.

Explain **fully**, in terms of structure and bonding, why silicon dioxide has a high melting point. **2**

2. (continued)

(d) The first and second reactions generate the nitrogen gas needed to inflate the air bag.

For every mole of sodium azide ($GFM = 65.0$ g) that reacts, a total of 1.6 moles of nitrogen gas is formed.

Calculate the volume of gas, in litres, formed when 80 g of sodium azide reacts. **2**

Take the volume of 1 mole of nitrogen to be 24 litres.

(e) The reactants used in air bags are fine powders to ensure the reactions are fast enough to inflate the air bag quickly.

Explain, in terms of collision theory, why fine powders ensure fast reactions in the air bag. **2**

[Turn over

3. Alcohols, aldehydes and ketones can be distinguished using chemical tests.

Using your knowledge of chemistry, describe the experimental procedures, including the expected results, that could be used to distinguish between alcohols, aldehydes and ketones. Any suitable chemicals and apparatus can be used. **3**

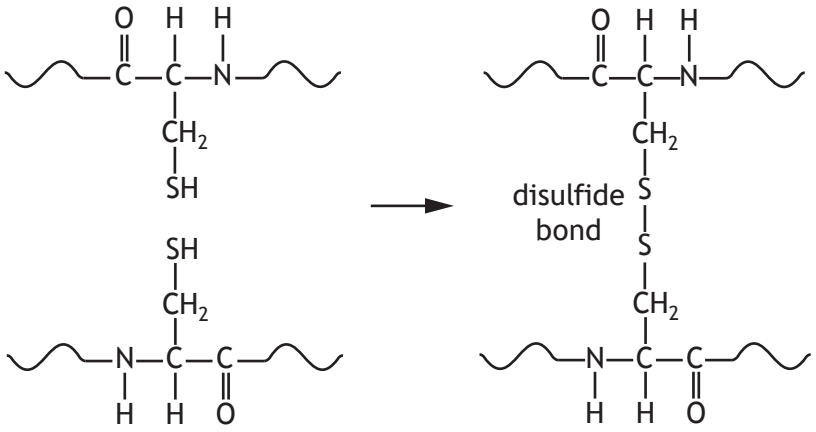
4. Hair and skin contain proteins.

(a) Most of the protein in human hair is keratin.

Approximately 25% of keratin is made from the amino acid cysteine.

(i) Name the other product formed when amino acids join to form proteins such as keratin. **1**

(ii) Cysteine gives keratin a very rigid structure, as it can form strong covalent links, called disulfide bonds, between adjacent keratin fibres.



Suggest a name for the type of reaction occurring when a disulfide bond is formed. **1**

(iii) Keratin also contains smaller percentages of the essential amino acids methionine and histidine.

State what is meant by an essential amino acid. **1**

[Turn over

4. (a) (continued)

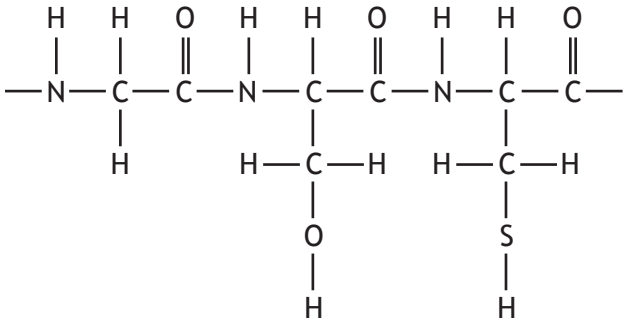
(iv) Hair can be straightened using chemicals.

(A) In the first stage of the straightening process, a chemical is applied that causes the protein fibres to change shape.

Name the process occurring when protein fibres change shape. **1**

(B) In the second stage of the process, short chains of keratin are applied to the hair.

The diagram shows the structure of a section of keratin.



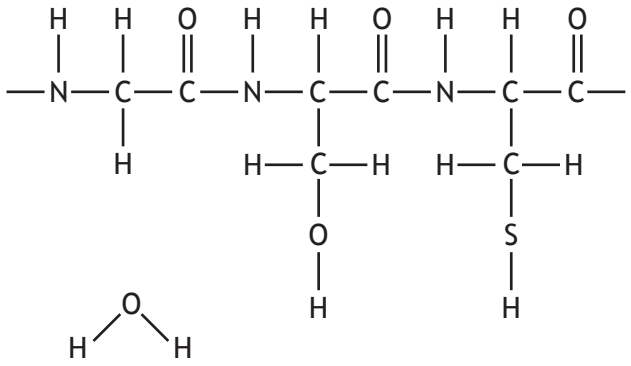
Draw one of the amino acids that make up this section of keratin. **1**

4. (a) (iv) (continued)

(C) The short chains of keratin absorb water, making the hair appear thicker. The water forms hydrogen bonds to parts of the keratin.

Complete the diagram to show where hydrogen bonding could occur between the keratin and a water molecule. **1**

(An additional diagram, if required, can be found on *page 36*.)



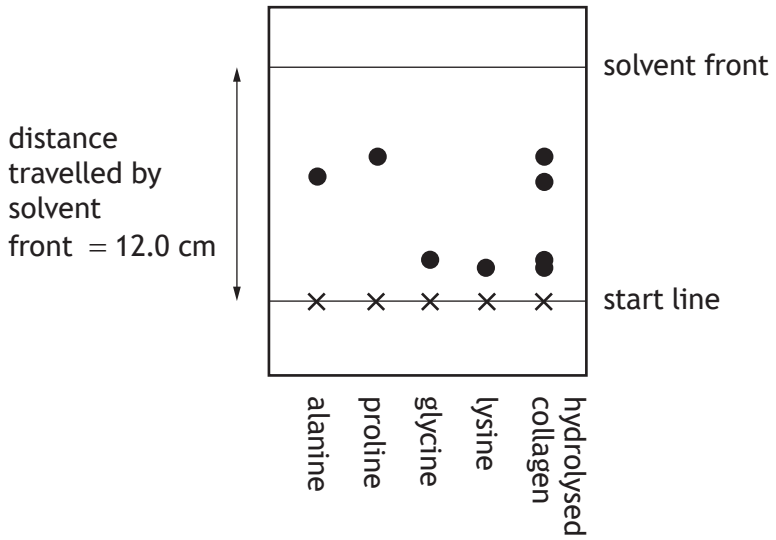
[Turn over

4. (continued)

(b) Collagen is a protein that is found in skin and blood vessels.

A sample of collagen was analysed using paper chromatography to determine its amino acid composition. Amino acid samples were spotted onto paper at the start line, along with a hydrolysed collagen sample. The spots represent the distances travelled by the amino acids carried by the solvent.

The chromatogram obtained is shown.



The R_f value for an amino acid is found using the following equation.

$$R_f = \frac{\text{distance travelled by amino acid}}{\text{distance travelled by solvent front}}$$

The table shows the distances travelled by the amino acids in this sample.

Amino acid	Distance travelled (cm)
Alanine	5.2
Proline	5.6
Glycine	1.8
Lysine	1.6

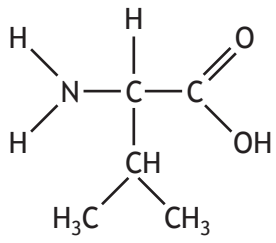
(i) Calculate the R_f value for proline. **1**

4. (b) (continued)

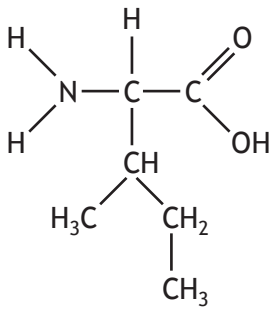
(ii) The sample of collagen contained all four amino acids.

Suggest why there are only three spots in the chromatogram from hydrolysed collagen. **1**

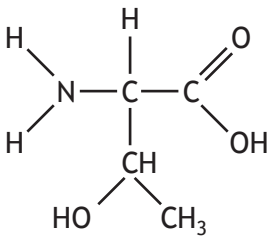
(iii) More polar amino acids are carried for shorter distances by the solvent, as they attach more firmly to the chromatography paper.



valine



isoleucine



threonine

Predict which of the amino acids shown would have the smallest R_f value. **1**

[Turn over

MARKS | DO NOT WRITE IN THIS MARGIN

5. Edible fats and oils form part of a balanced diet.

(a) State a reason why fats and oils form part of a balanced diet. **1**

(b) Fats and oils are formed using the alcohol glycerol.

(i) Draw a structural formula for glycerol. **1**

(ii) Name the functional group present in all fats and oils. **1**

5. (continued)

(c) Explain **fully** why fats have higher melting points than oils. **3**
In your answer you should refer to the structure of fats and oils.

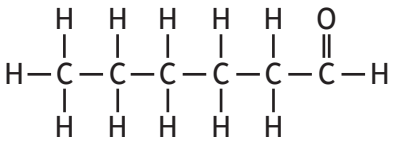
[Turn over

5. (continued)

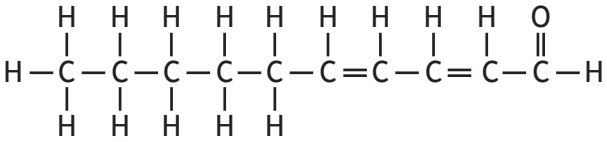
(d) Fats and oils can react with oxygen from the air.

(i) State the term used to describe the resulting unpleasant flavour. **1**

(ii) Two of the molecules responsible for the unpleasant flavour are shown below.



molecule A



molecule B

Describe a chemical test, and the expected result, that could be used to distinguish between these two molecules. **1**

5. (d) (continued)

(iii) (A) The reaction of fats and oils with oxygen takes place via a free radical chain reaction.

State what is meant by the term free radical. **1**

(B) The steps in the reaction are shown below, RH is used to represent a fat or oil molecule.

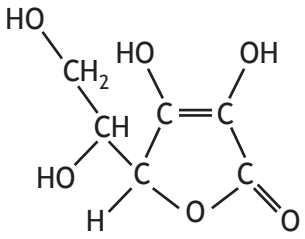
Step	Reactions
1	$RH \rightarrow R^{\bullet} + H^{\bullet}$
2	$R^{\bullet} + O_2 \rightarrow ROO^{\bullet}$ $ROO^{\bullet} + RH \rightarrow ROOH + R^{\bullet}$
3	$R^{\bullet} + R^{\bullet} \rightarrow R{-}R$ $ROO^{\bullet} + R^{\bullet} \rightarrow ROOR$

Name Step 1. **1**

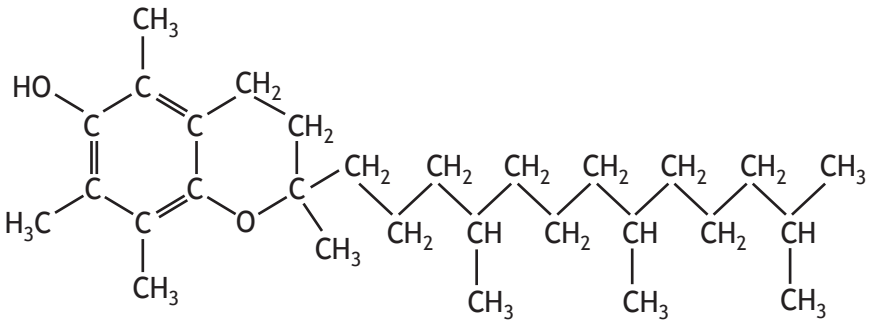
[Turn over

5. (d) (continued)

(iv) Vitamin C and vitamin E are antioxidants used in some foods.



vitamin C



vitamin E

(A) Antioxidants, such as vitamin C and E, can prevent oxidation by acting as free radical scavengers.

State how free radical scavengers prevent chain reactions from occurring. **1**

(B) Antioxidants are used as preservatives in oils and fatty foods.

Explain **fully** why vitamin E is more suitable than vitamin C as an antioxidant in oils and fatty foods. **2**

5. (continued)

(e) Soaps are formed by the alkaline hydrolysis of edible fats and oils.

(i) Soaps are salts of fatty acids. One common fatty acid used in soaps is myristic acid, $C_{13}H_{27}COOH$.

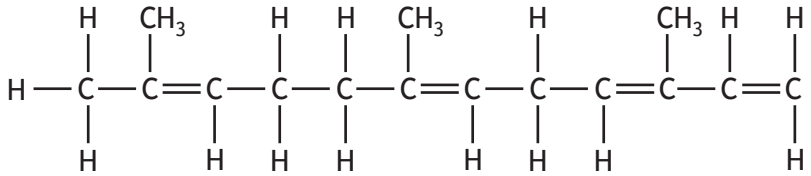
Write a formula for the sodium salt of myristic acid. **1**

(ii) Explain why soapless detergents are used in areas of hard water instead of soaps. **1**

(iii) (A) Essential oils containing terpenes can be added to soaps to provide fragrance. Terpenes are formed by joining together isoprene units.

State the systematic name for an isoprene unit. **1**

(B) Farnesene is a terpene often found in floral essential oils.



farnesene

State the number of isoprene units in farnesene. **1**

[Turn over

6. Biodiesel is a fuel that can be made from plant oils and is a mixture of compounds.

(a) Oils are converted into biodiesel in a process called trans-esterification.

$$\text{oil} + \text{methanol} \rightleftharpoons \text{biodiesel} + \text{glycerol}$$

(i) One of the compounds in biodiesel is the ester shown.

$$CH_3(CH_2)_6COOCH_3$$

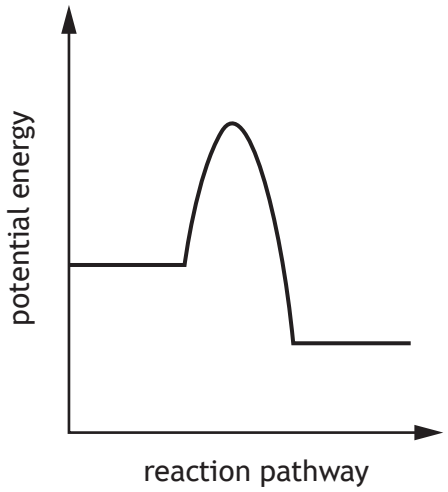
State the systematic name for this ester. **1**

(ii) Esters can be prepared in the laboratory by heating a carboxylic acid with an alcohol and a few drops of catalyst. Some of the reactants and products are volatile and flammable, so it is important to prevent vapours from escaping.

(A) Draw a labelled diagram to show apparatus suitable for preparing an ester. **2**

6. (a) (ii) (continued)

(B) The potential energy diagram below shows the change in potential energy for the reaction to make an ester using a catalyst.



Add a dotted line to the diagram to show the change in potential energy for the formation of an ester carried out **without** a catalyst. **1**

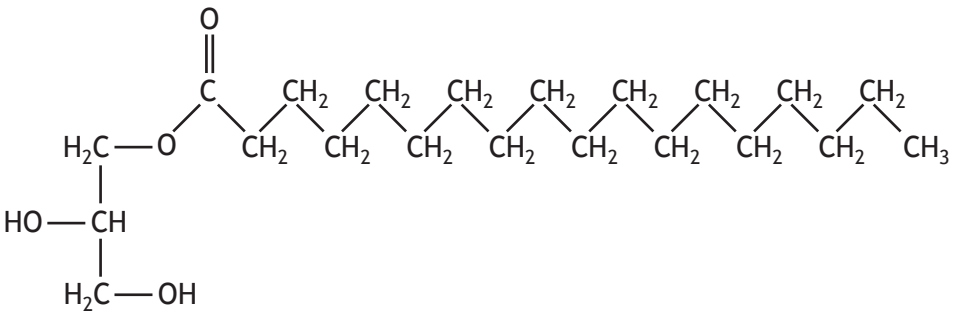
(An additional diagram, if required, can be found on *page 36*.)

[Turn over

6. (continued)

(b) Emulsifiers are added to biodiesel so that a stable emulsion can be formed with any water present.

An emulsifier molecule is shown.



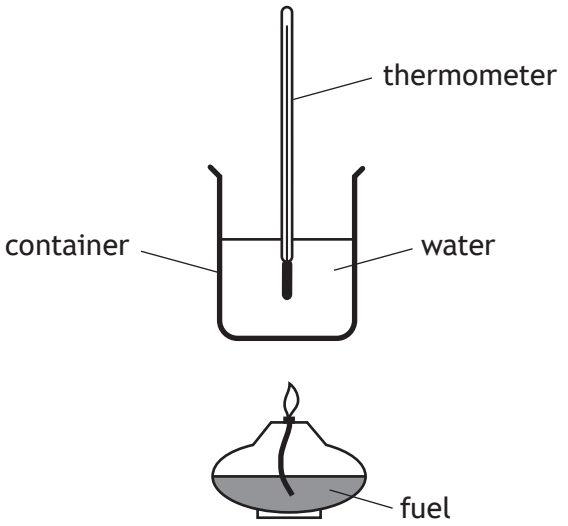
(i) Explain **fully** the reasons this molecule can act as an emulsifier. **2**

(ii) The emulsifiers used in biodiesel and foods can be made from edible oils.

State how emulsifiers are made from edible oils. **1**

6. (continued)

(c) The energy released per gram of fuel can be measured using the following apparatus.



A student wanted to compare the energy released by different types of biodiesel.

(i) Suggest one variable the student would need to keep the same in both experiments to make a fair comparison. **1**

(ii) Heat is lost to the surroundings during the experiment.

Suggest an improvement to the apparatus that would prevent heat loss to the surroundings. **1**

[Turn over

6. (c) (continued)

(iii) The student's results for one type of biodiesel are shown in the table.

Mass of biodiesel burned (g)	0.420
Volume of water (cm^3)	200
Initial temperature of water (°C)	17
Final temperature of water (°C)	38

Calculate the energy released, in $kJ\,g^{-1}$, when 1.0 g of the biodiesel was burned. **3**

(d) E20 is a mixture of 20% biodiesel and 80% regular diesel and is sold for use in diesel vehicles. Biodiesel costs £0.85 per litre.

Calculate the cost of the biodiesel used to produce 75 litres of E20. **1**

7. Seawater contains several dissolved salts.

(a) The salinity of seawater is a measure of the total amount of dissolved salts and can be calculated using the following formula.

Salinity (parts per thousand) = chloride ion concentration ($mg\,l^{-1}$) × 0.0018066

A sample of seawater had a salinity of 35 parts per thousand.

Calculate the chloride ion concentration, in $\mathbf{g\,l^{-1}}$, of this sample. **2**

(b) Another sample of seawater was analysed to determine the mass of chloride ions present. The sample was titrated with a standard solution of silver nitrate solution.

(i) (A) State what is meant by a standard solution. **1**

(B) Describe how a pipette should be prepared and used to accurately measure the sample of seawater. **2**

(ii) An indicator was added to the seawater in a conical flask.

State why an indicator is used. **1**

[Turn over

7. (b) (continued)

(iii) Three 10.0 cm^3 samples of seawater were titrated with a 0.5 mol l^{-1} silver nitrate solution.

The table shows the results obtained.

Sample	Volume of silver nitrate used (cm^3)
1	11.90
2	11.60
3	11.50

(A) Explain why the average titre value is 11.55 cm^3, not 11.70 cm^3. **1**

(B) In this titration, one mole of chloride ions, $Cl^-(aq)$, reacts with one mole of silver ions, $Ag^+(aq)$.

Using the average titre value, calculate the concentration, in mol l^{-1}, of chloride ions in the seawater. **3**

7. (b) (iii) (continued)

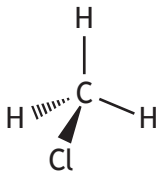
(C) Another sample of a different seawater contained 0.00492 moles of chloride ions in a 10 cm^3 sample.

Calculate the mass of chloride ions, in grams, in one litre of seawater. **2**

[Turn over

7. (continued)

(c) Some micro-organisms living in seawater convert chloride ions to chloromethane. The chloromethane then evaporates.



Explain **fully** why chloromethane has a lower boiling point than water. **2**

In your answer you should refer to the intermolecular forces involved.

8. Reversible reactions can cause challenges for chemists trying to maximise yields in chemical processes.

Using your knowledge of chemistry, suggest how yields can be maximised in reversible reactions. **3**

9. Titanium is a metal which is expensive to extract and purify.

It can be extracted from an ore in a two-step process called the Kroll process.

(a) The first step involves heating the ore with chlorine and carbon to convert titanium dioxide, TiO_2, to titanium chloride, $TiCl_4$.

$$TiO_2 + 2Cl_2 + C \rightarrow TiCl_4 + CO_2$$

$GFM = 79.9$ g $\quad GFM = 71.0$ g

The largest reactors can process 1600 kg of titanium dioxide per day.

(i) Calculate the mass of chlorine, in kg, required to react with 1600 kg of titanium dioxide, TiO_2. **2**

(ii) Titanium chloride is a liquid at room temperature.

Suggest the type of **bonding and structure** present in titanium chloride. **1**

9. (continued)

(b) In the second step, titanium chloride reacts with magnesium to produce titanium metal. The reaction is carried out in a sealed vessel, in an argon atmosphere, at 1500 °C.

$$\underset{GFM = 189.9\text{ g}}{TiCl_4} + \underset{GFM = 24.3\text{ g}}{2Mg} \rightarrow \underset{GFM = 47.9\text{ g}}{Ti} + \underset{GFM = 95.3\text{ g}}{2MgCl_2}$$

(i) Calculate the atom economy for the production of titanium in this reaction. **2**

(ii) Write the ion-electron equation for the oxidation of magnesium atoms. **1**

(iii) In a reaction, 1900 kg of titanium chloride was reacted with 750 kg of magnesium. Magnesium was the reactant in excess.

Calculate the number of moles of magnesium left unreacted. **3**

9. (b) (continued)

(iv) Suggest why the reaction is carried out in an argon atmosphere. **1**

(c) A new process for extraction of titanium has been developed.

An electric current is passed through molten titanium dioxide at a temperature of 900 °C. Titanium metal is produced and the only by-product is carbon dioxide.

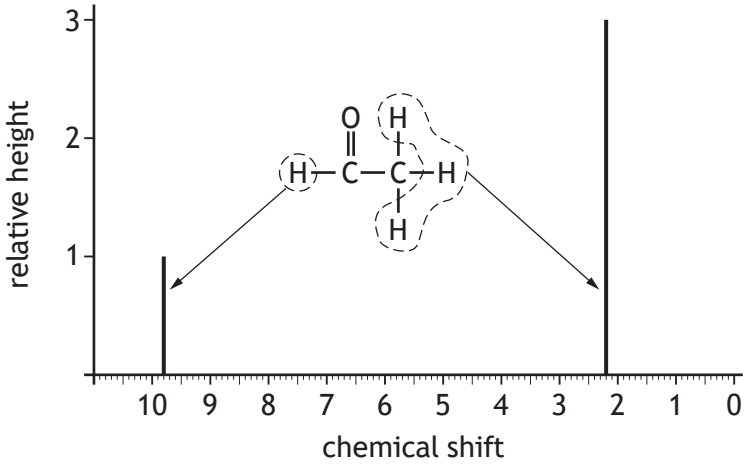
Suggest why this method could be considered preferable to the Kroll process. **1**

10. A hydrogen NMR spectrum can be used to help determine the structure of an organic molecule.

In a hydrogen NMR spectrum:

- a vertical line represents the hydrogen atom(s) in a specific structural environment in a molecule
- the position of the vertical line on the x-axis gives the 'chemical shift' value for the hydrogen atom(s) in that structural environment
- the height of the vertical line is related to the number of hydrogen atom(s) in that structural environment.

An example showing a hydrogen NMR spectrum for ethanal is shown.

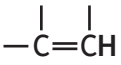


(a) Chemical shift values of hydrogen atoms in different structural environments fall within a range and are listed on *page 17* of the data booklet.

For example, the chemical shift value for the hydrogen atom in the aldehyde group shown falls within the range 10.0−9.4.



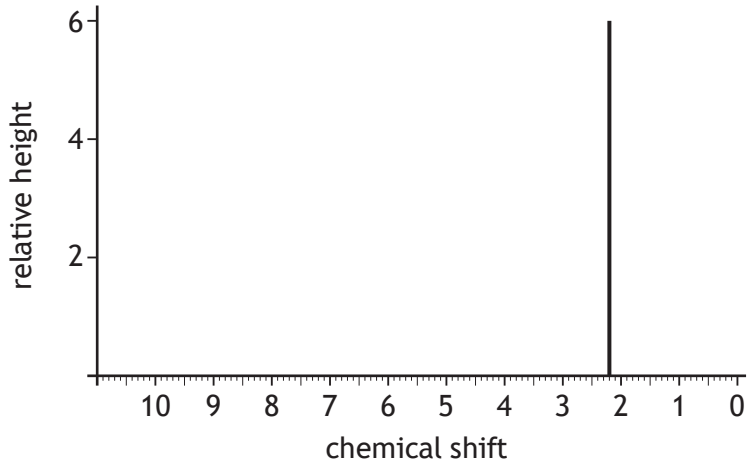
Use the data booklet to find the range in the chemical shift values for the hydrogen atom in the following structural environment. **1**



[Turn over

10. (continued)

(b) The hydrogen NMR spectrum for a compound with the molecular formula C_3H_6O is shown.



Suggest a name for this compound. **1**

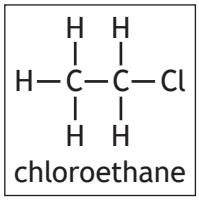
10. (continued)

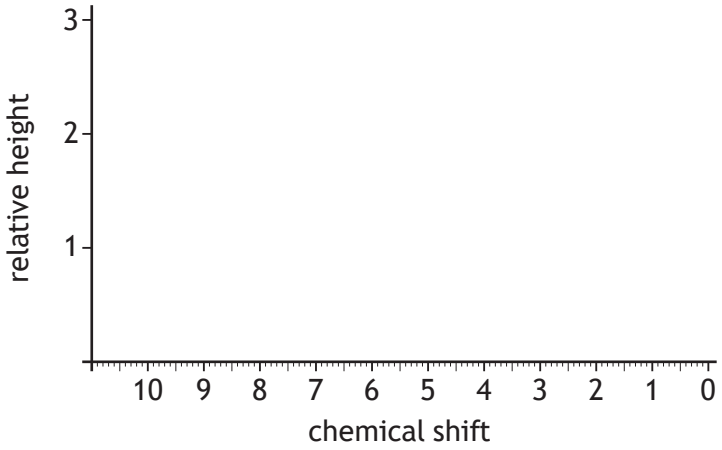
(c) The chemical shift values shown in the table are based on the range of values given in the data booklet.

Structural environments of hydrogen atoms	Chemical shift
$-CH_3$	1.0
$-CH_2Cl$	3.7

Using information from the table above, draw the hydrogen NMR spectrum that would be obtained for chloroethane. **1**

(An additional diagram, if required, can be found on *page 37*.)





[END OF QUESTION PAPER]

National
Qualifications
2023

X813/76/12

Chemistry
Paper 1 — Multiple choice

FRIDAY, 12 MAY
9:00 AM – 9:40 AM

Total marks — 25

Attempt ALL questions.

You may use a calculator.

Instructions for the completion of Paper 1 are given on *page 02* of your answer booklet X813/76/02.

Record your answers on the answer grid on *page 03* of your answer booklet.

You may refer to the Chemistry Data Booklet for Higher and Advanced Higher.

Space for rough work is provided at the end of this booklet.

Before leaving the examination room you must give your answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

X8137612

Total marks — 25

Attempt ALL questions

1. Which of the following compounds has the least ionic character?

A Sodium iodide

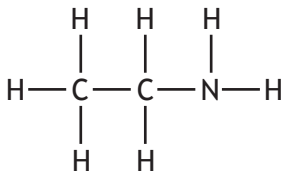
B Sodium fluoride

C Potassium iodide

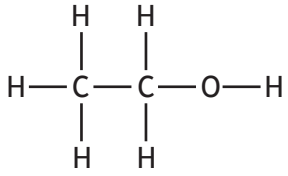
D Potassium fluoride

2. In which of the following compounds would hydrogen bonding **not** occur?

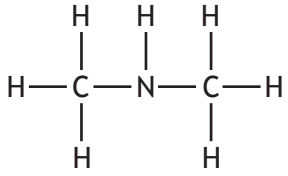
A



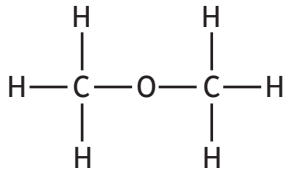
B



C



D



3. Fats are formed from glycerol molecules and fatty acid molecules.
The mole ratio of glycerol molecules to fatty acid molecules is

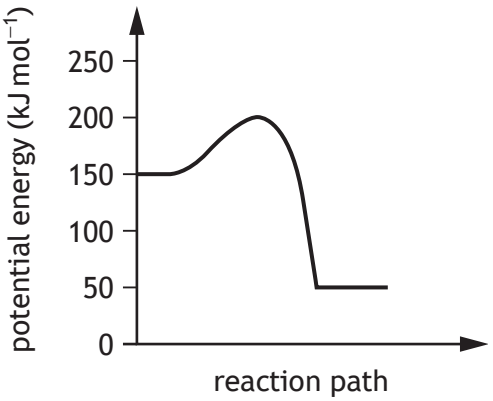
A 1 : 2

B 2 : 1

C 1 : 3

D 3 : 1

4. A reaction was carried out as shown in the energy diagram.



Which of the following has a value of 150 kJ mol^{-1}?

A Activation energy of the reverse reaction

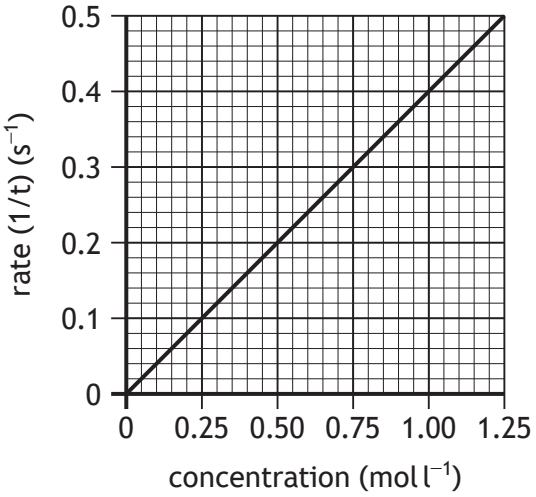
B Enthalpy change of the reverse reaction

C Activation energy of the forward reaction

D Enthalpy change of the forward reaction

[Turn over

5. The graph shows how the rate of a reaction varies with the concentration of one of the reactants.



What was the concentration, in mol l^{-1}, when the reaction time was 10 s?

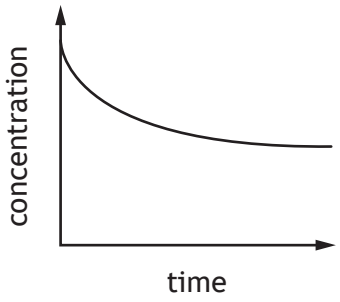
A 0.04

B 0.10

C 0.25

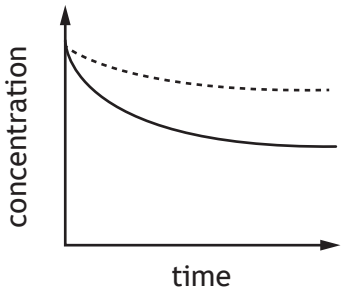
D 0.40

6. The diagram represents the change in concentration of a reactant against time during a reversible chemical reaction.

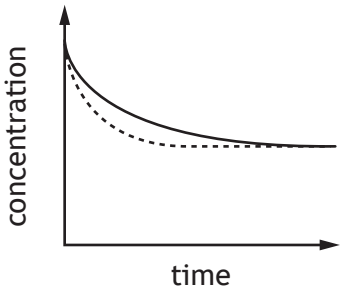


In which diagram below does the dotted line show the result of repeating the reaction using a catalyst?

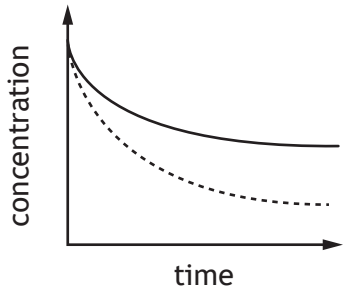
A



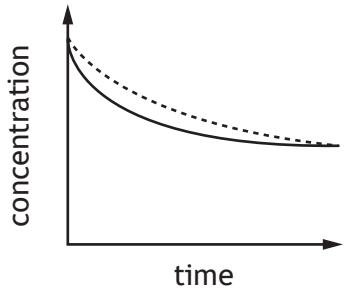
B



C



D



[Turn over

7. The enthalpy of combustion of methanol ($GFM = 32.0$ g) is -726 kJ mol^{-1}.
What mass of methanol has to be burned to produce 145.2 kJ?

A 3.2 g
B 6.4 g
C 32.0 g
D 160.0 g

8. Which of the following statements is true?

A The sodium atom is larger than the sodium ion.
B The chloride ion is smaller than the chlorine atom.
C The magnesium ion is larger than the magnesium atom.
D The oxygen atom is larger than the oxide ion.

9. Which of the following structures is **never** found in compounds?

A Covalent molecular
B Covalent network
C Monatomic
D Ionic

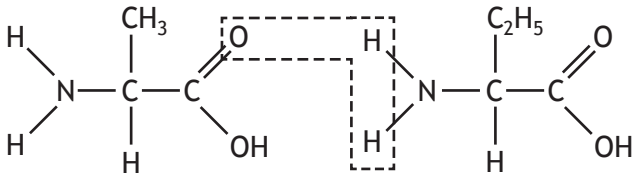
10. Which of the following carbon containing compounds is an isomer of hexanal?

A 2-methylbutanal
B 3-methylpentan-2-one
C 2,2-dimethylbutan-1-ol
D 3,3-dimethylpentanal

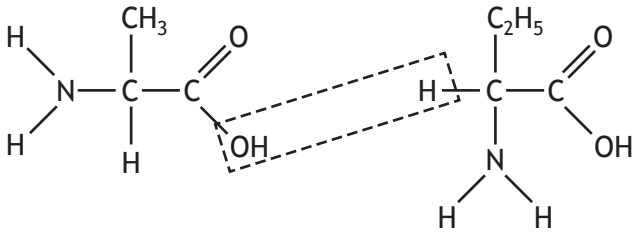
11. When two amino acids react in a condensation reaction, water is eliminated and a peptide link is formed.

Which of the following represents this process?

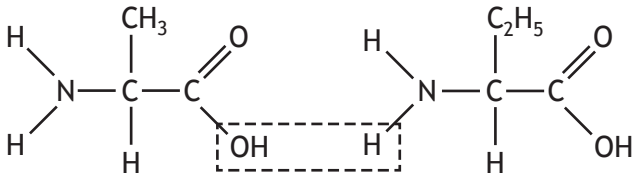
A



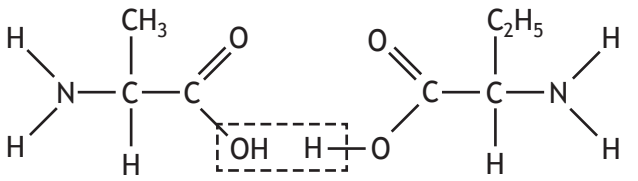
B



C



D



[Turn over

12. Which of the following equations represents an enthalpy of combustion?

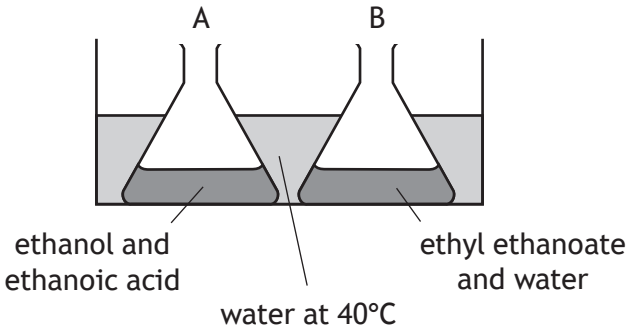
A $2CH_4(g) + 3O_2(g) \rightarrow 2CO(g) + 4H_2O(\ell)$

B $CH_4(g) + 1\frac{1}{2}O_2(g) \rightarrow CO(g) + 2H_2O(\ell)$

C $2C_2H_6(g) + 7O_2(g) \rightarrow 4CO_2(g) + 6H_2O(\ell)$

D $C_2H_6(g) + 3\frac{1}{2}O_2(g) \rightarrow 2CO_2(g) + 3H_2O(\ell)$

13. Two flasks, A and B, were placed in a water bath at 40°C.



After several days, the contents of the flasks were analysed.

Which results would be expected?

A Flask A contains ethyl ethanoate, water, ethanol and ethanoic acid; flask B is unchanged.

B Flask A contains only ethyl ethanoate and water; flask B is unchanged.

C Flask A contains only ethyl ethanoate and water; flask B contains only ethanol and ethanoic acid.

D Flask A and flask B contain ethyl ethanoate, water, ethanol and ethanoic acid.

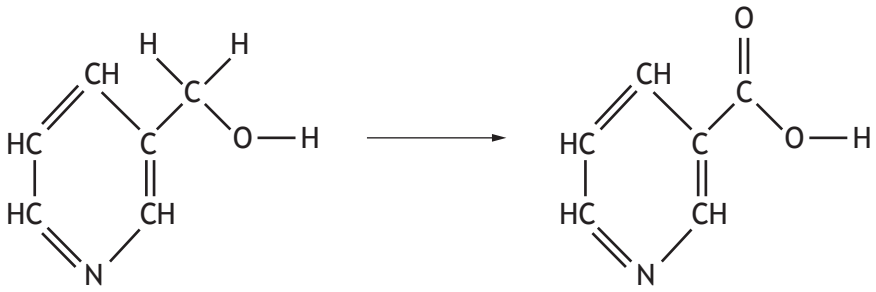
14. During a redox process in acid solution, iodate ions are converted into iodine.

$$2IO_3^-(aq) + 12H^+(aq) + xe^- \rightarrow I_2(aq) + 6H_2O(\ell)$$

What value of ***x*** is required to balance the equation?

A 12
B 11
C 10
D 6

15. A step in the synthesis of vitamin B_3 is shown.



What name is given to this type of reaction?

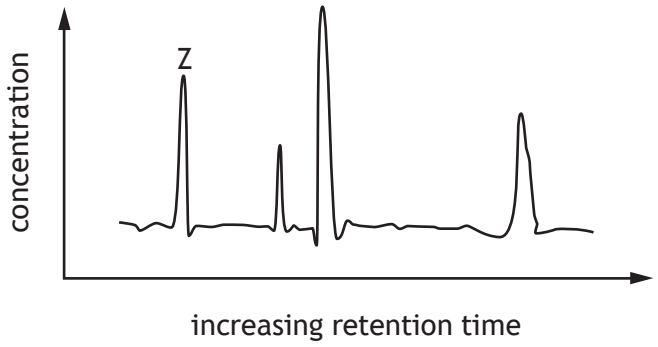
A Condensation
B Hydration
C Reduction
D Oxidation

[Turn over

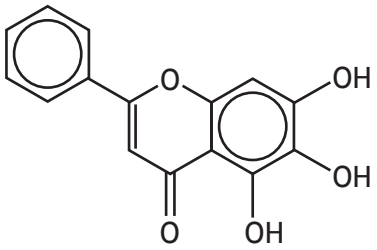
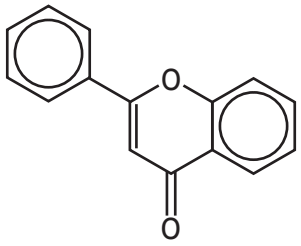
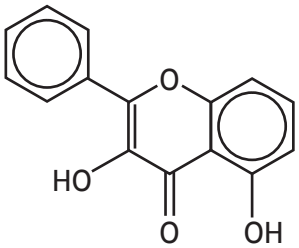
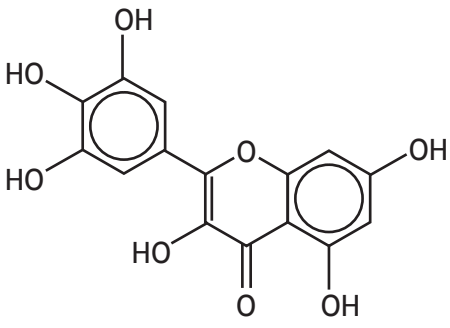
16. A chemist analysed a mixture of four dyes, A, B, C and D, using gas-liquid chromatography.

The time taken to travel through the column (retention time) depends on the polarity of the molecule. The more polar the molecule the longer the retention time.

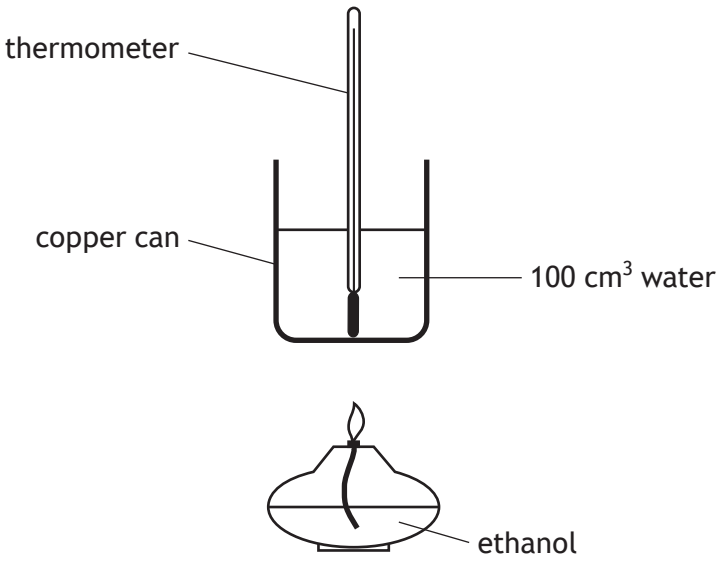
The following chromatogram was obtained.



Which of the following compounds corresponds to peak Z?

Dye	Structure
A	
B	
C	
D	

17. The apparatus was used to measure the enthalpy of combustion of ethanol.



Which of the following would **not** improve the accuracy of the result?

A Using a draught shield

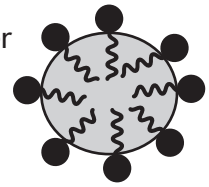
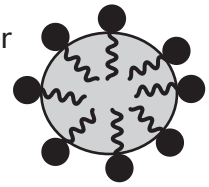
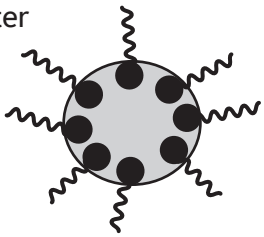
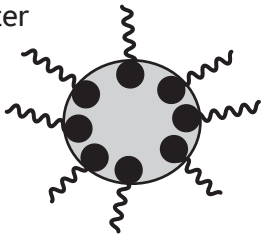
B Moving the thermometer

C Using a glass beaker instead of a copper can

D Stirring the water

[Turn over

18. Which line in the table best describes the ball-like structures formed when soap is added to an oil and water mixture?

	Diagram	Description
A	water 	non-polar head dissolves in water, ionic tail dissolves in oil droplet
B	water 	ionic head dissolves in water, non-polar tail dissolves in oil droplet
C	water 	non-polar head dissolves in oil droplet, ionic tail dissolves in water
D	water 	ionic head dissolves in oil droplet, non-polar tail dissolves in water

19. In an experiment, nickel oxide is added to sulfuric acid until no more nickel oxide reacts. The products are nickel sulfate and water.

The correct method to separate and collect a dry, pure sample of nickel sulfate is

A evaporation

B filtration

C filtration followed by evaporation

D evaporation followed by filtration.

20. Which of the following compounds would react with sodium hydroxide solution to form a salt?

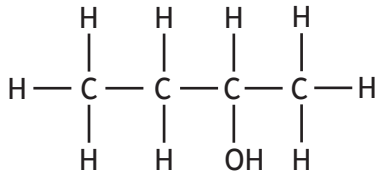
A CH_3CHO

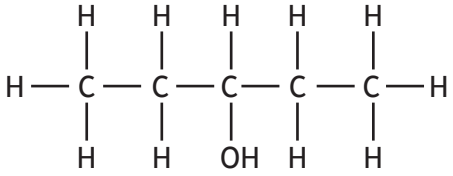
B CH_3COOH

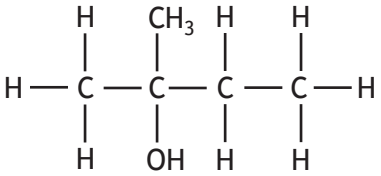
C CH_3COCH_3

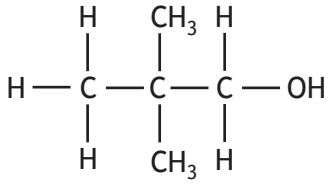
D CH_3CH_2OH

21. Which structural formula represents a primary alcohol?

A 

B 

C 

D 

22. Reduction of 4-methylpentan-2-one to the corresponding alcohol results in the molecule

A gaining 2 g per mole

B losing 2 g per mole

C losing 16 g per mole

D not changing in mass.

23. Which of the following gas samples has the same volume as 16.0 g of oxygen?
(All volumes are measured at the same temperature and pressure)

A 21.0 g of carbon monoxide

B 44.0 g of carbon dioxide

C 46.0 g of nitrogen dioxide

D 46.0 g of dinitrogen tetroxide

[Turn over

24. The number of moles of positive ions in 0.25 moles of aluminium sulfate is

A 0.5

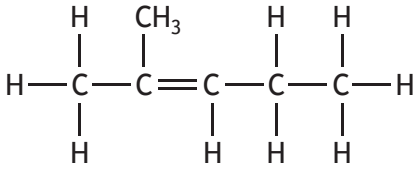
B 1.0

C 2.0

D 3.0

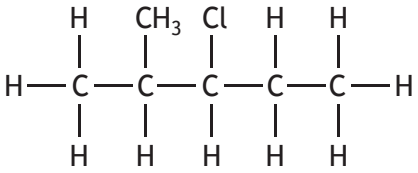
25. Addition of hydrogen chloride, HCl, to an alkene can give a mixture of two products. The product produced in the greatest amount in the reaction is called the major product.

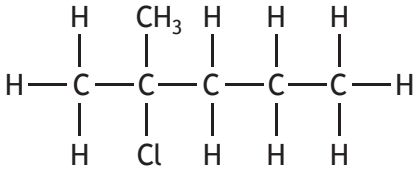
The major product is formed when the hydrogen atom of HCl attaches to the carbon atom of the double bond that has the greatest number of hydrogen atoms attached.

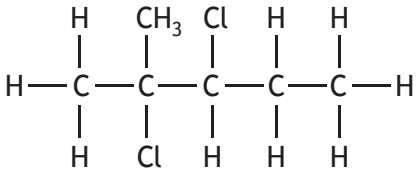


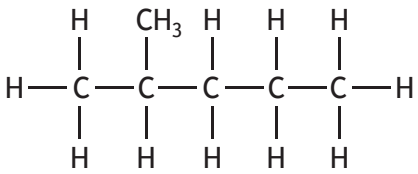
2-methylpent-2-ene

The major product in the reaction of HCl with the 2-methylpent-2-ene is

A 

B 

C 

D 

[END OF QUESTION PAPER]

FOR OFFICIAL USE

National Qualifications 2023

Mark

X813/76/01

Chemistry Paper 2

FRIDAY, 12 MAY

10:10 AM – 12:30 PM

X8137601

Fill in these boxes and read what is printed below.

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth

Day | Month | Year

Scottish candidate number

Total marks — 95

Attempt ALL questions.

You may use a calculator.

You may refer to the Chemistry Data Booklet for Higher and Advanced Higher.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers and rough work is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting. Any rough work must be written in this booklet. Score through your rough work when you have written your final copy.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

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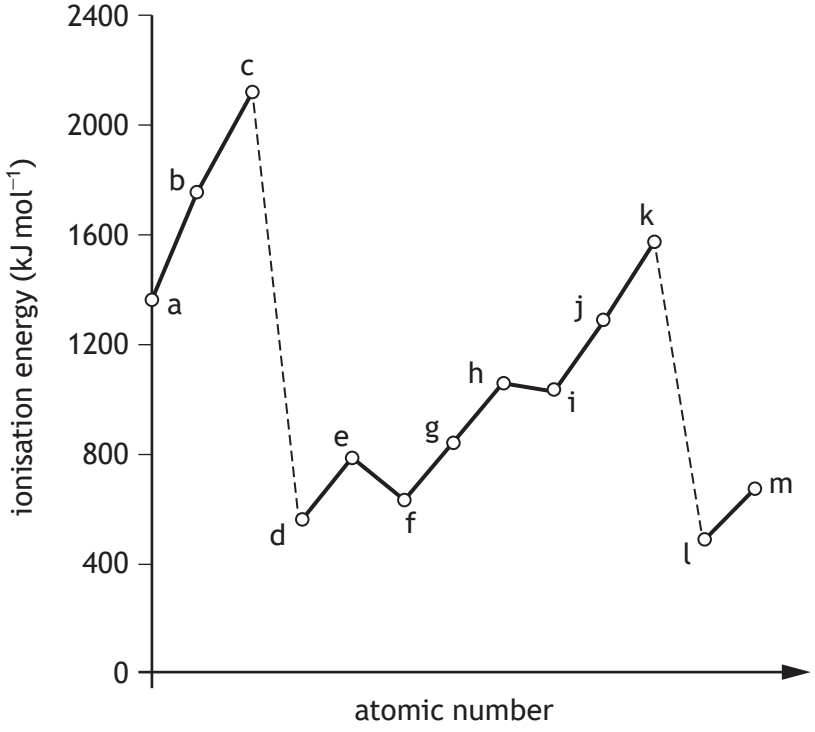
Total marks — 95

Attempt ALL questions

1. Elements are arranged in the periodic table in order of increasing atomic number. Many physical and chemical properties of the elements show periodic trends.

(a) First ionisation energy is a property that has a periodic trend.

The diagram shows part of a graph of first ionisation energy against atomic number for some elements in the periodic table.



(i) Explain why there is an increase in first ionisation energy from elements **d** to **k** in the diagram. **1**

(ii) State an element from **a** to **m** in the diagram that represents an element from group 7. **1**

1. (a) (continued)

(iii) The table shows four ionisation energies of sodium.

Ionisation energy ($kJ\,mol^{-1}$)			
First	Second	Third	Fourth
496	4562	6910	9543

(A) Explain **fully** the large increase between the first and second ionisation energies of sodium. **2**

(B) Use the information in the table to determine the enthalpy change, in $kJ\,mol^{-1}$, for the following reaction. **1**

$$Na^{+}(g) \rightarrow Na^{3+}(g) + 2e^{-}$$

[Turn over

1. (continued)

(b) Electronegativity is another property that has a periodic trend.

(i) State what is meant by the term electronegativity. **1**

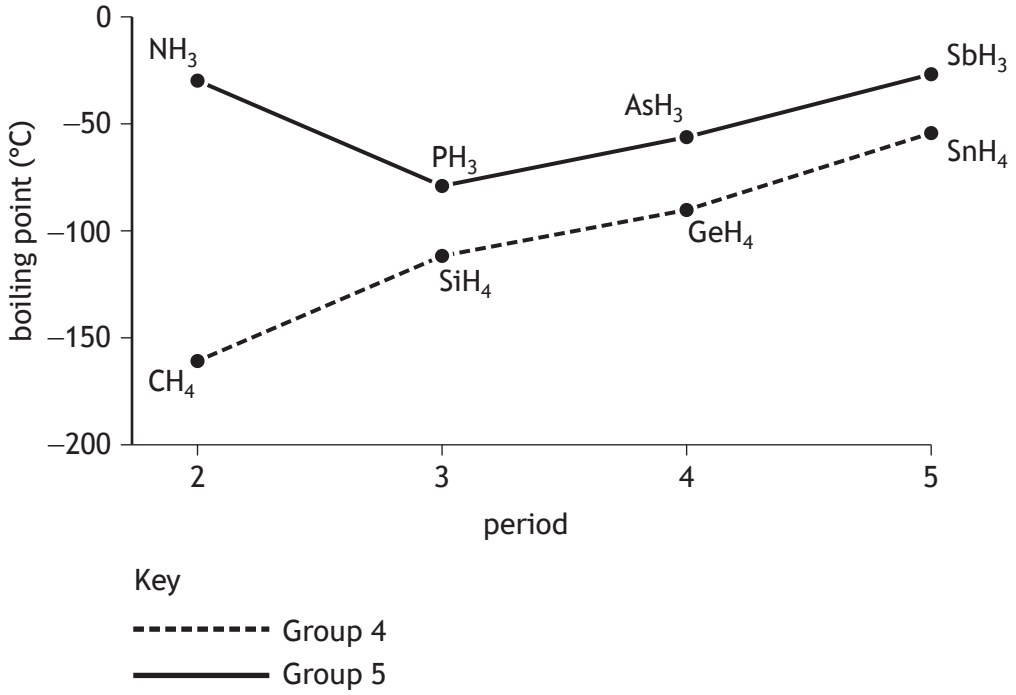
(ii) Explain **fully** why electronegativity decreases going down a group. **2**

(iii) Suggest which of the group 2 elements is the best reducing agent. **1**

2. Hydrides are compounds containing hydrogen and one other element.

MARKS DO NOT WRITE IN THIS MARGIN

(a) The graph shows the boiling points of group 4 and group 5 hydrides.



(i) Explain **fully** why the boiling points of the group 4 hydrides increase going down the group.

In your answer you should refer to the intermolecular forces involved. **3**

(ii) Name the type of intermolecular force that is responsible for the anomalous boiling point of ammonia, NH_3. **1**

[Turn over

2. (continued)

(b) (i) Silicon hydride, SiH_4, can be formed by reacting silicon with hydrogen.

$$Si(s) + 2H_2(g) \rightarrow SiH_4(g)$$

Calculate the enthalpy change, in kJ mol^{-1}, for this reaction using the following information. **2**

$$SiH_4(g) + 2O_2(g) \rightarrow SiO_2(s) + 2H_2O(\ell) \quad \Delta H = -1517 \text{ kJ mol}^{-1}$$

$$Si(s) + O_2(g) \rightarrow SiO_2(s) \quad \Delta H = -911 \text{ kJ mol}^{-1}$$

$$H_2(g) + \tfrac{1}{2}O_2(g) \rightarrow H_2O(\ell) \quad \Delta H = -286 \text{ kJ mol}^{-1}$$

2. (b) (continued)

(ii) One method of preparing silicon hydride involves reacting magnesium silicide, Mg_2Si, with hydrochloric acid, HCl.

15.32 g of magnesium silicide was reacted with excess hydrochloric acid. 2.56 g of silicon hydride was produced.

$$4HCl + Mg_2Si \rightarrow SiH_4 + 2MgCl_2$$

$$GFM = 76.7 \text{ g} \qquad GFM = 32.1 \text{ g}$$

Calculate the percentage yield of silicon hydride. **2**

(iii) The table shows the melting points of silicon hydride, SiH_4, and silicon oxide, SiO_2.

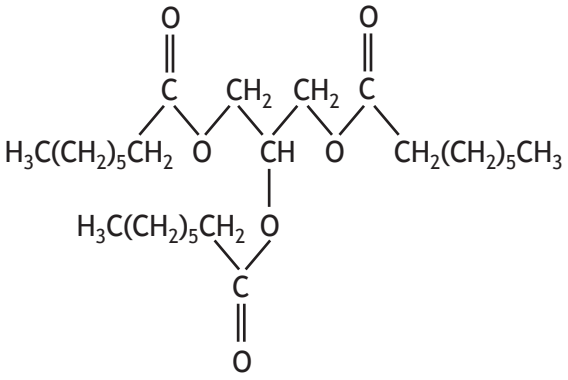
	Melting point (°C)
SiH_4	−185
SiO_2	1710

Explain **fully** why silicon oxide has a much higher melting point than silicon hydride. **3**

[Turn over

3. Cheese is a complex substance containing a wide variety of chemicals.

(a) The structure of a fat found in cheese is shown below.



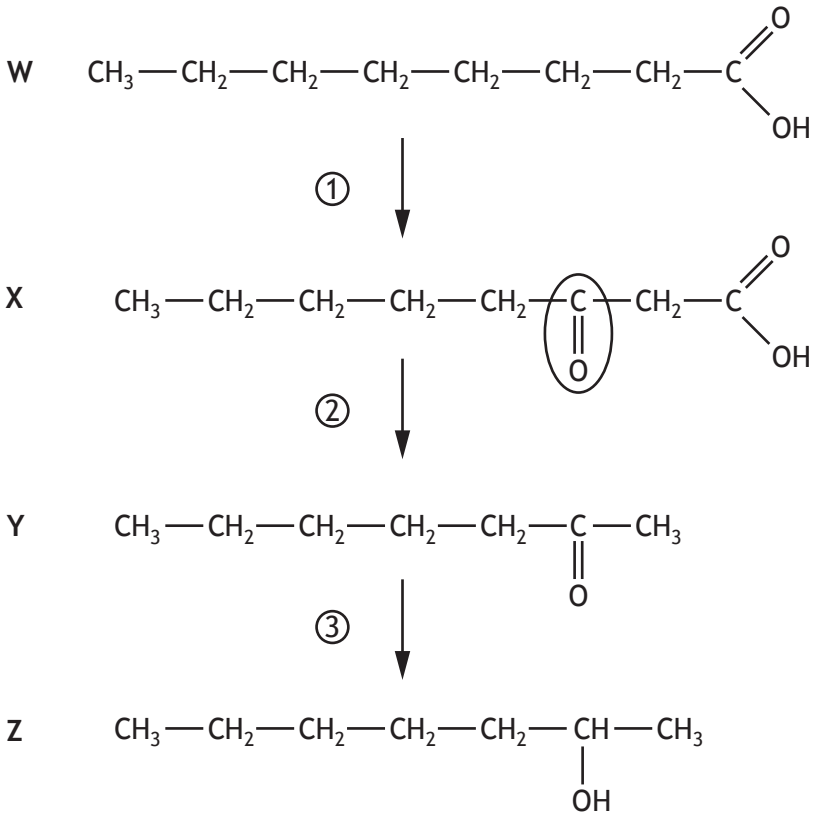
(i) (A) The alcohol needed to form fat molecules is glycerol.

State the systematic name for glycerol. **1**

(B) Name the type of reaction used to form fat molecules from fatty acids and glycerol. **1**

3. (a) (continued)

(ii) Fatty acid **W** reacts as shown.



(A) Identify the functional group circled in molecule **X**. **1**

(B) Name molecule **Y**. **1**

(C) Identify the type of reaction used to convert molecule **Y** into molecule **Z**. **1**

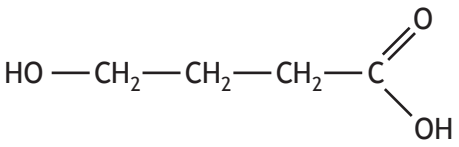
(D) State which of the reactions, ①, ② or ③, results in an increase in the oxygen to hydrogen ratio. **1**

[Turn over

3. (a) (continued)

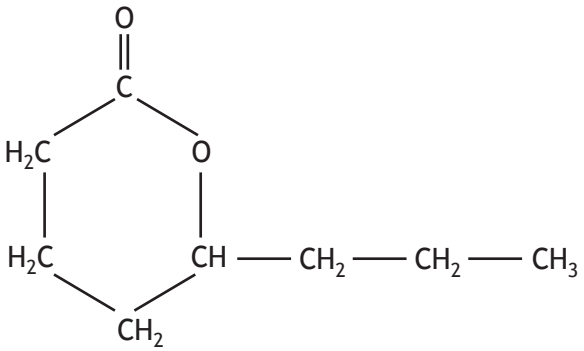
(iii) Fatty acids can be converted into hydroxycarboxylic acids.

An example of a hydroxycarboxylic acid is shown.



The two functional groups in a hydroxycarboxylic acid react together to form a cyclic ester.

An example of a cyclic ester is shown.



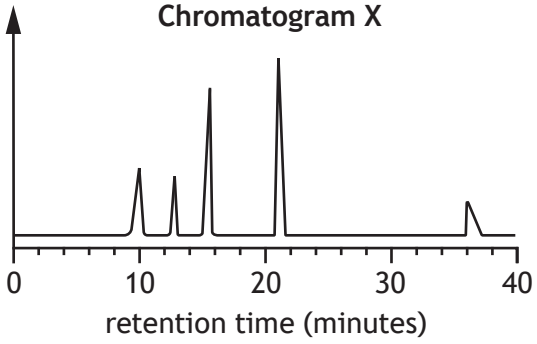
Draw a structural formula for the hydroxycarboxylic acid that can be used to produce this cyclic ester. **1**

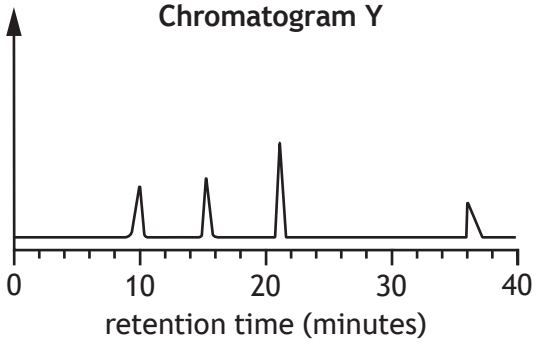
3. (a) (continued)

(iv) The flavour of cheese changes over time as the concentrations of flavour molecules change.

Gas chromatography can be used to analyse the concentrations of flavour molecules.

(A) Chromatograms for two samples of cheese are shown below.



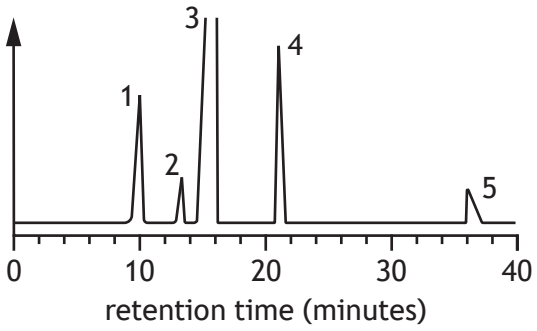


Determine the retention time, in minutes, of the peak in **Chromatogram X** that is missing in **Chromatogram Y**. **1**

[Turn over

3. (a) (iv) (continued)

(B) The following chromatogram was obtained from another sample of cheese. The concentration of a flavour molecule in cheese can be determined by calculating the area under the peak that corresponds to that molecule.



The concentration of flavour molecule 3 cannot be determined from this chromatogram.

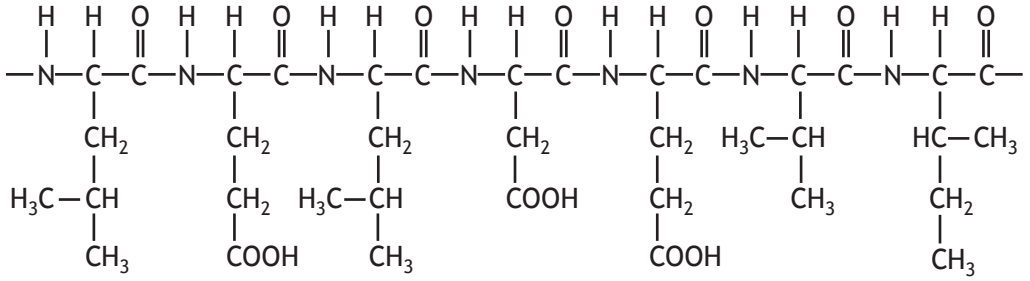
Suggest what would need to be done to the sample to allow the concentration of flavour molecule 3 to be determined. **1**

MARKS

DO NOT WRITE IN THIS MARGIN

(b) The main protein in cheese is called casein.

The diagram shows part of the structure of a casein molecule.



The table shows the relative proportions of the amino acids found in this section of protein.

Amino acid	Relative proportion
Aspartic acid	1
Glutamic acid	2
Isoleucine	1
Leucine	2
Valine	1

(i) Leucine and valine are amino acids that must be obtained through the diet.

State the term for this type of amino acid. **1**

(ii) Using information from the diagram and the table, draw a structural formula for glutamic acid. **1**

[Turn over

3. (b) (continued)

(iii) When cheese is heated, the proteins change shape.

State the term used to describe this process. **1**

(c) Processed cheese is made from cheese, soluble milk proteins, water and an emulsifier.

State the function of an emulsifier. **1**

(d) A compound added to cheese as a mould inhibitor has the formula $Ca^{2+}(CH_3CH_2COO^-)_2$.

Name this compound. **1**

4. Volumetric analysis involves using a standard solution in a reaction with a well-defined end point to determine the concentration of another substance. Correct use of the appropriate apparatus and solutions is essential to ensure accurate determination of concentration by titration.

Using your knowledge of chemistry, describe the experimental procedures, including equipment, used to accurately determine the concentration of a substance by volumetric analysis. **3**

[Turn over

5. Gin is made by flavouring a mixture of ethanol and water with plant extracts.

(a) (i) The mixture of ethanol and water is made by fermentation followed by distillation.

In fermentation, enzymes in yeast convert glucose, $C_6H_{12}O_6$, into ethanol and carbon dioxide.

The equation for fermentation is shown.

$$\underset{GFM = 180\text{ g}}{C_6H_{12}O_6(aq)} \rightarrow \underset{GFM = 46\text{ g}}{2CH_3CH_2OH(aq)} + \underset{GFM = 44\text{ g}}{2CO_2(g)}$$

(A) A 50.0 cm^3 sample of glucose solution contained 5.79 g of glucose.

Calculate the volume, in litres, of carbon dioxide gas that would be produced if 16 litres of this glucose solution was fermented. **3**

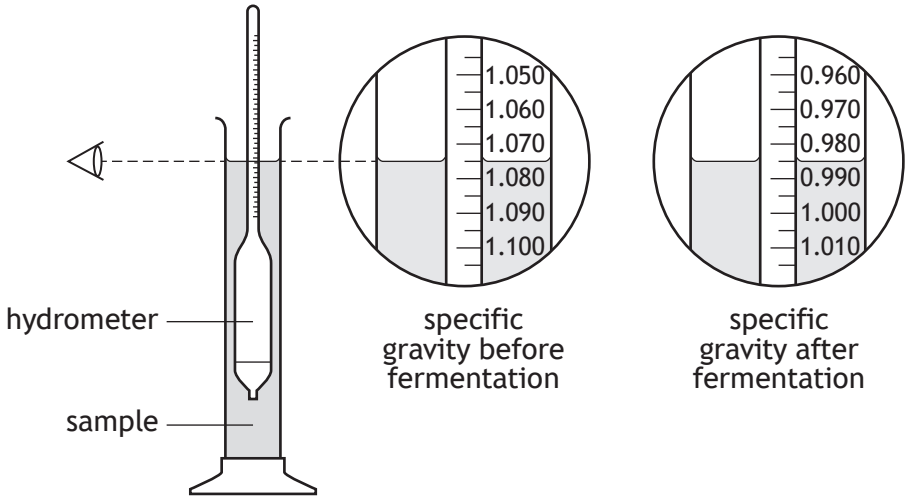
Take the volume of 1 mole of carbon dioxide gas to be 24 litres.

(B) Calculate the atom economy for the production of ethanol. **2**

5. (a) (continued)

(ii) The percentage of alcohol by volume can be calculated by measuring the specific gravity of samples taken before and after fermentation.

The specific gravity is measured using a hydrometer. The level of the sample on the hydrometer scale, read at eye level, is the specific gravity.



The % alcohol by volume can be calculated using the formula

$$\% \text{ alcohol by volume} = \left(\frac{\text{change in specific gravity}}{0.7362} \right) \times 100$$

Calculate the % alcohol by volume for this sample. **2**

[Turn over

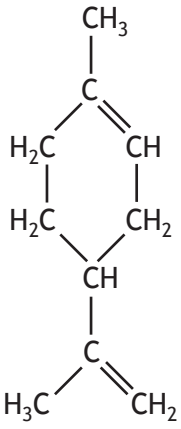
5. (continued)

(b) Plant material is used to flavour the ethanol and water mixture. The mixture is then distilled.

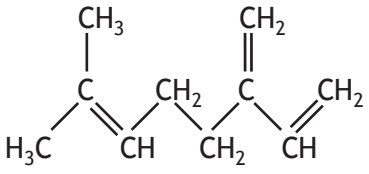
(i) The first fraction from the distillation contains toxic methanol and propan-2-one and is discarded.

Describe a chemical test, with the expected result for **both** compounds, that could be used to distinguish between methanol and propan-2-one. **2**

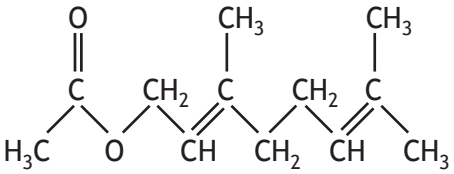
(ii) The second fraction from the distillation is collected for bottling as gin. Some of the plant compounds that give gin its flavour are shown.



limonene



myrcene



geranyl acetate

(A) Name the class of compounds to which unsaturated hydrocarbons such as limonene and myrcene belong. **1**

(B) Circle an isoprene unit on the limonene structure above. **1**

(An additional structure, if required, can be found on *page 40*.)

5. (b) (ii) (continued)

(C) Geranyl acetate can undergo hydrolysis to produce an alcohol and another product.

Name the other product. **1**

(c) Gin is often mixed with tonic water before drinking. Tonic water contains quinine, a bitter tasting compound. Historically quinine was used to treat malaria.

To treat malaria an intake of 10.0 mg of quinine per kilogram of body weight is required every 8 hours.

Calculate the mass of quinine required by a 70 kg adult in one day. **2**

[Turn over

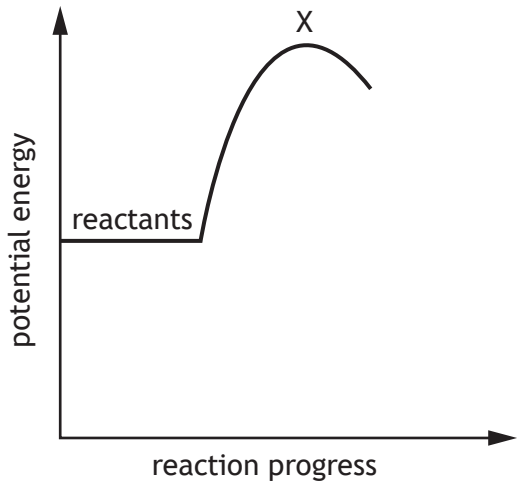
6. Ammonium nitrate is a commonly used fertiliser.

(a) (i) (A) Ammonium nitrate is made industrially by adding nitric acid, HNO_3, to ammonia, NH_3.

$$HNO_3(aq) + NH_3(g) \rightarrow NH_4NO_3(aq) \quad \Delta H = -ve$$

Complete the diagram to show the shape of the enthalpy diagram for this reaction. **1**

(An additional diagram, if required, can be found on *page 40*.)



(B) State the term for the unstable arrangement of atoms formed at the point labelled X on the potential energy diagram above. **1**

(ii) 1316 litres of nitric acid of concentration 9.5 mol l^{-1} was reacted with 220 kg of ammonia ($GFM = 17$ g) to produce ammonium nitrate.

$$HNO_3(aq) + NH_3(g) \rightarrow NH_4NO_3(aq) \quad \Delta H = -ve$$

Show, by calculation, which reactant was in excess. **2**

6. (a) (continued)

(iii) Ammonium nitrate can also be produced by the reaction

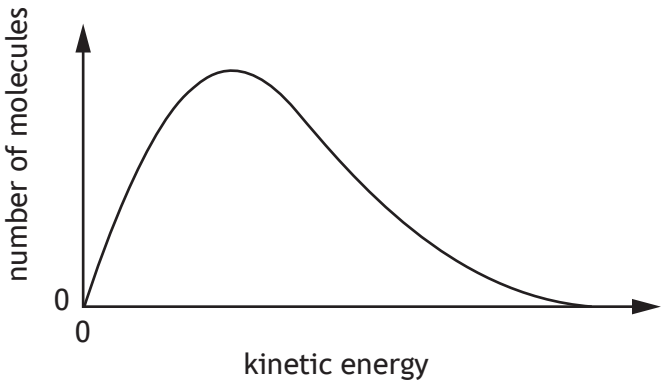
$$Ca(NO_3)_2(s) + 2NH_3(g) + CO_2(g) + H_2O(\ell) \rightarrow 2NH_4NO_3(aq) + CaCO_3(s) \quad \Delta H = -ve$$

Suggest a reason why the method shown in part (a) (ii) is the preferred industrial route. **1**

(b) The rate of reaction can be altered by changing the temperature or using a catalyst.

(i) **Graph 1** shows the distribution of kinetic energies of molecules in a gas mixture.

Graph 1



(A) Suggest what is represented by the area under the curve in **Graph 1**. **1**

(B) Add a second curve to **Graph 1** to show the distribution of kinetic energies at a higher temperature. **1**

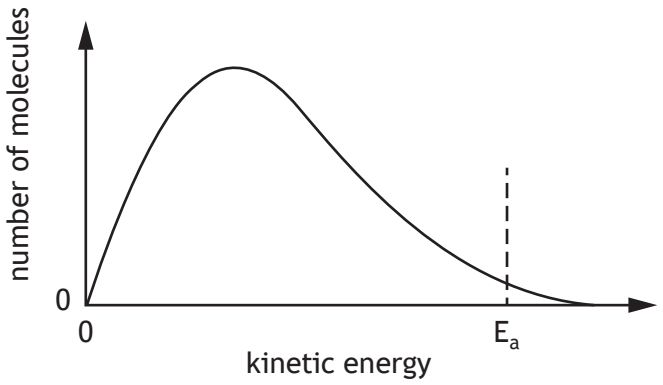
(An additional graph, if required, can be found on *page 41*.)

[Turn over

6. (b) (continued)

(ii) In **Graph 2**, the activation energy for the reaction, E_a, is marked on the x-axis.

Graph 2



Draw a line on **Graph 2** to show how a catalyst affects the activation energy, E_a. **1**

(An additional graph, if required, can be found on *page 41*.)

6. (continued)

(c) Ammonia used to produce ammonium nitrate can be made by the Haber process.

$$3H_2(g) + N_2(g) \rightleftharpoons 2NH_3(g) \qquad \Delta H = -92 \text{ kJ mol}^{-1}$$

(i) Reversible reactions, such as the Haber process, can reach a state of dynamic equilibrium in a closed system.

State what is meant by the term dynamic equilibrium. **1**

(ii) (A) The ammonia produced is continuously removed.

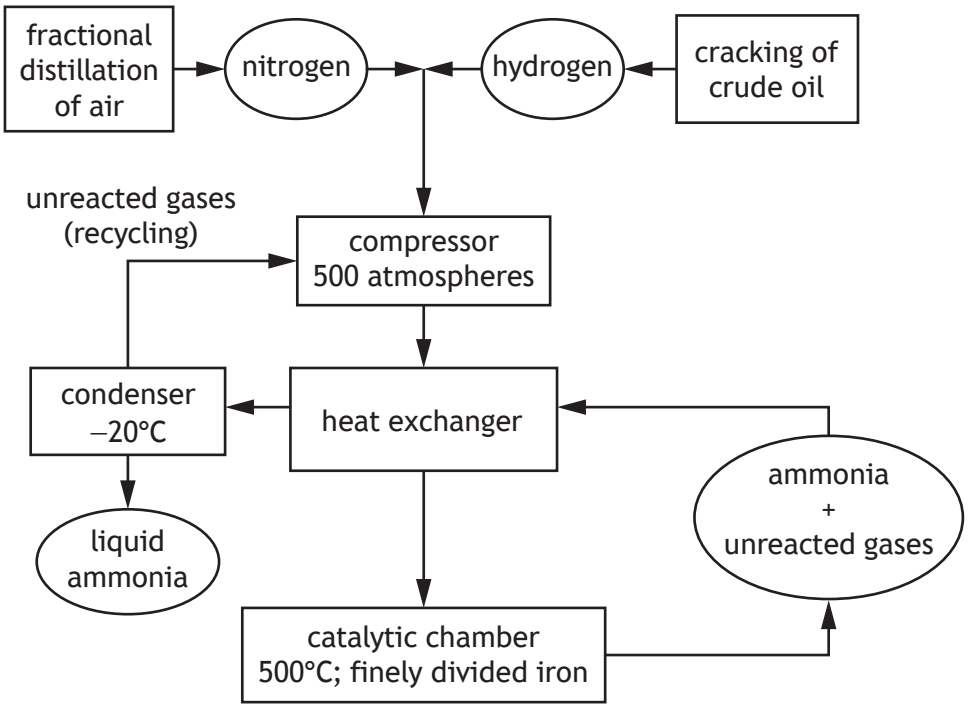
Explain how this will affect the production of ammonia. **2**

[Turn over

6. (c) (ii) (continued)

(B) A flow diagram of the Haber process is shown.

One way to reduce costs in the process is to use a heat exchanger to transfer excess heat from one part of the process to use in another part of the process.



From the flow diagram, state another way that the manufacturing process maximises profit or minimises the impact on the environment. **1**

6. (continued)

(d) Ammonia is currently being investigated for use in fuel cells.

The reactions taking place at the electrodes are

$$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$$

$$2NH_3 + 6OH^- \rightarrow N_2 + 6H_2O + 6e^-$$

(i) Write the overall redox equation for the reaction taking place in the fuel cell. **1**

(ii) Identify the reducing agent in the reaction taking place in the fuel cell. **1**

[Turn over

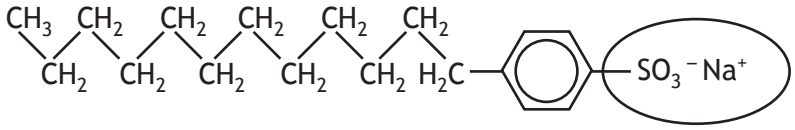
7. Tap water contains a number of dissolved chemicals.

(a) In some parts of the country, tap water contains a high level of dissolved metal salts.

(i) State the term used to describe this type of water. **1**

(ii) Soapless detergents are used with this type of water to prevent insoluble scum forming.

The structure of a typical soapless detergent is shown.



The circled region of the molecule is ionic and dissolves in water.

State the term used to describe this part of the molecule. **1**

7. (a) (continued)

(iii) Calcium ions are commonly found in tap water. The concentration of calcium ions in a tap water sample was determined by titrating with a chemical called EDTA, $C_{10}H_{12}N_2O_8^{4-}$.

A 50.0 cm^3 water sample was collected and reacted with a standard solution of EDTA, with a concentration of 0.0045 mol l^{-1}. The average titre volume was 9.3 cm^3.

$$Ca^{2+}(aq) + C_{10}H_{12}N_2O_8^{4-}(aq) \rightarrow [Ca(C_{10}H_{12}N_2O_8)]^{2-}(aq)$$

Calculate the concentration, in mol l^{-1}, of calcium ions in the tap water. **3**

[Turn over

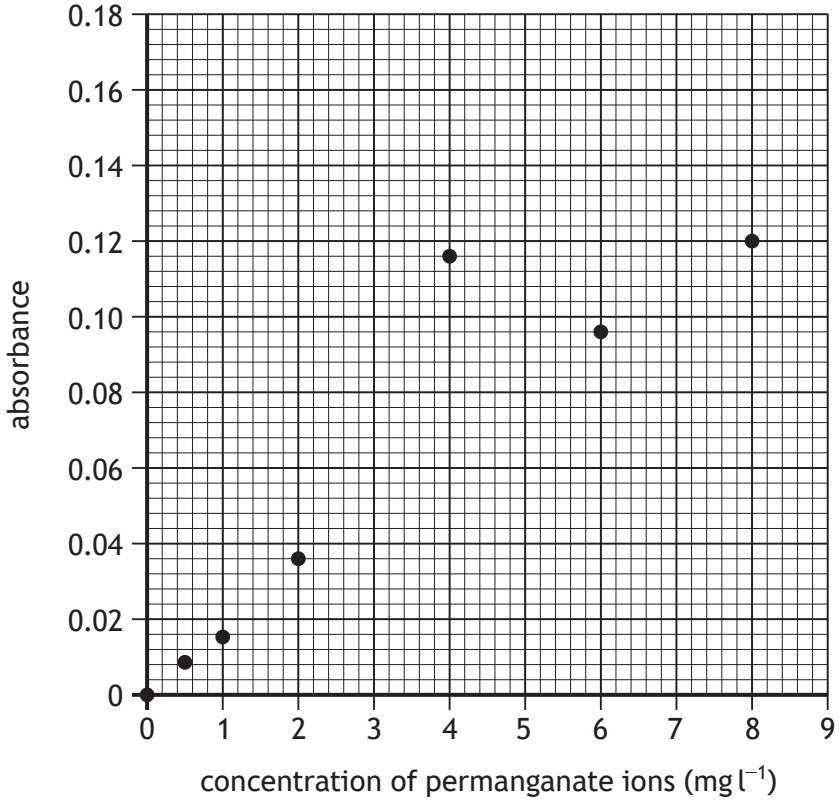
7. (continued)

(b) Another ion found in tap water is manganese(II), Mn^{2+}.

The manganese(II) ions are oxidised to purple permanganate ions, MnO_4^-. The concentration of permanganate ions can be determined by measuring how much light is absorbed by the solution.

The higher the concentration of permanganate ions in the solution, the more light is absorbed.

The absorbances of several standard solutions of permanganate were measured, and the results plotted.



A water sample had an absorbance of 0.08.

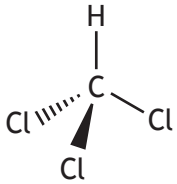
Estimate the concentration of permanganate ions, in mg l^{-1}, in this sample. **1**

7. (continued)

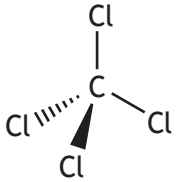
(c) Chlorine is added to tap water to make it safe to drink.

The chlorine can react with substances in the water to produce trichloromethane, $CHCl_3$.

(i) Trichloromethane is more soluble in water than tetrachloromethane due to the polarities of the molecules.



trichloromethane



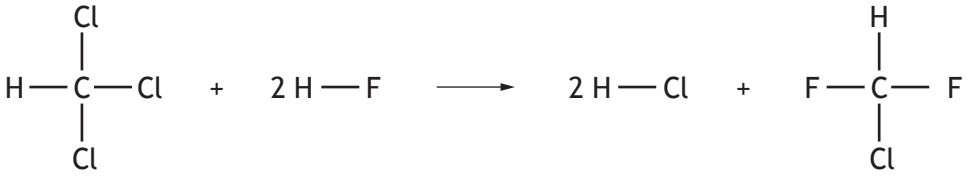
tetrachloromethane

Explain the difference in polarities of trichloromethane and tetrachloromethane molecules. **2**

[Turn over

7. (c) (continued)

(ii) Trichloromethane is used on an industrial scale to produce plastics. The first step in this reaction is to react it with hydrogen fluoride, HF.



Using bond enthalpies and mean bond enthalpies from the data booklet, calculate the enthalpy change, in $kJ\,mol^{-1}$, for the reaction of trichloromethane with hydrogen fluoride. **2**

8. Dog food contains many different compounds including fats, vegetable oils, fatty acids, flavour and aroma molecules, proteins, water, antioxidants and emulsifiers.

It is important that dog food has a long shelf life, retains its appearance and texture, as well as providing sufficient nutritional value.

Using your knowledge of chemistry, explain the role of different compounds in dog food. **3**

9. Haloalkanes are alkane molecules that contain at least one group 7 atom.

(a) The table shows information on the boiling points of some haloalkanes.

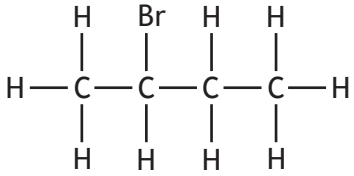
Haloalkane	Boiling point (°C)		
	X = Cl	**X = Br**	**X = I**
CH_3-X	−24.2	3.6	42.4
CH_3CH_2-X	12.3	38.4	72.3
$CH_3CH_2CH_2$-X	46.6	71.0	102.0

(i) Using the information in the table, describe **two** different trends in the boiling points. **2**

(ii) Name the strongest type of intermolecular forces broken when bromoethane, CH_3CH_2Br, boils. **1**

9. (continued)

(b) Haloalkanes can be classified as primary, secondary or tertiary depending on the position of the group 7 atom.



2-bromobutane

(i) State why 2-bromobutane can be classified as a secondary haloalkane. **1**

(ii) Draw a structural formula for an isomer of 2-bromobutane that is a tertiary haloalkane. **1**

[Turn over

9. (continued)

(c) Alkanes can react with group 7 molecules in free radical reactions to form haloalkanes.

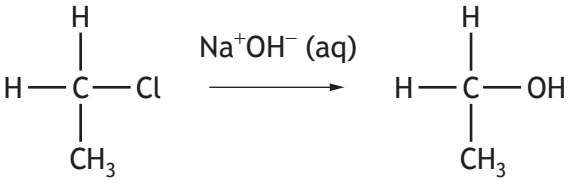
Reaction step	Name of step
$Br_2 \rightarrow 2Br\bullet$	Initiation
$Br\bullet + CH_4 \rightarrow HBr + \bullet CH_3$ $\bullet CH_3 + Br_2 \rightarrow CH_3Br + Br\bullet$	Propagation
	Termination

(i) State what is required for initiation to take place. **1**

(ii) Complete the table to show a possible termination step. **1**

9. (continued)

(d) Haloalkanes can react to form alcohols as shown.



Depending on the structure of the haloalkane used, the alcohol produced can be oxidised to form an aldehyde or ketone.

Compound **P** was converted to compound **R** in two steps.

$$\underset{\textbf{P}}{C_4H_9Br} \rightarrow \underset{\textbf{Q}}{C_4H_9OH} \rightarrow \underset{\textbf{R}}{C_4H_8O}$$

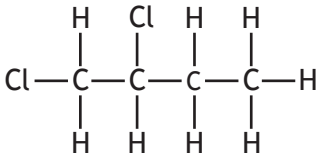
Compound **R** does not react with Tollens' reagent or Fehling's solution.

Draw a structural formula for compound **P**. **1**

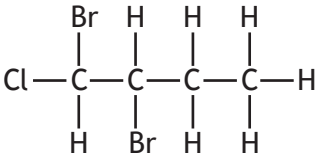
[Turn over

9. (continued)

(e) The structures of two haloalkanes are shown.



1,2-dichlorobutane



1,2-dibromo-1-chlorobutane

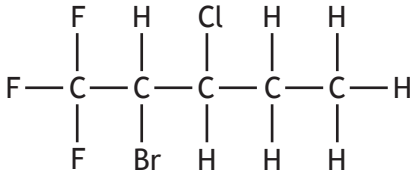
The names of haloalkanes are derived from their structures using the following rules.

1. The name is based on the longest chain of carbon atoms.
2. The presence of group 7 atoms is shown by shortening the name of the group 7 atom.

Group 7 atom	Shortened name
fluorine	fluoro-
chlorine	chloro-
bromine	bromo-

3. The chain is numbered to assign numbers to the group 7 atoms. The numbers should be assigned so the lowest possible numbers are used.
4. If two or more of the same group 7 atoms are present, use the prefixes di, tri or tetra.
5. The shortened name of the group 7 atoms attached to the chain are listed alphabetically (ignoring the prefixes di, tri and tetra for alphabetical purposes).

Using these rules, name this molecule. **1**



MARKS DO NOT WRITE IN THIS MARGIN

10. A student was given a sample known to be a mixture of magnesium oxide, MgO, and magnesium carbonate, $MgCO_3$, and asked to determine the mass of magnesium carbonate in the mixture.

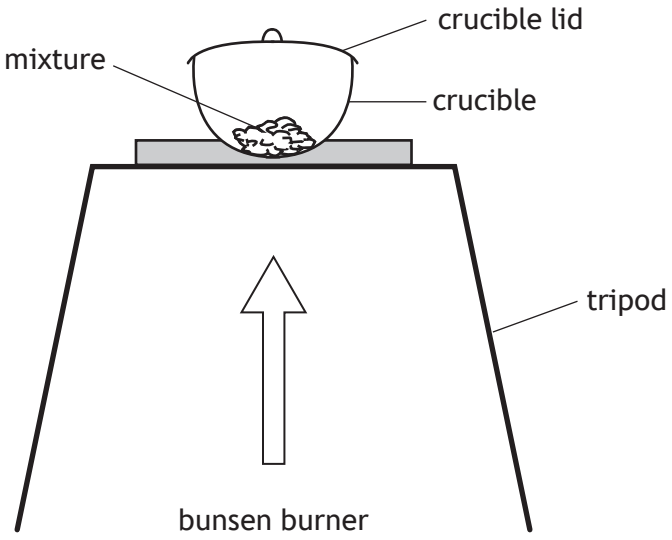
(a) The first step was to accurately measure a mass of 1.5 g of the mixture into a crucible, using a balance.

Describe **fully** how the student could carry out this step. **2**

(b) Metal carbonates decompose on heating to give metal oxides and carbon dioxide gas.

$$MgCO_3(s) \rightarrow MgO(s) + CO_2(g)$$

The apparatus was used to strongly heat the mixture of $MgCO_3$ and MgO. The student used a pair of tongs to lift the lid from time to time.



(i) Suggest why the student had to lift the lid from time to time. **1**

[Turn over

10. (b) (continued)

(ii) Explain why a Bunsen burner is a suitable method of heating this reaction. **1**

(c) Once the crucible had cooled, the final mass of the sample was determined. The mass of CO_2 lost is used to calculate the mass of $MgCO_3$ in the mixture.

The results are shown below.

Mass of sample before heating	1.598 g
Mass of sample after heating	1.294 g

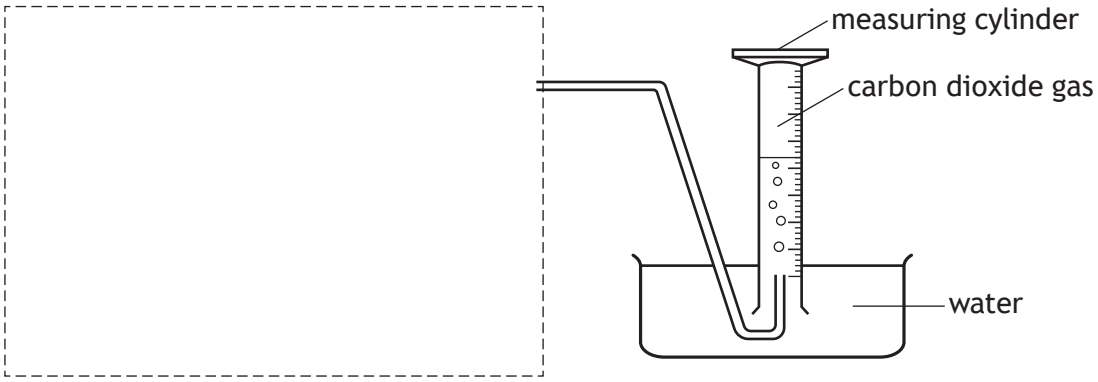
$$\underset{GFM = 84.3\text{ g}}{MgCO_3(s)} \rightarrow MgO(s) + \underset{GFM = 44.0\text{ g}}{CO_2(g)}$$

Calculate the mass, in g, of magnesium carbonate present in the mixture. **2**

10. (continued)

(d) Another experiment to determine the mass of magnesium carbonate in the mixture involves measuring the volume of CO_2 produced.

This experiment is carried out by reacting the mixture of magnesium oxide and magnesium carbonate with excess acid. The volume of carbon dioxide gas produced can be measured by collecting the gas over water as shown.



production of gas

collecting and measuring volume of gas

(i) Complete a labelled diagram to show an apparatus suitable for the production of gas. **1**

(An additional diagram, if required, can be found on *page 42*.)

(ii) Suggest why carbon dioxide can be collected over water. **1**

[END OF QUESTION PAPER]

National
Qualifications
2022

X813/76/12

Chemistry
Paper 1 — Multiple choice

FRIDAY, 29 APRIL
9:00 AM – 9:40 AM

Total marks — 25

Attempt ALL questions.

You may use a calculator.

Instructions for the completion of Paper 1 are given on *page 02* of your answer booklet X813/76/02.

Record your answers on the answer grid on *page 03* of your answer booklet.

You may refer to the Chemistry Data Booklet for Higher and Advanced Higher

Space for rough work is provided at the end of this booklet.

Before leaving the examination room you must give your answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

X8137612

Total marks — 25

Attempt ALL questions

1. An element contains covalent bonding and London dispersion forces.

The element could be:

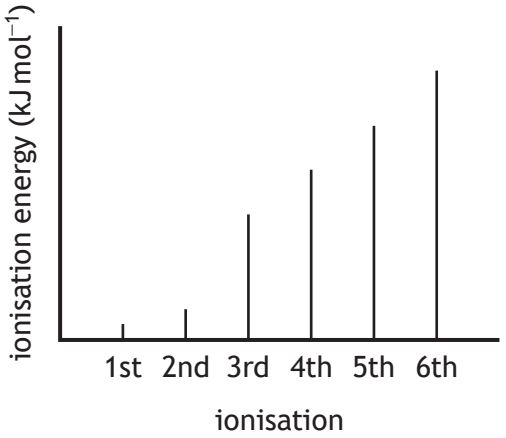
A boron

B neon

C sodium

D sulfur.

2. The graph below shows the relative quantities of energy equivalent to successive ionisation energies for an element.



The most stable ion formed from an atom of this element has a charge of:

A 2+

B 3+

C 2–

D 3–

3. HCl has a higher boiling point than H_2 because:

A the polar covalent bonds in HCl are stronger than the covalent bonds in H_2

B the polar covalent bonds in HCl are stronger than the van der Waals' forces in H_2

C the van der Waals' forces in HCl are stronger than the van der Waals' forces in H_2

D the van der Waals' forces in HCl are stronger than the covalent bonds in H_2.

4. Which line in the table would best describe elements that act as reducing agents?

	Gains or loses electrons	Electronegativity
A	gains	low
B	loses	low
C	gains	high
D	loses	high

5. The correct redox equation for the reaction of iron(II) ions with acidified dichromate ions is:

A $Cr_2O_7^{2-}(aq) + 14H^+(aq) + Fe^{2+}(aq) \rightarrow 2Cr^{3+}(aq) + 7H_2O(\ell) + Fe(s)$

B $Cr_2O_7^{2-}(aq) + 14H^+(aq) + Fe^{2+}(aq) \rightarrow 2Cr^{3+}(aq) + 7H_2O(\ell) + Fe^{3+}(aq)$

C $Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6Fe^{2+}(aq) \rightarrow 2Cr^{3+}(aq) + 7H_2O(\ell) + 6Fe(s)$

D $Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6Fe^{2+}(aq) \rightarrow 2Cr^{3+}(aq) + 7H_2O(\ell) + 6Fe^{3+}(aq)$

6. A mixture of magnesium bromide and magnesium sulfate is known to contain 3 moles of magnesium ions and 4 moles of bromide ions.

How many moles of sulfate ions are present?

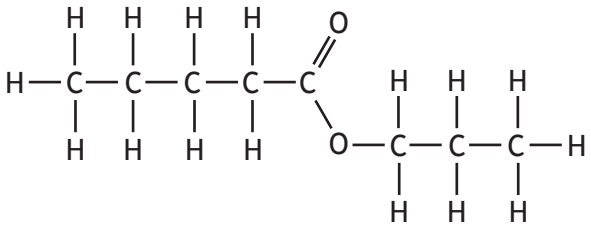
A 1

B 2

C 3

D 4

7.



The correct name for this ester is:

A butyl propanoate

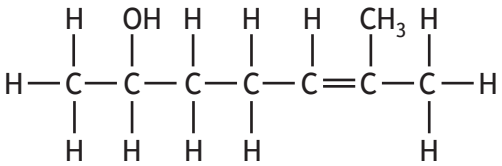
B propyl butanoate

C pentyl propanoate

D propyl pentanoate.

[Turn over

8. The structural formula for a compound is shown.



Which of the following is **not** an isomer of this compound?

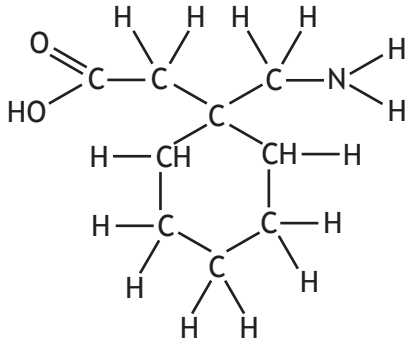
A octan-4-one

B 2-ethylhexanal

C 2-ethylhexan-1-ol

D 5-methylheptan-3-one

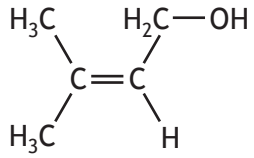
9. Gabapentin is a medicine that can be used to treat nerve pain.



Which line in the table shows the two functional groups present in this compound?

A	amine	carboxyl
B	amine	hydroxyl
C	hydroxyl	carboxyl
D	hydroxyl	carbonyl

10. Prenol is a compound that occurs naturally in citrus fruits.



Which line in the table correctly describes the reaction of prenol with bromine solution and with hot copper(II) oxide?

	Reaction with bromine solution	**Reaction with hot copper(II) oxide**
A	no reaction	no reaction
B	no reaction	brown solid formed
C	decolourises	brown solid formed
D	decolourises	no reaction

11. The iodine number of an oil is the mass of iodine, in grams, that will react with 100 g of oil and is a measure of the degree of saturation.

Olive oil has an iodine number of 84 and palm oil has an iodine number of 48.

Which of the following statements is correct?

A Palm oil is more saturated and has a lower melting point than olive oil.

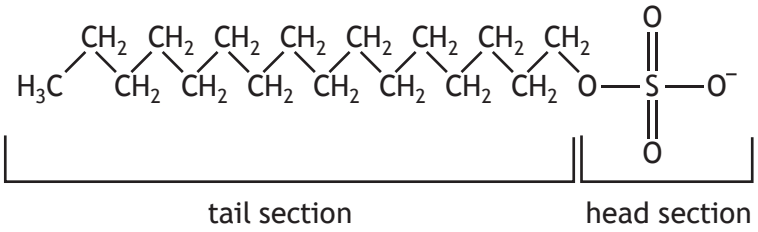
B Palm oil is more saturated and has a higher melting point than olive oil.

C Palm oil is less saturated and has a lower melting point than olive oil.

D Palm oil is less saturated and has a higher melting point than olive oil.

[Turn over

12. The structure of a soapless detergent molecule is given below.



Which line in the table describes a step in the cleansing action of a soapless detergent?

	Head section	Tail section
A	The hydrophobic head dissolves in water.	The hydrophilic tail dissolves in oil.
B	The hydrophilic head dissolves in water.	The hydrophobic tail dissolves in oil.
C	The hydrophobic head dissolves in oil.	The hydrophilic tail dissolves in water.
D	The hydrophilic head dissolves in oil.	The hydrophobic tail dissolves in water.

13. Which of the following is a secondary alcohol?

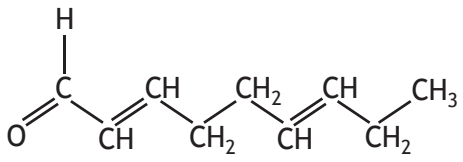
A 2-methylbutan-1-ol

B 2-methylbutan-2-ol

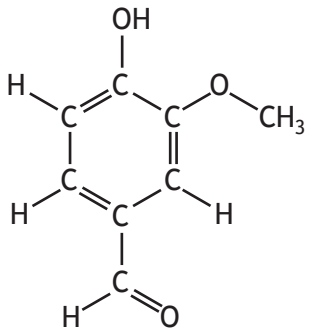
C butan-1-ol

D butan-2-ol

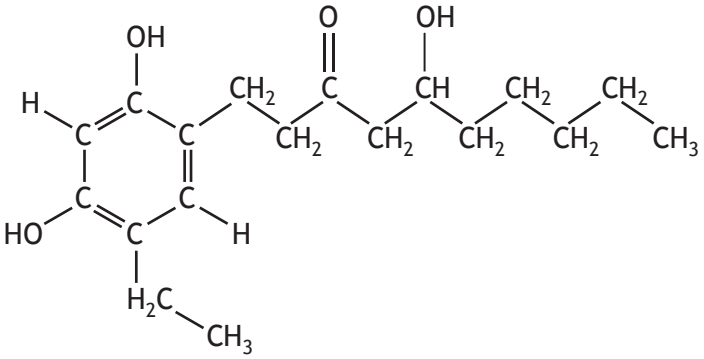
14. The compounds below are examples of flavour molecules found in some plants.



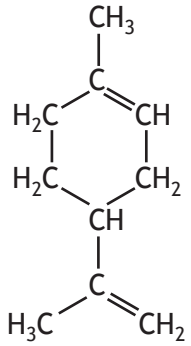
cucumber flavour



vanilla flavour



ginger flavour



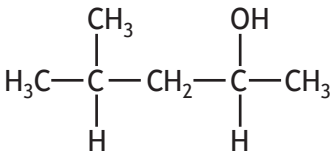
orange flavour

Which line in the table shows the solubilities of these compounds in water and in oil?

	Water soluble	**Oil soluble**
A	cucumber and ginger	orange and vanilla
B	cucumber and orange	ginger and vanilla
C	ginger and vanilla	cucumber and orange
D	orange and vanilla	cucumber and ginger

[Turn over

15. The structural formula for a compound is shown.



The product of oxidation of this compound is:

A 2-methylpentan-4-one

B 4-methylpentan-2-one

C 2-methylpentanal

D 4-methylpentanal.

16. Which of the following describes how to fill a burette with acid and take the initial reading in a titration?

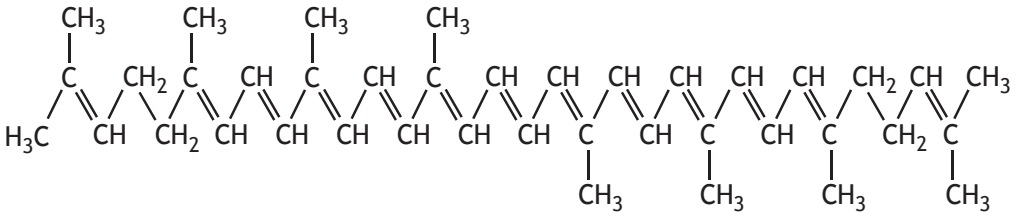
A Rinse the burette with the acid. Fill to above the scale with acid. Drain some of the acid and read from the top of the meniscus.

B Rinse the burette with deionised water. Fill to above the scale with acid. Drain some of the acid and read from the bottom of the meniscus.

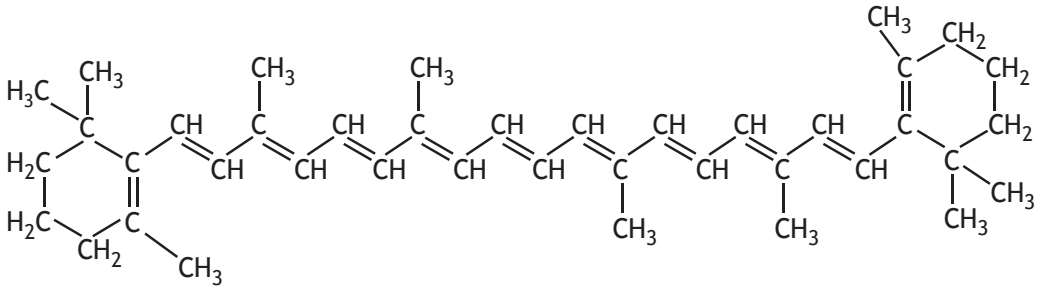
C Rinse the burette with the acid. Fill to above the scale with acid. Drain some of the acid and read from the bottom of the meniscus.

D Rinse the burette with deionised water. Fill to above the scale with acid. Drain some of the acid and read from the top of the meniscus.

17. Tomato juice contains a mixture of terpenes including lycopene and beta-carotene.
Terpenes can be separated using chromatography.



lycopene



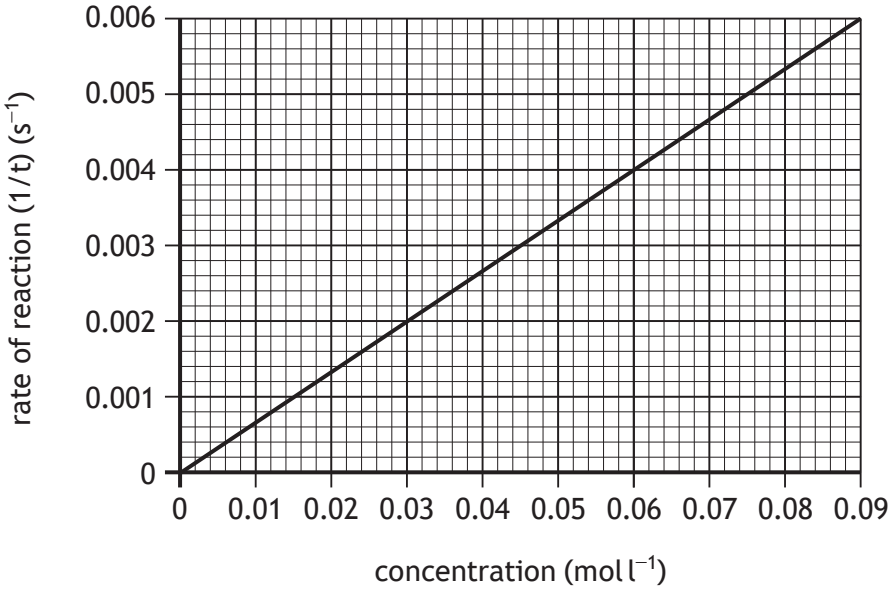
beta-carotene

Which of the following is the most suitable solvent to separate lycopene and beta-carotene?

A Ethanol
B Pentane
C Propanoic acid
D Water

[Turn over

18. The graph shows how the rate of a reaction varies with the concentration of a reactant.



When the concentration of the reactant is 0.06 mol l^{-1}, the reaction time is:

A 0.004 s

B 0.09 s

C 17 s

D 250 s.

19. Butene reacts with oxygen as shown.

$$C_4H_8(g) + 6O_2(g) \rightarrow 4CO_2(g) + 4H_2O(g)$$

100 cm^3 of butene was reacted with excess oxygen.

Compared with the total volume of gases before reaction, what would be the total volume of gases after complete reaction?

A 100 cm^3 more

B 100 cm^3 less

C 300 cm^3 more

D 300 cm^3 less

20. In aqueous solution ethanoic acid forms an equilibrium mixture with its ions.

$$CH_3COOH(aq) \rightleftharpoons H^+(aq) + CH_3COO^-(aq)$$

Which of the following solutions, when added to the equilibrium mixture, would favour the forward reaction?

A NaCl(aq)

B HCl(aq)

C NaOH(aq)

D $CH_3COONa(aq)$

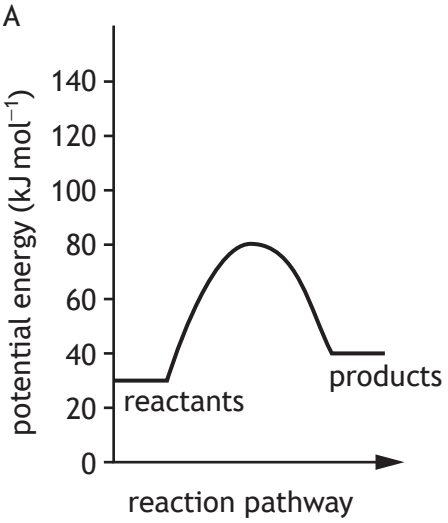
[Turn over

21. Some energy values associated with a chemical reaction are shown in the table.

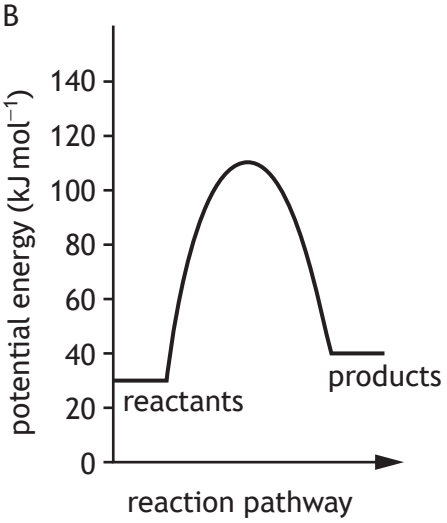
Enthalpy of reactants ($kJ\ mol^{-1}$)	Activation energy of forward reaction ($kJ\ mol^{-1}$)	Activation energy of reverse reaction ($kJ\ mol^{-1}$)
30	110	70

Which of the following correctly shows the potential energy diagram for the above conditions?

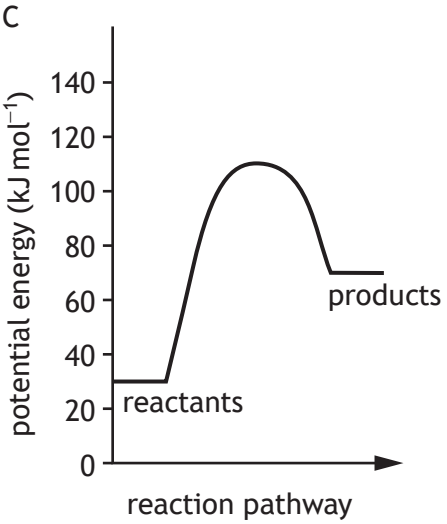
A



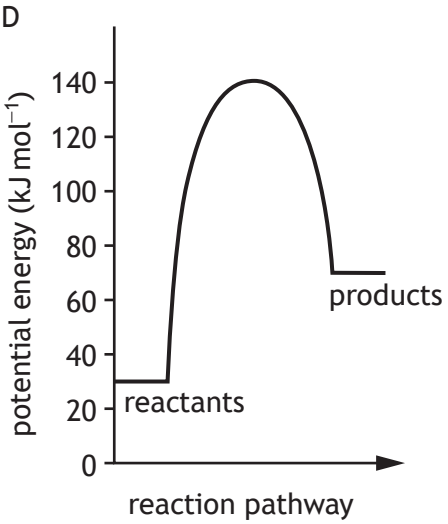
B



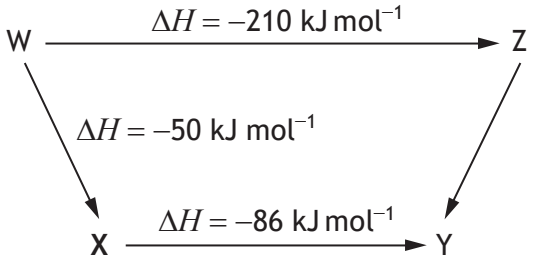
C



D



22. Consider the reaction pathway below.



According to Hess' law, the ΔH value, in kJ mol^{-1}, for reaction Z to Y is:

A −74

B +74

C −346

D +346

23. 50.0 cm^3 of 0.100 mol l^{-1} ammonia solution was transferred to a 250 cm^3 volumetric flask. The flask was made up to the mark with deionised water.

The final concentration, in mol l^{-1}, of the ammonia solution is:

A 2.0×10^{-2}

B 2.5×10^{-2}

C 4.0×10^{-2}

D 5.0×10^{-2}

24. An experiment involves reacting 0.02 moles of silver ions with ions of a group 7 element to form 2.868 g of precipitate.

Which of the following is the precipitate?

A Silver(I) fluoride

B Silver(I) chloride

C Silver(I) bromide

D Silver(I) iodide

[Turn over

25. A titration experiment was carried out to determine the concentration of vitamin C in orange juice.

A sample of the orange juice solution was pipetted into a flask and 10 cm^3 water was added to dilute the sample. Starch indicator was added to the flask. The mixture was then titrated in the flask using iodine solution of known concentration.

Which line in the table shows the most appropriate apparatus to use when carrying out this procedure?

	To add water	**Type of flask**
A	measuring cylinder	conical flask
B	beaker	conical flask
C	measuring cylinder	volumetric flask
D	beaker	volumetric flask

[END OF QUESTION PAPER]

FOR OFFICIAL USE

National Qualifications 2022

Mark

X813/76/01

Chemistry Paper 2

FRIDAY, 29 APRIL

10:10 AM – 12:30 PM

X8137601

Fill in these boxes and read what is printed below.

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth

Day | Month | Year

Scottish candidate number

Total marks — 95

Attempt ALL questions.

You may use a calculator.

You may refer to the Chemistry Data Booklet for Higher and Advanced Higher.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers and rough work is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting. Any rough work must be written in this booklet. Score through your rough work when you have written your final copy.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

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Total marks — 95

Attempt ALL questions

MARKS | DO NOT WRITE IN THIS MARGIN

1. Elements and compounds can exist as diatomic molecules.

(a) The seven elements that exist as diatomic molecules are shown in the periodic table below.

1 H																	2 He
3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og

(i) Explain why diatomic elements form non-polar molecules. **1**

(ii) Nitrogen, oxygen and fluorine are found in the second period of the periodic table.

Explain the decrease in covalent radius going from nitrogen to fluorine. **1**

(b) First ionisation energies decrease going down a group.

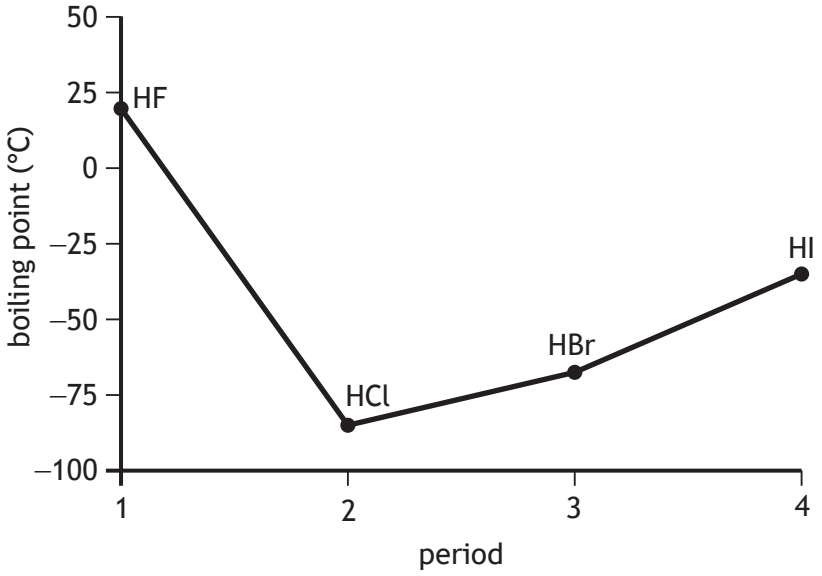
(i) State what is meant by the term *first ionisation energy*. **1**

1. (b) (continued)

(ii) Explain why the first ionisation energy of the group 7 elements decreases going down the group. **1**

(c) Hydrogen halides are diatomic molecules formed between hydrogen and the elements fluorine, chlorine, bromine and iodine.

The boiling points of the hydrogen halides are shown on the graph below.



(i) Hydrogen fluoride, HF, has the highest boiling point of the hydrogen halides.

State the name of the strongest type of intermolecular force found between hydrogen fluoride molecules and explain how this type of intermolecular force arises. **2**

[Turn over

1. (c) (continued)

(ii) The table shows the boiling points of hydrogen chloride, hydrogen bromide and hydrogen iodide.

Hydrogen halide	Boiling point (°C)
Hydrogen chloride	−85
Hydrogen bromide	−66
Hydrogen iodide	−35

Explain **fully** why the boiling point increases from hydrogen chloride to hydrogen iodide. **2**

2. Fireworks contain a range of chemicals including a fuel, oxidising agents and metal salts.

(a) One oxidising agent used in fireworks is potassium perchlorate, $KClO_4$. This reacts with aluminium metal and produces a bright flash.

The equation for the reaction is

$$KClO_4 \quad + \quad Al \quad \rightarrow \quad KCl \quad + \quad Al_2O_3$$

Balance this equation. **1**

(b) Fireworks were traditionally made using compounds containing the chlorate ion, ClO_3^-, as an oxidising agent.

(i) Chlorate ions release oxygen when they decompose.

Potassium chlorate, $KClO_3$, ($GFM = 122.6$ g) reacts as shown.

$$2KClO_3(s) \rightarrow 3O_2(g) + 2KCl(s)$$

Calculate the volume of oxygen produced, in litres, when 4.6 g of potassium chlorate decomposes. **2**

Take the volume of 1 mole of oxygen gas to be 24 litres.

[Turn over

2. (b) (continued)

(ii) The decomposition of potassium chlorate can be speeded up by the addition of a catalyst.

State the effect of adding a catalyst on the enthalpy change for this reaction. **1**

(iii) A firework containing 5.5 g of potassium perchlorate ($GFM = 138.6$ g) releases 103 kJ of energy.

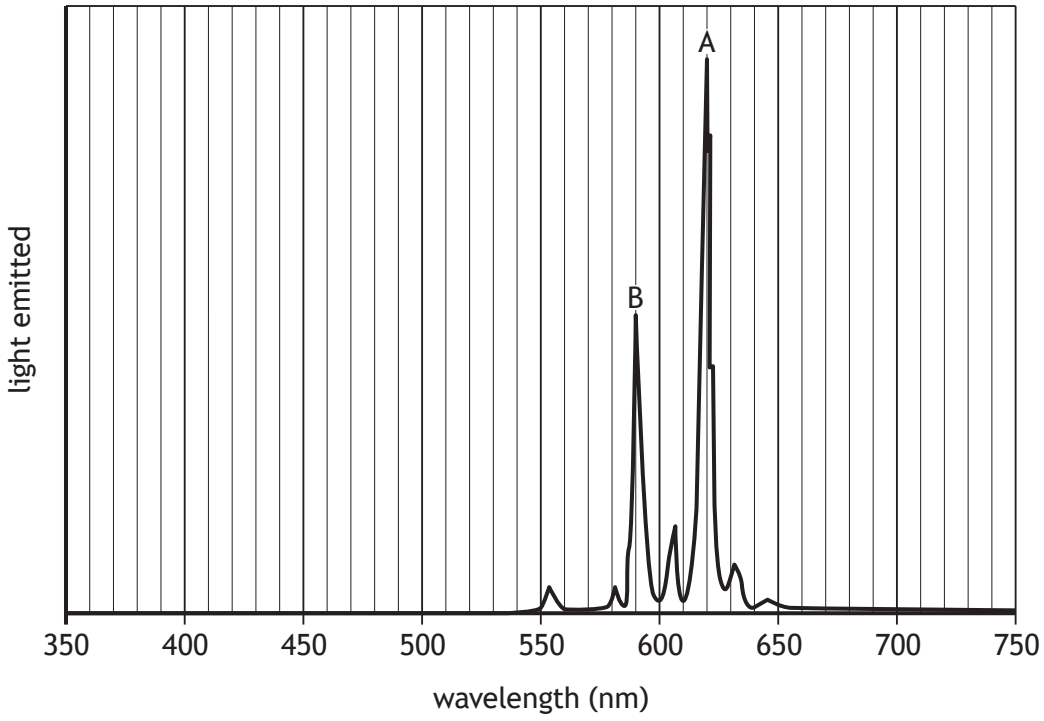
Calculate the energy, in kJ, released per mole of potassium perchlorate. **1**

2. (b) (continued)

(iv) Explain **fully** why increasing temperature increases the rate of a chemical reaction. **2**

(c) The different flame colours produced by metal salts are caused by different wavelengths of light being emitted. The flame colours associated with different wavelengths are given in the data booklet.

The following profile shows the colours emitted by one particular firework. Each peak represents a different colour of light.



Peak A has a wavelength of 620 nm, corresponding to red light.

Suggest the **metal** responsible for peak B on the spectrum. **1**

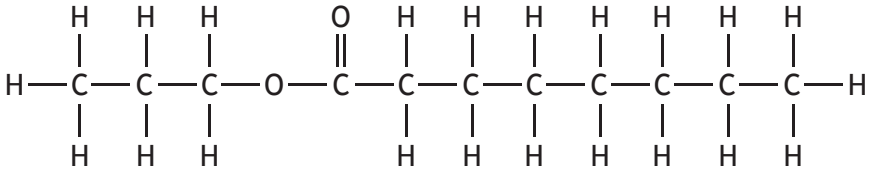
[Turn over

3. Atoms of different elements have different attractions for bonding electrons. Electronegativity is a measure of the attraction an atom involved in a bond has for the electrons in the bond.

Using your knowledge of chemistry, discuss the importance of electronegativity in bonding, structure and properties of compounds. **3**

4. Coconut oil contains a mixture of compounds.

(a) Propyl octanoate is a compound found in coconut oil.



(i) Name the functional group in propyl octanoate. **1**

(ii) Draw a structural formula for the carboxylic acid formed by hydrolysis of propyl octanoate. **1**

(iii) An isomer of propyl octanoate with the same functional group was hydrolysed. One of the products of this hydrolysis was butanoic acid.

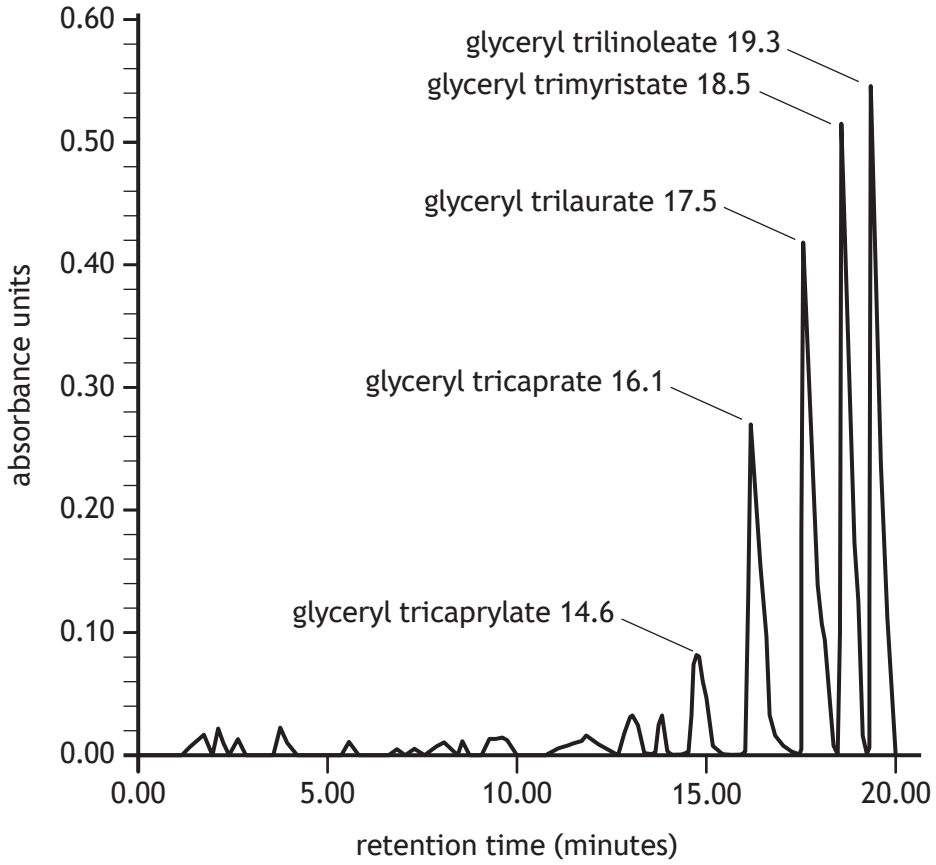
Suggest a name for the other product. **1**

[Turn over

4. (continued)

(b) Chromatography can be used to separate the fats and oils in coconut oil.

The result of a chromatography experiment is shown.



(i) Using the graph **and** the information in the table, predict the number of carbons in glyceryl trilaurate. **1**

Name	Molecular formula	Melting point (°C)
Glyceryl tricaprylate	$C_{27}H_{50}O_6$	10
Glyceryl tricaprate	$C_{33}H_{62}O_6$	31
Glyceryl trilinoleate	$C_{57}H_{98}O_6$	–5

4. (b) (continued)

(ii) Identify the compound listed in the table which is the most unsaturated. **1**

(c) Edible oils such as coconut oil can be used to make emulsifiers.

(i) State how emulsifiers are made from edible oils. **1**

(ii) Explain **fully** how emulsifiers prevent non-polar and polar liquids from separating into layers. **2**

[Turn over

5. Fusel oil is formed as a by-product during the production of bioethanol for fuel. It is a mixture of several alcohols.

(a) The shortened structural formula of one of the alcohols contained in fusel oil is shown.

$$CH_3CH(CH_3)CH_2CH_2OH$$

State the name of this alcohol. **1**

(b) Propan-1-ol is also found in fusel oil.

Propan-1-ol is reacted with an oxidising agent to produce propanal.

(i) Complete the ion-electron equation for the oxidation reaction. **1**

$$C_3H_7OH \rightarrow C_3H_6O$$

(ii) Acidified potassium dichromate can be used as the oxidising agent and reacts as shown below.

$$Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^- \rightarrow 2Cr^{3+}(aq) + 7H_2O(\ell)$$

Suggest why the potassium dichromate must be acidified. **1**

5. (b) (continued)

(iii) State the colour change that would be observed when propan-1-ol reacts with acidified potassium dichromate. **1**

(iv) The equation for the reduction of another oxidising agent that could be used to oxidise propan-1-ol is shown below.

$$Ag^{+}(aq) + e^{-} \rightarrow Ag(s)$$

Name the reagent that provides this oxidising agent. **1**

(v) State why 2-methylbutan-2-ol cannot be oxidised using these oxidising agents. **1**

(vi) In the reaction of butan-1-ol to butanal, oxidation can be identified by an increase in the oxygen to hydrogen ratio.

Complete the table to show the oxygen to hydrogen ratios in butan-1-ol and butanal. **1**

	Oxygen to hydrogen ratio
Butan-1-ol	
Butanal	

[Turn over

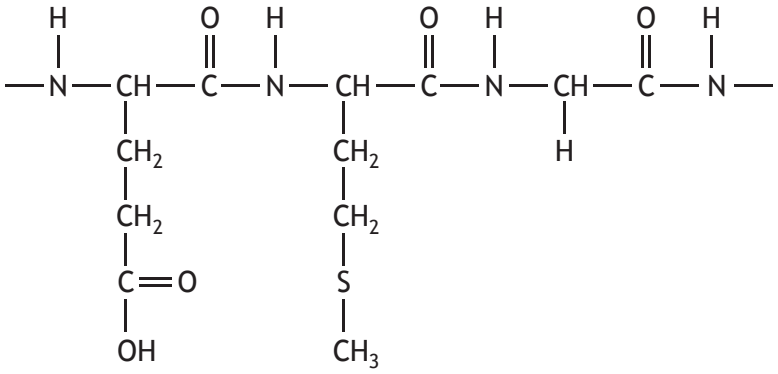
6. Sweet potatoes contain nutrients, including starch, vitamin C and proteins.

(a) Catalase is an enzyme contained in sweet potatoes that speeds up the breakdown of hydrogen peroxide.

(i) State what is meant by the term enzyme. **1**

(ii) Enzymes are a type of protein. Proteins are formed from smaller molecules called amino acids.

(A) A section of a protein is shown.



Circle a peptide link in the above structure. **1**

(An additional diagram can be found on *page 33*.)

(B) Draw a structural formula for one of the amino acids used to form this section of protein. **1**

6. (a) (ii) (continued)

(C) State what is meant by the term **essential** amino acid. **1**

(D) Name the type of reaction that takes place when amino acids join to form proteins. **1**

(iii) As sweet potatoes are cooked, the ability of catalase to break down hydrogen peroxide decreases.

Explain **fully** what happens to the enzyme structure to cause this reduction in activity. **2**

[Turn over

6. (a) (continued)

(iv) Hydrogen peroxide is broken down by catalase to produce water and oxygen.

An experiment was carried out to determine how much oxygen was produced from the breakdown of hydrogen peroxide by a sample of sweet potato.

Draw a diagram showing assembled apparatus that could be used to react hydrogen peroxide solution with sweet potato and measure the volume of oxygen produced.

Your diagram should include labels showing the names and positions of the reactants and the collected product. **3**

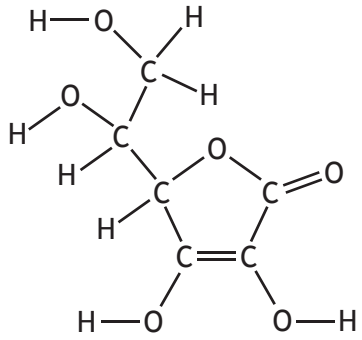
(b) Sweet potatoes are a good source of the antioxidant vitamin C.

(i) Antioxidants like vitamin C are added to food.

Explain why antioxidants are added to food. **1**

6. (b) (continued)

(ii) The structure of vitamin C is shown.



Explain **fully** why vitamin C is soluble in water. **2**

(c) Unlike sweet potatoes, white potatoes contain the chemical solanine, that can be toxic to humans in large doses. A dose of 3 mg per kg of body weight can cause toxic symptoms.

A typical white potato can contain 0.2 mg per g of solanine.

Calculate the mass of white potato that could produce a toxic dose to an adult weighing 65 kg. **3**

[Turn over

7. Natural gas is a source of methane.

(a) Methane, CH_4, can be used as a fuel.

In an experiment, methane was burned to raise the temperature of 100 cm^3 of water by 27 °C.

Using the enthalpy of combustion of methane (891 $kJ\,mol^{-1}$), calculate the mass of methane, in g, burned in this experiment. **3**

(b) The equation for the combustion of methane is shown.

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$$

Bond enthalpies can be used to calculate a theoretical enthalpy change for this reaction.

Using bond enthalpies from the data booklet, calculate the enthalpy change, in $kJ\,mol^{-1}$, for the combustion of methane. **2**

7. (continued)

(c) Methane reacts with steam to produce hydrogen.

$$\underset{GFM = 16\text{ g}}{CH_4(g)} + \underset{GFM = 18\text{ g}}{H_2O(g)} \rightarrow \underset{GFM = 28\text{ g}}{CO(g)} + \underset{GFM = 2\text{ g}}{3H_2(g)}$$

Calculate the atom economy for the formation of hydrogen. **2**

(d) Another naturally occurring gas is nitrogen dioxide, NO_2.

Nitrogen dioxide exists in equilibrium with dinitrogen tetroxide, N_2O_4.

$$2NO_2(g) \rightleftharpoons N_2O_4(g) \qquad \Delta H = -58\text{ kJ mol}^{-1}$$

Complete the table to show the conditions that would maximise the yield of **nitrogen dioxide**. **1**

Condition	High/Low
Temperature	
Pressure	

[Turn over

7. (continued)

(e) (i) In the United States Space Shuttle, dinitrogen tetroxide was reacted with methylhydrazine.

$$4CH_3NHNH_2(\ell) + 5N_2O_4(\ell) \rightarrow 4CO_2(g) + 12H_2O(g) + 9N_2(g)$$

Calculate the enthalpy of this reaction, in kJ, by using the data shown below. **2**

$C(s) + 3H_2(g) + N_2(g)$	$\rightarrow$	$CH_3NHNH_2(\ell)$	$\Delta H = +54$ kJ mol^{-1}
$N_2(g) + 2O_2(g)$	$\rightarrow$	$N_2O_4(\ell)$	$\Delta H = -20$ kJ mol^{-1}
$C(s) + O_2(g)$	$\rightarrow$	$CO_2(g)$	$\Delta H = -394$ kJ mol^{-1}
$H_2(g) + \frac{1}{2}O_2(g)$	$\rightarrow$	$H_2O(\ell)$	$\Delta H = -286$ kJ mol^{-1}
$H_2O(l)$	$\rightarrow$	$H_2O(g)$	$\Delta H = +41$ kJ mol^{-1}

(ii) Draw the full structural formula for methylhydrazine, CH_3NHNH_2. **1**

8. Fizzy drinks are made by adding carbon dioxide gas, preservative, colouring and flavouring to water.

(a) Carbon dioxide for fizzy drinks can be produced using the water-gas shift reaction.

$$CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$$

(i) A catalyst for this reaction is copper(II) oxide.

Complete the table by circling one option on each line to show the effect of copper(II) oxide on the reaction. **2**

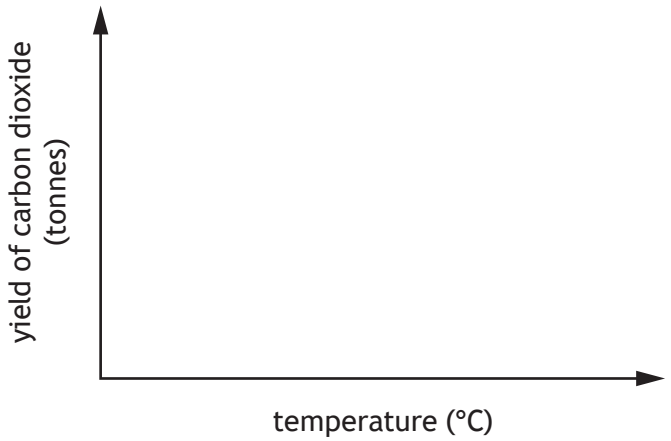
(An additional table can be found on *page 33*.)

Feature of reaction	Effect of catalyst
Rate of forward reaction	increase/decrease/no effect
Rate of reverse reaction	increase/decrease/no effect
Position of equilibrium	moves to right/moves to left/no effect

(ii) The water-gas shift reaction is exothermic.

Draw a line on the axes below to show how the yield of carbon dioxide would vary with increasing temperature. **1**

(An additional diagram can be found on *page 33*.)



[Turn over

8. (continued)

(b) A preservative added to some fizzy drinks is made by reacting sorbic acid and potassium hydroxide.

In an experiment, 7 g of sorbic acid, $C_6H_8O_2$, is reacted with 250 cm^3 of potassium hydroxide solution, concentration 0.5 mol l^{-1}.

$$C_6H_8O_2(s) + KOH(aq) \rightarrow H_2O(\ell) + C_6H_7O_2K(aq)$$

$GFM = 112$ g

Show, by calculation, that sorbic acid is the limiting reactant. **2**

8. (continued)

(c) Ammonium ferric citrate (GFM = 261.8 g) gives some drinks a characteristic orange colour. A typical drink contains 0.002% of ammonium ferric citrate.

A 1% solution contains 1 g made up to 100 cm^3 of solution.

Calculate the number of moles of ammonium ferric citrate required to make 330 cm^3 of this fizzy drink. **2**

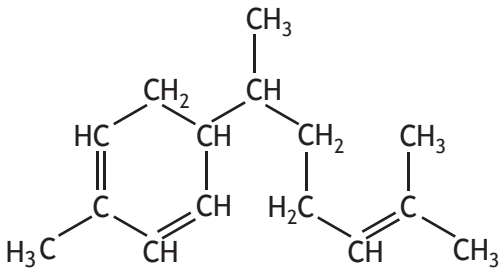
[Turn over

8. (continued)

(d) Ginger root is used as a flavouring for some fizzy drinks.

(i) Ginger oil is an essential oil obtained from ginger root.

Zingiberene is one of the main components in this essential oil.



zingiberene

(A) State one property of an essential oil. **1**

(B) Zingiberene is formed from isoprene units.

(I) Name the type of compound formed when isoprene units join together. **1**

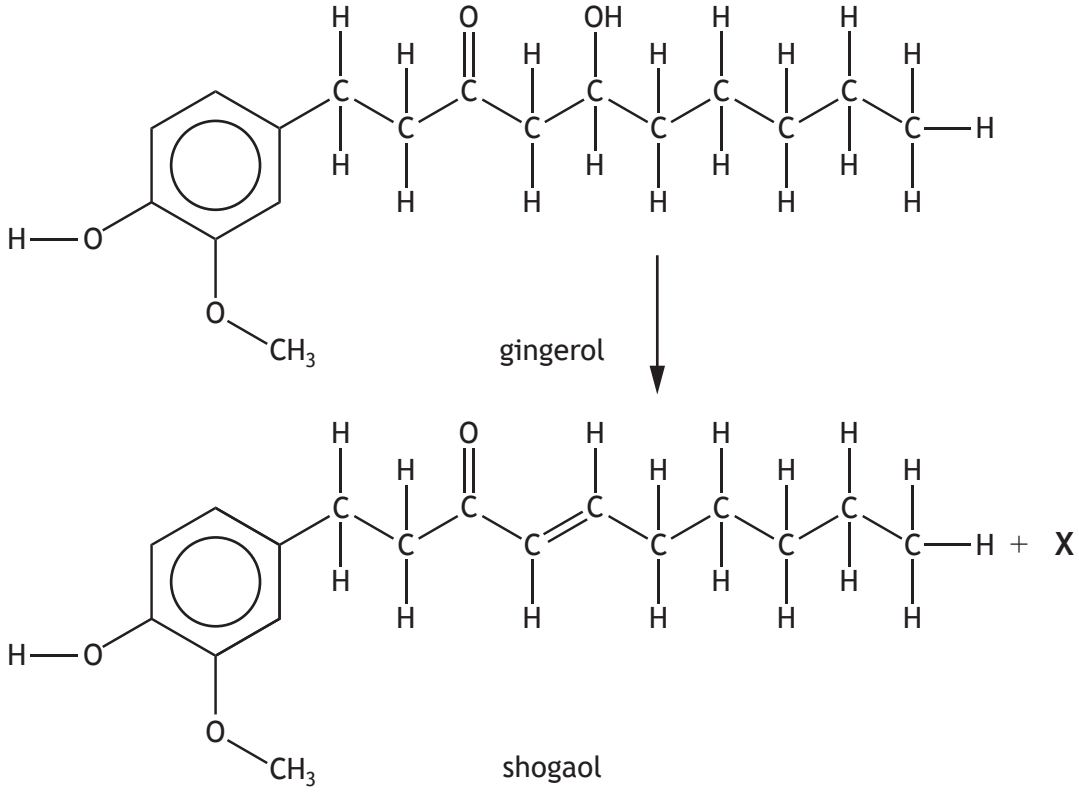
(II) Isoprene is also called 2-methyl-1,3-butadiene.

Draw a structural formula for isoprene. **1**

(III) State the number of isoprene units in a zingiberene molecule. **1**

8. (d) (continued)

(ii) Gingerol is another compound found in ginger root. Gingerol can form the compound shogaol as shown.



(A) Name product **X**. **1**

(B) Name two functional groups present in gingerol and shogaol that are **not** present in zingiberene. **1**

[Turn over

9. For a particular set of reaction conditions, the actual yield is the quantity of desired product made in a reaction.

Some examples of reactions with their desired products are shown.

Equation	Desired product
$Ba(NO_3)_2(aq) + Na_2SO_4(aq) \rightarrow BaSO_4(s) + 2NaNO_3(aq)$	$BaSO_4(s)$
$CH_3OH(\ell) + C_2H_5COOH(\ell) \rightleftharpoons C_2H_5COOCH_3(\ell) + H_2O(\ell)$	$C_2H_5COOCH_3(\ell)$
$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$	$H_2(g)$

Using your knowledge of chemistry, describe how the actual yield in a reaction could be determined.

Your answer should include experimental procedures that could be used to determine the quantity of product made in reactions such as the examples shown in the table. **3**

10. A refrigerant is a chemical used in cooling processes. Some refrigerant compounds can damage the ozone layer.

(a) The ozone depletion potential (ODP) of a refrigerant compound is the relative amount of damage that it can cause to the ozone layer. The higher the number, the greater the damage.

	Refrigerant compound	Ozone depletion potential
1	$C_2F_4Br_2$	6.00
2	CF_2ClBr	3.00
3	C_2FCl_5	1.00
4	$C_2F_3Cl_3$	0.85
5	$C_2F_4Cl_2$	0.58
6	$C_2H_3Cl_3$	0.16

(i) Describe a relationship between the formulae of refrigerant compounds 3, 4 and their ODP. **1**

(ii) Identify which pair of compounds should be used to show the effect of replacing chlorine atoms with bromine atoms in refrigerant compounds. **1**

(iii) The refrigerants carbon dioxide, CO_2, and ammonia, NH_3, have ODP values of 0.00.

Suggest why this is the case. **1**

10. (continued)

(b) The compound difluoromethane, CH_2F_2, is also used as a refrigerant. It is made by reacting fluorine gas with fluoromethane, CH_3F, in a free radical chain reaction.

(i) State what is meant by a free radical. **1**

(ii) The first step in the reaction involves splitting a fluorine molecule to produce two fluorine radicals.

$$F_2 \rightarrow 2F\bullet$$

(A) State the name given to this step. **1**

(B) Write an equation for a possible propagation step in this reaction. **1**

(c) Household fridges use coolants made from refrigerant compounds. A common coolant is made from 50% difluoromethane, CH_2F_2, ($GFM = 52$ g) and 50% pentafluoroethane, CF_3CHF_2, ($GFM = 120$ g).

A typical fridge contains 0.05 kg of coolant.

Calculate the number of moles of pentafluoroethane required to make this mass of coolant. **1**

[Turn over

11. Spinach is a leafy green vegetable.

(a) Fertilisers containing copper(II) ethanoate are used to supply spinach with copper ions.

Copper(II) ethanoate can be made by reacting copper(II) carbonate with ethanoic acid.

(i) Name the other products of this reaction. **1**

(ii) Write the ionic formula of copper(II) ethanoate. **1**

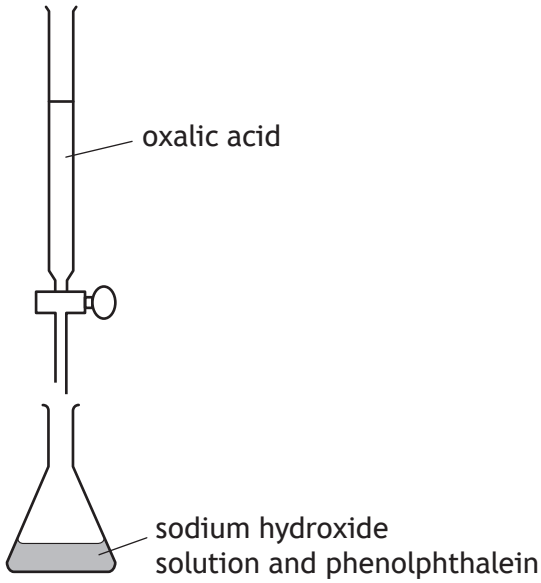
(b) Spinach is a source of oxalic acid.

A standard solution of oxalic acid can be used to determine the accurate concentration of a sodium hydroxide solution.

Given an accurately known mass of oxalic acid, describe **fully** how 250 cm^3 of a standard solution of oxalic acid could be prepared. **3**

11. (continued)

(c) The concentration of sodium hydroxide can be determined by titration with oxalic acid.



An accurate volume of sodium hydroxide solution is measured into a conical flask using a pipette.

(i) Draw a diagram of a pipette suitable for measuring an accurate volume. **1**

(ii) The indicator used in this titration is phenolphthalein. Phenolphthalein is colourless in acidic and neutral solutions but is pink in alkaline conditions.

State the colour change that would be observed at the end point in this titration. **1**

[Turn over

11. (c) (continued)

(iii) The titration was repeated until results were obtained that were within 0.2 cm^3 of each other.

State the term used to describe titre volumes within 0.2 cm^3 of each other. **1**

(d) The equation for the reaction of oxalic acid and sodium hydroxide is shown.

$$H_2C_2O_4(aq) + 2NaOH(aq) \rightarrow Na_2C_2O_4(aq) + 2H_2O(\ell)$$

The concentration of sodium hydroxide solution was determined by titrating 25.0 cm^3 samples with 0.126 mol l^{-1} oxalic acid solution.

The average volume of oxalic acid solution required in the titration was 26.75 cm^3.

Calculate the concentration, in mol l^{-1}, of the sodium hydroxide. **3**

[END OF QUESTION PAPER]

National Qualifications

X813/76/12

Chemistry
Paper 1 — Multiple choice

Duration — 40 minutes

Total marks — 25

Attempt ALL questions.

You may use a calculator.

Instructions for the completion of Paper 1 are given on *page 02* of your answer booklet X813/76/02.

Record your answers on the answer grid on *page 03* of your answer booklet.

You may refer to the Chemistry Data Booklet for Higher and Advanced Higher.

Space for rough work is provided at the end of this booklet.

Before leaving the examination room you must give your answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

X8137612

Total marks — 25

Attempt ALL questions

1. Aluminium carbonate can be produced by the following reaction.

$$2AlCl_3(aq) + 3K_2CO_3(aq) \rightarrow Al_2(CO_3)_3(s) + 6KCl(aq)$$

The most suitable method for separating the aluminium carbonate from the mixture is

A filtration

B distillation

C evaporation

D collection over water.

2. The difference in the size of sodium and chlorine atoms is mainly due to the difference in the

A mass of each atom

B number of electrons

C number of neutrons

D number of protons.

3. Solid carbon dioxide is known as 'dry ice'. It changes directly to a gas when it is heated.

$$CO_2(s) \rightarrow CO_2(g)$$

The strongest bonds broken in this process are

A polar covalent bonds

B London dispersion forces

C non-polar covalent bonds

D permanent dipole-permanent dipole interactions.

4. Which of the following statements is correct?

A Elements with high electronegativities tend to be reduced

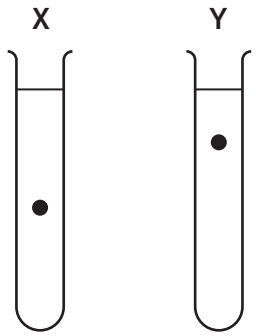
B Elements with high electronegativities tend to act as reducing agents

C Elements with low electronegativities tend to gain electrons

D Elements with low electronegativities tend to act as oxidising agents

5. The viscosities of two liquids, **X** and **Y**, were investigated by dropping a metal ball into a tube of each liquid.

The diagram shows the position of the metal balls after 10 seconds.



Which line in the table correctly describes the viscosity and relative strengths of the van der Waals forces in liquids **X** and **Y**?

	X	**Y**
A	most viscous	strongest van der Waals forces
B	least viscous	weakest van der Waals forces
C	least viscous	strongest van der Waals forces
D	most viscous	weakest van der Waals forces

6. What is the enthalpy change, in kJ mol^{-1}, for the following reaction?

$$Be(g) \rightarrow Be^{2+}(g) + 2e^{-}$$

A 900

B 1757

C 2657

D 3514

7. Which of the following is an isomer of pentanoic acid?

A 2-methylpropanoic acid

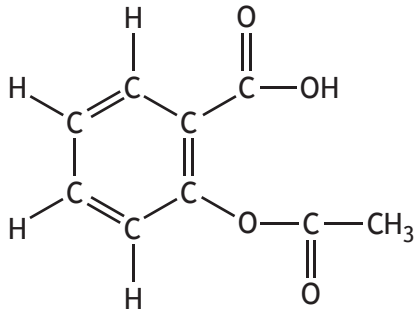
B propyl methanoate

C 2-ethylbutanoic acid

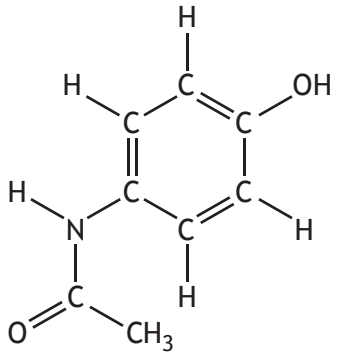
D ethyl propanoate

[Turn over

8. The structures of two common painkillers are shown below.



aspirin



paracetamol

Which of the following is true?

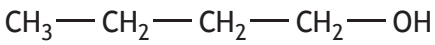
A Both painkillers are ketones

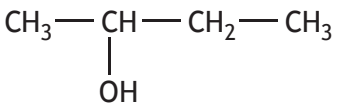
B Aspirin contains a carboxyl group and an ester link

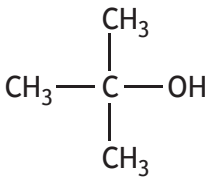
C Paracetamol contains a hydroxyl group and a carboxyl group

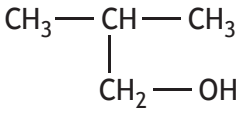
D Neither painkiller contains an amide link

9. Which two isomers would each produce an acid when warmed with acidified potassium dichromate solution?

1 

2 

3 

4 

A 1 and 2

B 1 and 4

C 2 and 3

D 3 and 4

10. When an aldehyde is converted into the corresponding alcohol a reduction reaction takes place.

Reduction of 2-methylbutanal ($GFM = 86$) produces a compound with a GFM of

A 70

B 84

C 88

D 102.

11. Which of the following compounds would react with sodium hydroxide solution to form the salt sodium propanoate?

A $HCOOC_2H_5$

B CH_3COOCH_3

C C_2H_5COOH

D C_3H_7COOH

12. When a protein is denatured

A it is broken into amino acids

B hydrogen bonds are broken

C peptide links are hydrolysed

D water molecules are eliminated.

13. Compared with oils, fats are

A less saturated and have higher melting points

B less saturated and have lower melting points

C more saturated and have higher melting points

D more saturated and have lower melting points.

14. Vitamin C is an antioxidant used to preserve food and lengthen shelf-life.

Which of the following does **not** describe an antioxidant?

A Electron donor

B Oxidising agent

C Reducing agent

D Free-radical scavenger

[Turn over

15. On exposure to UV light, methane and chlorine undergo a chain reaction.

Which of the following is a propagation step in this reaction?

A $\cdot CH_3 + Cl\cdot \rightarrow CH_3Cl$

B $Cl_2 \rightarrow 2Cl\cdot$

C $H\cdot + Cl\cdot \rightarrow HCl$

D $CH_4 + Cl\cdot \rightarrow \cdot CH_3 + HCl$

16. A reaction was carried out at four different temperatures. The table shows the times taken for the reaction to occur.

Temperature (°C)	20	30	40	50
Time (s)	60	30	14	5

The results show that

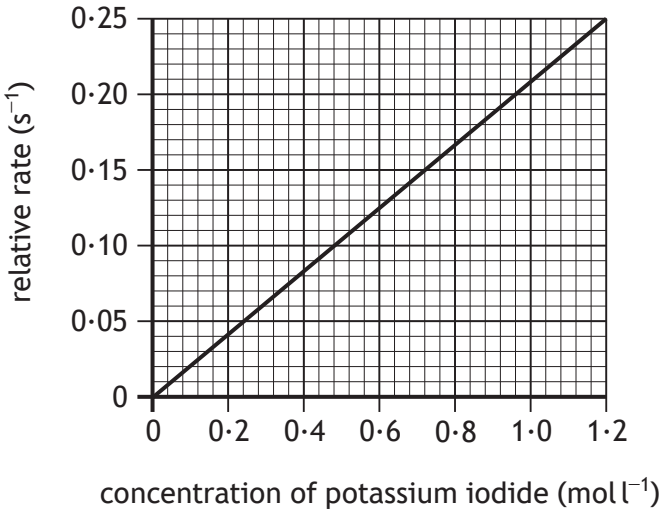
A a small rise in temperature results in a large increase in the rate of the reaction

B the activation energy increases with increasing temperature

C doubling the temperature of the reaction doubles the rate of the reaction

D the reaction is slowing down with increasing temperature.

17. The graph shows the effect of increasing the concentration of potassium iodide solution on reaction rate.



What was the concentration, in mol l^{-1}, of the potassium iodide solution used in a reaction that took 5 s to complete?

A 0·04

B 0·20

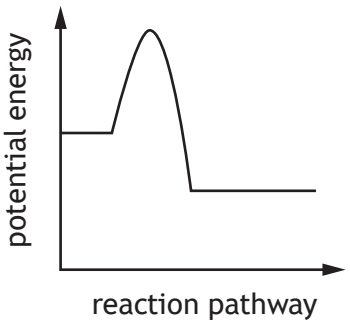
C 0·24

D 0·96

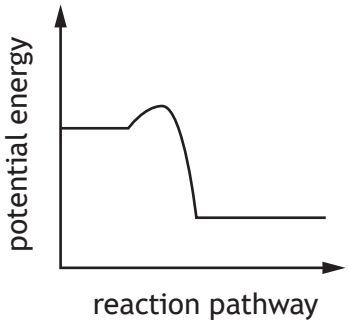
[Turn over

18. Which of the following diagrams represents an exothermic reaction that is more likely to take place at room temperature?

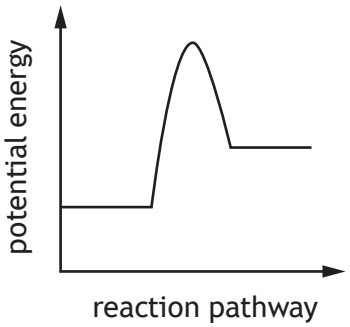
A



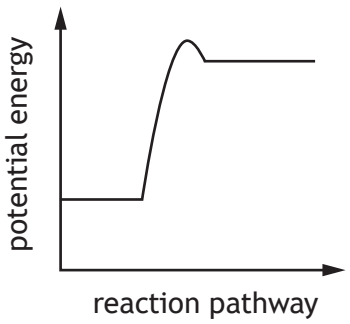
B



C



D



19. Which of the following will result in the volume of the products being half the volume of the reactants?

A $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$

B $C(s) + O_2(g) \rightarrow CO_2(g)$

C $C_2H_4(g) + 3O_2(g) \rightarrow 2CO_2(g) + 2H_2O(\ell)$

D $C(s) + H_2O(g) \rightarrow H_2(g) + CO(g)$

20. The ester ethyl ethanoate is produced by the following reaction.

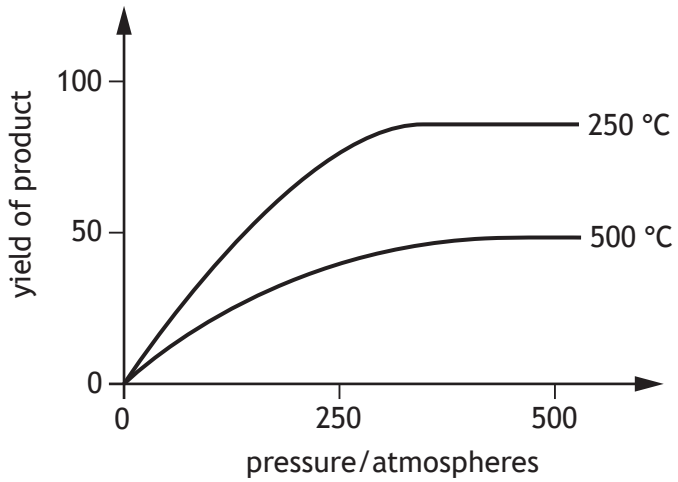
$$CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$$

Which of the following mixtures would produce 0·8 moles of ester if the yield was 80%?

	moles of CH_3COOH	**moles of C_2H_5OH**
A	0·4	0·4
B	0·5	0·5
C	0·8	0·8
D	1·0	1·0

[Turn over

21. The graph shows how the yield of product in a reversible reaction varies with pressure at two different temperatures.



From this information it can be concluded that

A the reaction is exothermic

B all reactants are converted to products at 250 °C and 300 atmospheres

C increasing the temperature increases the yield

D increasing the pressure above 200 atmospheres has no effect on yield.

22. $2KOH(aq) + H_2SO_4(aq) \rightarrow K_2SO_4(aq) + 2H_2O(\ell)$

How many moles of potassium hydroxide, KOH, neutralise 50 cm^3 of 0·2 mol l^{-1} sulfuric acid, H_2SO_4?

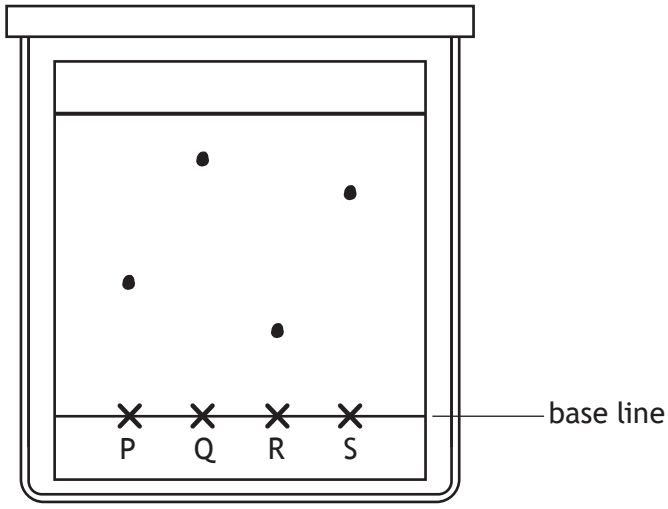
A 0·01

B 0·02

C 0·10

D 0·40

23. Four amino acids, P, Q, R and S were analysed by chromatography.
Larger molecules travel a shorter distance from the base line.
Less polar molecules travel a greater distance from the base line.



Which of the following statements is correct?

A P is less polar than S
B Q is a larger molecule than P
C R is more polar than P
D S is a smaller molecule than Q

24. A student produced the following results for a redox titration.

Sample	Volume of solution added (cm^3)
1	21·0
2	20·3
3	20·7
4	20·4

The volume of solution, in cm^3, that should be used in the titration calculation is

A 20·35
B 20·50
C 20·55
D 20·60

[Turn over for next question

25. An equilibrium mixture of NO_2 and N_2O_4 in a sealed gas syringe has a pale brown colour.

$$2NO_2(g) \rightleftharpoons N_2O_4(g)$$

Increasing the pressure causes the mixture to become paler.

Increasing the temperature causes the mixture to become darker.

Which line in the table correctly identifies the colour of NO_2 and the enthalpy change for the forward reaction?

	Colour of NO_2	**Enthalpy change for the forward reaction**
A	brown	exothermic
B	brown	endothermic
C	colourless	exothermic
D	colourless	endothermic

[END OF QUESTION PAPER]

FOR OFFICIAL USE

National Qualifications

Mark

X813/76/01

Chemistry Paper 2

Duration — 2 hours 20 minutes

X8137601

Fill in these boxes and read what is printed below.

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth

Day Month Year

Scottish candidate number

Total marks — 95

Attempt ALL questions.

You may use a calculator.

You may refer to the Chemistry Data Booklet for Higher and Advanced Higher.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers and rough work is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting. Any rough work must be written in this booklet. Score through your rough work when you have written your final copy.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

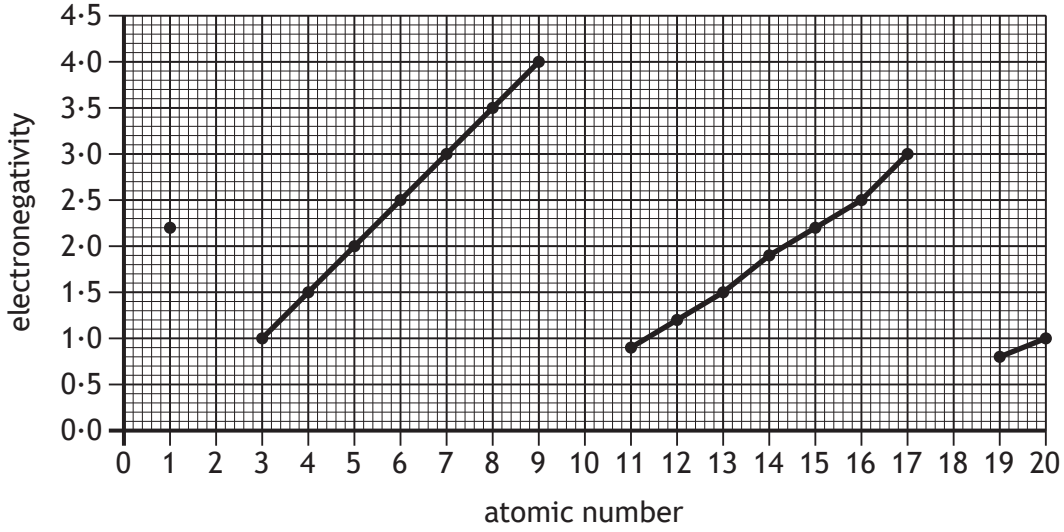
X8137601

Total marks — 95

Attempt ALL questions

1. Electronegativity is a measure of the attraction an atom involved in a bond has for the electrons of the bond.

The graph shows the trend in electronegativity for the first 20 elements.



(a) (i) State the trend in electronegativity as you go across period 2 from lithium to fluorine. **1**

(ii) No electronegativity values are shown for the elements with atomic numbers 2, 10 and 18.

Suggest why no values are provided for these elements. **1**

1. (a) (continued)

(iii) On descending Group 7 from fluorine to iodine, the electronegativity of the elements decreases.

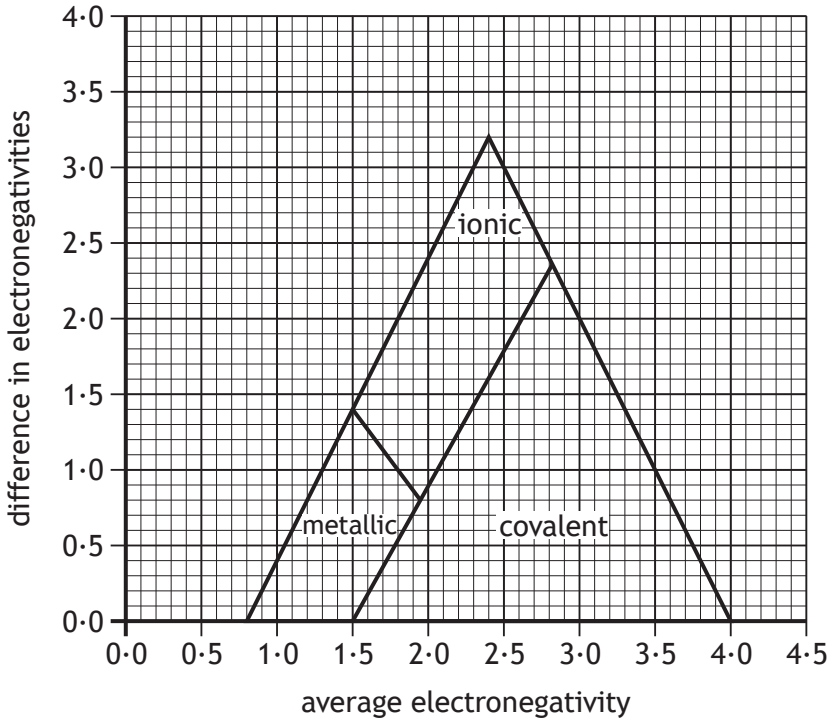
Explain why the electronegativity of the elements decreases as you go down the group. **1**

[Turn over

1. (continued)

(b) Electronegativity values can be used to predict the type of bonding present in substances.

The type of bonding between two elements can be predicted using the diagram below.



(i) Using the information in the diagram, state the highest average electronegativity found in **ionic** compounds. **1**

MARKS

DO NOT WRITE IN THIS MARGIN

(ii) The electronegativity values of magnesium and nitrogen are shown.

Electronegativity of magnesium = 1·2
Electronegativity of nitrogen = 3·0

Draw an X on the diagram on *page 04* to show the position of magnesium nitride.

Show your calculations clearly. **2**

(An additional diagram, if required, can be found on *page 38*.)

(iii) Compounds with a difference in electronegativities of 1·5 can have ionic or covalent properties.

(A) The electronegativity difference between the elements in lithium sulfide is 1·5.

Write an **ionic** formula for lithium sulfide. **1**

(B) A compound contains two non-metal elements with an electronegativity difference of 1·5.

Suggest names for the two **non-metal** elements. **1**

(c) Fluorine has a greater attraction for bonding electrons than hydrogen.

State the term used to describe the type of **covalent** bond in hydrogen fluoride. **1**

2. Carbon and its compounds are important in the chemical industry.

(a) Carbon can exist in multiple forms. Two of these are fullerenes and diamond.

(i) Name another form of carbon. **1**

(ii) Both diamond and the fullerene, C_{60}, can change directly from a solid to a gas. This is called sublimation. For diamond this occurs at 3825 °C. The fullerene changes from a solid to a gas at approximately 550 °C. Complete the table below to show the strongest type of attraction broken when diamond and the fullerene sublime. **2**

Form of carbon	Strongest attraction broken
diamond	
fullerene	

(iii) The fullerene, C_{60}, reacts with bromine solution in an addition reaction to produce the bromofullerene, $C_{60}Br_{24}$.

Determine the number of double bonds present in a molecule of C_{60}. **1**

(b) Carbon can combine with oxygen to make carbon monoxide, CO. Carbon monoxide is used in the production of iron from iron(III) oxide.

$$Fe_2O_3(s) + 3CO(g) \rightarrow 2Fe(\ell) + 3CO_2(g)$$

$GFM = 159{\cdot}6$ g $\quad GFM = 28{\cdot}0$ g $\quad GFM = 55{\cdot}8$ g $\quad GFM = 44{\cdot}0$ g

Calculate the atom economy for the production of iron. **2**

2. (continued)

(c) Carbon monoxide can be produced by the reaction of methane and steam.

$$CH_4(g) + H_2O(g) \rightarrow CO(g) + 3H_2(g)$$

$CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g)$ $\quad \Delta H = -283 \text{ kJ mol}^{-1}$

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(g)$ $\quad \Delta H = -286 \text{ kJ mol}^{-1}$

$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$ $\quad \Delta H = -891 \text{ kJ mol}^{-1}$

Calculate the enthalpy change, in kJ mol^{-1}, for this reaction. **2**

[Turn over

3. A teacher told a class that collisions were the key to chemical reactions.

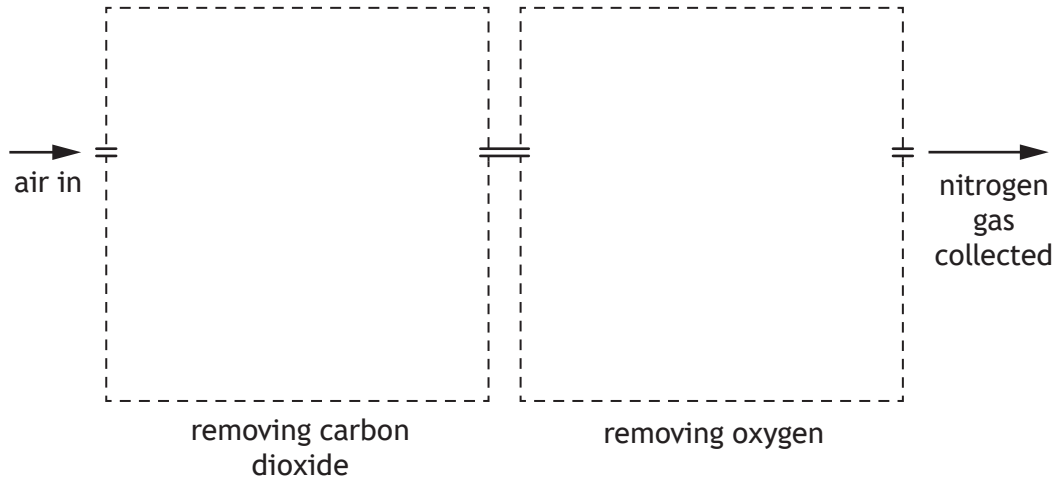
Using your knowledge of chemistry, discuss this statement. **3**

4. Nitrogen is a stable element and only reacts with a few other elements.

(a) Explain, using bond enthalpy values, why the element nitrogen is so unreactive. **1**

(b) A laboratory preparation of nitrogen gas involves separating nitrogen from all of the other gases in the air and then collecting it.

Air is bubbled through potassium hydroxide solution to remove carbon dioxide. The remaining gases are passed over heated copper to remove oxygen before the remaining nitrogen is collected.



Complete a labelled diagram for the preparation of a sample of nitrogen by drawing suitable apparatus for carrying out this preparation. **2**

(An additional diagram, if required, can be found on *page 38*.)

[Turn over

4. (continued)

MARKS | DO NOT WRITE IN THIS MARGIN

(c) Nitrogen can react with lithium at room temperature to form the compound lithium nitride, Li_3N.

(i) A scientist prepared a sample of lithium nitride by reacting 0·9 litres of nitrogen gas with 0·5 g of lithium.

$$6Li(s) + N_2(g) \rightarrow 2Li_3N(s)$$

$$GFM = 6{\cdot}9 \text{ g}$$

Determine, by calculation, which of the reactants was in excess. **3**

Take the volume of 1 mole of nitrogen gas to be 24 litres.

(Clearly show your working for the calculation.)

(ii) Lithium nitride reduces copper(I) ions to copper atoms.

Write the ion-electron equation to show the reduction of copper(I) ions. **1**

(iii) Lithium nitride is ionic.

State the term used to describe the structure of solid ionic compounds like lithium nitride. **1**

MARKS

DO NOT WRITE IN THIS MARGIN

(d) During thunderstorms, nitrogen can react with oxygen to form different compounds.

Nitrogen and oxygen can react to form the free radical molecule nitrogen monoxide, ·N=O.

(i) State what is meant by the term free radical. **1**

(ii) The equation for this reaction is

$$N{\equiv}N + O{=}O \longrightarrow \begin{matrix} \cdot N{=}O \\ \cdot N{=}O \end{matrix}$$

The enthalpy change, ΔH, for this reaction is +91 kJ mol^{-1}.

Use this data and the bond enthalpy values shown in the data booklet to calculate the bond enthalpy, in kJ mol^{-1}, of the nitrogen to oxygen double bond in nitrogen monoxide. **2**

(iii) Nitrogen monoxide free radicals can react with hydroxyl free radicals to form a molecule of nitrous acid, HNO_2.

(A) Name the type of reaction that occurs when two free radicals join together. **1**

(B) Draw a possible structure for the HNO_2 molecule. **1**

4. (continued)

(e) Nitrogen can also react with hydrogen in the Haber process to form ammonia.

(i) The equation to produce ammonia is shown.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \qquad \Delta H = -92 \text{ kJ mol}^{-1}$$

Explain fully the effect of decreasing the reaction temperature on the yield of ammonia. **2**

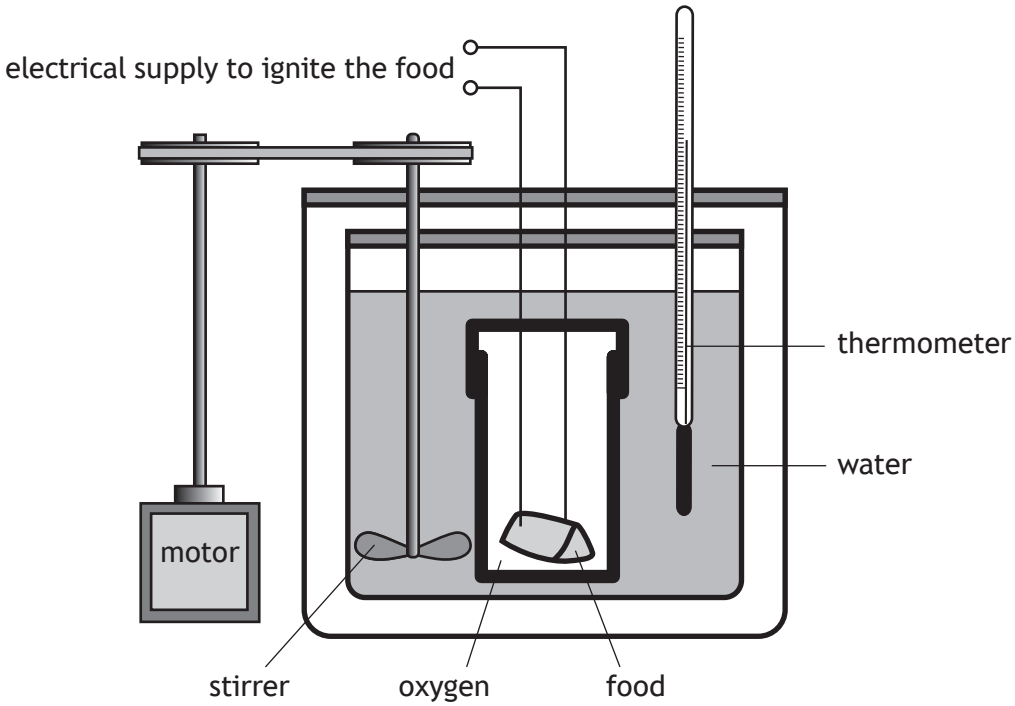
(ii) Ammonia can be reacted to produce the explosive nitroglycerin. A small shock or physical bump can make the nitroglycerin explode. The reaction that takes place is shown.

$$C_3H_5N_3O_9 \rightarrow CO_2 + H_2O + N_2 + O_2$$

(A) Balance this equation. **1**

(B) Suggest why a small shock or bump can cause nitroglycerin to react. **1**

5. A bomb calorimeter is used in the food industry to determine the energy released by foods when they are burned.



Bomb calorimeter

(a) Identify a feature of the bomb calorimeter and explain how this allows the accurate determination of the energy released by burning foods. **1**

[Turn over

5. (continued)

(b) Fats and oils release a large amount of energy when they are burned.

(i) A 1·00 g sample of the oil, triolein (GFM = 884 g) was burned in a bomb calorimeter.

The temperature rise in the 775 cm^3 of water was 11·9 °C.

Calculate the enthalpy of combustion, in kJ mol^{-1}, of triolein. **3**

(ii) Foods with a lower respiratory quotient are better for people who find it difficult to obtain energy from food.

The respiratory quotient, RQ, is the ratio of carbon dioxide, CO_2, produced to the oxygen, O_2, consumed when a food is burned in the body.

$$\text{Respiratory quotient} = \frac{CO_2 \text{ produced}}{O_2 \text{ consumed}}$$

The equation for the combustion of triolein, $C_{57}H_{104}O_6$, is shown.

$$C_{57}H_{104}O_6(\ell) + 80\ O_2(g) \rightarrow 57\ CO_2(g) + 52\ H_2O(\ell)$$

Determine the respiratory quotient for triolein. **1**

5. (continued)

(c) Tristearin, $C_{57}H_{110}O_6$, is a saturated fat.

The table shows the viscosity of different saturated fats at 70 °C.

Fat	Molecular formula	Viscosity at 70 °C (units)
Tributyrin	$C_{15}H_{26}O_6$	3·0
Tricaproin	$C_{21}H_{38}O_6$	5·9
Tricaprylin	$C_{27}H_{50}O_6$	8·8
Tricaprin	$C_{33}H_{62}O_6$	11·7
Trilaurin	$C_{39}H_{74}O_6$	14·6

(i) Predict the viscosity of tristearin at 70 °C. **1**

(ii) Edible fats and oils are molecules that contain three ester links.

Explain why glycerol is able to form fats and oils. **1**

[Turn over

6. Seaweed can contain high levels of iodine.

(a) One type of seaweed contains 0·133 g of iodine per kilogram of seaweed.

The World Health Organisation recommends a daily intake of iodine of 0·15 mg.

Calculate the mass of seaweed that would provide the recommended daily intake. **2**

(b) An experiment was carried out to determine the quantity of iodine in a sample of dried seaweed.

(i) The first step in this process involves burning an accurately known mass of dried seaweed to ash.

Describe the steps involved in measuring the mass of the seaweed by difference. **1**

(ii) The seaweed ash contains iodide ions. These react with hydrogen peroxide in acid conditions.

$$H_2O_2(aq) + 2I^-(aq) + 2H^+(aq) \rightarrow 2H_2O(\ell) + I_2(aq)$$

Identify the reducing agent in this reaction. **1**

6. (b) (continued)

(iii) The released iodine reacts with sodium thiosulfate, $Na_2S_2O_3$.

A standard solution of sodium thiosulfate, $Na_2S_2O_3$, is required to react with the released iodine.

State what is meant by a standard solution. **1**

(iv) For this sample of seaweed, 0·00026 moles of sodium thiosulfate were required to fully react with the released iodine, I_2.

$$I_2(aq) + 2Na_2S_2O_3(aq) \rightarrow 2NaI(aq) + Na_2S_4O_6(aq)$$

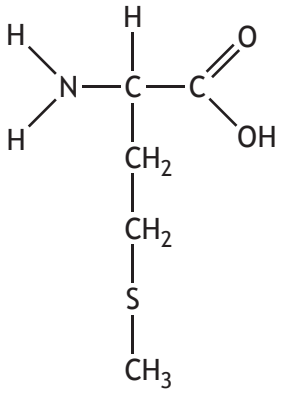
(A) Calculate the number of moles of iodine required to react with 0·00026 moles of sodium thiosulfate. **1**

(B) Using your answer to part (A), calculate the mass, in g, of iodine in the seaweed sample. **1**

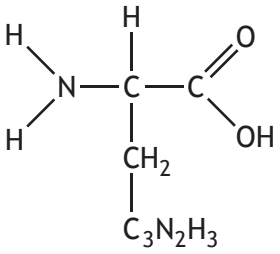
6. (continued)

(c) Seaweed is a major component of the diet of sheep living on the island of North Ronaldsay.

Sheep wool is made mainly of a protein. This protein contains the essential amino acids methionine and histidine.



methionine



histidine

(i) State what is meant by an essential amino acid. **1**

(ii) When two amino acids are joined together by a peptide link, a dipeptide is formed.

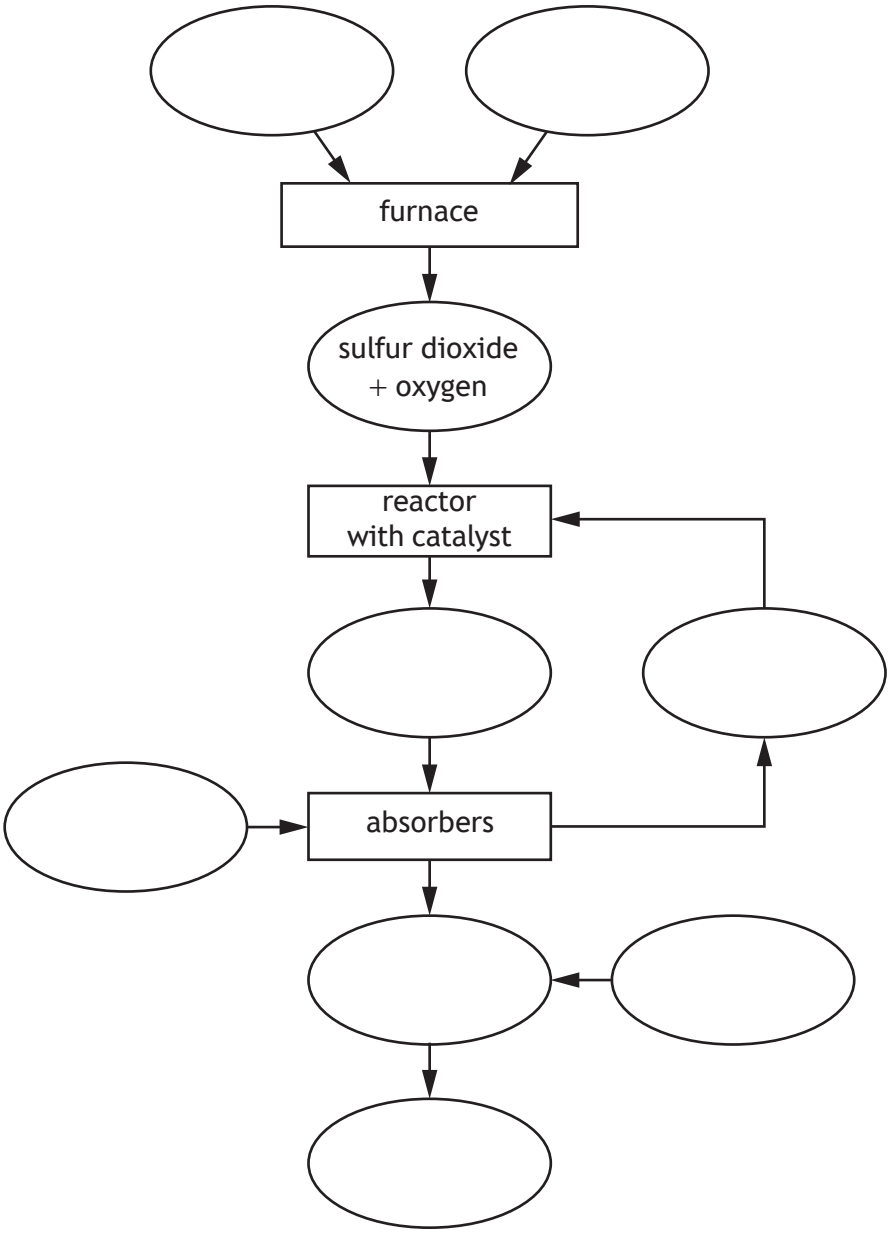
Draw a structural formula for the dipeptide formed from methionine and histidine. **1**

7. Sulfuric acid is an important chemical with many uses in industry. The main process used to make sulfuric acid is the Contact Process.

(a) The Contact Process starts when sulfur is burned in a furnace with excess air. This forms a gas mixture that contains sulfur dioxide and oxygen. The gas mixture is cooled and then passed into the reactor which contains a catalyst. Sulfur trioxide is formed and then passed into absorbers where it is absorbed into concentrated sulfuric acid to form a product called oleum. The oleum is diluted with water to give sulfuric acid. Unreacted sulfur dioxide is not absorbed and is recycled back into the reactor.

Use the information above to complete the flow diagram for the Contact Process. **2**

(An additional diagram, if required, can be found on *page 39*.)



7. (continued)

(b) The reactions involved in the Contact Process are highly exothermic.

State a disadvantage of industrial processes that involve reactions that are highly exothermic. **1**

(c) Sulfur and oxygen contain London dispersion forces.

(i) Explain how London dispersion forces arise. **2**

(ii) Explain fully why the London dispersion forces in sulfur are stronger than those in oxygen. **2**

7. (continued)

(d) The reaction that occurs in the reactor of the Contact Process is

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g) \qquad \Delta H = -192 \text{ kJ mol}^{-1}$$

Circle the correct statement in each column of the table to show the effect of using a catalyst in the reaction. **1**

Effect of catalyst on enthalpy change	Effect of catalyst on activation energy
increase stay the same decrease	increase stay the same decrease

(e) One use of sulfuric acid is in the production of soapless detergents.

(i) State the advantage that soapless detergents have over soap when used with hard water. **1**

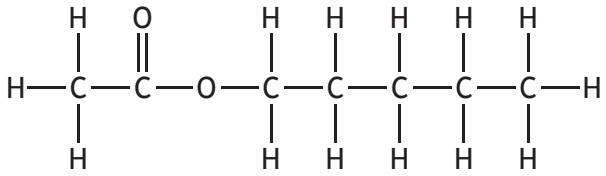
(ii) Describe the key structural features of a soapless detergent molecule. **1**

[Turn over

8. Sweets contain a wide variety of chemicals.

(a) Many sweets contain esters.

(i) The structure of an ester used to produce a pear flavour in some sweets is



Name this ester. **1**

(ii) Name the type of reaction used to form esters. **1**

(b) Fizzy sweets can contain citric acid, $C_6H_8O_7$, and sodium bicarbonate, $NaHCO_3$.

When the sweets dissolve, the citric acid and sodium bicarbonate react together to make carbon dioxide gas.

(i) To calculate the mass of citric acid in a sweet, 5 sweets were dissolved in water and the resulting carbon dioxide was collected and measured.

Suggest why carbon dioxide can be collected over water. **1**

8. (b) (continued)

(ii) In one experiment, 5 sweets were dissolved, and 55 cm^3 of carbon dioxide gas was produced.

$C_6H_8O_7(aq) + 3NaHCO_3(aq) \rightarrow 3CO_2(g) + 3H_2O(\ell) + C_6H_5O_7Na_3(aq)$

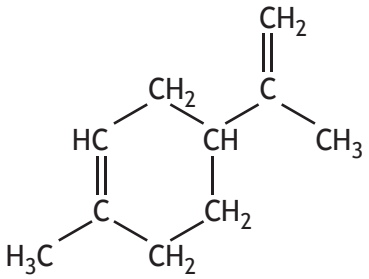
$GFM = 192$ g

Calculate the mass of citric acid, in g, in one sweet. **2**

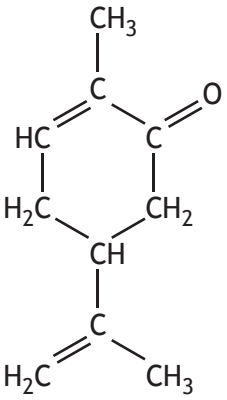
Take the volume of 1 mole of carbon dioxide to be 24 litres.

8. (continued)

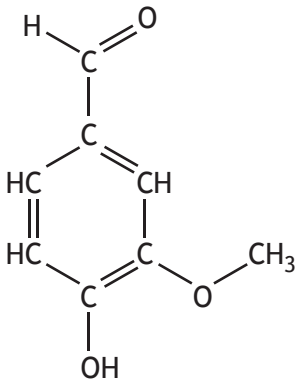
(c) The distinctive smell of some sweets is due to molecules such as limonene, carvone and vanillin.



limonene



carvone



vanillin

(i) Vanillin is an aldehyde and carvone is a ketone.

(A) State the colour change that would be observed when aldehydes react with acidified potassium dichromate. **1**

(B) Suggest a different chemical that could be used to distinguish aldehydes from ketones. **1**

(ii) Limonene is a terpene consisting of joined isoprene units.

State the number of isoprene units in a limonene molecule. **1**

8. (c) (continued)

(iii) 1 kg of natural vanillin costs £1050. To make a packet of sweets, 5 cm^3 of vanillin solution is used. This contains 0·184 g of vanillin per 100 cm^3 of solution.

Calculate the cost, in pence, of the natural vanillin required to make the packet of sweets. **2**

[Turn over

9. A brand of antiseptic mouthwash contains hydrogen peroxide along with several other chemicals such as water, flavourings and colouring.

(a) 100 cm^3 of mouthwash contains 1·5 g of 35% hydrogen peroxide solution.

A 35% hydrogen peroxide solution contains 35 g of hydrogen peroxide in 100 cm^3 of solution.

(i) Calculate the mass of hydrogen peroxide, in g, present in a 300 cm^3 bottle of the mouthwash. **1**

(ii) Enzymes in saliva act as catalysts in the decomposition of hydrogen peroxide.

Name the family of compounds to which enzymes belong. **1**

(iii) The concentration of hydrogen peroxide can be determined by a titration with a solution of potassium permanganate.

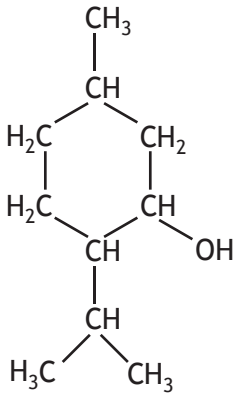
Name the **two** pieces of equipment that would be required to accurately measure the volumes of hydrogen peroxide and potassium permanganate used in the titration. **1**

(b) Volatile, non-water-soluble compounds obtained from plants can be used to provide the minty aroma of the mouthwash.

State the term used to describe the mixture of volatile, non-water-soluble aroma compounds obtained from plants. **1**

9. (continued)

(c) The mouthwash also contains menthol.



menthol

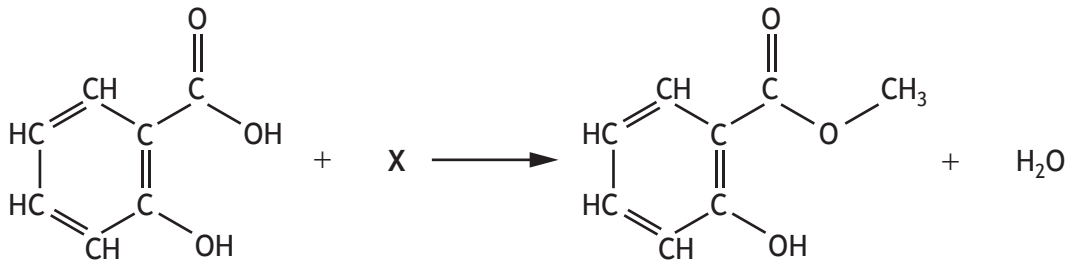
(i) Menthol is based on isoprene units.
State the systematic name for isoprene. **1**

(ii) Menthol can be oxidised to form a mint flavoured compound.
State the type of compound formed when menthol is oxidised. **1**

[Turn over

9. (continued)

(d) Methyl salicylate is an ester found in the mouthwash that can be formed from salicylic acid.



salicylic acid methyl salicylate

(i) Name reactant **X**. **1**

(ii) A scientist prepared a sample of methyl salicylate using 28·3 g salicylic acid and an excess of reactant **X**.

$C_7H_6O_3$ + **X** → $C_8H_8O_3$ + H_2O

salicylic acid methyl salicylate

$GFM = 138$ g $GFM = 152$ g

The scientist produced 24·7 g of methyl salicylate.

Calculate the percentage yield of methyl salicylate. **2**

9. (d) (continued)

(iii) Methyl salicylate can be toxic to humans at 0·14 g per kg of body mass. It can be obtained from a plant substance called oil of wintergreen. 5·0 cm^3 of oil of wintergreen contains 7·0 g of methyl salicylate.

Calculate the minimum volume, in cm^3, of oil of wintergreen that would provide a toxic dose to a human with body mass of 65 kg. **2**

[Turn over

10. Cow's milk is mostly made up of water, with small amounts of fats and oils, proteins, sugars and other compounds. Milk is white because it contains small droplets of fats and oils that are dispersed in the water. It also contains small droplets of proteins.

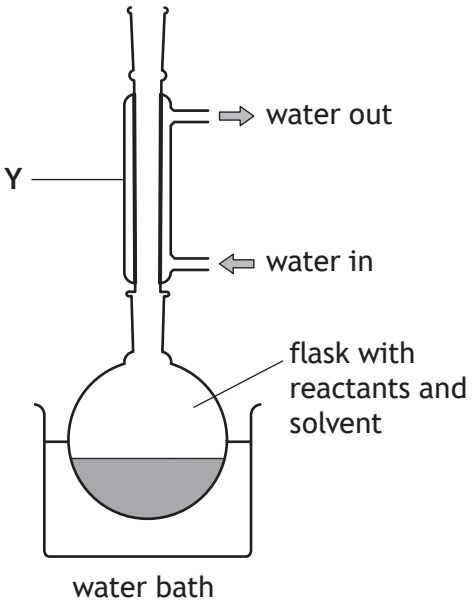
The fats in the milk depend on what the cow eats — in summer the fats obtained from the milk have a higher melting point than those obtained from winter milk.

Milk goes off because of reactions involving the hydroxyl groups on the sugar molecules in milk.

Using your knowledge of chemistry, discuss the chemistry of cow's milk. **3**

11. A Grignard reaction involves reacting compounds containing a carbonyl group with a Grignard reagent to make alcohols.

(a) The first step in the reaction involves heating the reactants in a solvent to form a Grignard reagent.



(i) Suggest why a water bath is used to heat the reaction. **1**

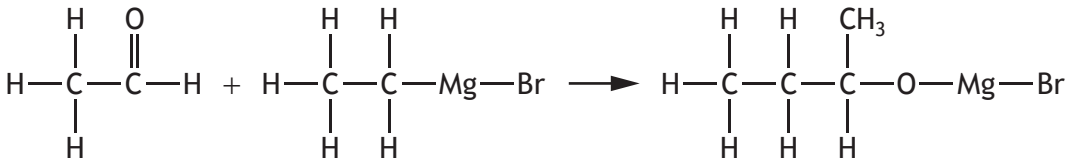
(ii) Name the piece of apparatus labelled **Y** in the diagram. **1**

[Turn over

11. (continued)

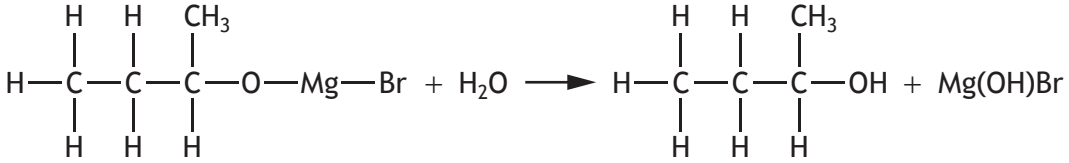
(b) Ethanal reacts with a Grignard reagent in two steps.

step 1



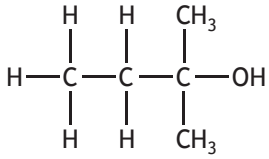
Grignard reagent

step 2



(i) Suggest a name for the reaction shown in step 1. **1**

(ii) The alcohol that is made when propanone reacts with the same Grignard reagent is shown.



Name this alcohol. **1**

11. (b) (continued)

(iii) Draw a structural formula for the alcohol formed when the Grignard reagent CH_3CH_2MgBr reacts with pentan-2-one. **1**

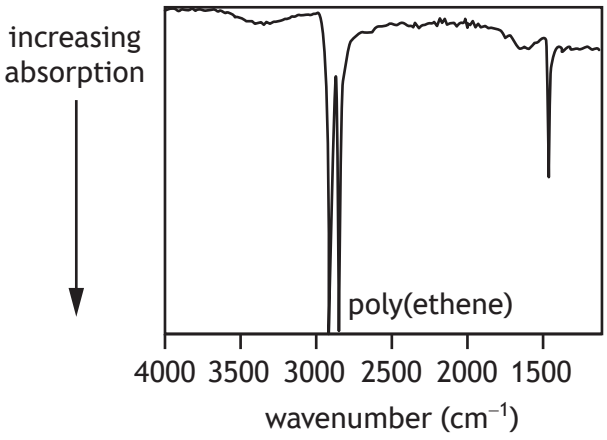
[Turn over

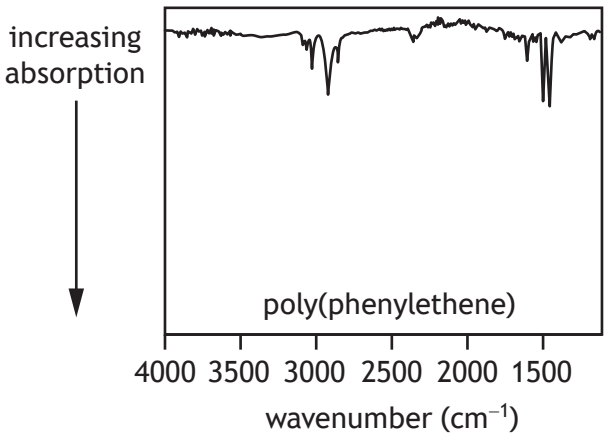
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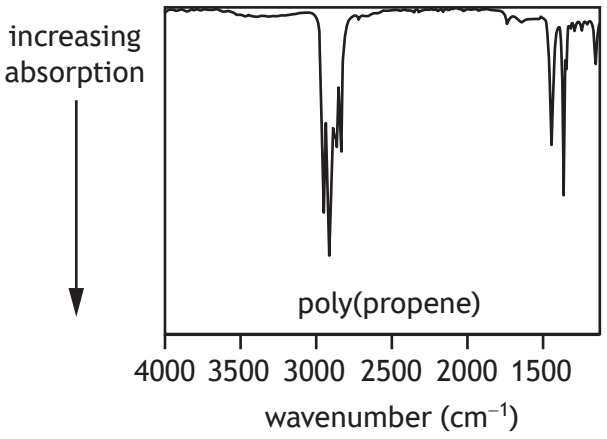
12. One of the problems with recycling plastics is identifying the type of plastic.

Infrared spectroscopy is a technique that can be used to identify the bonds present in plastics. A spectrum is produced for each sample analysed. The same bond always absorbs infrared radiation in the same range of wavenumbers, even in different molecules. For example C-H bonds absorb in the wavenumber range 2700–3300 cm^{-1}.

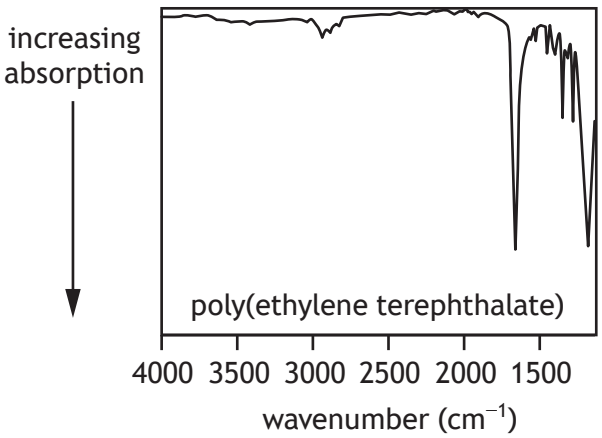
Four different types of plastic were analysed using infrared spectroscopy and the spectra produced are shown.



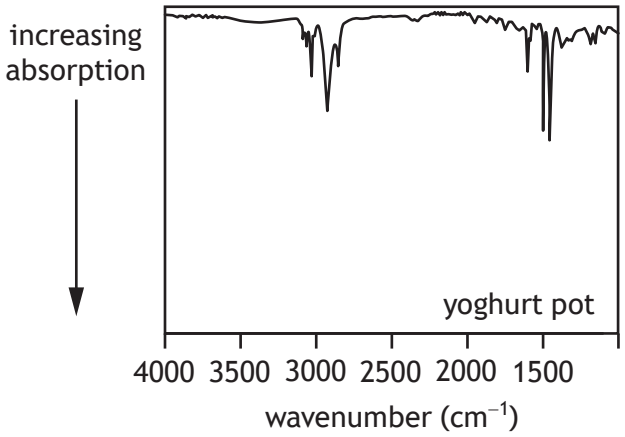




12. (continued)



(a) The spectrum obtained from the analysis of the plastic used to make a yoghurt pot is shown.



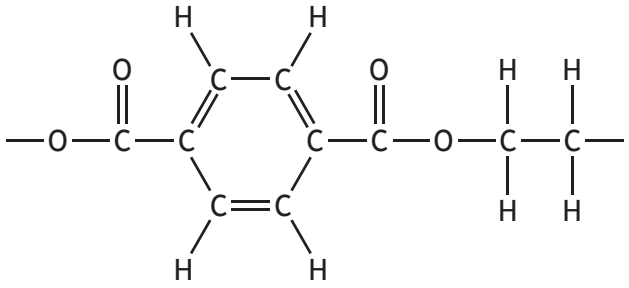
Identify the type of plastic used to make the yoghurt pot. **1**

[Turn over

12. (continued)

(b) The spectrum produced from poly(ethylene terephthalate) contains an absorption at a wavenumber of 1720 cm^{-1}.

Part of the structure of poly(ethylene terephthalate) is shown.

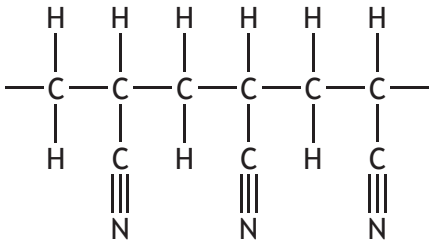


Using the information on page 14 of the data booklet, circle the bond in poly(ethylene terephthalate) that is responsible for this absorption. **1**

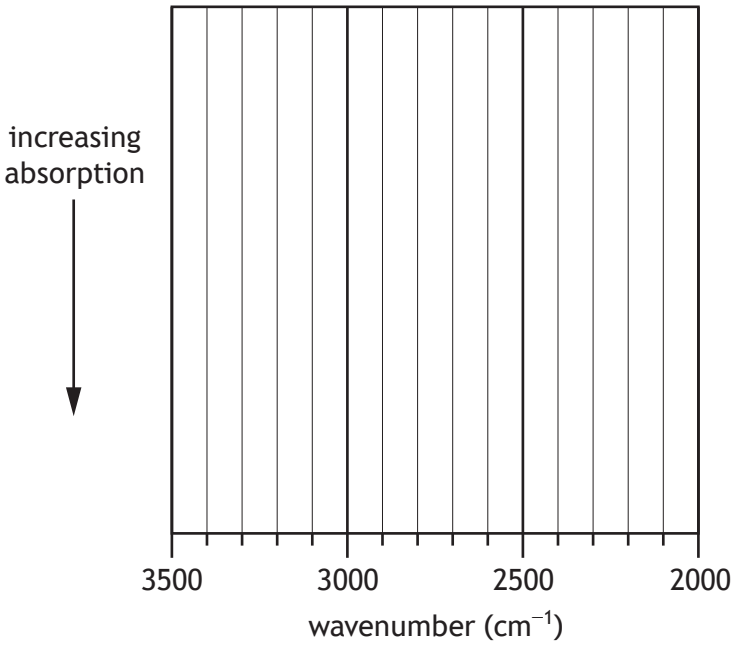
(An additional diagram, if required, can be found on *page 40*.)

12. (continued)

(c) Polyacrylonitrile plastic has the following structure.



Using the information on page 14 of the data booklet, sketch the infrared spectrum you would predict for polyacrylonitrile, showing only the absorptions within the range 3500–2000 cm^{-1}. **1**



(An additional diagram, if required, can be found on *page 40*.)

[END OF QUESTION PAPER]

National
Qualifications
2019

X813/76/12

Chemistry
Paper 1 — Multiple choice

FRIDAY, 10 MAY
9:00 AM – 9:40 AM

Total marks — 25

Attempt ALL questions.

You may use a calculator.

Instructions for the completion of Paper 1 are given on *page 02* of your answer booklet X813/76/02.

Record your answers on the answer grid on *page 03* of your answer booklet.

You may refer to the Chemistry Data Booklet for Higher and Advanced Higher.

Space for rough work is provided at the end of this booklet.

Before leaving the examination room you must give your answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

X8137612

Total marks — 25

Attempt ALL questions

1. Hydrogen will form a non-polar covalent bond with an element that has an electronegativity value of

A 0·9

B 1·5

C 2·2

D 2·5.

2. Which of the following is a polar molecule?

A CCl_4

B NH_3

C CO_2

D CH_4

3. Which of the following is most likely to act as a reducing agent?

A CO

B MnO_4^-

C H_2O_2

D $Cr_2O_7^{2-}$

4. The following reactions take place when nitric acid is added to zinc.

$$NO_3^-(aq) + 4H^+(aq) + 3e^- \rightarrow NO(g) + 2H_2O(\ell)$$

$$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$$

How many moles of Zn(s) are oxidised by one mole of NO_3^- (aq)?

A 0·67

B 1·0

C 1·5

D 2·0

5. Which of the following compounds is a tertiary alcohol?

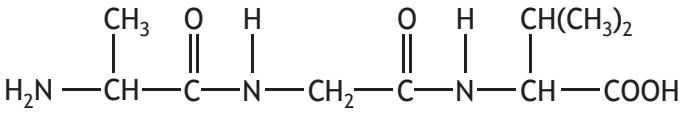
A 2,2-dimethylpropan-1-ol

B 2-methylbutan-2-ol

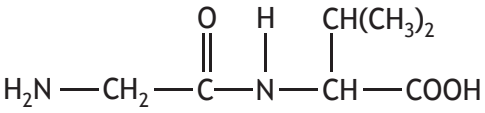
C pentan-3-ol

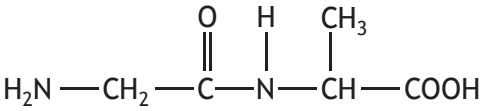
D 3-methylbutan-2-ol

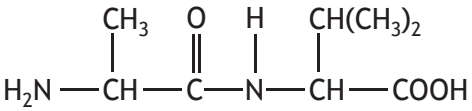
6. Molecule **X** has the structure

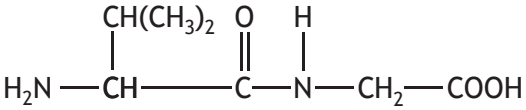


Which of the following could be produced by partial hydrolysis of **X**?

A 

B 

C 

D 

7. A compound with molecular formula $C_6H_{12}O_2$ could be

A pentyl ethanoate

B hexan-2-one

C 3-methylpentan-2-ol

D hexanoic acid.

[Turn over

8. Compound X reacted with hot copper(II) oxide and the resulting product did **not** give a colour change when heated with Fehling's solution.

Compound X could be

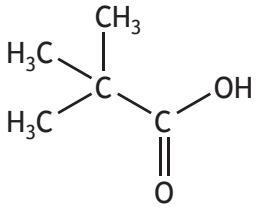
A pentan-1-ol

B pentan-2-ol

C pentan-3-one

D pentanoic acid.

9. The structure of pivalic acid is shown.



Which of the following is the correct systematic name of pivalic acid?

A pentanoic acid

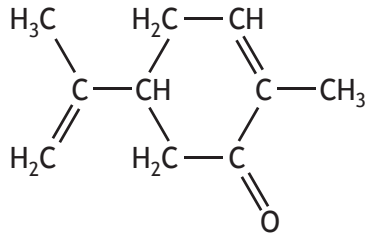
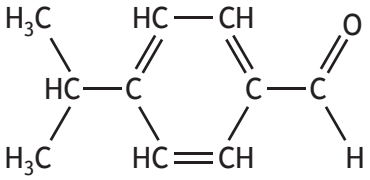
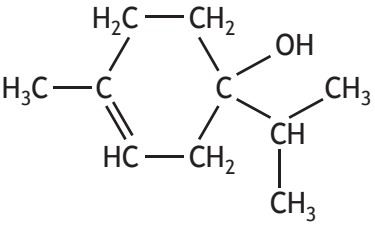
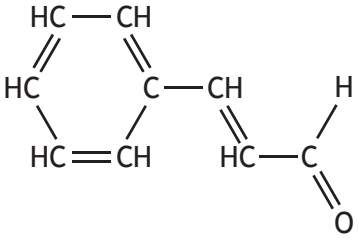
B 2,2,2-trimethylethanoic acid

C 2-ethylpropanoic acid

D 2,2-dimethylpropanoic acid

10. The table shows four compounds that contribute to the aroma of spices.

Which compound is **not** derived from a terpene?

	Structural formula	Molecular formula
A		$C_{10}H_{14}O$
B		$C_{10}H_{12}O$
C		$C_{10}H_{18}O$
D		C_9H_8O

11. Which reaction can be classified as reduction?

A methanol → methanoic acid

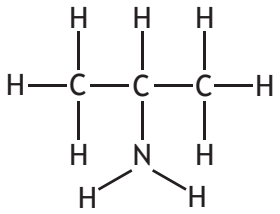
B propanal → propanoic acid

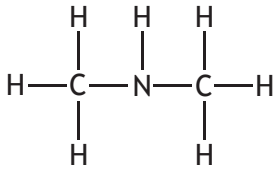
C butan-2-one → butan-2-ol

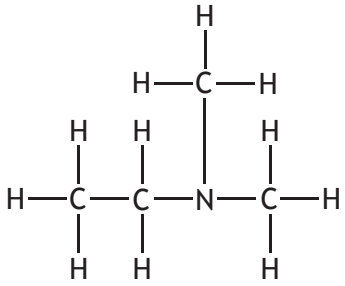
D propan-2-ol → propanone

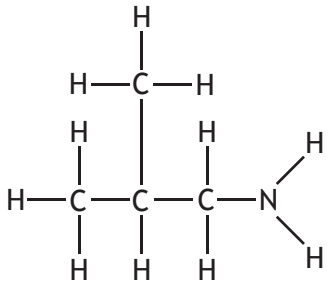
[Turn over

12. A secondary amine has two carbon atoms directly bonded to the nitrogen atom.
Which of the following is a secondary amine?

A 

B 

C 

D 

13. The number of moles of ions in 1 mol of copper(II) phosphate is

A 1

B 2

C 3

D 5.

14. Which of the following gas samples has the same volume as 4·0 g of methane, CH_4?
(All volumes are measured at the same temperature and pressure.)

A 1·0 g of helium
B 1·0 g of hydrogen
C 3·5 g of nitrogen
D 35·5 g of chlorine

15. Magnesium carbonate reacts with nitric acid.

$MgCO_3(s) + 2HNO_3(aq) \rightarrow Mg(NO_3)_2(aq) + H_2O(\ell) + CO_2(g)$

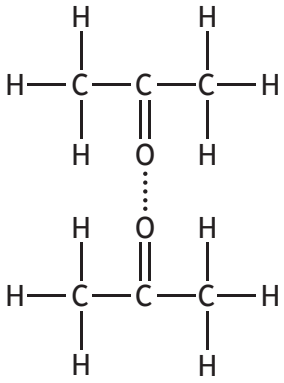
0·05 mol of magnesium carbonate was added to a solution containing 0·06 mol of nitric acid.
Which of the following statements is true?

A 0·05 mol of carbon dioxide is produced
B 0·06 mol of magnesium nitrate is produced
C Magnesium carbonate is in excess by 0·02 mol
D Nitric acid is in excess by 0·01 mol

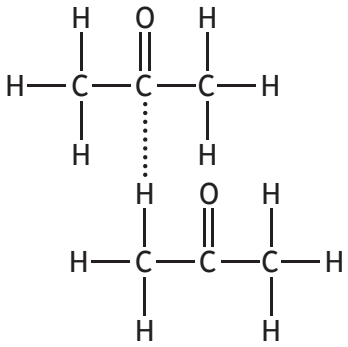
[Turn over

16. In which of the following diagrams does the dotted line represent a permanent dipole-permanent dipole interaction between propanone molecules?

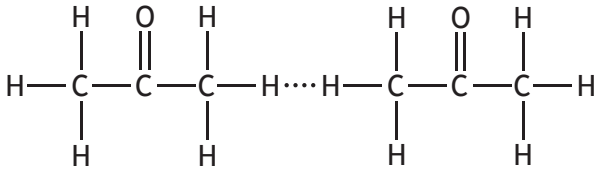
A



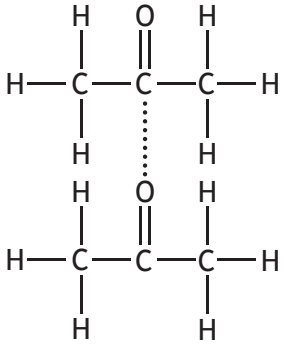
B



C



D



17. Iron can be produced from iron(III) oxide.

$$2Fe_2O_3(s) + 3C(s) \rightarrow 4Fe(s) + 3CO_2(g)$$

GFM = 159·6 g GFM = 12·0 g GFM = 55·8 g GFM = 44·0 g

The atom economy for the production of iron is

A 69·9%

B 62·8%

C 58·2%

D 32·5%.

18. 100 cm^3 of propane is mixed with 600 cm^3 of oxygen and the mixture is ignited.

$$C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(\ell)$$

At the end of the reaction, the total volume of gas would be

A 300 cm^3

B 400 cm^3

C 700 cm^3

D 800 cm^3.

19. A two-step reaction is shown below.

$$A \xrightarrow[\text{step 1}]{} B \xrightarrow[\text{step 2}]{} C$$

The first step gave a yield of 60% and the second step a yield of 90%.
The overall yield would be

A 30%

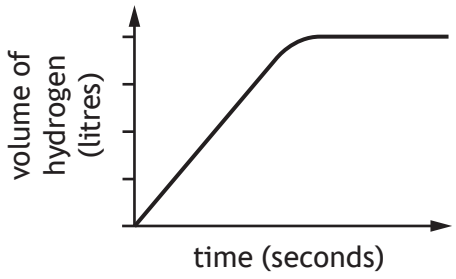
B 54%

C 67%

D 150%.

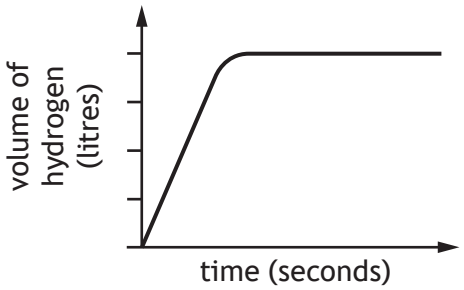
[Turn over

20. The volume of hydrogen gas given off against time when an excess of zinc lumps is added to 100 cm^3 of 1 mol l^{-1} hydrochloric acid is shown.

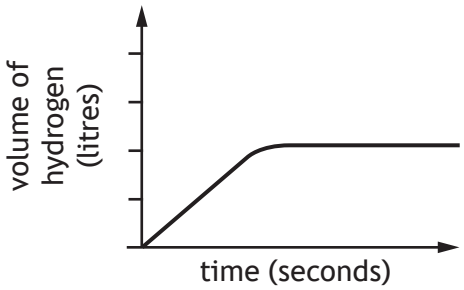


Which of the following graphs would show the volume of hydrogen gas given off when an excess of zinc powder was added to 50 cm^3 of 1 mol l^{-1} hydrochloric acid?

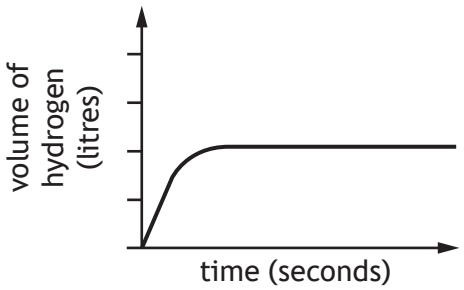
A



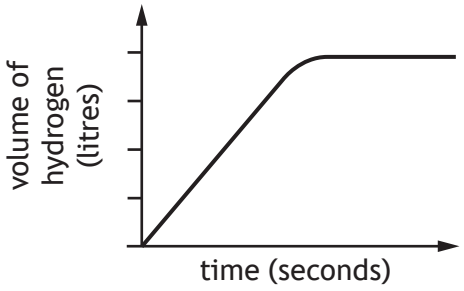
B



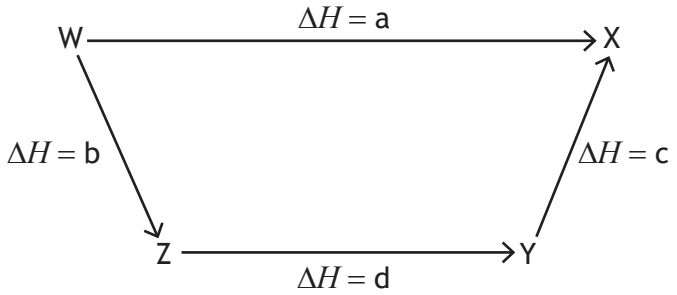
C



D



21. Consider the reaction pathway shown below.



According to Hess's Law

A $b = a - c - d$

B $b = a + c + d$

C $b = d - c + a$

D $b = d + c - a$.

22. Which of the following is **not** a factor that affects the rate of a reaction?

A Activation energy

B Kinetic energies of reactant molecules

C Concentration of reactants

D Enthalpy change of reaction

23. In which of the following reactions would the yield of product be increased by lowering the pressure?

A $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

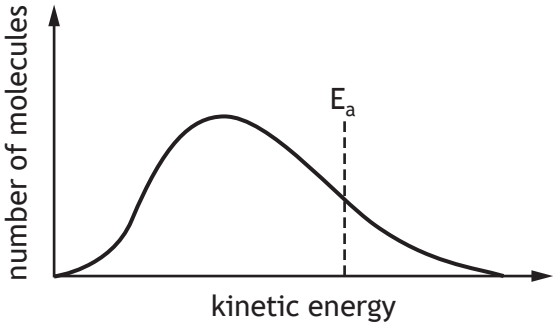
B $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

C $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

D $CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$

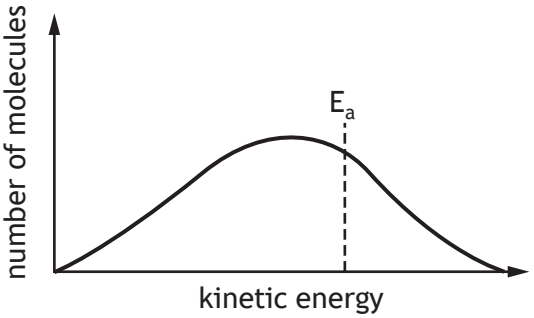
[Turn over

24. The graph shows the distribution of kinetic energies for a reaction involving two gases.

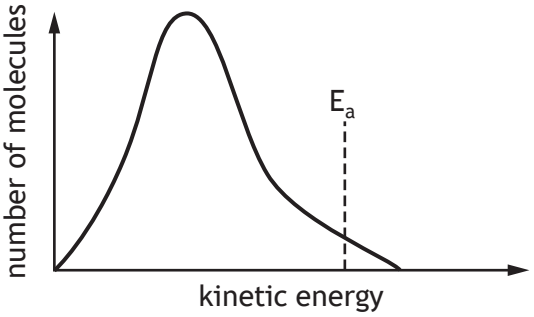


Which graph would show the effect of increasing temperature?

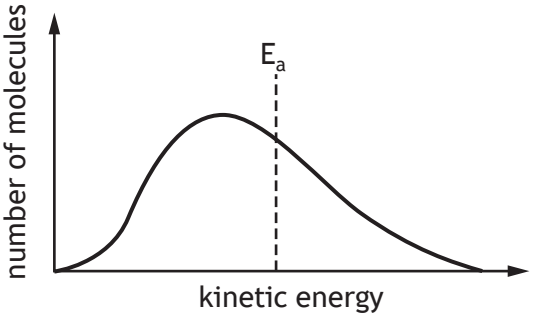
A



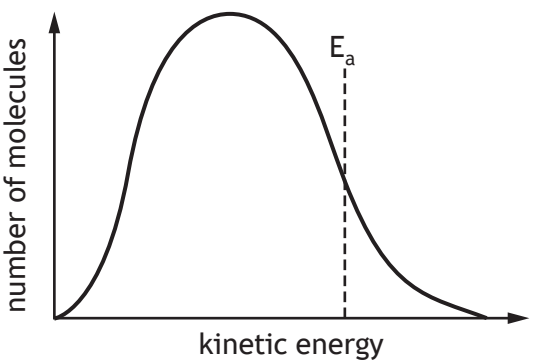
B



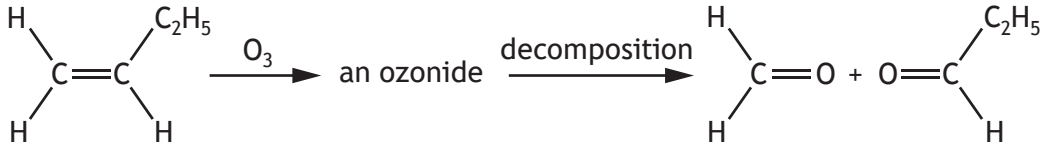
C



D



25. Alkenes react with ozone, O_3, to form ozonides which can be decomposed to give carbonyl compounds.



Which of the following alkenes would produce a mixture of ethanal and propanone?

A $CH_3CH=CHCH_2CH_3$

B $CH_3CH=CHCH_3$

C $CH_3C(CH_3)=CH_2$

D $CH_3CH=C(CH_3)CH_3$

[END OF QUESTION PAPER]

FOR OFFICIAL USE

National Qualifications 2019

Mark

X813/76/01

Chemistry Paper 2

FRIDAY, 10 MAY

10:10 AM – 12:30 PM

X8137601

Fill in these boxes and read what is printed below.

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth

Day | Month | Year

Scottish candidate number

Total marks — 95

Attempt ALL questions.

You may use a calculator.

You may refer to the Chemistry Data Booklet for Higher and Advanced Higher.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers and rough work is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting. Any rough work must be written in this booklet. Score through your rough work when you have written your final copy.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

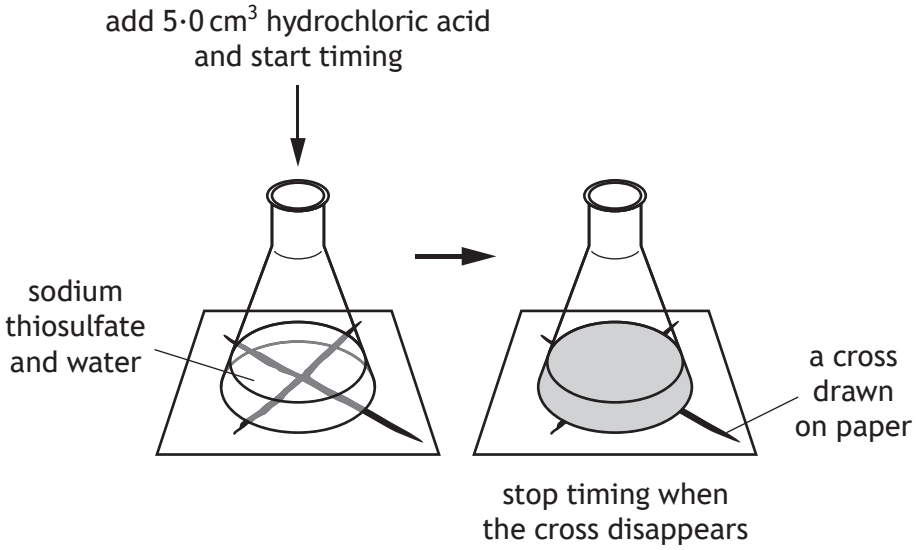
X8137601 01

Total marks — 95

Attempt ALL questions

1. Sodium thiosulfate, $Na_2S_2O_3$, can be used to investigate the effect of reaction conditions on the rate of reaction.

(a) Sodium thiosulfate solution reacts with hydrochloric acid to form a precipitate of solid sulfur. By placing the reaction mixture in a conical flask over a cross and recording the time taken for the cross to disappear, the effect of changing the reaction conditions can be investigated.



(i) The equation for the reaction is

$$Na_2S_2O_3(aq) + HCl(aq) \rightarrow S(s) + SO_2(g) + NaCl(aq) + H_2O(\ell)$$

Balance the equation. **1**

1. (a) (continued)

(ii) In one set of experiments, the effect of varying the concentration of sodium thiosulfate was investigated.

Experiment	Volume of 0·15 mol l^{-1} $Na_2S_2O_3$ (cm^3)	Volume of water (cm^3)	Rate (s^{-1})
A	50	0	0·0454
B	40		0·0370
C	30		0·0285
D	20		0·0169
E	10	40	0·0063

(A) Complete the table to show the volumes of water that would have been used to vary the concentration of sodium thiosulfate. **1**

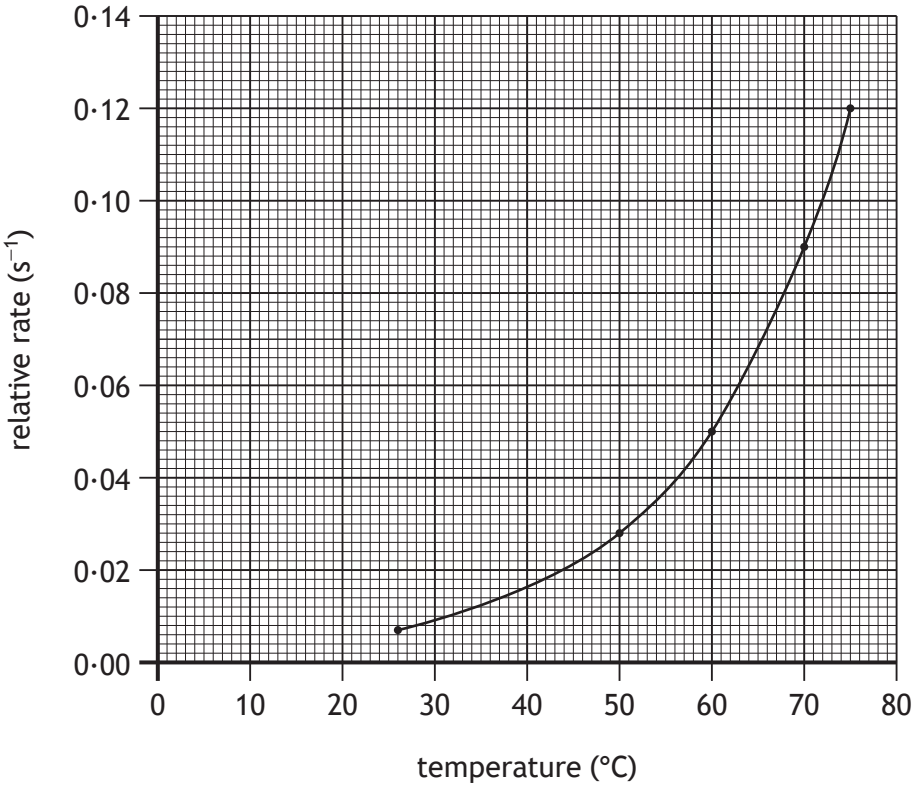
(B) Calculate the time, in seconds, for the cross to disappear in experiment C. **1**

[Turn over

1. (a) (continued)

(iii) The reaction can also be used to investigate the effect of changing temperature on the rate of reaction.

The results from an investigation are shown in the graph below.



Use the graph to determine the temperature rise, in °C, required to double the rate of the reaction. **1**

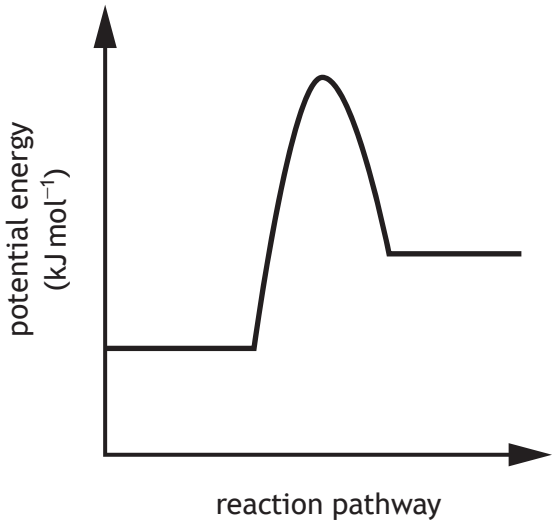
(b) Collision theory states that for particles to react they must first collide with each other.

State **two** conditions necessary for the collisions to result in the formation of products. **2**

1. (continued)

(c) Sodium thiosulfate also reacts with iron(III) nitrate.

The potential energy diagram below shows the change in potential energy during the reaction carried out without a catalyst.



(i) Draw an **X** on the potential energy diagram above to show where the activated complex is formed. **1**

(An additional diagram, if required, can be found on *page 41*).

(ii) Cu^{2+} ions catalyse the reaction.

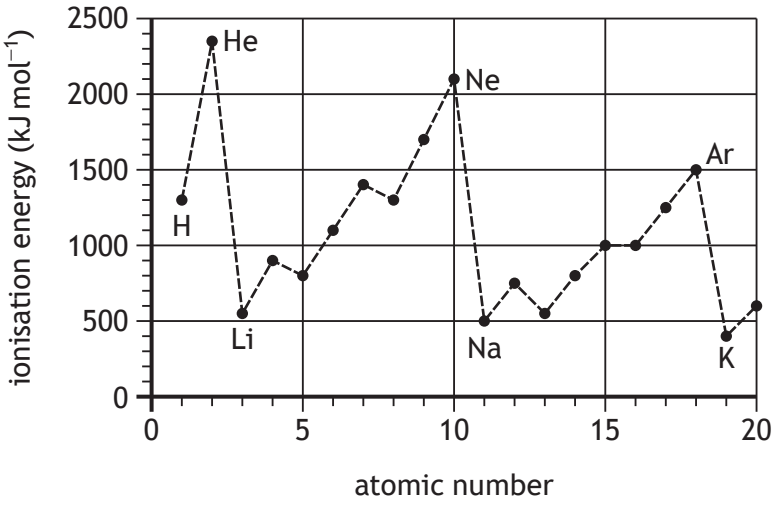
Add a dotted line to the diagram to show the change in potential energy with the catalyst. **1**

(An additional diagram, if required, can be found on *page 41*).

[Turn over

2. 2019 is the 150th anniversary of the periodic table's creation by Dmitri Mendeleev. The patterns identified by Mendeleev form the basis of the modern periodic table. The major periodic trends include ionisation energy and covalent radius.

(a) The first ionisation energies of elements with atomic number 1 to 20 are shown in the graph.

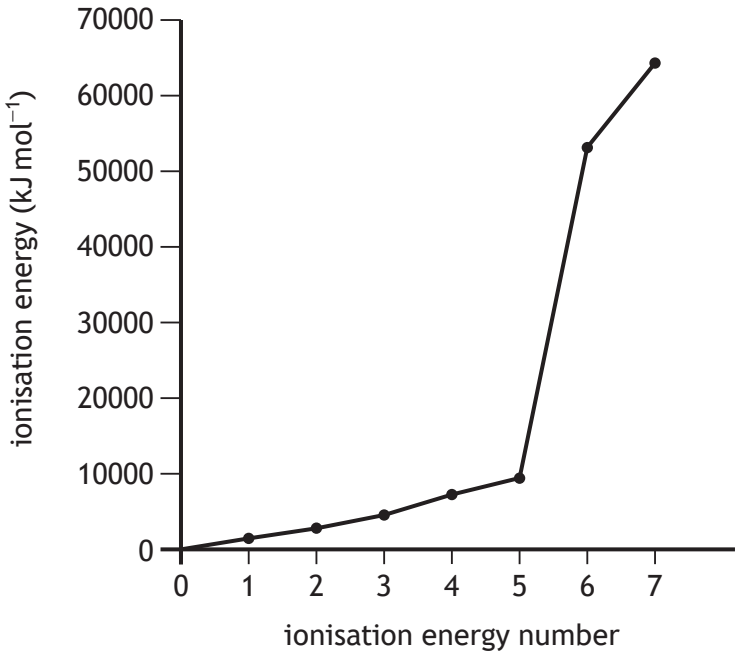


(i) Explain why the first ionisation energy shows an increase going from lithium to neon. **1**

(ii) Explain why the first ionisation energy of potassium is less than the first ionisation energy of lithium. **1**

2. (continued)

(b) A graph showing the ionisation energies for nitrogen is shown.



(i) Write the equation for the second ionisation energy of nitrogen. **1**

(ii) Explain **fully** the increase between the 5th and 6th ionisation energies of nitrogen. **2**

[Turn over

2. (continued)

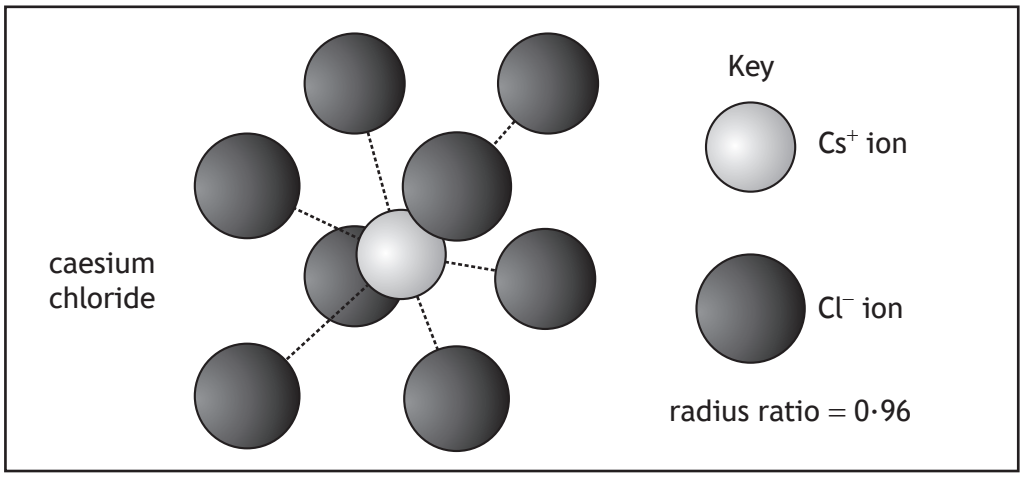
(c) Ionic radius is a measure of the size of an ion.

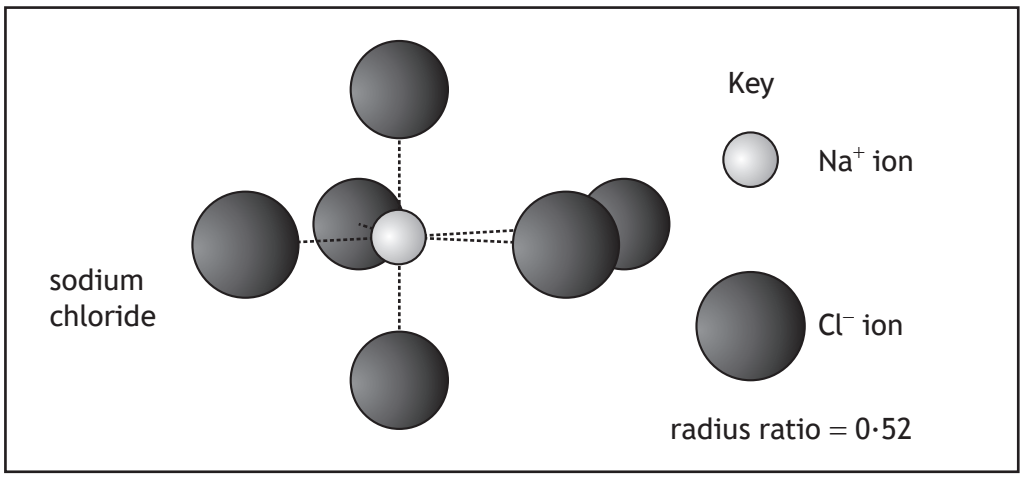
Explain **fully** why the ionic radius of phosphorus is greater than the ionic radius of aluminium. **2**

2. (continued)

(d) The structure of an ionic compound consists of a giant lattice of oppositely charged ions. The arrangement of ions is determined by the 'radius ratio' of the ions involved.

$$\text{radius ratio} = \frac{\text{radius of positive ion}}{\text{radius of negative ion}}$$





By using the table of ionic radii on *page 17* of the data booklet, predict whether the structure of barium oxide, BaO, is similar to caesium chloride or sodium chloride.

Your answer **must** include a calculated radius ratio. **1**

3. The melting point of **non-metal elements** depends on structure and bonding.
Using your knowledge of chemistry, comment on this statement. **3**

4. Cider is made from apples in a process that involves crushing and pressing the apples, converting the sugars into alcohol, maturing and bottling.

(a) Brewers add yeast, which contains a mixture of enzymes to convert the sugars in the apples into alcohol and carbon dioxide.

(i) State what is meant by the term enzyme. **1**

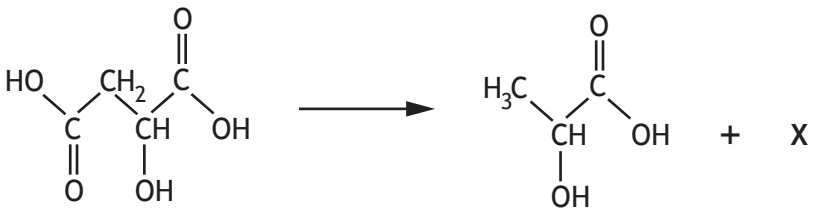
(ii) The % mass of alcohol in the cider can be calculated using the formula

$$\%\,\text{mass of alcohol} = \frac{\text{mass of alcohol}}{\text{mass of cider}} \times 100$$

A 50·0 cm^3 sample of cider was found to contain 3·05 g of alcohol. 1·0 cm^3 of the cider weighed 1·36 g.

Calculate the % mass of alcohol in the cider. **1**

(b) During the maturing process malic acid is converted to lactic acid and another product.



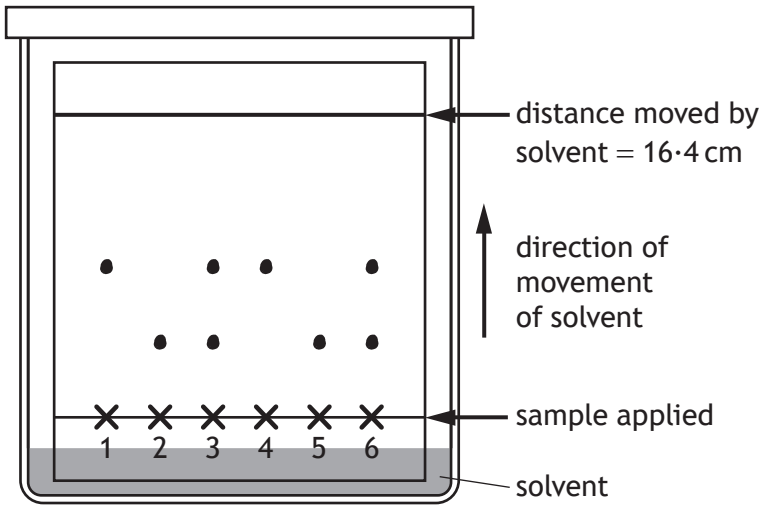
malic acid lactic acid

(i) Name compound **X**. **1**

4. (b) (continued)

(ii) The maturing process in cider samples can be monitored using thin layer chromatography.

Samples of lactic acid, malic acid and ciders A, B, C, and D are spotted on a silica plate and the solvent allowed to travel up the plate. The chromatogram obtained is shown below.



Number	Sample applied	Distance moved by spot(s) (cm)
1	lactic acid	8·2
2	malic acid	4·1
3	cider A	4·1, 8·2
4	cider B	8·2
5	cider C	4·1
6	cider D	4·1, 8·2

4. (b) (ii) (continued)

The retention factor, R_f, for a substance can be a useful method of identifying the substance.

$$R_f = \frac{\text{distance moved by the substance}}{\text{distance moved by the solvent}}$$

(A) Calculate the R_f value of malic acid. **1**

(B) The maturing process is complete when all of the malic acid has been converted to lactic acid. The cider is now ready to be bottled.

Use the chromatogram to determine which cider is ready to be bottled. **1**

[Turn over

4. (continued)

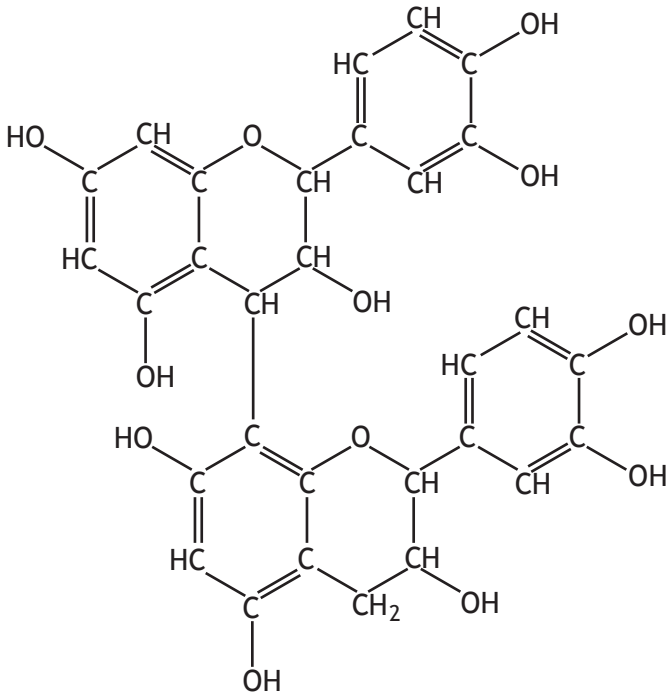
MARKS DO NOT WRITE IN THIS MARGIN

(c) Glycerol can be added to cider before bottling to produce a sweeter tasting cider.

State the systematic name for glycerol. **1**

(d) Cider contains many naturally occurring compounds that affect taste and aroma.

(i) Procyanidin B2 provides a bitter taste to cider.



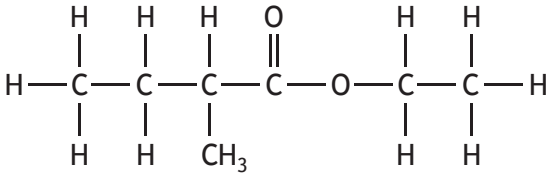
procyanidin B2

Explain **fully** why procyanidin B2 is water soluble. **2**

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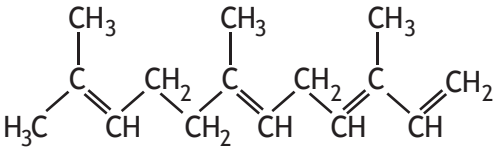
(ii) Cider smells of apples because it contains ethyl 2-methylbutanoate.



ethyl 2-methylbutanoate

Name the carboxylic acid used to make ethyl 2-methylbutanoate. **1**

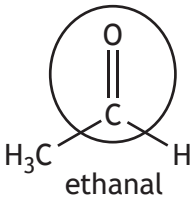
(iii) Farnesene is a terpene responsible for the ripe apple aroma of cider.



farnesene

Name the molecule on which terpenes are based. **1**

(e) Ethanol in cider can be oxidised to ethanal, spoiling the aroma.



ethanal

(i) Name the functional group circled in the ethanal molecule. **1**

(ii) Further oxidation of ethanal can produce another product that spoils the flavour of cider.

Name this product. **1**

5. The combustion reactions of methane and heptane can be studied in different ways.

(a) The combustion of methane produces carbon dioxide and water vapour when carried out at temperatures above 100 °C.

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$$

(i) Using bond enthalpies and mean bond enthalpies from the data booklet, calculate the enthalpy change, in kJ mol^{-1}, for this reaction. **2**

(ii) Explain the difference between bond enthalpy and mean bond enthalpy. **1**

5. (a) (continued)

(iii) Calculate the mass, in g, of carbon dioxide produced by combustion of 200 cm^3 methane in excess oxygen. **2**

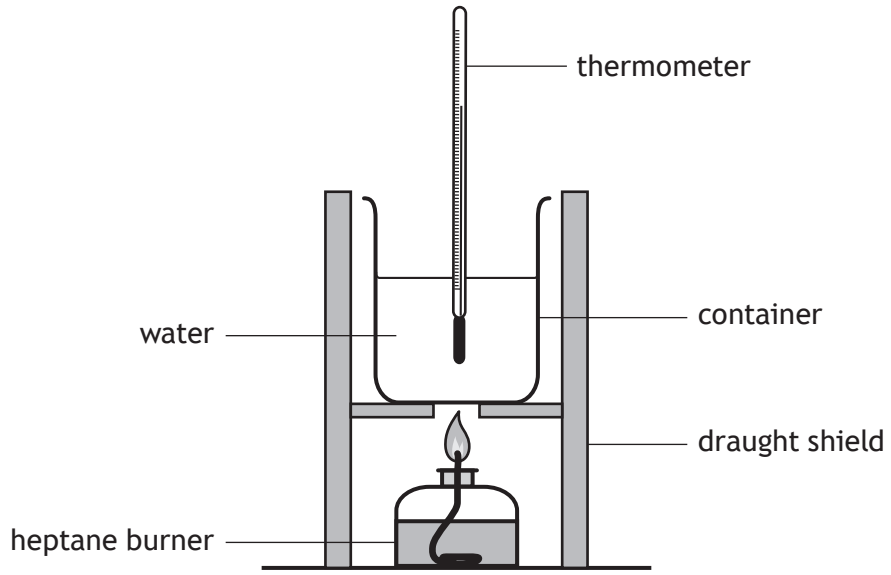
Take the volume of 1 mole of methane gas to be 24 litres.

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$$

GFM = 44·0 g

5. (continued)

(b) The enthalpy of combustion of heptane, C_7H_{16}, can be determined using a calorimeter.



The following results were obtained.

Mass of heptane burned (g)	1·1
Mass of 1 mole of heptane (g)	100·0
Volume of water used (cm^3)	400
Initial temperature of water (°C)	26
Final temperature of water (°C)	49

(i) State the measurements required to calculate the mass of heptane burned in this experiment. **1**

5. (b) (continued)

(ii) Calculate the enthalpy of combustion, in kJ mol^{-1}, for heptane from the experimental results given. **3**

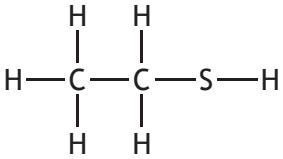
(iii) The theoretical value for the enthalpy of combustion of heptane is significantly higher than the experimental value.

Suggest why the experimental value is different to the theoretical value. **1**

[Turn over

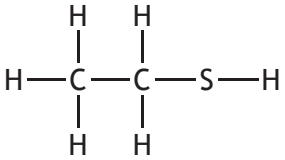
6. Thiols are compounds that contain an –SH functional group. They often have very strong, unpleasant odours.

(a) Ethanethiol is used to add a smell to gaseous fuels in order to give warnings of gas leaks.

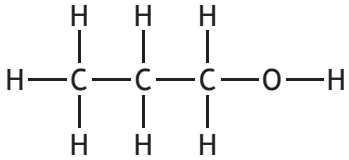


ethanethiol

(i) A student used the boiling points of ethanethiol and propan-1-ol to compare the strength of intermolecular forces.



ethanethiol
boiling point = 35 °C



propan-1-ol
boiling point = 97 °C

(A) State the reason why propan-1-ol was a suitable alcohol to compare with ethanethiol. **1**

(B) Explain why propan-1-ol has a higher boiling point than ethanethiol. Your answer should include the names of the intermolecular forces broken when each liquid boils. **2**

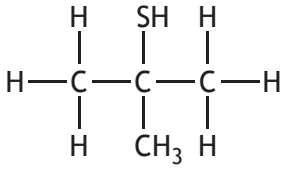
6. (a) (continued)

(ii) Name the thiol that contains only one carbon atom. **1**

(iii) The minimum concentration of ethanethiol in air that can be detected by humans is $2{\cdot}7 \times 10^{-7}$ mg per cm^3 of air.

Calculate the minimum mass of ethanethiol that needs to be present in a room containing 43 900 litres of air in order for it to be detected. **2**

(b) 2-methyl-2-propanethiol is also used to add a smell to gaseous fuels.



2-methyl-2-propanethiol

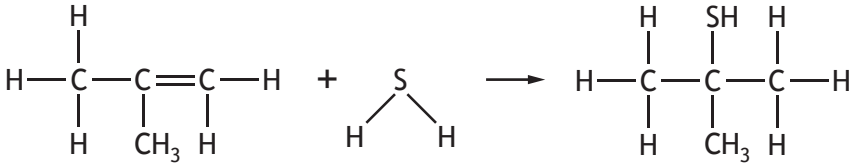
(i) Suggest why 2-methyl-2-propanethiol is classified as a tertiary thiol. **1**

[Turn over

6. (b) (continued)

(ii) Thiols can be made by the addition of hydrogen sulfide to alkenes.

2-methyl-2-propanethiol can be made by the addition reaction shown.



2-methylpropene
GFM = 56·0 g

2-methyl-2-propanethiol
GFM = 90·1 g

(A) Draw the structure for the other isomer formed in this addition reaction. **1**

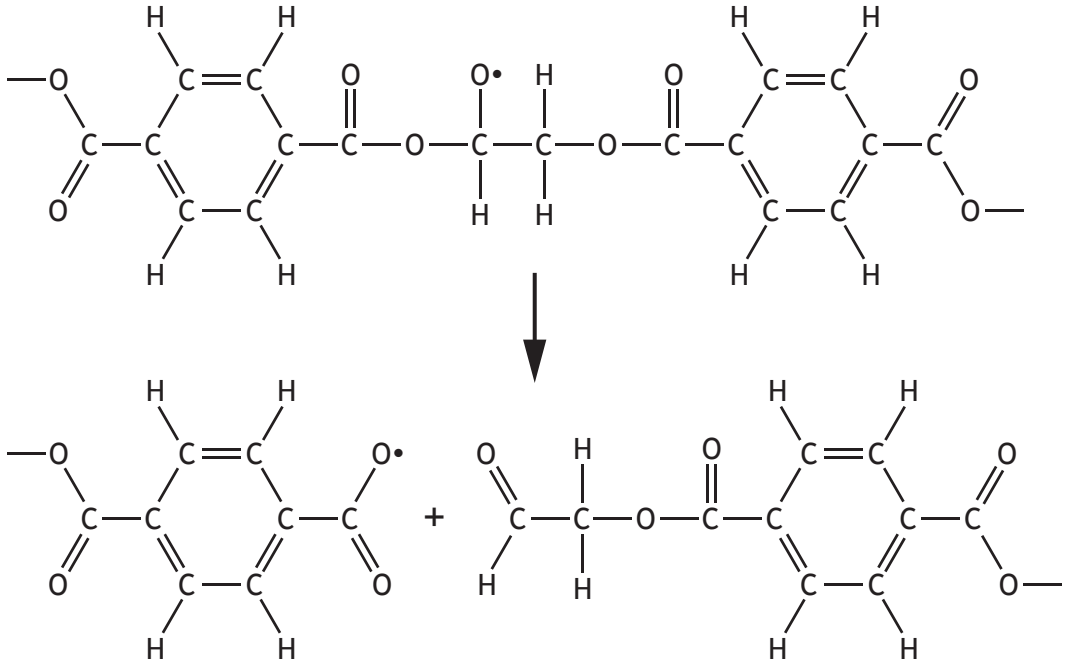
(B) A chemist obtained an 84% yield of 2-methyl-2-propanethiol after starting with 30·5 g of 2-methylpropene.

Calculate the mass, in g, of 2-methyl-2-propanethiol made by the chemist. **2**

7. Esters can be synthetic or natural.

(a) The synthetic polyester PET, poly(ethylene terephthalate), has many ester links. PET can break down by a free radical reaction.

One of the steps involved in breaking down PET is shown.



(i) State the name for this step. **1**

(ii) Name the component of sunlight that can cause plastics such as PET to break down. **1**

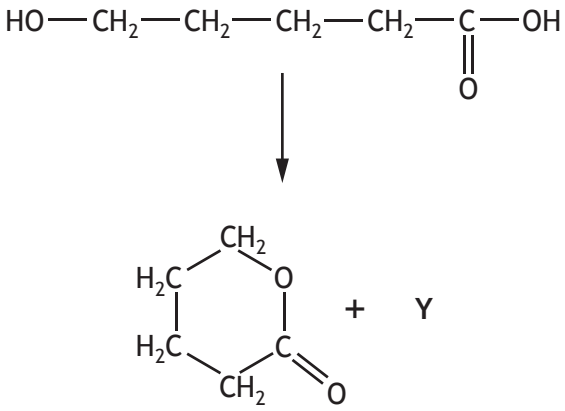
(iii) Name the type of substance that can be added to plastics to prevent them breaking down in this way. **1**

[Turn over

7. (continued)

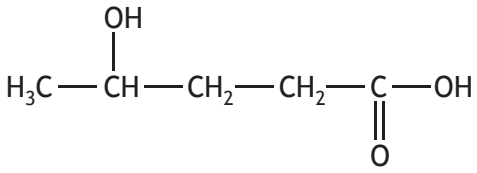
(b) (i) Natural cyclic esters called lactones can be formed from hydroxycarboxylic acids.

5-hydroxypentanoic acid is a hydroxycarboxylic acid that when heated, with dilute acid, will form a cyclic ester.



Name product **Y** in this reaction. **1**

(ii) Draw the structure for the cyclic compound formed when 4-hydroxypentanoic acid is heated with dilute acid. **1**

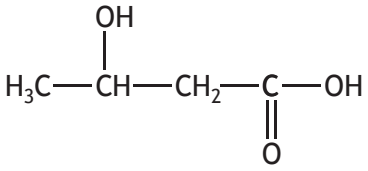


4-hydroxypentanoic acid



7. (b) (continued)

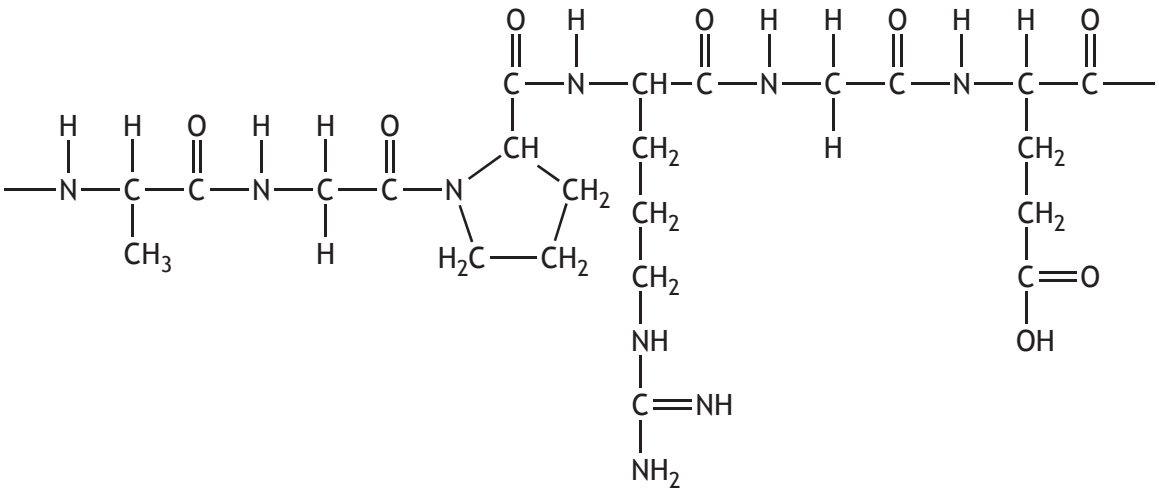
(iii) Name the hydroxycarboxylic acid shown below. **1**



[Turn over

8. Gelatin is a soluble protein that can be added to different food products.

(a) A structure for a section of a protein chain in gelatin is shown.



(i) State the number of amino acids that joined together to form the section of the protein chain shown. **1**

(ii) Name the weakest van der Waals' force between water and gelatin molecules. **1**

(b) A student was investigating the viscosity of different concentrations of gelatin solution.

(i) The student was asked to prepare a 2% gelatin solution, which is a solution that contains 2 g of gelatin per 100 cm^3 of solution.

The student prepared this solution by adding 100 cm^3 of distilled water into a volumetric flask, then adding 2 g of gelatin.

Describe how the student **should** have made up the solution. **3**

8. (b) (continued)

(ii) The results obtained from the student's viscosity experiment are shown.

Concentration of gelatin solution (%)	Viscosity (units)
2·0	1·0
4·0	2·0
6·0	4·0
8·0	7·0
10·0	

Predict the student's result for the viscosity, in units, of a 10·0% gelatin solution. **1**

(c) Bromelain is a mixture of enzymes found in pineapple that aid digestion.

(i) Adding raw pineapple to gelatin results in the gelatin molecules being hydrolysed. The rate of hydrolysis is reduced if the pineapple is cooked.

Explain why the rate of hydrolysis is reduced. **1**

(ii) Bromelain can be purchased as tablets that contain 500 mg of bromelain. The flesh from a pineapple contains 13·2 mg of bromelain per gram.

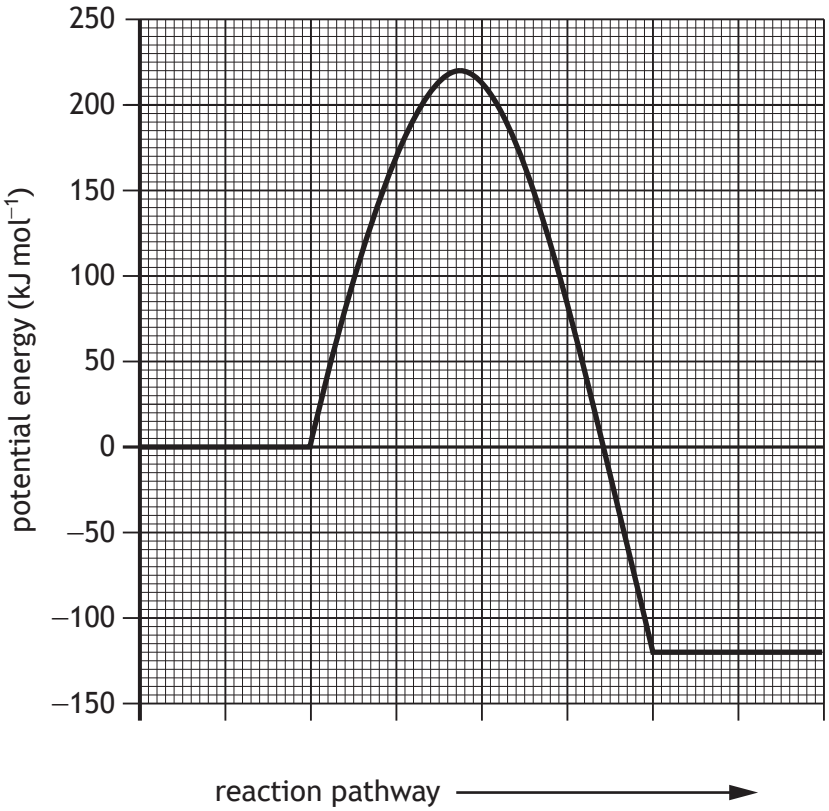
Calculate the mass, in g, of this pineapple that would be needed to provide 500 mg of bromelain. **1**

[Turn over

9. Chlorine is used in the production of many other chemicals.

(a) Chlorine can be produced by the reaction of hydrogen chloride with air using the Deacon process.

$$4HCl(g) + O_2(g) \rightleftharpoons 2Cl_2(g) + 2H_2O(g)$$

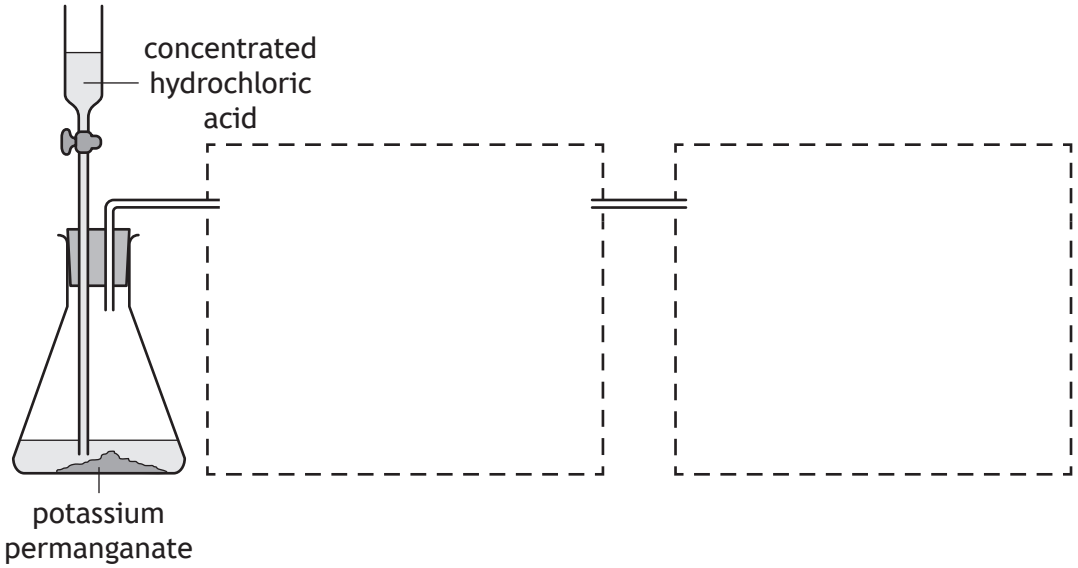


(i) Using the potential energy diagram, determine the activation energy, in kJ mol^{-1}, for the **forward** reaction. **1**

(ii) Explain why increasing the temperature in the Deacon process results in less chlorine being produced. **1**

9. (continued)

(b) One laboratory method for the preparation of chlorine gas involves adding concentrated hydrochloric acid to potassium permanganate. The chlorine gas produced also contains small amounts of hydrogen chloride gas. To remove the hydrogen chloride gas the gases are bubbled through water. Finally, insoluble chlorine gas is collected.



Complete a labelled diagram to show an apparatus suitable for carrying out this preparation. **2**

(An additional diagram, if required, can be found on *page 41*)

[Turn over

9. (continued)

(c) Carbon tetrachloride, CCl_4, is prepared by the reaction of chlorine gas, Cl_2, with methane, CH_4.

$$CH_4(g) + 4Cl_2(g) \rightarrow CCl_4(g) + 4HCl(g)$$

Calculate the enthalpy change, in kJ mol^{-1}, for this reaction using the following information. **2**

$C(s) + 2H_2(g) \rightarrow CH_4(g) \quad \Delta H = -75$ kJ mol^{-1}

$C(s) + 2Cl_2(g) \rightarrow CCl_4(g) \quad \Delta H = -98$ kJ mol^{-1}

$\frac{1}{2}H_2(g) + \frac{1}{2}Cl_2(g) \rightarrow HCl(g) \quad \Delta H = -92$ kJ mol^{-1}

10. A student investigated the purity of a sample of magnesium chloride, $MgCl_2$. The sample was dissolved in water and then an excess of silver nitrate, $AgNO_3$, was added to produce a precipitate of silver chloride, AgCl. The precipitate was collected, dried and weighed.

$$MgCl_2(aq) + 2AgNO_3(aq) \rightarrow 2AgCl(s) + Mg(NO_3)_2(aq)$$

(a) The student prepared the magnesium chloride solution by dissolving 2·503 g of impure magnesium chloride in water.

Explain why the student should use distilled or deionised water, rather than tap water, when preparing the solution. **1**

(b) (i) Complete the table to show the **most appropriate** piece of apparatus that could be used to measure the required volumes. **2**

Measurement	Apparatus
20·0 cm^3 **(accurately)**	
35 cm^3 **(approximately)**	

(ii) The steps required to collect, dry and weigh the precipitate are listed below. However, the steps are in the **wrong order**.

A. Weigh the precipitate and the filter paper

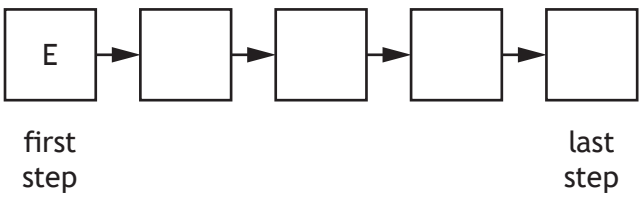
B. Wash the precipitate with water to remove any impurities

C. Filter the precipitate

D. Dry the precipitate in an oven

E. Weigh the filter paper

Complete the flow chart below to show the correct order of steps the student should carry out to collect, dry and weigh the precipitate. **1**



(An additional diagram, if required, can be found on *page 41*)

10. (b) (continued)

(iii) 1·393 g of silver chloride precipitate was produced from the magnesium chloride solution.

$$MgCl_2(aq) + 2AgNO_3(aq) \rightarrow 2AgCl(s) + Mg(NO_3)_2(aq)$$

GFM = 95·3 g GFM = 143·4 g

Calculate the mass of magnesium chloride, in g, present in the magnesium chloride solution. **2**

(c) The average mass of magnesium chloride in 2·503 g of the original impure sample was calculated to be 2·403 g.

Calculate the % of magnesium chloride present in the original sample. **1**

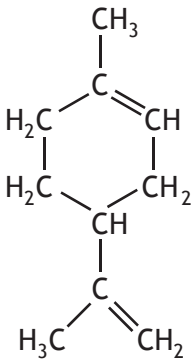
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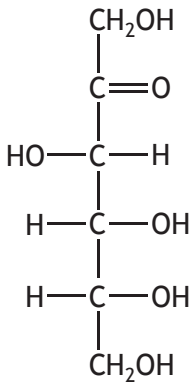
11. Differences in physical and chemical properties can be used to distinguish one compound from another.

The compounds extracted from orange juice include antioxidants, flavour molecules, essential oils, aroma molecules and coloured molecules.

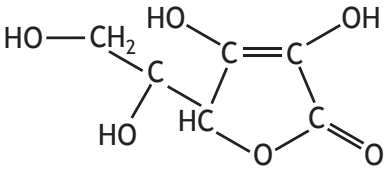
Some examples of these are shown below.



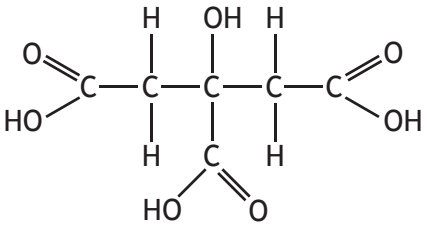
limonene



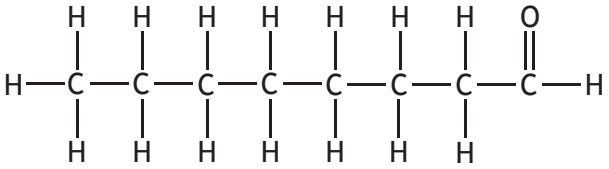
fructose



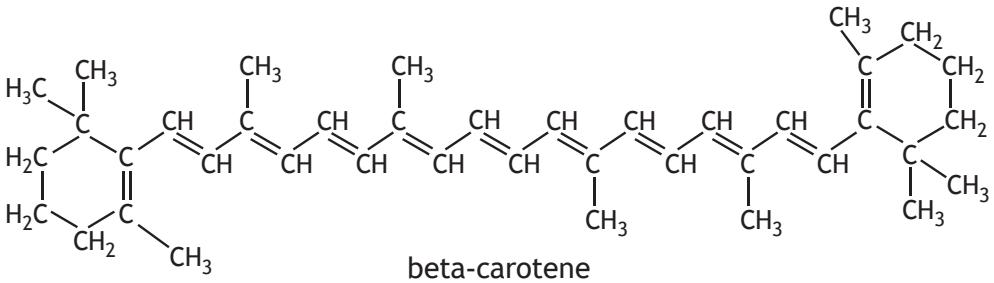
vitamin C



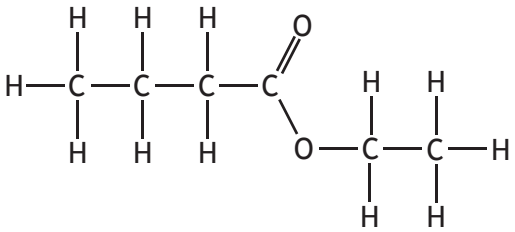
citric acid



octanal



beta-carotene



ethyl butanoate

11. (continued)

Using your knowledge of chemistry, comment on how the differences in physical and chemical properties can be used to distinguish between the compounds extracted from orange juice. **3**

[Turn over

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12. The label from a bottle of pine fresh bleach cleaner is shown.

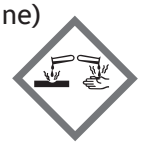
PINE FRESH BLEACH CLEANER

Formulated to kill germs and remove stains

Ingredients:
aqua, sodium hypochlorite, sodium hydroxide, less than 5% anionic surfactants, non-ionic surfactants, soap, perfume

WARNING!
Do not use together with other products. May release dangerous gases (chlorine)

DANGER
Keep out of reach of children

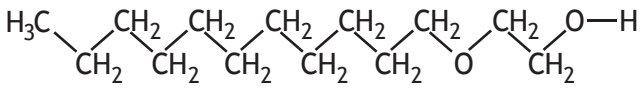
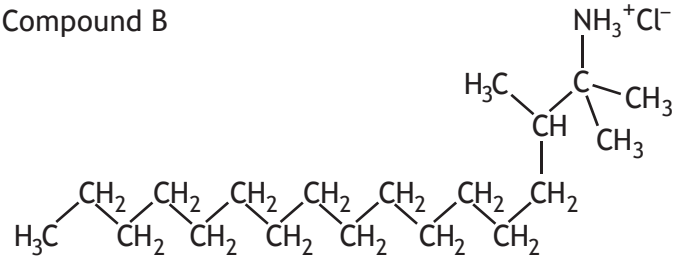
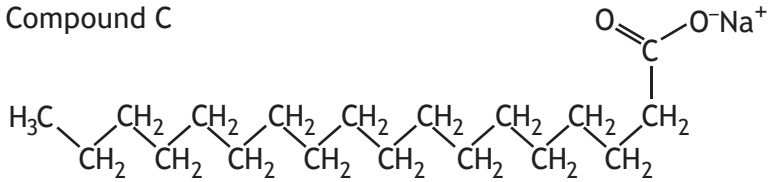


CORROSIVE

6 000129 7103487 >

(a) Surfactant molecules are added to bleach cleaner to act as detergents, soaps or emulsifiers.

Information on three of the surfactants in the bleach cleaner is shown in the table.

Surfactant structure	Type of surfactant	Head group
Compound A 	non-ionic	polar
Compound B 		
Compound C 	ionic	negatively charged

12. (a) (continued)

(i) Complete the table for compound B. **1**

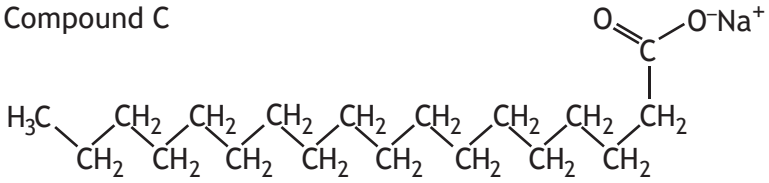
(ii) Compound C is a soap molecule.

(A) Soaps can be made from fats and oils.

Name the reaction used to make soaps from fats and oils. **1**

(B) Soap molecules allow oil to mix with water.

Compound C



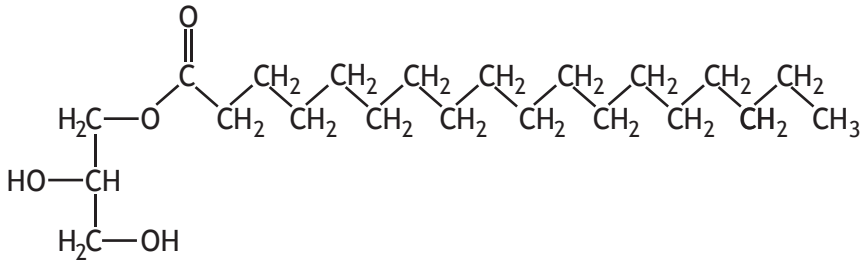
Explain fully the cleaning action of compound C. **3**

You may wish to use diagrams to illustrate your answers.

[Turn over

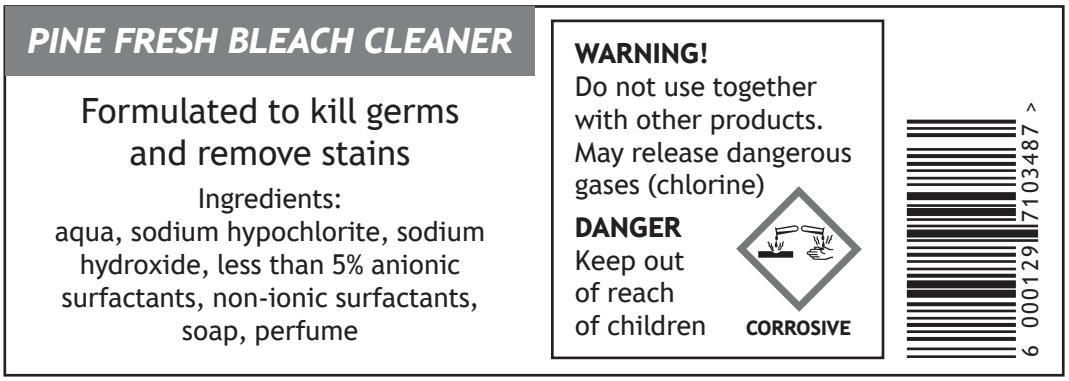
12. (a) (continued)

(iii) The structure of an emulsifier molecule is shown below.



State how emulsifiers are made from edible oils. **1**

(b) Sodium hypochlorite, Na^+OCl^-, is the main active compound in bleach.



Sodium hypochlorite, Na^+OCl^-, is produced by reacting chlorine with sodium hydroxide solution.

$$Cl_2(g) + 2Na^+OH^-(aq) \rightarrow Na^+OCl^-(aq) + Na^+Cl^-(aq) + H_2O(\ell)$$

(i) A chlorine molecule has a pure covalent bond.

Explain what is meant by a pure covalent bond. **1**

12. (b) (continued)

(ii) When the chlorine is reacted with sodium hydroxide solution an excess of sodium hydroxide is used.

Suggest why an excess of sodium hydroxide is used. **1**

(c) In the bleach cleaner an equilibrium exists.

$$2H^+(aq) + OCl^-(aq) + Cl^-(aq) \rightleftharpoons Cl_2(g) + H_2O(\ell)$$

The label warns that the bleach cleaner should not be used with other products as it may release chlorine gas.

Explain clearly why mixing the bleach with an acid would shift the equilibrium to the right, resulting in the release of chlorine gas from the bleach cleaner. **2**

[Turn over for next question

MARKS

DO NOT WRITE IN THIS MARGIN

(d) The concentration of hypochlorite, OCl^-, in bleach can be determined by a redox reaction that involves two steps.

Step 1

An excess of acidified potassium iodide is added to the bleach. This converts the iodide ions into iodine.

$$OCl^-(aq) + 2I^-(aq) + 2H^+(aq) \rightarrow I_2(aq) + Cl^-(aq) + H_2O(\ell)$$

Step 2

The iodine produced in step 1 is titrated with sodium thiosulfate, $Na_2S_2O_3$.

$$I_2(aq) + 2Na_2S_2O_3(aq) \rightarrow 2NaI(aq) + Na_2S_4O_6(aq)$$

(i) Write the ion-electron equation for the reduction reaction taking place in **Step 1**. **1**

(ii) A 25 cm^3 sample of a diluted bleach was transferred into a conical flask and excess acidified potassium iodide added. The iodine produced was titrated with 0·098 $mol\,l^{-1}$ $Na_2S_2O_3$, requiring an average volume of 9·0 cm^3 to reach the end point.

Calculate the concentration, in $mol\,l^{-1}$, of sodium hypochlorite in the diluted bleach. **3**

[END OF QUESTION PAPER]

National
Qualifications
2018

X713/76/02

Chemistry
Section 1 — Questions

MONDAY, 21 MAY
9:00 AM – 11:30 AM

Instructions for the completion of Section 1 are given on *page 02* of your question and answer booklet X713/76/01.

Record your answers on the answer grid on *page 03* of your question and answer booklet.

You may refer to the Chemistry Data Booklet for Higher and Advanced Higher.

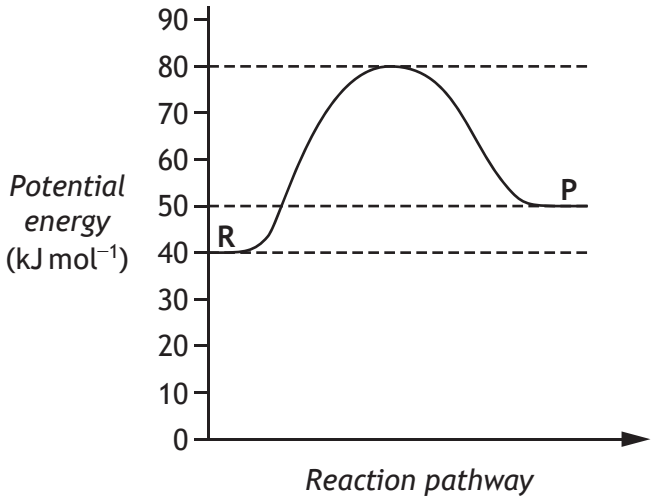
Before leaving the examination room you must give your question and answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

SECTION 1 — 20 marks

Attempt ALL questions

1. The potential energy diagram below refers to the reversible reaction involving reactants **R** and products **P**.



What is the enthalpy change, in kJ mol^{-1}, for the **reverse** reaction?

A −40

B −10

C +10

D +30

2. The relative rate of a reaction which reached completion in 1 minute 40 seconds is

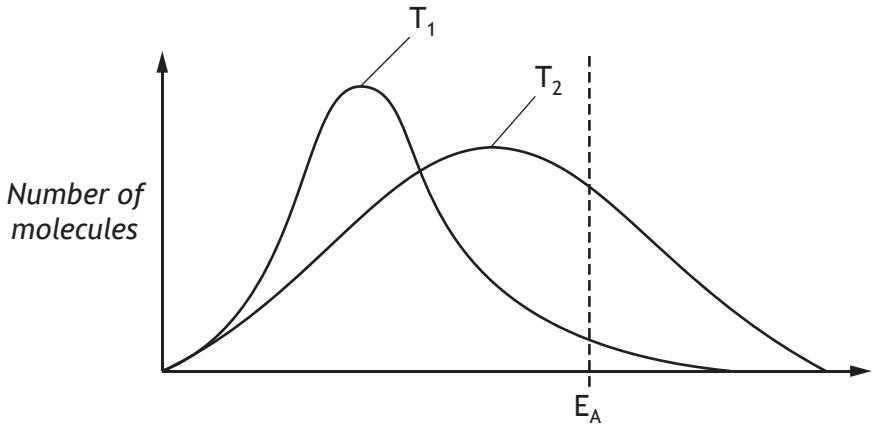
A 0·010 s^{-1}

B 0·714 s^{-1}

C 0·010 min^{-1}

D 0·714 min^{-1}.

3.



Which of the following is the correct interpretation of the above energy distribution diagram for a reaction as the temperature **decreases** from T_2 to T_1?

	Activation energy (E_A)	*Number of successful collisions*
A	remains the same	increases
B	decreases	decreases
C	decreases	increases
D	remains the same	decreases

4. The table shows the first three ionisation energies of aluminium.

Ionisation energy (kJ mol^{-1})		
First	*Second*	*Third*
578	1817	2745

Using this information, what is the enthalpy change, in kJ mol^{-1}, for the following reaction?

$$Al^{+}(g) \rightarrow Al^{3+}(g) + 2e^{-}$$

A 1817

B 2395

C 4562

D 5140

[Turn over

5. An element contains covalent bonding and London dispersion forces.

The element could be

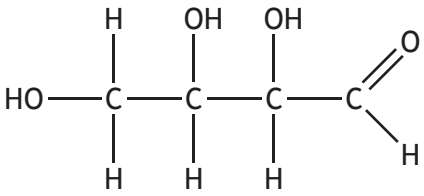
A boron

B neon

C sodium

D sulfur.

6. Erythrose is a chemical that is known to kill cancer cells.



erythrose

The two functional groups present in erythrose are

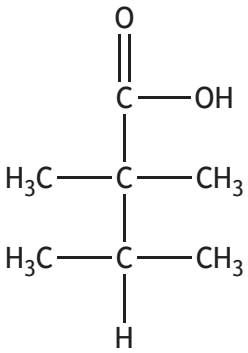
A carboxyl and ester

B carbonyl and ester

C carbonyl and hydroxyl

D carboxyl and hydroxyl.

7.



The name of the above compound is

A 2,2,3-trimethylbutanoic acid

B 2,3,3-trimethylbutanoic acid

C 1,1,2,2-tetramethylpropanoic acid

D 2,2,3,3-tetramethylpropanoic acid.

8. Which of the following is an isomer of pentan-3-ol?

A $CH_3CH_2CH(OH)CH_2CH_3$

B $CH_3CHCHCH_2CH_2OH$

C $CH_3CHCHCH(OH)CH_3$

D $CH_3CH(CH_3)CH_2CH_2OH$

9. Oxidation of 4-methylpentan-2-ol to the corresponding ketone results in the alcohol

A losing 2 g per mole

B gaining 2 g per mole

C losing 16 g per mole

D gaining 16 g per mole.

10. Essential amino acids are defined as the amino acids which

A are necessary for building proteins

B humans must acquire through their diet

C plants cannot synthesise for themselves

D are produced when any protein is hydrolysed.

11. A mixture of carbon monoxide and hydrogen can be converted into water and a mixture of hydrocarbons.

$\mathbf{n}\ CO + (\mathbf{2n} + \mathbf{1})\ H_2 \rightarrow \mathbf{n}\ H_2O +$ hydrocarbons

What is the general formula for the hydrocarbons produced?

A C_nH_{2n-2}

B C_nH_{2n}

C C_nH_{2n+1}

D C_nH_{2n+2}

12. A mixture of sodium chloride and sodium sulfate is known to contain 0·6 mol of chloride ions and 0·2 mol of sulfate ions.

How many moles of sodium ions are present?

A 0·4

B 0·5

C 0·8

D 1·0

13. Under the same conditions of temperature and pressure, which of the following gases would occupy the largest volume?

A 0·20 g of hydrogen

B 0·44 g of carbon dioxide

C 0·60 g of neon

D 0·80 g of argon

14. $3CuO + 2NH_3 \rightarrow 3Cu + N_2 + 3H_2O$

What volume of gas, in cm^3, would be obtained by reaction between 100 cm^3 of ammonia gas and excess copper(II) oxide?

All volumes are measured at atmospheric pressure and 20 °C.

A 50

B 100

C 200

D 400

15. $Cl_2(g) + H_2O(\ell) \rightleftharpoons Cl^-(aq) + ClO^-(aq) + 2H^+(aq)$

The addition of which of the following substances would move the above equilibrium to the right?

A Hydrogen

B Hydrogen chloride

C Sodium chloride

D Sodium hydroxide

16. When 3·6 g of butanal (mass of one mole = 72 g) was burned, 124 kJ of energy was released.

What is the enthalpy of combustion of butanal, in kJ mol^{-1}?

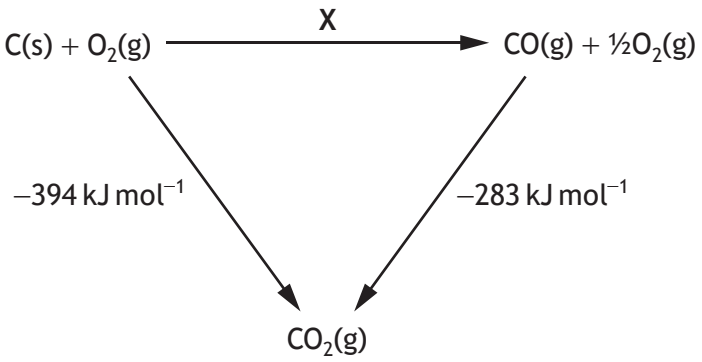
A −6·2

B +6·2

C −2480

D +2480

17. Consider the reaction pathways shown below.



According to Hess's Law, the enthalpy change, in kJ mol^{-1}, for reaction **X** is

A +111

B −111

C −677

D +677.

18. $SO_3^{2-}(aq) + H_2O(\ell) \rightarrow SO_4^{2-}(aq) + 2H^+(aq) + 2e^-$

Which of the following ions could be used to oxidise sulfite ions to sulfate ions?

A $Cr^{3+}(aq)$

B $Al^{3+}(aq)$

C $Fe^{3+}(aq)$

D $Sn^{4+}(aq)$

19. During a redox reaction nitrate ions, NO_3^-, are converted to nitrogen monoxide, NO.

$$NO_3^- \rightarrow NO$$

Which line in the table correctly completes the ion-electron equation?

	Reactants	*Products*
A	$6H^+ + 5e^-$	$3H_2O$
B	$4H^+ + 3e^-$	$2H_2O$
C	$6H^+$	$3H_2O + 5e^-$
D	$4H^+$	$2H_2O + 3e^-$

20. $ICl(\ell) + Cl_2(g) \rightleftharpoons ICl_3(s) \quad \Delta H = -106 kJ\,mol^{-1}$

Which line in the table identifies correctly the changes that will cause the greatest increase in the proportion of solid in the above equilibrium mixture?

	Temperature	*Pressure*
A	decrease	decrease
B	decrease	increase
C	increase	decrease
D	increase	increase

[END OF SECTION 1. NOW ATTEMPT THE QUESTIONS IN SECTION 2 OF YOUR QUESTION AND ANSWER BOOKLET.]

FOR OFFICIAL USE

National Qualifications 2018

Mark

X713/76/01

Chemistry
Section 1 — Answer Grid and Section 2

MONDAY, 21 MAY
9:00 AM – 11:30 AM

X7137601

Fill in these boxes and read what is printed below.

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth

Day		Month		Year		Scottish candidate number								

Total marks — 100

SECTION 1 — 20 marks

Attempt ALL questions.

Instructions for the completion of Section 1 are given on *page 02*.

SECTION 2 — 80 marks

Attempt ALL questions.

You may refer to the Chemistry Data Booklet for Higher and Advanced Higher.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers and rough work is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting. Any rough work must be written in this booklet. You should score through your rough work when you have written your final copy.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

X7137601 01

SECTION 2 — 80 marks

Attempt ALL questions

1. The elements of group 7 in the periodic table are known as the halogens.

(a) Going down group 7 the electronegativity of the halogens decreases.

(i) State what is meant by the term *electronegativity*. **1**

(ii) Explain why electronegativity values decrease going down group 7. **1**

(b) Explain **fully** why the boiling points of the halogens increase going down group 7.

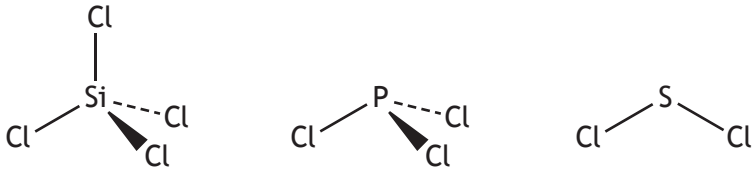
In your answer you should name the intermolecular forces involved. **3**

2. The elements sodium to argon form the third period of the periodic table.

(a) Explain the decrease in atom size going across the third period from sodium to argon. **1**

(b) Elements in the third period of the periodic table form chlorides.

The structures of three of these chlorides are shown.



(i) Circle the structure of the molecule above that contains **bonds** with the lowest polarity. **1**

(An additional diagram, if required, can be found on *page 37*).

(ii) Explain **fully** why, of these three chlorides, silicon tetrachloride is the most soluble in hexane. **2**

[Turn over

2. (continued)

(c) Silicon tetrachloride can be used to make silicon nitride (Si_3N_4), a compound found in many cutting tools.

(i) Silicon nitride has a melting point of 1900 °C and does not conduct electricity when molten.

Explain **fully**, in terms of structure and bonding, why silicon nitride has a high melting point. **2**

(ii) An equation for the formation of silicon nitride is shown.

$3SiCl_4$	+	$16NH_3$	→	Si_3N_4	+	$12NH_4Cl$
mass of one mole = 170·1 g		mass of one mole = 17·0 g		mass of one mole = 140·3 g		mass of one mole = 53·5 g

Calculate the atom economy for the formation of silicon nitride. **2**

2. (continued)

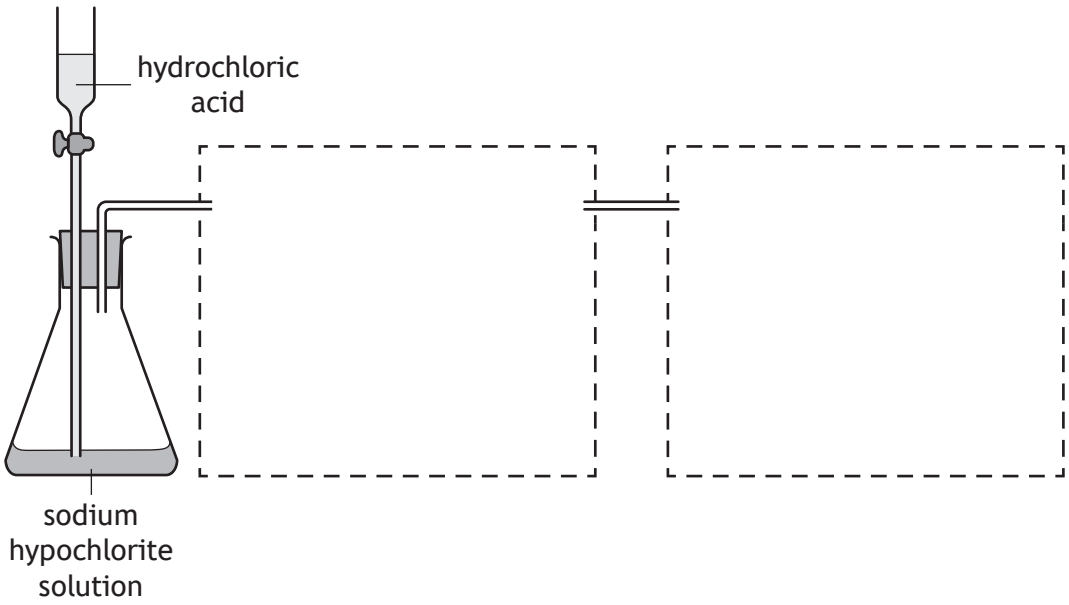
(d) Aluminium, another element in the third period, also forms a chloride. Aluminium chloride is prepared by reacting aluminium metal and chlorine gas.

Chlorine gas is produced by the reaction between hydrochloric acid and sodium hypochlorite. The chlorine is then passed over heated aluminium foil, forming aluminium chloride as a hot gas. The hot aluminium chloride gas and unreacted chlorine gas are passed into a flask where the aluminium chloride cools to a fine white powder.

For safety it is important that any unreacted chlorine gas can escape from the flask.

(i) Complete a labelled diagram to show an apparatus suitable for carrying out this preparation. **2**

(An additional diagram, if required, can be found on *page 37*).

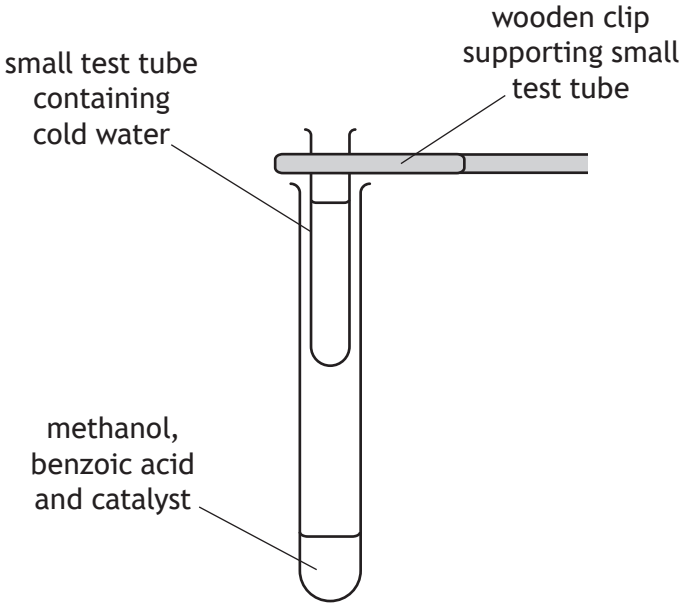


(ii) Explain why the aluminium foil needs to be heated at the start of the preparation, despite the reaction being highly exothermic. **1**

[Turn over

3. Methyl benzoate is commonly added to perfumes as it has a pleasant smell.

A student carries out a reaction to produce methyl benzoate using the following apparatus.



(a) The reaction mixture needs to be heated.

Describe a safe method of heating a flammable mixture. **1**

(b) Suggest a reason why there is a small test tube filled with cold water in the neck of the tube containing the reaction mixture. **1**

3. (continued)

(c) The chemical reaction involved in the experiment is shown.

$C_6H_5COOH(s)$	+	$CH_3OH(\ell)$	→	$C_6H_5COOCH_3(\ell)$	+	**X**
benzoic acid				methyl benzoate		
mass of one mole = 122 g		mass of one mole = 32 g		mass of one mole = 136 g		

(i) Name product **X**. **1**

(ii) In a laboratory experiment, a student used 5·0 g of benzoic acid and 2·5 g of methanol to produce methyl benzoate.

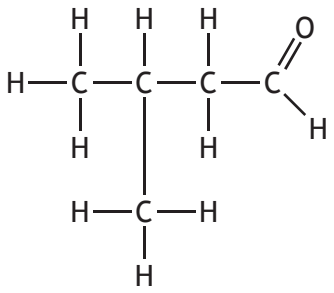
Explain why benzoic acid is the limiting reactant.

You must include calculations in your answer. **2**

(iii) The student produced 3·1 g of methyl benzoate from 5·0 g of benzoic acid. Benzoic acid costs £39·80 for 500 g.

Calculate the cost, in £, of the benzoic acid needed to make 100 g of methyl benzoate using the student's method. **2**

4. 3-Methylbutanal is a compound that is found in low concentrations in many types of food. The structure of 3-methylbutanal is shown.



(a) Draw a structural formula for a ketone that is an isomer of 3-methylbutanal. **1**

(b) Name a reagent which could be used to distinguish between 3-methylbutanal and a ketone. **1**

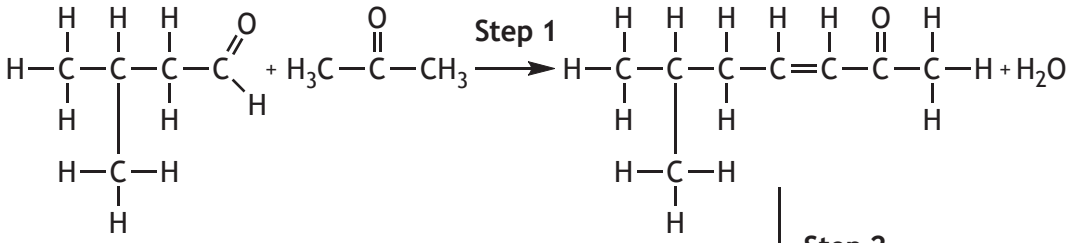
(c) Name the strongest intermolecular force that occurs between 3-methylbutanal molecules. **1**

4. (continued)

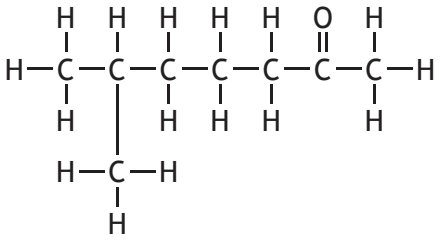
(d) 3-Methylbutanal is found in olive oil. **2**

Explain **fully** what can happen to 3-methylbutanal that will cause the olive oil to develop an unpleasant taste.

(e) 3-Methylbutanal can be used as a reactant in the production of other compounds. One reaction scheme involving 3-methylbutanal is shown.



3-methylbutanal propanone



product A

(i) Explain why **step 1** is described as a condensation reaction. **1**

(ii) Give the systematic name for **product A**. **1**

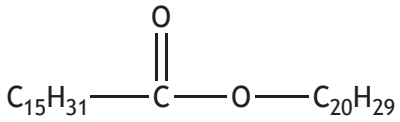
5. Many chemical compounds are related to each other by their structural features, the way they are made and how they are used.

Using your knowledge of chemistry, describe the relationships between fats, oils, detergents, soaps and emulsifiers. **3**

6. Skin creams contain many different chemicals.

(a) Retinol (vitamin A) promotes cell regeneration.

One method of supplying retinol to the skin is by using a skin cream containing the compound retinyl palmitate.



retinyl palmitate

Retinyl palmitate is absorbed into the skin and then broken down to form retinol.

(i) Name the type of reaction that occurs when retinyl palmitate is broken down to form retinol. **1**

(ii) Write a molecular formula for retinol. **1**

(b) Skin creams often contain vitamin E to prevent damage to the skin caused by free radicals.

(i) Describe how free radicals are formed. **1**

6. (b) (continued)

(ii) Hydroxyl free radicals (•OH) can attack fatty acids present in cell membranes. One step in the chain reaction is shown below.

$$C_{18}H_{31}O_2 \quad + \quad \bullet OH \quad \rightarrow \quad C_{18}H_{30}O_2\bullet \quad + \quad H_2O$$

State the name given to this step in the chain reaction. **1**

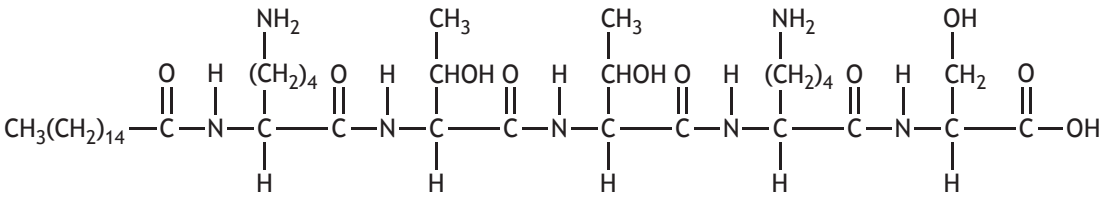
(iii) The antioxidant vitamin E is a free radical scavenger.

State how free radical scavengers prevent further chain reactions. **1**

[Turn over

6. (continued)

(c) Palmitoyl pentapeptide-4 is also used in skin creams.



(i) Circle a peptide link in the above structure. **1**

(An additional diagram, if required, can be found on *page 37*).

(ii) Palmitoyl pentapeptide-4 is formed from palmitic acid and three different amino acids.

Molecule	*Number of molecules used to form one molecule of palmitoyl pentapeptide-4*
palmitic acid	1
threonine	2
serine	1
lysine	2

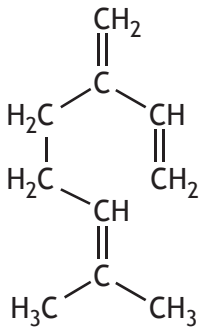
Draw a structural formula for the amino acid serine. **1**

7. Terpenes consist of joined isoprene units (2-methylbuta-1,3-diene). They are classified by the number of isoprene units in the molecule.

Class of terpene	*Number of isoprene units*
hemiterpene	1
monoterpene	2
sesquiterpene	3
diterpene	4
triterpene	6

(a) Myrcene and humulene are terpenes present in hops which give beer its characteristic flavour and aroma.

(i) Circle an isoprene unit on the myrcene structure below. **1**



(An additional diagram, if required, can be found on *page 38*).

(ii) Humulene has the molecular formula $C_{15}H_{24}$.

Name the class of terpene to which humulene belongs. **1**

[Turn over

7. (continued)

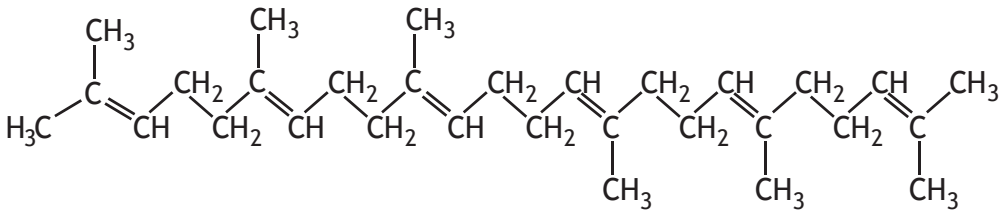
(b) (i) Squalene, a triterpene, is included in some flu vaccines to enhance the body's immune response. A single dose of flu vaccine contains 10·69 mg of squalene.

Calculate the mass of squalene required to produce a batch of 500 000 doses of flu vaccine.

Your answer must be given in kg. **2**

(ii) Squalane is a fully saturated hydrocarbon used in skin moisturising cream.

Squalane can be made by the reaction of squalene with hydrogen.

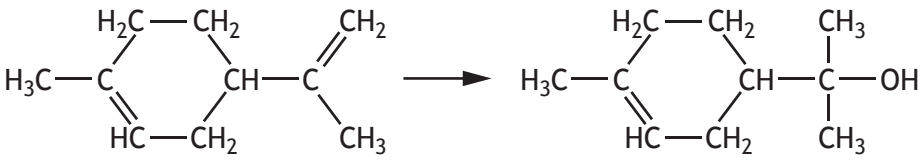


squalene

State the number of moles of hydrogen needed to fully saturate one mole of squalene to produce one mole of squalane. **1**

7. (continued)

(c) The monoterpene limonene, found in lemon oil, can be converted into the alcohol, terpineol.



limonene terpineol

(i) Name the type of reaction taking place. **1**

(ii) When terpineol is heated with copper(II) oxide, no reaction takes place.

Explain why no reaction takes place. **1**

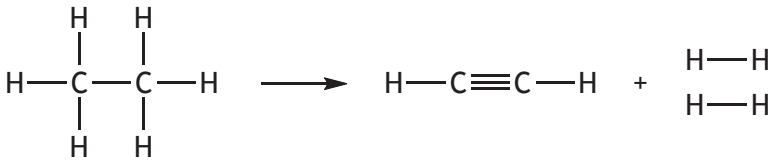
[Turn over

8. The alkynes are a homologous family of hydrocarbons.

(a) The simplest member of the family is ethyne, C_2H_2, used in welding torches.



Ethyne can be produced from ethane.



Using bond enthalpies and mean bond enthalpies from the data book, calculate the enthalpy change, in kJ mol^{-1}, for this reaction. **2**

(b) Hess's Law can be used to calculate the enthalpy change for reactions that do not normally take place, such as the formation of propyne from its elements.

$$3C(s) + 2H_2(g) \rightarrow C_3H_4(g)$$

Calculate the enthalpy change, in kJ mol^{-1}, for this reaction using the following information. **2**

$C(s) + O_2(g) \rightarrow CO_2(g)$ $\quad \Delta H = -394$ kJ mol^{-1}

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(\ell)$ $\quad \Delta H = -286$ kJ mol^{-1}

$C_3H_4(g) + 4O_2(g) \rightarrow 3CO_2(g) + 2H_2O(\ell)$ $\quad \Delta H = -1939$ kJ mol^{-1}

8. (continued)

(c) Propyne, C_3H_4 (1 mole = 40 g), has been suggested as a possible rocket fuel.

(i) The enthalpy of combustion of propyne is $-1939\ kJ\ mol^{-1}$.

Calculate the energy released, in kJ, when 1 kg of propyne is burned completely. **1**

(ii) The mass of air required to burn 1 g of fuel can be calculated using the relationship shown.

Mass of air, in g = 4·3 × mass of oxygen, in g, for complete combustion of 1 g of fuel

Calculate the mass of air, in g, required to burn 1 g of propyne. **2**

$$C_3H_4(g) + 4O_2(g) \rightarrow 3CO_2(g) + 2H_2O(\ell)$$

8. (c) (continued)

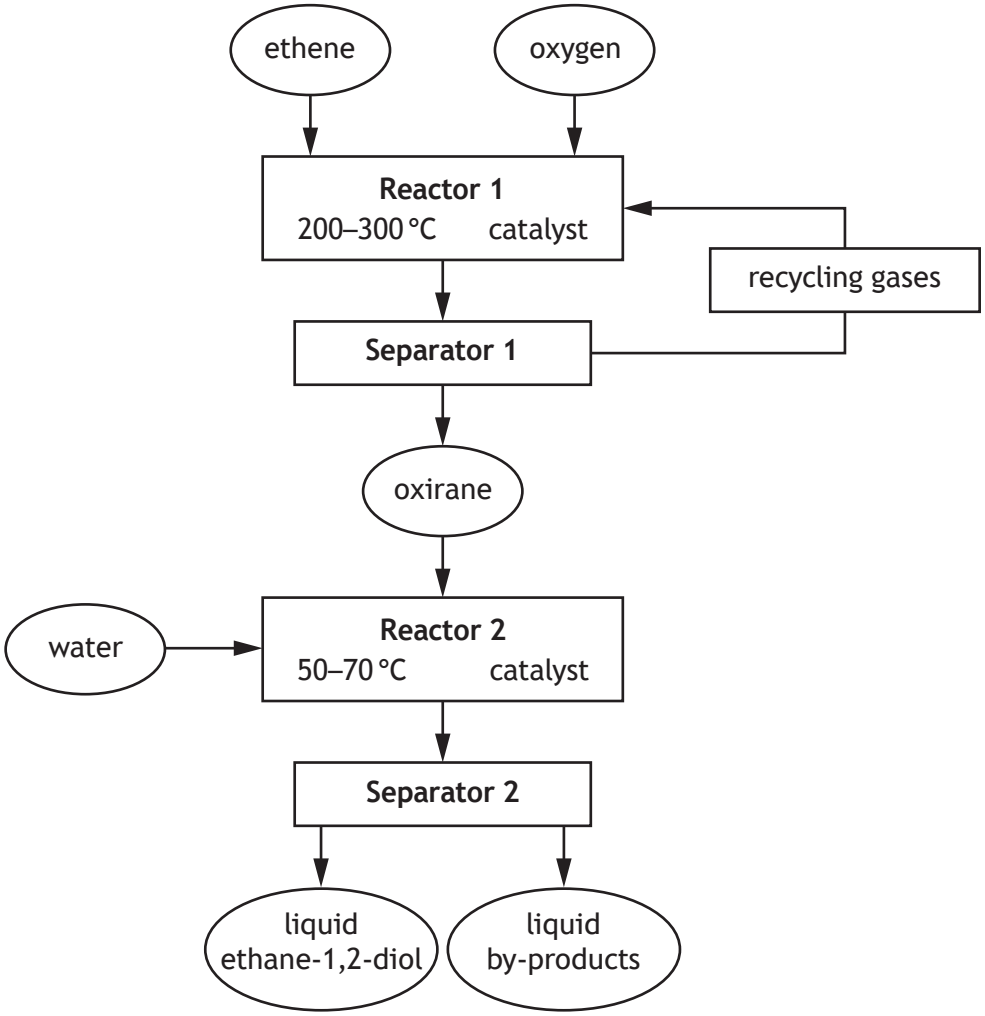
(iii) The table shows the mass of air required to burn 1 g of different fuels.

Fuel	*Mass of 1 mole (g)*	*Mass of air required to burn 1 g*
ethane	30	16·1
propane	44	15·6
methanol	32	6·5
ethanol	46	9·0

Suggest why methanol and ethanol, compared to the other fuels, require less air to burn 1 g. **1**

9. Ethane-1,2-diol can be made from ethene.

(a) The flow chart of an industrial process to produce ethane-1,2-diol is shown.



(i) Industrial processes are designed to maximise profit.

Using the flowchart, suggest two ways to maximise profit in this industrial process. **2**

9. (a) (continued)

(ii) Name the process used in **Separator 2** to separate ethane-1,2-diol from the larger liquid by-products. **1**

(b) Explain fully why ethane-1,2-diol is more viscous than propan-1-ol. **2**

(c) Draw a structural formula for a diol that contains three carbon atoms. **1**

[Turn over

9. (continued)

(d) Ethane-1,2-diol has been found to be harmful to animals. Treatment for affected animals involves using a 20% ethanol solution.

(i) The 20% ethanol solution is prepared by accurately measuring 20 cm^3 of ethanol and then making up to exactly 100 cm^3 with water.

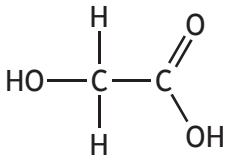
Describe the procedure which should be used to prepare 100 cm^3 of the 20% ethanol solution. **2**

(ii) An affected animal must be treated with 9 doses of 20% ethanol solution. Each dose contains 5 cm^3 of the ethanol solution for every kilogram body mass of the animal.

Calculate the total volume, in cm^3, of the 20% ethanol solution needed to treat a 3·5 kg animal. **1**

9. (d) (continued)

(iii) Ethane-1,2-diol is harmful because it is oxidised in the body to form glycolic acid.



glycolic acid

(A) Draw a structural formula for another possible product of oxidation of ethane-1,2-diol. **1**

(B) Glycolic acid can be neutralised by sodium hydroxide to form sodium glycolate.

Give a formula for sodium glycolate. **1**

[Turn over

10. The molar volume (in units of litres per mole) is the same for all gases at the same temperature and pressure.

Using your knowledge of chemistry, suggest how the molar volume of gases could be measured and compared. Any suitable chemicals and apparatus can be used. Some suggested chemicals and apparatus are given below. **3**

Chemicals	*Apparatus*
hydrochloric acid	gas syringe
zinc	measuring cylinder
magnesium	delivery tube
calcium	stoppers
water	500 cm^3 flask
sodium carbonate	vacuum pump
calcium carbonate	balance
cylinder of nitrogen	cork ring
cylinder of hydrogen	burette
cylinder of carbon dioxide	filter funnel

11. Iodine is required for a healthy diet. Food grown in certain parts of the world is low in iodine. To prevent iodine deficiency in people's diets, table salt can be 'iodised' by the addition of very small quantities of potassium iodide, KI.

The number of moles of iodide in a sample of salt can be determined by the following procedure.

Step 1

Prepare a standard salt solution by dissolving an accurately weighed sample of iodised salt (50·0 g) in water to give a final volume of 250 cm^3.

Step 2

Transfer 50 cm^3 of salt solution to a conical flask and add excess bromine solution to convert the iodide ions to iodine.

Step 3

Titrate the iodine (I_2) released with sodium thiosulfate solution ($Na_2S_2O_3$).

(a) Describe a procedure to accurately weigh out a 50·0 g sample of iodised table salt. **1**

(b) The overall equation for the reaction of bromine solution with iodide ions is shown.

$$2I^-(aq) + Br_2(aq) \rightarrow I_2(aq) + 2Br^-(aq)$$

Write the ion-electron equation for the oxidation reaction. **1**

11. (continued)

(c) Three samples were prepared as described in **step 2**. Each sample was titrated with 0·0010 mol l^{-1} sodium thiosulfate solution.

The results are shown below.

Sample	*Volume of sodium thiosulfate* (cm^3)
1	10·0
2	9·4
3	9·6

(i) Calculate the average volume, in cm^3, of sodium thiosulfate solution that should be used to determine the number of moles of iodine released. **1**

(ii) Calculate the number of moles of iodine released from 50 cm^3 of the standard salt solution. **2**

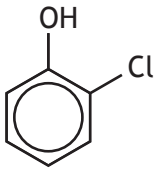
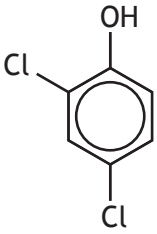
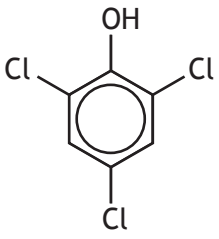
$$I_2(aq) + 2Na_2S_2O_3(aq) \rightarrow 2NaI(aq) + Na_2S_4O_6(aq)$$

[Turn over

12. Many modern antiseptics are based on phenol. The table shows the germ-killing power of some phenol compounds.

DO NOT WRITE IN THIS MARGIN

(a)

Compound	*Structure*	*Germ-killing power (relative to phenol)*
phenol		1·0
4-methylphenol		2·5
2-chlorophenol		3·6
4-ethylphenol		7·5
2,4-dichlorophenol		13·0
4-propylphenol		20·0
2,4,6-trichlorophenol		23·0

12. (a) (continued)

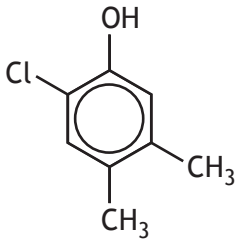
(i) Suggest two ways in which structural features increase germ-killing power of phenol compounds. **2**

(ii) The names of the phenol compounds in the table are derived from their structures using the following rules.

Phenol is used as the parent name for the compound.

1. The –OH functional group is assigned as being on carbon 1 of the ring.
2. The ring can be numbered clockwise or anticlockwise to assign numbers to the other atoms or groups. The numbers should be assigned so that the lowest possible numbers are used.
3. If two or more identical atoms or groups are present, use one of the prefixes di, tri or tetra.
4. The names of the atoms or groups attached to the ring are listed alphabetically (ignoring the prefixes for alphabetical purposes).

Using these rules, name this molecule.



1

[Turn over

12. (continued)

(b) There are different methods of producing phenol.

(i) In the early 1900s, phenol was produced by the following reaction.

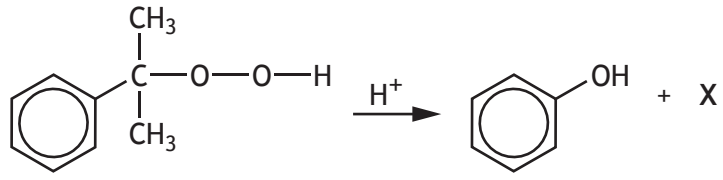
$$\underset{\text{benzene}}{C_6H_6} + H_2SO_4 + 2NaOH \rightarrow \underset{\text{phenol}}{C_6H_5OH} + Na_2SO_3 + 2H_2O$$

mass of one mole = 78·0 g

mass of one mole = 94·0 g

Calculate the mass of phenol, in kg, produced from 117 kg of benzene if the percentage yield is 90%. **2**

(ii) Phenol is now usually produced by the Cumene Process.



cumene hydroperoxide

phenol

Name the other product, **X**, formed in the Cumene Process. **1**

[END OF QUESTION PAPER]

National
Qualifications
2017

X713/76/02

Chemistry
Section 1 — Questions

MONDAY, 8 MAY

9:00 AM – 11:30 AM

Instructions for the completion of Section 1 are given on *Page 02* of your question and answer booklet X713/76/01.

Record your answers on the answer grid on *Page 03* of your question and answer booklet.

You may refer to the Chemistry Data Booklet for Higher and Advanced Higher.

Before leaving the examination room you must give your question and answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

SECTION 1 — 20 marks

Attempt ALL questions

1. Which of the following bonds is the **least** polar?

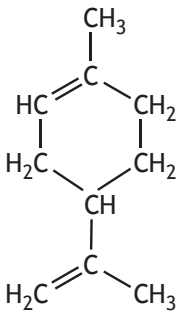
A C — I

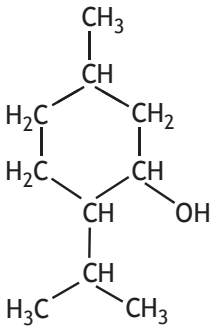
B C — F

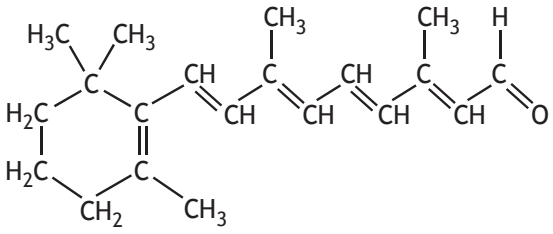
C C — Cl

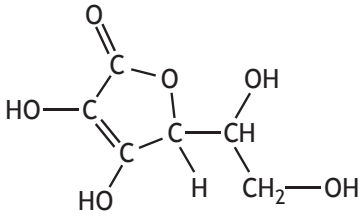
D C — Br

2. Which of the following compounds would be the **most water** soluble?

A 

B 

C 

D 

3. Which of the following atoms has the greatest attraction for bonding electrons?

A Sulfur

B Silicon

C Nitrogen

D Hydrogen

4. Which type of structure is found in phosphorus?

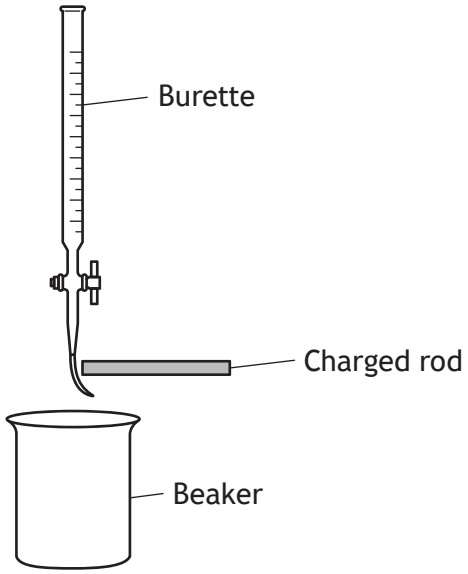
A Covalent network

B Covalent molecular

C Monatomic

D Metallic lattice

5. The polarity of molecules can be investigated using a charged rod. The charged rod will attract a stream of polar liquid flowing from a burette.



Which of the following liquids would **not** be attracted?

A Water

B Propanone

C Propanol

D Hexane

[Turn over

6. $x P_2H_4 + y O_2 \rightarrow P_4O_{10} + z H_2O$

The equation is balanced when

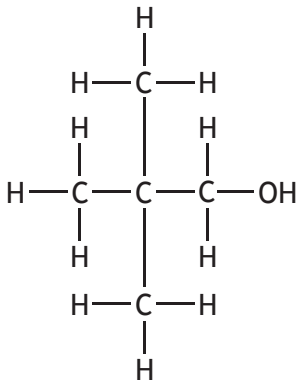
A $x = 1, y = 5, z = 4$

B $x = 4, y = 6, z = 2$

C $x = 2, y = 7, z = 4$

D $x = 2, y = 5, z = 2$

7. What is the systematic name for the compound below?



A 2,2,2-trimethylethanol

B 2,2-dimethylpropan-1-ol

C 2,2-dimethylpropan-3-ol

D 2,2-dimethylpentan-1-ol

8. Which of the following fatty acids is the most unsaturated?

A $C_{15}H_{29}COOH$

B $C_{15}H_{31}COOH$

C $C_{17}H_{31}COOH$

D $C_{17}H_{35}COOH$

9. Which of the following is **not** a step in a free radical chain reaction?

A Activation
B Initiation
C Propagation
D Termination

10. Which of the following is an isomer of ethyl propanoate ($CH_3CH_2COOCH_2CH_3$)?

A Methyl propanoate
B Pentan-2-one
C Pentanoic acid
D Pentane-1,2-diol

11. Essential oils are

A non-water soluble, non-volatile compounds
B non-water soluble, volatile compounds
C water soluble, non-volatile compounds
D water soluble, volatile compounds.

12. The enthalpy of combustion of a hydrocarbon is the enthalpy change when

A one mole of a hydrocarbon burns to give one mole of water
B one mole of a hydrocarbon burns to give one mole of carbon dioxide
C one mole of a hydrocarbon burns completely in oxygen
D one mole of a hydrocarbon burns in one mole of oxygen.

13. Which of the following is the strongest reducing agent?

A Fluorine
B Lithium
C Calcium
D Iodine

[Turn over

14.

$TiCl_4$	+	2Mg	→	$2MgCl_2$	+	Ti
mass of one mole = 189·9 g		mass of one mole = 24·3 g		mass of one mole = 95·3 g		mass of one mole = 47·9 g

The atom economy for the production of titanium in the above equation is equal to

A $\dfrac{47{\cdot}9}{189{\cdot}9+24{\cdot}3}\times 100$

B $\dfrac{47{\cdot}9}{189{\cdot}9+(2\times 24{\cdot}3)}\times 100$

C $\dfrac{95{\cdot}3+47{\cdot}9}{189{\cdot}9+24{\cdot}3}\times 100$

D $\dfrac{(2\times 47{\cdot}9)}{189{\cdot}9+24{\cdot}3}\times 100$

15. The vitamin C content of a carton of orange juice was determined by four students. Each student carried out the experiment three times.

	Experiment 1 (mg/100 cm^3)	*Experiment 2* (mg/100 cm^3)	*Experiment 3* (mg/100 cm^3)
Student A	30·0	29·0	28·0
Student B	26·4	26·6	26·8
Student C	26·9	27·0	26·9
Student D	26·9	26·5	26·9

The most reproducible results were obtained by

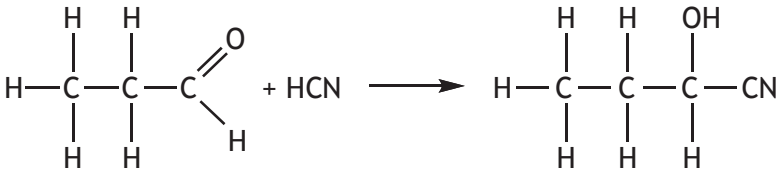
A Student A

B Student B

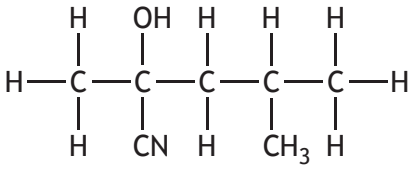
C Student C

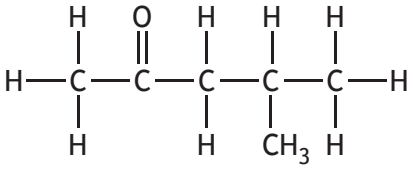
D Student D.

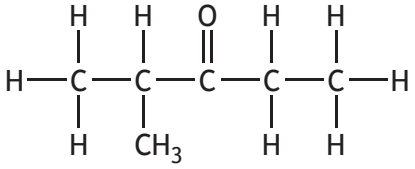
16. Cyanohydrin compounds can be made from carbonyl compounds by reacting the carbonyl compound with hydrogen cyanide (HCN).

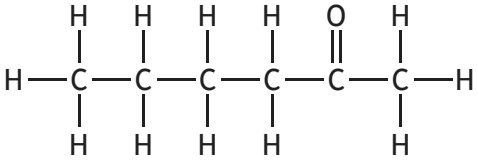


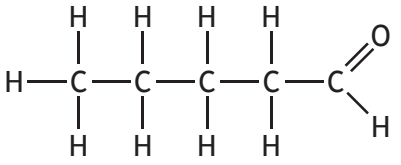
Which carbonyl compound would react with hydrogen cyanide (HCN) to form the following compound?



A 

B 

C 

D 

[Turn over

17. Chemical reactions are in a state of dynamic equilibrium only when

A the reaction involves no enthalpy change

B the concentrations of reactants and products are equal

C the activation energies of the forward and backward reactions are equal

D the rate of the forward reaction equals that of the backward reaction.

18. Bromine and hydrogen react together to form hydrogen bromide.

$$H_2(g) + Br_2(g) \rightarrow 2HBr(g)$$

Bonds broken	Bonds made
H—H	2 × H—Br
Br—Br	

Bond	Bond enthalpy (kJ mol^{-1})
H—H	436
Br—Br	194
H—Br	366

The enthalpy change for this reaction, in kJ mol^{-1}, is

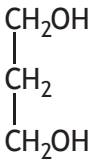
A −102

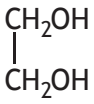
B +102

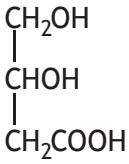
C −264

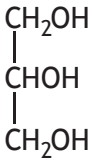
D +264.

19. Which of the following is a structural formula for glycerol?

A


B


C


D


20. Which line in the table best describes the effect of adding a catalyst to the following reaction?

$4NH_3(g) + 5O_2(g) \rightleftharpoons 4NO(g) + 6H_2O(g) \qquad \Delta H = -ve$

	Position of equilibrium	*Rate of forward reaction*
A	unchanged	unchanged
B	unchanged	increased
C	moves to right	unchanged
D	moves to right	increased

[END OF SECTION 1. NOW ATTEMPT THE QUESTIONS IN SECTION 2 OF YOUR QUESTION AND ANSWER BOOKLET.]

FOR OFFICIAL USE

National Qualifications 2017

Mark

X713/76/01

Chemistry
Section 1 — Answer Grid and Section 2

MONDAY, 8 MAY

9:00 AM – 11:30 AM

X7137601

Fill in these boxes and read what is printed below.

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth

Day | Month | Year

Scottish candidate number

Total marks — 100

SECTION 1 — 20 marks

Attempt ALL questions.

Instructions for the completion of Section 1 are given on *Page 02*.

SECTION 2 — 80 marks

Attempt ALL questions.

You may refer to the Chemistry Data Booklet for Higher and Advanced Higher.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers and rough work is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting. Any rough work must be written in this booklet. You should score through your rough work when you have written your final copy.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

SQA ©

X7137601 01

SECTION 2 — 80 marks

Attempt ALL questions

1. The elements sodium to argon make up the third period of the Periodic Table.

Na	Mg	Al	Si	P	S	Cl	Ar

(a) Name the element from the third period that exists as a covalent network. **1**

(b) Ionisation energy changes across the period.

(i) Explain why the first ionisation energy increases across the period. **1**

(ii) Write an equation, including state symbols, for the **second** ionisation energy of magnesium. **1**

(iii) The table shows the values for the first four ionisation energies of aluminium.

Ionisation energies (kJ mol^{-1})			
First	Second	Third	Fourth
578	1817	2745	11 577

Explain why there is a large difference between the third and fourth ionisation energies. **1**

1. (continued)

(c) The boiling point of chlorine is much higher than that of argon.

Explain **fully**, in terms of structure and the type of van der Waals forces present, why the boiling point of chlorine is higher than that of argon. **3**

2. Reactions involving iodine are commonly used to investigate rates of reaction.

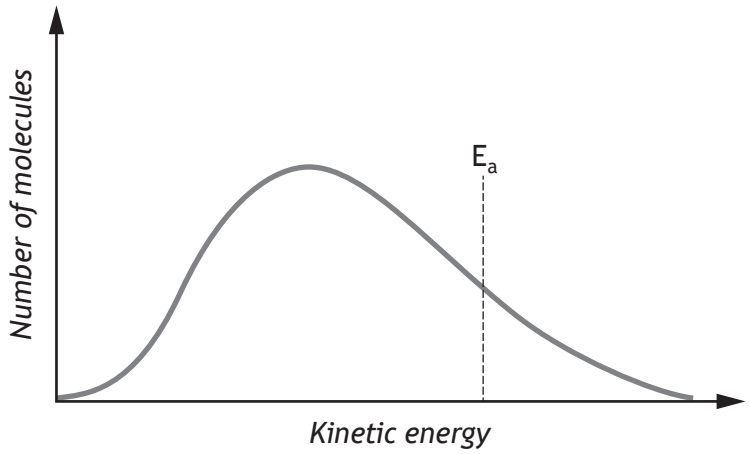
(a) One reaction involves hydrogen and iodine reacting together to form hydrogen iodide.

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

(i) This reaction is thought to occur by initially breaking bonds in one of the reactants.

Explain, using bond enthalpies, which bond is more likely to break first during this reaction. **1**

(ii) The graph shows the distribution of kinetic energies of reactant molecules in the gas mixture at 300 °C.



Add a second curve to the graph to show the distribution of kinetic energies at 400 °C. **1**

(An additional graph, if required, can be found on *Page 35*)

2. (a) (continued)

(iii) The reaction to produce hydrogen iodide is exothermic.

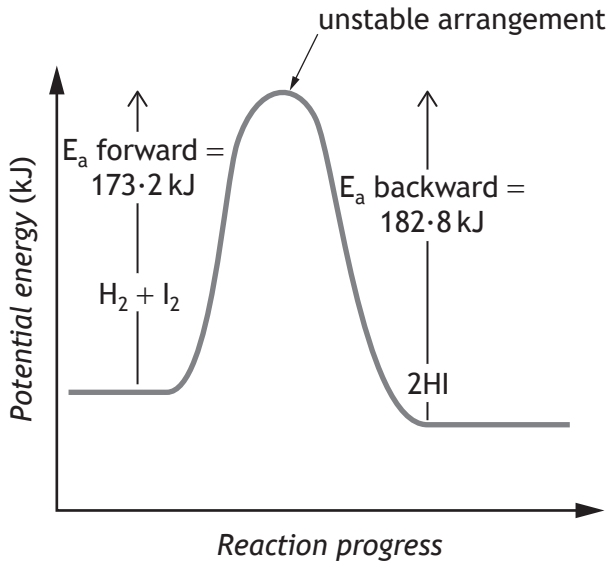
$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

(A) State the effect of increasing temperature on the position of equilibrium. **1**

(B) State why changing the pressure has no effect on this equilibrium reaction. **1**

2. (a) (continued)

(iv) The potential energy diagram for the reaction between hydrogen and iodine is shown.



(A) State the term for the unstable arrangement of atoms. **1**

(B) Calculate the enthalpy change, in kJ, for the forward reaction. **1**

(C) Platinum can be used as a catalyst for this reaction.

State the effect that platinum would have on the activation energy for the reaction. **1**

2. (continued)

(b) The reaction between iodide ions, $I^-(aq)$, and persulfate ions, $S_2O_8^{2-}(aq)$, is used to investigate the effect of changing concentration on rate of reaction. The relative rate of the reaction is determined by mixing the reactants in a beaker and recording the time taken for the mixture to change colour.

The results of the investigation are shown in the table.

Experiment	*Concentration of $I^-(aq)$* (mol l^{-1})	*Concentration of $S_2O_8^{2-}(aq)$* (mol l^{-1})	*Time* (s)	*Relative rate* (s^{-1})
1	0·04	0·05	241	0·00415
2	0·06	0·05	180	0·00556
3	0·08	0·05		0·00819
4	0·1	0·05	103	0·00971

(i) The instructions state that a dry beaker must be used for each experiment.

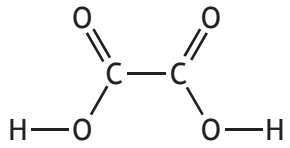
Suggest a reason why the beaker should be dry. **1**

(ii) Calculate the time, in seconds, for the reaction in experiment 3. **1**

(iii) Explain why decreasing the concentration of iodide ions lowers the reaction rate. **1**

3. The leaves of the rhubarb plant are considered poisonous because they contain high levels of oxalic acid.

Oxalic acid is a white, water-soluble solid. It is a dicarboxylic acid that has the structural formula shown.



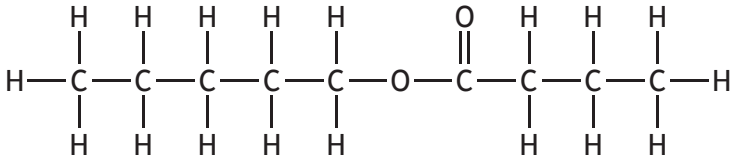
Oxalic acid reacts with bases to form salts.

It can also be oxidised by strong oxidising agents to form carbon dioxide gas. The oxidation equation for oxalic acid is shown.

$$H_2C_2O_4 \quad \rightarrow \quad 2CO_2 + 2e^- + 2H^+$$

Using your knowledge of chemistry, comment on how the mass of oxalic acid in a rhubarb leaf could be determined. **3**

4. Pentyl butanoate is responsible for some of the flavour in apricots and strawberries.



(a) Hydrolysis of pentyl butanoate using sodium hydroxide produces an alcohol and the salt of the carboxylic acid.

(i) Name the alcohol that would be formed when pentyl butanoate is hydrolysed. **1**

(ii) Draw a structural formula for the sodium salt of the carboxylic acid that would be formed. **1**

(b) Fats and oils belong to the same class of compounds as pentyl butanoate.

(i) Name this class of compounds. **1**

(ii) When a fat is hydrolysed using sodium hydroxide, sodium salts of fatty acids are produced.

State a use for sodium salts of fatty acids. **1**

4. (b) (continued)

(iii) Hydrolysis of fats using hydrochloric acid produces fatty acids. Stearic acid is a fatty acid that can be made from hydrolysis of beef fat. It is a fuel sometimes found in fireworks.

During combustion, stearic acid ($C_{17}H_{35}COOH$) produces 623 kJ of energy **per mole of CO_2 produced.**

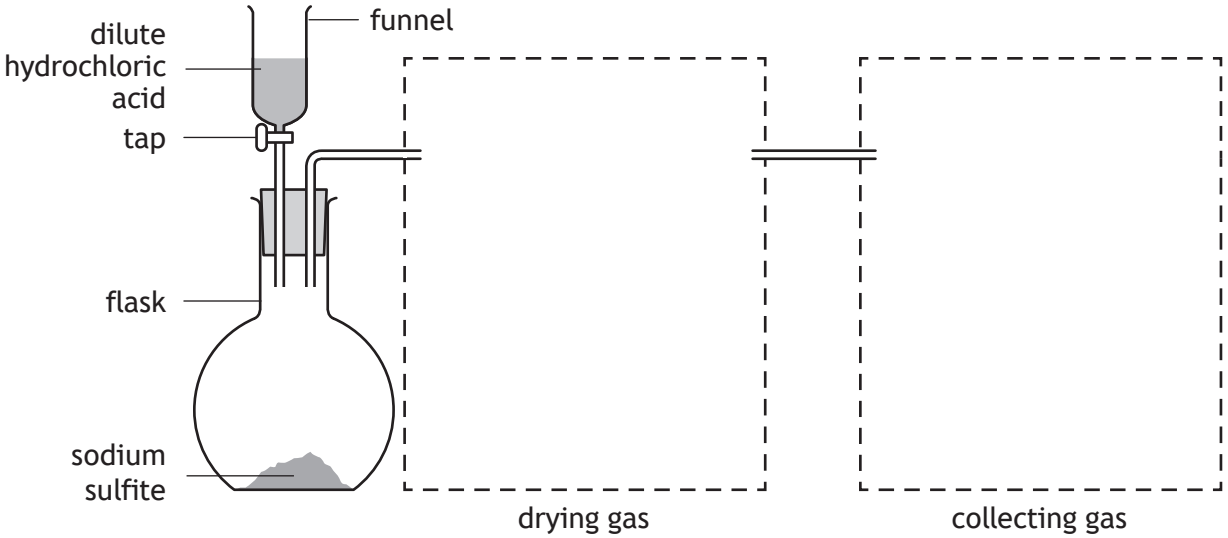
$$C_{17}H_{35}COOH + 26O_2 \rightarrow 18CO_2 + 18H_2O$$

mass of one mole = 284 g

Calculate the energy released, in kJ, by combustion of 10 grams of stearic acid. **2**

5. Sulfur dioxide is a colourless, toxic gas that is soluble in water and more dense than air.

(a) One laboratory method for preparation of sulfur dioxide gas involves adding dilute hydrochloric acid to solid sodium sulfite. The sulfur dioxide gas produced is dried by bubbling the gas through concentrated sulfuric acid. The sulfur dioxide gas can then be collected.



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(i) Complete the diagram by drawing:

in the first box, apparatus suitable for drying the sulfur dioxide gas;

in the second box, apparatus suitable for collecting the gas. **2**

(An additional diagram, if required, can be found on *Page 35*)

5. (a) (continued)

(ii) 0·40 g of sodium sulfite, Na_2SO_3, is reacted with 50 cm^3 of dilute hydrochloric acid, concentration 1·0 mol l^{-1}.

$$Na_2SO_3(s) + 2HCl(aq) \rightarrow H_2O(\ell) + 2NaCl(aq) + SO_2(g)$$

mass of one mole = 126·1 g

Show, by calculation, that sodium sulfite is the limiting reactant. **2**

(b) Another reaction that produces sulfur dioxide gas involves combustion of carbon disulfide in the reaction shown.

$$CS_2(\ell) + 3O_2(g) \rightarrow CO_2(g) + 2SO_2(g)$$

Calculate the enthalpy change, in kJ mol^{-1}, for this reaction using the following information. **2**

$C(s) + O_2(g) \rightarrow CO_2(g)$ $\quad \Delta H = -393{\cdot}5$ kJ mol^{-1}

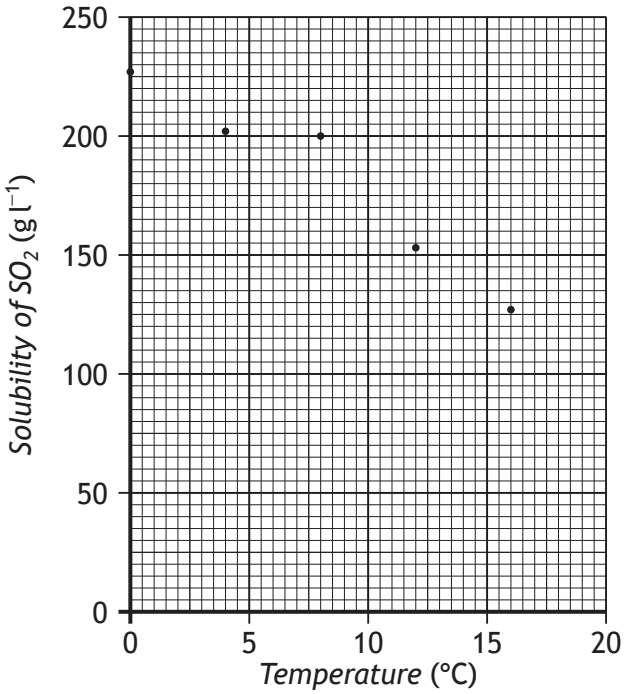
$S(s) + O_2(g) \rightarrow SO_2(g)$ $\quad \Delta H = -296{\cdot}8$ kJ mol^{-1}

$C(s) + 2S(s) \rightarrow CS_2(\ell)$ $\quad \Delta H = +87{\cdot}9$ kJ mol^{-1}

MARKS

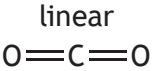
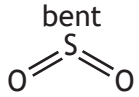
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(c) The graph shows results for an experiment to determine the solubility of sulfur dioxide in water.



(i) Determine the solubility of sulfur dioxide, in $g\ l^{-1}$, in water at 10 °C. **1**

(ii) Information about sulfur dioxide and carbon dioxide is shown in the table.

	Shape of molecule	*Electronegativity difference between elements*	*Solubility in water at 25 °C* ($g\ l^{-1}$)
carbon dioxide	linear 	1·0	1·45
sulfur dioxide	bent 	1·0	94

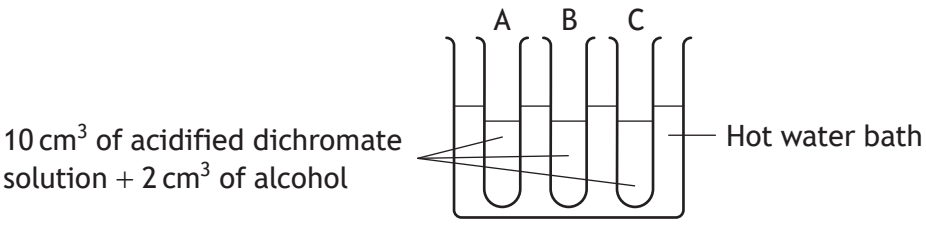
Explain **fully** why carbon dioxide is much less soluble in water than sulfur dioxide is in water. **2**

6. A student was carrying out an investigation into alcohols, aldehydes and ketones.

(a) The student was given three alcohols labelled as **A**, **B** and **C**. These alcohols were all isomers with the formula C_4H_9OH.

(i) Draw a structural formula for the secondary alcohol with the formula C_4H_9OH. **1**

(ii) The student set up the following experiment.



Alcohol	*Observation*
A	Colour change
B	No change
C	Colour change

(A) Suggest why a water bath is an appropriate method of heating the reaction mixture. **1**

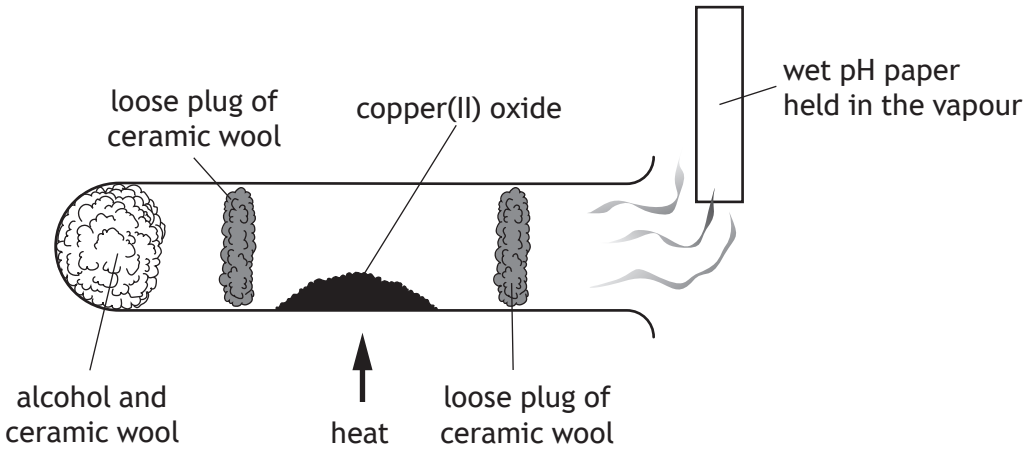
(B) Describe the colour change that would have been observed with alcohols **A** and **C**. **1**

(C) Alcohol **B** is not oxidised.

State the **type** of alcohol which cannot be oxidised by acidified dichromate solution. **1**

6. (a) (continued)

(iii) The student set up a second experiment with alcohol **A**.



Hot copper(II) oxide is an oxidising agent.

(A) When alcohol **A** (C_4H_9OH) is oxidised the product turns the pH paper red.

Suggest a name for the product. **1**

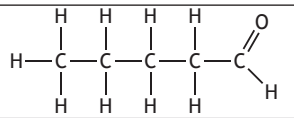
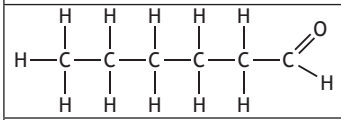
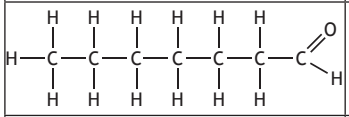
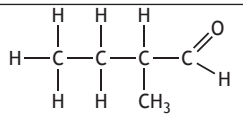
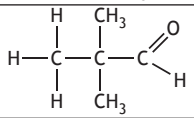
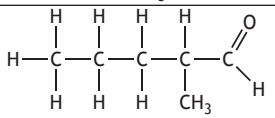
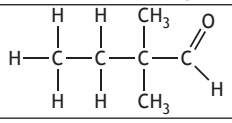
(B) Complete the ion-electron equation for the oxidation reaction. **1**

$C_4H_9OH \rightarrow C_4H_8O_2$

6. (continued)

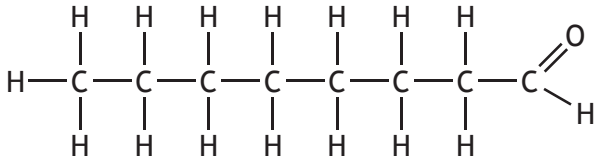
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(b) The student found the following information about the boiling points of some aldehydes.

Aldehyde	*Molecular formula*	*Boiling point* (°C)
	$C_5H_{10}O$	102
	$C_6H_{12}O$	130
	$C_7H_{14}O$	153
	$C_5H_{10}O$	95
	$C_5H_{10}O$	75
	$C_6H_{12}O$	119
	$C_6H_{12}O$	111

(i) Name the aldehyde that has a boiling point of 119 °C. **1**

(ii) Predict the boiling point, in °C, of the following molecule. **1**



6. (b) (continued)

(iii) Using information from the table, describe one way in which differences in structure affect the boiling point of **isomeric** aldehydes. **1**

(iv) State what would be observed when an aldehyde is gently heated with Tollens' reagent. **1**

(c) Ketones contain a carbonyl group.

Name the type of intermolecular interaction between ketone molecules. **1**

7. Some people take iron tablets as a dietary supplement. Iron tablets may contain iron(II) sulfate.

(a) A student was investigating the iron(II) content of iron tablets. A work card gave the following instructions for preparing an iron tablet solution.

1. Add five iron tablets to about 50 cm^3 of dilute sulfuric acid in a small beaker and stir to dissolve.
2. Transfer quantitatively to a 100 cm^3 volumetric flask.
3. Make up the solution to the graduation mark on the volumetric flask.
4. Stopper the flask and then invert it to mix the solution.

To 'transfer quantitatively' means that **all** of the iron tablet solution must be transferred into the volumetric flask.

Describe how this is carried out in practice. **1**

(b) The concentration of iron(II) ions (Fe^{2+}) in this iron tablet solution can be determined by a redox titration with permanganate (MnO_4^-) solution.

$$5Fe^{2+}(aq) + 8H^+(aq) + MnO_4^-(aq) \rightarrow 5Fe^{3+}(aq) + Mn^{2+}(aq) + 4H_2O(\ell)$$

(i) Suggest why it is **not** necessary to add an indicator to this titration. **1**

7. (b) (continued)

(ii) Suggest why the titration must be carried out under acidic conditions. **1**

(iii) Three 25·0 cm^3 samples of the iron tablet solution were titrated with a standard solution of 0·020 mol l^{-1} permanganate (MnO_4^-). The results are shown below.

Sample	*Volume of permanganate* (cm^3)
1	14·9
2	14·5
3	14·6

(A) State why the volume of permanganate used in the calculation was taken to be 14·55 cm^3, although this is not the average of the three titres in the table. **1**

(B) Calculate the concentration, in mol l^{-1}, of iron(II) ions in the iron tablet solution. **3**

$$5Fe^{2+}(aq) + 8H^+(aq) + MnO_4^-(aq) \rightarrow 5Fe^{3+}(aq) + Mn^{2+}(aq) + 4H_2O(\ell)$$

7. (b) (iii) (continued)

(C) State what is meant by the term **standard solution**. **1**

(D) Name an appropriate piece of apparatus which could be used to measure 25·0 cm^3 samples of iron tablet solution. **1**

(c) In a different experiment, five iron tablets were found to contain 0·00126 moles of iron(II) ions.

Calculate the average mass, in **mg**, of iron present in **one** tablet. **1**

(d) It is recommended an adult female takes in 14·8 mg of iron per day.

100 g of a breakfast cereal contains 12·0 mg of iron.

Calculate the percentage of the recommended daily amount of iron provided for an adult female by a 30 g serving. **2**

8. Skin care products contain a mixture of polar covalent, non-polar covalent and ionic compounds. The compounds need to stay mixed within the product.

Skin care products also need to spread easily and remain on the skin for a period of time, as well as provide physical and chemical protection from the sun. In order to do this, skin care products contain a range of chemicals including water, fats and oils, antioxidants, minerals and sun block.

Using your knowledge of chemistry, explain the role of different chemicals in skin care products. **3**

9. Dishwasher tablets contain chemicals which remove dirt from dishes.

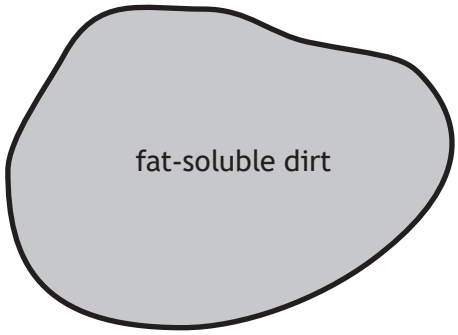
(a) Dishwasher tablets include detergents. These molecules act like soaps to allow mixing of fat-soluble dirt and water.

(i) During the cleaning process, the detergent molecules combine with fat-soluble dirt.

A simplified diagram of a detergent molecule is shown.



Complete the diagram below to show how detergent molecules combine with fat-soluble dirt. **1**



(An additional diagram, if required, can be found on *Page 36*)

(ii) State the term used to describe the non-polar, hydrocarbon tail of a detergent molecule. **1**

9. (continued)

(b) Dishwasher tablets produce the bleach hydrogen peroxide, H_2O_2. One action of this oxidising agent is to oxidise food.

(i) Suggest another action of the bleach produced by the dishwasher tablets. **1**

(ii) Hydrogen peroxide decomposes to form water and oxygen.

$$2H_2O_2(\ell) \rightarrow 2H_2O(\ell) + O_2(g)$$

A dishwasher tablet produces 0·051 g of hydrogen peroxide (mass of one mole = 34 g).

Calculate the volume of oxygen that would be produced when 0·051 g of hydrogen peroxide decomposes. **3**

Take the volume of 1 mole of oxygen gas to be 24 litres.

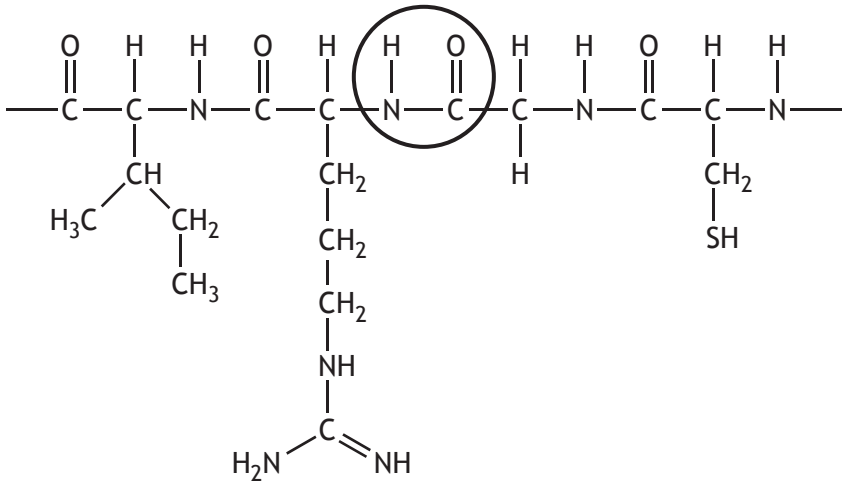
(c) Enzymes are commonly added to dishwasher tablets. These are used to break down proteins.

(i) The proteins are broken down into small, water-soluble molecules.

Name the small, water-soluble molecules made when proteins are broken down completely. **1**

9. (c) (continued)

(ii) The structure of a section of protein chain found in egg white is shown.



(A) Name the functional group circled. **1**

(B) Draw a structural formula for **one** of the molecules that would be made when this section of egg white protein chain is completely broken down. **1**

(iii) As part of the program in the dishwasher, the conditions in the dishwasher change so that the enzyme molecules no longer work because they change shape.

(A) State the term used to describe the change in shape of enzyme molecules. **1**

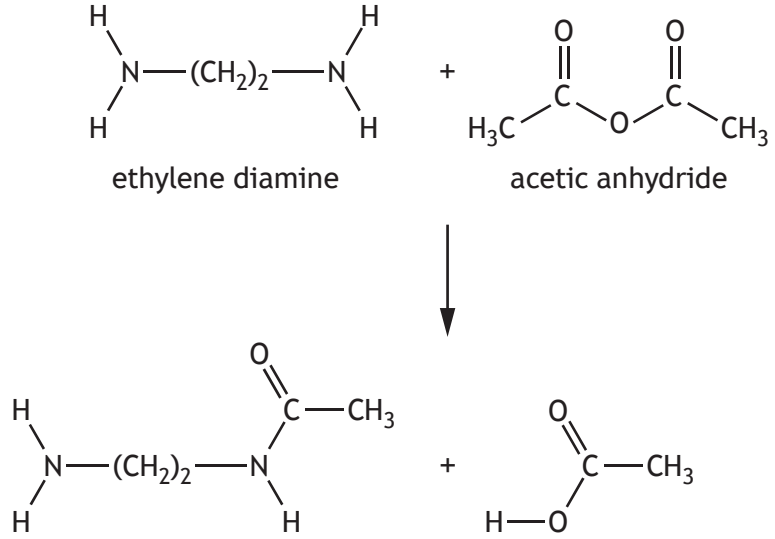
(B) Suggest a change in conditions which would cause the enzyme molecules to change shape. **1**

9. (continued)

(d) A bleach activator is frequently added to dishwasher tablets to speed up the bleaching reaction. One common bleach activator is TAED.

TAED could be produced in a process which involves a number of stages.

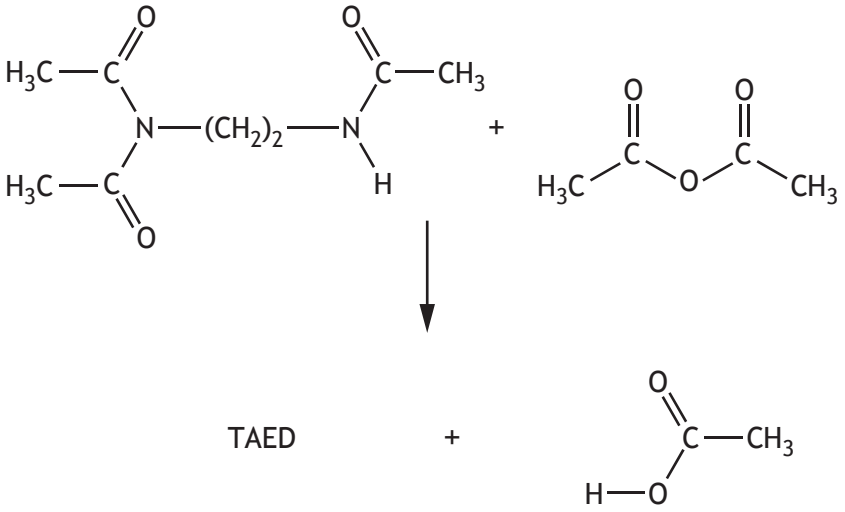
(i) The first stage in producing TAED is shown below.



Suggest a name for this type of reaction. **1**

9. (d) (continued)

(ii) The final stage in the process producing TAED is shown below.



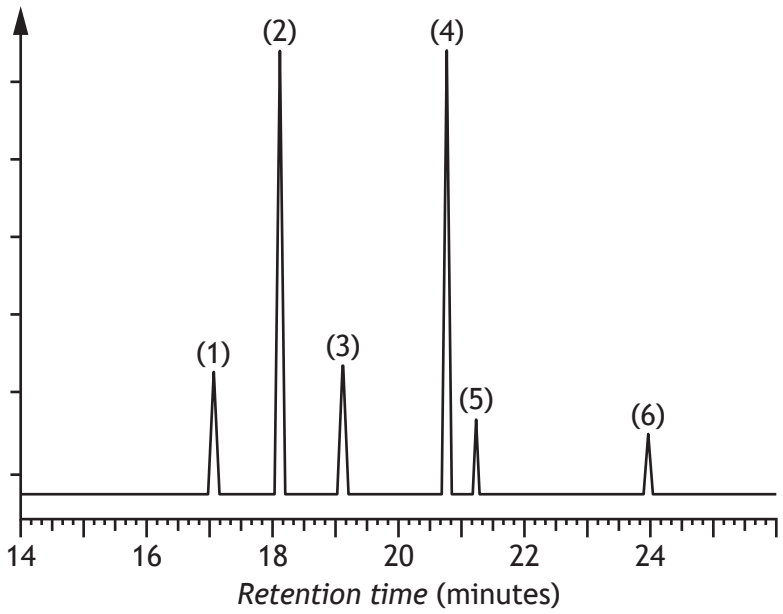
Draw a structural formula for TAED. **1**

10. Essential oils from the lavender plant are used in aromatherapy.

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(a) Gas chromatography can be used to separate and identify the organic compounds in lavender oils.

Chromatogram 1 - Lavender oil **A**



Peak	*Component*	*Component peak area*
1	1,8-cineole	7432
2	linalool	31 909
3	camphor	7518
4	linalyl acetate	27 504
5	geranyl acetate	3585
6	farnesene	1362

Total peak area = 79 310

The relative concentration of each component can be calculated using the following formula.

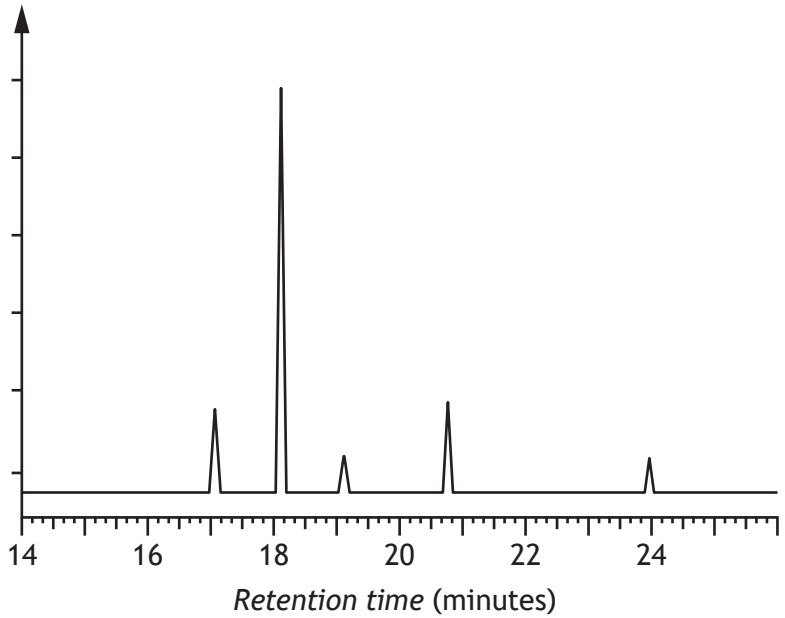
$$\text{Relative concentration} = \frac{\text{Component peak area}}{\text{Total peak area}} \times 100\ (\%)$$

(i) Calculate the relative concentration of linalool in lavender oil **A**. **1**

10. (a) (continued)

(ii) Different varieties of lavender oils have different compositions.

Chromatogram 2 – Lavender oil **B**



Identify the component found in lavender oil **A** that is missing from lavender oil **B**. **1**

(b) A brand of mouthwash contains the component 1,8-cineole at a concentration of 0·92 mg per cm^3. The cost of 1 kg of 1,8-cineole is £59·10.

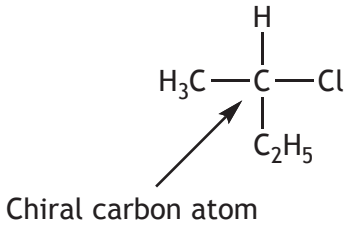
Calculate the cost, in pence, of 1,8-cineole that is present in a 500 cm^3 bottle of this brand of mouthwash. **2**

10. (continued)

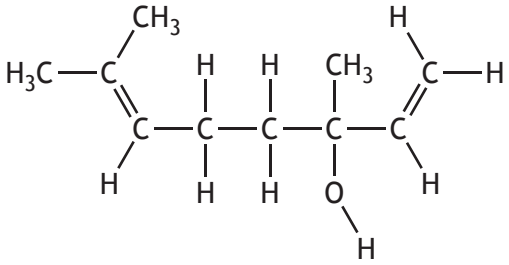
(c) The component molecules found in lavender oils are terpenes or terpenoids.

(i) A chiral carbon is a carbon atom attached to **four** different atoms or groups of atoms.

An example is shown below.



A molecule of the terpenoid linalool has a chiral carbon. Linalool has the following structure.



Circle the chiral carbon atom in the linalool structure. **1**

(An additional diagram, if required, can be found on *Page 36*)

(ii) Farnesene is a terpene consisting of **three** isoprene units (2-methylbuta-1,3-diene) joined together.

Write the molecular formula of farnesene. **1**

[END OF QUESTION PAPER]

National
Qualifications
2016

X713/76/02

Chemistry
Section 1 — Questions

WEDNESDAY, 18 MAY

9:00 AM – 11:30 AM

Instructions for the completion of Section 1 are given on *Page 02* of your question and answer booklet X713/76/01.

Record your answers on the answer grid on *Page 03* of your question and answer booklet.

Reference may be made to the Chemistry Higher and Advanced Higher Data Booklet.

Before leaving the examination room you must give your question and answer booklet to the Invigilator; if you do not you may lose all the marks for this paper.

SQA ©

SECTION 1 — 20 marks

Attempt ALL questions

1. Particles with the same electron arrangement are said to be isoelectronic.

Which of the following compounds contains ions which are isoelectronic?

A Na_2S

B $MgCl_2$

C KBr

D $CaCl_2$

2. Which line in the table is correct for the polar covalent bond in hydrogen chloride?

	Relative position of bonding electrons	*Dipole notation*
A	H —:— Cl	$\overset{\delta+}{H}—\overset{\delta-}{Cl}$
B	H :— Cl	$\overset{\delta+}{H}—\overset{\delta-}{Cl}$
C	H —:— Cl	$\overset{\delta-}{H}—\overset{\delta+}{Cl}$
D	H :— Cl	$\overset{\delta-}{H}—\overset{\delta+}{Cl}$

3. Which of the following compounds has the greatest ionic character?

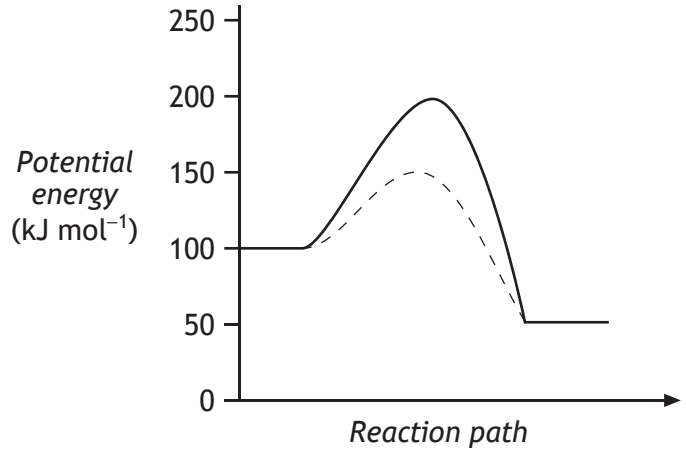
A Caesium fluoride

B Caesium iodide

C Sodium fluoride

D Sodium iodide

4. The diagram below shows the energy profiles for a reaction carried out with and without a catalyst.



What is the enthalpy change, in kJ mol^{-1}, for the catalysed reaction?

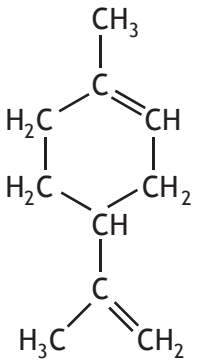
A −100

B −50

C +50

D +100

5. Limonene is a terpene molecule present in lemons.



limonene

How many isoprene units are joined together in a limonene molecule?

A 1

B 2

C 3

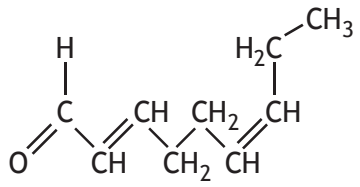
D 4

[Turn over

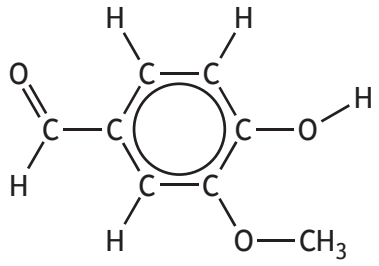
6. The following molecules give flavour to food.

Which of the following flavour molecules would be most likely to be retained in the food when the food is cooked in water?

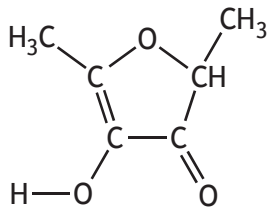
A



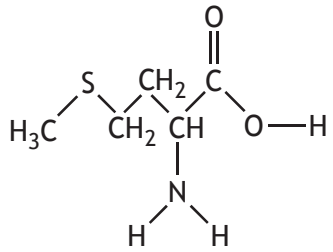
B



C



D



7. vegetable oil ⟶ vegetable fat

Which of the following reactions brings about the above change?

A Hydrolysis

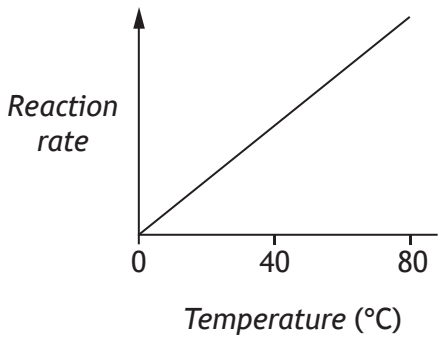
B Condensation

C Hydrogenation

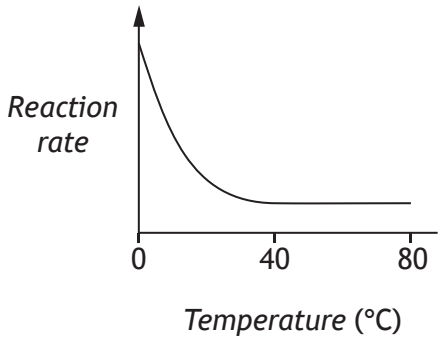
D Dehydrogenation

8. The rate of hydrolysis of protein, using an enzyme, was studied at different temperatures. Which of the following graphs would be obtained?

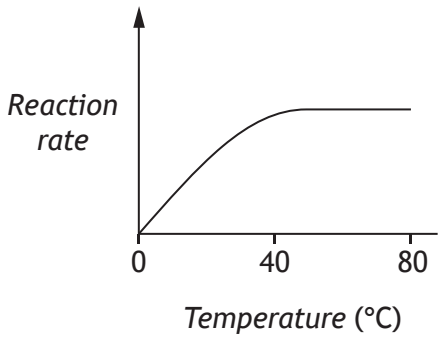
A



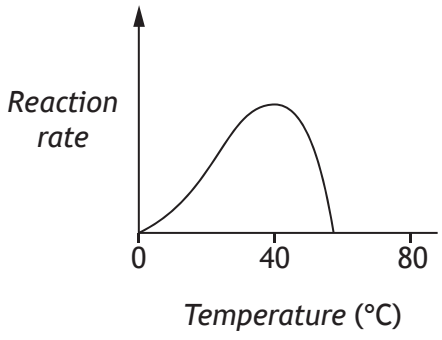
B



C



D



[Turn over

9. Which of the following is the salt of a long-chain fatty acid?

A Fat
B Oil
C Soap
D Glycerol

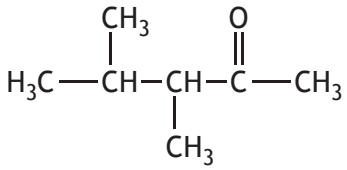
10. Emulsifiers for use in food are commonly made by reacting edible oils with

A esters
B glycerol
C fatty acids
D amino acids.

11. The equation for the reduction reaction taking place when ethanal reacts with Tollens' reagent is

A $Cu^{2+}(aq) + e^- \rightarrow Cu^+(aq)$
B $Ag^+(aq) + e^- \rightarrow Ag(s)$
C $Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^- \rightarrow 2Cr^{3+}(aq) + 7H_2O(\ell)$
D $MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightarrow Mn^{2+}(aq) + 4H_2O(\ell)$

12. The name of the compound with structure:



is

A 2,3-dimethylpentan-4-one
B 2,3-dimethylpentan-2-al
C 3,4-dimethylpentan-2-one
D 3,4-dimethylpentan-2-al.

13. $CaCO_3(s) + 2HNO_3(aq) \rightarrow Ca(NO_3)_2(aq) + CO_2(g) + H_2O(\ell)$

Mass of 1 mol = 100 g

Mass of 1 mol = 164 g

2·00 g of calcium carbonate ($CaCO_3$) was reacted with 200 cm^3 of 0·1 mol l^{-1} nitric acid (HNO_3).

Take the volume of 1 mole of carbon dioxide to be 24 litres.

In the reaction

A $CaCO_3$ is the limiting reactant

B an excess of 0·1 mol of nitric acid remains at the end of the reaction

C 1·64 g of calcium nitrate is produced by the reaction

D 480 cm^3 of carbon dioxide is produced by the reaction.

14. The mean bond enthalpy of a C – F bond is 484 kJ mol^{-1}.

In which of the processes is ΔH approximately equal to +1936 kJ mol^{-1}?

A $CF_4(g) \rightarrow C(s) + 2F_2(g)$

B $CF_4(g) \rightarrow C(g) + 4F(g)$

C $CF_4(g) \rightarrow C(g) + 2F_2(g)$

D $CF_4(g) \rightarrow C(s) + 4F(g)$

15. In a reversible reaction, equilibrium is reached when

A molecules of reactants cease to change into molecules of products

B the concentrations of reactants and products are equal

C the concentrations of reactants and products are constant

D the activation energy of the forward reaction is equal to that of the reverse reaction.

16. Which of the following equations represents the enthalpy of combustion of propane?

A $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(\ell)$

B $C_3H_8(g) + \frac{7}{2}O_2(g) \rightarrow 3CO(g) + 4H_2O(\ell)$

C $C_3H_8(g) + 3O_2(g) \rightarrow 3CO_2(g) + 4H_2(g)$

D $C_3H_8(g) + \frac{3}{2}O_2(g) \rightarrow 3CO(g) + 4H_2(g)$

[Turn over

17. An oxidising agent

A gains electrons and is oxidised

B loses electrons and is oxidised

C gains electrons and is reduced

D loses electrons and is reduced.

18. During a redox process in acid solution, chlorate ions, $ClO_3^-(aq)$, are converted into chlorine, $Cl_2(g)$.

$$ClO_3^-(aq) \rightarrow Cl_2(g)$$

The numbers of $H^+(aq)$ and $H_2O(\ell)$ required to balance the ion-electron equation for the formation of 1 mol of $Cl_2(g)$ are, respectively

A 3 and 6

B 6 and 3

C 6 and 12

D 12 and 6.

19. Which of the following ions could be used to oxidise iodide ions to iodine?

$$2I^-(aq) \rightarrow I_2(s) + 2e^-$$

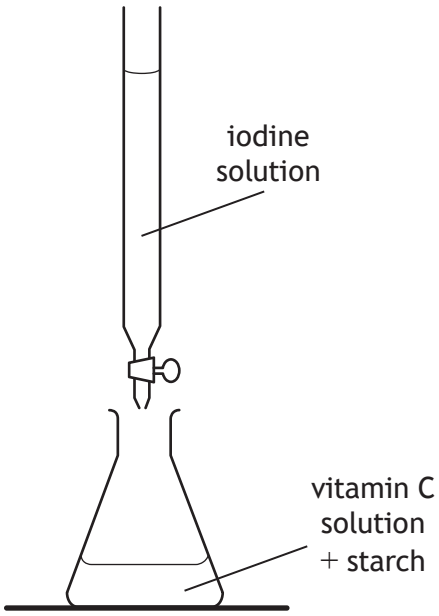
A $SO_4^{2-}(aq)$

B $SO_3^{2-}(aq)$

C $Cr^{3+}(aq)$

D $Cr_2O_7^{2-}(aq)$

20.



A student was carrying out a titration to establish the concentration of vitamin C using iodine solution.

Which of the following would help the student achieve a precise end-point?

A Placing a white tile underneath the conical flask

B Using the bottom of the meniscus when reading the burette

C Repeating titrations

D Carrying out a rough titration first

[END OF SECTION 1. NOW ATTEMPT THE QUESTIONS IN SECTION 2 OF YOUR QUESTION AND ANSWER BOOKLET.]

FOR OFFICIAL USE

National Qualifications 2016

Mark

X713/76/01

Chemistry
Section 1 — Answer Grid and Section 2

WEDNESDAY, 18 MAY
9:00 AM – 11:30 AM

X7137601

Fill in these boxes and read what is printed below.

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth
Day Month Year

Scottish candidate number

Total marks — 100

SECTION 1 — 20 marks

Attempt ALL questions.

Instructions for completion of Section 1 are given on *Page 02*.

SECTION 2 — 80 marks

Attempt ALL questions

Reference may be made to the Chemistry Higher and Advanced Higher Data Booklet.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers and rough work is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting. Any rough work must be written in this booklet. You should score through your rough work when you have written your final copy.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not you may lose all the marks for this paper.

SQA ©

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SECTION 2 — 80 marks

Attempt ALL questions

1. Hydrogen peroxide gradually decomposes into water and oxygen, according to the following equation.

$$2H_2O_2(aq) \rightarrow 2H_2O(\ell) + O_2(g)$$

(a) At room temperature, the reaction is very slow. It can be speeded up by heating the reaction mixture.

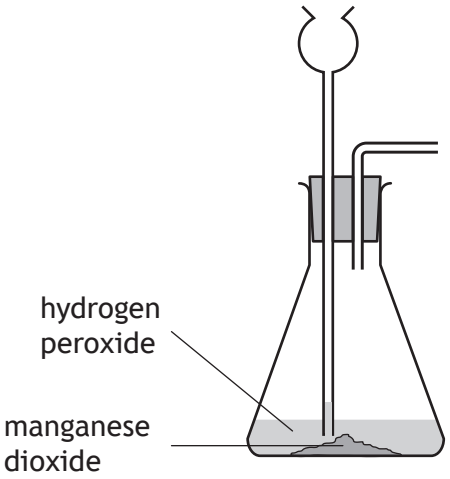
State why increasing the temperature causes an increase in reaction rate. **1**

(b) (i) The reaction can also be speeded up by adding a catalyst, such as manganese dioxide.

To determine the rate of the reaction, the volume of gas produced in a given time can be measured.

Complete the diagram below to show how the gas produced can be collected and measured. **1**

(An additional diagram, if required, can be found on *Page 38*).



1. (b) (continued)

(ii) The concentration of hydrogen peroxide is often described as a volume strength. This relates to the volume of oxygen that can be produced from a hydrogen peroxide solution.

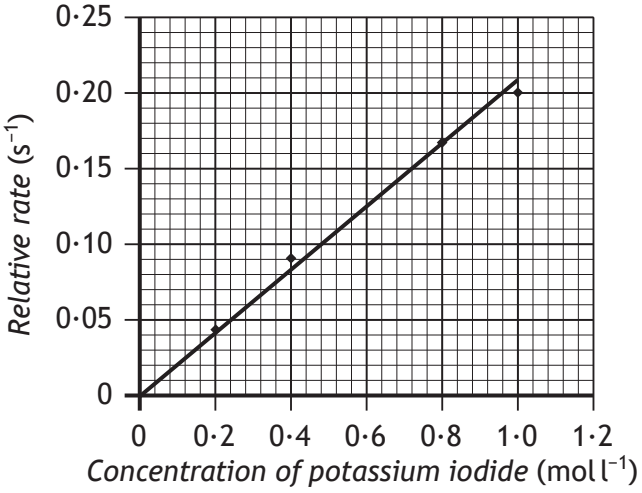
volume of oxygen produced	=	volume strength	×	volume of hydrogen peroxide solution

In an experiment, 74 cm^3 of oxygen was produced from 20 cm^3 of hydrogen peroxide solution.

Calculate the volume strength of the hydrogen peroxide. **1**

(c) Hydrogen peroxide can react with potassium iodide to produce water and iodine.

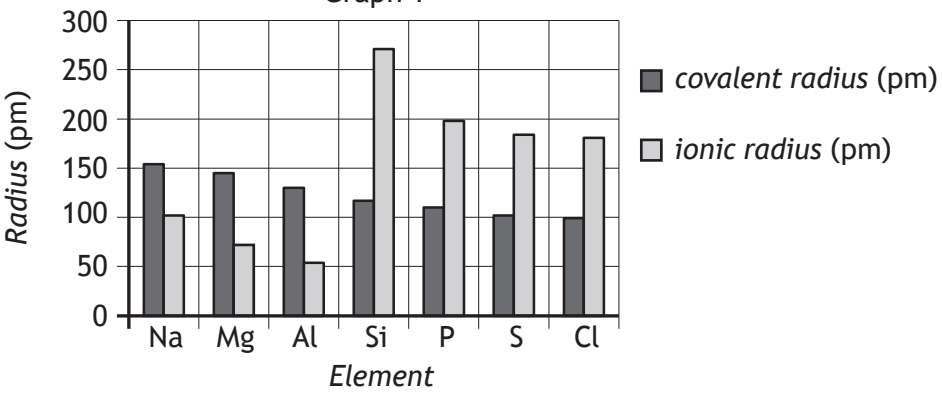
A student carried out an experiment to investigate the effect of changing the concentration of potassium iodide on reaction rate. The results are shown below.



Calculate the time taken, in s, for the reaction when the concentration of potassium iodide used was 0·6 mol l^{-1}. **1**

2. (a) Graph 1 shows the sizes of atoms and ions for elements in the third period of the Periodic Table.

Graph 1



The covalent radius is a measure of the size of an atom.

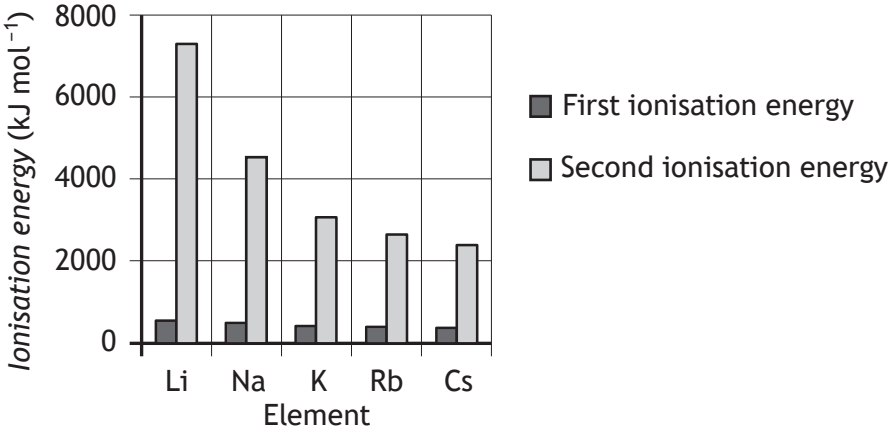
(i) Explain why covalent radius decreases across the period from sodium to chlorine. **1**

(ii) Explain **fully** why the covalent radius of sodium is larger than the ionic radius of sodium. **2**

2. (continued)

(b) Graph 2 shows the first and second ionisation energies of elements in Group 1 of the Periodic Table.

Graph 2



(i) Explain why the first ionisation energy decreases going down Group 1. **1**

(ii) Explain **fully** why the second ionisation energy is much greater than the first ionisation energy for Group 1 elements. **2**

2. (continued)

(c) The lattice enthalpy is the energy needed to completely separate the ions in one mole of an ionic solid.

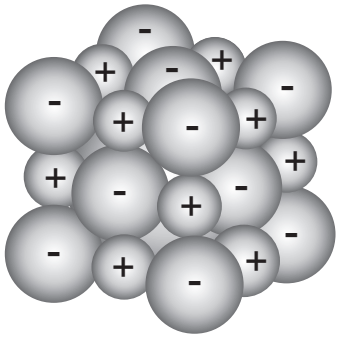


Table 1 shows the size of selected ions.

Table 1

Ion	Li^+	Na^+	K^+	Rb^+	F^-	Cl^-
Ionic radius (pm)	76	102	138	152	133	181

Table 2 shows the lattice enthalpies, in $kJ\,mol^{-1}$, for some Group 1 halides.

Table 2

Ions	F^-	Cl^-
Li^+	1030	834
Na^+	910	769
K^+	808	701
Rb^+		658

(i) Predict the lattice enthalpy, in $kJ\,mol^{-1}$, for rubidium fluoride. **1**

(ii) Write a general statement linking lattice enthalpy to ionic radii. **1**

3. Phosphine (PH_3) is used as an insecticide in the storage of grain.

Phosphine can be produced by the reaction of water with aluminium phosphide

$$AlP(s) + 3H_2O(\ell) \longrightarrow PH_3(g) + Al(OH)_3(aq)$$

(a) State the type of bonding and structure in phosphine. **1**

(b) 2·9 kg of aluminium phosphide were used in a phosphine generator.

Calculate the volume of phosphine gas, in litres, that would have been produced.

(Take the volume of 1 mole of phosphine to be 24 litres). **2**

(c) Carbon dioxide is fed into the phosphine generator to keep the phosphine concentration less than 2·6%. Above this level phosphine can ignite due to the presence of diphosphane, $P_2H_4(g)$, as an impurity.

Draw a structural formula for diphosphane. **1**

4. The viscosity of alcohols depends on a number of factors:

- the strength of intermolecular forces
- the size of the molecule
- temperature

These factors can be investigated using alcohols and apparatus from the lists below.

Alcohols	*Apparatus*
methanol	beakers
ethanol	funnels
propan-1-ol	burettes
ethane-1,2-diol	measuring cylinders
butan-1-ol	plastic syringes
propane-1,3-diol	glass tubing
pentan-1-ol	stoppers
propane-1,2,3-triol	timer
	metre stick
	ball bearing
	clamp stands
	kettle
	thermometer

Using your knowledge of chemistry, identify the alcohols and apparatus that you would select and describe how these could be used to investigate one, or more, of the factors affecting the viscosity of alcohols. **3**

5. When fats and oils are hydrolysed, mixtures of fatty acids are obtained.

(a) Name the other product obtained in this reaction. **1**

(b) The table below shows the percentage composition of the fatty acid mixtures obtained by hydrolysis of coconut oil and olive oil.

Class of fatty acids produced on hydrolysis	*Name of oil*	
	Coconut oil	Olive oil
Saturated	91	14
Monounsaturated	6	72
Polyunsaturated	3	14

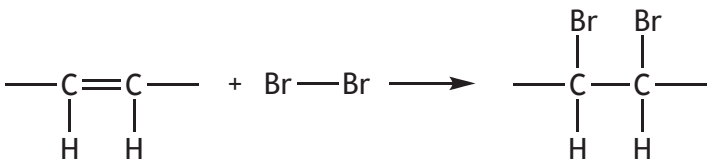
(i) One of the fatty acids produced by the hydrolysis of olive oil is linoleic acid, $C_{17}H_{31}COOH$.

State the class of fatty acid to which linoleic acid belongs. **1**

(ii) Hydrolysed coconut oil contains the fatty acid, caprylic acid, with the formula $CH_3(CH_2)_6COOH$.

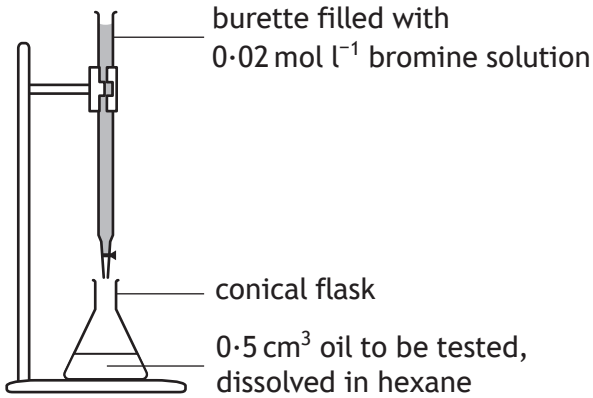
State the systematic name for caprylic acid. **1**

(c) The degree of unsaturation of oil can be tested by adding drops of bromine solution to the oil. Bromine adds across carbon to carbon double bonds in the fatty acid chains.



5. (c) (continued)

The following apparatus can be used to compare the degree of unsaturation of different oils.



(i) Describe how this apparatus could be used to show that olive oil has a greater degree of unsaturation than coconut oil. **2**

(ii) Suggest why hexane is used as the solvent, rather than water. **1**

(iii) Coconut oil has a melting point of 25 °C. Olive oil has a melting point of −6 °C.

Give two reasons why coconut oil has a higher melting point than olive oil. **2**

6. Peptide molecules can be classified according to the number of amino acid units joined by peptide bonds in the molecule.

Type of peptide	*Example of amino acid sequence*
dipeptide	aspartic acid-phenylalanine
tripeptide	isoleucine-proline-proline
tetrapeptide	lysine-proline-proline-arginine
pentapeptide	serine-glycine-tyrosine-alanine-leucine
	alanine-glycine-valine-proline-tyrosine-serine
polypeptide	many amino acids

(a) Complete the table to identify the type of peptide with the following amino acid sequence

alanine-glycine-valine-proline-tyrosine-serine **1**

(b) Partial hydrolysis of another pentapeptide molecule gave a mixture of three smaller peptide molecules with the following amino acid sequences.

leucine-glycine-valine

isoleucine-leucine

glycine-valine-serine

Write the amino acid sequence for the original pentapeptide molecule. **1**

..... – ..... – ..... – ..... – .....

(c) Some amino acids needed to form polypeptides cannot be produced in the human body.

State the term used to describe amino acids that the body cannot make. **1**

6. (continued)

(d) Paper chromatography is often used to analyse the mixtures of amino acids produced when peptides are broken down.

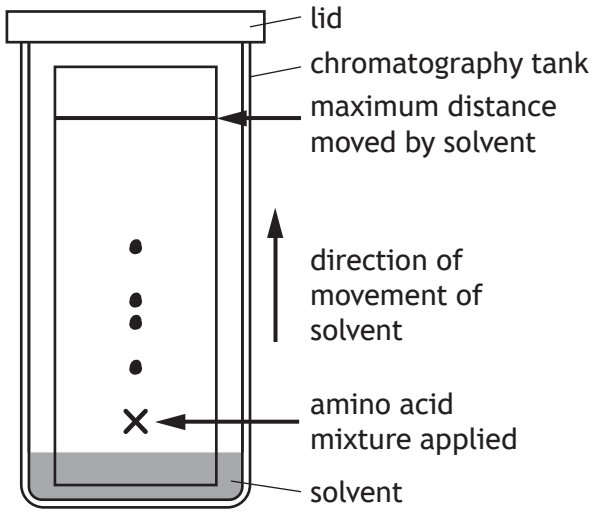
On a chromatogram, the retention factor R_f, for a substance can be a useful method of identifying the substance.

$$R_f = \frac{\text{distance moved by the substance}}{\text{maximum distance moved by the solvent}}$$

The structure of the pentapeptide methionine enkephalin was investigated.

A sample of the pentapeptide was completely hydrolysed into its constituent amino acids and this amino acid mixture was applied to a piece of chromatography paper and placed in a solvent.

The chromatogram obtained is shown below.



(i) Suggest why only four spots were obtained on the chromatogram of the hydrolysed pentapeptide. **1**

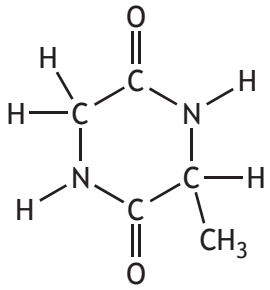
(ii) It is known that this amino acid mixture contains the amino acid methionine. The R_f value for methionine in this solvent is 0·40.

Draw a circle around the spot on the chromatogram that corresponds to methionine. **1**

6. (continued)

(e) Over the last decade several families of extremely stable peptide molecules have been discovered, where the peptide chain forms a ring.

(i) A simple cyclic dipeptide is shown.



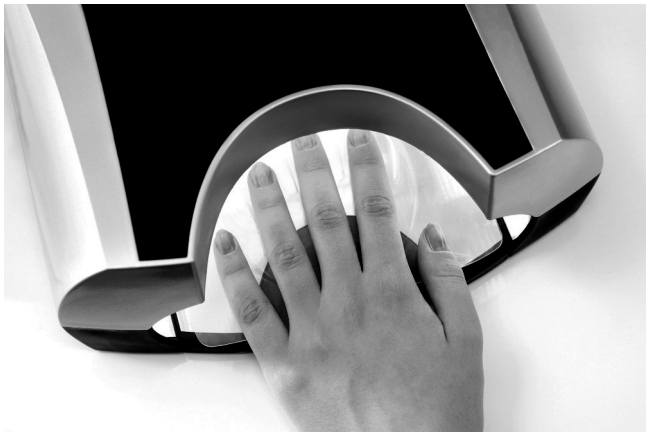
Draw a structural formula for one of the amino acids that would be formed on complete hydrolysis of the above cyclic dipeptide. **1**

(ii) Alpha-amanitin is a highly toxic cyclic peptide found in death cap mushrooms. The lethal dose for humans is 100 mg per kg of body mass.

1·0 g of death cap mushrooms contains 250 mg of alpha-amanitin.

Calculate the minimum mass of death cap mushrooms that would contain the lethal dose for a 75 kg adult. **2**

7. Modern shellac nail varnishes are more durable and so last longer than traditional nail polish.



The shellac nail varnish is applied in thin layers to the nails and then the fingers are placed under a UV lamp.

(a) The Skin Care Foundation has recommended that a sun-block is applied to the fingers and hand before using the lamp.

Suggest why the Skin Care Foundation makes this recommendation. **1**

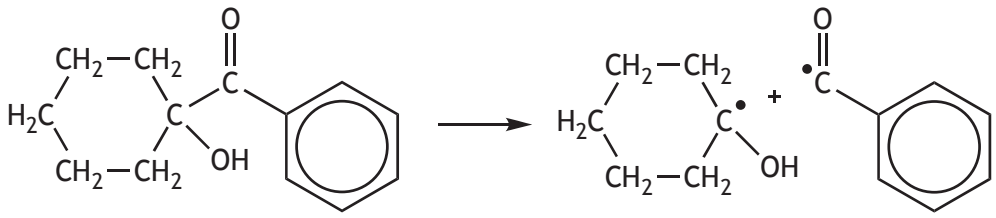
(b) A *free radical* chain reaction takes place and the varnish hardens.

(i) State what is meant by the term *free radical*. **1**

7. (b) (continued)

(ii) The shellac nail varnish contains a mixture of ingredients that take part in the free radical chain reaction.

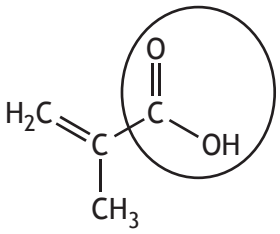
One of the steps in the free radical chain reaction is:



State the term used to describe this type of step in a free radical chain reaction. **1**

(iii) During the free radical chain reaction small molecules join to form large chain molecules.

One example of a small molecule used is



Name the functional group circled above. **1**

(iv) Alcohol wipes are used to finish the varnishing treatment. Alcohol wipes contain the alcohol propan-2-ol.

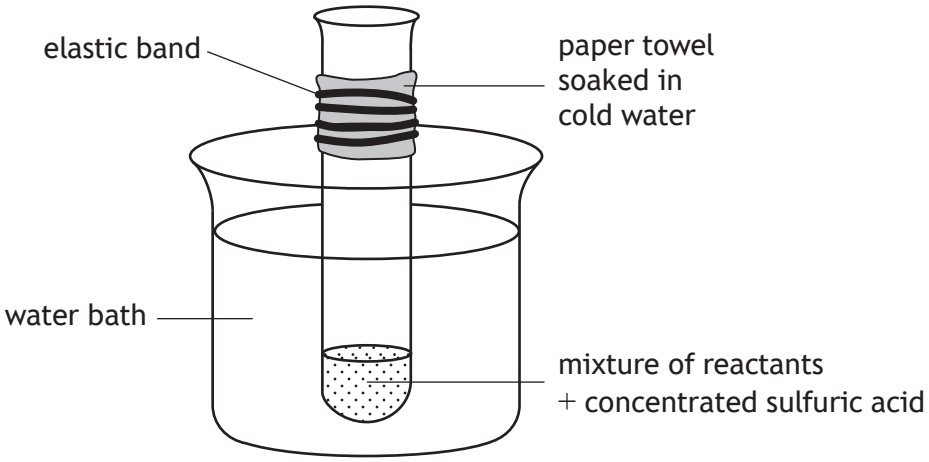
State why propan-2-ol can be described as a secondary alcohol. **1**

7. (continued)

(c) Traditional nail varnishes use ethyl ethanoate and butyl ethanoate as solvents.

(i) Draw a structural formula for butyl ethanoate. **1**

(ii) Ethyl ethanoate can be made in the laboratory using the following apparatus.



Suggest why a wet paper towel is wrapped around the test tube. **1**

7. (c) (continued)

(iii) A student used 2·5 g of ethanol and a slight excess of ethanoic acid to produce 2·9 g of ethyl ethanoate.

ethanol	+	ethanoic acid	⇌	ethyl ethanoate	+	water
mass of one mole = 46·0 g				mass of one mole = 88·0 g		

(One mole of ethanol reacts with one mole of ethanoic acid to produce one mole of ethyl ethanoate.)

Calculate the percentage yield of ethyl ethanoate. **2**

(iv) Name the type of reaction that takes place during the formation of ethyl ethanoate. **1**

[Turn over

8. Methanol (CH_3OH) is an important chemical in industry.

(a) Methanol is produced from methane in a two-step process.

In step 1, methane is reacted with steam as shown.

Step 1: $CH_4(g) + H_2O(g) \rightleftharpoons 3H_2(g) + CO(g) \quad \Delta H = +210\,kJ\,mol^{-1}$

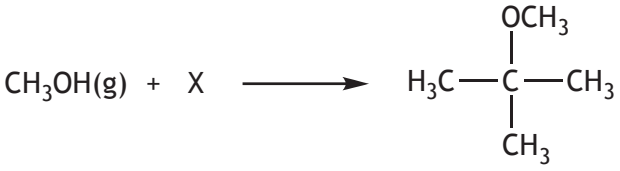
In step 2, hydrogen reacts with carbon monoxide.

Step 2: $2H_2(g) + CO(g) \rightleftharpoons CH_3OH(g) \quad \Delta H = -91\,kJ\,mol^{-1}$

Complete the table to show the most favourable conditions to maximise the yield for each step. **2**

	Temperature (High/Low)	*Pressure* (High/Low)
Step 1		
Step 2		

(b) Methanol reacts with compound X, in an addition reaction, to form methyl tertiary-butyl ether, an additive for petrol.



methyl tertiary-butyl ether

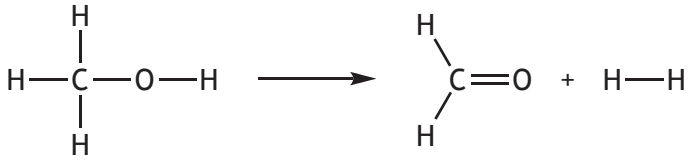
(i) Suggest a structure for compound X. **1**

(ii) The atom economy of this reaction is 100%.

Explain what this means. **1**

8. (continued)

(c) Methanol can be converted to methanal as shown.



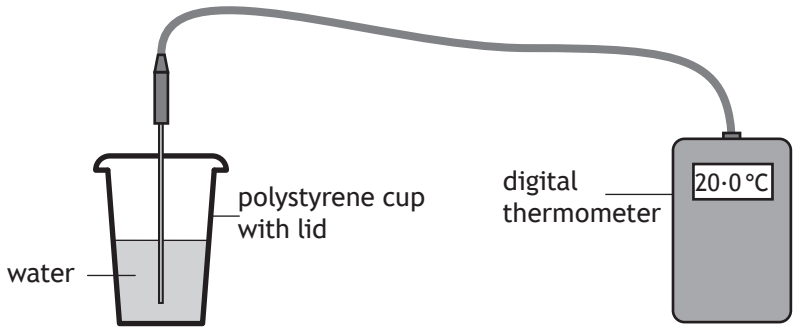
Using bond enthalpy and mean bond enthalpy values from the data booklet, calculate the enthalpy change, in kJ mol^{-1}, for the reaction. **2**

[Turn over

DO NOT WRITE IN THIS MARGIN

9. A group of students carried out an investigation into the energy changes that take place when metal hydroxides dissolve in water.

The following apparatus was used as a simple calorimeter to determine the change in temperature.



The experiment was carried out as follows.

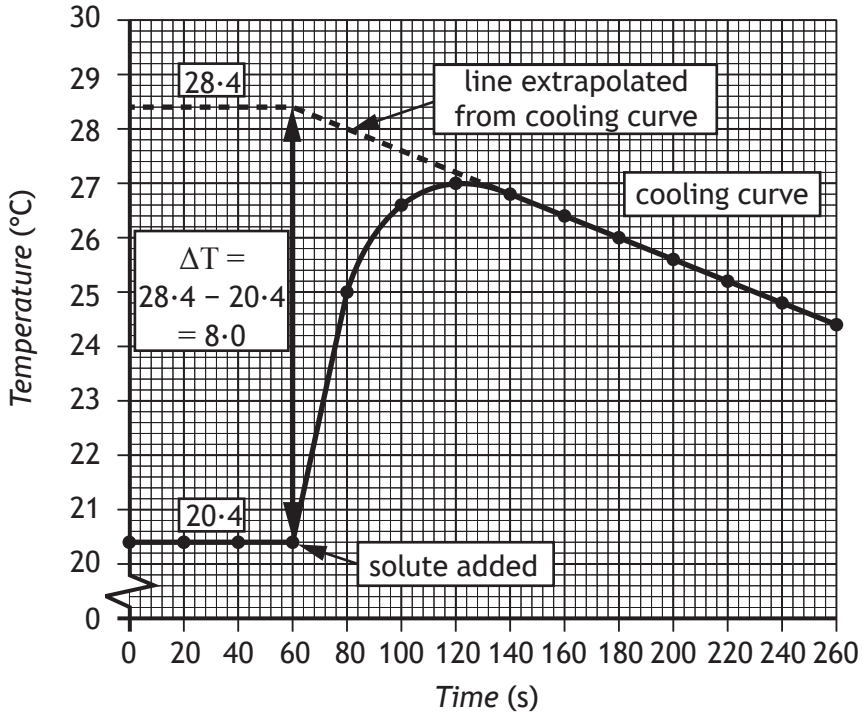
Step 1: 100 cm^3 of deionised water was added to the cup.

Step 2: The stop-clock was started, the water stirred continuously and the temperature recorded every 20 seconds.

Step 3: After 60 seconds, an accurately weighed mass of the metal hydroxide was added to the water and the temperature recorded every 20 seconds.

Graph 1 shows the group's results for lithium hydroxide.

Graph 1 – Lithium hydroxide



The heat energy transferred to the water can be calculated as shown.

$$E_h = cm\Delta T$$
$$= 4{\cdot}18 \times 0{\cdot}10 \times 8{\cdot}0$$
$$= 3{\cdot}3\ \text{kJ}$$

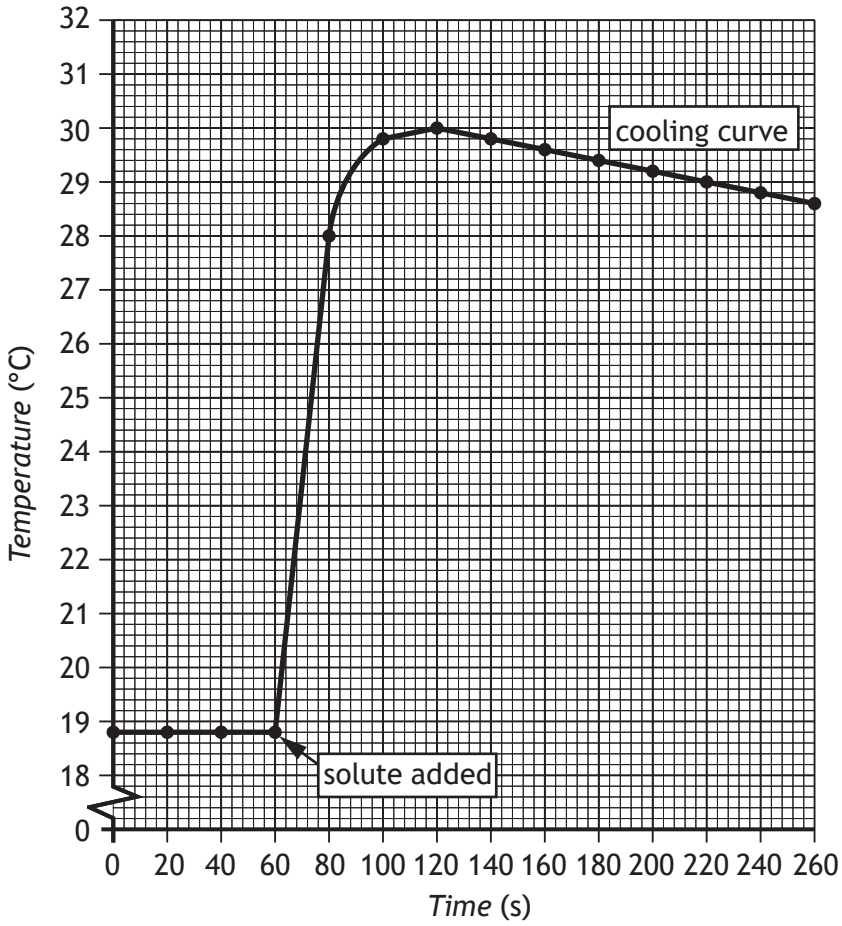
9. (continued)

MARKS DO NOT WRITE IN THIS MARGIN

(a) The experiment was repeated using sodium hydroxide.

Graph 2 shows the results of this experiment.

Graph 2 – Sodium Hydroxide



(i) Using Graph 2 calculate the heat energy transferred to the water, in kJ, when the sodium hydroxide dissolved. **2**

(ii) Suggest why the experiment was carried out in a polystyrene cup with a lid. **1**

9. (a) (continued)

(iii) **In another experiment** the students found that 5·61 g of potassium hydroxide (KOH) released 5·25 kJ of heat energy on dissolving.

Use this information to calculate the energy released, in kJ mol^{-1}, when one mole of potassium hydroxide dissolves in water. **1**

(b) Calcium hydroxide solution can be formed by adding calcium metal to excess water.

Solid calcium hydroxide would form if the exact molar ratio of calcium to water is used. The equation for the reaction is

$$Ca(s) + 2H_2O(\ell) \rightarrow Ca(OH)_2(s) + H_2(g)$$

Calculate the enthalpy change, in kJ mol^{-1}, for the reaction above by using the data shown below.

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(\ell)$ $\quad \Delta H = -286$ kJ mol^{-1}

$Ca(s) + O_2(g) + H_2(g) \rightarrow Ca(OH)_2(s)$ $\quad \Delta H = -986$ kJ mol^{-1} **2**

10. The chemical industry creates an immense variety of products which impact on virtually every aspect of our lives. Industrial scientists, including chemical engineers, production chemists and environmental chemists, carry out different roles to maximise the efficiency of industrial processes.

Using your knowledge of chemistry, comment on what industrial scientists can do to maximise profit from industrial processes and minimise impact on the environment. **3**

11. Soft drinks contain a variety of sugars. A student investigated the sugar content of a soft drink.

(a) The density of the soft drink can be used to estimate its total sugar concentration. Solutions of different sugars, with the same concentration, have similar densities.

The first experiment was to determine the total sugar concentration of the soft drink by comparing the density of the drink with the density of standard sucrose solutions.

(i) This firstly involved producing standard sucrose solutions of different concentrations.

The standard sucrose solutions were made up in volumetric flasks.

Draw a diagram of a volumetric flask. **1**

(ii) The density of each standard sucrose solution was then determined. In order to determine the density of each solution, the student accurately measured the mass of 10·0 cm^3 of each sucrose solution.

Describe **fully** a method that the student could have used to accurately measure the mass of 10·0 cm^3 of each sucrose solution. **2**

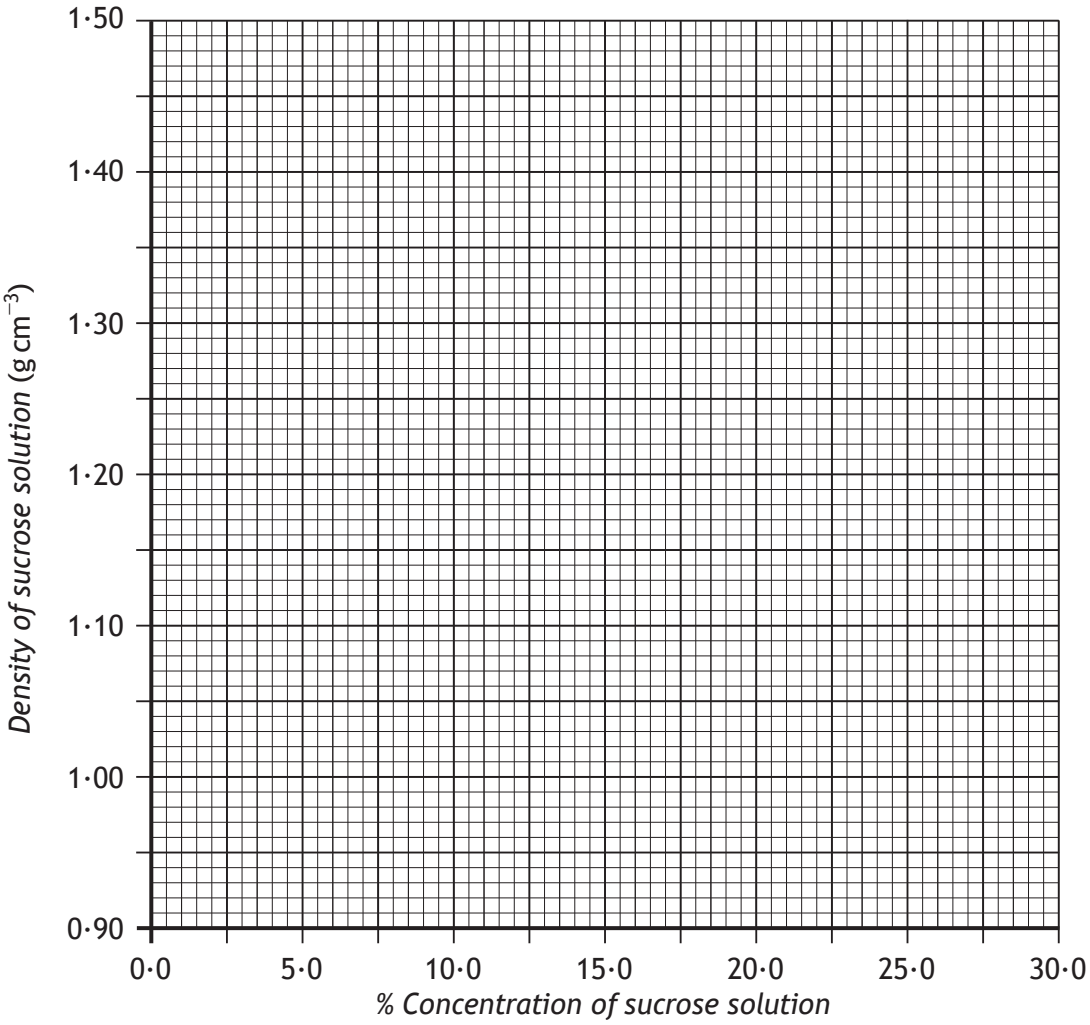
11. (a) (continued)

(iii) The results that the student obtained for the density of the standard solutions of sucrose are shown in the table.

% Concentration of sucrose solution	*Density of sucrose solution* ($g\ cm^{-3}$)
0·0	1·00
5·0	1·10
10·0	1·19
15·0	1·31
20·0	1·41

Draw a line graph using the student's results. **2**

(Additional graph paper, if required, can be found on *Page 38*.)



11. (a) (continued)

(iv) The student used the line graph to obtain the relationship between the concentration of sugars in solution and the density of the solution.

This equation shows the relationship.

$$\text{density of sugar in g cm}^{-3} = (0{\cdot}0204 \times \%\text{ concentration of sugars in solution }) + 1{\cdot}00$$

The student then determined the density of a soft drink. In order to ensure that the drink was flat, all the gas had been allowed to escape.

(A) Suggest a reason why the soft drink needed to be flat before its density was determined. **1**

(B) The soft drink tested had a density of $1{\cdot}07\text{ g cm}^{-3}$.

Using the equation, calculate the % concentration of sugars present in the soft drink. **1**

(v) A different soft drink is found to contain 10·6 grams of sugar in 100 cm^3.

Calculate the total mass of sugar present, in grams, in a 330 cm^3 can of this soft drink. **1**

(b) The second experiment in the investigation was to determine the concentration of specific types of sugar called reducing sugars. This was carried out by titration with Fehling's solution.

(i) Reducing sugars contain an aldehyde functional group.

Draw this functional group. **1**

11. (b) (continued)

(ii) The overall reaction that occurs with Fehling's solution and a reducing sugar is shown.

$$C_6H_{12}O_6 + 2Cu^{2+} + H_2O \rightarrow C_6H_{12}O_7 + 2Cu^{+} + 2H^{+}$$

reducing sugar — Fehling's solution

Write the ion-electron equation for the oxidation reaction. **1**

(iii) State the colour change that would be observed when reducing sugars are reacted with Fehling's solution. **1**

(iv) For the titrations, the student diluted the soft drink to improve the accuracy of results.

$25{\cdot}0\,cm^3$ samples of the diluted soft drink were titrated with Fehling's solution which had a Cu^{2+} concentration of $0{\cdot}0250\,mol\,l^{-1}$.

The average volume of Fehling's solution used in the titrations was $19{\cdot}8\,cm^3$.

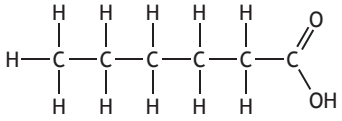
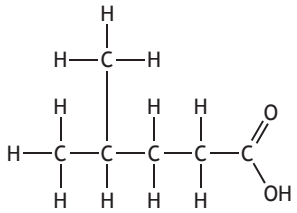
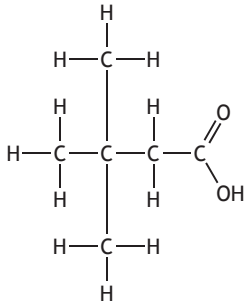
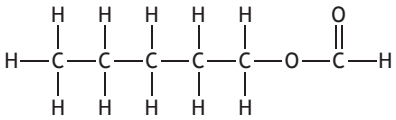
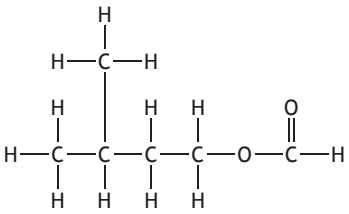
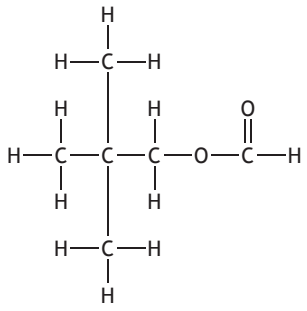
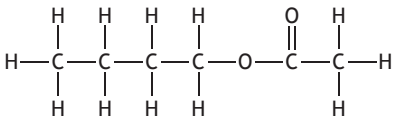
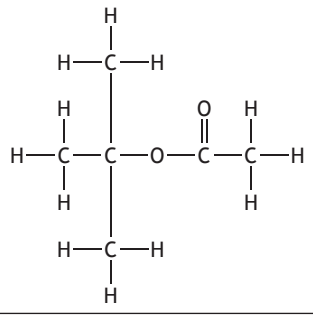
$$C_6H_{12}O_6 + 2Cu^{2+} + H_2O \rightarrow C_6H_{12}O_7 + 2Cu^{+} + 2H^{+}$$

reducing sugar — Fehling's solution

Calculate the concentration, in $mol\,l^{-1}$, of reducing sugars present in the diluted sample of the soft drink. **3**

12. (a) The table shows the boiling points and structures of some isomers with molecular formula $C_6H_{12}O_2$.

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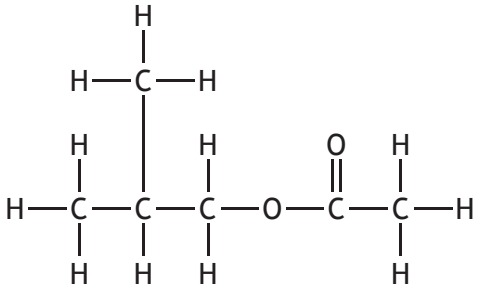
Isomer	*Structure*	*Boiling point* (°C)
1		205
2		201
3		187
4		132
5		125
6		119
7		126
8		98

12. (a) (continued)

(i) Name the intermolecular force which accounts for the higher boiling points of isomers 1, 2 and 3. **1**

(ii) Using the information in the table, describe **two** ways in which differences in structure affect the boiling points of isomeric esters 4–8. **2**

(iii) Predict the boiling point, in °C, for the isomer shown below. **1**



[Turn over

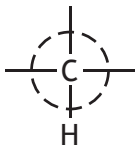
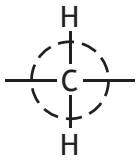
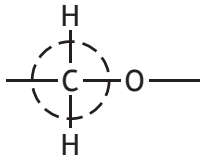
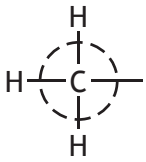
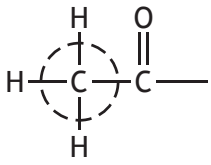
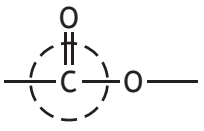
12. (continued)

(b) Carbon-13 NMR spectroscopy is a technique that can be used in chemistry to determine the structure of organic molecules such as esters.

In a carbon-13 NMR spectrum, a carbon atom in a molecule is identified by its **chemical shift**. This value depends on the other atoms bonded to the carbon atom, which is known as the "chemical environment" of the carbon-13 atom.

Carbon-13 chemical shift values are shown in the table below.

The carbon-13 atom in each chemical environment has been circled.

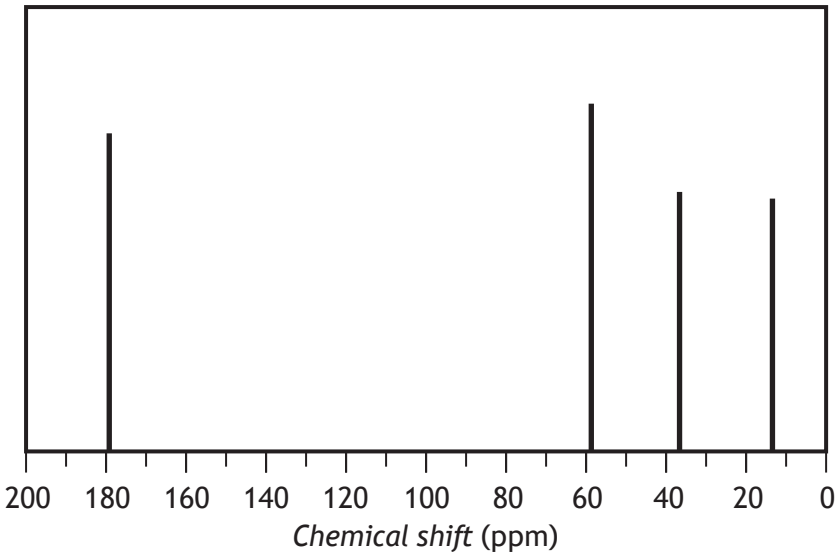
Chemical environment	*Chemical shift* (ppm)
	25–35
	16–25
	50–90
	10–15
	20–50
	170–185

The **number** of peaks in a carbon-13 NMR spectrum corresponds to the number of carbon atoms in different chemical environments within the molecule.

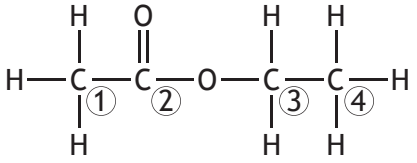
The **position** of a peak (the chemical shift) indicates the type of carbon atom.

12. (b) (continued)

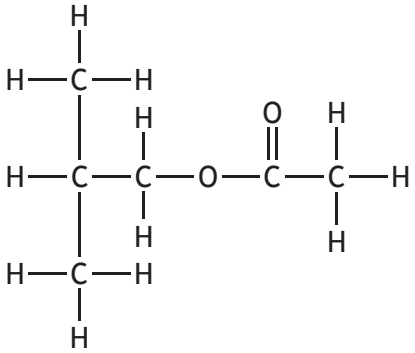
The spectrum for ethyl ethanoate is shown below.



(i) Label each peak in the ethyl ethanoate spectrum with a number to match the carbon atom in ethyl ethanoate, shown below. **1**



(ii) Determine the number of peaks that would be seen in the carbon-13 NMR spectrum for the ester shown below. **1**



Number of peaks in carbon-13 NMR spectrum

= _____

[END OF QUESTION PAPER]

National
Qualifications
2015

X713/76/02

Chemistry
Section 1 — Questions

THURSDAY, 28 MAY
1:00 PM – 3:30 PM

Instructions for the completion of Section 1 are given on *Page two* of your question and answer booklet X713/76/01.

Record your answers on the answer grid on *Page three* of your question and answer booklet.

Reference may be made to the Chemistry Higher and Advanced Higher Data Booklet.

Before leaving the examination room you must give your question and answer booklet to the Invigilator; if you do not you may lose all the marks for this paper.

SQA ©

SECTION 1 — 20 marks

Attempt ALL questions

1. The elements nitrogen, oxygen, fluorine and neon

A can form negative ions

B are made up of diatomic molecules

C have single bonds between the atoms

D are gases at room temperature.

2. Which of the following equations represents the first ionisation energy of fluorine?

A $F^-(g) \rightarrow F(g) + e^-$

B $F^-(g) \rightarrow \frac{1}{2}F_2(g) + e^-$

C $F(g) \rightarrow F^+(g) + e^-$

D $\frac{1}{2}F_2(g) \rightarrow F^+(g) + e^-$

3. Which of the following atoms has least attraction for bonding electrons?

A Carbon

B Nitrogen

C Phosphorus

D Silicon

4. Which of the following is **not** an example of a van der Waals' force?

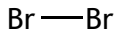
A Covalent bond

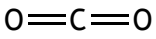
B Hydrogen bond

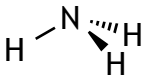
C London dispersion force

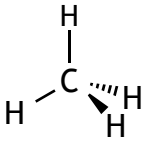
D Permanent dipole – permanent dipole attraction

5. Which of the following has more than one type of van der Waals' force operating between its molecules in the liquid state?

A 

B 

C 

D 

6. Oil molecules are more likely to react with oxygen in the air than fat molecules.

During the reaction the oil molecules

A are reduced

B become rancid

C are hydrolysed

D become unsaturated.

7. Which of the following mixtures will form when NaOH(aq) is added to a mixture of propanol and ethanoic acid?

A Propanol and sodium ethanoate

B Ethanoic acid and sodium propanoate

C Sodium hydroxide and propyl ethanoate

D Sodium hydroxide and ethyl propanoate

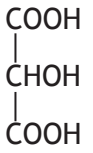
8. Oils contain carbon to carbon double bonds which can undergo addition reactions with iodine.

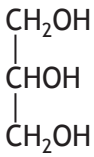
The iodine number of an oil is the mass of iodine in grams that will react with 100 g of oil.

Which line in the table shows the oil that is likely to have the lowest melting point?

	Oil	*Iodine number*
A	Corn	123
B	Linseed	179
C	Olive	81
D	Soya	130

9. When an oil is hydrolysed, which of the following molecules is always produced?

A 

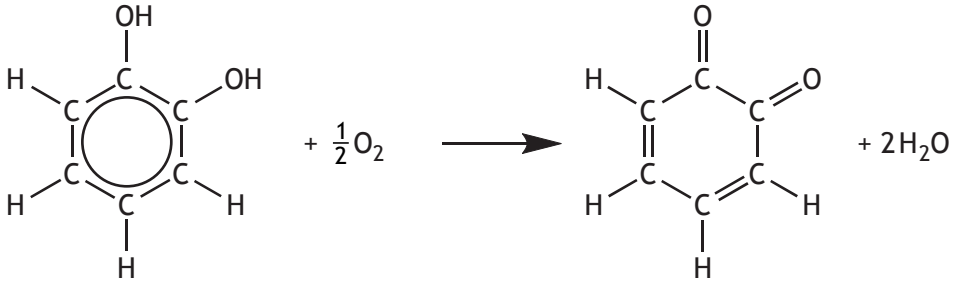
B 

C $C_{17}H_{35}COOH$

D $C_{17}H_{33}COOH$

10. Enzymes are involved in the browning of cut fruit.

One reaction taking place is:



Which of the following correctly describes the above reaction?

A Oxidation

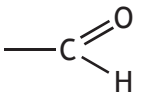
B Reduction

C Hydrolysis

D Condensation

11. Which of the following statements is correct for ketones?

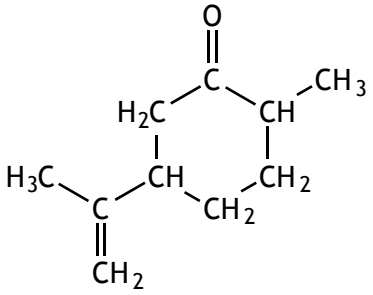
A They are formed by oxidation of tertiary alcohols.

B They contain the group  .

C They contain a carboxyl group.

D They will not react with Fehling's solution.

12. Carvone is a natural product that can be extracted from orange peel.

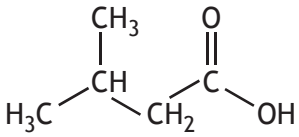


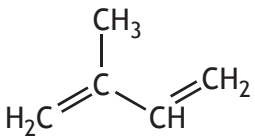
Carvone

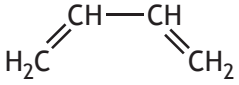
Which line in the table correctly describes the reaction of carvone with bromine solution and with acidified potassium dichromate solution?

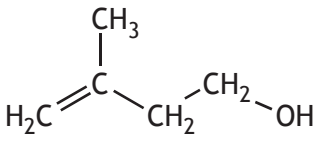
	Reaction with bromine solution	*Reaction with acidified potassium dichromate solution*
A	no reaction	no reaction
B	no reaction	orange to green
C	decolourises	orange to green
D	decolourises	no reaction

13. The structure of isoprene is

A 

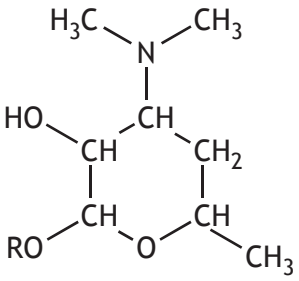
B 

C 

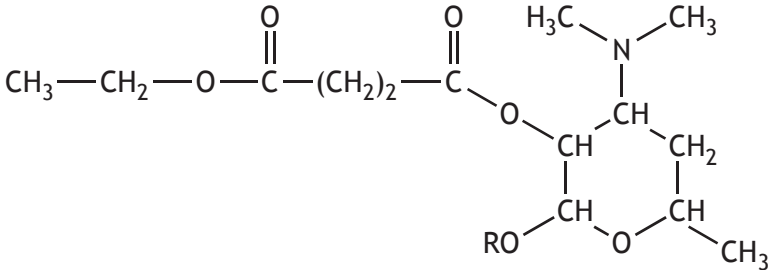
D 

[Turn over

14. The antibiotic, erythromycin, has the following structure.



To remove its bitter taste, the erythromycin is reacted to give the compound with the structure shown below.



Which of the following types of compound has been reacted with erythromycin to produce this compound?

A Alcohol

B Aldehyde

C Carboxylic acid

D Ketone

15. Which of the following is an isomer of 2,2-dimethylpentan-1-ol?

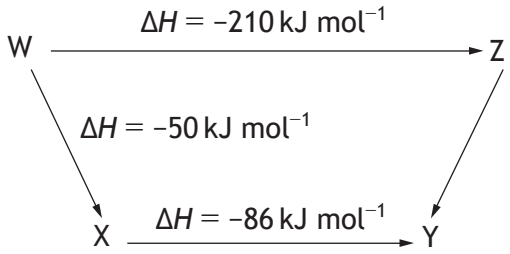
A $CH_3CH_2CH_2CH(CH_3)CH_2OH$

B $(CH_3)_3CCH(CH_3)CH_2OH$

C $CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_2OH$

D $(CH_3)_2CHC(CH_3)_2CH_2CH_2OH$

16. Consider the reaction pathway shown below.



According to Hess's Law, the ΔH value, in kJ mol^{-1}, for reaction Z to Y is

A +74

B −74

C +346

D −346.

[Turn over

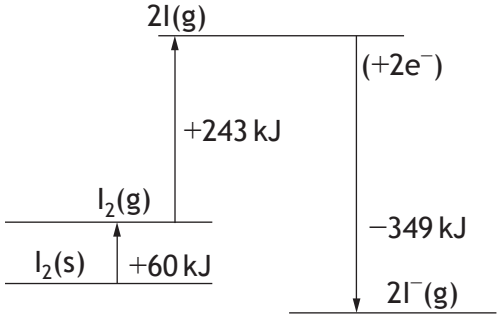
17. $I_2(s) \rightarrow I_2(g)$ $\Delta H = +60\ kJ\ mol^{-1}$

$I_2(g) \rightarrow 2I(g)$ $\Delta H = +243\ kJ\ mol^{-1}$

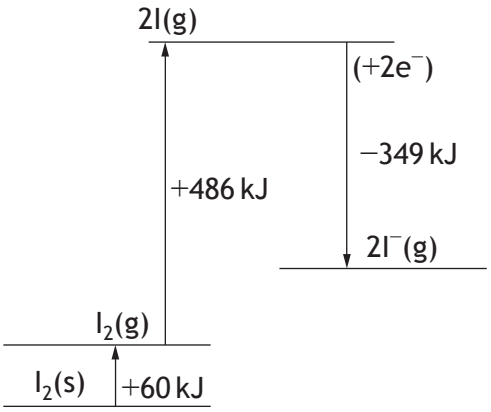
$I(g) + e^- \rightarrow I^-(g)$ $\Delta H = -349\ kJ\ mol^{-1}$

Which of the following would show the energy diagram for $I_2(s) + 2e^- \rightarrow 2I^-(g)$?

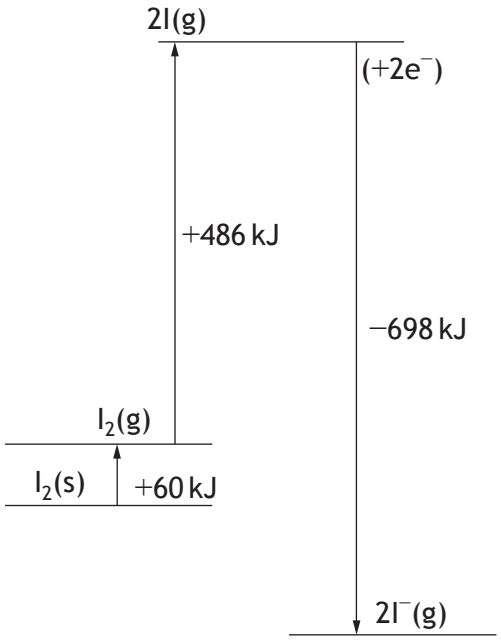
A



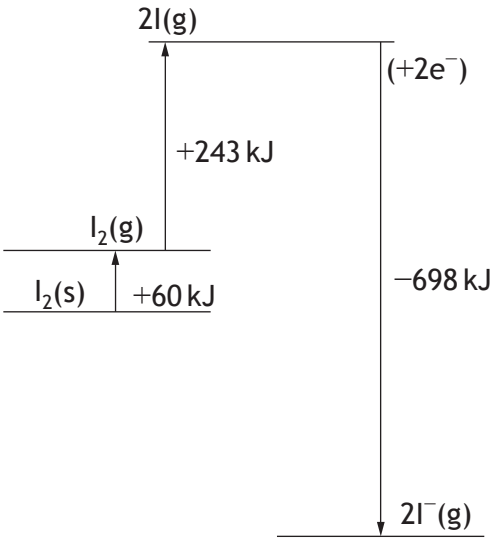
B



C



D



18. Which of the following statements regarding a chemical reaction at equilibrium is always correct?

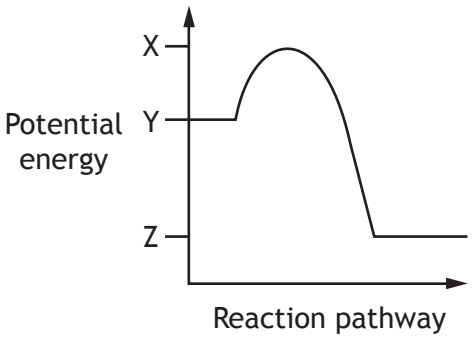
A The rates of the forward and reverse reactions are equal.

B The concentration of reactants and products are equal.

C The forward and reverse reactions have stopped.

D The addition of a catalyst changes the position of the equilibrium.

19. A reaction has the following potential energy diagram.



The activation energy for the forward reaction is

A X – Y

B Y – X

C Y – Z

D Z – Y.

20. Which of the following will react with Br_2 but **not** with I_2?

A OH^-

B SO_3^{2-}

C Fe^{2+}

D Mn^{2+}

[END OF SECTION 1. NOW ATTEMPT THE QUESTIONS IN SECTION 2 OF YOUR QUESTION AND ANSWER BOOKLET.]

FOR OFFICIAL USE

National Qualifications 2015

Mark

X713/76/01

Chemistry
Section 1 — Answer Grid and Section 2

THURSDAY, 28 MAY
1:00 PM – 3:30 PM

X7137601

Fill in these boxes and read what is printed below.

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth
Day Month Year

Scottish candidate number

Total marks — 100

SECTION 1 — 20 marks

Attempt ALL questions.

Instructions for completion of Section 1 are given on *Page two*.

SECTION 2 — 80 marks

Attempt ALL questions

Reference may be made to the Chemistry Higher and Advanced Higher Data Booklet.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers and rough work is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting. Any rough work must be written in this booklet. You should score through your rough work when you have written your final copy.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not you may lose all the marks for this paper.

SQA ©

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MARKS | DO NOT WRITE IN THIS MARGIN

SECTION 2 — 80 marks

Attempt ALL questions

1. Volcanoes produce a variety of molten substances, including sulfur and silicon dioxide.

(a) Complete the table to show the strongest type of attraction that is broken when each substance melts.

Substance	*Melting point* (°C)	*Strongest type of attraction broken when substance melts*
sulfur	113	
silicon dioxide	1610	

2

(b) Volcanic sulfur can be put to a variety of uses. One such use involves reacting sulfur with phosphorus to make a compound with formula P_4S_3.

(i) Draw a possible structure for P_4S_3. **1**

(ii) Explain why the covalent radius of sulfur is smaller than that of phosphorus. **1**

1. (b) (continued)

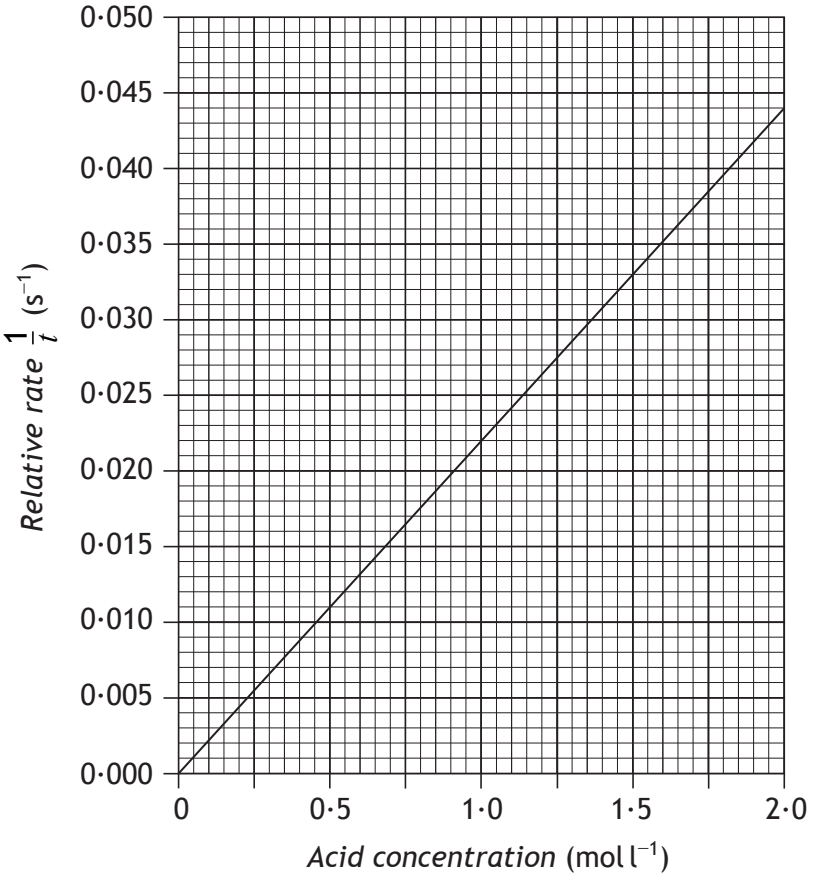
(iii) The melting point of sulfur is much higher than that of phosphorus.

Explain fully, in terms of the structures of sulfur and phosphorus molecules and the intermolecular forces between molecules of each element, why the melting point of sulfur is much higher than that of phosphorus. **3**

[Turn over

2. (a) A student investigated the effect of changing acid concentration on reaction rate. Identical strips of magnesium ribbon were dropped into different concentrations of excess hydrochloric acid and the time taken for the magnesium to completely react recorded.

A graph of the student's results is shown below.

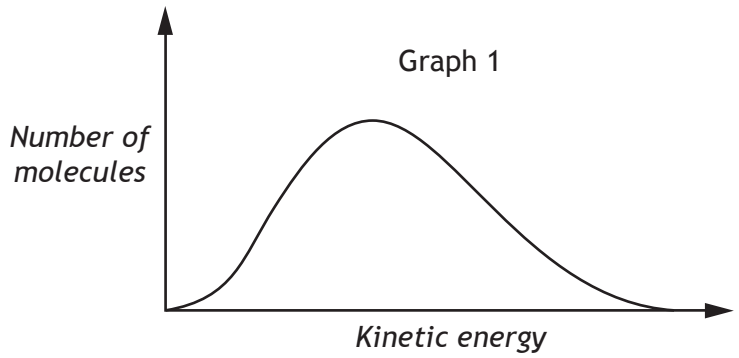


Use information from the graph to calculate the reaction time, in seconds, when the concentration of the acid was 1·0 mol l^{-1}. **1**

2. (continued)

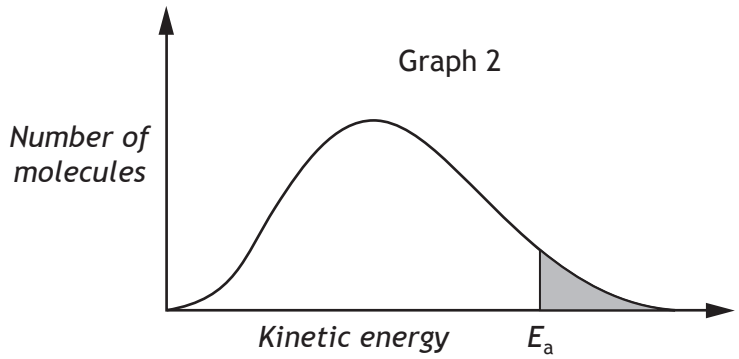
(b) The rate of reaction can also be altered by changing the temperature or using a catalyst.

(i) Graph 1 shows the distribution of kinetic energies of molecules in a gas at 100 °C.



Add a second curve to graph 1 to show the distribution of kinetic energies at 50 °C. **1**

(ii) In graph 2, the shaded area represents the number of molecules with the required activation energy, E_a.



Draw a line to show how a catalyst affects the activation energy. **1**

[Turn over

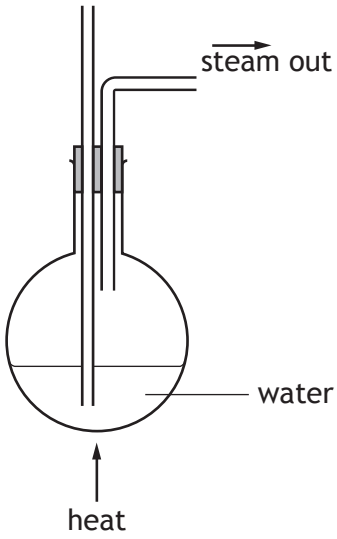
MARKS | DO NOT WRITE IN THIS MARGIN

3. (a) Methyl cinnamate is an ester used to add strawberry flavour to foods. It is a naturally occurring ester found in the essential oil extracted from the leaves of strawberry gum trees.

To extract the essential oil, steam is passed through shredded strawberry gum leaves. The steam and essential oil are then condensed and collected.

(i) Complete the diagram to show an apparatus suitable for carrying out this extraction. **2**

(An additional diagram, if required, can be found on *Page thirty-seven*).



(ii) The essential oil extracted is a mixture of compounds.

Suggest a technique that could be used to separate the mixture into pure compounds. **1**

(b) A student prepared a sample of methyl cinnamate from cinnamic acid and methanol.

cinnamic acid	+	methanol	→	methyl cinnamate	+	water
mass of one mole = 148 g		mass of one mole = 32 g		mass of one mole = 162 g		

6·5 g of cinnamic acid was reacted with 2·0 g of methanol.

3. (b) (continued)

(i) Show, by calculation, that cinnamic acid is the limiting reactant. (One mole of cinnamic acid reacts with one mole of methanol.) **2**

(ii) (A) The student obtained 3·7 g of methyl cinnamate from 6·5 g of cinnamic acid.

Calculate the percentage yield. **2**

(B) The student wanted to scale up the experiment to make 100 g of methyl cinnamate.

Cinnamic acid costs £35·00 per 250 g.

Calculate the cost of cinnamic acid needed to produce 100 g of methyl cinnamate. **2**

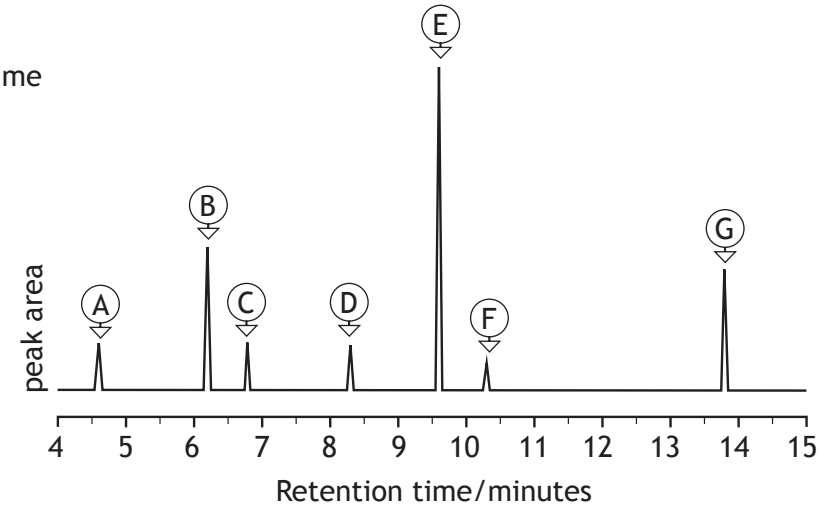
[Turn over

4. Up to 10% of perfumes sold in the UK are counterfeit versions of brand name perfumes.

One way to identify if a perfume is counterfeit is to use gas chromatography. Shown below are gas chromatograms from a brand name perfume and two different counterfeit perfumes. Some of the peaks in the brand name perfume have been identified as belonging to particular compounds.

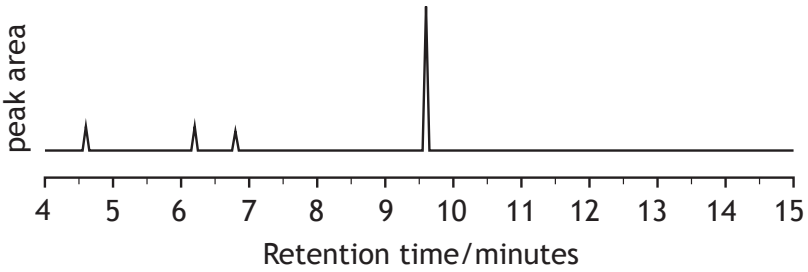
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Brand name perfume

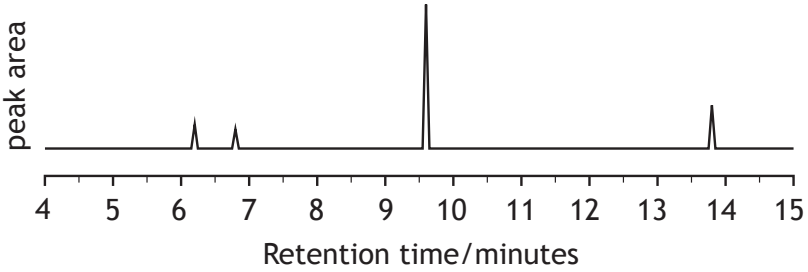


(A) linalool
(B) citronellol
(C) geraniol
(D) eugenol
(E) anisyl alcohol
(F) coumarin
(G) benzyl salicylate

Counterfeit A



Counterfeit B



4. (continued)

(a) Identify one compound present in the brand name perfume that appears in both counterfeit perfumes. **1**

(b) Some compounds in the brand name perfume are not found in the counterfeit perfumes. State another difference that the chromatograms show between the counterfeit perfumes and the brand name perfume. **1**

(c) The gas used to carry the perfume sample along the chromatography column is helium.

(i) Suggest why helium is used. **1**

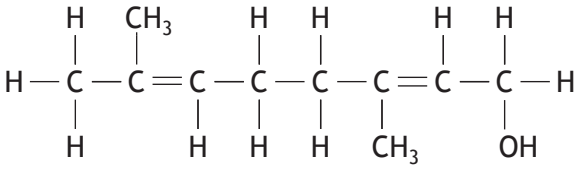
(ii) Apart from the polarity of the molecules, state another factor that would affect the retention time of molecules during gas chromatography. **1**

4. (continued)

(d) Many of the compounds in perfumes are molecules consisting of joined isoprene units.

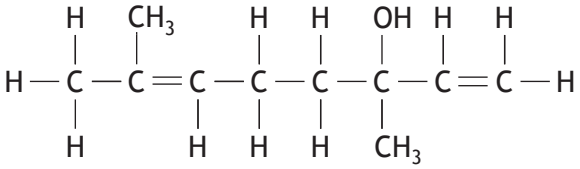
(i) State the name that is given to molecules consisting of joined isoprene units. **1**

(ii) Geraniol is one of the compounds found in perfume. It has the following structural formula and systematic name.



3,7-dimethylocta-2,6-dien-1-ol

Linalool can also be present. Its structural formula is shown.



(A) State the systematic name for linalool. **1**

(B) Explain why linalool can be classified as a tertiary alcohol. **1**

4. (continued)

(e) Coumarin is another compound found in the brand name perfume. It is present in the spice cinnamon and can be harmful if eaten in large quantities.

The European Food Safety Authority gives a tolerable daily intake of coumarin at 0·10 mg per kilogram of body weight.

1·0 kg of cinnamon powder from a particular source contains 4·4 g of coumarin. Calculate the mass of cinnamon powder, in g, which would need to be consumed by an adult weighing 75 kg to reach the tolerable daily intake. **2**

[Turn over

MARKS | DO NOT WRITE IN THIS MARGIN

5. Patterns in the Periodic Table

The Periodic Table is an arrangement of all the known elements in order of increasing atomic number. The reason why the elements are arranged as they are in the Periodic Table is to fit them all, with their widely diverse physical and chemical properties, into a logical pattern.

Periodicity is the name given to regularly-occurring similarities in physical and chemical properties of the elements.

Some Groups exhibit striking similarity between their elements, such as Group 1, and in other Groups the elements are less similar to each other, such as Group 4, but each Group has a common set of characteristics.

Adapted from Royal Society of Chemistry, Visual Elements (rsc.org)

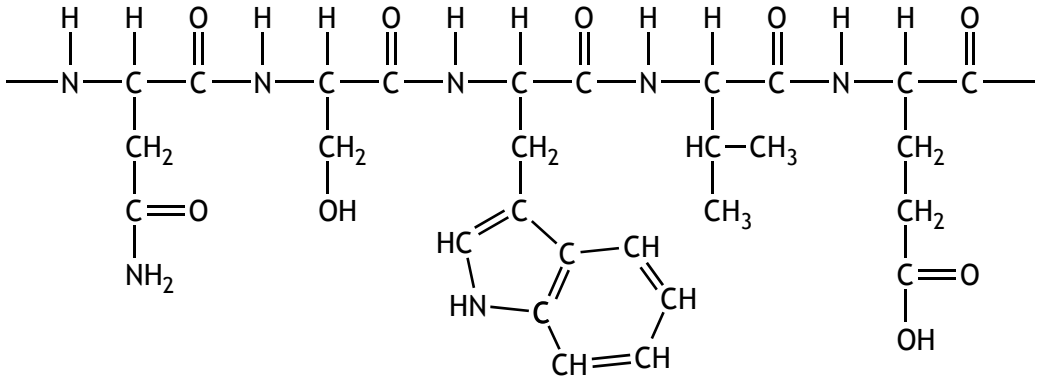
Using your knowledge of chemistry, comment on similarities and differences in the patterns of physical and chemical properties of elements in both Group 1 and Group 4. **3**

6. Uncooked egg white is mainly composed of dissolved proteins. During cooking processes, the proteins become denatured as the protein chains unwind, and the egg white solidifies.

(a) Explain why the protein chains unwind. **1**

(b) The temperature at which the protein becomes denatured is called the melting temperature. The melting temperature of a protein can be determined using fluorescence. In this technique, the protein is mixed with a dye that gives out visible light when it attaches to hydrophobic parts of the protein molecule. The hydrophobic parts of the structure are on the inside of the protein and the dye has no access to them unless the protein unwinds.

(i) Ovalbumin is a protein found in egg white. Part of the structure of unwound ovalbumin is shown below.

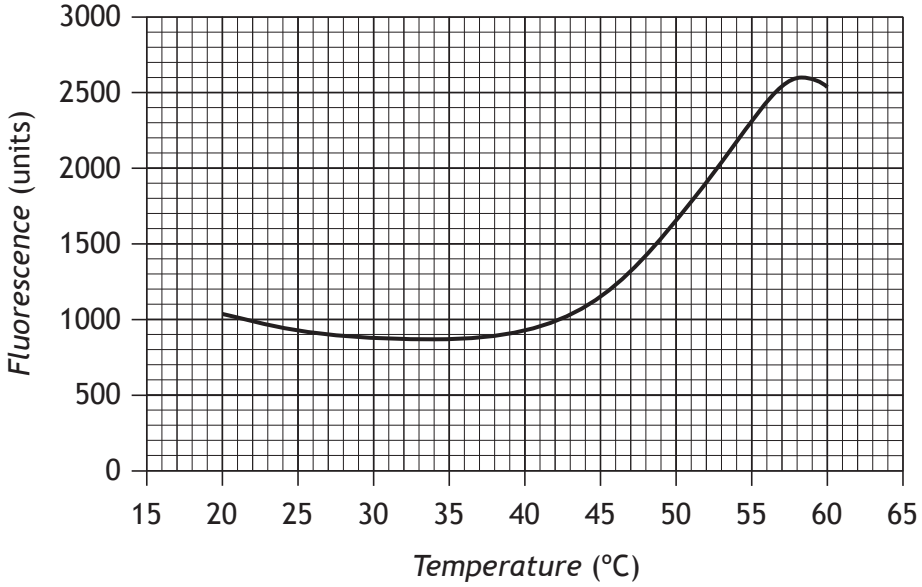


Circle the part of the structure to which the hydrophobic dye is most likely to attach. **1**

[Turn over

6. (b) (continued)

(ii) Another protein in egg white is conalbumin. The temperature of a conalbumin/dye mixture is gradually increased. The fluorescence is measured and a graph is produced.



The melting temperature is the temperature at which the fluorescence is halfway between the highest and lowest fluorescence values.

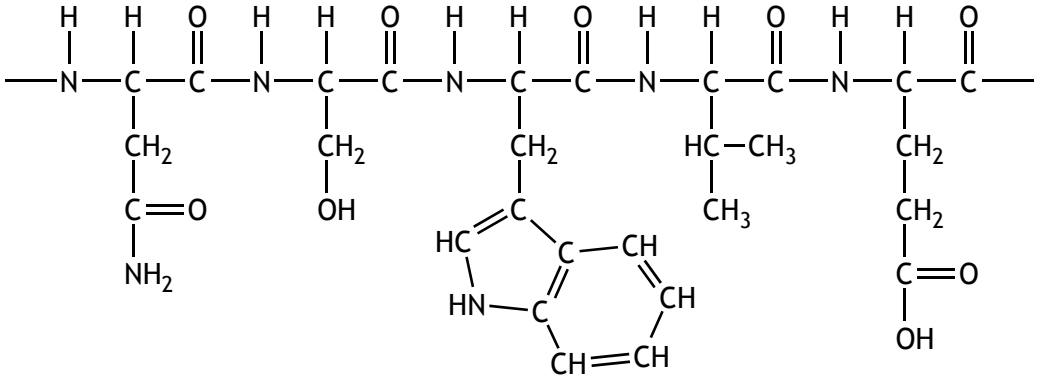
Determine the melting temperature, in °C, for this protein. **1**

6. (continued)

(c) Once cooked and eaten, the digestive system breaks the protein chains into amino acids with the help of enzymes.

(i) State the name of the digestion process where enzymes break down proteins into amino acids. **1**

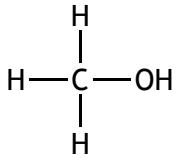
(ii)



(A) State how many amino acid molecules joined to form this section of protein. **1**

(B) Draw the structure of one amino acid that would be produced when this section of the protein chain is broken down. **1**

7. Methanol can be used as a fuel, in a variety of different ways.



(a) An increasingly common use for methanol is as an additive in petrol.

Methanol has been tested as an additive in petrol at 118 g per litre of fuel.

Calculate the volume of carbon dioxide, in litres, that would be released by combustion of 118 g of methanol.

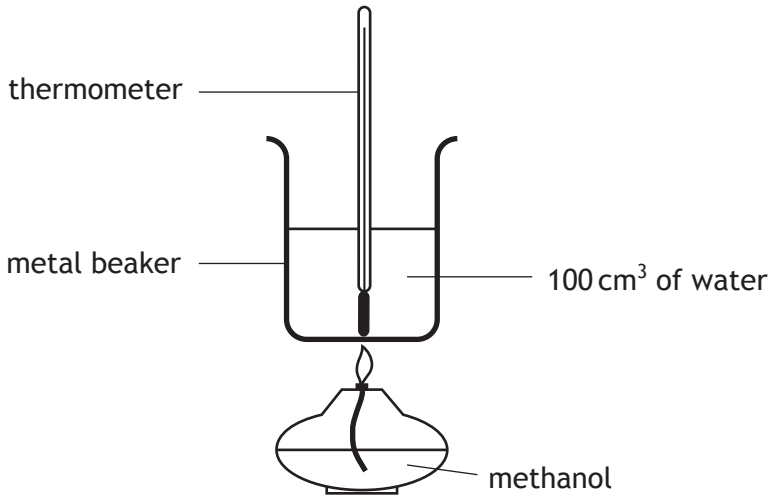
$$2CH_3OH(\ell) + 3O_2(g) \rightarrow 2CO_2(g) + 4H_2O(\ell)$$

(Take the molar volume of carbon dioxide to be 24 litres mol^{-1}). **2**

7. (continued)

(b) A student investigated the properties of methanol and ethanol.

(i) The student carried out experiments to determine the enthalpy of combustion of the alcohols.



(A) The student carried out the first experiment as shown, but was told to repeat the experiment as the thermometer had been placed in the wrong position.

Suggest why the student's placing of the thermometer was incorrect. **1**

(B) The student always used 100 cm^3 of water.

State another variable that the student should have kept constant. **1**

[Turn over

7. (b) (i) (continued)

(C) The student burned 1·07 g of methanol and recorded a temperature rise of 23 °C.

Calculate the enthalpy of combustion, in kJ mol^{-1}, for methanol using the student's results. **3**

(ii) The student determined the density of the alcohols by measuring the mass of a volume of each alcohol.

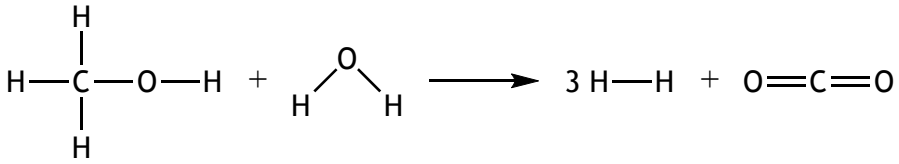
The student's results are shown below.

	Methanol	*Ethanol*
Volume of alcohol (cm^3)	25·0	25·0
Mass of alcohol (g)	19·98	20·05
Density of alcohol (g cm^{-3})		0·802

Calculate the density, in g cm^{-3}, of methanol. **1**

7. (continued)

(c) Methanol is used as a source of hydrogen for fuel cells. The industrial process involves the reaction of methanol with steam.



(i) State why it is important for chemists to predict whether reactions in an industrial process are exothermic or endothermic. **1**

(ii) Using bond enthalpies from the data booklet, calculate the enthalpy change, in $kJ\,mol^{-1}$, for the reaction of methanol with steam. **2**

[Turn over

8. Sodium carbonate is used in the manufacture of soaps, glass and paper as well as the treatment of water.

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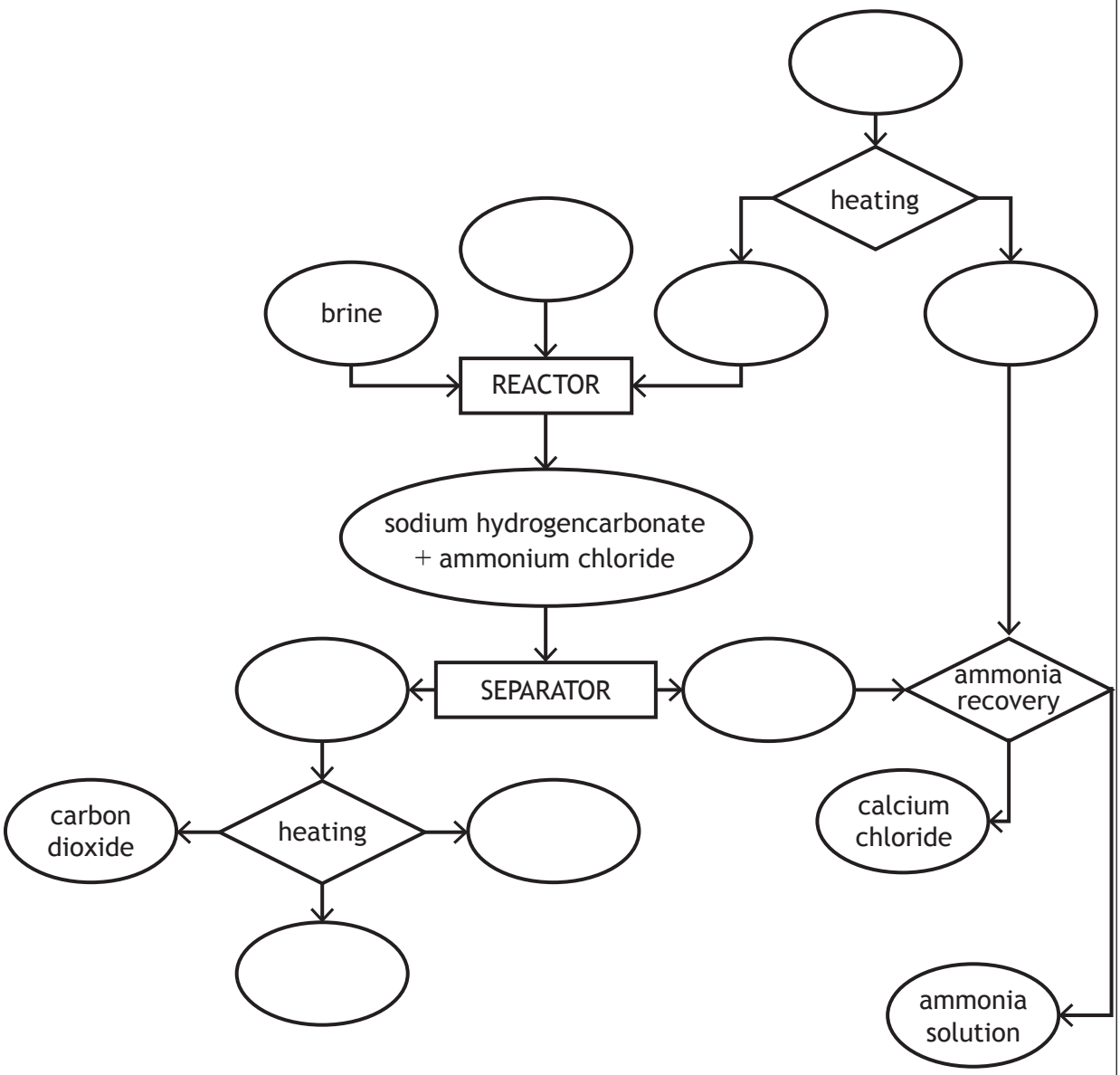
One industrial process used to make sodium carbonate is the Solvay process.

The Solvay process involves several different chemical reactions.

It starts with heating calcium carbonate to produce carbon dioxide, which is transferred to a reactor where it reacts with ammonia and brine. The products of the reactor are solid sodium hydrogencarbonate and ammonium chloride which are passed into a separator.

The sodium hydrogencarbonate is heated to decompose it into the product sodium carbonate along with carbon dioxide and water. To recover ammonia the ammonium chloride from the reactor is reacted with calcium oxide produced by heating the calcium carbonate. Calcium chloride is a by-product of the ammonia recovery process.

(a) Using the information above, complete the flow chart by adding the names of the chemicals involved. **2**



8 (continued)

(b) The reaction that produces the solid sodium hydrogencarbonate involves the following equilibrium:

$$HCO_3^-(aq) + Na^+(aq) \rightleftharpoons NaHCO_3(s)$$

Brine is a concentrated sodium chloride solution.

Explain fully why using a concentrated sodium chloride solution encourages production of sodium hydrogencarbonate as a solid. **2**

[Turn over

MARKS | DO NOT WRITE IN THIS MARGIN

9. Occasionally, seabirds can become contaminated with hydrocarbons from oil spills. This causes problems for birds because their feathers lose their waterproofing, making the birds susceptible to temperature changes and affecting their buoyancy. If the birds attempt to clean themselves to remove the oil, they may swallow the hydrocarbons causing damage to their internal organs.

Contaminated seabirds can be cleaned by rubbing vegetable oil into their feathers and feet before the birds are rinsed with diluted washing-up liquid.

Using your knowledge of chemistry, comment on the problems created for seabirds by oil spills and the actions taken to treat affected birds. **3**

MARKS | DO NOT WRITE IN THIS MARGIN

10. Plants require trace metal nutrients, such as zinc, for healthy growth. Zinc ions are absorbed from soil through the plant roots.

The zinc ion concentration in a solution can be found by adding a compound which gives a blue colour to the solution with zinc ions. The concentration of zinc ions is determined by measuring the absorption of light by the blue solution. The higher the concentration of zinc ions in a solution, the more light is absorbed.

A student prepared a stock solution with a zinc ion concentration of $1\,g\,l^{-1}$. Samples from this were diluted to produce solutions of known zinc ion concentration.

(a) The stock solution was prepared by adding 1·00 g of zinc metal granules to $20\,cm^3$ of 2 mol l^{-1} sulfuric acid in a $1000\,cm^3$ standard flask.

$$Zn(s) + H_2SO_4(aq) \rightarrow ZnSO_4(aq) + H_2(g)$$

The flask was left for 24 hours, without a stopper. The solution was then diluted to 1000 cm^3 with water.

(i) **Explain fully** why the flask was left for 24 hours, without a stopper. **2**

(ii) Explain why the student should use deionised water or distilled water, rather than tap water, when preparing the stock solution. **1**

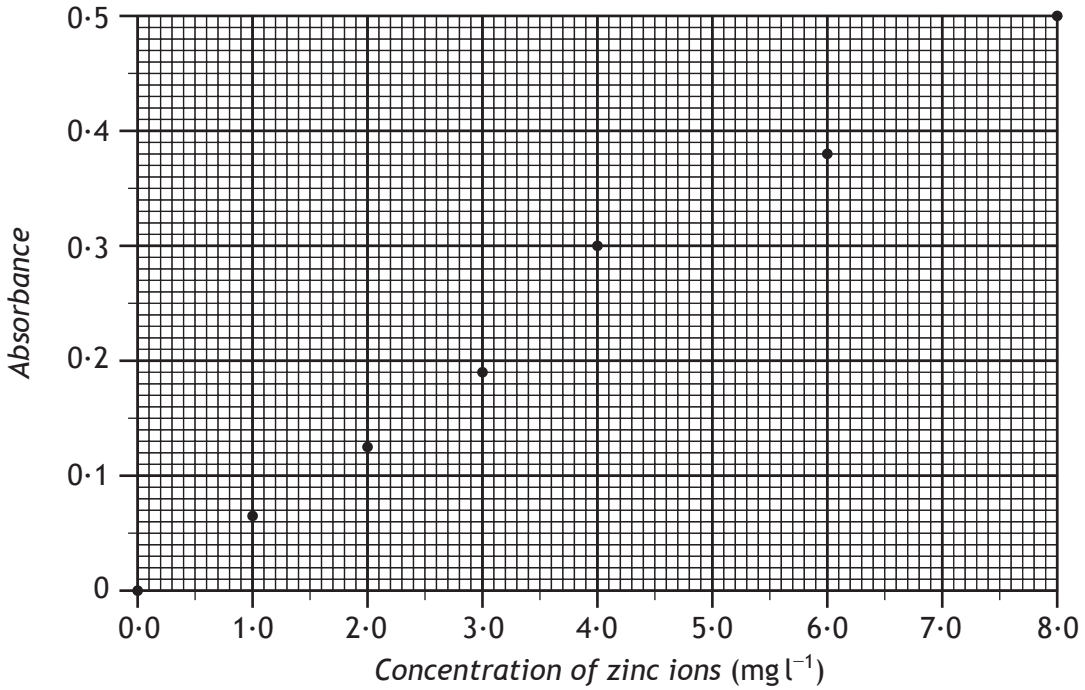
(b) Solutions of known zinc ion concentration were prepared by transferring accurate volumes of the stock solution to standard flasks and diluting with water.

(i) Name the piece of apparatus which should be used to transfer $10\,cm^3$ of stock solution to a standard flask. **1**

10. (b) (continued)

(ii) Calculate the concentration, in mg l^{-1}, of the solution prepared by transferring 10 cm^3 of the 1 g l^{-1} stock solution to a 1000 cm^3 standard flask and making up to the mark. **1**

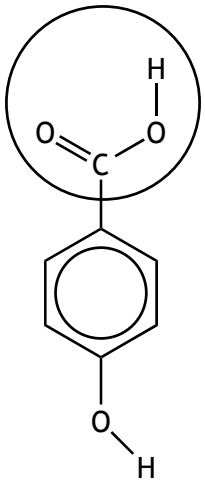
(c) The light absorbance of different solutions was measured and the results plotted.



A solution prepared from a soil sample was tested to determine the concentration of zinc ions. The solution had an absorbance of 0·3.

Determine the concentration, in mg l^{-1}, of zinc ions in the solution. **1**

11.



4-hydroxybenzoic acid

4-hydroxybenzoic acid can react with alcohols to form compounds known as parabens.

(a) Name the functional group circled in the structure of 4–hydroxybenzoic acid. **1**

(b) Name the type of reaction taking place when parabens are formed. **1**

(c) Draw the paraben formed when 4–hydroxybenzoic acid reacts with ethanol. **1**

11. (continued)

(d) Parabens can be used as preservatives in cosmetics and toiletries.

Parabens are absorbed into the body through the skin. The following table indicates the absorption of some parabens.

Paraben	*Absorption* ($\mu g\,cm^{-2}$)
Methyl	32·50
Ethyl	20·74
Propyl	11·40
Butyl	7·74
Hexyl	1·60

State a conclusion that can be drawn from the information in the table. **1**

[Turn over

MARKS | DO NOT WRITE IN THIS MARGIN

12. (a) The concentration of sodium hypochlorite in swimming pool water can be determined by redox titration.

Step 1

A 100·0 cm^3 sample from the swimming pool is first reacted with an excess of acidified potassium iodide solution forming iodine.

$$NaOCl(aq) + 2I^-(aq) + 2H^+(aq) \rightarrow I_2(aq) + NaCl(aq) + H_2O(\ell)$$

Step 2

The iodine formed in step 1 is titrated using a standard solution of sodium thiosulfate, concentration 0·00100 mol l^{-1}. A small volume of starch solution is added towards the endpoint.

$$I_2(aq) + 2Na_2S_2O_3(aq) \rightarrow 2NaI(aq) + Na_2S_4O_6(aq)$$

(i) Describe in detail how a burette should be prepared and set up, ready to begin the titration. **3**

(ii) Write the ion-electron equation for the oxidation reaction occurring in step 1. **1**

12. (a) (continued)

(iii) Calculate the concentration, in mol l^{-1}, of sodium hypochlorite in the swimming pool water, if an average volume of 12·4 cm^3 of sodium thiosulfate was required. **3**

(b) The level of hypochlorite in swimming pools needs to be maintained between 1 and 3 parts per million (1 – 3 ppm).

400 cm^3 of a commercial hypochlorite solution will raise the hypochlorite level of 45 000 litres of water by 1 ppm.

Calculate the volume of hypochlorite solution that will need to be added to an Olympic-sized swimming pool, capacity 2 500 000 litres, to raise the hypochlorite level from 1 ppm to 3 ppm. **2**

[Turn over

12. (continued)

(c) The familiar chlorine smell of a swimming pool is not due to chlorine but compounds called chloramines. Chloramines are produced when the hypochlorite ion reacts with compounds such as ammonia, produced by the human body.

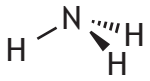
$OCl^-(aq) + NH_3(aq) \rightarrow NH_2Cl(aq) + OH^-(aq)$
monochloramine

$OCl^-(aq) + NH_2Cl(aq) \rightarrow NHCl_2(aq) + OH^-(aq)$
dichloramine

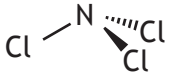
$OCl^-(aq) + NHCl_2(aq) \rightarrow NCl_3(aq) + OH^-(aq)$
trichloramine

Chloramines are less soluble in water than ammonia due to the polarities of the molecules, and so readily escape into the atmosphere, causing irritation to the eyes.

(i) Explain the difference in polarities of ammonia and trichloramine molecules.



ammonia



trichloramine

2

12. (c) (continued)

(ii) Chloramines can be removed from water using ultraviolet light treatment.

One step in the process is the formation of free radicals.

$$NH_2Cl \xrightarrow{UV} \bullet NH_2 + \bullet Cl$$

State what is meant by the term free radical. **1**

(iii) Another step in the process is shown below.

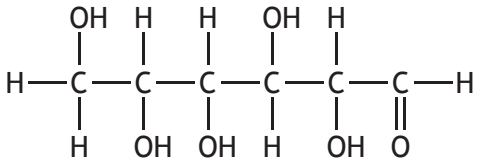
$$NH_2Cl + \bullet Cl \longrightarrow \bullet NHCl + HCl$$

State the name for this type of step in a free radical reaction. **1**

[Turn over for Question 13 on *Page thirty-six*

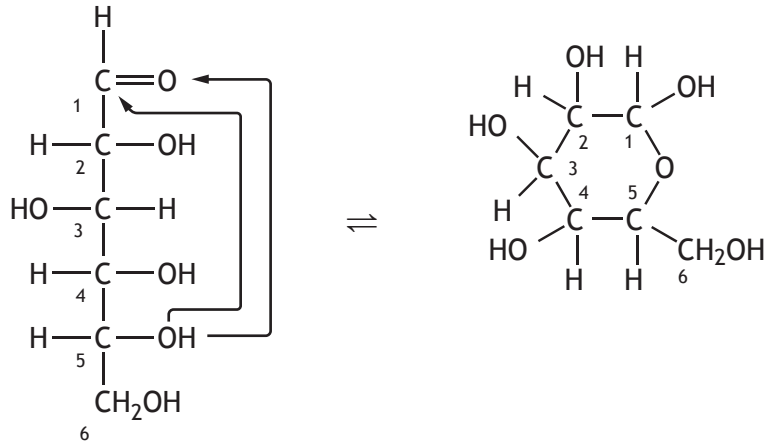
13. (a) One test for glucose involves Fehling's solution.

Circle the part of the glucose molecule that reacts with Fehling's solution. **1**



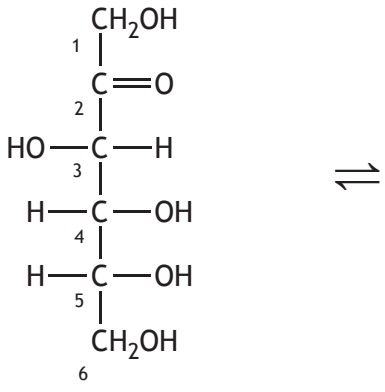
(b) In solution, sugar molecules exist in an equilibrium in straight-chain and ring forms.

To change from the straight-chain form to the ring form, the oxygen of the hydroxyl on carbon number 5 joins to the carbonyl carbon. This is shown below for glucose.



glucose

Draw the structure of a ring form for fructose. **1**



fructose

[END OF QUESTION PAPER]

National
Qualifications
2026

2026 Chemistry

Higher – Paper 1

Question Paper Finalised Marking Instructions

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Marking instructions for each question

Question	Answer	Mark
1.	D	**1**
2.	C	**1**
3.	B	**1**
4.	C	**1**
5.	A	**1**
6.	D	**1**
7.	B	**1**
8.	B	**1**
9.	D	**1**
10.	D	**1**
11.	C	**1**
12.	A	**1**
13.	A	**1**
14.	B	**1**
15.	D	**1**
16.	C	**1**
17.	B	**1**
18.	A	**1**
19.	D	**1**
20.	B	**1**
21.	A	**1**
22.	C	**1**
23.	C	**1**
24.	D	**1**
25.	C	**1**

[END OF MARKING INSTRUCTIONS]

National
Qualifications
2026

2026 Chemistry

Higher – Paper 2

Question Paper Finalised Marking Instructions

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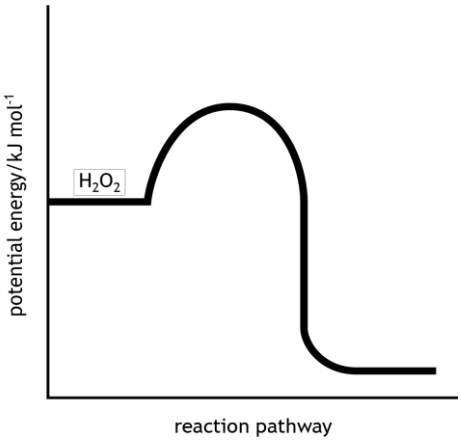
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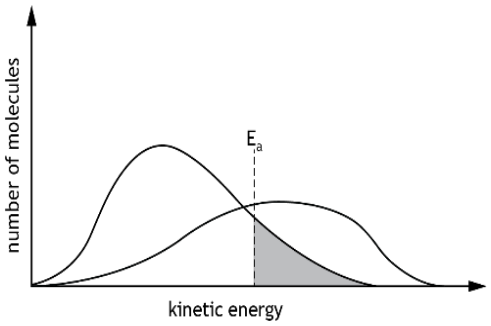
Marking instructions for each question

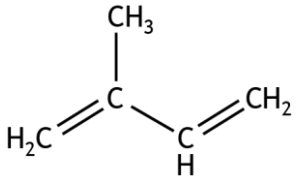
Question			Expected response	Max mark	Additional guidance
1.	(a)	(i)	Lattice	**1**	Accept network. Covalent or metallic cancels.
		(ii) (A)	Attraction between positive and negative (ions/particles/species). **OR** Attraction between opposite charges.	**1**	Do not accept mention of positive or negative atoms or molecules.
		(ii) (B)	There is a large difference in electronegativity (between sodium and chlorine). **OR** The electronegativity difference is 2.3.	**1**	Answer must make reference to difference.
	(b)	(i)	3	**1**	
		(ii)	100.714/100.71/100.7/101 (°C) Partial marks --- Calculating $\Delta T_b = 0.714$. **(1 mark)** **OR** Evidence of adding an incorrectly calculated value of ΔT_b to 100. **(1 mark)**	**2**	Units are not required but must be correct if given. (Wrong units would only be penalised once in any paper).

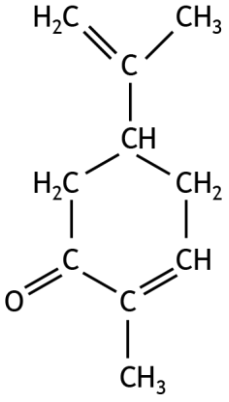
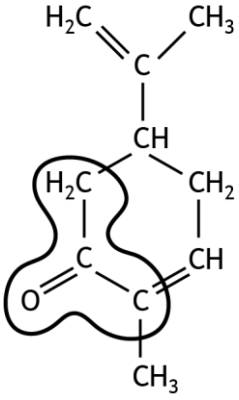
Question			Expected response	Max mark	Additional guidance
2.	(a)		**1 mark** for description of LDFs as forces of attraction **between** temporary dipoles (and induced dipoles). **1 mark** for an explanation of the cause of temporary dipoles in terms of uneven distribution of electrons/ electron wobble/movement of electrons (in the molecule).	**2**	Attraction involving permanent dipoles cancels this mark.
	(b)		They have the same number of electrons. **OR** The strengths of the London dispersion forces are the same.	**1**	
	(c)		Water contains polar bonds. **(1 mark)** The shape of the molecule means that the dipoles are not cancelled out. **OR** Has an overall dipole. **(1 mark)**	**2**	

Question			Expected response	Max mark	Additional guidance
3.			**Award 1 mark** where the candidate has demonstrated, at an appropriate level, a limited understanding of the chemistry involved. They have made some statement(s) which are relevant to the situation, showing that they have understood at least a little of the chemistry within the problem. **Award 2 marks** where the candidate has demonstrated, at an appropriate level, a reasonable understanding of the chemistry involved. They make some statement(s) which are relevant to the situation, showing that they have understood the problem. **Award 3 marks** where the candidate has demonstrated, at an appropriate level, a good understanding of the chemistry involved. They show a good comprehension of the chemistry of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be 'excellent' or 'complete' for the candidate to gain full marks. **Award 0 marks** where the candidate has not demonstrated, at an appropriate level, an understanding of the chemistry involved. There is no evidence that they have recognised the area of chemistry involved, or they have not given any statement of a relevant chemistry principle. Award this mark also if the candidate merely restates the chemistry given in the question.	**3**	

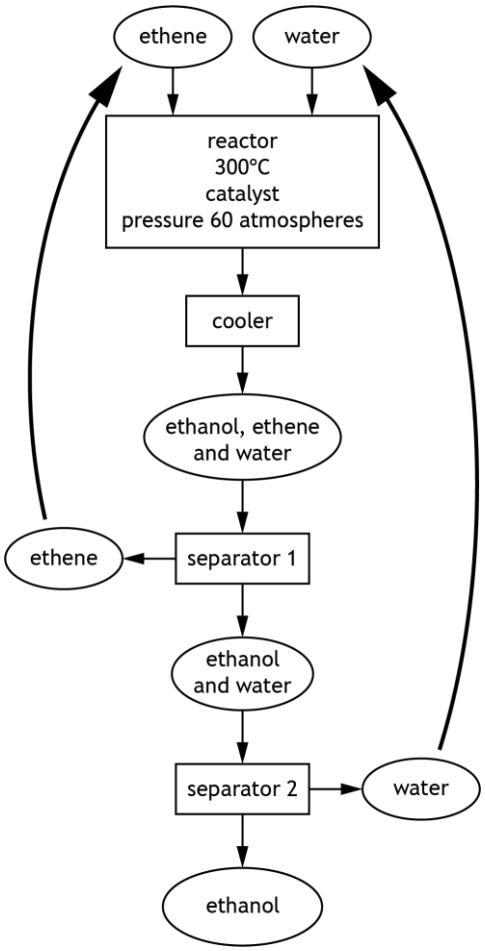
Question			Expected response	Max mark	Additional guidance
4.	(a)	(i)		**1**	The activation energy should be less than the enthalpy change. Shown by correct scale of diagram or correct numerical labels.
		(ii)	173 (kJ mol^{-1})	**1**	Units are not required but must be correct if given. (Wrong units would only be penalised once in any paper).
	(b)		A catalyst provides an alternative reaction pathway **(1 mark)** with a lower activation energy. **OR** Greater number of particles with energy greater than the activation energy. **(1 mark)**	**2**	
	(c)		0.17(g) Partial marks --- No of moles oxygen 2.5x10^{-3}. **(1 mark)** **OR** Correct calculation of mass from an incorrect number of moles of oxygen (with mole ratio applied correctly). **(1 mark)** **OR** By proportion 68 grams ◄► 24 litres. **(1 mark)**	**2**	
	(d)	(i)	Atoms/molecules with unpaired electron(s).	**1**	Accept species in place of atoms or molecules. If reference is made to a singular atom, electron must also be singular.

Question			Expected response	Max mark	Additional guidance
4.	(d)	(ii)	•OH **OR** Hydroxyl	**1**	Accept OH•. Hydroxide/OH/OH^- not accepted.
	(e)	(i)	(The number of) molecules with energy greater than (or equal to) activation energy.	**1**	Accept (the number of) molecules with sufficient energy to form an activated complex/to allow a successful collision/to react.
		(ii)	 Second line displaced to right of original. Peak of curve should be further to the right and no higher than the original peak. **(1 mark)**	**1**	

Question			Expected response	Max mark	Additional guidance
5.	(a)		Non-water soluble **OR** Volatile	**1**	Any contradictory answer cancels.
	(b)	(i)		**1**	
		(ii)	2	**1**	
	(c)	(i)	Oils/limonene are non-polar. **AND** Hexane is non-polar.	**1**	
		(ii)	-4547.84 (kJ mol^{-1}) **(3 marks)** **If incorrect answer given, a maximum of two partial marks can be awarded for demonstration of the two concepts:** **1 mark** for a demonstration of the correct use of the relationship $Eh=cm\Delta T$ as shown by (4.18 × (an order of magnitude of 0.5) × 56) (ignore units for this partial mark). **1 mark** for evidence of the knowledge that enthalpy of combustion relates to 1 mole, evidenced by the scaling up of a calculated value of energy released. • correct scaling of calculated Eh from 3.5 to 136 g • 3.5 g ↔ Eh 136 g ↔ $\frac{136 \times Eh}{3.5}$ • $\frac{3.5}{136}$ (0.0257) ↔ Eh 1 ↔ **X**	**3**	4547.84 (kJ mol^{-1}) **(2 marks)** No units required. If wrong unit is given (where no unit required, wrong units would only be penalised once in any paper). kJ is acceptable in place of kJ mol^{-1} (KJ or Kj or KJ mol^{-1} or Kj mol^{-1} accepted).

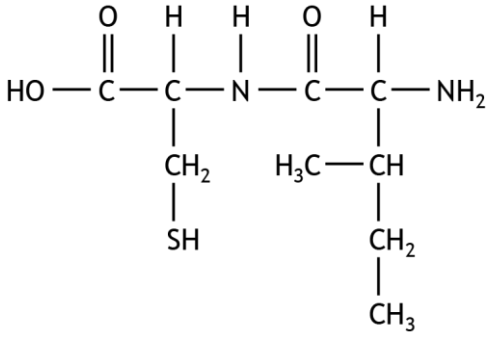
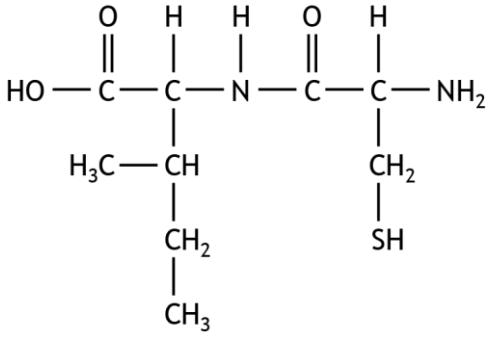
Question			Expected response	Max mark	Additional guidance
5.	(d)	(i)	The carbon connected to the functional group/hydroxyl (group) is connected to two carbon atoms. **OR** The carbon connected to the functional group/hydroxyl is connected to one hydrogen atom.	**1**	
		(ii)		**1**	 Connectivity must be correct in the circled area.

Question			Expected response	Max mark	Additional guidance
6.	(a)	(i)	3-methylbutan-1-ol	**1**	
		(ii)	Brown (solid forms)	**1**	
	(b)		1314.78/1314.8/1315 g **(2 marks)** Partial marks --- Calculating moles of ethene required (46.96). **(1 mark)** **OR** 4% taken of an incorrect number of moles (eg 27/23) and multiplied by 28 (g). **OR** 32,869.6 (g).	**2**	
	(c)	(i)	Proteins	**1**	
		(ii)	Intermolecular bonds/Van der Waals forces/Hydrogen bonds (within the protein) break. **(1 mark)** The protein changes shape/chains unwind. **(1 mark)**	**2**	

Question			Expected response	Max mark	Additional guidance
7.	(a)	(i)		**1**	Either arrow is accepted.
		(ii)	Distillation	**1**	
		(iii) (A)	Addition	**1**	
		(iii) (B)	Increasing pressure favours side with fewer moles of gas/gas molecules (or converse). **(1 mark)** Rate of forward reaction is increased/speeds up (by increasing pressure). **(1 mark)**	**2**	

Question			Expected response	Max mark	Additional guidance
7.	(a)	(iii) (C)	-41 (kJ mol^{-1}) Partial marks --- The five relevant values for bond enthalpies of the C=C, C-H, O-H, C-O and C-C (or multiples thereof) are retrieved from the data booklet (614, 412, 463, 360 and 346 – ignore signs). **(1 mark)** **OR** If only four values are retrieved, the candidate recognises that bond breaking is endothermic and bond formation is exothermic and correctly manipulates the bond enthalpy values they have used to give their answer. **(1 mark)**	**2**	(+)41 (kJ mol^{-1}) would be awarded 1 mark. No units required. Only 1 mark can be awarded for the correct answer if wrong unit is given. kJ is acceptable in place of kJ mol^{-1} (KJ or Kj or KJ mol^{-1} or Kj mol^{-1} accepted). (Wrong units would only be penalised once in any paper).
		(iii) (D)	No by-products/no waste products/ only one product. **OR** All the atoms in the reactants are converted into the product. **OR** Mass of product is equal to mass of reactants. **OR** The proportion of the total mass of all starting materials successfully converted into the desired product is 100%.	**1**	
	(b)	(i)	Increase in oxygen to hydrogen ratio/decrease in hydrogen to oxygen ratio.	**1**	
		(ii)	Ethanoic acid	**1**	

Question			Expected response	Max mark	Additional guidance
7.	(b)	(iii) (A)	Reactants or products are flammable/could catch fire.	**1**	
		(iii) (B)	Orange to green/blue-green/blue	**1**	
		(iii) (C)	$C_2H_5OH + H_2O \rightarrow CH_3COOH + 4H^+ + 4e^-$	**1**	Negative charge on the electron is not required.
		(iii) (D)	To supply the hydrogen ions (for the reaction).	**1**	
		(iii) (E)	Tertiary	**1**	
		(iv)	Fehling's (solution) **OR** Tollens' (reagent)	**1**	Accept Benedict's (solution).
	(c)	(i)	Diagram shows a workable method for the passing of ethanol vapour over heated aluminium oxide. Aluminium oxide must be labelled. **(1 mark)** Diagram allows ethene to be collected over water. **(1 mark)**	**2**	
		(ii)	Unsaturated.	**1**	

Question			Expected response	Max mark	Additional guidance
8.	(a)	(i)	Condensation	**1**	
		(ii)	 **OR** 	**1**	Ignore incorrect connectivity in the side chains.
	(b)	(i) (A)	Hydrophobic	**1**	
		(i) (B)	The **side chain** contains a carboxyl/COOH group.	**1**	No mark can be awarded unless side chain is mentioned or indicated on structure.
		(ii)	Hydrogen bonds	**1**	
	(c)		(Strong) covalent bonds must be broken.	**1**	

Question			Expected response	Max mark	Additional guidance
9.			**Award 1 mark** where the candidate has demonstrated, at an appropriate level, a limited understanding of the chemistry involved. They have made some statement(s) which are relevant to the situation, showing that they have understood at least a little of the chemistry within the problem. **Award 2 marks** where the candidate has demonstrated, at an appropriate level, a reasonable understanding of the chemistry involved. They make some statement(s) which are relevant to the situation, showing that they have understood the problem. **Award 3 marks** where the candidate has demonstrated, at an appropriate level, a good understanding of the chemistry involved. They show a good comprehension of the chemistry of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be 'excellent' or 'complete' for the candidate to gain full marks. **Award 0 marks** where the candidate has not demonstrated, at an appropriate level, an understanding of the chemistry involved. There is no evidence that they have recognised the area of chemistry involved, or they have not given any statement of a relevant chemistry principle. Award this mark also if the candidate merely restates the chemistry given in the question.	**3**	

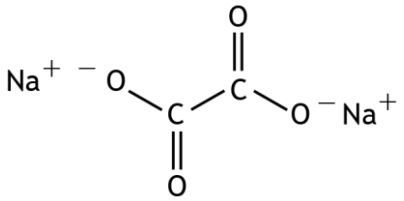
<table>
<tr><th colspan="3">Question</th><th>Expected response</th><th>Max mark</th><th>Additional guidance</th></tr>
<tr><td>10.</td><td>(a)</td><td>(i)</td><td>
<table>
<tr><th>Condition</th><th>Increase/Decrease</th></tr>
<tr><td>Pressure</td><td>increase</td></tr>
<tr><td>Temperature</td><td>decrease</td></tr>
<tr><td>Concentration of ammonia in the reaction mixture</td><td>decrease</td></tr>
</table>
</td><td>1</td><td></td></tr>
<tr><td></td><td></td><td>(ii)</td><td>No effect</td><td>1</td><td></td></tr>
<tr><td></td><td></td><td>(iii)</td><td>66.71/66.7/67 (%)
Partial marking

Theoretical yield = 607 kg.
OR
Allow follow through from incorrect calculation of theoretical yield.</td><td>2</td><td></td></tr>
<tr><td></td><td>(b)</td><td></td><td>(+) 206(kJ mol^{-1})
[(+283) + (726) + (-803)] = (+) 206 (kJ mol^{-1})
Partial marks

Treat as two concepts. Either would be acceptable for 1 mark.
Evidence of understanding of reversal of first and second enthalpy values e.g. +283 and + 242 (or multiples of +242) must be seen. The other enthalpy value (regardless of value) must be negative, or this partial mark cannot be awarded.
OR
Evidence of understanding of multiplying the second enthalpy value by 3. Ignore the enthalpy sign associated with these numbers. The other enthalpy values must not be multiplied, or this partial mark cannot awarded.</td><td>2</td><td></td></tr>
</table>

Question			Expected response	Max mark	Additional guidance
10.	(c)	(i)	19/19.2/19.17 (%) Partial marks --- Correct use of atom economy relationship ignoring stoichiometry (or with incorrect use of stoichiometry) (working must be shown). **(1 mark)** Correct working with no correct answer given. **(1 mark)** 0.19166 **(1 mark)**	**2**	
		(ii)	283.3 (hours) (**2 marks)** Partial marks --- 17000 (minutes) **OR** 1020000 (seconds) **(1 mark)**	**2**	

Question			Expected response	Max mark	Additional guidance
11.	(a)	(i)	Standard (solution)	**1**	
		(ii)	1.125, 1.13, 1.1(g)	**1**	
		(iii)	**2 marks** from any of the following: • rinse beaker and transfer the rinsings/washings into the flask **OR** • transfer quantitatively into the flask **(1 mark)** • use deionised/distilled water (to make up to the mark) **(1 mark)** • make up so that the bottom of the meniscus meets the line **(1 mark)**	**2**	
	(b)	(i)	To determine the end point (of the reaction). **OR** The reaction is not self-indicating.	**1**	
		(ii)	**1 mark** for any of the following: • tap opened/solution drained to ensure no air bubbles • solution run onto the scale	**1**	
		(iii) (A)	**1 mark** for any of the following: • the first result is not concordant • it is the average of concordant results • or description of concordancy	**1**	

Question			Expected response	Max mark	Additional guidance
11.	(b)	(iii) (B)	0.111 (mol l^{-1}) **Partial marks** can be awarded using a scheme of two "concept" marks, and one "arithmetic" mark. **1 mark** for knowledge of the relationship between moles, concentration and volume. This could be shown by one of the following steps: • calculation of moles oxalic acid solution e.g. 0.0517 x 0.02695 = 0.00139 • calculation of concentration of sodium hydroxide e.g. 0.002787 ÷ 0.025 • insertion of correct pairings of values for concentration and volume in a valid titration formula If the relationship between moles, concentration and volume is used more than once, it must be used correctly every time. **1 mark** for the relationship between a calculated number of moles of oxalic acid and sodium hydroxide. **OR** Insertion of correct stoichiometric values in a valid titration formula. **1 mark** is awarded for correct arithmetic through the calculation. This mark can only be awarded if both concept marks have been awarded.	**3**	No units required. Only **2 marks** can be awarded for the correct answer if wrong unit is given. (Wrong units would only be penalised once in any paper).
	(c)		 **OR** Equivalent without charges as long as no covalent bond shown between Na and O.	**1**	

Question			Expected response	Max mark	Additional guidance
12.	(a)	(i)	Sodium and Cadmium Na and Cd	**1**	
		(ii)	Green	**1**	
	(b)	(i)	4.4 (mg l^{-1})	**1**	No units required. No mark can be awarded for correct answer if wrong unit is given (where no unit required, wrong units would only be penalised once in any paper). For any response other than 4.4, a line of best fit must be drawn and concentration at 40 checked with half box tolerance applied.
		(ii)	Hard (water)	**1**	
	(c)	(i)	$He(g) \rightarrow He^{+}(g) + e^{-}$	**1**	State symbols must be shown. Negative charge on the electron is not required.
		(ii)	Helium (only) has two electrons. **OR** No third electron in helium (atom).	**1**	

[END OF MARKING INSTRUCTIONS]

General marking principles for Higher Chemistry

Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.

(a) Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.

(b) If a candidate response does not seem to be covered by either the principles or detailed marking instructions, and you are uncertain how to assess it, you must seek guidance from your team leader.

(c) Do not award half marks.

(d) Where a candidate makes an error at an early stage in a multi-stage calculation, award marks for correct follow-on working in subsequent stages, unless the error significantly reduces the complexity of the remaining stages. Apply the same principle for questions that require several stages of non-mathematical reasoning. The exception to this rule is where the marking instructions for a numerical question assign separate 'concept marks' and an 'arithmetic mark'. In such situations, the marking instructions will give clear guidance on the assignment or partial marks.

(e) Unless a numerical question specifically requires evidence of working to be shown, award full marks for a correct final response (including units) on its own.

(f) Candidates may fully access larger mark allocations whether their responses are in continuous prose, linked statements, or a series of developed bullet points.

(g) Do not deduct marks for inaccurate or unconventional spelling or vocabulary as long as the meaning of the word(s) is conveyed. **For example**, responses that include 'distilling' for 'distillation', or 'it gets hotter' for 'the temperature rises', are acceptable.

(h) In many questions, the unit in which the answer is to be expressed is given. In these questions, the candidate does not need to state a unit in their answer; but if they do, the unit must be correct. The full mark allocation cannot be awarded if an incorrect unit is shown. In these questions, incorrect units would only be penalised once in any paper.

(i) If a correct response is followed by a wrong response, award no marks. **For example** in response to the question, 'State the colour seen when blue Fehling's solution is warmed with an aldehyde', do not award marks for the response 'red green'. However, if a correct response is followed by additional information which does not conflict with that, ignore the additional information, whether correct or not. **For example** in response to a question concerned with melting point, 'State why the tube should not be made of copper', the response 'Copper has a low melting point and is coloured grey' **would** gain marks.

(j) Award full marks for the correct response to a calculation without working. Award partial marks, as shown in the detailed marking instructions, when working is given but the final response is incorrect. An exception is when candidates are asked to 'Find, by calculation' - do not award full marks for the correct response without working.

(k) Ignore the omission of one H atom from a full structural formula provided the bond is shown.

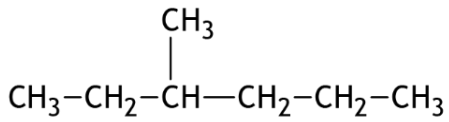
(l) Award marks for a symbol or correct formula in place of a name **unless stated otherwise in the detailed marking instructions.**

(m) When formulae of ionic compounds are given as responses, candidates only need to show ion charges if these have been specifically asked for. However, if ion charges are shown, they must be correct. If incorrect charges are shown, do not award marks.

(n) If an answer comes directly from the text of the question, do not award marks. **For example**, in response to the question, 'A student found that 0·05 mol of propane, C_3H_8 burned to give 82·4 kJ of energy. $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(\ell)$. Name the kind of enthalpy change that the student measured', do not award marks for 'burning' since the word 'burned' appears in the text.

(o) A guiding principle in marking is to give credit for correct elements of a response rather than to look for reasons not to give marks.

Example 1: The structure of a hydrocarbon found in petrol is shown below.



Name the hydrocarbon

- Award the full mark for '3, methyl-hexane', although the punctuation is not correct.

Example 2: A student measured the pH of four carboxylic acids to find out how their strength is related to the number of chlorine atoms in the molecule. The results are shown.

CH_3COOH	1.65
$CH_2ClCOOH$	1.27
$CHCl_2COOH$	0.90
CCl_3COOH	0.51

Describe the relationship between the number of chlorine atoms in the molecule and the strengths of the acids.

- Award the full mark for a response such as 'the more Cl_2, the stronger the acid', although not completely correct.

(p) Unless the question is clearly about a non-chemistry issue, for example costs in an industrial chemical process, do not award marks for a non-chemical response.
For example, in response to the question, 'Why does the (catalytic) converter have a honeycomb structure?', do not award a mark for 'To make it work'. This response may be correct but it is not a chemical response.

(q) Only award marks for a valid response to the question asked. Where candidates are asked to:

- **identify, name, give** or **state**, they must only name or present in brief form.
- **describe**, they must provide a statement or structure of characteristics and/or features.
- **explain**, they must relate cause and effect and/or make relationships between things clear.
- **compare**, they must demonstrate knowledge and understanding of the similarities and/or differences between things.
- **complete**, they must finish a chemical equation or fill in a table with information.
- **determine** or **calculate**, they must determine a number from given facts, figures or information.
- **draw**, they must draw a diagram or structural formula, for example 'Draw a diagram to show the part of a poly(propene) molecule formed from two propene molecules.'
- **estimate**, they must determine an approximate value for something.
- **predict**, they must suggest what may happen based on available information.
- **evaluate**, they must make a judgement based on criteria.
- **suggest**, they must apply their knowledge and understanding of chemistry to a new situation. A number of responses are acceptable: award marks for any suggestions that are supported by knowledge and understanding of chemistry.
- **use their knowledge of chemistry or aspect of chemistry to comment on**, they must apply their skills, knowledge and understanding to respond appropriately to the problem/situation presented (for example by making a statement of principle(s) involved and/or a relationship or equation, and applying these to respond to the problem/situation). Candidates gain marks for the breadth and/or depth of their conceptual understanding.
- **write**, they must complete a chemical or word equation, for example 'Write the word equation for the complete combustion of ethanol.'

2026 Higher Chemistry Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	D	5	2	31	**62**	0	0
2	C	5	5	**81**	10	0	0
3	B	9	**77**	7	7	0	0
4	C	4	12	**81**	2	0	0
5	A	**48**	21	13	17	1	0
6	D	13	9	15	**62**	0	0
7	B	11	**74**	7	8	0	0
8	B	8	**71**	15	5	0	0
9	D	11	4	17	**67**	0	0
10	D	6	11	13	**70**	0	0
11	C	18	10	**69**	3	0	0
12	A	**86**	11	1	1	0	0
13	A	**55**	10	21	14	0	0
14	B	32	**57**	3	8	0	0
15	D	5	47	10	**38**	0	0
16	C	10	19	**49**	22	0	0
17	B	21	**41**	18	20	0	0
18	A	**45**	19	21	14	1	0
19	D	11	2	1	**85**	0	0
20	B	3	**81**	9	8	0	0
21	A	**31**	58	6	5	0	0
22	C	2	16	**80**	1	0	0
23	C	19	11	**56**	14	0	0
24	D	7	5	19	**68**	0	0
25	C	8	18	**57**	16	0	0

2026 Higher Chemistry Section 2 Marks Data Analysis

2026 Item	Max Mark	Average Mark	% Average Mark	Discrimination	% Not Attempted
Q1ai	1	0.8	84.8%	31	0.5%
Q1aiiA	1	0.3	30.7%	31	2.4%
Q1aiiB	1	0.6	64.5%	29	1.4%
Q1bi	1	0.5	51.0%	48	4.2%
Q1bii	2	1.3	62.5%	49	3.1%
Q2a	2	0.8	38.1%	53	3.0%
Q2b	1	0.1	13.0%	35	1.5%
Q2c	2	0.3	16.7%	37	1.6%
Q3	3	1.5	49.5%	57	2.3%
Q4ai	1	0.5	49.3%	31	1.8%
Q4aii	1	0.6	56.5%	50	4.7%
Q4b	2	0.9	46.5%	47	1.2%
Q4c	2	1.3	63.5%	64	2.0%
Q4di	1	0.6	59.0%	47	2.6%
Q4dii	1	0.1	14.8%	33	5.0%
Q4ei	1	0.4	36.5%	46	1.8%
Q4eii	1	0.6	55.7%	39	1.1%
Q5a	1	0.5	50.2%	52	3.0%
Q5bi	1	0.4	40.9%	56	3.8%
Q5bii	1	0.9	92.5%	38	1.3%
Q5ci	1	0.6	56.9%	54	3.7%
Q5cii	3	1.8	59.8%	65	2.7%
Q5di	1	0.8	76.9%	49	1.1%
Q5dii	1	0.6	57.5%	57	8.1%
Q6ai	1	0.6	56.6%	53	0.7%
Q6aii	1	0.4	41.1%	37	4.3%
Q6b	2	1.3	63.4%	61	4.3%
Q6ci	1	0.6	64.3%	39	2.7%
Q6cii	2	1.1	56.2%	58	1.9%
Q7ai	1	0.8	78.7%	30	2.9%
Q7aii	1	0.5	45.8%	46	1.5%
Q7aiiiA	1	0.6	56.9%	33	1.1%
Q7aiiiB	2	0.6	28.9%	53	4.4%
Q7aiiiC	2	1.1	53.7%	55	2.9%
Q7aiiiD	1	0.5	51.7%	39	2.1%
Q7bi	1	0.5	54.8%	46	2.0%
Q7bii	1	0.9	86.2%	42	0.9%
Q7biiiA	1	0.7	73.7%	46	1.1%
Q7biiiB	1	0.7	70.1%	50	2.0%
Q7biiiC	1	0.4	40.4%	53	7.7%
Q7biiiD	1	0.4	40.4%	52	9.1%
Q7biiiE	1	0.8	81.0%	46	3.3%
Q7biv	1	0.5	50.1%	41	4.8%
Q7ci	2	0.8	41.2%	50	4.4%
Q7cii	1	0.8	81.3%	46	3.3%
Q8ai	1	0.7	70.3%	54	3.5%
Q8aii	1	0.4	43.4%	54	6.0%
Q8biA	1	0.6	56.8%	51	2.4%
Q8biB	1	0.3	26.3%	35	2.7%
Q8bii	1	0.9	87.4%	31	0.9%
Q8c	1	0.1	12.0%	30	7.7%
Q9	3	1.5	49.1%	66	5.8%
Q10ai	1	0.5	51.9%	57	0.9%
Q10aii	1	0.8	80.6%	44	1.2%
Q10aiii	2	1.3	66.6%	60	1.7%
Q10b	2	1.3	65.3%	54	4.7%
Q10ci	2	1.5	75.6%	57	2.1%
Q10cii	2	1.1	54.0%	54	3.7%
Q11ai	1	0.9	89.1%	43	3.8%
Q11aii	1	0.8	79.9%	51	3.0%
Q11aiii	2	1.1	55.1%	53	2.8%
Q11bi	1	0.7	65.1%	49	1.6%
Q11bii	1	0.1	13.1%	22	5.6%
Q11biiiA	1	0.7	72.9%	46	3.2%
Q11biiiB	3	1.9	64.4%	63	6.5%
Q11biiiC	1	0.1	14.7%	34	8.6%
Q12ai	1	0.7	74.4%	28	2.4%
Q12aii	1	1	96.8%	15	1.5%
Q12bi	1	0.6	55.4%	41	3.4%
Q12bii	1	0.8	78.8%	45	3.3%
Q12ci	1	0.8	78.3%	30	1.1%
Q12cii	1	0.7	73.8%	39	1.6%
Paper	95	53.3	56.1%		

2026 Higher Chemistry Section 2 Frequency of Marks Analysis

2026 Item	0	1	2	3
Q1ai	15	85		
Q1aiiA	69	31		
Q1aiiB	35	65		
Q1bi	49	51		
Q1bii	13	48	39	
Q2a	47	31	23	
Q2b	87	13		
Q2c	72	24	5	
Q3	11	38	42	9
Q4ai	51	49		
Q4aii	43	57		
Q4b	31	44	24	
Q4c	25	22	52	
Q4di	41	59		
Q4dii	85	15		
Q4ei	63	37		
Q4eii	44	56		
Q5a	50	50		
Q5bi	59	41		
Q5bii	8	92		
Q5ci	43	57		
Q5cii	19	20	22	38
Q5di	23	77		
Q5dii	42	58		
Q6ai	43	57		
Q6aii	59	41		
Q6b	29	15	56	
Q6ci	36	64		
Q6cii	19	49	32	
Q7ai	21	79		
Q7aii	54	46		
Q7aiiiA	43	57		
Q7aiiiB	48	47	5	
Q7aiiiC	29	35	36	
Q7aiiiD	48	52		
Q7bi	45	55		
Q7bii	14	86		
Q7biiiA	26	74		
Q7biiiB	30	70		
Q7biiiC	60	40		
Q7biiiD	60	40		
Q7biiiE	19	81		
Q7biv	50	50		
Q7ci	40	38	22	
Q7cii	19	81		
Q8ai	30	70		
Q8aii	57	43		
Q8biA	43	57		
Q8biB	74	26		
Q8bii	13	87		
Q8c	88	12		
Q9	13	37	40	10
Q10ai	48	52		
Q10aii	19	81		
Q10aiii	28	11	61	
Q10b	21	28	51	
Q10ci	15	19	66	
Q10cii	39	13	47	
Q11ai	11	89		
Q11aii	20	80		
Q11aiii	26	37	37	
Q11bi	35	65		
Q11bii	87	13		
Q11biiiA	27	73		
Q11biiiB	23	13	12	52
Q11biiiC	85	15		
Q12ai	26	74		
Q12aii	3	97		
Q12bi	45	55		
Q12bii	21	79		
Q12ci	22	78		
Q12cii	26	74		

2026 Higher Chemistry Section 3 Assignment Marks Analysis

Item	Max Mark	Average Mark	% Average Mark	Discrimination
1	1	1	98.5%	12
2	3	1.7	56.6%	46
3a	1	0.4	37.0%	33
3b	1	1	95.7%	10
3c	1	0.6	58.0%	29
3d	1	0.6	64.0%	32
3e	1	1	96.5%	15
3f	1	0.6	58.4%	33
4a	1	1	96.4%	21
4b	1	0.9	88.6%	21
4c	1	0.9	87.6%	28
4d	1	0.6	63.1%	27
5	1	0.3	30.5%	37
6	1	0.6	62.8%	27
7	3	0.9	30.6%	45
8	1	1	98.6%	13
Paper	20	13	64.9%	

Item	Max Mark	0	1	2	3
1	1	2	98		
2	3	3	38	47	13
3a	1	63	37		
3b	1	4	96		
3c	1	42	58		
3d	1	36	64		
3e	1	3	97		
3f	1	42	58		
4a	1	4	96		
4b	1	11	89		
4c	1	12	88		
4d	1	37	63		
5	1	70	30		
6	1	37	63		
7	3	36	41	18	5
8	1	1	99		
Paper	20				

Item	Max Mark	Item Description
1	1	An aim that describes clearly the purpose of the investigation
2	3	Underlying chemistry
3a	1	A brief summary of the approach used to collect experimental data.
3b	1	Sufficient raw data from the candidate's experiment.
3c	1	Data presented in correctly produced table(s) and correct units shown for calculated values
3d	1	Values calculated correctly using a chemical relationship
3e	1	Data relevant to the experiment obtained from an internet/literature source
3f	1	A citation and reference for a source of the internet/literature data
4a	1	An appropriate format from the options of scatter graph, line graph or bar graph
4b	1	The axis/axes of the graph has/have suitable scale(s).
4c	1	The axes of the graph have suitable labels and units
4d	1	Accurately plotted data points and, where appropriate, a line of best fit.
5	1	A valid comparison of the experimental data with data from the internet/literature source
6	1	A valid conclusion that relates to the aim and is supported by all the data in the report.
7	3	A valid conclusion that relates to the aim and is supported by all the data in the report.
8	1	A valid conclusion that relates to the aim and is supported by all the data in the report.

National
Qualifications
2025

2025 Chemistry

Higher – Paper 1

Question Paper Finalised Marking Instructions

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Marking instructions for each question

Question	Answer	Mark
1.	A	**1**
2.	C	**1**
3.	B	**1**
4.	D	**1**
5.	B	**1**
6.	A	**1**
7.	B	**1**
8.	D	**1**
9.	C	**1**
10.	B	**1**
11.	D	**1**
12.	B	**1**
13.	C	**1**
14.	A	**1**
15.	A	**1**
16.	D	**1**
17.	D	**1**
18.	B	**1**
19.	D	**1**
20.	C	**1**
21.	D	**1**
22.	A	**1**
23.	C	**1**
24.	C	**1**
25.	A	**1**

[END OF MARKING INSTRUCTIONS]

National
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2025

2025 Chemistry

Higher – Paper 2

Question Paper Finalised Marking Instructions

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Marking instructions for each question

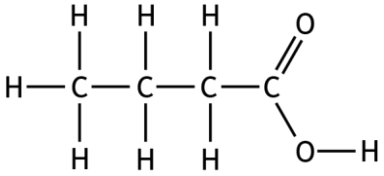
Question			Expected response	Max mark	Additional guidance
1.	(a)	(i)	$Na(g) \longrightarrow Na^{+}(g) + e^{-}$	**1**	State symbols must be shown, negative sign on electron not required.
		(ii)	1735 ($kJ\ mol^{-1}$)	**1**	No units required. No mark can be awarded for correct answer if wrong unit is given (where no unit required, wrong units would only be penalised once in any paper). kJ is acceptable in place of kJ mol−1 (KJ or Kj or KJ mol−1 or Kj mol−1 accepted).
		(iii)	The second ionisation energy (in sodium) involves removal of an electron from an (electron) shell that is inner/full/(more) stable/closer to the nucleus. **OR** Second electron is removed from an (electron) shell that is inner/full/(more) stable/closer to the nucleus. **(1 mark)** The second electron is less screened/the second (electron) shell is less screened. **OR** The second electron is more strongly attracted to/pulled towards the nucleus. **(1 mark)**	**2**	Correct statements made about the 2nd ionisation energy/electron of magnesium can also be credited as long as magnesium is named. Stating that the 2nd electron requires more energy than the 1st electron is not sufficient on its own. Shielding is acceptable in place of screening. Energy level is acceptable in place of electron shell. Increased attraction of the electron for the nucleus would be considered cancelling.
		(iv)	(They have the same number of occupied electron shells but) there are more protons/increased nuclear charge of the Mg^{2+} ion.	**1**	

Question			Expected response	Max mark	Additional guidance
1.	(b)	(i)	(Ionic bonds are) the electrostatic attraction between positive and negative ions. **OR** (An ionic bond is) the electrostatic attraction between oppositely charged ions.	**1**	Accept reference to a single ionic bond being the electrostatic attraction between a positive ion and a negative ion.
		(ii)	6 bananas (if working shown must be correct) **Partial marks** **----------** Mass of potassium in 7 day dose= $8 \times 7 \times 0.0012 \times 39.1 = 2.62752$ (g) **(1 mark)** Any calculated mass of potassium divided by 0.450. (**1 mark**)	**2**	2627.52 (mg) accepted for partial mark.

Question			Expected response	Max mark	Additional guidance
2.	(a)		**Bonding** **metallic** covalent **covalent** covalent **(1 mark)** **Structure** lattice **(covalent) network** molecular **(discrete) molecular/molecule** **(1 mark)**	**2**	**Covalent** lattice is accepted for (covalent) network.
	(b)	(i)	(The covalent radius) decreases.	**1**	
		(ii)	Argon does not form (covalent) bonds.	**1**	Argon is unreactive/monatomic is not accepted on its own.
	(c)		(van der Waals) forces increase/get stronger (going down the group). **(1 mark)** LDFs are the forces (broken between the atoms). **(1 mark)** The number of electrons increases (and so the stronger the forces). **(1 mark)**	**3**	Mention of increasing/stronger covalent/ionic/metallic bonds – cancels the first mark. This mark can only be awarded if no other forces are mentioned (as being broken). Mention of (breaking) bonds/forces between **molecules** cancels one mark.

Question			Expected response	Max mark	Additional guidance
3.			**Award 1 mark** where the candidate has demonstrated, at an appropriate level, a limited understanding of the chemistry involved. They have made some statement(s) which are relevant to the situation, showing that they have understood at least a little of the chemistry within the problem. **Award 2 marks** where the candidate has demonstrated, at an appropriate level, a reasonable understanding of the chemistry involved. They make some statement(s) which are relevant to the situation, showing that they have understood the problem. **Award 3 marks** where the candidate has demonstrated, at an appropriate level, a good understanding of the chemistry involved. They show a good comprehension of the chemistry of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be 'excellent' or 'complete' for the candidate to gain full marks. **Award 0 marks** where the candidate has not demonstrated, at an appropriate level, an understanding of the chemistry involved. There is no evidence that they have recognised the area of chemistry involved, or they have not given any statement of a relevant chemistry principle. Award this mark also if the candidate merely restates the chemistry given in the question.	**3**	

Question			Expected response	Max mark	Additional guidance
4.	(a)	(i)	$C_3H_7OH + 4\frac{1}{2}\ O_2 \rightarrow 3CO_2 + 4H_2O$	**1**	Multiples not accepted.
		(ii)	-1050.04 (kJ mol^{-1}) **If incorrect answer given, a maximum of two partial marks can be awarded for demonstration of the two concepts:** 1 mark for a demonstration of the correct use of the relationship Eh = cmΔT as shown by (4.18 × (an order of magnitude 150) × 13.9). (ignore units for this mark). 1 mark for evidence of scaling up of a calculated value of energy released to one mole. This can be demonstrated by: • correct scaling of calculated Eh from 0.498 to 60 g • 0.498 g ⟷ Eh 60 g ⟷ $\frac{60 \times Eh}{0.498}$ • $\frac{0.498}{60}$ (0.0083) ⟷ Eh 1 ⟷ X	**3**	1050.04 (kJ mol^{-1}) **(2 marks)** No units required. if wrong unit is given (where no unit required, wrong units would only be penalised once in any paper). kJ is acceptable in place of kJ mol^{-1} (KJ or Kj or KJ mol^{-1} or Kj mol^{-1} accepted).
		(iii)	• balance • (alcohol) burner • (copper) can/beaker • thermometer • measuring cylinder/volume measurement • clamp stand/tripod/can suspension method Any four for 2 marks. Any two for 1 mark.	**2**	Bomb calorimeter worth maximum of 1 mark.
		(iv)	Heat/energy loss to the surroundings **OR** Incomplete combustion. **OR** Evaporation of alcohol/fuel.	**1**	Evaporation on its own is not sufficient.
	(b)	(i)	1:3 **OR** 2:6	**1**	Hydrogen:oxygen ratio accepted if labelled.

Question			Expected response	Max mark	Additional guidance
4.	(b)	(ii) (A)	Acidified (potassium/sodium) dichromate. **OR** Hot copper(II)oxide.	**1**	Copper(II)oxide without mention of heating is insufficient for the mark.
		(ii) (B)		**1**	Correct structure of 2-methylpropanoic acid is also accepted.
		(ii) (C)	Butanone/butan-2-one.	**1**	Structure not accepted.
	(c)	(i)	Volumetric/standard flask.	**1**	
		(ii) (A)	Rinse with sodium hydroxide (before filling).	**1**	
		(ii) (B)	20.4 (cm^3)	**1**	Units are not required but must be correct if given. (wrong units would only be penalised once in any paper).

Question			Expected response	Max mark	Additional guidance
4.	(c)	(ii) (C)	0.08316 ($moll^{-1}$) **If incorrect answer given, partial marks can be awarded using a scheme of two "concept" marks, and one "arithmetic" mark.** 1 mark for knowledge of the relationship between moles, concentration and volume. This could be shown by one of the following steps: Calculation of moles sodium hydroxide solution e.g. 0·105 x 0.0198 = 2.079×10^{-3} moles. **OR** Calculation of concentration of ethanoic acid e.g. $2.079 \times 10^{-3} \div 0.025$ **OR** Insertion of correct pairings of values for concentration and volume in a valid titration formula If the relationship between moles, concentration and volume is used more than once, it must be used correctly every time. 1 mark for knowledge of relationship between moles of sodium hydroxide and ethanoic acid. This could be shown by one of the following steps: Calculation of moles ethanoic acid from moles sodium hydroxide i.e. calculated moles of sodium hydroxide used again to calculate concentration of ethanoic acid. **OR** Insertion of correct stoichiometric values in a valid titration formula. 1 mark is awarded for correct arithmetic through the calculation. This mark can only be awarded if both concept marks have been awarded.	**3**	Units are not required but must be correct if given. (wrong units would only be penalised once in any paper)

Question			Expected response	Max mark	Additional guidance
4.	(c)	(iii) (A)	$CH_3COO^-K^+$	**1**	$K^+CH_3COO^-$ is also acceptable.
		(iii) (B)	5.4 (g)	**1**	Units are not required but must be correct if given. (wrong units would only be penalised once in any paper)

Question			Expected response	Max mark	Additional guidance
5.	(a)		(Sucrose) is polar due to its hydroxyl groups. **OR** (Sucrose) can form hydrogen bonds due to its hydroxyl groups. **(1 mark)** An explanation which links solubility of the molecule (sucrose) to the polarity of water/hydrogen bonding of water. **(1 mark)**	**2**	(-)OH accepted as hydroxyl. "Like dissolves like" on its own is not sufficient for the second mark.
	(b)	(i)	Tyr-Pro-Phe-Pro-Gly-Pro-Iso. **OR** Iso-Pro-Gly-Pro-Phe-Pro-Tyr.	**1**	
		(ii)	Essential (amino acid)	**1**	
	(c)	(i)	Ester(s)	**1**	Triglyceride accepted
		(ii) (A)	Glycerol/propane-1,2,3-triol	**1**	propan-1,2,3-triol accepted
		(ii) (B)	Palmitic acid	**1**	
		(iii)	Addition	**1**	Reduction accepted
	(d)	(i)	Ethanal	**1**	
		(ii)	Carbonyl	**1**	
		(iii) (A)	Diterpenoid	**1**	
		(iii) (B)	A correctly circled isoprene unit.	**1**	Including the OH is not accepted.
	(e)		3	**1**	

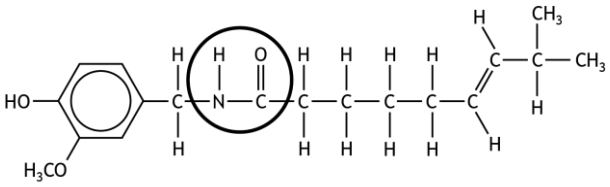
Question			Expected response	Max mark	Additional guidance
6.	(a)		$H_2O_2(aq) + 2H^+(aq) + 2e^- \rightarrow 2H_2O(\ell)$	**1**	Ignore state symbols. Charge on electron not required.
	(b)		$2S_2O_3^{2-}(aq) + I_2(aq) \rightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$	**1**	Ignore state symbols.
	(c)	(i)	10	**1**	
		(ii)	To keep the concentration (of the reactants) constant. **OR** Adding water will change/affect/ dilute/decrease the concentration (of the reactants).	**1**	
	(d)	(i)	(As concentration doubles the rate of reaction) doubles.	**1**	
		(ii)	16.129 (s), 16.13(s)	**1**	Units are not required but must be correct if given. (wrong units would only be penalised once in any paper)
		(iii)	The more particles/ions (in a given volume). **(1 mark)** More successful collisions (and the faster the reaction). **(1 mark)**	**2**	

Question			Expected response	Max mark	Additional guidance
7.	(a)		1129.4 (kJ mol^{-1}) **Partial marks** ---------- Correctly retrieving 3 bond enthalpies from data book (243, 804, 338). **(1 mark)** Correct manipulation of incorrect bond enthalpies (must include the given ΔH = -107.6 kJ mol^{-1}). **(1 mark)**	**2**	Retrieval of a fourth bond enthalpy (ie 360 for CO) is not considered cancelling for the first mark. Units not required, but if stated must be correct. Incorrect units would only be penalised once per paper. kJ is acceptable in place of kJ mol^{-1} (KJ or Kj or KJ mol^{-1} or Kj mol^{-1} accepted).
	(b)		(Cooling favours) exothermic reaction. **(1 mark)** Cooling below 100 °C removes $H_2O(g)$ from equilibrium as $H_2O(\ell)$ **(1 mark)**	**2**	
	(c)	(i)	1 mark for workable apparatus showing reaction of hydrochloric acid with calcium carbonate. 1 mark for showing gas being passed through sealed container of sodium hydroxide.	**2**	Treat each mark and associated labels separately.
		(ii)	16.92/16.9/17(%) **Partial marks** ---------- Correct use of atom economy relationship without correct use of stoichiometry (working must be shown). **(1 mark)** Correct working with no correct answer given. **(1 mark)** Incorrect use of stoichiometry. Answer and working must be shown. **(1 mark)** 0·1692 **(1 mark)**	**2**	

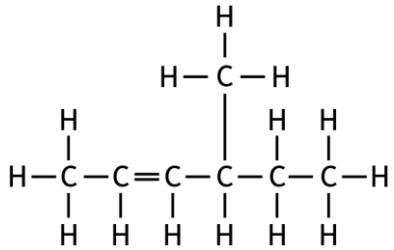
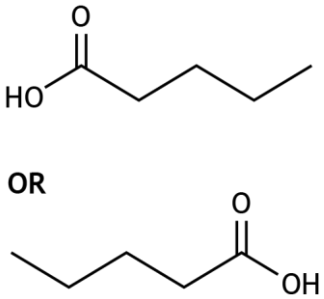
Question			Expected response	Max mark	Additional guidance
8.	(a)	(i)	198.4 (g) **Partial marks** --- Calculating number of moles of SnO_2 required (1.316). **(1 mark)** **OR** Calculating the theoretical mass of tin required to produce 100 g at 64% yield (156.25). **(1 mark)** **OR** Calculating mass of SnO_2 required for 100% yield (126.96). **(1 mark)** **OR** Follow through from an incorrectly calculated 100% yield. **(1 mark)**	**2**	
		(ii)	Carbon is in excess by 0.083 (moles). **Partial marks** --- Moles of SnO_2 = 0.167 AND moles of C = 0.25 **(1 mark)** **OR** Allow for follow through from correctly applying mole ratio and subtraction to calculated number of moles **(1 mark)** --- By proportion 25.2 ◄► 2.007 grams **(1 mark)** **OR** Allow for follow through from calculated mass of excess carbon converted to moles. **(1 mark)**	**2**	A **maximum of 1 mark** can be awarded if carbon is not stated as being the reactant in excess. **1 mark** can be awarded for identification of tin(IV) oxide as the reactant in excess if supported by candidates working.

Question			Expected response	Max mark	Additional guidance
8.	(a)	(iii)	49.92 (g) **Partial marks** --- Calculating moles of CO_2 **(1 mark)** Correctly applying mole ratio to work out a mass for an incorrect number of moles. **(1 mark)** --- By proportion 150.7 g ◄► 80 litres **(1 mark)**	**2**	
	(b)	(i)	Filtration	**1**	
		(ii)	Seawater is a free/cheap raw material. **OR** Calcium chloride can be sold as a by product.	**1**	

Question			Expected response	Max mark	Additional guidance
9.	(a)	(i)	Propagation	**1**	
		(ii)	An (unpaired) electron	**1**	Do not accept electrons.
	(b)		Denaturing	**1**	
	(c)		Cosmetics/food products/plastics	**1**	
	(d)	(i)	Oxygen/gas is escaping	**1**	
		(ii)	Two from: • mass/volume/surface area/size of potato discs • concentration of H_2O_2 • temperature (of H_2O_2 or pH solution)	**2**	Either from • type/variety of potato, or • (discs from the) same potato Accepted for 1 mark Weight is accepted for mass (of potato discs).
		(iii)	As the pH increases the rate (of reaction) increases up to pH 10/ reaches a maximum/optimum at pH 10 (and then decreases). **OR** As the pH increases the rate (of reaction) increases and then decreases.	**1**	Rate must be mentioned.

Question			Expected response	Max mark	Additional guidance
10.	(a)	(i)	 capsaicin (CAP)	**1**	
		(ii)	Bromine solution would quickly/rapidly decolourise with CAP/capsaicin. **OR** Bromine solution will decolourise with CAP/capsaicin but not with DHC/ dihydrocapsaicin. **OR** Bromine solution will not decolourise with DHC/dihydrocapsaicin but will with CAP/capsaicin.	**1**	
	(b)		One from: • Capsaicinoids do not dissolve/are (relatively) insoluble in water • Capsaicinoids are hydrophobic • Capsaicinoids are non-polar	**1**	An answer referring to a named capsaicinoid is accepted. It/they is taken as referring to capsaicinoid.
	(c)		135 (s)	**1**	2.25 minutes is not accepted Sec(s) is not accepted as unit.
	(d)		3(p) **Partial marks** --- Mass of CAP in cream = 0.015 (g). **OR** Evidence of a calculated mass of CAP × (£)1.93/193(p). **OR** 60 g of CAP costs £115.82 **OR** 0.025% of a calculated cost of 60 g CAP **OR** $\frac{60}{1000}$ x £48.26	**2**	£0.03 is accepted. Rounding of the final answer to the nearest correct monetary unit.
	(e)		$C_6H_8O_6 \rightarrow C_6H_6O_6 + 2H^+ + 2e^-$	**1**	State symbols are not required, but if shown must be correct. Negative sign on electron not required.

Question			Expected response	Max mark	Additional guidance
11.			**Award 1 mark** where the candidate has demonstrated, at an appropriate level, a limited understanding of the chemistry involved. They have made some statement(s) which are relevant to the situation, showing that they have understood at least a little of the chemistry within the problem. **Award 2 marks** where the candidate has demonstrated, at an appropriate level, a reasonable understanding of the chemistry involved. They make some statement(s) which are relevant to the situation, showing that they have understood the problem. **Award 3 marks** where the candidate has demonstrated, at an appropriate level, a good understanding of the chemistry involved. They show a good comprehension of the chemistry of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be 'excellent' or 'complete' for the candidate to gain full marks. **Award 0 marks** where the candidate has not demonstrated, at an appropriate level, an understanding of the chemistry involved. There is no evidence that they have recognised the area of chemistry involved, or they have not given any statement of a relevant chemistry principle. Award this mark also if the candidate merely restates the chemistry given in the question.	**3**	

Question			Expected response	Max mark	Additional guidance
12.	(a)		2,7-dimethyloctane	**1**	
	(b)			**1**	Any correct structure for 4-methylhex-2-ene
	(c)		 **OR**	**1**	

[END OF MARKING INSTRUCTIONS]

General marking principles for Higher Chemistry

Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.

(a) Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.

(b) If a candidate response does not seem to be covered by either the principles or detailed marking instructions, and you are uncertain how to assess it, you must seek guidance from your team leader.

(c) Do not award half marks.

(d) Where a candidate makes an error at an early stage in a multi-stage calculation, award marks for correct follow-on working in subsequent stages, unless the error significantly reduces the complexity of the remaining stages. Apply the same principle for questions that require several stages of non-mathematical reasoning. The exception to this rule is where the marking instructions for a numerical question assign separate 'concept marks' and an 'arithmetic mark'. In such situations, the marking instructions will give clear guidance on the assignment or partial marks.

(e) Unless a numerical question specifically requires evidence of working to be shown, award full marks for a correct final response (including units) on its own.

(f) Candidates may fully access larger mark allocations whether their responses are in continuous prose, linked statements, or a series of developed bullet points.

(g) Do not deduct marks for inaccurate or unconventional spelling or vocabulary as long as the meaning of the word(s) is conveyed. **For example**, responses that include 'distilling' for 'distillation', or 'it gets hotter' for 'the temperature rises', are acceptable.

(h) In many questions, the unit in which the answer is to be expressed is given. In these questions, the candidate does not need to state a unit in their answer; but if they do, the unit must be correct. The full mark allocation cannot be awarded if an incorrect unit is shown. In these questions, incorrect units would only be penalised once in any paper.

(i) If a correct response is followed by a wrong response, award no marks. **For example** in response to the question, 'State the colour seen when blue Fehling's solution is warmed with an aldehyde', do not award marks for the response 'red green'. However, if a correct response is followed by additional information which does not conflict with that, ignore the additional information, whether correct or not. **For example** in response to a question concerned with melting point, 'State why the tube should not be made of copper', the response 'Copper has a low melting point and is coloured grey' **would** gain marks.

(j) Award full marks for the correct response to a calculation without working. Award partial marks, as shown in the detailed marking instructions, when working is given but the final response is incorrect. An exception is when candidates are asked to 'Find, by calculation' - do not award full marks for the correct response without working.

(k) Ignore the omission of one H atom from a full structural formula provided the bond is shown.

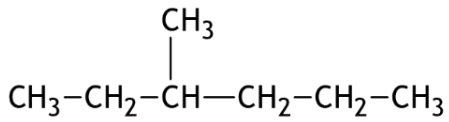
(l) Award marks for a symbol or correct formula in place of a name **unless stated otherwise in the detailed marking instructions.**

(m) When formulae of ionic compounds are given as responses, candidates only need to show ion charges if these have been specifically asked for. However, if ion charges are shown, they must be correct. If incorrect charges are shown, do not award marks.

(n) If an answer comes directly from the text of the question, do not award marks. **For example**, in response to the question, 'A student found that 0·05 mol of propane, C_3H_8 burned to give 82·4 kJ of energy. $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(\ell)$. Name the kind of enthalpy change that the student measured', do not award marks for 'burning' since the word 'burned' appears in the text.

(o) A guiding principle in marking is to give credit for correct elements of a response rather than to look for reasons not to give marks.

Example 1: The structure of a hydrocarbon found in petrol is shown below.



Name the hydrocarbon

- Award the full mark for '3, methyl-hexane', although the punctuation is not correct.

Example 2: A student measured the pH of four carboxylic acids to find out how their strength is related to the number of chlorine atoms in the molecule. The results are shown.

CH_3COOH	1.65
$CH_2ClCOOH$	1.27
$CHCl_2COOH$	0.90
CCl_3COOH	0.51

Describe the relationship between the number of chlorine atoms in the molecule and the strengths of the acids.

- Award the full mark for a response such as 'the more Cl_2, the stronger the acid', although not completely correct.

(p) Unless the question is clearly about a non-chemistry issue, for example costs in an industrial chemical process, do not award marks for a non-chemical response.
For example, in response to the question, 'Why does the (catalytic) converter have a honeycomb structure?', do not award a mark for 'To make it work'. This response may be correct but it is not a chemical response.

(q) Only award marks for a valid response to the question asked. Where candidates are asked to:

- **identify, name, give** or **state**, they must only name or present in brief form.
- **describe**, they must provide a statement or structure of characteristics and/or features.
- **explain**, they must relate cause and effect and/or make relationships between things clear.
- **compare**, they must demonstrate knowledge and understanding of the similarities and/or differences between things.
- **complete**, they must finish a chemical equation or fill in a table with information.
- **determine** or **calculate**, they must determine a number from given facts, figures or information.
- **draw**, they must draw a diagram or structural formula, for example 'Draw a diagram to show the part of a poly(propene) molecule formed from two propene molecules.'
- **estimate**, they must determine an approximate value for something.
- **predict**, they must suggest what may happen based on available information.
- **evaluate**, they must make a judgement based on criteria.
- **suggest**, they must apply their knowledge and understanding of chemistry to a new situation. A number of responses are acceptable: award marks for any suggestions that are supported by knowledge and understanding of chemistry.
- **use their knowledge of chemistry or aspect of chemistry to comment on**, they must apply their skills, knowledge and understanding to respond appropriately to the problem/situation presented (for example by making a statement of principle(s) involved and/or a relationship or equation, and applying these to respond to the problem/situation). Candidates gain marks for the breadth and/or depth of their conceptual understanding.
- **write**, they must complete a chemical or word equation, for example 'Write the word equation for the complete combustion of ethanol.'

2025 Higher Chemistry Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	A	**87**	5	4	4	0	0
2	C	2	10	**77**	12	0	0
3	B	15	**72**	8	4	1	0
4	D	4	38	4	**53**	0	0
5	B	2	**90**	7	1	0	0
6	A	**93**	2	4	2	0	0
7	B	6	**69**	17	8	0	0
8	D	12	13	11	**64**	1	0
9	C	6	12	**74**	7	1	0
10	B	5	**72**	7	15	0	0
11	D	11	2	5	**82**	0	0
12	B	7	**84**	7	2	0	0
13	C	3	10	**79**	8	0	0
14	A	**71**	3	18	8	0	0
15	A	**64**	29	4	3	0	0
16	D	18	16	8	**57**	0	0
17	D	3	16	3	**78**	0	0
18	B	2	**74**	4	20	1	0
19	D	5	7	6	**82**	0	0
20	C	17	28	**46**	8	0	0
21	D	14	20	22	**43**	1	0
22	A	**62**	18	3	17	0	0
23	C	22	19	**36**	21	1	0
24	C	20	6	**62**	12	0	0
25	A	**69**	26	3	2	0	0

2025 Higher Chemistry Section 2 Marks Data Analysis

2025 Item	Max Mark	Average Mark	% Average Mark	Discrimination	% Not Attempted
Q1ai	1	0.8	82.4%	29	0.4%
Q1aii	1	0.8	75.5%	40	1.2%
Q1aiii	2	0.7	36.1%	43	0.9%
Q1aiv	1	0.6	58.2%	45	2.5%
Q1bi	1	0.1	14.3%	29	2.7%
Q1bii	2	1.3	64.1%	47	1.2%
Q2a	2	1.4	70.1%	38	0.6%
Q2bi	1	0.9	89.2%	31	0.3%
Q2bii	1	0.5	48.4%	38	1.5%
Q2c	3	2.2	74.3%	54	1.8%
Q3	3	1.2	39.5%	62	9.3%
Q4ai	1	0.4	42.0%	35	1.3%
Q4aii	3	1.7	56.4%	62	2.2%
Q4aiii	2	1	47.9%	40	1.0%
Q4aiv	1	0.8	75.2%	49	2.6%
Q4bi	1	0.8	80.4%	48	2.0%
Q4biiA	1	0.4	45.0%	42	3.7%
Q4biiB	1	0.7	68.3%	59	8.3%
Q4biiC	1	0.5	52.5%	55	4.5%
Q4ci	1	0.5	45.7%	39	0.5%
Q4ciiA	1	0.7	65.9%	40	1.7%
Q4ciiB	1	0.7	68.3%	35	0.5%
Q4ciiC	3	2.3	78.1%	54	2.3%
Q4ciiiA	1	0.4	43.8%	42	8.0%
Q4ciiiB	1	0.8	79.7%	47	1.8%
Q5a	2	1.1	54.6%	56	1.6%
Q5bi	1	0.3	26.7%	40	8.3%
Q5bii	1	0.9	92.9%	40	2.3%
Q5ci	1	0.6	59.4%	37	3.4%
Q5ciiA	1	0.8	79.2%	43	1.3%
Q5ciiB	1	0.7	71.4%	33	2.1%
Q5ciii	1	0.6	59.5%	44	1.6%
Q5di	1	0.7	67.6%	52	1.8%
Q5dii	1	0.8	83.5%	48	0.8%
Q5diiiA	1	0.8	81.9%	39	1.2%
Q5diiiB	1	0.6	60.8%	38	6.5%
Q5e	1	0.9	85.5%	30	1.7%
Q6a	1	0.5	47.4%	50	7.1%
Q6b	1	0.8	76.9%	37	2.9%
Q6ci	1	1	98.3%	10	0.6%
Q6cii	1	0.3	31.0%	26	0.4%
Q6di	1	0.7	65.3%	34	0.7%
Q6dii	1	0.7	65.5%	42	0.9%
Q6diii	2	1.2	61.5%	43	1.4%
Q7a	2	1.3	63.7%	53	2.9%
Q7b	2	0.7	33.9%	54	4.1%
Q7ci	2	0.8	40.5%	50	8.1%
Q7cii	2	1.8	88.5%	41	1.4%
Q8ai	2	0.9	44.1%	55	6.1%
Q8aii	2	1.5	77.3%	55	3.4%
Q8aiii	2	1.5	76.6%	59	5.8%
Q8bi	1	0.5	54.8%	34	3.7%
Q8bii	1	0.3	25.5%	27	8.5%
Q9ai	1	0.9	92.1%	40	3.2%
Q9aii	1	0.5	50.3%	41	3.1%
Q9b	1	0.9	92.1%	37	2.8%
Q9c	1	0.8	84.0%	31	2.9%
Q9di	1	0.5	48.1%	44	3.1%
Q9dii	2	1.4	70.0%	30	1.0%
Q9diii	1	0.1	7.6%	20	2.8%
Q10ai	1	0.9	86.8%	36	1.4%
Q10aii	1	0.5	46.7%	50	6.0%
Q10b	1	0.5	47.3%	13	4.3%
Q10c	1	0.5	53.6%	32	1.6%
Q10d	2	1	48.2%	45	3.1%
Q10e	1	0.5	52.0%	54	8.1%
Q11	3	0.7	23.4%	55	20.1%
Q12a	1	0.6	61.1%	41	1.9%
Q12b	1	0.7	66.2%	28	2.1%
Q12c	1	0.2	22.9%	21	3.2%
Paper	95	56.5	59.5%		

2025 Higher Chemistry Section 2 Frequency of Marks Analysis

2025 Item	0	1	2	3
Q1ai	18	82		
Q1aii	24	76		
Q1aiii	41	46	13	
Q1aiv	42	58		
Q1bi	86	14		
Q1bii	24	23	52	
Q2a	12	36	52	
Q2bi	11	89		
Q2bii	52	48		
Q2c	9	12	27	53
Q3	26	38	27	9
Q4ai	58	42		
Q4aii	21	21	25	33
Q4aiii	28	49	23	
Q4aiv	25	75		
Q4bi	20	80		
Q4biiA	55	45		
Q4biiB	32	68		
Q4biiC	47	53		
Q4ci	54	46		
Q4ciiA	34	66		
Q4ciiB	32	68		
Q4ciiC	11	11	10	68
Q4ciiiA	56	44		
Q4ciiiB	20	80		
Q5a	27	37	36	
Q5bi	73	27		
Q5bii	7	93		
Q5ci	41	59		
Q5ciiA	21	79		
Q5ciiB	29	71		
Q5ciii	41	59		
Q5di	32	68		
Q5dii	17	83		
Q5diiiA	18	82		
Q5diiiB	39	61		
Q5e	15	85		
Q6a	53	47		
Q6b	23	77		
Q6ci	2	98		
Q6cii	69	31		
Q6di	35	65		
Q6dii	35	65		
Q6diii	20	38	43	
Q7a	22	30	49	
Q7b	36	60	4	
Q7ci	46	27	27	
Q7cii	8	7	85	
Q8ai	35	43	23	
Q8aii	13	19	68	
Q8aiii	18	10	71	
Q8bi	45	55		
Q8bii	74	26		
Q9ai	8	92		
Q9aii	50	50		
Q9b	8	92		
Q9c	16	84		
Q9di	52	48		
Q9dii	12	36	52	
Q9diii	92	8		
Q10ai	13	87		
Q10aii	53	47		
Q10b	53	47		
Q10c	46	54		
Q10d	30	44	26	
Q10e	48	52		
Q11	57	23	14	7
Q12a	39	61		
Q12b	34	66		
Q12c	77	23		

2025 Higher Chemistry Section 3 Assignment Marks Analysis

Item	Max Mark	Average Mark	% Average Mark	Discrimination
1	1	1	98.3%	14
2	3	1.7	55.1%	43
3a	1	0.3	34.3%	30
3b	1	1	95.6%	10
3c	1	0.6	64.1%	31
3d	1	0.6	60.8%	30
3e	1	1	97.1%	19
3f	1	0.6	57.5%	31
4a	1	1	95.9%	23
4b	1	0.9	88.0%	21
4c	1	0.9	87.3%	26
4d	1	0.7	68.4%	29
5	1	0.3	25.5%	35
6	1	0.6	63.3%	21
7	3	1	32.7%	44
8	1	1	97.6%	13
Paper	20	13	64.8%	

Item	Max Mark	0	1	2	3
1	1	2	98		
2	3	3	42	43	12
3a	1	66	34		
3b	1	4	96		
3c	1	36	64		
3d	1	39	61		
3e	1	3	97		
3f	1	42	58		
4a	1	4	96		
4b	1	12	88		
4c	1	13	87		
4d	1	32	68		
5	1	75	25		
6	1	37	63		
7	3	33	41	20	6
8	1	2	98		
Paper	20				

Item	Max Mark	Item Description
1	1	An aim that describes clearly the purpose of the investigation
2	3	Underlying chemistry
3a	1	A brief summary of the approach used to collect experimental data.
3b	1	Sufficient raw data from the candidate's experiment.
3c	1	Data presented in correctly produced table(s) and correct units shown for calculated values
3d	1	Values calculated correctly using a chemical relationship
3e	1	Data relevant to the experiment obtained from an internet/literature source
3f	1	A citation and reference for a source of the internet/literature data
4a	1	An appropriate format from the options of scatter graph, line graph or bar graph
4b	1	The axis/axes of the graph has/have suitable scale(s).
4c	1	The axes of the graph have suitable labels and units
4d	1	Accurately plotted data points and, where appropriate, a line of best fit.
5	1	A valid comparison of the experimental data with data from the internet/literature source
6	1	A valid conclusion that relates to the aim and is supported by all the data in the report.
7	3	A valid conclusion that relates to the aim and is supported by all the data in the report.
8	1	A valid conclusion that relates to the aim and is supported by all the data in the report.

National
Qualifications
2024

2024 Chemistry

Higher – Paper 1

Question Paper Finalised Marking Instructions

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Marking instructions for each question

Question	Answer	Mark
1.	C	**1**
2.	D	**1**
3.	C	**1**
4.	C	**1**
5.	B	**1**
6.	A	**1**
7.	C	**1**
8.	A	**1**
9.	B	**1**
10.	A	**1**
11.	B	**1**
12.	C	**1**
13.	D	**1**
14.	C	**1**
15.	A	**1**
16.	D	**1**
17.	D	**1**
18.	B	**1**
19.	A	**1**
20.	D	**1**
21.	A	**1**
22.	B	**1**
23.	C	**1**
24.	B	**1**
25.	B	**1**

[END OF MARKING INSTRUCTIONS]

National
Qualifications
2024

2024 Chemistry

Higher – Paper 2

Question Paper Finalised Marking Instructions

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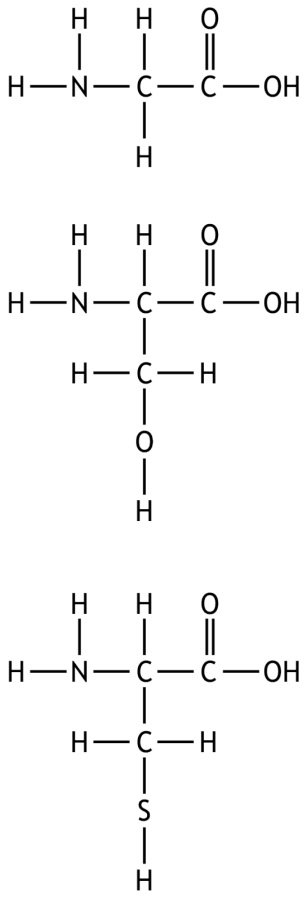
Marking instructions for each question

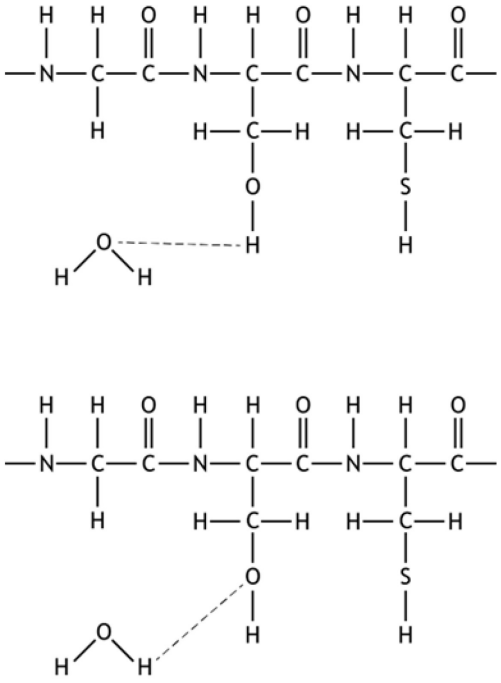
Question			Expected response	Max mark	Additional guidance
1.	(a)	(i)	(The first ionisation energy is) the energy required to remove one mole of electrons from one mole of gaseous atoms.	**1**	
		(ii)	$P(g) \rightarrow P^{+}(g) + e^{-}$	**1**	State symbols must be shown. Negative sign on electron not required.
	(b)		Stronger intermolecular/Van der Waals forces between phosphorus compared with nitrogen. **(1 mark)** London dispersion forces/LDFs are the intermolecular forces present. **(1 mark)** There are more electrons in phosphorus/P_4 compared to nitrogen/N_2. **(1 mark)**	**3**	Any mention of breaking covalent/ionic/metallic bonds is cancelling.
	(c)		230 (kJ mol^{-1}) **Partial marks:** (6 x 201) + (2 x (-)488) **(1 mark)** **OR** recognising that bond breaking is endothermic and bond making is exothermic. **(1 mark)**	**2**	-230 (kJ mol^{-1}) 1 mark No units required. (Wrong units would only be penalised once in any paper). (KJ mol^{-1} or Kj mol^{-1} or KJ mol^{-1} accepted).
	(d)	(i)	77/77.1(%) **Partial marks:** Calculates theoretical mass = 149.85 (g) **(1 mark)** **OR** Correctly calculates the number of moles of reactant (2.42) and product (1.21). **(1 mark)** **OR** Follow through from incorrectly calculated theoretical mass using relationship correctly. **(1 mark)**	**2**	
		(ii)	Reducing agent	**1**	

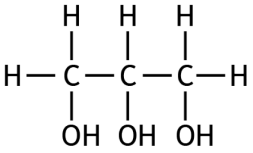
Question			Expected response	Max mark	Additional guidance
1.	(d)	(iii)	145 (litres) **Partial marks:** 280/28 = 10 **(1 mark)** **OR** 1450 divided by incorrectly calculated ratio **(1 mark)** **OR** 1450/280 = 5.18 **(1 mark)** **OR** Incorrectly calculated ratio x 28 **(1 mark)** **OR** by proportion 280 (g) = 1450 (l) **(1 mark)**	**2**	No units required. (Wrong units would only be penalised once in any paper).

Question			Expected response	Max mark	Additional guidance
2.	(a)	(i)	Activation energy	**1**	
		(ii)	(-)52.43, (-)52.4, (-)52 (kJ) Negative sign is not required	**1**	No units required. (Wrong units would only be penalised once in any paper). (KJ or Kj or KJ accepted). kJ mol^{-1} is not acceptable
		(iii)	-/1-/-1	**1**	The word negative is not accepted on its own.
	(b)		**10**Na + **2**KNO_3 → K_2O + **5**Na_2O + N_2	**1**	Or correct multiples
	(c)		Silicon dioxide is a covalent network **(1 mark)** (Strong) covalent bonds are broken **(1 mark)**	**2**	'Covalent lattice' is acceptable, but lattice on its own is not. Naming silicon as the covalent network is cancelling for first mark. Covalent network molecule is cancelling for first mark. A diagram of a molecule is cancelling for the first mark.
	(d)		47.26, 47.3, 47 (litres) OR 47260, 47300, 47000 ($cm^{3)}$ **(2 marks)** **Partial marks:** Correctly calculated number of moles of sodium azide multiplied by 1.6 to give number of moles of nitrogen eg moles of nitrogen = 1.969 **(1 mark)** **OR** An incorrectly calculated number of moles nitrogen x 24 **(1 mark)** **OR** by proportion 65 g → 38.4 l **(1 mark)** Allow follow through from incorrect multiple of 24 l **(1 mark)**	**2**	No units required. Only 1 mark can be awarded for the correct answer if wrong unit is given. (wrong units would only be penalised once in any paper)
	(e)		(Powders) have larger/higher surface area **(1 mark)** More successful collisions **(1 mark)**	**2**	

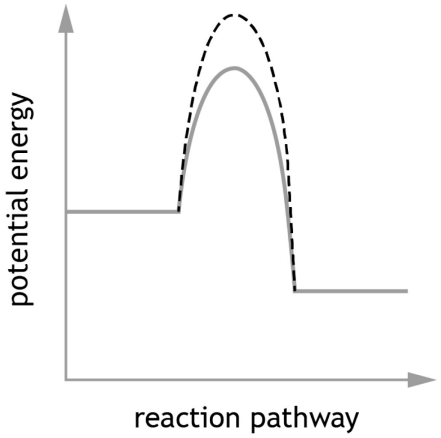
Question			Expected response	Max mark	Additional guidance
3.			**Award 1 mark** where the candidate has demonstrated, at an appropriate level, a limited understanding of the chemistry involved. They have made some statement(s) which are relevant to the situation, showing that they have understood at least a little of the chemistry within the problem. **Award 2 marks** where the candidate has demonstrated, at an appropriate level, a reasonable understanding of the chemistry involved. They make some statement(s) which are relevant to the situation, showing that they have understood the problem. **Award 3 marks** where the candidate has demonstrated, at an appropriate level, a good understanding of the chemistry involved. They show a good comprehension of the chemistry of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be 'excellent' or 'complete' for the candidate to gain full marks. **Award 0 marks** where the candidate has not demonstrated, at an appropriate level, an understanding of the chemistry involved. There is no evidence that they have recognised the area of chemistry involved, or they have not given any statement of a relevant chemistry principle. Award this mark also if the candidate merely restates the chemistry given in the question.	**3**	

Question			Expected response	Max mark	Additional guidance
4.	(a)	(i)	Water	**1**	
		(ii)	Oxidation/condensation	**1**	Accept dehydrogenation
		(iii)	An amino acid that must be/is obtained from the diet/cannot be made in the body	**1**	
		(iv) (A)	Denaturing	**1**	Accept denature(s)/denatured.
		(iv) (B)	Any of the three correct amino acid structures 	**1**	All bonds and atoms must be correct to award this mark.

Question			Expected response	Max mark	Additional guidance
4.	(a)	(iv) (C)		**1**	If more than one hydrogen bond indicated, all must be correct. If partial charges are shown they must be correct.
	(b)	(i)	0.467/0.47	**1**	Ignore any units.
		(ii)	The spots for glycine and lysine overlap/are close together. **OR** The R_f values are similar/close together for glycine and lysine.	**1**	
		(iii)	Threonine	**1**	

Question			Expected response	Max mark	Additional guidance
5.	(a)		Fats are a concentrated source of energy **OR** Fats are essential for the transport/storage of fat-soluble vitamins in the body	**1**	
	(b)	(i)		**1**	Accept correct shortened structural formula.
		(ii)	Ester (link)	**1**	
	(c)		Fats are more saturated (with more regular shape). **(1 mark)** This means they can pack together more closely/more compact structure. **(1 mark)** Stronger LDF/forces of attraction between the molecules. **(1 mark)**	**3**	Accept correct answer in terms of oils.
	(d)	(i)	Rancid	**1**	
		(ii)	Bromine solution will quickly/rapidly decolourise with molecule B. **OR** Bromine solution will decolourise with molecule B but not molecule A **OR** Bromine solution will not decolourise with molecule A but will with molecule B.	**1**	
		(iii) (A)	Atoms or molecules with unpaired electron(s).	**1**	Accept species in place of atoms or molecules. If reference is made to a singular atom, electron must also be singular.
		(iii) (B)	Initiation	**1**	

Question			Expected response	Max mark	Additional guidance
5.	(d)	(iv) (A)	Can react with free radicals forming stable molecules.	**1**	Accept: Donates/provides electron(s) (to pair with an unpaired electron) **OR** Acts as a reducing agent
		(iv) (B)	Vitamin E is less polar (than vitamin C) **(1 mark)** An explanation that links solubility of vitamin E to the polarity of fats and oils **(1 mark)**	**2**	Accept response in terms of vitamin C.
	(e)	(i)	$C_{13}H_{27}COONa$ **OR** $C_{13}H_{27}COO^-Na^+$ **OR** $(C_{13}H_{27}COO^-)(Na^+)$	**1**	Accept structural formula. Bond between O and Na is cancelling.
		(ii)	To prevent scum forming	**1**	Accept an answer in terms of soap
		(iii) (A)	2-methylbuta-1,3-diene	**1**	Apply GMP (o).
		(iii) (B)	3	**1**	

Question			Expected response	Max mark	Additional guidance
6.	(a)	(i)	Methyl octanoate	**1**	
		(ii) (A)	Diagram shows a labelled safe method of heating (eg water/sand bath or heating mantle) and labelled reaction mixture. **(1 mark)** Diagram shows a labelled method of preventing the vapours escaping e.g. wet paper towel/small test tube of cold water/condenser at neck of test tube. **(1 mark)**	**2**	A closed system/stoppered test tube is not acceptable for the second mark.
		(ii) (B)		**1**	The peak drawn shows a higher activation energy. The line drawn by the candidate must start and finish at the same potential energies and positions on the reaction pathway.
	(b)	(i)	Correctly identifies that the molecule has two parts with different polarities. **(1 mark)** Correctly identifies that the polar part/hydroxyl groups dissolve in water/polar liquids/are hydrophillic whilst the non-polar part/hydrocarbon chain dissolve in biodiesel/non-polar liquids/are hydrophobic **(1 mark)** **OR** Correctly identifies that the molecule has two parts that are hydrophillic and hydrophobic. **(1 mark)** Correctly identifies that the hydrophillic part dissolves in water/polar liquids and the hydrophobic part dissolves in biodiesel/non-polar liquids **(1 mark)**	**2**	
		(ii)	Reacting (edible oils) with glycerol	**1**	

Question			Expected response	Max mark	Additional guidance
6.	(c)	(i)	**1 from:** Distance between burner/wick/flame and container Material/type of container Stirring water in both Size of wick Use same draught shield Indication of same insulation in both experiments	**1**	Distance from fuel to container is not accepted. Same mass of fuel is not accepted Same mass/volume of water is not accepted.
		(ii)	**1 from:** Use a draught shield Move flame closer to container Fit a lid on the container Insulate container	**1**	
		(iii)	(-)41.8 (kJ g^{-1}) (3 marks) If final answer is wrong a maximum of 2 marks for the following may be awarded: 1 mark for a demonstration of the correct use of the relationship Eh = cmΔT as shown by (4·18 x (an order of magnitude of 2) x 21) (ignore units for this mark). 1 mark for evidence of scaling up of a calculated value of energy released to 1 gram.	**3**	No units required. A maximum of 2 marks can be awarded for correct answer if wrong unit is given (where no unit required, wrong units would only be penalised once in any paper). kJ is acceptable in place of kJ g^{-1} (KJ or Kj or KJ g^{-1} or Kj g^{-1} accepted).
	(d)		£12.75	**1**	

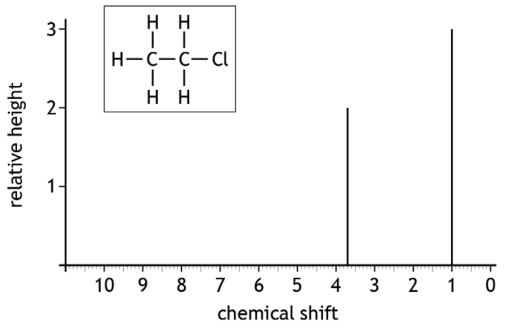
Question			Expected response	Max mark	Additional guidance
7.	(a)		19/19.4/19.37 (g l^{-1}) **Partial marks:** 19373.4 **(1 mark)** **OR** Correct conversion of incorrectly calculated value to g l^{-1} **(1 mark)**	**2**	Answer must be given in g $l^{-1.}$
	(b)	(i) (A)	Solution of accurately/precisely/ exactly known concentration	**1**	
		(i) (B)	Rinse with seawater/sample/solution **(1 mark)** Fill above the mark and allow liquid to drop down to the mark (and read from the bottom of the meniscus) **(1 mark)** **OR** Fill to mark, read from the bottom of the meniscus **(1 mark)**	**2**	Rinse with water is not accepted.
		(ii)	To identify the end point.	**1**	
		(iii) (A)	Only concordant values are used/(the titre for) sample 1 is not concordant.	**1**	
		(iii) (B)	0.5775 (mol l^{-1}) If incorrect answer given, a maximum of two marks can be awarded for demonstration of knowledge of the two concepts: 1 mark for knowledge of the relationship between moles, concentration and volume. 1 mark for application of mole ratio.	**3**	No units required. Only two marks can be awarded if wrong unit is given. (wrong units would only be penalised once in any paper).

Question			Expected response	Max mark	Additional guidance
7.	(b)	(iii) (C)	17.47 g **Partial marking:** Calculation of mass of chloride in 10 cm^3 sample = 0.1747 (g) **(1 mark)** **OR** multiplying calculated mass of chloride ions by 100 **(1 mark)** **OR** Calculation of total number of moles in 1 litre = 0.492 **(1 mark)** **OR** For multiplying calculated number of moles by 35.5 **(1 mark)**	**2**	No units required. Only one mark can be awarded if wrong unit is given. (wrong units would only be penalised once in any paper).
	(c)		Chloromethane has permanent dipole to permanent dipole interactions whereas water has hydrogen bonding. **(1 mark)** Hydrogen bonding is stronger than permanent dipole to permanent dipole interactions. **(1 mark)**	**2**	

Question			Expected response	Max mark	Additional guidance
8.			**Award 1 mark** where the candidate has demonstrated, at an appropriate level, a limited understanding of the chemistry involved. They have made some statement(s) which are relevant to the situation, showing that they have understood at least a little of the chemistry within the problem. **Award 2 marks** where the candidate has demonstrated, at an appropriate level, a reasonable understanding of the chemistry involved. They make some statement(s) which are relevant to the situation, showing that they have understood the problem. **Award 3 marks** where the candidate has demonstrated, at an appropriate level, a good understanding of the chemistry involved. They show a good comprehension of the chemistry of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be 'excellent' or 'complete' for the candidate to gain full marks. **Award 0 marks** where the candidate has not demonstrated, at an appropriate level, an understanding of the chemistry involved. There is no evidence that they have recognised the area of chemistry involved, or they have not given any statement of a relevant chemistry principle. Award this mark also if the candidate merely restates the chemistry given in the question.	**3**	

Question			Expected response	Max mark	Additional guidance
9.	(a)	(i)	2843.55 (kg) **Partial marking:** 1 mark for moles Cl_2 required = 40050 moles **(1 mark)** **OR** Follow through for an arithmetically correct answer derived by multiplying a wrongly calculated number of moles **(1 mark)** **OR** **By proportion** 79.9(g) = (71 x 2) 142 (g) **(1 mark)** **OR** A correctly calculated mass of chlorine from an incorrect mole ratio **OR** Correct application of the mole ratio to an incorrectly calculated number of moles of TiO_2.	**2**	No units required. Only one mark can be awarded if wrong unit is given. (wrong units would only be penalised once in any paper).
		(ii)	Covalent molecular.	**1**	Accept discrete covalent molecular.
	(b)	(i)	20%/20.1%/20.08 % **Partial marking:** Correct use of atom economy relationship without correct use of stoichiometry (ie using 1 mole of Mg) (working must be shown) **(1 mark)** **OR** Correct use of atom economy relationship with correct use of stoichiometry (working must be shown) for magnesium chloride **(1 mark)** **OR** Correct retrieval of GFMs and use of stoichiometry. **(1 mark)** **OR** Correct working with no correct answer given. **(1 mark)**	**2**	

Question			Expected response	Max mark	Additional guidance
9.	(b)	(ii)	$Mg(s) \rightarrow Mg^{2+}(aq) + 2e^-$	**1**	Ignore state symbols. Negative sign on e not required. Accept double headed arrow. Accept correct multiples.
		(iii)	10854 (moles of magnesium) **Partial marking:** **Using moles:** Correct calculation of moles of titanium chloride = 10005 **and** moles magnesium = 30864 **(1 mark)** **OR** Correct application of mole ratio. eg 20010 moles magnesium requires 10005 moles titanium chloride (or any order of magnitude eg 20 or 10). **(1 mark)** Allow for follow through from calculated number of moles of magnesium minus a calculated number of moles derived from titanium chloride **(1 mark)**	**3**	
		(iv)	Argon is inert/unreactive	**1**	
	(c)		Can be carried out at lower temperature/requires less energy/ has a higher atom economy	**1**	

Question			Expected response	Max mark	Additional guidance
10.	(a)		6.0 - 4.5	**1**	
	(b)		Propanone/acetone	**1**	
	(c)			**1**	

[END OF MARKING INSTRUCTIONS]

General marking principles for Higher Chemistry

Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.

(a) Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.

(b) If a candidate response does not seem to be covered by either the principles or detailed marking instructions, and you are uncertain how to assess it, you must seek guidance from your team leader.

(c) Do not award half marks.

(d) Where a candidate makes an error at an early stage in a multi-stage calculation, award marks for correct follow-on working in subsequent stages, unless the error significantly reduces the complexity of the remaining stages. Apply the same principle for questions that require several stages of non-mathematical reasoning. The exception to this rule is where the marking instructions for a numerical question assign separate 'concept marks' and an 'arithmetic mark'. In such situations, the marking instructions will give clear guidance on the assignment or partial marks.

(e) Unless a numerical question specifically requires evidence of working to be shown, award full marks for a correct final response (including units) on its own.

(f) Candidates may fully access larger mark allocations whether their responses are in continuous prose, linked statements, or a series of developed bullet points.

(g) Do not deduct marks for inaccurate or unconventional spelling or vocabulary as long as the meaning of the word(s) is conveyed. **For example**, responses that include 'distilling' for 'distillation', or 'it gets hotter' for 'the temperature rises', are acceptable.

(h) In many questions, the unit in which the answer is to be expressed is given. In these questions, the candidate does not need to state a unit in their answer; but if they do, the unit must be correct. The full mark allocation cannot be awarded if an incorrect unit is shown. In these questions, incorrect units would only be penalised once in any paper.

(i) If a correct response is followed by a wrong response, award no marks. **For example** in response to the question, 'State the colour seen when blue Fehling's solution is warmed with an aldehyde', do not award marks for the response 'red green'. However, if a correct response is followed by additional information which does not conflict with that, ignore the additional information, whether correct or not. **For example** in response to a question concerned with melting point, 'State why the tube should not be made of copper', the response 'Copper has a low melting point and is coloured grey' **would** gain marks.

(j) Award full marks for the correct response to a calculation without working. Award partial marks, as shown in the detailed marking instructions, when working is given but the final response is incorrect. An exception is when candidates are asked to 'Find, by calculation' - do not award full marks for the correct response without working.

(k) Ignore the omission of one H atom from a full structural formula provided the bond is shown.

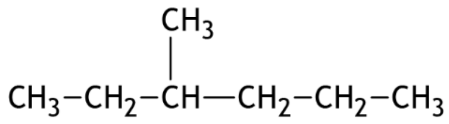
(l) Award marks for a symbol or correct formula in place of a name **unless stated otherwise in the detailed marking instructions.**

(m) When formulae of ionic compounds are given as responses, candidates only need to show ion charges if these have been specifically asked for. However, if ion charges are shown, they must be correct. If incorrect charges are shown, do not award marks.

(n) If an answer comes directly from the text of the question, do not award marks. **For example**, in response to the question, 'A student found that 0·05 mol of propane, C_3H_8 burned to give 82·4 kJ of energy. $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(\ell)$. Name the kind of enthalpy change that the student measured', do not award marks for 'burning' since the word 'burned' appears in the text.

(o) A guiding principle in marking is to give credit for correct elements of a response rather than to look for reasons not to give marks.

Example 1: The structure of a hydrocarbon found in petrol is shown below.



Name the hydrocarbon

- Award the full mark for '3, methyl-hexane', although the punctuation is not correct.

Example 2: A student measured the pH of four carboxylic acids to find out how their strength is related to the number of chlorine atoms in the molecule. The results are shown.

CH_3COOH	1.65
$CH_2ClCOOH$	1.27
$CHCl_2COOH$	0.90
CCl_3COOH	0.51

Describe the relationship between the number of chlorine atoms in the molecule and the strengths of the acids.

- Award the full mark for a response such as 'the more Cl_2, the stronger the acid', although not completely correct.

(p) Unless the question is clearly about a non-chemistry issue, for example costs in an industrial chemical process, do not award marks for a non-chemical response.
For example, in response to the question, 'Why does the (catalytic) converter have a honeycomb structure?', do not award a mark for 'To make it work'. This response may be correct but it is not a chemical response.

(q) Only award marks for a valid response to the question asked. Where candidates are asked to:

- **identify, name, give** or **state**, they must only name or present in brief form.
- **describe**, they must provide a statement or structure of characteristics and/or features.
- **explain**, they must relate cause and effect and/or make relationships between things clear.
- **compare**, they must demonstrate knowledge and understanding of the similarities and/or differences between things.
- **complete**, they must finish a chemical equation or fill in a table with information.
- **determine** or **calculate**, they must determine a number from given facts, figures or information.
- **draw**, they must draw a diagram or structural formula, for example 'Draw a diagram to show the part of a poly(propene) molecule formed from two propene molecules.'
- **estimate**, they must determine an approximate value for something.
- **predict**, they must suggest what may happen based on available information.
- **evaluate**, they must make a judgement based on criteria.
- **suggest**, they must apply their knowledge and understanding of chemistry to a new situation. A number of responses are acceptable: award marks for any suggestions that are supported by knowledge and understanding of chemistry.
- **use their knowledge of chemistry or aspect of chemistry to comment on**, they must apply their skills, knowledge and understanding to respond appropriately to the problem/situation presented (for example by making a statement of principle(s) involved and/or a relationship or equation, and applying these to respond to the problem/situation). Candidates gain marks for the breadth and/or depth of their conceptual understanding.
- **write**, they must complete a chemical or word equation, for example 'Write the word equation for the complete combustion of ethanol.'

2024 Higher Chemistry Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	C	44	1	**53**	2	0	0
2	D	8	7	36	**48**	0	0
3	C	17	7	**62**	13	0	0
4	C	6	31	**53**	11	0	0
5	B	13	**52**	16	19	0	0
6	A	**56**	31	7	6	0	0
7	C	20	4	**70**	6	0	0
8	A	**90**	3	5	2	0	0
9	B	8	**83**	4	5	0	0
10	A	**74**	7	5	14	0	0
11	B	1	**61**	36	3	0	0
12	C	16	17	**59**	8	0	0
13	D	10	14	22	**54**	0	0
14	C	14	4	**72**	10	0	0
15	A	**67**	18	12	3	0	0
16	D	22	19	14	**44**	0	0
17	D	6	5	46	**43**	0	0
18	B	8	**75**	13	4	0	0
19	A	**47**	20	19	13	2	0
20	D	8	2	6	**83**	0	0
21	A	**79**	9	8	4	0	0
22	B	16	**42**	24	18	1	0
23	C	1	34	**41**	25	0	0
24	B	6	**86**	4	4	0	0
25	B	23	**48**	12	16	0	0

2024 Higher Chemistry Section 2 Marks Data Analysis

2024 Item	Max Mark	Average Mark	% Average Mark	Discrimination	% Not Attempted
Q1ai	1	0.5	45.5%	42	1.0%
Q1aii	1	0.8	76.5%	25	0.8%
Q1b	3	1.8	59.0%	57	1.7%
Q1c	2	1.4	68.7%	50	1.8%
Q1di	2	1.4	68.0%	57	1.2%
Q1dii	1	0.3	26.7%	34	7.8%
Q1diii	2	1.5	75.8%	54	5.1%
Q2ai	1	0.9	91.3%	33	1.9%
Q2aii	1	0.5	45.6%	47	7.6%
Q2aiii	1	0.3	26.9%	31	2.6%
Q2b	1	0.7	67.3%	30	1.0%
Q2c	2	0.8	40.5%	51	3.1%
Q2d	2	1.3	65.4%	56	3.9%
Q2e	2	1.1	57.4%	43	1.6%
Q3	3	1.2	39.9%	66	10.2%
Q4ai	1	0.7	68.8%	56	5.6%
Q4aii	1	0.6	58.7%	35	2.8%
Q4aiii	1	0.8	82.7%	45	1.7%
Q4aivA	1	0.8	79.2%	42	8.0%
Q4aivB	1	0.5	54.2%	63	1.7%
Q4aivC	1	0.3	25.7%	37	4.4%
Q4bi	1	0.6	63.3%	37	0.5%
Q4bii	1	0.2	22.0%	20	6.1%
Q4biii	1	0.7	66.8%	30	0.6%
Q5a	1	0.2	21.2%	39	5.8%
Q5bi	1	0.7	74.8%	51	4.0%
Q5bii	1	0.4	39.1%	45	1.8%
Q5c	3	1.4	47.4%	59	3.6%
Q5di	1	0.7	68.4%	44	4.1%
Q5dii	1	0.3	34.7%	46	8.1%
Q5diiiA	1	0.5	48.6%	47	4.3%
Q5diiiB	1	0.8	84.0%	46	2.9%
Q5divA	1	0.4	38.6%	45	6.1%
Q5divB	2	0.3	17.1%	45	8.2%
Q5ei	1	0.2	18.6%	44	13.5%
Q5eii	1	0.7	66.8%	47	4.8%
Q5eiiiA	1	0.2	24.9%	44	9.7%
Q5eiiiB	1	0.8	84.2%	30	1.0%
Q6ai	1	0.4	38.7%	53	5.1%
Q6aiiA	2	0.8	41.7%	50	8.7%
Q6aiiB	1	0.2	19.6%	28	0.6%
Q6bi	2	0.9	46.6%	58	7.5%
Q6bii	1	0.3	35.0%	52	13.9%
Q6ci	1	0.2	23.1%	23	0.5%
Q6cii	1	0.8	75.1%	25	1.1%
Q6ciii	3	1.7	57.8%	62	2.8%
Q6d	1	0.7	69.2%	31	2.1%
Q7a	2	1.3	62.9%	49	4.4%
Q7biA	1	0.6	59.8%	47	4.6%
Q7biB	2	0.7	32.9%	45	5.3%
Q7bii	1	0.5	46.0%	43	2.0%
Q7biiiA	1	0.6	56.1%	42	2.2%
Q7biiiB	3	1.5	50.9%	64	6.2%
Q7biiiC	2	1.1	54.3%	52	6.5%
Q7c	2	0.9	46.6%	56	4.1%
Q8	3	1	33.1%	71	11.7%
Q9ai	2	1.4	68.5%	52	5.7%
Q9aii	1	0.3	27.5%	28	2.4%
Q9bi	2	1.5	75.3%	55	2.5%
Q9bii	1	0.6	64.6%	39	7.8%
Q9biii	3	1.3	43.1%	58	8.3%
Q9biv	1	0.3	30.9%	30	8.5%
Q9c	1	0.3	35.0%	37	6.6%
Q10a	1	0.8	79.6%	31	6.9%
Q10b	1	0.2	20.8%	25	5.9%
Q10c	1	0.4	44.0%	42	9.1%
Paper	95	48.7	51.2%		

2024 Higher Chemistry Section 2 Frequency of Marks Analysis

2024 Item	0	1	2	3
Q1ai	54	46		
Q1aii	23	77		
Q1b	24	14	22	39
Q1c	16	31	53	
Q1di	22	20	58	
Q1dii	73	27		
Q1diii	14	21	65	
Q2ai	9	91		
Q2aii	54	46		
Q2aiii	73	27		
Q2b	33	67		
Q2c	48	23	29	
Q2d	24	21	55	
Q2e	21	43	36	
Q3	29	34	27	11
Q4ai	31	69		
Q4aii	41	59		
Q4aiii	17	83		
Q4aivA	21	79		
Q4aivB	46	54		
Q4aivC	74	26		
Q4bi	37	63		
Q4bii	78	22		
Q4biii	33	67		
Q5a	79	21		
Q5bi	25	75		
Q5bii	61	39		
Q5c	24	26	32	17
Q5di	32	68		
Q5dii	65	35		
Q5diiiA	51	49		
Q5diiiB	16	84		
Q5divA	61	39		
Q5divB	74	18	8	
Q5ei	81	19		
Q5eii	33	67		
Q5eiiiA	75	25		
Q5eiiiB	16	84		
Q6ai	61	39		
Q6aiiA	39	39	22	
Q6aiiB	80	20		
Q6bi	37	33	30	
Q6bii	65	35		
Q6ci	77	23		
Q6cii	25	75		
Q6ciii	26	21	7	46
Q6d	31	69		
Q7a	16	41	42	
Q7biA	40	60		
Q7biB	49	35	15	
Q7bii	54	46		
Q7biiiA	44	56		
Q7biiiB	34	18	11	38
Q7biiiC	29	33	38	
Q7c	35	38	28	
Q8	42	26	21	10
Q9ai	21	22	58	
Q9aii	73	27		
Q9bi	15	20	66	
Q9bii	35	65		
Q9biii	36	20	23	21
Q9biv	69	31		
Q9c	65	35		
Q10a	20	80		
Q10b	79	21		
Q10c	56	44		

2024 Higher Chemistry Section 3 Assignment Marks Analysis

Item	Max Mark	Average Mark	% Average Mark	Discrimination
1	1	1	97.8%	14
2	3	1.7	58.0%	44
3a	1	0.4	35.5%	27
3b	1	1	95.1%	12
3c	1	0.6	57.8%	33
3d	1	0.6	60.7%	28
3e	1	1	95.6%	20
3f	1	0.5	53.6%	34
4a	1	1	98.1%	19
4b	1	0.9	88.5%	23
4c	1	0.9	85.3%	27
4d	1	0.7	65.1%	28
5	1	0.3	25.7%	33
6	1	0.6	59.2%	28
7	3	0.9	30.1%	41
8	1	1	97.4%	17
Paper	20	12.8	64.0%	

Item	Max Mark	0	1	2	3
1	1	2	98		
2	3	4	37	41	19
3a	1	65	35		
3b	1	5	95		
3c	1	42	58		
3d	1	39	61		
3e	1	4	96		
3f	1	46	54		
4a	1	2	98		
4b	1	11	89		
4c	1	15	85		
4d	1	35	65		
5	1	74	26		
6	1	41	59		
7	3	38	38	18	5
8	1	3	97		
Paper	20				

Item	Max Mark	Item Description
1	1	An aim that describes clearly the purpose of the investigation
2	3	Underlying chemistry
3a	1	A brief summary of the approach used to collect experimental data.
3b	1	Sufficient raw data from the candidate's experiment.
3c	1	Data presented in correctly produced table(s) and correct units shown for calculated values
3d	1	Values calculated correctly using a chemical relationship
3e	1	Data relevant to the experiment obtained from an internet/literature source
3f	1	A citation and reference for a source of the internet/literature data
4a	1	An appropriate format from the options of scatter graph, line graph or bar graph
4b	1	The axis/axes of the graph has/have suitable scale(s).
4c	1	The axes of the graph have suitable labels and units
4d	1	Accurately plotted data points and, where appropriate, a line of best fit.
5	1	A valid comparison of the experimental data with data from the internet/literature source
6	1	A valid conclusion that relates to the aim and is supported by all the data in the report.
7	3	A valid conclusion that relates to the aim and is supported by all the data in the report.
8	1	A valid conclusion that relates to the aim and is supported by all the data in the report.

National
Qualifications
2023

2023 Chemistry

Higher

Finalised Marking Instructions

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Marking instructions for each question

Question	Answer	Mark
1.	A	**1**
2.	D	**1**
3.	C	**1**
4.	A	**1**
5.	C	**1**
6.	B	**1**
7.	B	**1**
8.	A	**1**
9.	C	**1**
10.	B	**1**
11.	C	**1**
12.	D	**1**
13.	D	**1**
14.	C	**1**
15.	D	**1**
16.	B	**1**
17.	C	**1**
18.	B	**1**
19.	C	**1**
20.	B	**1**
21.	D	**1**
22.	A	**1**
23.	D	**1**
24.	A	**1**
25.	B	**1**

[END OF MARKING INSTRUCTIONS]

National
Qualifications
2023

2023 Chemistry

Higher Paper 2

Finalised Marking Instructions

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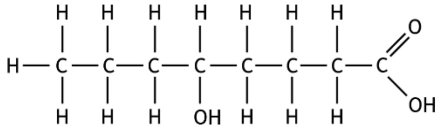
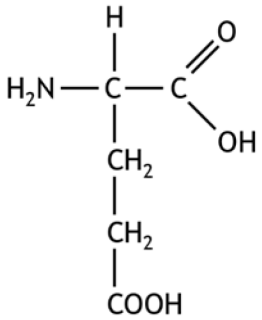
Marking instructions for each question

Question			Expected response	Max mark	Additional guidance
1.	(a)	(i)	Increasing greater/stronger/larger nuclear charge (holds electrons more tightly). **OR** Increasing number of/more protons.	**1**	Increased nuclear pull is not accepted on its own. Mention must be made of nuclear charge or number of protons. Increased attraction of electron for the nucleus would be considered cancelling.
		(ii)	b or j	**1**	A correct letter must be shown.
		(iii) (A)	Second ionisation energy involves removal of an electron **from an electron shell** that is inner/ full(whole)/(more) stable/closer to the nucleus. **OR** Second electron is removed **from an electron shell** that is inner/ full(whole)/(more) stable/closer to the nucleus. **(1 mark)** The second electron is less screened/the second electron shell is less screened. **OR** The second electron is more strongly attracted to/pulled towards the nucleus. **(1 mark)**	**2**	Correct statements made about the 1st ionisation energy/electron can also be credited. Stating that the 2nd electron requires more energy than the 1st electron is not sufficient on its own. Shielding is acceptable in place of screening. Increased attraction of the electron for the nucleus would be considered cancelling.
		(iii) (B)	11472 (kJ mol^{-1})	**1**	No units required. No mark can be awarded for correct answer if wrong unit is given (where no unit required, wrong units would only be penalised once in any paper). kJ is acceptable in place of kJ mol^{-1} (KJ or Kj or KJ mol^{-1} or Kj mol^{-1} accepted).

Question			Expected response	Max mark	Additional guidance
1.	(b)	(i)	Elec**tronegativity is the (measure of) attraction an atom/nucleus has for the electrons in a bond/shared** electrons. **(1 mark)**	**1**	
		(ii)	(More shells so) increased screening/more screening. **(1 mark)** (Covalent radius increases/atom size increases/more shells so) attraction of the nucleus/protons for the (outer/shared) electron(s) decreases. **(1 mark)**	**2**	Shielding is acceptable in place of screening. Increased attraction of the electron for the nucleus would be considered cancelling.
		(iii)	Barium **OR** Radium **OR** Strontium	**1**	Correct symbols accepted.

Question			Expected response	Max mark	Additional guidance
2.	(a)	(i)	(Intermolecular/van der Waals) forces increase (going down the group). **(1 mark)** LDFs are the forces (broken between the molecules). **(1 mark)** The more electrons the stronger the LDFs. **(1 mark)**	**3**	
		(ii)	Hydrogen bonding	**1**	
	(b)	(i)	(+) 34 (kJ mol^{-1}) [(+1517) + (-911) + (-572)] = (+) 34 (kJ mol^{-1}) Partial marks Treat as two concepts. Either would be acceptable for 1 mark. Evidence of understanding of reversal of first enthalpy value ie +1517 or 1517 must be seen. The other two enthalpy values (regardless of value) must be negative, or this partial mark cannot be awarded. **OR** Evidence of understanding of multiplying the third enthalpy value by 2. Ignore the enthalpy sign associated with these numbers.	**2**	If answer given is −34, maximum of 1 mark can be awarded. No units required. Only 1 mark can be awarded for the correct answer if wrong unit is given. (wrong units would only be penalised once in any paper). kJ is acceptable in place of kJ mol^{-1} (KJ or Kj or KJ mol^{-1} or Kj mol^{-1} accepted).
		(ii)	39.9/ 40 (%) **(2 marks)** Partial marks Calculates theoretical mass = 6.4 (g) **(1 mark)** **OR** Correctly calculates no of moles reactant (0.2) and product (0.08) **(1 mark)**	**2**	

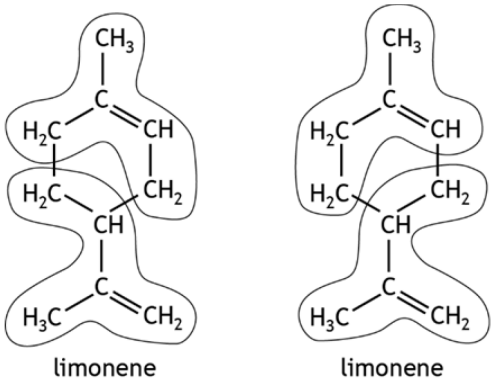
Question			Expected response	Max mark	Additional guidance
2.	(b)	(iii)	In SiO_2 covalent bonds are broken. **(1 mark)** In SiH_4 Van der Waals/LDFs/intermolecular forces are broken. **(1 mark)** Covalent bonds need more energy to break than van der Waals/LDFs/ intermolecular forces. **OR** Covalent bonds are stronger than van der Waals/LDFs/intermolecular forces. **(1 mark)**	**3**	A correct description of the relative strength of covalent bonds and van der Waals/LDFs/intermolecular forces is accepted for this mark.

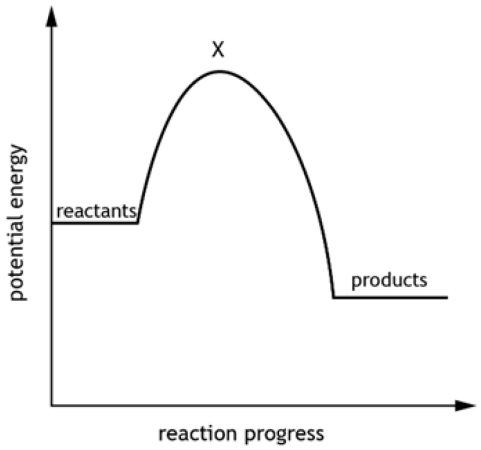
Question			Expected response	Max mark	Additional guidance
3.	(a)	(i) (A)	propane-1,2,3-triol	**1**	propan-1,2,3-triol is accepted Apply general marking principle (o)
		(i) (B)	Condensation/ esterification	**1**	
		(ii) (A)	Carbonyl	**1**	
		(ii) (B)	heptan-2-one	**1**	
		(ii) (C)	Reduction	**1**	Accept hydrogenation
		(ii) (D)	(reaction) 1	**1**	
		(iii)		**1**	
		(iv) (A)	12 – 13 (minutes)	**1**	No units required. Only 1 mark can be awarded for the correct answer if wrong unit is given. (wrong units would only be penalised once in any paper).
		(iv) (B)	dilute sample/use less of sample	**1**	
	(b)	(i)	Essential (Amino acids)	**1**	
		(ii)		**1**	
		(iii)	(The protein) denatures/denatured	**1**	
	(c)		To prevent non-polar and polar liquids separating (into layers)	**1**	
	(d)		Calcium propanoate	**1**	

Question			Expected response	Max mark	Additional guidance
4.			**Award 1 mark** where the candidate has demonstrated, at an appropriate level, a limited understanding of the chemistry involved. They have made some statement(s) which are relevant to the situation, showing that they have understood at least a little of the chemistry within the problem. **Award 2 marks** where the candidate has demonstrated, at an appropriate level, a reasonable understanding of the chemistry involved. They make some statement(s) which are relevant to the situation, showing that they have understood the problem. **Award 3 marks** where the candidate has demonstrated, at an appropriate level, a good understanding of the chemistry involved. They show a good comprehension of the chemistry of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be 'excellent' or 'complete' for the candidate to gain full marks. **Award 0 marks** where the candidate has not demonstrated, at an appropriate level, an understanding of the chemistry involved. There is no evidence that they have recognised the area of chemistry involved, or they have not given any statement of a relevant chemistry principle. Award this mark also if the candidate merely restates the chemistry given in the question.	**3**	

Question			Expected response	Max mark	Additional guidance
5.	(a)	(i) (A)	494.08, 494.1, 494 (litres) 494080, 494100 cm^3 Partial marking Calculating the mass of glucose in 16L ie 1852.8 (g). **(1 mark)** **OR** Number of moles of glucose in 16L 10.29 (moles) (or correctly calculated number of moles from incorrectly calculated mass of glucose). **(1 mark)** An incorrectly calculated number of moles of glucose × (2 × 24). **(1 mark)** **OR** by proportion 180g ↔ 48L **(1 mark)** Calculated mass of glucose ↔ Calculated mass of glucose × $\frac{48}{180}$ **(1 mark)** Failure to scale up to 16 litres of glucose solution but correct use of 50cm^3 and 5.79 g and 1:2 mole ratio will give an answer of 1.544 litres of carbon dioxide. This would be awarded **2 marks**. (working must be shown)	**3**	No units required. Only 2 marks can be awarded for the correct answer if wrong unit is given. (Wrong units would only be penalised once in any paper).

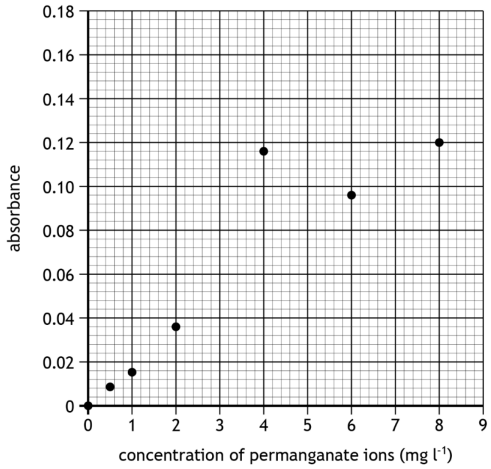
Question			Expected response	Max mark	Additional guidance
5.	(a)	(i) (B)	51, 51.1 (%) Partial mark for correct use of atom economy relationship without correct use of stoichiometry (working must be shown). $\frac{46}{180} \times 100 = 25.6$ **(1 mark)** **OR** Partial mark for correct working with no correct answer given. $\frac{(2 \times 46)}{180} \times 100$ 0.51 **(1 mark)** **OR** Partial mark for correct use of atom economy relationship with correct use of stoichiometry (working must be shown) for carbon dioxide $2 \times \frac{44}{180} \times 100$ = 48.9 % **(1 mark)**	**2**	No units required. Only 1 mark can be awarded for the correct answer if wrong unit is given. (Wrong units would only be penalised once in any paper).

Question			Expected response	Max mark	Additional guidance
5.	(a)	(ii)	12, 12.2, 12.22 (% / abv) Partial mark for correctly calculated change in specific gravity. 1.075-0.985 or 0.09. **(1 mark)** Partial mark for correctly calculated value using an incorrect value for change in specific gravity. **(1 mark)**	**2**	Ignore any units.
	(b)	(i)	**Acidified** dichromate changes from orange to green (blue-green/ blue) with methanol AND no colour change would be observed with propan-2-one. **(2 marks)** **OR** Using **hot** copper oxide a brown solid forms/copper forms/colour change from black to brown would be observed with methanol AND no colour change would be observed with propan-2-one. **(2 marks)**	**2**	Acidified dichromate **(1 mark)** Hot copper oxide **(1 mark)**
		(ii) (A)	Terpene(s)	**1**	
		(ii) (B)	 limonene limonene Only one correct unit needed.	**1**	
		(ii) (C)	Ethanoic acid/acetic acid	**1**	
	(c)		2100 **(1 mark)** mg **(1 mark)** **OR** 2·1 **(1 mark)** g (**1 mark)** **OR** 0-0021 **(1 mark)** kg (**1 mark)**	**2**	If an incorrect mass is calculated but the units are appropriate to the calculation then 1 mark would be awarded. If the candidate's working is unclear in terms of what is being worked out then the mark for units cannot be awarded.

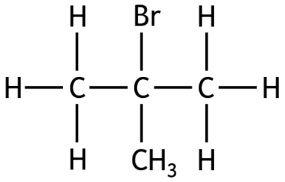
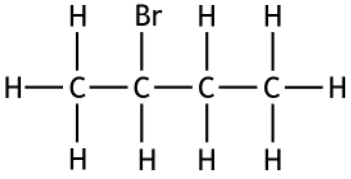
Question			Expected response	Max mark	Additional guidance
6.	(a)	(i) (A)		**1**	
		(i) (B)	Activated complex	**1**	Accept activation complex or transition state.
		(ii)	n HNO_3 = 1316 x 9.5 = 12502 moles **AND** n NH_3 = 220x10^3 / 17 = 12941 **(1 mark)** 1:1 ratio therefore NH_3 is in excess **(1 mark)** **OR** 12502 HNO_3 moles requires 12502 moles NH_3 but have 12941 moles therefore NH_3 in excess. **(2 marks)** --- 1316 l of HNO_3 needs 212.5 kg NH_3 **(1 mark)** **OR** 220 kg NH_3 needs 1362.2 l HNO_3 **(1 mark)** **OR** 12502 HNO_3 moles requires 12502 moles NH_3 but have 12941 moles **(1 mark)** **OR** 12941 moles NH_3 requires 12941 moles HNO_3 but only have 12502 moles HNO_3 **(1 mark)** Therefore, NH_3 is in excess **(1 mark)**	**2**	Accept correct statement for incorrectly calculated values using 1:1 ratio.

Question			Expected response	Max mark	Additional guidance
6.	(a)	(iii)	It has 100% atom economy **OR** Only has 1 product	**1**	
	(b)	(i) (A)	The total number of molecules/ particles.	**1**	
		(i) (B)	Second line displaced to right of original. Peak of curve should be further to the right and no higher than the original peak.	**1**	
		(ii)	A vertical line drawn at a lower kinetic energy than the original E_a shown on graph.	**1**	
	(c)	(i)	The rate of the forward reaction equals the rate of the reverse reaction.	**1**	
		(ii) (A)	(Removal of ammonia/product will) shift equilibrium to right hand side. **(1 mark)** **OR** Increases the yield of ammonia. **(1 mark)** Decreases the rate of the reverse reaction. **(1 mark)** **OR** The rate of the forward reaction is then greater than the rate of the reverse reaction. **(1 mark)**	**2**	
		(ii) (B)	Recycles unreacted gases/ reactants. **OR** Uses a catalyst (to reduce energy costs). **OR** Air is a low cost/free resource.	**1**	
	(d)	(i)	$4NH_3 + 3O_2 \rightarrow 2N_2 + 6H_2O$	**1**	Any correct multiple is accepted. Ignore any state symbols if given.
		(ii)	Ammonia/NH_3	**1**	

Question			Expected response	Max mark	Additional guidance
7.	(a)	(i)	Hard (water)	**1**	
		(ii)	Hydrophilic	**1**	Polar is not accepted.
		(iii)	0.000837 (mol l^{-1}) (3 marks) Partial marks can be awarded using a scheme of two "concept" marks, and one "arithmetic" mark. **1 mark** for knowledge of the relationship between moles, concentration and volume. This could be shown by one of the following steps: • Calculation of moles EDTA solution eg 0·0045 × 0·0093 = 0·00004185 • Calculation of concentration of calcium ions eg 0·00004185 ÷ 0.05 • Insertion of correct pairings of values for concentration and volume in a valid titration formula. If the relationship between moles, concentration and volume is used more than once, it must be used correctly every time. **1 mark** for the relationship between a calculated number of moles of EDTA and calcium ions. **OR** Insertion of correct stoichiometric values in a valid titration formula. **1 mark** is awarded for correct arithmetic through the calculation. This mark can only be awarded if both concept marks have been awarded.	**3**	No units required. Only 2 marks can be awarded for the correct answer if wrong unit is given. (Wrong units would only be penalised once in any paper).

Question			Expected response	Max mark	Additional guidance
7.	(b)		 4.4 to 5.4 **OR** Accurate reading from candidate's drawn line of best fit if value given is out with the range 4.4 to 5.4.	**1**	No units required. No mark can be awarded for the correct answer if wrong unit is given. (Wrong units would only be penalised once in any paper).
	(c)	(i)	Trichloromethane is polar and tetrachloromethane is non-polar. **(1 mark)** Trichloromethane has a permanent dipole and tetrachloromethane does not. **(1 mark)**	**2**	Correct description of a permanent dipole is accepted.
		(ii)	-14 (kJ mol^{-1}) Partial marks: Evidence of the use of all the correct bond enthalpies (or correct multiples thereof) (338, 412, 431, 484, 570 (ignore signs). **(1 mark)** **OR** If only four values are retrieved, the candidate recognises that bond breaking is endothermic and bond formation is exothermic and correctly manipulates the bond enthalpy values they have used to give their answer. **(1 mark)**	**2**	(+)14 (kJ mol^{-1}) would be awarded 1 mark. No units required. Only 1 mark can be awarded for the correct answer if wrong unit is given. kJ is acceptable in place of kJ mol^{-1} (KJ or Kj or KJ mol^{-1} or Kj mol^{-1} accepted). (Wrong units would only be penalised once in any paper).

Question			Expected response	Max mark	Additional guidance
8.			**Award 1 mark** where the candidate has demonstrated, at an appropriate level, a limited understanding of the chemistry involved. They have made some statement(s) which are relevant to the situation, showing that they have understood at least a little of the chemistry within the problem. **Award 2 marks** where the candidate has demonstrated, at an appropriate level, a reasonable understanding of the chemistry involved. They make some statement(s) which are relevant to the situation, showing that they have understood the problem. **Award 3 marks** where the candidate has demonstrated, at an appropriate level, a good understanding of the chemistry involved. They show a good comprehension of the chemistry of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be 'excellent' or 'complete' for the candidate to gain full marks. **Award 0 marks** where the candidate has not demonstrated, at an appropriate level, an understanding of the chemistry involved. There is no evidence that they have recognised the area of chemistry involved, or they have not given any statement of a relevant chemistry principle. Award this mark also if the candidate merely restates the chemistry given in the question.	**3**	

Question			Expected response	Max mark	Additional guidance
9.	(a)	(i)	The more carbons/longer carbon chain the higher the boiling point. **(1 mark)** The further down group 7/the halogen is the higher the boiling point. **(1 mark)**	**2**	Reference to group 7 molecule is cancelling.
		(ii)	Permanent dipole- Permanent dipole interactions.	**1**	Allow permanent dipole-dipole interaction. Accept pd-pd i's.
	(b)	(i)	The halogen/bromine (atom) is attached to a carbon that is attached to two other carbons. **OR** The halogen/bromine (atom) is attached to a carbon that has only one hydrogen attached.	**1**	
		(ii)		**1**	Accept shortened structural formula
	(c)	(i)	Ultraviolet/UV (radiation/light)	**1**	
		(ii)	1 mark for one of the following $\bullet CH_3 + Br\bullet \rightarrow CH_3Br$ $\bullet CH_3 + \bullet CH_3 \rightarrow CH_3CH_3$ $Br\bullet + Br\bullet \rightarrow Br_2$	**1**	
	(d)			**1**	Accept shortened structural formula
	(e)		2-bromo-3-chloro-1,1,1-trifluoropentane	**1**	Apply general marking principle (o)

Question			Expected response	Max mark	Additional guidance
10.	(a)		Tare the balance with the crucible. **(1 mark)** Transfer 1·5 g (into the crucible). **(1 mark)** **OR** Tare a weighing boat, transfer the 1.5 g onto the weighing boat. Record mass. Transfer into the crucible. **(1 mark)** Reweigh the weighing boat and record the mass/calculate the difference. **(1 mark)** **OR** Weigh mixture and weighing boat, record the mass. Transfer mixture into the crucible. **(1 mark)** Reweigh the weighing boat and record the mass/calculate the difference. **(1 mark)** Weigh the crucible (empty) and then with (1.5 g of) mixture. **(1 mark)** Subtract the mass of crucible from mass of crucible and mixture. **(1 mark)**	**2**	Accept use of a filter paper/paper towel/watch glass or other appropriate container instead of a weighing boat.
	(b)	(i)	Allow CO_2/gas to escape.	**1**	
		(ii)	Reactants/products/mixture are not flammable.	**1**	

Question			Expected response	Max mark	Additional guidance
10.	(c)		0.582, 0.58 (g) **(2 marks)** Partial marking Calculating number of moles of CO_2 (0.0069). (**1 mark)** Calculating mass of $MgCO_3$ from incorrectly calculated number of moles of CO_2. **(1 mark)**	**2**	
	(d)	(i)	Diagram shows a workable method of generating gas with mixture in contact with acid. Labels required for mixture/powder/magnesium oxide + magnesium carbonate and (excess) acid.	**1**	
		(ii)	Carbon dioxide is (relatively) insoluble/has very low solubility (in water).	**1**	

[END OF MARKING INSTRUCTIONS]

General marking principles for Higher Chemistry

Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.

(a) Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.

(b) If a candidate response does not seem to be covered by either the principles or detailed marking instructions, and you are uncertain how to assess it, you must seek guidance from your team leader.

(c) Do not award half marks.

(d) Where a candidate makes an error at an early stage in a multi-stage calculation, award marks for correct follow-on working in subsequent stages, unless the error significantly reduces the complexity of the remaining stages. Apply the same principle for questions that require several stages of non-mathematical reasoning. The exception to this rule is where the marking instructions for a numerical question assign separate 'concept marks' and an 'arithmetic mark'. In such situations, the marking instructions will give clear guidance on the assignment or partial marks.

(e) Unless a numerical question specifically requires evidence of working to be shown, award full marks for a correct final response (including units) on its own.

(f) Candidates may fully access larger mark allocations whether their responses are in continuous prose, linked statements, or a series of developed bullet points.

(g) Do not deduct marks for inaccurate or unconventional spelling or vocabulary as long as the meaning of the word(s) is conveyed. **For example**, responses that include 'distilling' for 'distillation', or 'it gets hotter' for 'the temperature rises', are acceptable.

(h) In many questions, the unit in which the answer is to be expressed is given. In these questions, the candidate does not need to state a unit in their answer; but if they do, the unit must be correct. The full mark allocation cannot be awarded if an incorrect unit is shown. In these questions, incorrect units would only be penalised once in any paper.

(i) If a correct response is followed by a wrong response, award no marks. **For example** in response to the question, 'State the colour seen when blue Fehling's solution is warmed with an aldehyde', do not award marks for the response 'red green'. However, if a correct response is followed by additional information which does not conflict with that, ignore the additional information, whether correct or not. **For example** in response to a question concerned with melting point, 'State why the tube should not be made of copper', the response 'Copper has a low melting point and is coloured grey' **would** gain marks.

(j) Award full marks for the correct response to a calculation without working. Award partial marks, as shown in the detailed marking instructions, when working is given but the final response is incorrect. An exception is when candidates are asked to 'Find, by calculation' - do not award full marks for the correct response without working.

(k) Ignore the omission of one H atom from a full structural formula provided the bond is shown.

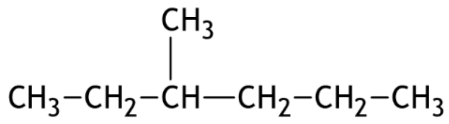
(l) Award marks for a symbol or correct formula in place of a name **unless stated otherwise in the detailed marking instructions.**

(m) When formulae of ionic compounds are given as responses, candidates only need to show ion charges if these have been specifically asked for. However, if ion charges are shown, they must be correct. If incorrect charges are shown, do not award marks.

(n) If an answer comes directly from the text of the question, do not award marks. **For example**, in response to the question, 'A student found that 0·05 mol of propane, C_3H_8 burned to give 82·4 kJ of energy. $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(\ell)$. Name the kind of enthalpy change that the student measured', do not award marks for 'burning' since the word 'burned' appears in the text.

(o) A guiding principle in marking is to give credit for correct elements of a response rather than to look for reasons not to give marks.

Example 1: The structure of a hydrocarbon found in petrol is shown below.



Name the hydrocarbon

- Award the full mark for '3, methyl-hexane', although the punctuation is not correct.

Example 2: A student measured the pH of four carboxylic acids to find out how their strength is related to the number of chlorine atoms in the molecule. The results are shown.

CH_3COOH	1.65
$CH_2ClCOOH$	1.27
$CHCl_2COOH$	0.90
CCl_3COOH	0.51

Describe the relationship between the number of chlorine atoms in the molecule and the strengths of the acids.

- Award the full mark for a response such as 'the more Cl_2, the stronger the acid', although not completely correct.

(p) Unless the question is clearly about a non-chemistry issue, for example costs in an industrial chemical process, do not award marks for a non-chemical response.
For example, in response to the question, 'Why does the (catalytic) converter have a honeycomb structure?', do not award a mark for 'To make it work'. This response may be correct but it is not a chemical response.

(q) Only award marks for a valid response to the question asked. Where candidates are asked to:

- **identify, name, give** or **state**, they must only name or present in brief form.
- **describe**, they must provide a statement or structure of characteristics and/or features.
- **explain**, they must relate cause and effect and/or make relationships between things clear.
- **compare**, they must demonstrate knowledge and understanding of the similarities and/or differences between things.
- **complete**, they must finish a chemical equation or fill in a table with information.
- **determine** or **calculate**, they must determine a number from given facts, figures or information.
- **draw**, they must draw a diagram or structural formula, for example 'Draw a diagram to show the part of a poly(propene) molecule formed from two propene molecules.'
- **estimate**, they must determine an approximate value for something.
- **predict**, they must suggest what may happen based on available information.
- **evaluate**, they must make a judgement based on criteria.
- **suggest**, they must apply their knowledge and understanding of chemistry to a new situation. A number of responses are acceptable: award marks for any suggestions that are supported by knowledge and understanding of chemistry.
- **use their knowledge of chemistry or aspect of chemistry to comment on**, they must apply their skills, knowledge and understanding to respond appropriately to the problem/situation presented (for example by making a statement of principle(s) involved and/or a relationship or equation, and applying these to respond to the problem/situation). Candidates gain marks for the breadth and/or depth of their conceptual understanding.
- **write**, they must complete a chemical or word equation, for example 'Write the word equation for the complete combustion of ethanol.'

2023 Higher Chemistry Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	A	**71**	7	10	12	0	0
2	D	3	3	5	**89**	0	0
3	C	4	3	**85**	8	0	0
4	A	**69**	13	13	5	0	0
5	C	2	5	**91**	2	0	0
6	B	5	**74**	7	13	0	0
7	B	5	**80**	4	11	1	0
8	A	**83**	5	5	6	0	0
9	C	3	5	**87**	5	0	0
10	B	6	**72**	17	5	0	0
11	C	9	3	**83**	5	0	0
12	D	14	20	15	**50**	1	0
13	D	10	31	31	**28**	0	0
14	C	6	8	**70**	16	0	0
15	D	11	5	6	**78**	0	0
16	B	36	**46**	9	9	0	0
17	C	6	13	**59**	22	0	0
18	B	7	**85**	4	4	0	0
19	C	21	7	**53**	18	1	0
20	B	9	**76**	7	8	0	0
21	D	3	1	2	**94**	0	0
22	A	**68**	16	7	9	1	0
23	D	10	33	14	**42**	1	0
24	A	**45**	20	23	11	1	0
25	B	30	**56**	6	8	0	0

2023 Higher Chemistry Section 2 Marks Data Analysis

2023 Item	Max Mark	Average Mark	% Average Mark	Discrimination	% Not Attempted
Q1ai	1	0.7	65.1%	46	1.2%
Q1aii	1	0.4	40.4%	28	1.1%
Q1aiiiA	2	0.8	40.5%	45	1.1%
Q1aiiiB	1	0.5	52.0%	49	3.8%
Q1bi	1	0.5	48.9%	44	2.2%
Q1bii	2	0.8	38.1%	42	1.4%
Q1biii	1	0.5	53.8%	30	1.2%
Q2ai	3	1.8	60.9%	57	3.4%
Q2aii	1	0.9	87.5%	32	0.7%
Q2bi	2	1.6	81.6%	50	4.1%
Q2bii	2	1.5	75.2%	55	1.4%
Q2biii	3	0.5	17.1%	46	4.0%
Q3aiA	1	0.7	72.3%	51	4.1%
Q3aiB	1	0.7	72.0%	39	2.1%
Q3aiiA	1	0.8	77.2%	50	1.3%
Q3aiiB	1	0.6	57.0%	52	2.3%
Q3aiiC	1	0.6	64.2%	46	1.0%
Q3aiiD	1	0.8	78.3%	30	0.8%
Q3aiii	1	0.3	25.7%	50	6.3%
Q3aivA	1	0.8	77.5%	32	1.9%
Q3aivB	1	0.5	45.3%	36	8.4%
Q3bi	1	0.9	91.0%	39	1.8%
Q3bii	1	0.3	34.1%	53	5.3%
Q3biii	1	0.8	84.4%	33	3.6%
Q3c	1	0.1	12.7%	26	2.9%
Q3d	1	0.3	29.2%	39	11.0%
Q4	3	1.2	39.6%	65	10.5%
Q5aiA	3	1.5	51.0%	62	4.0%
Q5aiB	2	1.3	66.7%	54	4.4%
Q5aii	2	1.9	93.2%	24	0.9%
Q5bi	2	0.4	21.0%	41	9.9%
Q5biiA	1	0.5	49.5%	37	3.6%
Q5biiB	1	0.8	77.3%	49	1.2%
Q5biiC	1	0.2	17.8%	44	2.2%
Q5c	2	1.7	84.1%	34	1.6%
Q6aiA	1	0.6	63.3%	23	25.7%
Q6aiB	1	0.8	82.7%	45	3.4%
Q6aii	2	1.1	54.2%	58	5.4%
Q6aiii	1	0.2	16.0%	30	2.7%
Q6biA	1	0.1	6.5%	23	3.9%
Q6biB	1	0.5	49.6%	35	1.0%
Q6bii	1	0.8	81.0%	36	0.8%
Q6ci	1	0.5	45.9%	46	3.6%
Q6ciiA	2	0.6	31.1%	48	4.5%
Q6ciiB	1	0.6	58.6%	24	3.7%
Q6di	1	0.2	17.4%	33	5.1%
Q6dii	1	0.3	28.1%	24	3.4%
Q7ai	1	0.8	76.3%	37	2.5%
Q7aii	1	0.7	65.5%	25	2.0%
Q7aiii	3	1.6	53.2%	59	8.7%
Q7b	1	0.6	56.8%	36	2.3%
Q7ci	2	0.7	36.9%	43	2.6%
Q7cii	2	1.4	71.5%	49	3.8%
Q8	3	1.4	46.7%	64	8.0%
Q9ai	2	1	48.7%	43	1.7%
Q9aii	1	0.2	19.1%	26	1.2%
Q9bi	1	0.7	70.2%	40	1.6%
Q9bii	1	0.8	75.2%	48	3.0%
Q9ci	1	0.5	54.7%	50	4.5%
Q9cii	1	0.5	46.0%	51	8.9%
Q9d	1	0.4	41.6%	28	2.4%
Q9e	1	0.2	21.0%	35	3.3%
Q10a	2	0.7	36.8%	34	9.5%
Q10bi	1	0.6	59.7%	27	2.0%
Q10bii	1	0.4	41.8%	37	3.2%
Q10c	2	1.2	58.7%	53	8.1%
Q10di	1	0.4	37.4%	43	8.4%
Q10dii	1	0.6	55.4%	44	5.6%
Paper	95	49.7	52.3%		

2023 Higher Chemistry Section 2 Frequency of Marks Analysis

2023 Item	0	1	2	3
Q1ai	35	65		
Q1aii	60	40		
Q1aiiiA	39	42	20	
Q1aiiiB	48	52		
Q1bi	51	49		
Q1bii	39	46	15	
Q1biii	46	54		
Q2ai	22	15	22	41
Q2aii	12	88		
Q2bi	11	14	75	
Q2bii	22	6	72	
Q2biii	70	18	4	9
Q3aiA	28	72		
Q3aiB	28	72		
Q3aiiA	23	77		
Q3aiiB	43	57		
Q3aiiC	36	64		
Q3aiiD	22	78		
Q3aiii	74	26		
Q3aivA	22	78		
Q3aivB	55	45		
Q3bi	9	91		
Q3bii	66	34		
Q3biii	16	84		
Q3c	87	13		
Q3d	71	29		
Q4	31	33	23	13
Q5aiA	32	16	18	34
Q5aiB	29	10	62	
Q5aii	3	8	89	
Q5bi	74	9	16	
Q5biiA	51	49		
Q5biiB	23	77		
Q5biiC	82	18		
Q5c	8	16	76	
Q6aiA	37	63		
Q6aiB	17	83		
Q6aii	29	34	37	
Q6aiii	84	16		
Q6biA	94	6		
Q6biB	50	50		
Q6bii	19	81		
Q6ci	54	46		
Q6ciiA	39	60	1	
Q6ciiB	41	59		
Q6di	83	17		
Q6dii	72	28		
Q7ai	24	76		
Q7aii	34	66		
Q7aiii	26	26	9	39
Q7b	43	57		
Q7ci	34	57	8	
Q7cii	13	31	56	
Q8	19	36	30	15
Q9ai	31	41	28	
Q9aii	81	19		
Q9bi	30	70		
Q9bii	25	75		
Q9ci	45	55		
Q9cii	54	46		
Q9d	58	42		
Q9e	79	21		
Q10a	53	21	26	
Q10bi	40	60		
Q10bii	58	42		
Q10c	37	9	54	
Q10di	63	37		
Q10dii	45	55		

National
Qualifications
2022

2022 Chemistry

Higher Paper 1 – Multiple choice

Finalised Marking Instructions

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Marking instructions for each question

Question	Response	Mark
1.	D	**1**
2.	A	**1**
3.	C	**1**
4.	B	**1**
5.	D	**1**
6.	A	**1**
7.	D	**1**
8.	C	**1**
9.	A	**1**
10.	C	**1**
11.	B	**1**
12.	B	**1**
13.	D	**1**
14.	C	**1**
15.	B	**1**
16.	C	**1**
17.	B	**1**
18.	D	**1**
19.	A	**1**
20.	C	**1**
21.	D	**1**
22.	B	**1**
23.	A	**1**
24.	B	**1**
25.	A	**1**

[END OF MARKING INSTRUCTIONS]

National
Qualifications
2022

2022 Chemistry

Higher Paper 2

Finalised Marking Instructions

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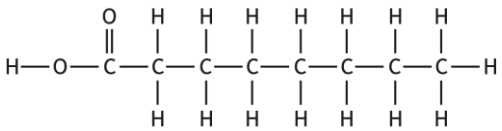
Marking instructions for each question

Question			Expected response	Max mark	Additional guidance
1.	(a)	(i)	(The atoms/nuclei) have the same attraction for the bonding electrons. **OR** (The atoms have) same electronegativity/electronegativity values given **OR** Bonding electrons shared equally (between the atoms).	**1**	Stating that it is a 'pure covalent bond' on its own is not sufficient. An unlabelled diagram on its own not sufficient
		(ii)	Increasing/greater/stronger/larger nuclear charge (holds electrons more tightly). **OR** Increasing number of/more protons.	**1**	Increased nuclear pull is not accepted on its own. Mention must be made of nuclear charge or number of protons. Increased attraction of the electron for the nucleus would be considered cancelling.
	(b)	(i)	The energy required to remove one mole of electrons from one mole of gaseous atoms.	**1**	
		(ii)	(More shells) so increased/more screening/ shielding. **OR** Covalent radius increases/atom size increases/ more shells so attraction of the nucleus/ protons for the outer electron(s) decreases.	**1**	'Shielding effect' by itself is not acceptable.

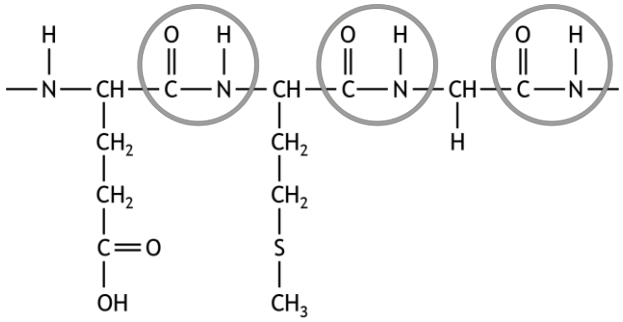
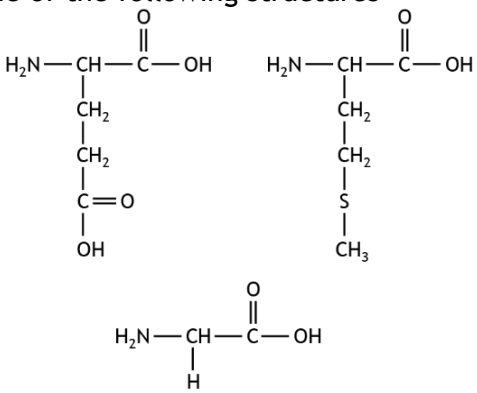
Question			Expected response	Max mark	Additional guidance
	(c)	(i)	Hydrogen bonds/bonding **(1 mark)** Hydrogen bonds occurs between molecules that have hydrogen: • bonded to N, O or F • bonded to a strongly electronegative element • bonded to an element with a large d difference in electronegativity compared to hydrogen • in a highly polar bond. **OR** Answers relating to the attraction between the opposite ends of the permanent dipole in molecules containing hydrogen atoms and the atoms of elements with high electronegativity/large difference in electronegativity **(1 mark)**	**2**	The first mark should not be cancelled by an incorrect explanation of how hydrogen bonding arises or an explanation of another type of bonding/interaction. All three of N, O and F must be listed.
		(ii)	Correctly identify that the London dispersion forces become stronger/increase (in moving from HCl to HI). **(1 mark)** The number of electrons in the molecules increases (from HCl to HI). **(1 mark)**	**2**	

Question			Expected response	Max mark	Additional guidance
2.	(a)		$3KClO_4 + 8Al \rightarrow 3KCl + 4Al_2O_3$	**1**	Correct multiples accepted.
	(b)	(i)	1.4/1.35/1.351 (litres) Partial marking Moles of $KClO_3$ = 0.03752 moles **(1 mark)** A 2:3 ratio applied to an incorrectly calculated number of moles and multiplied by 24 **(1 mark)** **OR** by proportion 245.2 g $\rightarrow$ 72 litres **(1 mark)** Follow through from incorrect multiples of 122.6 or 24 **(1 mark)**	**2**	No units required. Only 1 mark can be awarded for the correct answer if wrong unit is given. (Wrong units would only be penalised once in any paper).
		(ii)	None/no effect	**1**	
		(iii)	2596/2595.6 (kJ)	**1**	Accept -2596/2595.6 Accept kJ per mole ($kJmol^{-1}$) KJ/Kj is acceptable in place of kJ No units required. No mark can be awarded for the correct answer if wrong unit is given. (Wrong units would only be penalised once in any paper).
		(iv)	Increases the number of particles with energy equal to or greater than the activation energy **OR** Increases the number of particles with (sufficient) energy to form an activated complex/to react **(1 mark)** More successful collisions **(1 mark)**	**2**	'Activation' complex accepted.
	(c)		Sodium	**1**	

Question			Expected response	Max mark	Additional guidance
3.			**This is an open-ended question.** **1 mark:** The candidate has demonstrated, at an appropriate level, a limited understanding of the chemistry involved. They have made some statement(s) that are relevant to the situation, showing that they have understood at least a little of the chemistry within the problem. **2 marks:** The candidate has demonstrated, at an appropriate level, a reasonable understanding of the chemistry involved. They make some statement(s) that are relevant to the situation, showing that they have understood the problem. **3 marks:** The maximum available mark would be awarded to a candidate who has demonstrated, at an appropriate level, a good understanding, of the chemistry involved. The candidate shows a good comprehension of the chemistry of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be 'excellent' or 'complete' for the candidate to gain full marks.	**3**	**Award 0 marks** where the candidate has not demonstrated, at an appropriate level, an understanding of the chemistry involved. There is no evidence that they have recognised the area of chemistry involved, or they have not given any statement of a relevant chemistry principle. Award zero marks also if the candidate merely restates the chemistry given in the question.

Question			Expected response	Max mark	Additional guidance
4.	(a)	(i)	Ester (link)	**1**	
		(ii)		**1**	Accept correct shortened structural formula.
		(iii)	Any correctly named alcohol with 7 carbons.	**1**	Position of OH must be given in name.
	(b)	(i)	35 – 45 inclusive	**1**	
		(ii)	Glyceryl trilinoleate	**1**	
	(c)	(i)	By reacting with glycerol.	**1**	Adding/combining/joining to glycerol not accepted.
		(ii)	Correctly identifying that the emulsifier has two parts with different polarities or two parts that are hydrophobic/hydrophilic. **(1 mark)** Hydrophobic part/hydrocarbon chain/fatty acid chain/non-polar part dissolves in non-polar liquids whilst the hydrophilic part/hydroxyl groups/polar part dissolve in polar liquids. **(1 mark)**	**2**	Reference to heads and tails as parts of molecule is accepted.

<table>
<tr><th colspan="3">Question</th><th>Expected response</th><th>Max mark</th><th>Additional guidance</th></tr>
<tr><td>5.</td><td>(a)</td><td></td><td>3-methylbutan-1-ol</td><td>1</td><td></td></tr>
<tr><td></td><td>(b)</td><td>(i)</td><td>$C_3H_7OH \rightarrow C_3H_6O + 2H^+ + 2e^-$</td><td>1</td><td>Ignore state symbols.
Charge on electron not required.</td></tr>
<tr><td></td><td></td><td>(ii)</td><td>To provide hydrogen ions</td><td>1</td><td>Ignore state symbols.</td></tr>
<tr><td></td><td></td><td>(iii)</td><td>Orange to green</td><td>1</td><td>Accept blue or blue/green</td></tr>
<tr><td></td><td></td><td>(iv)</td><td>Tollens’ (reagent)</td><td>1</td><td></td></tr>
<tr><td></td><td></td><td>(v)</td><td>It is a tertiary alcohol.</td><td>1</td><td>Correct description of tertiary alcohol accepted.</td></tr>
<tr><td></td><td></td><td>(vi)</td><td><table><tr><th></th><th>Oxygen to hydrogen ratio</th></tr><tr><td>Butan-1-ol</td><td>1:10</td></tr><tr><td>Butanal</td><td>1:8</td></tr></table></td><td>1</td><td></td></tr>
</table>

Question			Expected response	Max mark	Additional guidance
6.	(a)	(i)	Biological catalyst	**1**	Biochemical catalyst/protein that acts as a catalyst are accepted.
		(ii) (A)		**1**	Any one correctly circled peptide link from those shown. Accept the C-N circled. Either the bond alone or including the C and N.
		(ii) (B)	any one of the following structures 	**1**	Ignore incorrect connectivity in the side chains only.
		(ii) (C)	An amino acid that cannot be made in the body/must be obtained through diet.	**1**	
		(ii) (D)	Condensation	**1**	Condensation polymerisation accepted.
		(iii)	Enzyme becomes denatured/ enzyme changes shape **(1 mark)** Intermolecular/hydrogen bonds are broken **(1 mark)**	**2**	
		(iv)	Diagram showing closed reaction vessel with reactants in contact with each other **(1 mark)** A means of measuring and collecting gas from the closed vessel **(1 mark)** Correct labelling of hydrogen peroxide, sweet potato and oxygen. Oxygen must be labelled inside the gas collection apparatus. **(1 mark)**	**3**	Addition of catalase/water labels are regarded as non-cancelling. Graduations must be shown on gas collection apparatus. First mark not awarded if delivery tube passes through the side of a measuring cylinder.

Question			Expected response	Max mark	Additional guidance
6.	(b)	(i)	To prevent unwanted oxidation/to oxidise in place of the compounds they have been added to protect/to stop (oxidation of edible oils) food acquiring a rancid flavour.	**1**	To 'prevent oxidation' on its own is not sufficient. To stop food going rancid is accepted.
		(ii)	Vitamin C molecule is polar due to its hydroxyl groups. **OR** Vitamin C can form hydrogen bonds due to its hydroxyl groups. **(1 mark)** An explanation which links solubility of vitamin C to the polarity of water/hydrogen bonding of water. **(1 mark)**	**2**	Accept (-OH) for hydroxyl. 'like dissolves like' not sufficient on its own for 1 mark
	(c)		975 (2) g (1) Partial marking 195 (mg) (of solanine) **(1 mark)** **OR** 15 (g) **(1 mark)** **OR** Correct scaling of a calculated mass of solanine to a mass of potato **(1 mark)**	**3**	Allow alternative units of mass as long as these match the numerical answer eg 0.975 kg Units mark can only be awarded with wrong numerical answer as long as evidence of scaling is shown.

Question			Expected response	Max mark	Additional guidance
7.	(a)		0.2/0.20/0.203 g Partial marking Using cmΔT with c = 4·18, m = a factor of 10 of 0.1 and ΔT = 27 (**1 mark**) Applying the concept that the combustion of 1 mole (16 g) of methane burns to produce 891 kJ (**1 mark**)	**3**	No units required. Only 2 marks can be awarded for the correct answer if wrong unit is given. (Wrong units would only be penalised once in any paper).
	(b)		-816 (kJ mol^{-1}) Partial marking 1 mark is available if either of the following operations is correctly executed: **Either** The four relevant values for bond enthalpies of the C-H, O=O, C=O, and O-H (or multiples thereof) are retrieved from the data booklet (412, 498, 804, 463 – ignore signs). **OR** If only three correct values are retrieved, the candidate recognises that bond breaking is endothermic and bond forming is exothermic and have correctly manipulated the bond enthalpies and multiples that they have used with working shown.	**2**	+816 would qualify for 1 mark Bond breaking $(4 \times 412) + (2 \times 498) = 2644$ Bond forming $[(2 \times 804) + (4 \times 463)] = -3460)$ No units required. Only 1 mark can be awarded for the correct answer if wrong unit is given. (Wrong units would only be penalised once in any paper) kJ is acceptable in place of kJ mol^{-1} (KJ or Kj or KJ mol^{-1} or Kj mol^{-1} accepted). If less than three bond enthalpies are retrieved then no mark can be awarded.

<table>
<tr><th colspan="3">Question</th><th>Expected response</th><th>Max mark</th><th>Additional guidance</th></tr>
<tr><td>7.</td><td>(c)</td><td></td><td>17.647/17.65/17.6/18 (%)

Partial mark for correct use of atom economy relationship without correct use of stoichiometry (working must be shown).

Partial marking

Correct working with no correct answer given.

$\frac{(3\times 2)}{(16+18)}\times 100$

$\frac{6}{(16+18)}\times 100$

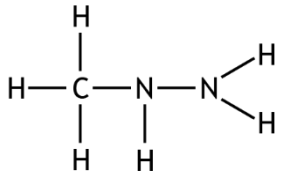
OR

Incorrect use of stoichiometry.

$\frac{2}{(16+18)}\times 100 = 5.88$

Answer and working must be shown.

0.176 (1 mark)</td><td>2</td><td>No units required. Only 1 mark can be awarded for the correct answer if wrong unit is given. (Wrong units would only be penalised once in any paper).</td></tr>
<tr><td></td><td>(d)</td><td></td><td><table><tr><th>Condition</th><th>High/Low</th></tr><tr><td>Temperature</td><td>high</td></tr><tr><td>Pressure</td><td>low</td></tr></table></td><td>1</td><td></td></tr>
</table>

Question			Expected response	Max mark	Additional guidance
7.	(e)	(i)	-4632 (kJ) [(-216) + (+100) + (-1576) + (-3432) + (+492)] = -4632 (kJ) Partial marking Either would be acceptable for 1 mark. Evidence of understanding of reversal of first enthalpy value (ie –54 or -216 must be seen) and the second enthalpy value (ie +20 or +100 must be seen). The other three enthalpy values (regardless of value) must not be reversed, or this partial mark cannot be awarded. **OR** Evidence of understanding of multiplying the first enthalpy value by 4 (+/- 216) **and** the second enthalpy value by 5 (+/1 100) **and** the third enthalpy value by 4 (+/- 1576) **and** the fourth enthalpy value by 12 (+/- 3432) **and** the fifth enthalpy value by 12 (+/- 492). Ignore the enthalpy signs associated with these numbers.	**2**	
		(ii)		**1**	

<table>
<tr><th colspan="3">Question</th><th>Expected response</th><th>Max mark</th><th>Additional guidance</th></tr>
<tr><td>8.</td><td>(a)</td><td>(i)</td><td>One mark for effect on rate (both responses needed), one mark for effect on equilibrium position.

<table><tr><th>Feature of reaction</th><th>Effect of catalyst</th></tr><tr><td>Rate of forward reaction</td><td>Increase</td></tr><tr><td>Rate of reverse reaction</td><td>Increase</td></tr><tr><td>Position of equilibrium</td><td>No effect</td></tr></table></td><td>2</td><td></td></tr>
<tr><td></td><td></td><td>(ii)</td><td>Graph should show decrease in yield as temperature increases.</td><td>1</td><td></td></tr>
<tr><td></td><td>(b)</td><td></td><td>Correctly calculates number of moles of:
Sorbic acid = 0.0625
Potassium hydroxide = 0.125

OR

Working out that 14 g of sorbic acid would be needed to react with potassium hydroxide (1 mark)

Statement demonstrating understanding of limiting reactant:

E.g. that there are fewer moles of sorbic acid therefore it is the limiting reactant

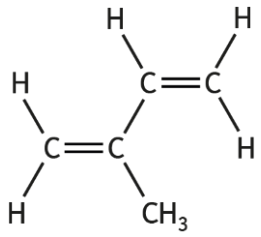
OR

there are more moles of potassium hydroxide therefore it is in excess

OR

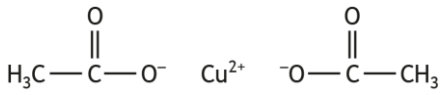
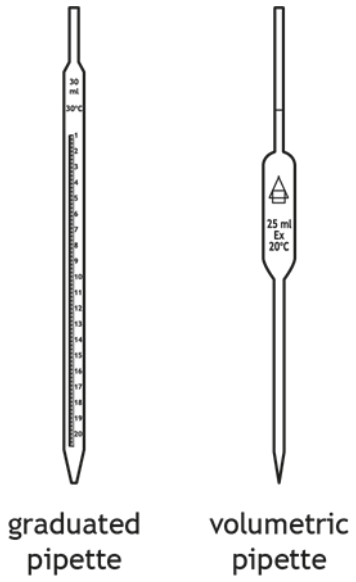
that 0.125 moles of potassium hydroxide would require 0.125 moles of sorbic acid (1 mark)</td><td>2</td><td></td></tr>
</table>

Question			Expected response	Max mark	Additional guidance
8.	(c)		2.52×10^{-5} (moles) Partial marking 1/261.8 (0.00382) in 100 cm^3 0.0126 in 330 cm^3 **(1 mark)** Follow through from incorrectly calculated number of moles multiplied by 0.002 **(1 mark)** **OR** $0.002 \times 3.3 = 0.0066$ **(1 mark)** Follow through from incorrectly calculated mass divided by 261.8 **(1 mark)**	**2**	

Question			Expected response	Max mark	Additional guidance
8.	(d)	(i) (A)	Volatile/non-water soluble/aroma	**1**	
		(B) (I)	Terpene	**1**	
		(B) (II)		**1**	
		(B) (III)	3	**1**	
		(ii) (A)	Water/H_20	**1**	
		(ii) (B)	hydroxyl AND carbonyl	**1**	Formulae -OH/-C=O not accepted. Ether or benzene ring accepted.

Question			Expected response	Max mark	Additional guidance
9.			**This is an open-ended question.** **1 mark:** The candidate has demonstrated, at an appropriate level, a limited understanding of the chemistry involved. They have made some statement(s) that are relevant to the situation, showing that they have understood at least a little of the chemistry within the problem. **2 marks:** The candidate has demonstrated, at an appropriate level, a reasonable understanding of the chemistry involved. They make some statement(s) that are relevant to the situation, showing that they have understood the problem. **3 marks:** The maximum available mark would be awarded to a candidate who has demonstrated, at an appropriate level, a good understanding, of the chemistry involved. The candidate shows a good comprehension of the chemistry of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be 'excellent' or 'complete' for the candidate to gain full marks.	**3**	**Award 0 marks** where the candidate has not demonstrated, at an appropriate level, an understanding of the chemistry involved. There is no evidence that they have recognised the area of chemistry involved, or they have not given any statement of a relevant chemistry principle. Award zero marks also if the candidate merely restates the chemistry given in the question.

Question			Expected response	Max mark	Additional guidance
10.	(a)	(i)	Less fluorine(s) (atoms) give a higher ODP. **OR** More fluorine(s) (atoms) gives a lower ODP. **OR** Less chlorine(s) (atoms) give a lower ODP. **OR** More chlorine(s) (atoms) gives a higher ODP.	**1**	Symbols F/Cl accepted in place of names. Use of fluorine/ chlorine 'molecules' would be considered cancelling.
		(ii)	1 and 5	**1**	$C_2F_4Br_2$ (1) and $C_2F_4Cl_2$ (5)
		(iii)	Carbon dioxide and ammonia do not contain halogen (atoms)/do not damage the ozone layer.	**1**	Accept names of halogens or group 7 elements Use of 'molecules' would be considered as cancelling.
	(b)	(i)	Species (atoms/molecules/particles) with unpaired electrons.	**1**	Use of 'element(s)' would be considered as cancelling.
		(ii) A	initiation	**1**	
		(ii) B	Any correct propagation step accepted, e.g. $F\bullet + CH_3F \rightarrow \bullet CH_2F + HF$ **OR** $F_2 + \bullet CH_2F \rightarrow CH_2F_2 + F\bullet$ **OR** $F_2 + \bullet CH_3 \rightarrow CH_3F + F\bullet$ **OR** $\bullet CH_2F + HF \rightarrow CH_2F_2 + H\bullet$	**1**	
	(c)		0.208/0.21/0.2 (moles)	**1**	No units required. No mark can be awarded for correct answer if wrong unit is given (Where no unit required, wrong units would only be penalised once in any paper).

Question			Expected response	Max mark	Additional guidance
11.	(a)	(i)	water and carbon dioxide	**1**	Formulae accepted.
		(ii)	$(CH_3COO^-)_2Cu^{2+}$ 	**1**	Brackets are required.
	(b)		Dissolve oxalic acid (in small volume of water) **(1 mark)** transfer quantitatively/with rinsings/washings **(1 mark)** fill/make up to the mark/line in a volumetric/standard flask. **(1 mark)**	**3**	If a solvent other than water is mentioned a maximum of 2 marks can be awarded.
	(c)	(i)	Diagram of graduated or bulb pipette. NOT dropping (Pasteur) pipette  graduated pipette volumetric pipette	**1**	Volumetric mark must be shown. End of pipette must narrow to a point.
		(ii)	pink to colourless	**1**	'Clear' is not accepted for colourless.
		(iii)	concordant	**1**	

Question			Expected response	Max mark	Additional guidance
11.	(d)		0.27/0.3 (mol l^{-1}) average titre = 26.75cm^3 n (oxalic acid) = 0.02675 × 0.126 = 3.3705 × 10^{-3} moles n (sodium hydroxide) = 2 × 3.3705 × 10^{-3} =6.741 × 10^{-3} moles concentration = 6.741 × 10^{-3} ÷ 0.025 = 0.27 (0.26964)$moll^{-1}$ **Partial marks can be awarded using a scheme of two "concept" marks, and one "arithmetic" mark.** **1 mark** for knowledge of the relationship between moles, concentration and volume. This could be shown by one of the following steps: Calculation of moles oxalic acid solution e.g. 0·02675 × 0·126 = 3.3705 × 10^{-3} moles **OR** calculation of concentration of sodium hydroxide e.g. 6.741 × 10^{-3} ÷ 0.025 **OR** Insertion of correct pairings of values for concentration and volume in a valid titration formula **1 mark** for knowledge of relationship between moles of oxalic acid and sodium hydroxide. This could be shown by one of the following steps: Calculation of moles sodium hydroxide from moles oxalic acid eg 2 × 3.3705 × 10^{-3} =6.741 × 10^{-3} moles **OR** Insertion of correct stoichiometric values in a valid titration formula **1 mark** is awarded for correct arithmetic through the calculation. This mark can only be awarded if both concept marks have been awarded.	**3**	No units required. Only 2 marks can be awarded for the correct answer if wrong unit is given. (Wrong units would only be penalised once in any paper).

[END OF MARKING INSTRUCTIONS]

General marking principles for Higher Chemistry

Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.

(a) Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.

(b) If a candidate response does not seem to be covered by either the principles or detailed marking instructions, and you are uncertain how to assess it, you must seek guidance from your team leader.

(c) Do not award half marks.

(d) Where a candidate makes an error at an early stage in a multi-stage calculation, award marks for correct follow-on working in subsequent stages, unless the error significantly reduces the complexity of the remaining stages. Apply the same principle for questions that require several stages of non-mathematical reasoning. The exception to this rule is where the marking instructions for a numerical question assign separate 'concept marks' and an 'arithmetic mark'. In such situations, the marking instructions will give clear guidance on the assignment or partial marks.

(e) Unless a numerical question specifically requires evidence of working to be shown, award full marks for a correct final response (including units) on its own.

(f) Candidates may fully access larger mark allocations whether their responses are in continuous prose, linked statements, or a series of developed bullet points.

(g) Do not deduct marks for inaccurate or unconventional spelling or vocabulary as long as the meaning of the word(s) is conveyed. **For example**, responses that include 'distilling' for 'distillation', or 'it gets hotter' for 'the temperature rises', are acceptable.

(h) In many questions, the unit in which the answer is to be expressed is given. In these questions, the candidate does not need to state a unit in their answer; but if they do, the unit must be correct. The full mark allocation cannot be awarded if an incorrect unit is shown. In these questions, incorrect units would only be penalised once in any paper.

(i) If a correct response is followed by a wrong response, award no marks. **For example** in response to the question, 'State the colour seen when blue Fehling's solution is warmed with an aldehyde', do not award marks for the response 'red green'. However, if a correct response is followed by additional information which does not conflict with that, ignore the additional information, whether correct or not. **For example** in response to a question concerned with melting point, 'State why the tube should not be made of copper', the response 'Copper has a low melting point and is coloured grey' **would** gain marks.

(j) Award full marks for the correct response to a calculation without working. Award partial marks, as shown in the detailed marking instructions, when working is given but the final response is incorrect. An exception is when candidates are asked to 'Find, by calculation' - do not award full marks for the correct response without working.

(k) Ignore the omission of one H atom from a full structural formula provided the bond is shown.

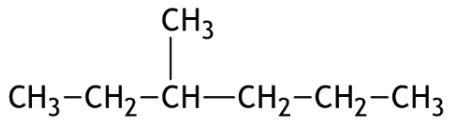
(l) Award marks for a symbol or correct formula in place of a name **unless stated otherwise in the detailed marking instructions.**

(m) When formulae of ionic compounds are given as responses, candidates only need to show ion charges if these have been specifically asked for. However, if ion charges are shown, they must be correct. If incorrect charges are shown, do not award marks.

(n) If an answer comes directly from the text of the question, do not award marks. **For example**, in response to the question, 'A student found that 0·05 mol of propane, C_3H_8 burned to give 82·4 kJ of energy. $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(\ell)$. Name the kind of enthalpy change that the student measured', do not award marks for 'burning' since the word 'burned' appears in the text.

(o) A guiding principle in marking is to give credit for correct elements of a response rather than to look for reasons not to give marks.

Example 1: The structure of a hydrocarbon found in petrol is shown below.



Name the hydrocarbon

- Award the full mark for '3, methyl-hexane', although the punctuation is not correct.

Example 2: A student measured the pH of four carboxylic acids to find out how their strength is related to the number of chlorine atoms in the molecule. The results are shown.

CH_3COOH	1.65
$CH_2ClCOOH$	1.27
$CHCl_2COOH$	0.90
CCl_3COOH	0.51

Describe the relationship between the number of chlorine atoms in the molecule and the strengths of the acids.

- Award the full mark for a response such as 'the more Cl_2, the stronger the acid', although not completely correct.

(p) Unless the question is clearly about a non-chemistry issue, for example costs in an industrial chemical process, do not award marks for a non-chemical response.
For example, in response to the question, 'Why does the (catalytic) converter have a honeycomb structure?', do not award a mark for 'To make it work'. This response may be correct but it is not a chemical response.

(q) Only award marks for a valid response to the question asked. Where candidates are asked to:

- **identify, name, give** or **state**, they must only name or present in brief form.
- **describe**, they must provide a statement or structure of characteristics and/or features.
- **explain**, they must relate cause and effect and/or make relationships between things clear.
- **compare**, they must demonstrate knowledge and understanding of the similarities and/or differences between things.
- **complete**, they must finish a chemical equation or fill in a table with information.
- **determine** or **calculate**, they must determine a number from given facts, figures or information.
- **draw**, they must draw a diagram or structural formula, for example 'Draw a diagram to show the part of a poly(propene) molecule formed from two propene molecules.'
- **estimate**, they must determine an approximate value for something.
- **predict**, they must suggest what may happen based on available information.
- **evaluate**, they must make a judgement based on criteria.
- **suggest**, they must apply their knowledge and understanding of chemistry to a new situation. A number of responses are acceptable: award marks for any suggestions that are supported by knowledge and understanding of chemistry.
- **use their knowledge of chemistry or aspect of chemistry to comment on**, they must apply their skills, knowledge and understanding to respond appropriately to the problem/situation presented (for example by making a statement of principle(s) involved and/or a relationship or equation, and applying these to respond to the problem/situation). Candidates gain marks for the breadth and/or depth of their conceptual understanding.
- **write**, they must complete a chemical or word equation, for example 'Write the word equation for the complete combustion of ethanol.'

2022 Higher Chemistry Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	**D**	16	3	7	**74**	0	0
2	**A**	**67**	15	15	2	0	0
3	**C**	17	30	**48**	4	1	0
4	**B**	16	**63**	11	10	0	0
5	**D**	17	18	19	**45**	1	0
6	**A**	**49**	21	18	11	1	0
7	**D**	2	5	9	**84**	0	0
8	**C**	16	20	**52**	11	1	0
9	**A**	**82**	5	7	6	0	0
10	**C**	1	6	**73**	20	0	0
11	**B**	10	**48**	26	15	0	0
12	**B**	3	**86**	8	3	0	0
13	**D**	3	8	2	**87**	0	0
14	**C**	12	6	**77**	5	0	0
15	**B**	8	**76**	7	9	0	0
16	**C**	8	26	**54**	12	0	0
17	**B**	14	**58**	12	15	1	0
18	**D**	11	1	2	**86**	0	0
19	**A**	**63**	11	22	3	1	0
20	**C**	6	15	**58**	19	1	0
21	**D**	1	9	15	**75**	1	0
22	**B**	15	**78**	5	2	0	0
23	**A**	**62**	18	6	14	1	0
24	**B**	14	**48**	15	21	2	0
25	**A**	**47**	5	39	7	0	1

2022 Higher Chemistry Section 2 Marks Data Analysis

2022 Item	Max Mark	Average Mark	% Average Mark	Discrimination	% Not Attempted
Q1ai	1	0.71	71%	46	2%
Q1aii	1	0.79	79%	43	1%
Q1bi	1	0.36	36%	39	1%
Q1bii	1	0.54	54%	34	1%
Q1ci	2	0.95	47%	36	1%
Q1cii	2	0.96	48%	51	3%
Q2a	1	0.83	83%	23	1%
Q2bi	2	1.41	71%	58	2%
Q2bii	1	0.70	70%	46	1%
Q2biii	1	0.67	67%	50	3%
Q2biv	2	0.95	47%	50	1%
Q2c	1	0.77	77%	15	1%
Q3	3	1.34	45%	65	4%
Q4ai	1	0.72	72%	47	1%
Q4aii	1	0.73	73%	53	2%
Q4aiii	1	0.31	31%	40	3%
Q4bi	1	0.73	73%	16	4%
Q4bii	1	0.79	79%	33	0%
Q4ci	1	0.32	32%	45	10%
Q4cii	2	0.83	41%	52	10%
Q5a	1	0.69	69%	42	0%
Q5bi	1	0.45	45%	54	9%
Q5bii	1	0.33	33%	48	9%
Q5biii	1	0.68	68%	48	2%
Q5biv	1	0.70	70%	48	6%
Q5bv	1	0.77	77%	50	6%
Q5bvi	1	0.62	62%	55	3%
Q6ai	1	0.81	81%	39	2%
Q6aiiA	1	0.87	87%	25	5%
Q6aiiB	1	0.57	57%	62	4%
Q6aiiC	1	0.88	88%	41	1%
Q6aiiD	1	0.69	69%	53	4%
Q6aiii	2	1.23	62%	57	4%
Q6aiv	3	1.89	63%	46	6%
Q6bi	1	0.40	40%	33	2%
Q6bii	2	1.14	57%	61	2%
Q6c	3	2.01	67%	50	2%
Q7a	3	1.60	53%	66	4%
Q7b	2	1.25	62%	58	3%
Q7c	2	1.66	83%	48	1%
Q7d	1	0.54	54%	30	1%
Q7ei	2	0.92	46%	47	4%
Q7eii	1	0.81	81%	30	1%
Q8ai	2	1.41	71%	44	1%
Q8aii	1	0.44	44%	47	2%
Q8b	2	1.52	76%	54	4%
Q8c	2	0.80	40%	51	8%
Q8diA	1	0.50	50%	44	3%
Q8diBi	1	0.82	82%	47	4%
Q8diBii	1	0.55	55%	50	4%
Q8diBiii	1	0.87	87%	36	2%
Q8diiA	1	0.91	91%	30	1%
Q8diiB	1	0.73	73%	48	1%
Q9	3	0.82	27%	59	16%
Q10ai	1	0.64	64%	27	3%
Q10aii	1	0.73	73%	37	3%
Q10aiii	1	0.38	38%	24	7%
Q10bi	1	0.61	61%	50	4%
Q10biiA	1	0.88	88%	43	2%
Q10biiB	1	0.15	15%	34	7%
Q10c	1	0.37	37%	40	8%
Q11ai	1	0.40	40%	41	5%
Q11aii	1	0.19	19%	38	12%
Q11b	3	1.18	39%	60	14%
Q11ci	1	0.22	22%	18	6%
Q11cii	1	0.52	52%	44	2%
Q11ciii	1	0.73	73%	43	7%
Q11d	3	1.77	59%	59	6%
Paper	95	55.02	58%		

2022 Higher Chemistry Section 2 Frequency of Marks Analysis

2022 Item	0	1	2	3
Q1ai	29	71		
Q1aii	21	79		
Q1bi	64	36		
Q1bii	46	54		
Q1ci	15	75	10	
Q1cii	42	19	39	
Q2a	17	83		
Q2bi	15	29	56	
Q2bii	30	70		
Q2biii	33	67		
Q2biv	29	48	23	
Q2c	23	77		
Q3	19	39	31	11
Q4ai	28	72		
Q4aii	27	73		
Q4aiii	69	31		
Q4bi	27	73		
Q4bii	21	79		
Q4ci	68	32		
Q4cii	43	31	26	
Q5a	31	69		
Q5bi	55	45		
Q5bii	67	33		
Q5biii	32	68		
Q5biv	30	70		
Q5bv	23	77		
Q5bvi	38	62		
Q6ai	19	81		
Q6aiiA	13	87		
Q6aiiB	43	57		
Q6aiiC	12	88		
Q6aiiD	31	69		
Q6aiii	12	54	35	
Q6aiv	14	17	35	34
Q6bi	60	40		
Q6bii	25	36	39	
Q6c	16	17	18	49
Q7a	31	20	7	42
Q7b	21	33	46	
Q7c	11	12	77	
Q7d	46	54		
Q7ei	34	40	26	
Q7eii	19	81		
Q8ai	17	25	58	
Q8aii	56	44		
Q8b	14	21	65	
Q8c	39	43	18	
Q8diA	50	50		
Q8diBi	18	82		
Q8diBii	45	55		
Q8diBiii	13	87		
Q8diiA	9	91		
Q8diiB	27	73		
Q9	51	24	16	8
Q10ai	36	64		
Q10aii	27	73		
Q10aiii	62	38		
Q10bi	39	61		
Q10biiA	12	88		
Q10biiB	85	15		
Q10c	63	37		
Q11ai	60	40		
Q11aii	81	19		
Q11b	47	12	17	24
Q11ci	78	22		
Q11cii	48	52		
Q11ciii	27	73		
Q11d	24	20	12	45

National
Qualifications

X813/76/12

Chemistry
Paper 1 — Multiple choice

Marking Instructions

Please note that these marking instructions have not been standardised based on candidate responses. You may therefore need to agree within your centre how to consistently mark an item if a candidate response is not covered by the marking instructions.

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Marking instructions for each question

Question	Response	Mark
1.	A	**1**
2.	D	**1**
3.	B	**1**
4.	A	**1**
5.	C	**1**
6.	C	**1**
7.	D	**1**
8.	B	**1**
9.	B	**1**
10.	C	**1**
11.	C	**1**
12.	B	**1**
13.	C	**1**
14.	B	**1**
15.	D	**1**
16.	A	**1**
17.	D	**1**
18.	B	**1**
19.	C	**1**
20.	D	**1**
21.	A	**1**
22.	B	**1**
23.	C	**1**
24.	A	**1**
25.	A	**1**

[END OF MARKING INSTRUCTIONS]

Marking instructions for each question

Question			Expected response	Max mark	Additional guidance
1.	(a)	(i)	Increases (across period)	**1**	
		(ii)	They don't form (covalent) bonds	**1**	Accept: They (noble gases) are unreactive. They are noble gases is not acceptable.
		(iii)	Screening increases so less attraction (of nucleus/protons for the bonding/outer/shared electrons**)** **OR** Covalent radius/atomic size/number of shells increases so less attraction (of nucleus/protons for the bonding/outer/shared electrons**)**	**1**	Shielding is acceptable in place of screening. 'Screening/shielding effect' by itself is not acceptable. Information in brackets is not required but if included, the direction of attraction must be correct.
	(b)	(i)	2·8 ± 0·05	**1**	
		(ii)	Cross at (2·1, 1·8) on graph **2 marks** **Partial mark** For calculation of both average electronegativity (2·1) and difference (1·8) **(1 mark)** **OR** For correctly plotting the point for the values candidate has calculated **(1 mark)**	**2**	A point other than (2·1, 1·8) plotted with no calculated values is worth 0 marks.
		(iii) A	$(Li^+)_2S^{2-}$ **Both** charges must be shown. Brackets are required for Li^+.	**1**	
		B	Carbon fluorine Sulfur fluorine Boron oxygen	**1**	Accept correct symbols. If candidate states name of a compound then it must be correct. Accept germanium and oxygen.
	(c)		Polar (covalent)	**1**	

<table>
<tr><th colspan="3">Question</th><th>Expected response</th><th>Max mark</th><th>Additional guidance</th></tr>
<tr><td>2.</td><td>(a)</td><td>(i)</td><td>Graphite</td><td>1</td><td></td></tr>
<tr><td></td><td></td><td>(ii)</td><td><table><tr><th>Form of carbon</th><th>Strongest attraction broken</th></tr><tr><td>Diamond</td><td>covalent (network) bond(s) (1 mark)</td></tr><tr><td>fullerene</td><td>London dispersion forces (1 mark)</td></tr></table></td><td>2</td><td>Mention of molecular would cancel the mark for covalent bonds for diamond.

Accept LDFs for London dispersion forces.

Treat each mark separately.</td></tr>
<tr><td></td><td></td><td>(iii)</td><td>12</td><td>1</td><td></td></tr>
<tr><td></td><td>(b)</td><td></td><td>45·81/45·8/46(%) (2 marks)

A partial mark can be awarded for any one of the following:

Correct use of atom economy relationship without correct use of stoichiometry (working must be shown). (1 mark)

OR

Correct working with no correct final answer. (1 mark)</td><td>2</td><td>No units required. Only 1 mark can be awarded for the correct answer if wrong unit is given. (Wrong units would only be penalised once in any paper).</td></tr>
</table>

Question			Expected response	Max mark	Additional guidance
2.	(c)		+250 kJ mol^{-1} **(2 marks)** Partial marks Treat as two concepts. **Either** would be acceptable for **1 mark.** Evidence of understanding of reversal of first and second enthalpy values must be seen ie +283 and +286 (or positive multiples of either/both). The third enthalpy value (regardless of value) must be negative, or this partial mark cannot be awarded. **OR** Evidence of understanding of multiplying the second enthalpy value by 3 (shown as 3 x +/-286 or +/- 858) Multiplication of any other enthalpy value by any factor is taken as cancelling of this partial mark.	**2**	If correct answer is shown, award 2 marks. **Only 1 concept mark can be awarded if the final answer is incorrect.** If answer given is -250, maximum of 1 mark can be awarded. No units required. Only 1 mark can be awarded for the correct answer if wrong unit is given. (Wrong units would only be penalised once in any paper). kJ is acceptable in place of kJ mol^{-1} (KJ or Kj or KJ mol^{-1} or Kj mol^{-1} accepted).

Question			Expected response	Max mark	Additional guidance
3.			**Award 1 mark** where the candidate has demonstrated, at an appropriate level, a limited understanding of the chemistry involved. They have made some statement(s) which are relevant to the situation, showing that they have understood at least a little of the chemistry within the problem. **Award 2 marks** where the candidate has demonstrated, at an appropriate level, a reasonable understanding of the chemistry involved. They make some statement(s) which are relevant to the situation, showing that they have understood the problem. **Award 3 marks** where the candidate has demonstrated, at an appropriate level, a good understanding of the chemistry involved. They show a good comprehension of the chemistry of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be 'excellent' or 'complete' for the candidate to gain full marks. **Award 0** marks where the candidate has not demonstrated, at an appropriate level, an understanding of the chemistry involved. There is no evidence that they have recognised the area of chemistry involved, or they have not given any statement of a relevant chemistry principle. Award this mark also if the candidate merely restates the chemistry given in the question.	**3**	Zero marks should be awarded if: The student has demonstrated, at an appropriate level, no understanding, of the chemistry involved. There is no evidence that the student has recognised the area of chemistry involved or has given any statement of a relevant chemistry principle. This mark would also be given when the student merely restates the chemistry given in the question.

Question			Expected response	Max mark	Additional guidance
4.	(a)		Bond enthalpy is high (945)/has the highest bond enthalpy value (in the data booklet)	**1**	(Bond(s)) take(s) a lot of energy to break/are very strong would not be accepted on its own but would not be cancelling.
	(b)		Diagram shows a workable method for removing carbon dioxide – must include label for KOH (solution) **(1 mark)** Diagram shows a workable method for passing gas over heated copper – must include label for heated copper **(1 mark)**	**2**	Treat each mark separately.

<table>
<tr><th colspan="3">Question</th><th>Expected response</th><th>Max mark</th><th>Additional guidance</th></tr>
<tr><td>4.</td><td>(c)</td><td>(i)</td><td>2 marks for correct calculations AND 1 mark for a statement which follows on from the calculation stating that nitrogen is in excess or that lithium is the limiting reactant.

Partial marks

By calculating number of moles:

1 mark for the correct calculation of number of moles of Li and N_2.

1 mark for correct application of the mole ratio. This can be shown by dividing a calculated number of moles of Li by 6 or multiplying a number of moles of N_2 by 6.

1 mark awarded for a correct statement following on from the candidate's calculations.

OR

By proportion:

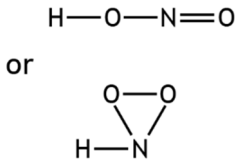
1 mark for 41·4 g $\rightarrow$ 24 litres

1 mark for follow through from incorrect multiples of 6·9g or 24 l

1 mark awarded for a correct statement following on from the candidate's calculations.</td><td>3</td><td>Correct calculation of moles of lithium = 0·07/0·072/0·0725 moles
and
nitrogen = 0·04/0·038/0·0375 moles

The values in the tables shown provide guidance as to whether 1 mark or 2 marks for calculations should be awarded.

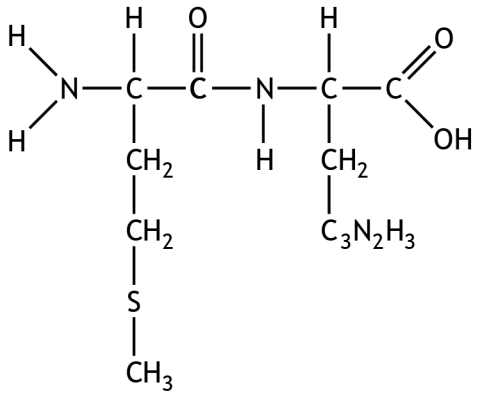
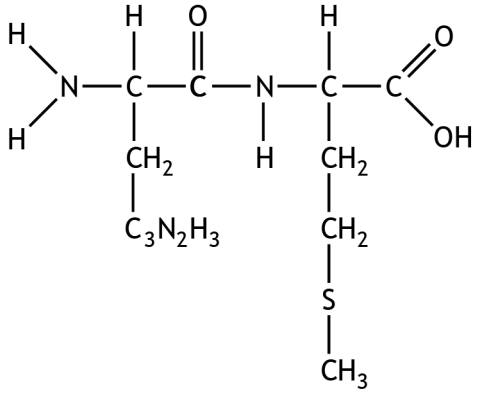
2 marks can be awarded for any one of the following paired values in the table obtained by applying the mole ratio
<table><tr><td>have</td><td>need</td></tr><tr><td>0·5 g Li</td><td>0·29 l N_2</td></tr><tr><td>0·9 l N_2</td><td>1·55 g Li</td></tr><tr><td>0·9 l N_2</td><td>0·29 l N_2</td></tr><tr><td>0·5 g Li</td><td>1·55 g Li</td></tr><tr><td>0·0725 mol Li</td><td>0·225 mol Li</td></tr><tr><td>0·0375 mol N_2</td><td>0·012 mol N_2</td></tr></table>
1 mark can be awarded for any one of the following paired values in the table obtained without applying mole ratio
<table><tr><td>have</td><td>need</td></tr><tr><td>0·5 g Li</td><td>1·74 l N_2</td></tr><tr><td>0·9 l N_2</td><td>0·26 g Li</td></tr><tr><td>0·9 l N_2</td><td>1·74 l N_2</td></tr><tr><td>0·5 g Li</td><td>0·26 g Li</td></tr></table>
This mark can only be awarded if the candidate shows appropriate calculations to justify the statement.</td></tr>
</table>

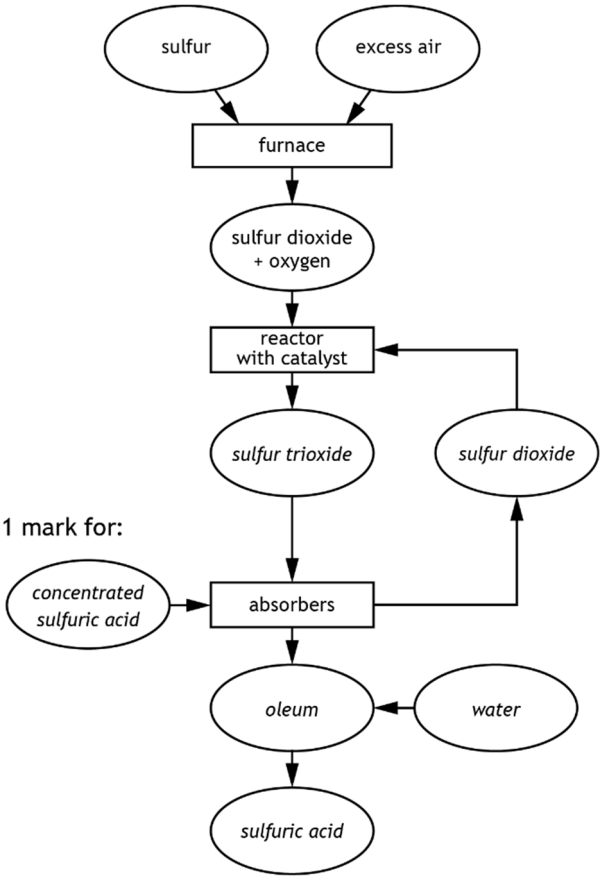
Question			Expected response	Max mark	Additional guidance
4.	(c)	(ii)	$Cu^{+}(aq) + e^{-} \rightarrow Cu(s)$	**1**	State symbols not required but if given must be correct. Accept electron without a negative sign.
		(iii)	(ionic) lattice/network	**1**	Covalent network is not acceptable
	(d)	(i)	atoms/molecules with an unpaired electron	**1**	
		(ii)	676 (kJ mol^{-1}) -676 would be worth **1 mark** Partial marking, **1 mark** can be awarded for candidate correctly retrieves both of the relevant bond enthalpy data (945, 498) and attempts to use this with 91 **OR** calculation carried out correctly with one error in retrieval of bond enthalpy	**2**	No units are required but award only 1 mark for correct answer if incorrect unit is given. (Wrong units would only be penalised once in any paper). kJ is acceptable in place of kJ mol^{-1} (KJ or Kj or KJ mol^{-1} or Kj mol^{-1} accepted)
		(iii) A	termination	**1**	
		B	 or 	**1**	

Question			Expected response	Max mark	Additional guidance
4.	(e)	(i)	Decreasing temperature favours the exothermic reaction/increasing temperature favours endothermic reactions **(1 mark)** Increases the yield of ammonia **(1mark)**	**2**	
		(ii) A	$C_3H_5N_3O_9 \rightarrow$ $3CO_2 + 2½H_2O + 1½N_2 + ¼ O_2$ **OR** correct multiples	**1**	
		B	The shock/bump provides the activation energy/E_A **OR** the shock/bump provides **sufficient/enough** energy to start the reaction **OR** the reaction has a low activation energy/E_A	**1**	If candidate uses "it" in response, this can be taken as "the shock/bump".

Question			Expected response	Max mark	Additional guidance
5.	(a)		**1 mark** for any correct feature and explanation as shown: • Contains oxygen to ensure complete combustion • Sample is surrounded by water so all energy transferred/reduce heat loss to surroundings • Sealed container prevents/ reduces heat loss to the surroundings • Stirring to ensure accurate temperature (measurement)	**1**	
	(b)	(i)	-34 078 (kJ mol^{-1}) Partial marks **1 mark** for a demonstration of the correct use of the relationship $E_h = cm\Delta T$ as shown by (4·18 × (an order of magnitude of 0·775) × 11·9) (ignore units for this partial mark) **1 mark** for evidence of the knowledge that enthalpy of combustion relates to 1 mole, evidenced by the scaling up of a calculated value of energy released. **1 mark** for correct arithmetic. This mark should be awarded if the candidate has obtained the 2 partial marks above but has applied correct early rounding within the calculation resulting in an answer that differs from –34 078).	**3**	Maximum of 2 marks can be awarded if negative enthalpy sign is not shown in final answer. Units not required. Only 2 marks can be awarded for the correct answer if wrong unit is given. (wrong units would only be penalised once in any paper). kJ is acceptable in place of kJ mol^{-1} (KJ or Kj or KJ mol^{-1} or Kj mol^{-1} accepted).
		(ii)	0·71/0.713/0.7125	**1**	
	(c)	(i)	23·3	**1**	Ignore any units
		(ii)	Glycerol has 3 hydroxyl groups.	**1**	Accept glycerol is propan(e)-1,2,3-triol/an alcohol with 3 hydroxyl (-OH) groups

Question			Expected response	Max mark	Additional guidance
6.	(a)		<table><tr><th>Mass **(1 mark)**</th><th>Unit **(1 mark)**</th></tr><tr><td>0·00113</td><td>kg</td></tr><tr><td>1·13</td><td>g</td></tr><tr><td>1130</td><td>mg</td></tr></table> Any correct line in the table (2 marks) Correctly calculated mass of iodine without units **(1)** Appropriate units **(1)**	**2**	As no unit is specified, the correct answer can be expressed with any appropriate unit (as shown in table). Any correct rounding accepted e.g. 1·1278/1·128/1·13 g Partial mark for appropriate units can only be awarded when the order of magnitude of an incorrectly calculated mass matches the unit. If the candidate's working is unclear then the mark for units cannot be awarded.
	(b)	(i)	Measuring the mass of container + seaweed/sample and subtracting the mass of the container	**1**	Taring the balance with container and then adding the seaweed/ sample is accepted "use of Tare function" on its own is not accepted.
		(ii)	I^-/iodide (ions)	**1**	
		(iii)	A solution of accurately/exactly/ precisely known concentration	**1**	
		(iv) A	0·00013 (moles)	**1**	Units not required, but if present, must be correct. Wrong units are only penalised once per paper.
		B	0·03299g/0·033 g/0·03 (g)	**1**	To award this mark, the candidates answer to part **B must** be correct for their use of their answer to part **A** (i.e. answer to part A x 253·8). Units not required, but if present, must be correct. Wrong units are only penalised once per paper.

Question			Expected response	Max mark	Additional guidance
6.	(c)	(i)	Amino acid that must be acquired/obtained from the diet.	**1**	Amino acid that cannot be made by the body is accepted.
		(ii)	Correctly drawn structure for dipeptide.  **OR** 	**1**	Shortened structural formula accepted. All atoms and bonds must be present as shown. However, **in the side chains only**, bonds drawn from C (or S) to H (or N) would not be penalised.

Question			Expected response	Max mark	Additional guidance
7.	(a)		1 mark for:  	**2**	1 mark for correct top half (4 responses) and 1 mark for correct bottom half (4 responses). Accept correct chemical formula in place of names. Accept "air" for "excess air".

<table>
<tr><th colspan="3">Question</th><th>Expected response</th><th>Max mark</th><th>Additional guidance</th></tr>
<tr><td>7.</td><td>(b)</td><td></td><td>Require heat to be removed (to prevent the temperature from rising).</td><td>1</td><td></td></tr>
<tr><td></td><td>(c)</td><td>(i)</td><td>1 mark for description of LDFs as forces of attraction between temporary dipoles (and induced dipoles)

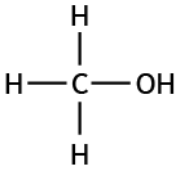
1 mark for an explanation of the cause of temporary dipoles in terms of uneven distribution of electrons/ electron wobble/movement of electrons in the molecule</td><td>2</td><td>Attraction involving permanent dipoles cancels this mark</td></tr>
<tr><td></td><td></td><td>(ii)</td><td>1 mark
Sulfur/S has more electrons than oxygen/O

1 mark
These forces are stronger due to sulfur structure being S_8 whereas oxygen is O_2</td><td>2</td><td>The structure of S_8 and O_2 must be evidenced in answer (molecular formula, structural formula) for the second mark to be awarded.

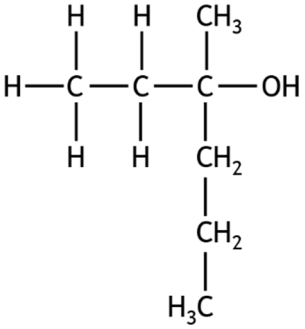
Correctly calculated number of electrons (16, 128) would also be evidence of structure.</td></tr>
<tr><td></td><td>(d)</td><td></td><td><table><tr><th>Effect of catalyst on enthalpy change</th><th>Effect of catalyst on activation energy</th></tr><tr><td>Increase
stay the same
decrease</td><td>increase
stay the same
decrease</td></tr></table></td><td>1</td><td></td></tr>
<tr><td></td><td>(e)</td><td>(i)</td><td>Do not form scum.</td><td>1</td><td></td></tr>
<tr><td></td><td></td><td>(ii)</td><td>Both parts of molecule must be described.

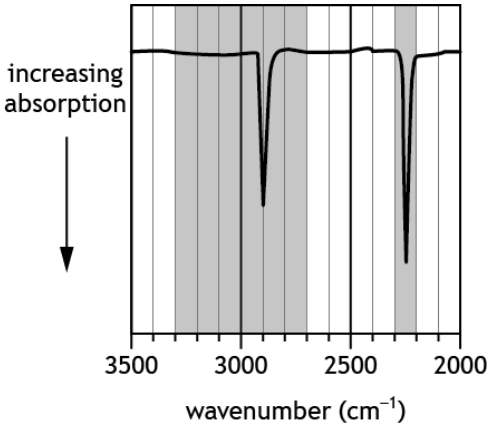
<table><tr><th>Head</th><th>Tail</th></tr><tr><td>Hydrophilic</td><td>Hydrophobic</td></tr><tr><td>Polar</td><td>Non-polar</td></tr><tr><td>Ionic</td><td>Non-polar</td></tr><tr><td>Water soluble</td><td>Fat soluble</td></tr></table>
Any term to describe the head from the table can be used with any term from the table to describe the tail.</td><td>1</td><td>If answer mentions head and/or tail, then it must be correct ie head is hydrophilic.</td></tr>
</table>

Question			Expected response	Max mark	Additional guidance
8.	(a)	(i)	pentyl ethanoate	**1**	
		(ii)	condensation/esterification	**1**	
	(b)	(i)	Carbon dioxide is (relatively) insoluble/has very low solubility	**1**	
		(ii)	0·029 (g) **2 marks** Partial marks Correctly calculated number of moles of carbon dioxide divided by 3 to give number of moles of citric acid e.g. moles of citric acid = 0·00076 **(1 mark)** Calculated number of moles of citric acid x 192 divided by 5 **(1 mark)** **OR** by proportion 192 g → 72 l (1 mark) 0·147 *g* → 0·055 l 0·147/5 = 0·029 (g) **(1 mark)** Allow follow through from incorrect multiple of 24 l	**2**	Units not required, but if present, must be correct. Wrong units are only penalised once per paper. Allow follow through from incorrect application of mole ratio or correct mole ratio applied to incorrectly calculated number of moles of CO_2
	(c)	(i) A	orange to green/blue-green/ blue	**1**	
		B	Tollens' reagent **OR** Fehling's solution	**1**	Benedict's/ Schiff's reagent would be accepted.
		(ii)	2	**1**	
		(iii)	1 pence/ £0·01 (2 marks) Partial marks 1 mark can be awarded for: cost per mg of vanillin = £0·00105 **OR** cost per g of vanillin = £1·05 **OR** cost of 0·184 g of vanillin = £0·1932 **OR** mass of vanillin needed = 0·0092 g	**2**	Candidates final answer must be in a correct monetary number.

Question			Expected response	Max mark	Additional guidance
9.	(a)	(i)	1·575/1·58/1·6(g)	**1**	Units not required. 0 marks should be awarded for the correct answer if wrong unit is given. (Wrong units would only be penalised once in any paper).
		(ii)	Protein(s)	**1**	
		(iii)	pipette and burette	**1**	Volumetric flask would not be taken as cancelling.
	(b)		essential oils	**1**	
	(c)	(i)	2-methylbuta-1,3-diene	**1**	
		(ii)	ketone	**1**	accept terpenoids
	(d)	(i)	methanol	**1**	A correct structure would also be accepted 
		(ii)	79·24/79·2/79 (%) (2 marks) Partial marking for **1 mark** Theoretical yield = 31·17 (g) **OR** Allow follow through from incorrect calculation of theoretical yield.	**2**	Units not required, but if present, must be correct. Wrong units are only penalised once per paper. Rounding of a calculated theoretical yield to 31 g gives 77·68/79·7/80 (%) which would be accepted.
		(iii)	6·5 (cm^3) Partial marking for **1 mark** For correct calculation of toxic dose = 9·1 g **OR** For calculation of the volume of oil of wintergreen containing 0·14 g of methyl salicylate = 0·1 cm^3	**2**	Units not required. Only 1 mark can be awarded for the correct answer if wrong unit is given. Wrong units would only be penalised once in any paper. Allow follow from incorrect first step of candidate's calculation.

Question			Expected response	Max mark	Additional guidance
10.			**Award 1 mark** where the candidate has demonstrated, at an appropriate level, a limited understanding of the chemistry involved. They have made some statement(s) which are relevant to the situation, showing that they have understood at least a little of the chemistry within the problem. **Award 2 marks** where the candidate has demonstrated, at an appropriate level, a reasonable understanding of the chemistry involved. They make some statement(s) which are relevant to the situation, showing that they have understood the problem. **Award 3 marks** where the candidate has demonstrated, at an appropriate level, a good understanding of the chemistry involved. They show a good comprehension of the chemistry of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be 'excellent' or 'complete' for the candidate to gain full marks. **Award 0** marks where the candidate has not demonstrated, at an appropriate level, an understanding of the chemistry involved. There is no evidence that they have recognised the area of chemistry involved, or they have not given any statement of a relevant chemistry principle. Award this mark also if the candidate merely restates the chemistry given in the question.	**3**	Zero marks should be awarded if: The student has demonstrated, at an appropriate level, no understanding, of the chemistry involved. There is no evidence that the student has recognised the area of chemistry involved or has given any statement of a relevant chemistry principle. This mark would also be given when the student merely restates the chemistry given in the question.

Question			Expected response	Max mark	Additional guidance
11.	(a)	(i)	Reactants/solvent is flammable/ catches fire with a flame	**1**	accept products
		(ii)	condenser	**1**	
	(b)	(i)	addition	**1**	
		(ii)	2-methylbutan-2-ol	**1**	
		(iii)	A correct structural formula for 3-methylhexan-3-ol e.g. a full structural formula or a shortened structural formula	**1**	A mixture of full and shortened structural formula is accepted, e.g. 

Question			Expected response	Max mark	Additional guidance
12.	(a)		Poly(phenylethene)	**1**	
	(b)		C=O circled	**1**	accept 'C=O stretch'
	(c)		2 single absorptions (peaks) required 1 for C-H within the range 2700-3300 cm^{-1} and 1 for C≡N within the range 2260-2215 cm^{-1}  	**1**	Ignore the intensity of the absorptions. Ignore the width of the absorptions so long as the peak maximum is defined and within the given range.

[END OF MARKING INSTRUCTIONS]

General marking principles for Higher Chemistry

Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.

(a) Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.

(b) If a candidate response does not seem to be covered by either the principles or detailed marking instructions, and you are uncertain how to assess it, you must seek guidance from your team leader.

(c) Do not award half marks.

(d) Where a candidate makes an error at an early stage in a multi-stage calculation, award marks for correct follow-on working in subsequent stages, unless the error significantly reduces the complexity of the remaining stages. Apply the same principle for questions that require several stages of non-mathematical reasoning. The exception to this rule is where the marking instructions for a numerical question assign separate 'concept marks' and an 'arithmetic mark'. In such situations, the marking instructions will give clear guidance on the assignment or partial marks.

(e) Unless a numerical question specifically requires evidence of working to be shown, award full marks for a correct final response (including units) on its own.

(f) Candidates may fully access larger mark allocations whether their responses are in continuous prose, linked statements, or a series of developed bullet points.

(g) Do not deduct marks for inaccurate or unconventional spelling or vocabulary as long as the meaning of the word(s) is conveyed. **For example**, responses that include 'distilling' for 'distillation', or 'it gets hotter' for 'the temperature rises', are acceptable.

(h) In many questions, the unit in which the answer is to be expressed is given. In these questions, the candidate does not need to state a unit in their answer; but if they do, the unit must be correct. The full mark allocation cannot be awarded if an incorrect unit is shown. In these questions, incorrect units would only be penalised once in any paper.

(i) If a correct response is followed by a wrong response, award no marks. **For example** in response to the question, 'State the colour seen when blue Fehling's solution is warmed with an aldehyde', do not award marks for the response 'red green'. However, if a correct response is followed by additional information which does not conflict with that, ignore the additional information, whether correct or not. **For example** in response to a question concerned with melting point, 'State why the tube should not be made of copper', the response 'Copper has a low melting point and is coloured grey' **would** gain marks.

(j) Award full marks for the correct response to a calculation without working. Award partial marks, as shown in the detailed marking instructions, when working is given but the final response is incorrect. An exception is when candidates are asked to 'Find, by calculation' - do not award full marks for the correct response without working.

(k) Ignore the omission of one H atom from a full structural formula provided the bond is shown.

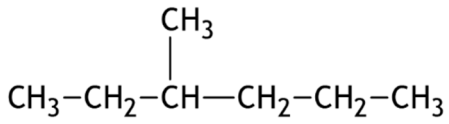
(l) Award marks for a symbol or correct formula in place of a name **unless stated otherwise in the detailed marking instructions.**

(m) When formulae of ionic compounds are given as responses, candidates only need to show ion charges if these have been specifically asked for. However, if ion charges are shown, they must be correct. If incorrect charges are shown, do not award marks.

(n) If an answer comes directly from the text of the question, do not award marks. **For example**, in response to the question, 'A student found that 0·05 mol of propane, C_3H_8 burned to give 82·4 kJ of energy. $C_3H_8(g) + 5O_2(g) = 3CO_2(g) + 4H_2O(\ell)$. Name the kind of enthalpy change that the student measured', do not award marks for 'burning' since the word 'burned' appears in the text.

(o) A guiding principle in marking is to give credit for correct elements of a response rather than to look for reasons not to give marks.

Example 1: The structure of a hydrocarbon found in petrol is shown below.



Name the hydrocarbon

- Award the full mark for '3, methyl-hexane', although the punctuation is not correct.

Example 2: A student measured the pH of four carboxylic acids to find out how their strength is related to the number of chlorine atoms in the molecule. The results are shown.

CH_3COOH	1·65
$CH_2ClCOOH$	1·27
$CHCl_2COOH$	0·90
CCl_3COOH	0·51

Describe the relationship between the number of chlorine atoms in the molecule and the strengths of the acids.

- Award the full mark for a response such as 'the more Cl_2, the stronger the acid', although not completely correct.

(p) Unless the question is clearly about a non-chemistry issue, for example costs in an industrial chemical process, do not award marks for a non-chemical response.
For example, in response to the question, 'Why does the (catalytic) converter have a honeycomb structure?', do not award a mark for 'To make it work'. This response may be correct but it is not a chemical response.

(q) Only award marks for a valid response to the question asked. Where candidates are asked to:

- **identify, name, give** or **state**, they must only name or present in brief form.
- **describe**, they must provide a statement or structure of characteristics and/or features.
- **explain**, they must relate cause and effect and/or make relationships between things clear.
- **compare**, they must demonstrate knowledge and understanding of the similarities and/or differences between things.
- **complete**, they must finish a chemical equation or fill in a table with information.
- **determine** or **calculate**, they must determine a number from given facts, figures or information.
- **draw**, they must draw a diagram or structural formula, for example 'Draw a diagram to show the part of a poly(propene) molecule formed from two propene molecules.'
- **estimate**, they must determine an approximate value for something.
- **predict**, they must suggest what may happen based on available information.
- **evaluate**, they must make a judgement based on criteria.
- **suggest**, they must apply their knowledge and understanding of chemistry to a new situation. A number of responses are acceptable: award marks for any suggestions that are supported by knowledge and understanding of chemistry.
- **use their knowledge of chemistry or aspect of chemistry to comment on**, they must apply their skills, knowledge and understanding to respond appropriately to the problem/situation presented (for example by making a statement of principle(s) involved and/or a relationship or equation, and applying these to respond to the problem/situation). Candidates gain marks for the breadth and/or depth of their conceptual understanding.
- **write**, they must complete a chemical or word equation, for example 'Write the word equation for the complete combustion of ethanol.'

National
Qualifications
2019

2019 Chemistry

Higher - Paper 1 – Multiple choice

Finalised Marking Instructions

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Marking instructions for each question

Question	Response	Mark
1.	C	**1**
2.	B	**1**
3.	A	**1**
4.	C	**1**
5.	B	**1**
6.	A	**1**
7.	D	**1**
8.	B	**1**
9.	D	**1**
10.	D	**1**
11.	C	**1**
12.	B	**1**
13.	D	**1**
14.	A	**1**
15.	C	**1**
16.	D	**1**
17.	B	**1**
18.	B	**1**
19.	B	**1**
20.	C	**1**
21.	A	**1**
22.	D	**1**
23.	C	**1**
24.	A	**1**
25.	D	**1**

[END OF MARKING INSTRUCTIONS]

National
Qualifications
2019

2019 Chemistry

Higher - Paper 2

Finalised Marking Instructions

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Marking instructions for each question

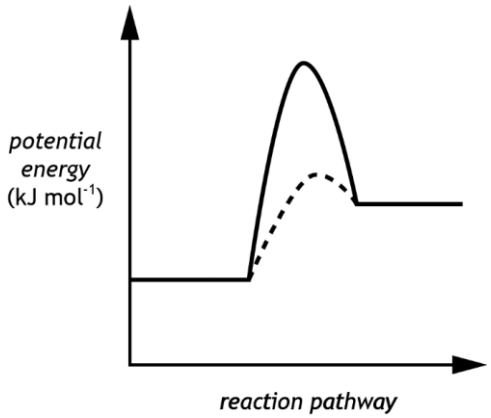
<table>
<tr><th colspan="3">Question</th><th>Expected response</th><th>Max mark</th><th>Additional guidance</th></tr>
<tr><td>1.</td><td>(a)</td><td>(i)</td><td>$Na_2S_2O_3$ + 2HCl $\rightarrow$S + SO_2 + 2NaCl+ H_2O</td><td>1</td><td>Correct multiples accepted.
Ignore state symbols if equation re-written.</td></tr>
<tr><td></td><td></td><td>(ii)
A</td><td><table><tr><th>Experiment</th><th>Volume of water (cm^3)</th></tr><tr><td>A</td><td>0</td></tr><tr><td>B</td><td>10</td></tr><tr><td>C</td><td>20</td></tr><tr><td>D</td><td>30</td></tr><tr><td>E</td><td>40</td></tr></table>All entries correct 1 Mark.</td><td>1</td><td></td></tr>
<tr><td></td><td></td><td>B</td><td>35/35·1/35·09 (seconds or s)</td><td>1</td><td>No units required. No mark can be awarded for correct answer if wrong unit is given (where no unit required, wrong units would only be penalised once in any paper).

sec or secs not accepted.</td></tr>
<tr><td></td><td></td><td>(iii)</td><td>12 +/- 1 (°C)</td><td>1</td><td>Award 0 marks if 10°C given.

No units required. No mark can be awarded for correct answer if wrong unit is given (where no unit required, wrong units would only be penalised once in any paper).

Degrees or ° not accepted</td></tr>
<tr><td></td><td>(b)</td><td></td><td>(Particles must have) sufficient or enough energy (to react).
OR
(Particles must have) energy equal to or greater than the activation energy or $E_{A.}$
OR
(Particles must have) minimum/ enough energy to form an activated complex. (1)
(Collision must occur with) suitable/correct/geometry/ orientation (1)</td><td>2</td><td>Do not accept ‘with correct speed’, ‘with high energy’ or “sufficient activation energy”.

“They”/”reactants” accepted for particles.</td></tr>
</table>

Question			Expected response	Max mark	Additional guidance
1.	(c)	(i)	X shown at peak of curve	**1**	Must be centred and within the y-axis. Clear indication of the correct position of the activated complex is accepted.
		(ii)		**1**	The peak drawn shows a lower activation energy. Line drawn does not need to be a dotted/dashed line. The line drawn by the candidate must start and finish at the same potential energies as on the original diagram.

Question			Expected response	Max mark	Additional guidance
2.	(a)	(i)	Increasing/greater/stronger/larger nuclear charge (holds electrons more tightly) **OR** Increasing number of protons	**1**	Increased nuclear pull is not accepted on its own. Mention must be made of nuclear charge **or** number of protons. Increased attraction of the electron **for** the nucleus would be considered cancelling.
		(ii)	(More shells) so increased/more screening/shielding. **OR** Covalent radius increases/atom size increases/more shells so attraction of the nucleus/protons for the **outer electron** decreases.	**1**	'Shielding effect' by itself is not acceptable. If candidate says 'it' assume the candidate is talking about potassium.
	(b)	(i)	$N^{+}(g) \rightarrow N^{2+}(g) + e^{-}$	**1**	State symbols must be shown. Negative charge on the electron is not required.
		(ii)	The 6th ionisation energy involves removing an electron from the shell which is inner/full (whole)/(more) stable/closer to the nucleus **OR** the 6th electron is removed from the electron shell which is inner/full (whole)/(more) stable/closer to the nucleus. **(1)** The 6th electron is less shielded **OR** the 6th electron is more strongly attracted to/pulled towards the nucleus. **(1)**	**2**	Correct statements made about the 5th ionisation energy/electron can also be credited. Stating that the 6th electron requires more energy than the 5th electron is not sufficient on its own.
	(c)		Al forms Al^{3+}/loses electrons to form an ion **and** P forms P^{3-}/gains electrons to form an ion. **(1)** Aluminium ion has one less energy level than phosphide/phosphorus ion. **OR** Phosphide/phosphorus ion has one more energy level than aluminium ion. **(1)**	**2**	A diagram or electron arrangement of **both** ions would be sufficient to gain this first mark.

Question			Expected response	Max mark	Additional guidance
2.	(d)		(Radius ratio =) 0·96, (hence) caesium chloride (or correct formula).	**1**	Calculated value (of 0·96) or 135/140 must be written for radius ratio required for mark.

Question			Expected response	Max mark	Additional guidance
3.			This is an open ended question **1 mark:** The student has demonstrated, at an appropriate level, a limited understanding of the chemistry involved. The student has made some statement(s) which is/are relevant to the situation, showing that at least a little of the chemistry within the problem is understood. **2 marks:** The student has demonstrated a reasonable understanding, at an appropriate level, of the chemistry involved. The student makes some statement(s) which is/are relevant to the situation, showing that the problem is understood. **3 marks:** The maximum available mark would be awarded to a student who has demonstrated a good understanding, at an appropriate level, of the chemistry involved. The student shows a good comprehension of the chemistry of the situation and has provided a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. This does not mean the answer has to be what might be termed an "excellent" answer or a "complete" one.	**3**	Zero marks should be awarded if: The student has demonstrated no understanding of the chemistry involved. There is no evidence that the student has recognised the area of chemistry involved or has given any statement of a relevant chemistry principle. This mark would also be given when the student merely restates the chemistry given in the question.

Question			Expected response	Max mark	Additional guidance
4.	(a)	(i)	biological catalyst	**1**	Biochemical catalyst/protein that acts as a catalyst accepted
		(ii)	4·5/4·49/4·485 (%)	**1**	No units required. No mark can be awarded for correct answer if wrong unit is given (where no unit required, wrong units would only be penalised once in any paper).
	(b)	(i)	Carbon dioxide/CO_2	**1**	
		(ii) A	0·25/0·3	**1**	Ignore any units.
		B	(Cider) B or (sample) 4	**1**	
	(c)		propan-1,2,3-triol **OR** propane-1,2,3-triol **OR** 1,2,3-propanetriol	**1**	
	(d)	(i)	(procyanidin B2) molecule is polar due to its **hydroxyl** groups **OR** (procyanidin B2) can form hydrogen bonds due to its **hydroxyl** groups **(1)** an explanation which links solubility of the molecule (procyanidin B2) to the polarity of water/hydrogen bonding of water **(1)**	**2**	(-)**OH** accepted as hydroxyl "Like dissolves like" on its own is not sufficient for 1 mark.
		(ii)	2-methylbutanoic acid	**1**	
		(iii)	Isoprene(s) **OR** 2-methylbuta-1,3-diene	**1**	Structure not accepted. Ignore any structure(s) drawn.
	(e)	(i)	carbonyl	**1**	
		(ii)	ethanoic acid/CH_3COOH	**1**	

Question			Expected response	Max mark	Additional guidance
5.	(a)	(i)	-694 (kJ mol^{-1}) Bond breaking (4 x 412) + (2 x 498) = 2644 Bond forming [(2 x 743) + (4 x 463)] = -3338 A single mark is available if either of the following operations is correctly executed **Either** The four relevant values for bond enthalpies of the C-H, O=O, C=O, and O-H (or multiples thereof) are retrieved from the data booklet (412, 498, 743, 463 – ignore signs). **OR** If only three correct values are retrieved, the candidate recognises that bond breaking is endothermic and bond forming is exothermic and have correctly manipulated the bond enthalpies and multiples that they have used with working shown.	**2**	+694 would qualify for 1 mark No units required. Only 1 mark can be awarded for the correct answer if wrong unit is given. (wrong units would only be penalised once in any paper) kJ is acceptable in place of kJ mol^{-1} (KJ or Kj or KJ mol^{-1} or Kj mol^{-1} accepted). If less than three bond enthalpies are retrieved then no mark can be awarded.
		(ii)	Mean bond enthalpy must refer to an average energy and to a number of compounds and bond enthalpy must relate to one compound/diatomic molecule.	**1**	

Question			Expected response	Max mark	Additional guidance
5.	(a)	(iii)	0·367/0·37/0·4 (g) **(2)** Partial marking $n = V/V_m = 0·2/24 = 0·008333….$ **(1)** an incorrectly calculated number of moles **based on gas volume** x 44 **(1)** or by proportion 24 l $\rightarrow$ 44 g **(1)** Follow through from incorrect multiples of 24 l **or** 44 g **(1)**	**2**	No units required. No mark can be awarded for correct answer if wrong unit is given (where no unit required, wrong units would only be penalised once in any paper). Working must be shown for incorrectly calculated number of moles based on gas volume.
	(b)	(i)	(Record the) mass/weight of the burner before and after (heating the water)	**1**	
		(ii)	-3496 (kJ mol^{-1}) (3 marks) **If final answer is wrong a maximum of 2 marks for the following concepts may be awarded** **1 mark** for a demonstration of the correct use of the relationship $E_h = cm\Delta T$ as shown by (4·18 x (an order of magnitude of 4) x 23) (ignore units for this mark). **1 mark** for evidence of the knowledge that enthalpy of combustion relates to 1 mole, evidenced by the scaling up of a calculated value of energy released.	**3**	An answer of +3496 would gain 2 marks. No units required. A maximum of 2 marks can be awarded for **correct answer** if wrong unit is given (where no unit required, wrong units would only be penalised once in any paper). kJ is acceptable in place of kJ mol^{-1} (KJ or Kj or KJ mol^{-1} or Kj mol^{-1} accepted).
		(iii)	**1 mark** for any of the following • Loss of heat/energy to the surroundings • Incomplete combustion (of heptane/alkane) • Loss (of heptane/alkane) by evaporation • No lid on container • No stirring • Absorption of heat glass/beaker or copper can	**1**	'Not using a bomb calorimeter' on its own would not be awarded a mark. "Loss of heat" on its own is not sufficient but would not be cancelling.

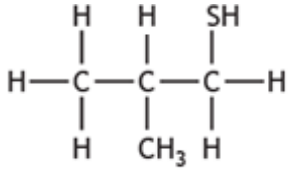
Question			Expected response	Max mark	Additional guidance
6.	(a)	(i) A	Same number of electrons (34) **(1)** **OR** same strength of London dispersion forces/LDFs **(1)**	**1**	
		(i) B	Propan-1-ol has stronger intermolecular/Van der Waal's forces than ethanethiol or vice versa. **OR** The intermolecular forces in propan-1-ol take more energy to break than those in ethanethiol or vice versa.**(1)** **1 mark** for identifying that the intermolecular forces in propan-1-ol are hydrogen bonds AND those in ethanethiol are permanent dipole – permanent dipole interactions/ attractions. **(1)**	**2**	If candidate says 'it/its' assume the candidate is talking about propan-1-ol. Any mention of breaking covalent/ ionic/metallic bonds is cancelling. Accept London Dispersion Forces/ LDFs in place of permanent dipole-permanent dipole interactions.
		(ii)	methanethiol	**1**	Accept methanthiol

<table>
<tr><th colspan="3">Question</th><th>Expected response</th><th>Max mark</th><th>Additional guidance</th></tr>
<tr><td></td><td></td><td>(iii)</td><td>11·853/11·85/11·9/12 mg – units required (2)

Correctly calculated mass of ethanethiol without units (1)

Appropriate units (1)
<table><tr><th>Mass (1)</th><th>Units (1)</th></tr><tr><td>12</td><td>mg</td></tr><tr><td>0·012</td><td>g</td></tr><tr><td>1·2 x 10^{-5}</td><td>kg</td></tr></table></td><td>2</td><td>If an incorrect mass is calculated but the units used are appropriate to the calculation then 1 mark would be awarded.

If the candidate's working is unclear then the mark for units cannot be awarded.</td></tr>
<tr><td></td><td>(b)</td><td>(i)</td><td>SH group is on a carbon connected to 3 other carbons/SH group is opposite the branch in a chain.

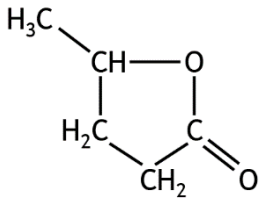
The SH group is attached to a carbon which has no hydrogens attached.</td><td>1</td><td></td></tr>
<tr><td></td><td></td><td>(ii)
A</td><td>  </td><td>1</td><td></td></tr>
<tr><td></td><td></td><td>(ii)
B</td><td>41·2 (g) (2)

OR

correct calculation of

Theoretical mass = 49·07 (g) (1)

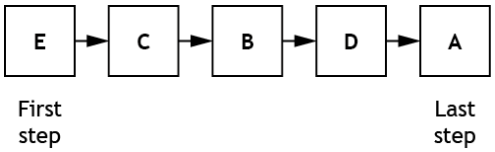
Allow follow on from incorrect calculation of theoretical mass for 1 mark</td><td>2</td><td>Award 0 marks if candidate gives 25·6 grams (84% of reactant mass)</td></tr>
</table>

Question			Expected response	Max mark	Additional guidance
7.	(a)	(i)	propagation	**1**	
		(ii)	UV/ultraviolet	**1**	
		(iii)	Anti-oxidant/free radical scavenger/reducing agent/electron donor	**1**	
	(b)	(i)	Water/H_2O	**1**	
		(ii)		**1**	
		(iii)	3-hydroxybutanoic acid	**1**	

Question			Expected response	Max mark	Additional guidance
8.	(a)	(i)	6	***1***	
		(ii)	London dispersion forces	**1**	Accept LDF
	(b)	(i)	dissolve the gelatin (in a small volume of water) **(1)** transfer quantitatively/with rinsings/washings **(1)** fill to the mark/line (of the volumetric flask) **(1)**	**3**	
		(ii)	11·0/11	**1**	
	(c)	(i)	Bromelain/enzyme changes shape or denatured. Bromelain/enzyme hydrogen bonds broken.	**1**	
		(ii)	38/37·9/37·88 (g)	**1**	No units required. No mark can be awarded for correct answer if wrong unit is given (where no unit required, wrong units would only be penalised once in any paper).

Question			Expected response	Max mark	Additional guidance
9.	(a)	(i)	(+) 220 (+/- 2) (kJ mol^{-1})	**1**	No units required. No mark can be awarded for correct answer if wrong unit is given (where no unit required, wrong units would only be penalised once in any paper). kJ is acceptable in place of kJ mol^{-1} (KJ or Kj or KJ mol^{-1} or Kj mol^{-1} accepted).
		(ii)	(Increasing temperature) favours the endothermic/reverse reaction **OR** The (forward) reaction is exothermic. **OR** The reverse reaction is endothermic.	**1**	
	(b)		Diagram shows a workable method for removal of HCl(g) allowing transfer of chlorine gas. Water/H_2O must be labelled. **(1)** Diagram shows a workable method of collecting a gas (eg a gas syringe). **(1)**	**2**	Treat each mark separately. Mark not awarded if delivery tube passes through the wall of measuring cylinder/test tube A closed system would not be a suitable method for collecting a gas.

Question			Expected response	Max mark	Additional guidance
	(c)		-391 (kJ mol^{-1}) **(2 marks)** Partial marks Treat as two concepts. Either would be acceptable for **1 mark**. Evidence of understanding of reversal of first enthalpy value. ie +75 must be seen. The other two enthalpy values (regardless of value) **must** be negative, or this partial mark cannot be awarded. **OR** Evidence of understanding of multiplying the third enthalpy value by 4(± 92). **OR** ± 368 Multiplication of any other enthalpy value by any factor is taken as cancelling of this partial mark.	**2**	**Only one concept mark can be awarded if the final answer is incorrect.** If answer given is +391, maximum of 1 mark can be awarded. No units required. Only 1 mark can be awarded for the correct answer if wrong unit is given. (wrong units would only be penalised once in any paper) kJ is acceptable in place of kJ mol^{-1} (KJ or Kj or KJ mol^{-1} or Kj mol^{-1} accepted).

Question			Expected response	Max mark	Additional guidance
10.	(a)		Chloride ions/magnesium ions/metal ions/salts may be present in tap water or not present/less in deionised/distilled water.	**1**	"ions" on its own is not sufficient Accept chlorine but not chlorine ion(s). If ions are shown, charges must be correct.
	(b)	(i)	<table><tr><th>Measurement</th><th>Apparatus</th></tr><tr><td>20·0 cm^3 (accurately)</td><td>Pipette</td></tr><tr><td>35 cm^3 (approximately)</td><td>Measuring cylinder</td></tr></table> **1 mark** for each correct entry	**2**	
		(ii)	 All correct for **1 mark**	**1**	
		(iii)	0·463/0·46/0·5 (g) Partial marking (**1 mark**) **1 mark** for knowledge of relationship between moles of silver chloride and magnesium chloride. This could be shown by a calculated number of moles of silver chloride correctly divided by 2 **OR** Incorrect mole ratio used but the relationship between moles and mass used correctly twice **OR** by proportion 95·3 ↔ 286·8 (2 x 143·4) **OR** Mole ratio not applied correctly but proportion used correctly	**2**	No units required. Only 1 mark can be awarded for correct answer if wrong unit is given (where no unit required, wrong units would only be penalised once in any paper).
		(c)	96/96·0 (%)	**1**	

Question			Expected response	Max mark	Additional guidance
11.			This is an open ended question **1 mark:** The student has demonstrated, at an appropriate level, a limited understanding of the chemistry involved. The student has made some statement(s) which is/are relevant to the situation, showing that at least a little of the chemistry within the problem is understood. **2 marks:** The student has demonstrated a reasonable understanding, at an appropriate level, of the chemistry involved. The student makes some statement(s) which is/are relevant to the situation, showing that the problem is understood. **3 marks:** The maximum available mark would be awarded to a student who has demonstrated a good understanding, at an appropriate level, of the chemistry involved. The student shows a good comprehension of the chemistry of the situation and has provided a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. This does not mean the answer has to be what might be termed an "excellent" answer or a "complete" one.	**3**	Zero marks should be awarded if: The student has demonstrated no understanding of the chemistry involved. There is no evidence that the student has recognised the area of chemistry involved or has given any statement of a relevant chemistry principle. This mark would also be given when the student merely restates the chemistry given in the question.

Question			Expected response	Max mark	Additional guidance
12	(a)	(i)	Ionic and positively charged Both needed	**1**	Accept positive/+ve/+
		(ii) A	alkaline hydrolysis/saponification	**1**	Do not accept "hydrolysis" on its own. Hydrolysis with named strong alkali would be acceptable.
		(ii) B	(Compound C) has an ionic/hydrophilic part **and** a non polar/hydrophobic part (or alternative wording/diagram showing knowledge of these parts of the molecule) **(1)** Correctly identifies the part of the molecule/head/COO^- dissolves in water/ is hydrophilic and the part of the molecule/tail/hydrocarbon chain dissolves in oil/hydrophobic. **(1)** Agitation separates oil from the surface/cause small oil droplets to form **OR** The (negatively-charged) ball-like structures repel each other (and the oil or grease is kept suspended in the water) **OR** Soaps/compound C allow(s) emulsions to form or break(s) oil into micelles. **(1)** Accept correct diagrams with annotations that show above.	**3**	
		(iii)	Reacting them (edible oils) with glycerol/ propan-1,2,3-triol/propane-1,2,3-triol/ 1,2,3-propanetriol/$C_3H_8O_3$	**1**	

Question			Expected response	Max mark	Additional guidance
12.	(b)	(i)	Both nuclei have the same attraction for the bonding electrons. **OR** Both atoms have same electronegativity/ or electronegativity values given **OR** Bonding electrons shared evenly.	**1**	Diagram on its own not sufficient
		(ii)	**1 mark** for any of the following • To ensure all chlorine is used up/to prevent chlorine being released • NaOH is the cheaper/less expensive reactant • To ensure that the bleach cleaner contains sodium hydroxide • Excess NaOH would neutralise any acid added to cleaner • Excess NaOH helps break up oil/grease	**1**	
	(c)		(Adding acid) adds/increases H^+ (ions) **(1)** Rate of forward reaction is increased/ speeds up (by addition of acid) **(1)**	**2**	
	(d)	(i)	$OCl^-_{(aq)} + 2H^+_{(aq)} + 2e^- \rightarrow Cl^-_{(aq)} + H_2O_{(l)}$	**1**	State symbols not required but **any** given must be correct. Charge on electron not required.

Question			Expected response	Max mark	Additional guidance
12	(d)	(ii)	$1{\cdot}76 \times 10^{-2}$ (mol l^{-1}) (3 marks) Partial marks can be awarded using a scheme of two "concept" marks, and one "arithmetic" mark **1 mark** for knowledge of the relationship between moles, concentration and volume. This could be shown by one of the following steps: Calculation of moles thiosulfate solution eg 0·098 × 0·009 = 0·000882 **OR** calculation of concentration of iodine solution eg 0·000441/0·025 **OR** Insertion of correct pairings of values for concentration and volume in a valid titration formula **1 mark** for knowledge of relationship between moles of thiosulfate and hypochlorite. This could be shown by one of the following steps: Calculation of moles hypochlorite from moles thiosulfate eg 0·000882/2 = 0·000441 **OR** Insertion of correct stoichiometric values in a valid titration formula **1 mark** is awarded for correct arithmetic through the calculation. This mark can only be awarded if both concept marks have been awarded.	**3**	No units required but only two marks can be awarded for correct answer if wrong unit is given. (wrong units would only be penalised once in the paper).

[END OF MARKING INSTRUCTIONS]

General marking principles for Higher Chemistry

Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.

(a) Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.

(b) If a candidate response does not seem to be covered by either the principles or detailed marking instructions, and you are uncertain how to assess it, you must seek guidance from your team leader.

(c) Do not award half marks.

(d) Where a candidate makes an error at an early stage in a multi-stage calculation, award marks for correct follow-on working in subsequent stages, unless the error significantly reduces the complexity of the remaining stages. Apply the same principle for questions that require several stages of non-mathematical reasoning. The exception to this rule is where the marking instructions for a numerical question assign separate 'concept marks' and an 'arithmetic mark'. In such situations, the marking instructions will give clear guidance on the assignment or partial marks.

(e) Unless a numerical question specifically requires evidence of working to be shown, award full marks for a correct final response (including units) on its own.

(f) Candidates may fully access larger mark allocations whether their responses are in continuous prose, linked statements, or a series of developed bullet points.

(g) Do not deduct marks for inaccurate or unconventional spelling or vocabulary as long as the meaning of the word(s) is conveyed. **For example**, responses that include 'distilling' for 'distillation', or 'it gets hotter' for 'the temperature rises', are acceptable.

(h) In many questions, the unit in which the answer is to be expressed is given. In these questions, the candidate does not need to state a unit in their answer; but if they do, the unit must be correct. The full mark allocation cannot be awarded if an incorrect unit is shown. In these questions, incorrect units would only be penalised once in any paper.

(i) If a correct response is followed by a wrong response, award no marks. **For example** in response to the question, 'State the colour seen when blue Fehling's solution is warmed with an aldehyde', do not award marks for the response 'red green'. However, if a correct response is followed by additional information which does not conflict with that, ignore the additional information, whether correct or not. **For example** in response to a question concerned with melting point, 'State why the tube should not be made of copper', the response 'Copper has a low melting point and is coloured grey' **would** gain marks.

(j) Award full marks for the correct response to a calculation without working. Award partial marks, as shown in the detailed marking instructions, when working is given but the final response is incorrect. An exception is when candidates are asked to 'Find, by calculation' - do not award full marks for the correct response without working.

(k) Ignore the omission of one H atom from a full structural formula provided the bond is shown.

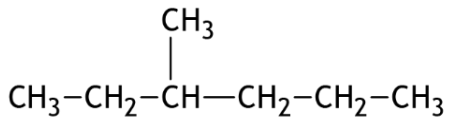
(l) Award marks for a symbol or correct formula in place of a name **unless stated otherwise in the detailed marking instructions.**

(m) When formulae of ionic compounds are given as responses, candidates only need to show ion charges if these have been specifically asked for. However, if ion charges are shown, they must be correct. If incorrect charges are shown, do not award marks.

(n) If an answer comes directly from the text of the question, do not award marks. **For example**, in response to the question, 'A student found that 0·05 mol of propane, C_3H_8 burned to give 82·4 kJ of energy. $C_3H_8(g) + 5O_2(g) = 3CO_2(g) + 4H_2O(\ell)$. Name the kind of enthalpy change that the student measured', do not award marks for 'burning' since the word 'burned' appears in the text.

(o) A guiding principle in marking is to give credit for correct elements of a response rather than to look for reasons not to give marks.

Example 1: The structure of a hydrocarbon found in petrol is shown below.



Name the hydrocarbon

- Award the full mark for '3, methyl-hexane', although the punctuation is not correct.

Example 2: A student measured the pH of four carboxylic acids to find out how their strength is related to the number of chlorine atoms in the molecule. The results are shown.

CH_3COOH	1·65
$CH_2ClCOOH$	1·27
$CHCl_2COOH$	0·90
CCl_3COOH	0·51

Describe the relationship between the number of chlorine atoms in the molecule and the strengths of the acids.

- Award the full mark for a response such as 'the more Cl_2, the stronger the acid', although not completely correct.

(p) Unless the question is clearly about a non-chemistry issue, for example costs in an industrial chemical process, do not award marks for a non-chemical response.
For example, in response to the question, 'Why does the (catalytic) converter have a honeycomb structure?', do not award a mark for 'To make it work'. This response may be correct but it is not a chemical response.

(q) Only award marks for a valid response to the question asked. Where candidates are asked to:

- **identify, name, give** or **state**, they must only name or present in brief form.
- **describe**, they must provide a statement or structure of characteristics and/or features.
- **explain**, they must relate cause and effect and/or make relationships between things clear.
- **compare**, they must demonstrate knowledge and understanding of the similarities and/or differences between things.
- **complete**, they must finish a chemical equation or fill in a table with information.
- **determine** or **calculate**, they must determine a number from given facts, figures or information.
- **draw**, they must draw a diagram or structural formula, for example 'Draw a diagram to show the part of a poly(propene) molecule formed from two propene molecules.'
- **estimate**, they must determine an approximate value for something.
- **predict**, they must suggest what may happen based on available information.
- **evaluate**, they must make a judgement based on criteria.
- **suggest**, they must apply their knowledge and understanding of chemistry to a new situation. A number of responses are acceptable: award marks for any suggestions that are supported by knowledge and understanding of chemistry.
- **use their knowledge of chemistry or aspect of chemistry to comment on**, they must apply their skills, knowledge and understanding to respond appropriately to the problem/situation presented (for example by making a statement of principle(s) involved and/or a relationship or equation, and applying these to respond to the problem/situation). Candidates gain marks for the breadth and/or depth of their conceptual understanding.
- **write**, they must complete a chemical or word equation, for example 'Write the word equation for the complete combustion of ethanol.'

2019 Higher Chemistry Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	**C**	4	2	**86**	8	0	0
2	**B**	4	**76**	15	3	0	1
3	**A**	**26**	23	20	28	0	2
4	**C**	15	24	**48**	12	1	1
5	**B**	9	**81**	7	3	0	1
6	**A**	**64**	7	22	7	0	1
7	**D**	8	8	4	**78**	0	1
8	**B**	6	**29**	53	11	0	1
9	**D**	1	10	1	**88**	0	0
10	**D**	5	2	22	**72**	0	1
11	**C**	6	8	**74**	11	0	1
12	**B**	5	**92**	2	1	0	1
13	**D**	24	21	18	**36**	0	1
14	**A**	**65**	6	26	2	1	0
15	**C**	12	7	**59**	21	1	1
16	**D**	10	19	9	**61**	0	1
17	**B**	2	**64**	6	27	0	1
18	**B**	32	**42**	15	10	1	1
19	**B**	12	**45**	18	23	1	1
20	**C**	16	28	**50**	5	0	1
21	**A**	**70**	2	4	24	0	1
22	**D**	10	3	1	**85**	0	1
23	**C**	6	14	**71**	7	0	1
24	**A**	**71**	6	17	5	0	1
25	**D**	39	5	8	**47**	0	1

National
Qualifications
2018

2018 Chemistry

Higher

Finalised Marking Instructions

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Marking Instructions for each question

Section 1

Question	Answer	Mark
1.	B	**1**
2.	A	**1**
3.	D	**1**
4.	C	**1**
5.	D	**1**
6.	C	**1**
7.	A	**1**
8.	D	**1**
9.	A	**1**
10.	B	**1**
11.	D	**1**
12.	D	**1**
13.	A	**1**
14.	A	**1**
15.	D	**1**
16.	C	**1**
17.	B	**1**
18.	C	**1**
19.	B	**1**
20	B	**1**

Section 2

Question			Expected response	Max mark	Additional guidance
1.	(a)	(i)	Electronegativity is the measure of attraction an atom/nucleus has for the electrons in a bond/shared electrons.	**1**	
		(ii)	(More shells) so increased shielding/more shielding. **OR** Covalent radius increases/atom size increases/more shells so attraction of the nucleus/protons for the (outer/shared) electrons decreases.	**1**	
	(b)		(Intermolecular) forces/bonds increase (going down the group). **(1)** LDFs are the forces (broken) between the molecules. **(1)** The more electrons the stronger the LDFs. **(1)**	**3**	Use of VdW is acceptable for first mark. Mention of **increasing/stronger** intramolecular/covalent/ionic/ metallic bonds – cancels the first mark. This mark can only be awarded if no other forces are mentioned (as being broken). Use of VdW does not cancel but is not sufficient for the second mark.

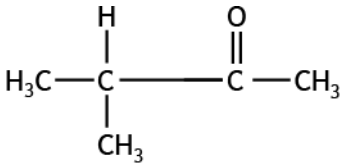
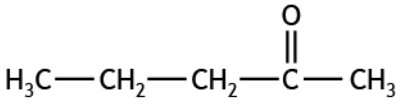
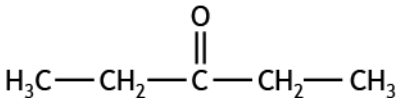
Question			Expected response	Max mark	Additional guidance
2.	(a)		Increasing number of protons (in the nucleus). **OR** Increasing/greater/stronger/higher nuclear charge.	**1**	Mention must be made of **nuclear charge or number of protons.** Increased nuclear pull is not accepted on its own. Increased attraction of the electron for the nucleus is not an acceptable answer.
	(b)	(i)	Sulfur chloride should be circled.	**1**	Accept other indications of correct structure, eg tick, arrow. Do not accept circling of single atoms or bond in SCl_2 molecule on its own.
		(ii)	Silicon tetrachloride and hexane are non-polar. **(1)** Silicon tetrachloride is non-polar due to its shape/dipoles/polarities cancelling out. **(1)**	**2**	Like dissolves like on its own – zero marks - answer must mention polarity.

Question			Expected response	Max mark	Additional guidance
2.	(c)	(i)	Silicon nitride is a (covalent) network. **(1)** (Strong) covalent bonds are broken. **(1)**	**2**	'Covalent lattice' is acceptable, but 'lattice' on its own is not. Mention of silicon is cancelling for the first mark. Covalent network molecule is cancelling.
		(ii)	17·934/17·93/17·9/18 (%) --- Partial mark for correct use of atom economy relationship without correct use of stoichiometry (working must be shown). Partial marks Correct working with no correct answer given. $\frac{140{\cdot}3}{(3\times170{\cdot}1)+(16\times17{\cdot}0)}\times100$ $\frac{140{\cdot}3}{510{\cdot}3+272{\cdot}0}\times100$ $\frac{140{\cdot}3}{782{\cdot}3}\times100$ Incorrect use of stoichiometry. $\frac{140{\cdot}3}{170{\cdot}1+17{\cdot}0}x100=74{\cdot}99$ $\frac{140{\cdot}3}{3\times170{\cdot}1+17{\cdot}0}x100=26{\cdot}6$ $\frac{140{\cdot}3}{170{\cdot}1+16\times17{\cdot}0}x100=31{\cdot}7$ Answer and working must be shown. --- 0·179 **(1)**	**2**	No units required. Only 1 mark can be awarded for the correct answer if wrong unit is given. (Wrong units would only be penalised once in any paper).

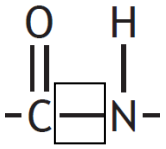
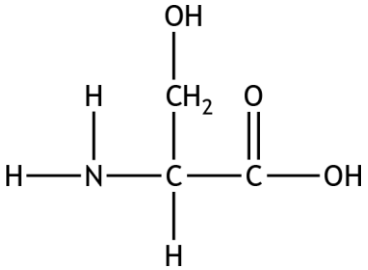
Question			Expected response	Max mark	Additional guidance
2.	(d)	(i)	Diagram shows a workable method for the passing of chlorine gas over heated aluminium. Aluminium must be labelled and there must be an indication of heat. Heated aluminium accepted. **(1)** Diagram allows aluminium chloride to be collected **in a flask** as a solid and chlorine gas to escape. **(1)**	**2**	Labels must be correct.
		(ii)	To provide (initial) activation energy/(sufficient) energy to form activated (activation) complex.	**1**	Accept activation energy is high.

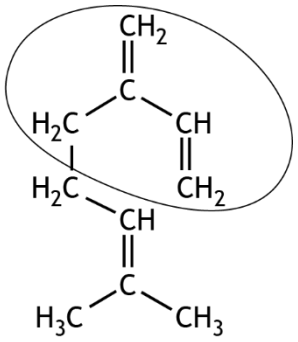
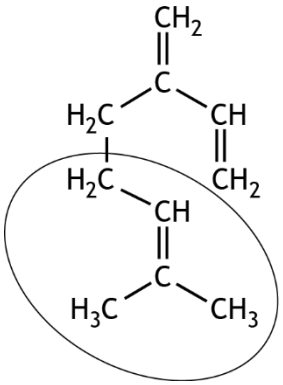
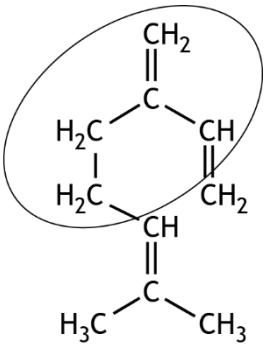
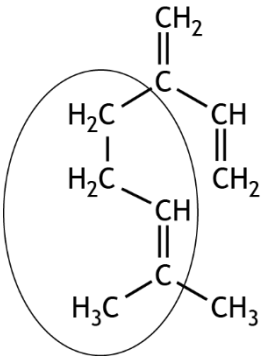
Question			Expected response	Max mark	Additional guidance
3.	(a)		Heating mantle or hot plate **OR** (hot) water bath.	**1**	Any mention of a naked flame would result in the loss of a mark.
	(b)		Condense reactants or products/acts as a condenser/to prevent escape of (volatile/gaseous) reactants or products/to prevent the escape of gas(es).	**1**	Accept correct description of condensation.
	(c)	(i)	Water	**1**	Accept formula H_2O
		(ii)	Correctly calculates number of moles of: Benzoic acid = 0·041 Methanol = 0·078. **OR** Working out that 1·31 g of methanol would be needed to react with 5 g of benzoic acid. **OR** Working out that 9·53 g of benzoic acid would be needed to react with 2·5 g of methanol. **(1)** Statement demonstrating understanding of limiting reactant eg there are less moles of benzoic acid therefore it is the limiting reactant. **OR** There are more moles of methanol therefore it is in excess. **OR** 0·078 moles of methanol would require 0·078 moles of benzoic acid. **(1)**	**2**	

Question			Expected response	Max mark	Additional guidance
3.	(c)	(iii)	(£)12·84 **(2)** Partial Marks Mass benzoic acid = 161·3(g). **OR** Cost to make 3·1g of methyl benzoate = (£) 0·398. **OR** Evidence of a calculated mass of benzoic acid × 7·96 or 8 (p). **(1)**	**2**	Accept 1284 p. Do not accept '1284' on its own; correct units are required. Allow follow through from an initial arithmetic error (for 1 mark). Rounding of final answer to the nearest penny is required.

Question			Expected response	Max mark	Additional guidance
4.	(a)		Correctly drawn structure of pentan- 2-one, pentan-3-one or 3- methylbutanone.   	**1**	Accept full or shortened structural formulae.
	(b)		Fehling's solution/Tollens' reagent/ **acidified** dichromate solution.	**1**	Accept Benedict's solution/Schiff's reagent.
	(c)		Permanent dipole-permanent dipole (interactions/attractions).	**1**	Allow permanent dipole-dipole interaction. Accept pd-pd i's.
	(d)		Will react with oxygen/undergo oxidation. **(1)** Forming a **carboxylic** acid (which has unpleasant taste). **(1)**	**2**	Rancid not acceptable on its own but is not cancelling. Acid not acceptable on its own.
	(e)	(i)	Because it has two molecules joining together with the loss of a small/water molecule.	**1**	Candidates must indicate joining together and loss.
		(ii)	6-methylheptan-2-one	**1**	General marking principle (p) applies.

Question			Expected response	Max mark	Additional guidance
5.			This is an open ended question **1 mark:** The student has demonstrated, at an appropriate level, a limited understanding of the chemistry involved. The candidate has made some statement(s) which is/are relevant to the situation, showing that at least a little of the chemistry within the problem is understood. **2 marks:** The student has demonstrated, at an appropriate level, a reasonable understanding of the chemistry involved. The student makes some statement(s) which is/are relevant to the situation, showing that the problem is understood. **3 marks:** The maximum available mark would be awarded to a student who has demonstrated, at an appropriate level, a good understanding of the chemistry involved. The student shows a good comprehension of the chemistry of the situation and has provided a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. This does not mean the answer has to be what might be termed an 'excellent' answer or a 'complete' one.	**3**	Zero marks should be awarded if: The student has demonstrated, at an appropriate level, no understanding, of the chemistry involved. There is no evidence that the student has recognised the area of chemistry involved or has given any statement of a relevant chemistry principle. This mark would also be given when the student merely restates the chemistry given in the question.

Question			Expected response	Max mark	Additional guidance
6.	(a)	(i)	(enzyme) hydrolysis	**1**	
		(ii)	$C_{20}H_{29}OH$ **OR** $C_{20}H_{30}O$	**1**	
	(b)	(i)	Bond breaking by UV (light) or example of initiation reaction (equation or diagram). eg chlorine splitting to give two free radicals is accepted, provided UV is shown.	**1**	UV on its own is not accepted. Bond breaking on its own is not accepted.
		(ii)	propagation	**1**	
		(iii)	Can react with free radicals forming stable molecules/free radicals (and prevent chain reactions). **OR** Donates electron(s). **OR** Acting as a reducing agent. **OR** Provide electrons to pair with an unpaired electron.	**1**	
	(c)	(i)	Circle any peptide link (CONH).	**1**	 Minimum acceptable structure identified.
		(ii)		**1**	Shortened structural formula accepted.

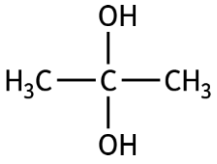
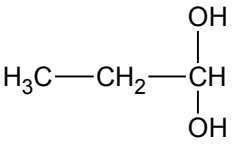
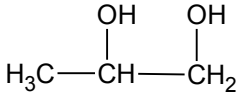
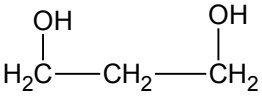
Question			Expected response	Max mark	Additional guidance
7.	(a)	(i)	 	**1**	0 marks awarded for  
		(ii)	Sesquiterpene	**1**	

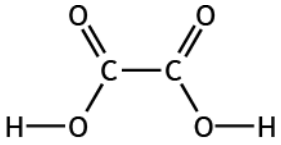
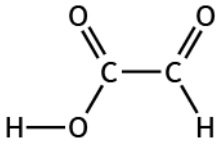
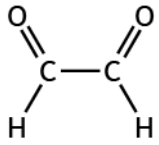
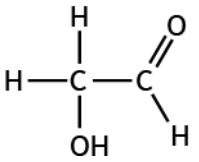
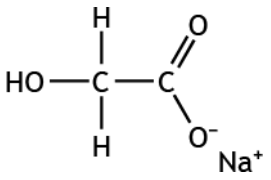
Question			Expected response	Max mark	Additional guidance
7.	(b)	(i)	5·345/5.35/5.3 (kg) **(2)** Partial marking Mass of squalene $= 10{\cdot}69 \times 500\,000$ $= 5\,345\,000$ (mg) **(1)** **OR** For incorrectly calculating mass in mg but correctly converting to kg. **(1)** **OR** For incorrectly calculating mass of squalene but correctly multiplying this by 500 000. **(1)** **OR** Conversion of 10·69 mg to kg ie $10{\cdot}69 \times 10^{-6}$.	**2**	Answer must be in kg to access both marks, 5345 g would be awarded 1 mark only.
		(ii)	6/six (moles)	**1**	
	(c)	(i)	Addition **OR** hydration.	**1**	Oxidation is not accepted.
		(ii)	(terpineol) is a tertiary alcohol (and cannot be oxidised).	**1**	'cannot be oxidised' on its own is awarded zero marks.

Question			Expected response	Max mark	Additional guidance
8.	(a)		286 (kJ mol^{-1}) Partial mark 1 mark Evidence of the use of **all** the correct bond enthalpies (or correct multiples thereof) (412, 348, 838, 436 (ignore signs)). **OR** If only three values are retrieved, the candidate recognises that bond breaking is endothermic and bond formation is exothermic and correctly manipulates the bond enthalpy values they have used to give their answer.	**2**	-286 (kJ mol^{-1}) 1 mark No units required. Only 1 mark can be awarded for the correct answer if wrong unit is given. (Wrong units would only be penalised once in any paper). kJ is acceptable in place of kJ mol^{-1} (KJ or Kj or KJ mol^{-1} or Kj mol^{-1} accepted). If fewer than three bond enthalpy values are retrieved then zero marks can be awarded.
	(b)		(+) 185 (kJ mol^{-1}) [(-1182) + (-572) + (+1939)] = (+) 185 (kJ mol^{-1}) Partial marks Treat as two concepts. Either would be acceptable for 1 mark. Evidence of understanding of reversal of third enthalpy value ie +1939 or 1939 must be seen. The other two enthalpy values (regardless of value) **must** be negative, or this partial mark cannot be awarded. **OR** Evidence of understanding of multiplying the first enthalpy value by 3 and the second enthalpy value by 2. Ignore the enthalpy signs associated with these numbers ie any combination of 3 (±394) **and** 2 (±286) **OR** ±1182 **and** ±572 Multiplication of the third enthalpy value by any factor is taken as cancelling of this partial mark.	**2**	If answer given is −185, maximum of 1 mark can be awarded. No units required. Only 1 mark can be awarded for the correct answer if wrong unit is given. (wrong units would only be penalised once in any paper) kJ is acceptable in place of kJ mol^{-1} (KJ or Kj or KJ mol^{-1} or Kj mol^{-1} accepted).

Question			Expected response	Max mark	Additional guidance
8.	(c)	(i)	48475 (kJ)	**1**	−48475 (kJ) is accepted. No units required. Zero marks can be awarded for the correct answer if wrong unit is given. kJ mol^{-1} is not an acceptable unit in this question. (Wrong units would only be penalised once in any paper).
		(ii)	13·76/13·8/14 (g) Partial mark - 1 mark Mass of oxygen required = 3·2 (g) **OR** 550·4 (g)	**2**	Accept correctly calculated answer in mg or kg, providing units are shown with answer. This is obtained from an incorrect mass of oxygen ($4 \times 32 = 128$) being calculated.
		(iii)	Methanol and ethanol contain oxygen in their structure, (so less additional oxygen is required).	**1**	Contains hydroxyl (groups) not sufficient on its own. Contains -OH is not considered sufficient on its own.

Question			Expected response	Max mark	Additional guidance
9.	(a)	(i)	**1 mark each** for any two of the following points. • recycle (waste) gases • use catalyst • low/reduce energy requirements • reactors are run at low temperatures/the temperatures in the reactors is lowered • inexpensive feedstocks • selling/using by-products	**2**	Recycle by-products is not accepted.
		(ii)	(fractional) distillation	**1**	Evaporation followed by condensation or a correct description of distillation are accepted.
	(b)		Propan-1-ol has fewer hydroxyl groups than ethane-1,2-diol / ethane-1,2-diol has more hydroxyl groups/propan-1-ol has 1 hydroxyl group and ethane-1,2-diol has 2. **(1)** Weaker/fewer hydrogen bonds between propan-1-ol molecules. **OR** Stronger/more hydrogen bonds between ethane-1,2-diol molecules. **(1)**	**2**	"It", if given in answer is taken to refer to ethane-1,2-diol.

Question			Expected response	Max mark	Additional guidance
9.	(c)		Structure of propane-1,2-diol or propane-1,3-diol or propane-1,1-diol or propane-2,2-diol.    	**1**	With structures involving an – OH group, no mark can be awarded if the 'O' is not bonded to a carbon, ie OH–CH_2.
	(d)	(i)	Pipette (used to measure 20cm^3 of ethanol.) **(1)** Statement of use of **volumetric/ standard** flask to make up to / fill to the mark/ to 100cm^3. **(1)**	**2**	Not acceptable to use burette.
		(ii)	157·5(cm^3)	**1**	No units required. No mark can be awarded for the correct answer if wrong unit is given. (Wrong units would only be penalised once in any paper).

Question			Expected response	Max mark	Additional guidance
9.	(d)	(iii) (A)	 **OR**  **OR**  **OR** 	**1**	Accept correct shortened structural formula. With structures involving an – OH group, no mark can be awarded if the 'O' is not bonded to a carbon, ie OH–CH_2. General marking instruction (l) is suspended here.
		(iii) (B)	Correct molecular formula ($NaC_2H_3O_3$) **OR** shortened structural formula ($HOCH_2COONa$) **OR** any full structural formula which shows the correct salt. 	**1**	Structures showing covalent bonds between Na and glycolate are not accepted. Charges are not required but if shown must both be shown and correct.

Question			Expected response	Max mark	Additional guidance
10.			This is an open ended question **1 mark:** The student has demonstrated, at an appropriate level, a limited understanding of the chemistry involved. The candidate has made some statement(s) which is/are relevant to the situation, showing that at least a little of the chemistry within the problem is understood. **2 marks:** The student has demonstrated, at an appropriate level, a reasonable understanding of the chemistry involved. The student makes some statement(s) which is/are relevant to the situation, showing that the problem is understood. **3 marks:** The maximum available mark would be awarded to a student who has demonstrated, at an appropriate level, a good understanding of the chemistry involved. The student shows a good comprehension of the chemistry of the situation and has provided a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. This does not mean the answer has to be what might be termed an 'excellent' answer or a 'complete' one.	**3**	Zero marks should be awarded if: The student has demonstrated, at an appropriate level, no understanding, of the chemistry involved. There is no evidence that the student has recognised the area of chemistry involved or has given any statement of a relevant chemistry principle. This mark would also be given when the student merely restates the chemistry given in the question.

Question			Expected response	Max mark	Additional guidance
11.	(a)		Correct description of weighing by difference. **OR** Correct description of use of the Tare function.	**1**	'weighing by difference' on its own is accepted.
	(b)		$2I^-(aq) \rightarrow I_2(aq) + 2e^-$ **OR** $2I^-(aq) \rightarrow I_2(s) + 2e^-$	**1**	State symbols not required. Negative sign on electron not required.
	(c)	(i)	$9{\cdot}5(cm^3)$	**1**	No units required but no mark is awarded if wrong unit is given. (Wrong units would only be penalised once in any paper).
		(ii)	$4{\cdot}75 \times 10^{-6}$ moles Partial mark for correct use of mole ratio. **OR** Determination of number of moles without using the mole ratio.	**2**	Allow follow through from c (i).

Question			Expected response	Max mark	Additional guidance
12.	(a)	(i)	More/adding chlorine(s). **(1)** More/adding carbon(s). **OR** Adding an alkyl/hydrocarbon chain/group. **OR** Longer/bigger carbon/hydrocarbon/ alkyl (chain/group). **(1)**	**2**	(More) branches is not accepted.
		(ii)	2-chloro-4,5-dimethylphenol	**1**	Apply general marking instruction (p).
	(b)	(i)	126·9/127(kg) Partial mark either for: Calculation of the theoretical yield 141 (no unit required) **(1)** **OR** for correctly calculating 90% of an incorrectly calculated theoretical yield. **(1)**	**2**	No units required but a maximum of 1 mark is awarded if wrong unit is given. (Wrong units would only be penalised once in any paper). Accept correctly calculated answer in g, providing units are shown with answer, eg 126900 g.
		(ii)	Propanone **OR** Acetone **OR** propan-2-one.	**1**	Structure is not acceptable on its own.

[END OF MARKING INSTRUCTIONS]

General marking principles for Higher Chemistry

This information is provided to help you understand the general principles you must apply when marking candidate responses to questions in this paper. These principles must be read in conjunction with the detailed marking instructions, which identify the key features required in candidate responses.

(a) Marks for each candidate response must always be assigned in line with these general marking principles and the detailed marking instructions for this assessment.

(b) Marking should always be positive, ie marks should be awarded for what is correct and not deducted for errors or omissions.

(c) If a specific candidate response does not seem to be covered by either the principles or detailed marking instructions, and you are uncertain how to assess it, you must seek guidance from your team leader.

(d) There are no half marks awarded.

(e) Where a candidate makes an error at an early stage in a multi stage calculation, credit should normally be given for correct follow on working in subsequent stages, unless the error significantly reduces the complexity of the remaining stages. The same principle should be applied in questions which require several stages of non-mathematical reasoning. The exception to this rule is where the marking instructions for a numerical question assign separate "concept marks" and an "arithmetic mark". In such situations, the marking instructions will give clear guidance on the assignment or partial marks.

(f) Unless a numerical question specifically requires evidence of working to be shown, full marks should be awarded for a correct final answer (including units) on its own.

(g) In many questions, the unit in which the answer is to be expressed is given. In these questions, the candidate does not need to state a unit in their answer; but if they do, the unit must be correct. The full mark allocation cannot be awarded if an incorrect unit is shown. In these questions, incorrect units would only be penalised once in any paper.

(h) Larger mark allocations may be fully accessed whether responses are provided in continuous prose, linked statements or a series of developed bullet points.

(i) Marks should not be deducted for inaccurate or unconventional spelling or vocabulary as long as the meaning of the word(s) is conveyed. **For example**, responses that include 'distilling' for 'distillation', or 'it gets hotter' for 'the temperature rises', should be accepted.

(j) If a correct answer is followed by a wrong answer, it should be treated as a cancelling error and no marks should be given. **For example**, in response to the question, 'State the colour seen when blue Fehling's solution is warmed with an aldehyde', the answer 'red green' gains no marks. However, if a correct answer is followed by additional information which does not conflict with that, the additional information should be ignored, whether correct or not. **For example**, in response to a question concerned with melting point, 'State why the tube should not be made of copper', the response 'Copper has a low melting point and is coloured grey' would **not** be treated as having a cancelling error.

(k) Full marks are usually awarded for the correct answer to a calculation without working and the partial marks shown in the detailed marking instructions are for use when working is given but the final answer is incorrect. An exception is when candidates are asked to 'Find, by calculation', when full marks cannot be awarded for the correct answer without working.

(l) Ignore the omission of one H atom from a full structural formula provided the bond is shown.

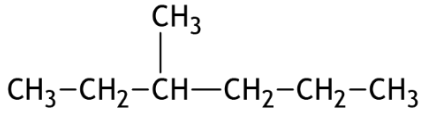
(m) A symbol or correct formula should be accepted in place of a name **unless stated otherwise in the detailed marking instructions.**

(n) When formulae of ionic compounds are given as answers it will only be necessary to show ion charges if these have been specifically asked for. However, if ion charges are shown, they must be correct. If incorrect charges are shown, no marks should be awarded.

(o) If an answer comes directly from the text of the question, no marks should be given. **For example**, in response to the question, ' A student found that 0·05 mol of propane, C_3H_8 burned togive 82·4 kJ of energy. $C_3H_8(g) + 5O_2(g) = 3CO_2(g) + 4H_2O(\ell)$. Name the kind of enthalpy change that the student measured', no marks should be given for "burning" since the word "burned" appears in the text.

(p) A guiding principle in marking is to give credit for correct elements of a response rather than to look for reasons not to give marks.

Example 1: The structure of a hydrocarbon found in petrol is shown below.



Name the hydrocarbon

- Although the punctuation is not correct '3, methyl-hexane' should gain the full mark.

Example 2: A student measured the pH of four carboxylic acids to find out how their strength is related to the number of chlorine atoms in the molecule. The results are shown.

Structural formula	*pH*
CH_3COOH	1·65
$CH_2ClCOOH$	1·27
$CHCl_2COOH$	0·90
CCl_3COOH	0·51

Describe the relationship between the number of chlorine atoms in the molecule and the strengths of the acids.

- Although not completely correct, an answer such as 'the more Cl_2, the stronger the acid' should gain the full mark.

(q) Unless the question is clearly about a non-chemistry issue, eg costs in an industrial chemical process, a non-chemical answer gains no marks.
For example, in response to the question, 'Why does the (catalytic) converter have a honeycomb structure?', 'to make it work' may be correct but it is not a chemical answer and the mark should not be given.

(r) Marks are awarded only for a valid response to the question asked. For example, in response to questions that ask candidates to:

- **identify, name, give** or **state**, they need only name or present in brief form;
- **describe**, they must provide a statement or structure of characteristics and/or features;
- **explain**, they must relate cause and effect and/or make relationships between things clear;
- **compare**, they must demonstrate knowledge and understanding of the similarities and/or differences between things;
- **complete**, they must finish a chemical equation or fill in a table with information
- **determine** or **calculate**, they must determine a number from given facts, figures or information;

- **draw**, they must draw a diagram or structural formula, eg “Draw a diagram to show the part of a poly(propene) molecule formed from two propene molecules”
- **estimate**, they must determine an approximate value for something;
- **predict**, they must suggest what may happen based on available information;
- **evaluate**, they must make a judgement based on criteria;
- **suggest**, they must apply their knowledge and understanding of [subject] to a new situation. A number of responses are acceptable; marks will be awarded for any suggestions that are supported by knowledge and understanding of [subject];
- **use your knowledge of [chemistry or aspect of chemistry] to comment on**, they must apply their skills, knowledge and understanding to respond appropriately to the problem/situation presented (for example by making a statement of principle(s) involved and/or a relationship or equation, and applying these to respond to the problem/situation). They will be rewarded for the breadth and/or depth of their conceptual understanding.
- **Write**, they must complete a chemical or word equation, eg “Write the word equation for the complete combustion of ethanol.”

2018 Higher Chemistry Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	**B**	2	**70**	24	3	0	0
2	**A**	**91**	4	3	2	0	0
3	**D**	9	12	8	**70**	0	0
4	**C**	10	17	**57**	16	1	0
5	**D**	21	6	7	**65**	0	0
6	**C**	2	1	**77**	20	0	0
7	**A**	**39**	8	8	44	0	0
8	**D**	23	6	6	**64**	0	0
9	**A**	**69**	9	12	9	1	0
10	**B**	5	**93**	1	1	0	0
11	**D**	8	11	37	**43**	1	0
12	**D**	26	8	32	**34**	1	0
13	**A**	**48**	22	9	19	1	0
14	**A**	**52**	14	25	8	1	0
15	**D**	15	26	15	**43**	1	0
16	**C**	11	7	**67**	15	1	0
17	**B**	9	**75**	14	3	0	0
18	**C**	24	15	**43**	18	1	0
19	**B**	23	**49**	10	18	1	0
20	**B**	13	**59**	13	15	0	0

National
Qualifications
2017

2017 Chemistry

Higher

Finalised Marking Instructions

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Marking instructions for each question

Section 1

Question	Answer	Max Mark
1.	A	**1**
2.	D	**1**
3.	C	**1**
4.	B	**1**
5.	D	**1**
6.	C	**1**
7.	B	**1**
8.	C	**1**
9.	A	**1**
10.	C	**1**
11.	B	**1**
12.	C	**1**
13.	B	**1**
14.	B	**1**
15.	C	**1**
16.	A	**1**
17.	D	**1**
18.	A	**1**
19.	D	**1**
20.	B	**1**

Section 2

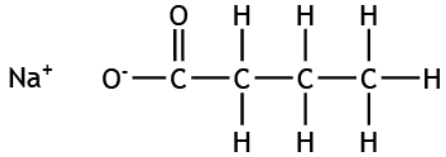
Question			Answer	Max mark	Additional guidance
1.	(a)		Silicon	**1**	Accept symbol
	(b)	(i)	Increasing/greater/stronger/higher nuclear charge (holds electron more tightly) **OR** Increasing number of protons	**1**	Increased nuclear pull is not accepted on its own. Mention must be made of nuclear charge **or** number of protons.
		(ii)	$Mg^{+}(g) \longrightarrow Mg^{2+}(g) + e^{-}$	**1**	State symbols required.
		(iii)	Fourth ionisation energy involves removal of an electron from an electron shell which is inner/full (whole)/(more) stable/closer to the nucleus **OR** fourth electron is removed from an electron shell which is inner/full (whole)/(more) stable/closer to the nucleus. **OR** removing third electron is taking from an outer/a part full shell **OR** taking an electron from a full shell requires more energy (than removing from a part full shell) **OR** taking an electron from a part full shell requires less energy (than removing from a full shell) **OR** fourth electron is less shielded than the third electron **OR** third electron is more shielded than the fourth electron	**1**	

Question			Answer	Max mark	Additional guidance
	(c)		1 mark - Correctly identify that there are stronger/more (Van der Waals) forces between chlorine (molecules) than between the argon (atoms) 1 mark - Correctly identifying that the van der Waals forces present in both these elements are London dispersion forces 1 mark Chlorine molecules (Cl_2) have more electrons than argon atoms (Ar).	**3**	This mark can only be awarded if no other forces are mentioned as being broken.

Question			Answer	Max mark	Additional guidance
2.	(a)	(i)	I-I bond is weaker/has a lower bond enthalpy value (so will break more easily) **OR** I_2 (151 kJ mol^{-1}) is less than H_2 (436 kJ mol^{-1}), (so will break more easily).	**1**	Iodine on its own is not acceptable.
		(ii)	Peak of curve should be further to the right and no higher than the original line.	**1**	
		(iii) (A)	Equilibrium will shift to the reactant side/left (hand side).	**1**	
		(iii) (B)	There are the same/equal volume/ number of moles/molecules (of gases) on each side (of the equation). **OR** Pressures of reactants and products are equal.	**1**	Ratios on their own, without an explanation, are not acceptable eg 1:1 or 2:2
		(iv) (A)	Activated complex	**1**	Accept activation complex or transition state.
		(iv) (B)	-9·6 (kJ) If candidate has calculated from graph values **OR** -9 (kJ) If candidate has calculated using bond enthalpies Answer must include the negative sign	**1**	No units required. No mark can be awarded for the correct answer if wrong unit is given. (wrong units would only be penalised once in any paper) kJ mol^{-1} is acceptable in place of kJ
		(iv) (C)	Decrease/lower it	**1**	

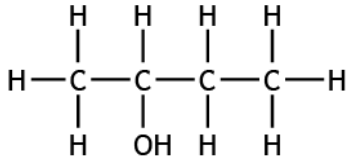
Question			Answer	Max mark	Additional guidance
	(b)	(i)	To keep the concentration (of the reactants) constant. **OR** Adding water will change/affect/ dilute/decrease the concentration (of the reactants) **OR** To keep the total volume constant.	**1**	
		(ii)	122·1 (accept 122) (s)	**1**	No units required. No mark can be awarded for the correct answer if wrong unit is given. (wrong units would only be penalised once in any paper)
		(iii)	The number of (successful) collisions will decrease. **OR** Less chance of (successful) collisions **OR** The frequency of (successful) collisions will decrease.	**1**	Any mention of time of collisions is unacceptable.

Question			Answer	Max mark	Additional guidance
3.			This is an open ended question **1 mark:** The student has demonstrated, at an appropriate level, a limited understanding of the chemistry involved. The candidate has made some statement(s) at which is/are relevant to the situation, showing that at least a little of the chemistry within the problem is understood. **2 marks:** The student has demonstrated, at an appropriate level, a reasonable understanding of the chemistry involved. The student makes some statement(s) which is/are relevant to the situation, showing that the problem is understood. **3 marks:** The maximum available mark would be awarded to a student who has demonstrated, at an appropriate level, a good understanding of the chemistry involved. The student shows a good comprehension of the chemistry of the situation and has provided a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. This does not mean the answer has to be what might be termed an 'excellent' answer or a 'complete' one.	**3**	Zero marks should be awarded if: The student has demonstrated, at an appropriate level, no understanding, of the chemistry involved. There is no evidence that the student has recognised the area of chemistry involved or has given any statement of a relevant chemistry principle. This mark would also be given when the student merely restates the chemistry given in the question.

Question			Answer	Max mark	Additional guidance
4.	(a)	(i)	Pentan-1-ol	**1**	Pentanol is not acceptable.
		(ii)		**1**	Candidates should be awarded zero marks if they draw a covalent bond between O and Na. Charges not required but if shown they must **both** be correct. Accept shortened structural formulae.
	(b)	(i)	ester	**1**	
		(ii)	soap	**1**	Accept emulsifier or detergent.
		(iii)	395 or -395 (kJ) Partial marking one mark can be awarded for **the correct application of number of moles of stearic acid** eg 10/284 x 623 or 0·0352 x 623 (=21.94) (=21.93) or 10/284 x 18 or 0·0352 x 18 (=0.634) (=0.634) **OR** **the correct application of the stoichiometry** eg the energy change for 1 mole of stearic acid as 623 x 18 = 11214 (kJ) or 284 g ←→ 623 x 18	**2**	No units required. Only 1 mark can be awarded for the correct answer if wrong unit is given. (wrong units would only be penalised once in any paper)

Question			Answer	Max mark	Additional guidance
5.	(a)	(i)	Diagram shows a workable method of bubbling through concentrated sulfuric acid. 1 mark Diagram for appropriate gas collection method i.e. using a gas syringe or upward displacement of air. 1 mark	**2**	
		(ii)	Calculating that 0·05 moles HCl would require 0·025 moles sodium sulfite and there are only 0.00317 moles of sodium sulfite **OR** Calculating that 0·00317 moles of sodium sulfite would require 0·00634 moles of HCl and there are 0.05 moles of HCl **OR** Calculating that 3·15 g sodium sulfite would be needed to react with 50 cm^3 hydrochloric acid and there are only 0.4 g of sodium sulfite **OR** Calculating that 6·3 cm^3 of (1 M) HCl would be needed to react with 0·4 g of sodium sulfite and there are 50 cm^3 (1M) HCl	**2**	General marking principle (j) applies. Partial marking 1 mark awarded for correct arithmetical calculation of moles of Na_2SO_3 (= 0·00317 mol) **AND** HCl = 0·05 mol) **OR** 1 mark awarded for correct arithmetical calculation of moles of acid (0.05) and correct application of stoichiometry to either reactant. **OR** 1 mark awarded for correct arithmetical calculation of moles of sodium sulfite (0.00317) and correct application of stoichiometry to either reactant.

Question			Answer	Max mark	Additional guidance
	(b)		-1075 (kJ mol^{-1}) Partial marks Treat as two concepts either would be acceptable for 1 mark Evidence of understanding of reversal for third equation only in order to achieve the target equation. Reversal of additional equations would be taken as cancelling **OR** evidence of understanding of multiplying for second equation by 2 in order to achieve the target equation. Multiplication of additional equations would be taken as cancelling.	**2**	If answer given is 1075 or +1075, maximum of 1 mark can be awarded. No units required. Only 1 mark can be awarded for the correct answer if wrong unit is given. (wrong units would only be penalised once in any paper) kJ is acceptable in place of kJ mol^{-1}
	(c)	(i)	163 -167 inclusive (g l^{-1})	**1**	No units required. No mark can be awarded for correct answer if wrong unit is given. (wrong units would only be penalised once in any paper)
		(ii)	1 mark for carbon dioxide is non-polar due to its shape/dipoles cancelling out **and** sulfur dioxide is polar due to its shape/dipoles not cancelling out 1 mark for an explanation which links polarity of CO_2 and SO_2 molecules to the polarity of water	**2**	

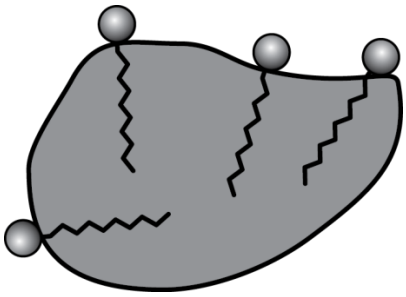
Question			Answer	Max mark	Additional guidance
6.	(a)	(i)		**1**	Accept a correct shortened structural formula
		(ii) (A)	Reactants or products are flammable/could catch fire.	**1**	
		(ii) (B)	orange to green/ blue-green/blue	**1**	
		(ii) (C)	Tertiary	**1**	
		(iii) (A)	Butanoic acid **OR** (2-)methylpropanoic acid	**1**	
		(iii) (B)	$C_4H_9OH + H_2O \rightarrow C_4H_8O_2 + 4H^+ + 4e^-$	**1**	Negative sign on electron not required. Positive sign on H^+ required.
	(b)	(i)	2-methylpentanal	**1**	2-methylpentan-1-al is not acceptable.
		(ii)	Any temperature between 166 and 181 (^oC)	**1**	No units required. No mark can be awarded if wrong unit is given. (wrong units would only be penalised once in any paper)
		(iii)	(More) branching lowers the boiling point (of isomeric aldehydes).	**1**	
		(iv)	Silver mirror/silver precipitate	**1**	Accept silver on its own
	(c)		(Permanent) dipole to (permanent) dipole	**1**	Accept London Dispersion Forces

Question			Answer	Max mark	Additional guidance
7.	(a)		Rinse beaker and transfer the rinsings/washings to the flask	**1**	
	(b)	(i)	The reaction is self-indicating. **OR** Potassium permanganate can act as its own indicator. **OR** Reaction changes colour.	**1**	
		(ii)	To provide H^+ ions for the reaction.	**1**	
		(iii) (A)	**1 mark** for any of the following • first titre is a rough (or approximate) result/practice • first titre is not accurate/not reliable/rogue • first titre is too far away from the others • you take average of concordant/close results	**1**	

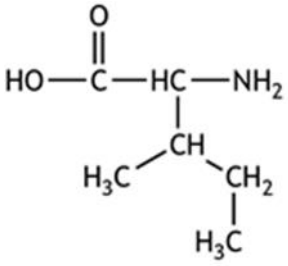
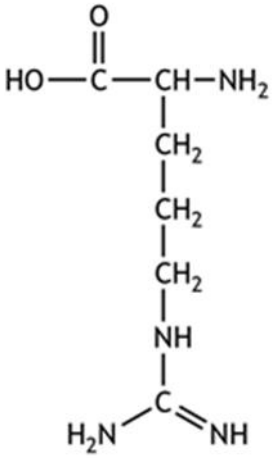
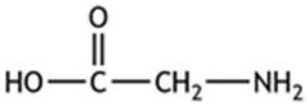
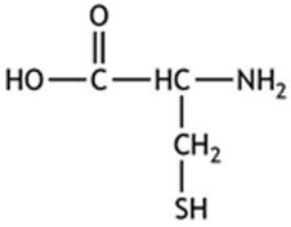
Question			Answer	Max mark	Additional guidance
		(iii) (B)	0·0582 (mol l^{-1}) **(3)** Partial marks can be awarded using a scheme of two "concept" marks. **1 mark** for knowledge of the relationship between moles, concentration and volume. This could be shown by any <u>one</u> of the following steps: • Calculation of moles MnO_4^- solution eg 0·02 × 0·01455 = 0·000291 • calculation of concentration of Fe^{2+} solution eg 0·001455/0·025 • Insertion of correct pairings of values for concentration and volume in a valid titration formula eg $\frac{0\cdot02\times14\cdot55}{n_1}=\frac{C_{Fe^{2+}}\times25\cdot0}{n_2}$ **1 mark** for knowledge of relationship between moles of MnO_4^- and Fe^{2+}. This could be shown by one of the following steps: • Calculation of moles Fe^{2+} from moles MnO_4^- – eg 0·000291x5 = 0·001455 • Insertion of correct stoichiometric values in a valid titration formula eg $\frac{0\cdot02\times14\cdot55}{1}=\frac{C_{Fe^{2+}}\times25\cdot0}{5}$	**3**	No units required but only two marks can be awarded for correct answer if wrong unit is given. (wrong units would only be penalised once in the paper) In terms of the mark for the stoichiometric relationship, the 5 should be associated with the unknown concentration of the Fe^{2+} and the 1 should be associated with the concentration of MnO_4^-
		(iii) (C)	A solution of accurately/exactly/ precisely known concentration	**1**	
		(iii) (D)	Pipette	**1**	Not: Burette Accept graduated pipette.

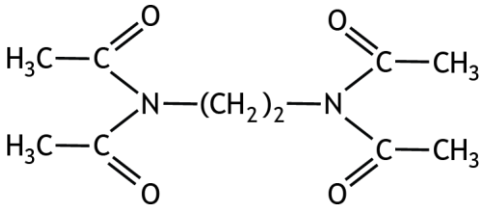
Question			Answer	Max mark	Additional guidance
	(c)		14mg, 14·06 mg, 0·01406 g	**1**	No units required. No mark can be awarded for correct answer if wrong unit is given. (wrong units would only be penalised once in the paper)
	(d)		24%, 24·3% Partial marks **1 mark** awarded for 30 g would contain 3·6 mg **1 mark** for $\frac{\text{any calculated mass}}{14\cdot8} \times 100$	**2**	

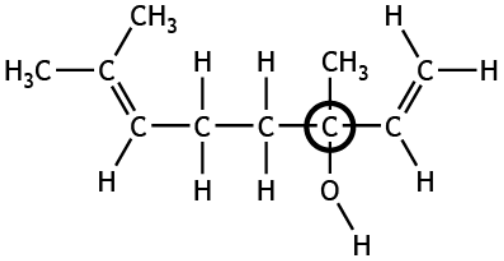
Question			Answer	Max mark	Additional guidance
8.			This is an open ended question **1 mark:** The student has demonstrated, at an appropriate level, a limited understanding of the chemistry involved. The candidate has made some statement(s) at which is/are relevant to the situation, showing that at least a little of the chemistry within the problem is understood. **2 marks:** The student has demonstrated, at an appropriate level, a reasonable understanding of the chemistry involved. The student makes some statement(s) which is/are relevant to the situation, showing that the problem is understood. **3 marks:** The maximum available mark would be awarded to a student who has demonstrated, at an appropriate level, a good understanding of the chemistry involved. The student shows a good comprehension of the chemistry of the situation and has provided a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. This does not mean the answer has to be what might be termed an 'excellent' answer or a 'complete' one.	**3**	Zero marks should be awarded if: The student has demonstrated, at an appropriate level, no understanding, of the chemistry involved. There is no evidence that the student has recognised the area of chemistry involved or has given any statement of a relevant chemistry principle. This mark would also be given when the student merely restates the chemistry given in the question.

Question			Answer	Max mark	Additional guidance
9.	(a)	(i)	A drawing similar to  Diagram shows **at least one** detergent molecule. All tails shown should be **fully** inside the fat-soluble dirt.	**1**	
		(ii)	hydrophobic	**1**	
	(b)	(i)	To break down coloured compounds/ removes stains/kill bacteria/kill fungi/inactivate viruses/germs	**1**	Stating only that oxidation occurs would be seen as restating the information from the stem. Discolour is not accepted.

Question			Answer	Max mark	Additional guidance
		(ii)	18 cm^3/0·018 litres **with correct unit** **Partial marking** 1 mark can be awarded for two of the three steps shown below correctly calculated: 1. number of moles of H_2O_2 2. mole ratio applied 3. calculated number of moles of O_2 multiplied by 24 (24000) If processed by proportion 68 g ←→ 24 l (24000 cm^3) 1 mark OR 0.051 g ←→ 0.036 l (36 cm^3) 1 mark **1 mark** for correct units.	**3**	If an incorrect volume is calculated but the units of volume used are appropriate to the calculation then 1 mark would be awarded for correct units. Accept L for litres

Question			Answer	Max mark	Additional guidance
	(c)	(i)	amino acids	**1**	
		(ii) (A)	Amide/amide link/peptide link	**1**	Amine/amino not acceptable
		(ii) (B)	Any of the shown amino acids:    	**1**	

Question			Answer	Max mark	Additional guidance
		(iii) (A)	Denaturing	**1**	
		(iii) (B)	Temperature increase/pH	**1**	Temperature on its own is not acceptable. High/higher/above optimum temperature are also accepted.
	(d)	(i)	Condensation	**1**	
		(ii)		**1**	

Question			Answer	Max mark	Additional guidance
10.	(a)	(i)	40·23/40·2/40 (%)	**1**	
		(ii)	geranyl acetate/peak 5	**1**	
	(b)		2·7p/ 3p Partial marking 1 mark can be awarded for Evidence of scaling up to 500cm^3 eg 460mg of 1,8-cineole **OR** Evidence for determining a correct cost for a calculated mass of 1,8-cineole eg 0·92mg costs 0.00544 pence	**2**	
	(c)	(i)		**1**	
		(ii)	$C_{15}H_{24}$ OR $(C_5H_8)_3$	**1**	

[END OF MARKING INSTRUCTIONS]

General marking principles for Higher Chemistry

This information is provided to help you understand the general principles you must apply when marking candidate responses to questions in this paper. These principles must be read in conjunction with the detailed marking instructions, which identify the key features required in candidate responses.

(a) Marks for each candidate response must always be assigned in line with these general marking principles and the detailed marking instructions for this assessment.

(b) Marking should always be positive, ie marks should be awarded for what is correct and not deducted for errors or omissions.

(c) If a specific candidate response does not seem to be covered by either the principles or detailed marking instructions, and you are uncertain how to assess it, you must seek guidance from your team leader.

(d) There are no half marks awarded.

(e) Where a candidate makes an error at an early stage in a multi stage calculation, credit should normally be given for correct follow on working in subsequent stages, unless the error significantly reduces the complexity of the remaining stages. The same principle should be applied in questions which require several stages of non-mathematical reasoning. The exception to this rule is where the marking instructions for a numerical question assign separate "concept marks" and an "arithmetic mark". In such situations, the marking instructions will give clear guidance on the assignment or partial marks.

(f) Unless a numerical question specifically requires evidence of working to be shown, full marks should be awarded for a correct final answer (including units) on its own.

(g) Larger mark allocations may be fully accessed whether responses are provided in continuous prose, linked statements or a series of developed bullet points.

(h) Marks should not be deducted for inaccurate or unconventional spelling or vocabulary as long as the meaning of the word(s) is conveyed. **For example**, responses that include 'distilling' for 'distillation', or 'it gets hotter' for 'the temperature rises', should be accepted.

(i) If a correct answer is followed by a wrong answer, it should be treated as a cancelling error and no marks should be given. **For example**, in response to the question, 'State the colour seen when blue Fehling's solution is warmed with an aldehyde', the answer 'red green' gains no marks.
However, if a correct answer is followed by additional information which does not conflict with that, the additional information should be ignored, whether correct or not. **For example**, in response to a question concerned with melting point, 'State why the tube should not be made of copper', the response 'Copper has a low melting point and is coloured grey' would **not** be treated as having a cancelling error.

(j) Full marks are usually awarded for the correct answer to a calculation without working and the partial marks shown in the detailed marking instructions are for use when working is given but the final answer is incorrect. An exception is when candidates are asked to 'Find, by calculation', when full marks cannot be awarded for the correct answer without working.

(k) Ignore the omission of one H atom from a full structural formula provided the bond is shown.

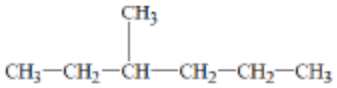
(l) A symbol or correct formula should be accepted in place of a name **unless stated otherwise in the detailed marking instructions.**

(m) When formulae of ionic compounds are given as answers it will only be necessary to show ion charges if these have been specifically asked for. However, if ion charges are shown, they must be correct. If incorrect charges are shown, no marks should be awarded.

(n) If an answer comes directly from the text of the question, no marks should be given. **For example**, in response to the question, ' A student found that 0·05 mol of propane, C_3H_8 burned to give 82·4 kJ of energy. $C_3H_8(g) + 5O_2(g) = 3CO_2(g) + 4H_2O(\ell)$. Name the kind of enthalpy change that the student measured', no marks should be given for "burning" since the word "burned" appears in the text.

(o) A guiding principle in marking is to give credit for correct elements of a response rather than to look for reasons not to give marks.

Example 1: The structure of a hydrocarbon found in petrol is shown below.



Name the hydrocarbon

- Although the punctuation is not correct '3, methyl-hexane' should gain the full mark.

Example 2: A student measured the pH of four carboxylic acids to find out how their strength is related to the number of chlorine atoms in the molecule. The results are shown.

Structural formula	*pH*
CH_3COOH	1·65
$CH_2ClCOOH$	1·27
$CHCl_2COOH$	0·90
CCl_3COOH	0·51

Describe the relationship between the number of chlorine atoms in the molecule and the strengths of the acids.

- Although not completely correct, an answer such as 'the more Cl_2, the stronger the acid' should gain the full mark.

(p) Unless the question is clearly about a non-chemistry issue, eg costs in an industrial chemical process, a non-chemical answer gains no marks.
For example, in response to the question, 'Why does the (catalytic) converter have a honeycomb structure?', 'to make it work' may be correct but it is not a chemical answer and the mark should not be given.

(q) Marks are awarded only for a valid response to the question asked. For example, in response to questions that ask candidates to:

- **identify, name, give** or **state**, they need only name or present in brief form;
- **describe**, they must provide a statement or structure of characteristics and/or features;
- **explain**, they must relate cause and effect and/or make relationships between things clear;
- **compare**, they must demonstrate knowledge and understanding of the similarities and/or differences between things;
- **complete**, they must finish a chemical equation or fill in a table with information
- **determine** or **calculate**, they must determine a number from given facts, figures or information;
- **draw**, they must draw a diagram or structural formula, eg "Draw a diagram to show the part of a poly(propene) molecule formed from two propene molecules"
- **estimate**, they must determine an approximate value for something;
- **predict**, they must suggest what may happen based on available information;
- **evaluate**, they must make a judgement based on criteria;

- **suggest**, they must apply their knowledge and understanding of [subject] to a new situation. A number of responses are acceptable; marks will be awarded for any suggestions that are supported by knowledge and understanding of [subject];
- **use your knowledge of [chemistry or aspect of chemistry] to comment on**, they must apply their skills, knowledge and understanding to respond appropriately to the problem/situation presented (for example by making a statement of principle(s) involved and/or a relationship or equation, and applying these to respond to the problem/situation). They will be rewarded for the breadth and/or depth of their conceptual understanding.
- **Write**, they must complete a chemical or word equation, eg "Write the word equation for the complete combustion of ethanol."

2017 Higher Chemistry Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	**A**	**81**	14	2	3	0	0
2	**D**	2	6	6	**86**	0	0
3	**C**	9	6	**76**	9	0	0
4	**B**	10	**86**	3	1	0	0
5	**D**	6	7	5	**82**	0	0
6	**C**	0	2	**84**	4	0	0
7	**B**	1	**95**	3	2	0	0
8	**C**	18	4	**65**	13	0	0
9	**A**	**98**	1	1	1	0	0
10	**C**	6	11	**68**	16	0	0
11	**B**	17	**61**	7	15	0	0
12	**C**	3	3	**85**	9	0	0
13	**B**	25	**67**	4	4	0	0
14	**B**	18	**79**	3	0	0	0
15	**C**	4	9	**84**	3	0	0
16	**A**	**83**	2	2	12	0	0
17	**D**	1	7	3	**89**	0	0
18	**A**	**76**	14	5	5	0	0
19	**D**	4	0	2	**93**	0	0
20	**B**	22	**0**	2	6	0	0

National
Qualifications
2016

2016 Chemistry

Higher

Finalised Marking Instructions

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Marking Instructions for each question

Section 1

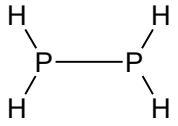
Question	Answer	Max Mark
1.	D	**1**
2.	A	**1**
3.	A	**1**
4.	B	**1**
5.	B	**1**
6.	A	**1**
7.	C	**1**
8.	D	**1**
9.	C	**1**
10.	B	**1**
11.	B	**1**
12.	C	**1**
13.	C	**1**
14.	B	**1**
15.	C	**1**
16.	A	**1**
17.	C	**1**
18.	D	**1**
19.	D	**1**
20.	A	**1**

Section 2

Question			Answer	Max Mark	Additional Guidance
1.	(a)		The number of **successful** collisions will increase/There will be a greater chance of successful collisions **(1 mark)** **OR** More reactant particles will have energy equal to or greater than the activation energy. **(1 mark)**	**1**	Also acceptable More particles have sufficient/enough energy to react.
	(b)	(i)	1 mark for drawing suitable experiment that will work **and** for indicating how **volume** will be measured eg collecting in a gas syringe or downward displacement of water from a measuring cylinder or similar.	**1**	Accept a recognisable/labelled gas syringe, as long as graduations are shown. Graduations must be shown if a measuring cylinder is used. Mark not awarded if delivery tube passes through the side of a measuring cylinder.
		(ii)	3·7 (volume strength).	**1**	Ignore any units. 3·7 must clearly be final answer.
	(c)		8 (s).	**1**	Accept 7·7 (1/0·13) to 8·3 (1/0·12) or correct answer to 2 decimal places. No units required but no mark is awarded if wrong unit is given. (wrong units would only be penalised once in any paper).

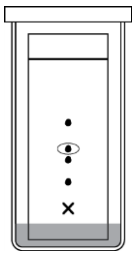
Question			Answer	Max Mark	Additional Guidance
2.	(a)	(i)	(The electron shells are pulled closer because) nuclear charge increases/the number of protons in the nucleus increases.	**1**	Mention must be made of nuclear charge or number of protons in candidate's answer giving the idea of pull by the nucleus. Increased nuclear pull is not accepted.
		(ii)	Two points are required Understanding that the atom loses an electron (when the ion is formed), **(1 mark)** **AND** the Sodium ion will only have two electron shells whereas the sodium atom has three electron shells **OR** the sodium ion will have fewer electron shells (than the sodium atom). **(1 mark)**	**2**	A diagram of the atom and ion or stating the electron arrangement of both would be sufficient to gain 1 mark but the second mark can only be awarded if this diagram or electron arrangement is accompanied with an explanatory statement.
	(b)	(i)	As you go down the group the outer electron is more shielded from the nuclear pull **OR** less strongly attracted by the nucleus.	**1**	
		(ii)	2nd ionisation energy involves removal from an electron shell which is inner/full (whole)/(more) stable/closer to the nucleus **OR** second electron is removed from an electron shell which is inner/full (whole)/(more) stable/closer to the nucleus. **(1 mark)** The electron is less shielded from, or, more strongly attracted to the nucleus. **(1 mark)**	**2**	Stating that more energy is required to remove the second electron would be seen as restating the information from the stem.

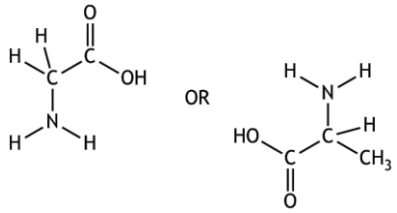
Question			Answer	Max Mark	Additional Guidance
2.	(c)	(i)	Any value in range 720–770 ($kJ\ mol^{-1}$).	**1**	Data value 760 $kJ\ mol^{-1}$ No units required but no mark is awarded if wrong unit is given. (wrong units would only be penalised once in any paper).
		(ii)	As the ionic radii (of the positive and/or the negative ion) increase, the lattice enthalpy decreases.	**1**	A general statement stating the greater the difference in ionic radii the greater the lattice enthalpy must be related to either chlorides or fluorides. The mark cannot be awarded for a statement such as, "As the lattice enthalpy decreases the ionic radii increases" which implies that ionic radius is a dependent variable.

Question			Answer	Max Mark	Additional Guidance
3.	(a)		Covalent molecular.	**1**	
	(b)		n =m/gfm =2900/58 =50 **(1 mark)** V = n × Vm = 50 × 24 = 1200 (litres) **(1 mark)** Or by Proportion 58 g ⟶ 24 l **(1 mark)** 2·9 kg ⟶ 24 × 2900/58ℓ = 1200 (ℓ) **(1 mark)**	**2**	Follow through applies in this question. No units required but only one mark can be awarded if wrong unit is given. (wrong units would only be penalised once in any paper) Accept L for ℓ.
	(c)			**1**	

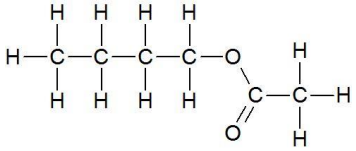
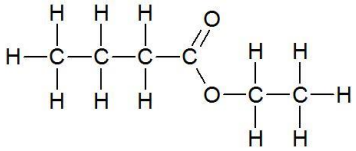
Question			Answer	Max Mark	Additional Guidance
4.			This is an open ended question **1 mark:** The student has demonstrated, at an appropriate level, a limited understanding of the chemistry involved. The student has made some statement(s) which is/are relevant to the situation, showing that at least a little of the chemistry within the problem is understood. **2 marks:** The student has demonstrated a reasonable understanding, at an appropriate level, of the chemistry involved. The student makes some statement(s) which is/are relevant to the situation, showing that the problem is understood. **3 marks:** The maximum available mark would be awarded to a student who has demonstrated a good understanding, at an appropriate level, of the chemistry involved. The student shows a good comprehension of the chemistry of the situation and has provided a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. This does not mean the answer has to be what might be termed an "excellent" answer or a "complete" one.	3	Zero marks should be awarded if: The student has demonstrated no understanding of the chemistry involved. There is no evidence that the student has recognised the area of chemistry involved or has given any statement of a relevant chemistry principle. This mark would also be given when the student merely restates the chemistry given in the question.

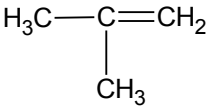
Question			Answer	Max Mark	Additional Guidance
5.	(a)		Glycerol or glycerine or propan(e)-1,2,3-triol.	**1**	
	(b)	(i)	Polyunsaturated.	**1**	
		(ii)	Octanoic acid.	**1**	
	(c)	(i)	Bromine solution is added to both until the bromine is no longer decolourised. (or reddish-brown colour remains) **(1 mark)** More bromine would be required for the more unsaturated/olive oil **OR** Less bromine would be required for the more saturated/coconut oil. **(1 mark)**	**2**	
		(ii)	Hexane is a non-polar (solvent)/ water is a polar (solvent).	**1**	Answers in terms of "Like dissolves like" are only acceptable if explanation in terms of polarity is given. Oil and water don't mix would not be awarded a mark.
		(iii)	Coconut oil molecules can pack more closely together **OR** Coconut oil has linear fatty acid chains/olive oil chains have bends/kinks (due to the double bonds) **(1 mark)** There are stronger/more **intermolecular** forces between the molecules in coconut oil **OR** The London dispersion forces (van der Waals' forces) between the molecules in coconut oil are stronger than in olive oils. **(1 mark)**	**2**	

Question			Answer	Max Mark	Additional Guidance
6.	(a)		Hexapeptide.	**1**	
	(b)		isoleucine-leucine-glycine-valine-serine **OR** serine-valine-glycine-leucine-isoleucine.	**1**	
	(c)		Essential.	**1**	
	(d)	(i)	The peptide molecule: must have contained an amino acid that is repeated in the sequence **OR** contained only four different amino acids (accept four different peptides). **OR** The peptide contains two amino acids: with the same R_f value **OR** that moved the same distance.	**1**	
		(ii)		**1**	

Question			Answer	Max Mark	Additional Guidance
6.	(e)	(i)	 OR	**1**	
		(ii)	30 g – units required **(2 marks)** Partial marks: Correctly calculated mass of mushrooms without units, **(1 mark)** Appropriate units **(1 mark)** Mass (1) / Units (1) 0·03 / kg 30 / g 30000 / mg	**2**	If an incorrect mass is calculated but the units used are appropriate to the calculation then 1 mark would be awarded. If the candidate has ended the calculation at the lethal dosage then the mark for units can be awarded if the unit is appropriate to the value calculated. If the candidate's working is unclear in terms of what is being worked out then the mark for units cannot be awarded.

Question			Answer	Max Mark	Additional Guidance
7.	(a)		Any reason from list UV light is damaging/harmful to skin. UV light causes sunburn. UV light can break bonds/molecules in skin. UV light damages collagen. UV light can cause skin cancer. UV light ages skin. UV light causes photo ageing. UV light creates free radicals/initiates free-radical chain reactions. Sunblocks contain free-radical scavengers.	**1**	
	(b)	(i)	Species (Atoms/molecules etc) with unpaired electrons.	**1**	
		(ii)	Initiation.	**1**	
		(iii)	Carboxylic acid **OR** carboxyl group.	**1**	
		(iv)	The hydroxyl/functional group is attached to a carbon that is attached to two other carbons **OR** The hydroxyl/functional group is attached to a carbon which has only one hydrogen attached.	**1**	The answer required is an answer based on structure not an answer based on a chemical property. ie, an answer stating it can be oxidised to give a ketone would not be awarded the mark.

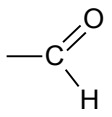
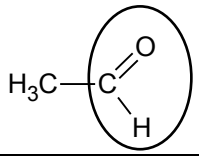
Question			Answer	Max Mark	Additional Guidance
7.	(c)	(i)	A correct structural formula for butyl ethanoate.	**1**	 ✓  X
		(ii)	To condense any vapours or reactants/products which evaporate **OR** To act as a condenser.	**1**	Must be indication that gases are being turned to liquid.
		(iii)	60·7/61 % **(2 marks)** **1 mark** Calculates theoretical mass = 4·78g **OR** correctly calculates no of moles reactant (0·054) and product (0·033) **1 mark** Calculating % yield using the actual and theoretical masses, or using the actual number of moles of products and actual number of moles of reactant.	**2**	Follow through applies.
		(iv)	Condensation **OR** Esterification.	**1**	

Question			Answer	Max Mark	Additional Guidance
8.	(a)	(i)	<table><tr><th></th><th>Temperature (High/Low)</th><th>Pressure (High/Low)</th></tr><tr><td>Step 1</td><td>High</td><td>Low</td></tr><tr><td>Step 2</td><td>Low</td><td>High</td></tr></table>	**2**	Temperature and pressure are dealt with separately as two single marks.
	(b)	(i)	Correct structure for 2-methylpropene.	**1**	 The name 2-methylpropene would be accepted on it's own. (on the basis that the candidate would have to process through the structure to obtain the name).
		(ii)	The proportion of the total mass of all starting materials successfully converted into the desired product is 100% **OR** All the atoms in the reactants are converted into the product you want./Mass of **product** is equal to mass of reactants **OR** No by-products/no waste products/ only one product is formed.	**1**	

Question			Answer	Max Mark	Additional Guidance
8	(c)		56 or + 56 **(2)** Bond breaking 412 + 412 + 412 + 360 + 463 = 2059 Bond forming 743 + 412 + 412 + 436 = 2003 2059 − 2003 = 56 ($kJ\,mol^{-1}$) A single mark is available if either of the following operations is correctly executed **Either** The five relevant values for the bond enthalpies of the C-H, H-H, C-O, O-H and C=O bonds (or multiples thereof) are retrieved from the data book 412,360,463,743,436 (ignore signs) **OR** If only four values are retrieved, the candidate recognises that bond breaking is endothermic and bond formation is exothermic and correctly manipulates the bond enthalpy values they have used to give their answer.	**2**	−56 would qualify for 1 mark. No units required but only one mark can be awarded for correct answer if wrong unit is given. (wrong units would only be penalised once in any paper). kJ is acceptable in place of $kJ\,mol^{-1}$. If less than four bond enthalpy values are retrieved then no mark can be awarded.

Question			Answer	Max Mark	Additional Guidance
9.	(a)	(i)	4·93 (kJ) (no units required) **1 mark** for extrapolating the graph and finding the corrected temperature difference ΔT = 11·8 (^oC) **1 mark** for use of the relationship $E_h = cm\Delta T$ to calculate the E_h = 4·93 (kJ) Maximum of **1 mark** for using ΔT = 11·2 which gives E_h = 4·68 (kJ)	**2**	▫T 11·6 - 12·0 acceptable 11·6 ⟶ 4·85 11·7 ⟶ 4·89 11·8 ⟶ 4·93 11·9 ⟶ 4·97 12·0 ⟶ 5·02 ▫T follow through applies No units required but only one mark can be awarded for correct answer if wrong unit is given. (wrong units would only be penalised once in any paper).
		(ii)	To prevent (minimise) heat loss to the surroundings **OR** polystyrene cup is a poor conductor of heat/insulator.	**1**	
		(iii)	52·5 (kJ or $kJmol^{-1}$)	**1**	accept −52·5 (kJ or kJ mol^{-1}). No units required. No mark can be awarded if wrong unit is given. (wrong units would only be penalised once in any paper).
	(b)		−414 kJ mol^{-1} (no units required) Partial marks Treat as two concepts either would be acceptable for **1 mark** Evidence of understanding of reversal of first equation in order to achieve the target equation. Reversal of both equations would be taken as cancelling **OR** evidence of understanding of multiplying first equation by 2 in order to achieve the target equation.	**2**	No units required but only one mark can be awarded for correct answer if wrong unit is given. (wrong units would only be penalised once in any paper).

Question			Answer	Max Mark	Additional Guidance
10.			This is an open ended question **1 mark:** The student has demonstrated, at an appropriate level, a limited understanding of the chemistry involved. The student has made some statement(s) which is/are relevant to the situation, showing that at least a little of the chemistry within the problem is understood. **2 marks:** The student has demonstrated a reasonable understanding, at an appropriate level, of the chemistry involved. The student makes some statement(s) which is/are relevant to the situation, showing that the problem is understood. **3 marks:** The maximum available mark would be awarded to a student who has demonstrated a good understanding, at an appropriate level, of the chemistry involved. The student shows a good comprehension of the chemistry of the situation and has provided a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. This does not mean the answer has to be what might be termed an "excellent" answer or a "complete" one.	**3**	Zero marks should be awarded if: The student has demonstrated no understanding of the chemistry involved. There is no evidence that the student has recognised the area of chemistry involved or has given any statement of a relevant chemistry principle. This mark would also be given when the student merely restates the chemistry given in the question.

Question			Answer		Max Mark	Additional Guidance
11.	(a)	(i)	A drawing that shows a flask with a long narrow neck and a single gradation mark which goes completely across, or is labelled, on the narrow neck.		**1**	
		(ii)	Accurate method for volume measurement eg uses pipette **(1 mark)** Describes weighing by difference or using a tared balance. **(1 mark)**		**2**	Burette would be accepted. Syringe not accepted for measuring volume accurately.
		(iii)	All points plotted correctly within tolerance of ½ a box **(1 mark)** Best fit line **(1 mark)**		**2**	
		(iv)	(A)	The dissolved gas/bubbles will affect the density/mass/ volume of solution.	**1**	"The gas would affect the results" would not be enough to be awarded the mark.
			(B)	3·43%.	**1**	If a rounded answer is given there must be evidence of the equation being used.
		(v)	34·98 (g)/35·0 (g).		**1**	No units required. No mark can be awarded if wrong unit is given. (wrong units would only be penalised once in any paper).
	(b)	(i)	Correctly drawn aldehyde group 		**1**	Functional group must show open bond. Aldehyde structures only acceptable if functional group is highlighted. 
		(ii)	$C_6H_{12}O_6 + H_2O \rightarrow C_6H_{12}O_7 + 2H + 2e$		**1**	
		(iii)	Blue to orange/(brick)red/brown/ yellow/green.		**1**	Blue-green is not acceptable as the original colour.

Question			Answer	Max Mark	Additional Guidance
11.	(b)	(iv)	0·0099 (mol l^{-1}) **(3 marks)** Partial marks can be awarded using a scheme of two 'concept' marks, and one 'arithmetic' mark. **1 mark** for knowledge of the relationship between moles, concentration and volume. This could be shown by any <u>one</u> of the following steps: • Calculation of number of moles of Cu^{2+} eg 0·025 × 0·0198 = 0·000495 • Calculation of conc of reducing sugars eg 0·0002475 ÷ 0·025 • Insertion of correct pairings of values of conc and volume in titration formula eg $\frac{0\cdot025\times19\cdot8}{n_1}=\frac{C_{RS}\times25\cdot0}{n_2}$ **1 mark** for knowledge of relationship between Cu^{2+} and reducing sugars. This could be shown by any <u>one</u> of the following steps: • Calculation of moles of reducing sugars from moles Cu^{2+} eg 0·000495 ÷ 2 = 0·0002475 • Insertion of correct stoichiometric values in titration formula e.g. $\frac{0\cdot025\times19\cdot8}{2}=\frac{C_{RS}\times25\cdot0}{1}$ **1 mark** is awarded for correct arithmetic throughout the calculation. This mark can only be awarded if both concept marks have been awarded.	**3**	No units required but only two marks can be awarded for correct answer if wrong unit is given. (wrong units would only be penalised once in any paper). 2 concept marks 1 mark for stoichiometric relationship being applied. 1 mark for correctly using n = c × v Candidates require to correctly apply relationship if used twice in their working. In terms of the mark for the stoichiometric relationship, the 1 should be associated with the unknown concentration of the sugar and the 2 with the with concentration of the Fehling's.

Question			Answer	Max Mark	Additional Guidance
12.	(a)	(i)	hydrogen bonding.	**1**	
		(ii)	(More) branching lowers the boiling point **(1 mark)** The shorter the alcohol, the lower the boiling point/The longer the carboxylic acid the lower the b.p. **OR** The nearer the ester link is to the right hand end (of the molecule) the higher the boiling point. **(1 mark)**	**2**	An answer relating to chain length must be specific to either the alcohol or carboxylic acid portion of the ester. Accept this positional trend if the candidate correctly identifies it by reference to the oxygen atom in the chain or the C=O but not if candidate refers to a carbonyl group.
		(iii)	Any temperature between 99 and 124 (°C) (data value 116°C)	**1**	No units required. No mark can be awarded if wrong unit is given. (wrong units would only be penalised once in any paper).
	(b)	(i)	Peaks labelled in order 2,3,1,4 $\longrightarrow$	**1**	
		(ii)	5.	**1**	

[END OF MARKING INSTRUCTIONS]

General Marking Principles for Higher Chemistry

This information is provided to help you understand the general principles you must apply when marking candidate responses to questions in this paper. These principles must be read in conjunction with the detailed marking instructions, which identify the key features required in candidate responses.

(a) Marks for each candidate response must always be assigned in line with these General Marking Principles and the Detailed Marking Instructions for this assessment.

(b) Marking should always be positive, ie marks should be awarded for what is correct and not deducted for errors or omissions.

(c) If a specific candidate response does not seem to be covered by either the principles or detailed Marking Instructions, and you are uncertain how to assess it, you must seek guidance from your Team Leader.

(d) There are no half marks awarded.

(e) Where a candidate makes an error at an early stage in a multi stage calculation, credit should normally be given for correct follow on working in subsequent stages, unless the error significantly reduces the complexity of the remaining stages. The same principle should be applied in questions which require several stages of non-mathematical reasoning. The exception to this rule is where the marking instructions for a numerical question assign separate "concept marks" and an "arithmetic mark". In such situations, the marking instructions will give clear guidance on the assignment or partial marks.

(f) Unless a numerical question specifically requires evidence of working to be shown, full marks should be awarded for a correct final answer (including units) on its own.

(g) Larger mark allocations may be fully accessed whether responses are provided in continuous prose, linked statements or a series of developed bullet points.

(h) Marks should not be deducted for inaccurate or unconventional spelling or vocabulary as long as the meaning of the word(s) is conveyed. **For example**, responses that include 'distilling' for 'distillation', or 'it gets hotter' for 'the temperature rises', should be accepted.

(i) If a correct answer is followed by a wrong answer, it should be treated as a cancelling error and no marks should be given. **For example**, in response to the question, 'State the colour seen when blue Fehling's solution is warmed with an aldehyde', the answer 'red green' gains no marks.
However, if a correct answer is followed by additional information which does not conflict with that, the additional information should be ignored, whether correct or not. **For example**, in response to a question concerned with melting point, 'State why the tube should not be made of copper', the response 'Copper has a low melting point and is coloured grey' would **not** be treated as having a cancelling error. If a candidate lists a number of possible answers it should not be for the marker to choose the correct answer from the list.

(j) Full marks are usually awarded for the correct answer to a calculation without working and the partial marks shown in the detailed marking instructions are for use when working is given but the final answer is incorrect. An exception is when candidates are asked to 'Find, by calculation', when full marks cannot be awarded for the correct answer without working.

(k) Ignore the omission of one H atom from a full structural formula provided the bond is shown.

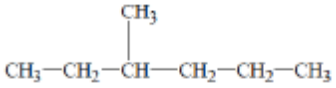
(l) A symbol or correct formula should be accepted in place of a name **unless stated otherwise in the detailed marking instructions.**

(m) When formulae of ionic compounds are given as answers it will only be necessary to show ion charges if these have been specifically asked for. However, if ion charges are shown, they must be correct. If incorrect charges are shown, no marks should be awarded.

(n) If an answer comes directly from the text of the question, no marks should be given. **For example**, in response to the question, ' A student found that 0·05 mol of propane, C_3H_8 burned to give 82·4 kJ of energy. $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(\ell)$. Name the kind of enthalpy change that the student measured', no marks should be given for "burning" since the word "burned" appears in the text.

(o) A guiding principle in marking is to give credit for correct elements of a response rather than to look for reasons not to give marks.

Example 1: The structure of a hydrocarbon found in petrol is shown below.



Name the hydrocarbon

- Although the punctuation is not correct '3, methyl-hexane' should gain the full mark.

Example 2: A student measured the pH of four carboxylic acids to find out how their strength is related to the number of chlorine atoms in the molecule. The results are shown.

Structural formula	*pH*
CH_3COOH	1·65
$CH_2ClCOOH$	1·27
$CHCl_2COOH$	0·90
CCl_3COOH	0·51

Describe the relationship between the number of chlorine atoms in the molecule and the strengths of the acids.

- Although not completely correct, an answer such as 'the more Cl_2, the stronger the acid' should gain the full mark.

(p) Unless the question is clearly about a non-chemistry issue, eg costs in an industrial chemical process, a non-chemical answer gains no marks.
For example, in response to the question, 'Why does the (catalytic) converter have a honeycomb structure?', 'to make it work' may be correct but it is not a chemical answer and the mark should not be given.

(q) Marks are awarded only for a valid response to the question asked. For example, in response to questions that ask candidates to:

- **identify, name, give** or **state**, they need only name or present in brief form;
- **describe**, they must provide a statement or structure of characteristics and/or features;
- **explain**, they must relate cause and effect and/or make relationships between things clear;

- **compare**, they must demonstrate knowledge and understanding of the similarities and/or differences between things;
- **complete**, they must finish a chemical equation or fill in a table with information
- **determine** or **calculate**, they must determine a number from given facts, figures or information;
- **draw**, they must draw a diagram or structural formula, eg "Draw a diagram to show the part of a poly(propene) molecule formed from two propene molecules"
- **estimate**, they must determine an approximate value for something;
- **predict**, they must suggest what may happen based on available information;
- **evaluate**, they must make a judgement based on criteria;
- **suggest**, they must apply their knowledge and understanding of [subject] to a new situation. A number of responses are acceptable; marks will be awarded for any suggestions that are supported by knowledge and understanding of [subject];
- **use your knowledge of [chemistry or aspect of chemistry] to comment on**, they must apply their skills, knowledge and understanding to respond appropriately to the problem/situation presented (for example by making a statement of principle(s) involved and/or a relationship or equation, and applying these to respond to the problem/situation). They will be rewarded for the breadth and/or depth of their conceptual understanding.
- **Write**, they must complete a chemical or word equation, eg "Write the word equation for the complete combustion of ethanol."

2016 Higher Chemistry Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D
1	**D**	10	11	23	**55**
2	**A**	**80**	10	7	2
3	**A**	**83**	5	8	4
4	**B**	7	**81**	9	2
5	**B**	3	**93**	3	1
6	**A**	**66**	8	9	16
7	**C**	11	5	**75**	8
8	**D**	5	5	19	**71**
9	**C**	3	2	**81**	14
10	**B**	21	**40**	25	13
11	**B**	10	**61**	16	12
12	**C**	8	2	**80**	9
13	**C**	27	10	**31**	31
14	**B**	13	**51**	16	20
15	**C**	2	12	**66**	20
16	**A**	**75**	9	12	4
17	**C**	3	8	**68**	21
18	**D**	5	43	3	**49**
19	**D**	19	13	25	**41**
20	**A**	**54**	16	15	16

National
Qualifications
2015

2015 Chemistry

New Higher

Finalised Marking Instructions

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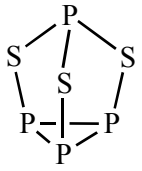
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Detailed Marking Instructions for each question

Section 1

Question	Answer	Max Mark
1.	D	**1**
2.	C	**1**
3.	D	**1**
4.	A	**1**
5.	C	**1**
6.	B	**1**
7.	A	**1**
8.	B	**1**
9.	B	**1**
10.	A	**1**
11.	D	**1**
12.	D	**1**
13.	B	**1**
14.	C	**1**
15.	B	**1**
16.	A	**1**
17.	D	**1**
18.	A	**1**
19.	A	**1**
20.	C	**1**

Section 2

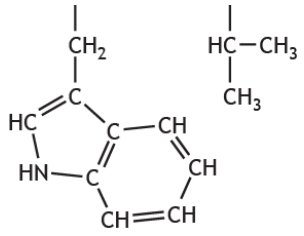
Question			Answer	Max Mark	Additional Guidance
1.	**(a)**		Sulfur – London dispersion forces / van der Waals / intermolecular forces **(1)** Silicon dioxide – (polar)covalent (network) bonds **(1)**	**2**	Accept LDF for London dispersion forces If candidate answers pure covalent, ignore pure.
	(b)	**(i)**	Any structure for P_4S_3 that obeys valency rules 	**1**	Only trivalent phosphorus structures accepted
		(ii)	Increased nuclear attraction for electrons / increased nuclear charge / sulfur has more protons in nucleus	**1**	0 marks awarded for increased attraction of electrons for nucleus
		(iii)	**1 mark** Correctly identify that the forces are stronger between sulfur (molecules) than between the phosphorus molecules **1 mark** Correctly identifying that there are London dispersion forces between the molecules of both these elements **1 mark** These forces are stronger due to sulfur structure being S_8 whereas phosphorus is P_4	**3**	This mark should only be awarded if no other forces are mentioned Must mention S_8 and P_4 (A-mark)
2.	**(a)**		From graph, rate = 0·022 t = 1/rate = 45 s accept answers in range 45 − 46 s	**1**	Units not required
	(b)	**(i)**	Second line displaced to left of original. Peak of curve should be to the left of the original peak	**1**	
		(ii)	A vertical line drawn at a lower kinetic energy that the original E_a shown on graph	**1**	

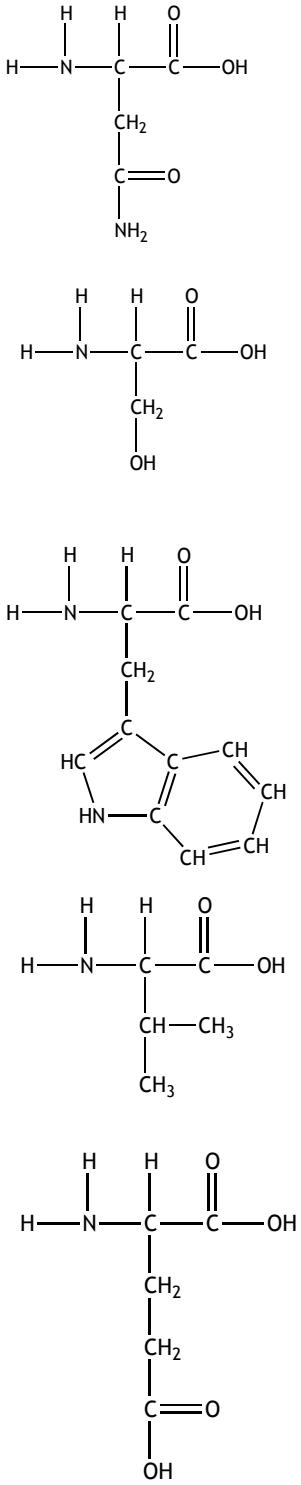
Question			Answer	Max Mark	Additional Guidance
3.	**(a)**	**(i)**	Workable apparatus for passing steam **through** strawberry gum leaves **(1)** Workable apparatus for condensing the steam and essential oil **(1)**	**2**	Treat both marks separately "Through" not "over" A closed system would not allow candidates to gain mark for condensation.
		(ii)	(Fractional) distillation or chromatography	**1**	
	(b)	**(i)**	**1 mark** awarded for correct arithmetical calculation of moles of acid = 0·044 **and** moles alcohol = 0·063 (no penalty for candidates who round to 0·04 and 0·06 etc) **or** working out that 9·25 g cinnamic acid would be needed to react with 2 g methanol or 6·5 g cinnamic acid would react with 1·41 g methanol **1 mark** awarded for statement demonstrating understanding of limiting reactant. eg there are less moles of cinnamic acid therefore it is the limiting reactant **or** 0·0625 moles methanol would require 0·625 moles cinnamic acid **or** methanol is in excess therefore cinnamic acid is the limiting reactant.	**2**	

Question			Answer	Max Mark	Additional Guidance
		(ii) (A)	52% **(2)** Partial Marking 1 mark is given for working out the theoretical yield ie 7·1 g or for working out **both** the moles of reactant used and product formed ie **both** 0·044 moles and 0·023 moles 1 mark is given for calculating the % yield, either using the actual and theoretical masses, or using the actual number of moles of products and actual number of moles of reactant	**2**	**0 marks awarded for** – 3·7/6·5 × 100 or 56·9%
		(ii) (B)	£24·59 **(2)** Partial marking for **1 mark** Award 1 mark for Evidence for costing to produce of 3·7 g (£0·91) **or** evidence of a calculated mass of cinnamic acid x 14p **or** evidence that 176 g of cinnamic acid required £12·80 would be using 100% yield	**2**	Assume units are £ unless otherwise stated Apply follow through from (b) (ii) (A)

Question			Answer	Max Mark	Additional Guidance
4.	**(a)**		Any one of the common compounds correctly identified ie citronellol / geraniol / anisyl alcohol	**1**	
	(b)		The concentration / volume of compounds (that are common to both/present in the counterfeit) is different are present in lower concentration in the counterfeit	**1**	Answer must relate to the perfume and not to the chromatogram.
	(c)	**(i)**	Inert / will not react with the molecules (being carried through the column)	**1**	
		(ii)	Size (mass) of molecules / temperature of column.	**1**	
	(d)	**(i)**	Terpenes	**1**	
		(ii) (A)	3,7-dimethylocta-1,6-dien-3-ol	**1**	
		(ii) (B)	Hydroxyl attached to C which is attached to 3 other C atoms **or** hydroxyl attached to a C that has no H atoms attached	**1**	
	(e)		1·7 g (units not required) **(2)** partial marking **for 1 mark** for evidence within candidate answer of calculating that 1 mg coumarin is obtained from 0·227 g cinnamon or tolerable daily intake =7·5mg for 75kg individual or evidence of multiplying DTI by 227 (multiplying by 1000 and dividing by 4·4)	**2**	

Question			Answer	Max Mark	Additional Guidance
5.			This is an open ended question **1 mark:** The student has demonstrated, at an appropriate level, a limited understanding of the chemistry involved. The student has made some statement(s) which is/are relevant to the situation, showing that at least a little of the chemistry within the problem is understood. **2 marks:** The student has demonstrated a reasonable understanding, at an appropriate level, of the chemistry involved. The student makes some statement(s) which is/are relevant to the situation, showing that the problem is understood. **3 marks:** The maximum available mark would be awarded to a student who has demonstrated a good understanding, at an appropriate level, of the chemistry involved. The student shows a good comprehension of the chemistry of the situation and has provided a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. This does not mean the answer has to be what might be termed an "excellent" answer or a "complete" one.	**3**	Zero marks should be awarded if: The student has demonstrated no understanding of the chemistry involved. There is no evidence that the student has recognised the area of chemistry involved or has given any statement of a relevant chemistry principle. This mark would also be given when the student merely restates the chemistry given in the question.

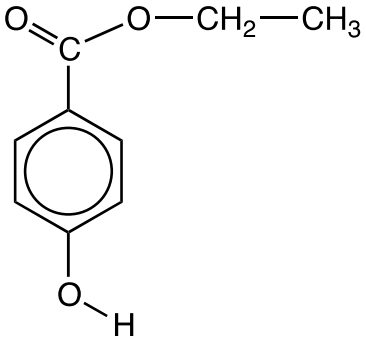
Question			Answer	Max Mark	Additional Guidance
6.	**(a)**		Heat breaks hydrogen bonds	**1**	
	(b)	**(i)**	Either of structures shown circled 	**1**	
		(ii)	50·5 ± 1 °C	**1**	
	(c)	**(i)**	Hydrolysis	**1**	
		(ii) (A)	5	**1**	

Question			Answer	Max Mark	Additional Guidance
		(ii) (B)	Correctly drawn amino acid structure 	**1**	Ignore bond positioning in side chains

Question			Answer		Max Mark	Additional Guidance
7.	**(a)**		118/32 or 3·69 mol CH_3OH **(1)** 3·69 × 24 = 88·5 litres **(1)**		**2**	Units not required but if given need to be appropriate to the calculation. eg 88 500 cm^3 Follow through applies. eg 1 mark can be awarded if use 64 for GFM giving an answer of 44·25 litres **or** for an arithmetically correct answer derived by multiplying a wrongly calculated number of moles (for which working shown) by 24
	(b)	**(i)**	(A)	Thermometer touching bottom or directly above flame or temperature rise recorded would be greater than expected.	**1**	
			(B)	Distance between flame and beaker **or** Height of wick in burner Same type of beaker (this needs to be qualified) Same draught proofing	**1**	
			(C)	2 concept marks + 1 arithmetic mark Concept marks Demonstration of the correct use of the relationship $E_h = cm\Delta T$ **(1)** eg 4·18 × 0·1 × 23 **or** 9·61 **and** Knowledge that enthalpy of combustion relates to 1 mol **(1)** evidenced by scaling up of energy released Correct arithmetic = −288 kJ mol^{-1} **(1)** Allow follow through of wrong GFM from part (a)	**3**	Maximum of 2 marks can be awarded if negative enthalpy sign is not shown in final answer. If candidate converts 1·07g to 0·033 mol, then candidate answer should be −291 kJ mol^{-1} Units not required

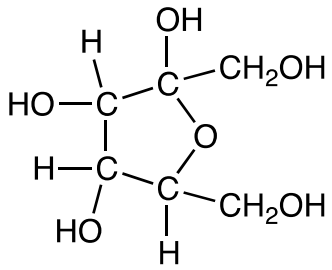
Question			Answer	Max Mark	Additional Guidance
		(ii)	0·799 (0·8)	**1**	Units not required
	(c)	**(i)**	If reactions are exothermic heat will need to be removed / If reactions are endothermic heat will need to be supplied **or** Chemists can create conditions to maximise yield	1	This is not a mark about safety
		(ii)	Answer = +191 kJ mol^{-1} **(2)** Partial mark 1 mark Evidence of the use of **al**l the correct bond enthalpies**.** **(1)** (412, 360, 463, 436, 743) **or** Correct use of incorrect bond enthalpy values	**2**	Positive sign does not need to be given in answer
8.	**(a)**		Calcium carbonate / carbon dioxide / ammonia / calcium oxide **all** correctly identified in flow diagram **(1)** Ammonium chloride / sodium hydrogen carbonate / sodium carbonate / water – all correctly identified in flow diagram **(1)**	**2**	
	(b)		(Adding brine) increases sodium ion concentration hence equilibrium shifts to right **(1)** Rate of forward reaction is increased (by addition of brine) **(1)**	**2**	

Question			Answer	Max Mark	Additional Guidance
9.			This is an open ended question **1 mark:** The student has demonstrated, at an appropriate level, a limited understanding of the chemistry involved. The candidate has made some statement(s) at which is/are relevant to the situation, showing that at least a little of the chemistry within the problem is understood. **2 marks:** The student has demonstrated, at an appropriate level, a reasonable understanding of the chemistry involved. The student makes some statement(s) which is/are relevant to the situation, showing that the problem is understood. **3 marks:** The maximum available mark would be awarded to a student who has demonstrated, at an appropriate level, a good understanding of the chemistry involved. The student shows a good comprehension of the chemistry of the situation and has provided a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. This does not mean the answer has to be what might be termed an 'excellent' answer or a 'complete' one.	**3**	Zero marks should be awarded if: The student has demonstrated, at an appropriate level, no understanding, of the chemistry involved. There is no evidence that the student has recognised the area of chemistry involved or has given any statement of a relevant chemistry principle. This mark would also be given when the student merely restates the chemistry given in the question.

Question			Answer	Max Mark	Additional Guidance
10.	**(a)**	**(i)**	24 hours allows time for all of the zinc to react **(1)** No stopper allows hydrogen gas to escape from the flask. **(1)**	**2**	
		(ii)	Zinc ions / impurities / metal ions / salts may be present in tap water	**1**	
	(b)	**(i)**	pipette	**1**	Do not accept measuring cylinder or syringe or burette
		(ii)	10 (Units not required, if given mg per litre, mg l^{-1})	**1**	Accept 0·01 g l^{-1}
	(c)		Answer in range 4·6 – 4·8 (mg per litre, mg l^{-1})	**1**	A-mark Use candidate best fit line, if provided, to check answer
11.	**(a)**		Carboxyl / carboxylic (acid) group	**1**	
	(b)		Esterification / condensation	**1**	
	(c)			**1**	
	(d)		As molecular size (no. of carbon atoms) increases, the absorption decreases.	**1**	

Question			Answer	Max Mark	Additional Guidance
12.	**(a)**	**(i)**	3 points **1 mark** for rinsing the burette – rinse the burette with the **thiosulfate / required solution** / with the solution to be put in it. **2 marks (1 mark each)** for any 2 of the following points fill burette above the scale with thiosulfate solution filter funnel used should be removed tap opened / some solution drained to ensure no air bubbles (thiosulfate) solution run into scale reading should be made from bottom of meniscus	**3**	
		(ii)	$2I^-(aq) \longrightarrow I_2(aq) + 2e^-$	**1**	Ignore state symbols

Question			Answer	Max Mark	Additional Guidance
		(iii)	0·000062 (mol l^{-1}) Partial marks can be awarded using a scheme of two "concept" marks, and one "arithmetic" mark **1 mark** for knowledge of the relationship between moles, concentration and volume. This could be shown by one of the following steps: Calculation of moles thiosulfate solution eg 0·001 × 0·0124 = 0·0000124 **or** calculation of concentration of iodine solution eg 0·0000062/0·1 **or** Insertion of correct pairings of values for concentration and volume in a valid titration formula **1 mark** for knowledge of relationship between moles of thiosulfate and hypochlorite. This could be shown by one of the following steps: Calculation of moles hypochlorite from moles thiosulfate – eg 0·0000124/2 = 0·0000062 **or** Insertion of correct stoichiometric values in a valid titration formula **1 mark** is awarded for correct arithmetic through the calculation. This mark can only be awarded if both concept marks have been awarded.	**3**	units not required

Question			Answer	Max Mark	Additional Guidance
	(b)		**1 mark** correct arithmetic either 44·4 (litres) or 44400 (cm^3) **1 mark** correct units	**2**	
	(c)	**(i)**	**1 mark** Ammonia is polar and trichloramine is non-polar. **1 mark** Explanation of this in terms of polarities of bonds or electronegativity differences of atoms in bonds	**2**	
		(ii)	Substances that have unpaired electrons	**1**	
		(iii)	Propagation	**1**	
13.	**(a)**		Aldehyde group correctly identified	**1**	
	(b)		Ring form correctly drawn 	**1**	

[END OF MARKING INSTRUCTIONS]

General Marking Principles for Higher Chemistry

This information is provided to help you understand the general principles you must apply when marking candidate responses to questions in this paper. These principles must be read in conjunction with the detailed marking instructions, which identify the key features required in candidate responses.

(a) Marks for each candidate response must always be assigned in line with these General Marking Principles and the Detailed Marking Instructions for this assessment.

(b) Marking should always be positive. This means that, for each candidate response, marks are accumulated for the demonstration of relevant skills, knowledge and understanding: they are not deducted from a maximum on the basis of errors or omissions.

(c) If a specific candidate response does not seem to be covered by either the principles or detailed Marking Instructions, and you are uncertain how to assess it, you must seek guidance from your Team Leader.

(d) There are no half marks awarded.

(e) Where a candidate makes an error at an early stage in a multi stage calculation, credit should normally be given for correct follow on working in subsequent stages, unless the error significantly reduces the complexity of the remaining stages. The same principle should be applied in questions which require several stages of non-mathematical reasoning. The exception to this rule is where the marking instructions for a numerical question assign separate "concept marks" and an "arithmetic mark". In such situations, the marking instructions will give clear guidance on the assignment or partial marks.

(f) Unless a numerical question specifically requires evidence of working to be shown, full marks should be awarded for a correct final answer (including units) on its own.

(g) Larger mark allocations may be fully accessed whether responses are provided in continuous prose, linked statements or a series of developed bullet points.

(h) Marks should not be deducted for inaccurate or unconventional spelling or vocabulary as long as the meaning of the word(s) is conveyed. **For example**, responses that include 'distilling' for 'distillation', or 'it gets hotter' for 'the temperature rises', should be accepted.

(i) If a correct answer is followed by a wrong answer, it should be treated as a cancelling error and no marks should be given. **For example**, in response to the question, 'State the colour seen when blue Fehling's solution is warmed with an aldehyde', the answer 'red green' gains no marks.
However, if a correct answer is followed by additional information which does not conflict with that, the additional information should be ignored, whether correct or not. **For example**, in response to a question concerned with melting point, 'State why the tube should not be made of copper', the response 'Copper has a low melting point and is coloured grey' would **not** be treated as having a cancelling error. **I**f a candidate lists a number of possible answers it should not be for the marker to choose the correct answer from the list

(j) Full marks are usually awarded for the correct answer to a calculation without working and the partial marks shown in the detailed marking instructions are for use when working is given but the final answer is incorrect. An exception is when candidates are asked to 'Find, by calculation', when full marks cannot be awarded for the correct answer without working.

(k) Ignore the omission of one H atom from a full structural formula provided the bond is shown or of one bond if the hydrogen is shown.
When structures involving an –OH group or an $-NH_2$ are asked for, a mark should only be deducted when a bond is drawn to the wrong atom, if understanding of the functional group structure is required.

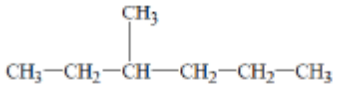
(l) A symbol or correct formula should be accepted in place of a name unless stated otherwise in the detailed marking instructions.

(m) When formulae of ionic compounds are given as answers it will only be necessary to show ion charges if these have been specifically asked for. However, if ion charges are shown, they must be correct. If incorrect charges are shown, no marks should be awarded.

(n) If an answer comes directly from the text of the question, no marks should be given. **For example**, in response to the question, ' A student found that 0·05 mol of propane, C_3H_8 burned to give 82·4 kJ of energy. $C_3H_8(g) + 5O_2(g) = 3CO_2(g) + 4H_2O(\ell)$. Name the kind of enthalpy change that the student measured', no marks should be given for "burning" since the word "burned" appears in the text.

(o) A guiding principle in marking is to give credit for correct elements of a response rather than to look for reasons not to give marks.

Example 1: The structure of a hydrocarbon found in petrol is shown below.



Name the hydrocarbon

- Although the punctuation is not correct '3, methyl-hexane' should gain the full mark.

Example 2: A student measured the pH of four carboxylic acids to find out how their strength is related to the number of chlorine atoms in the molecule. The results are shown.

Structural formula	*pH*
CH_3COOH	1·65
$CH_2ClCOOH$	1·27
$CHCl_2COOH$	0·90
CCl_3COOH	0·51

Describe the relationship between the number of chlorine atoms in the molecule and the strengths of the acids.

- Although not completely correct, an answer such as 'the more Cl_2, the stronger the acid' should gain the full mark.

(p) Unless the question is clearly about a non-chemistry issue, eg costs in an industrial chemical process, a non-chemical answer gains no marks.
For example, in response to the question, 'Why does the (catalytic) converter have a honeycomb structure?', 'to make it work' may be correct but it is not a chemical answer and the mark should not be given.

(q) Marks are awarded only for a valid response to the question asked. For example, in response to questions that ask candidates to:

- **identify, name, give** or **state**, they need only name or present in brief form;
- **describe**, they must provide a statement or structure of characteristics and/or features;
- **explain**, they must relate cause and effect and/or make relationships between

things clear;

- **compare**, they must demonstrate knowledge and understanding of the similarities and/or differences between things;
- **complete**, they must finish a chemical equation or fill in a table with information
- **determine** or **calculate**, they must determine a number from given facts, figures or information;
- **draw**, they must draw a diagram or structural formula, eg "Draw a diagram to show the part of a poly(propene) molecule formed from two propene molecules"
- **estimate**, they must determine an approximate value for something;
- **predict**, they must suggest what may happen based on available information;
- **evaluate**, they must make a judgement based on criteria;
- **suggest**, they must apply their knowledge and understanding of [subject] to a new situation. A number of responses are acceptable; marks will be awarded for any suggestions that are supported by knowledge and understanding of [subject];
- **use your knowledge of [chemistry or aspect of chemistry] to comment on**, they must apply their skills, knowledge and understanding to respond appropriately to the problem/situation presented (for example by making a statement of principle(s) involved and/or a relationship or equation, and applying these to respond to the problem/situation). They will be rewarded for the breadth and/or depth of their conceptual understanding.
- **Write**, they must complete a chemical or word equation, eg "Write the word equation for the complete combustion of ethanol."

2015 New Higher Chemistry Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D
1	**D**	6	7	2	**84**
2	**C**	2	2	**89**	6
3	**D**	6	9	9	**74**
4	**A**	**94**	3	1	2
5	**C**	2	10	**80**	8
6	**B**	14	**63**	11	11
7	**A**	**29**	7	57	7
8	**B**	1	**50**	45	3
9	**B**	9	**79**	7	4
10	**A**	**58**	4	3	35
11	**D**	5	14	13	**68**
12	**D**	8	10	34	**48**
13	**B**	9	**83**	2	6
14	**C**	5	9	**74**	11
15	**B**	12	**69**	4	14
16	**A**	**58**	29	3	9
17	**D**	15	6	10	**67**
18	**A**	**89**	6	2	3
19	**A**	**67**	20	7	5
20	**C**	18	25	**38**	18